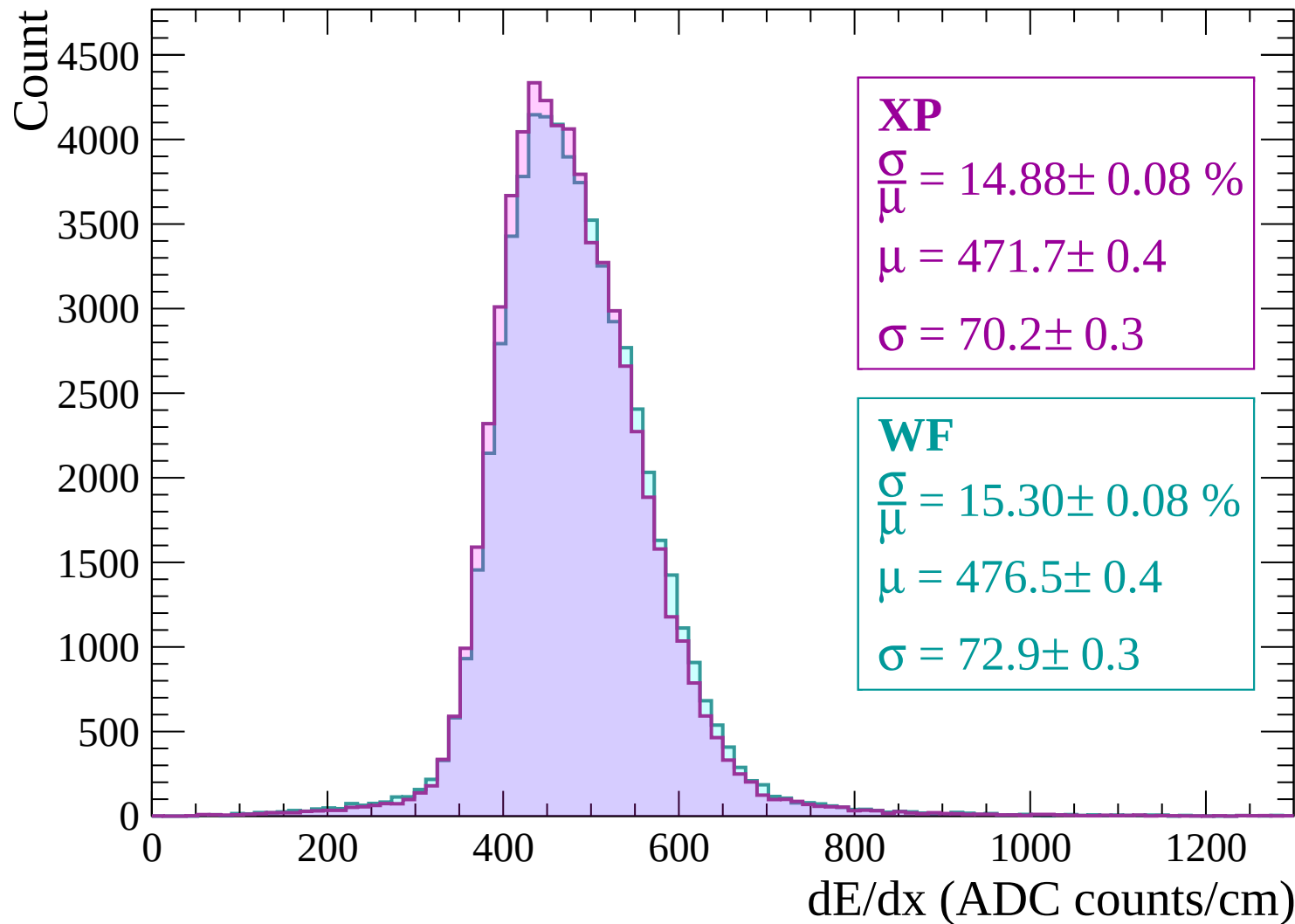
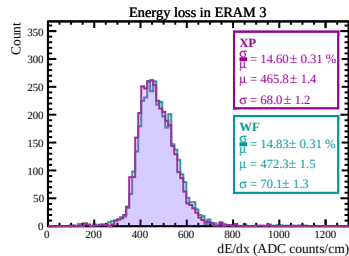
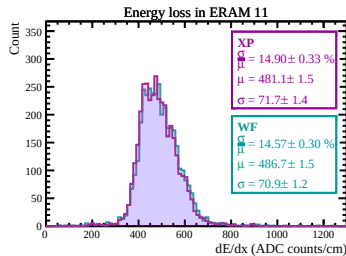
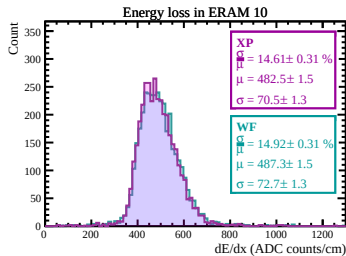
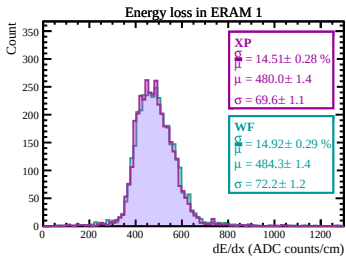
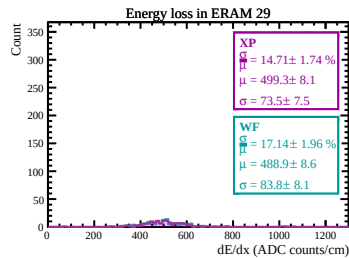
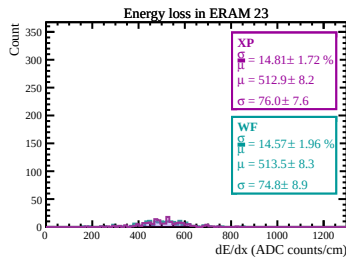
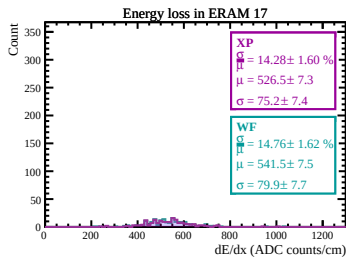
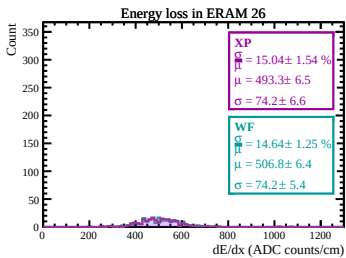
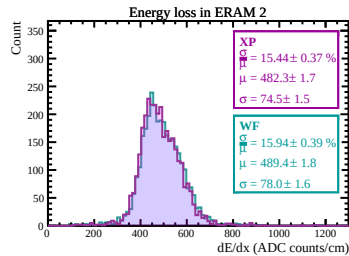
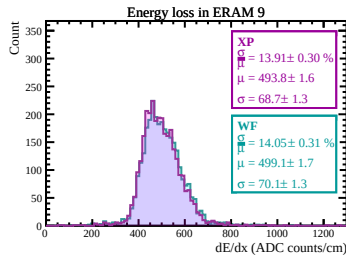
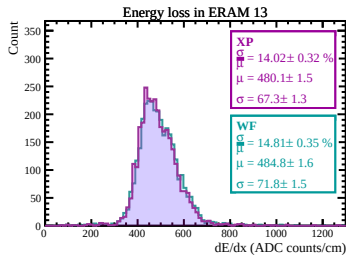
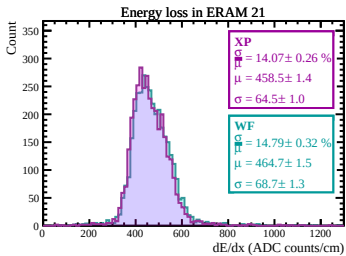
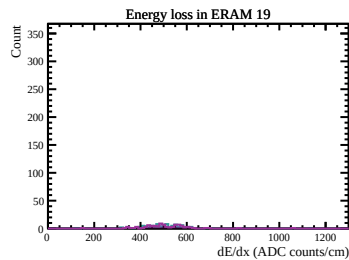
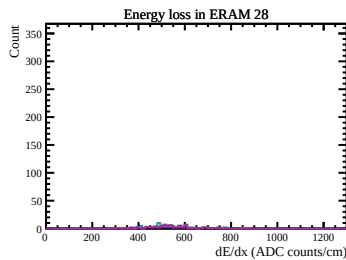
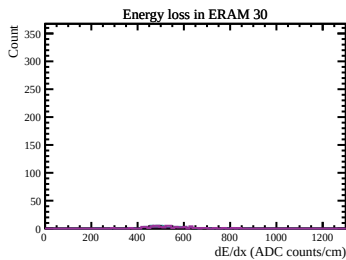
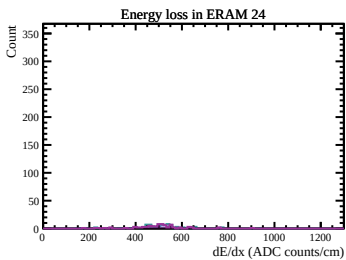
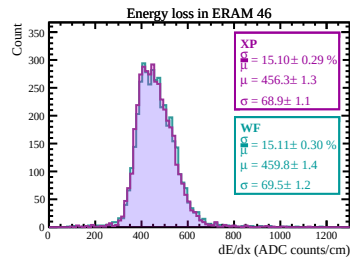
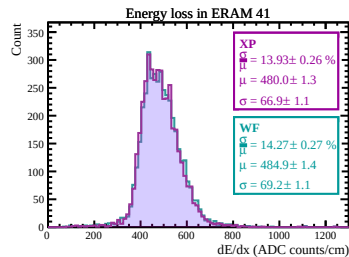
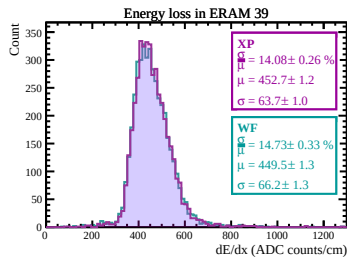
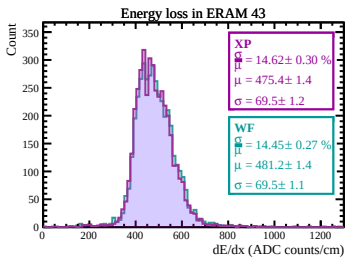
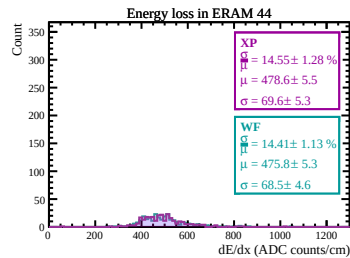
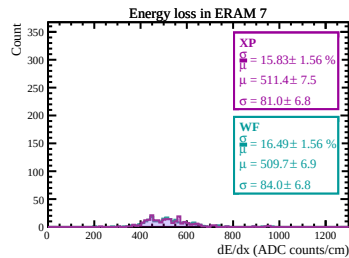
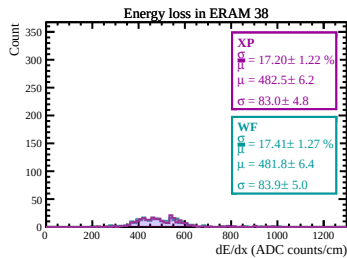
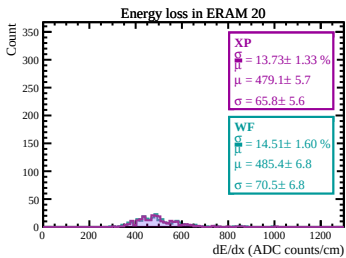
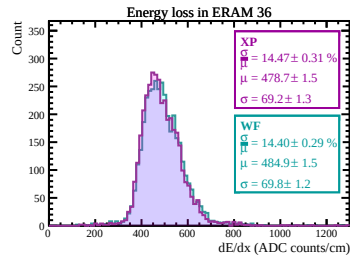
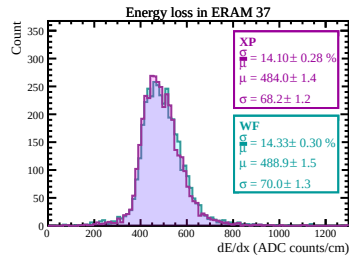
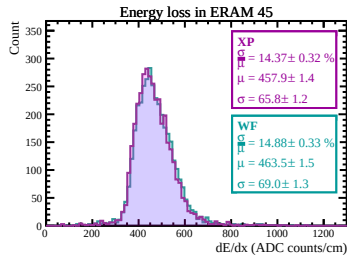
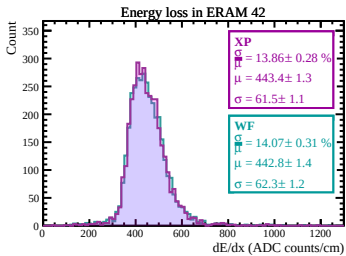
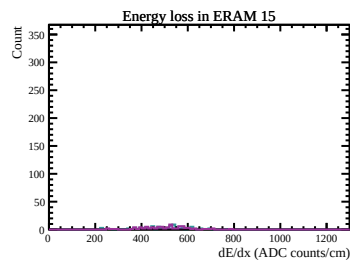
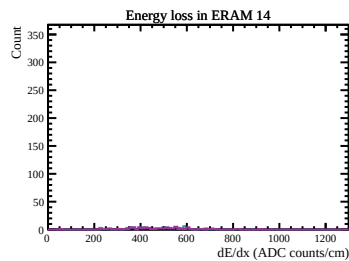
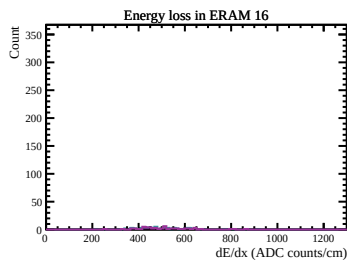
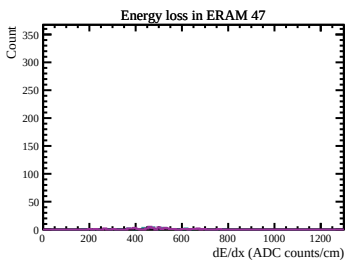
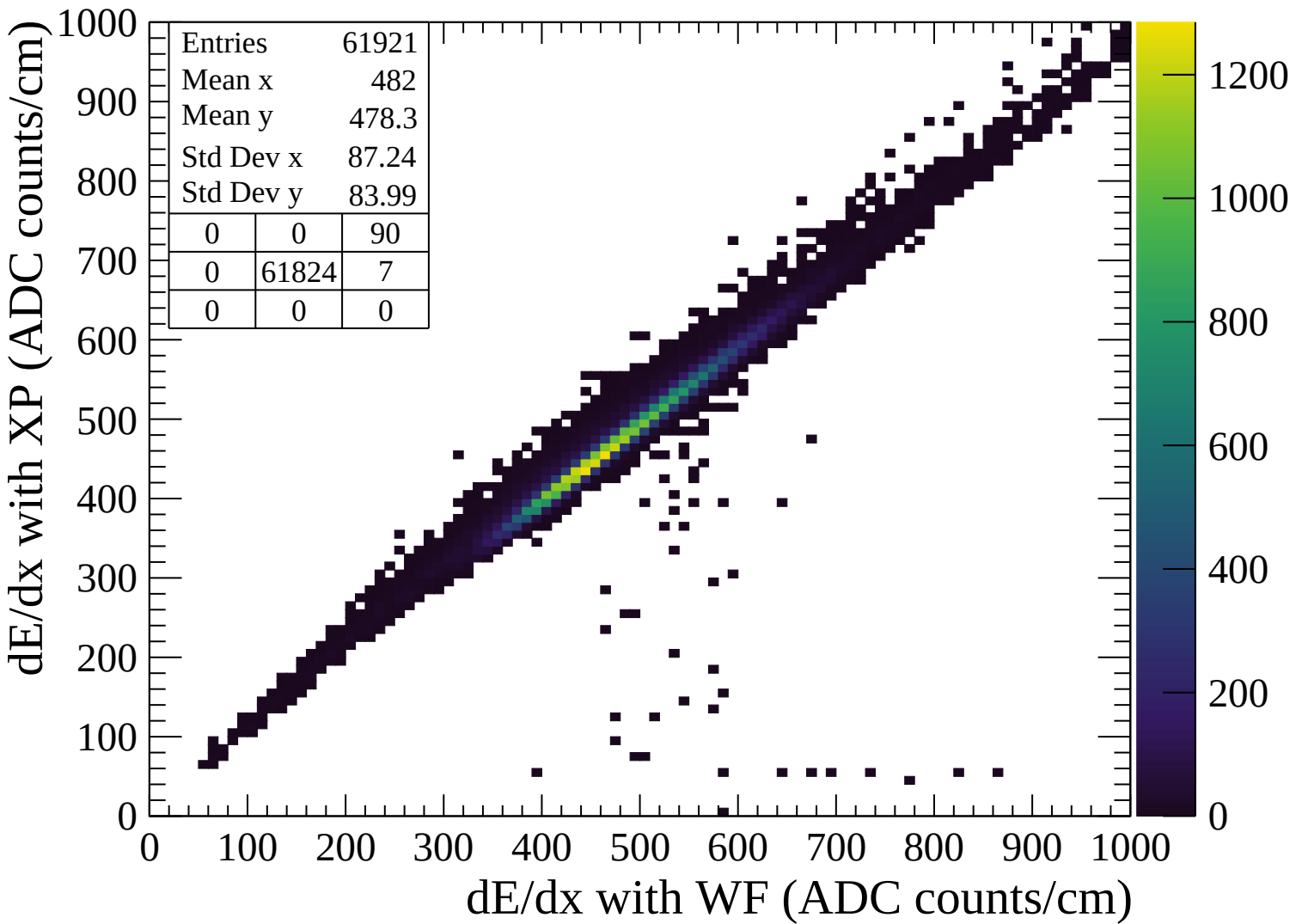
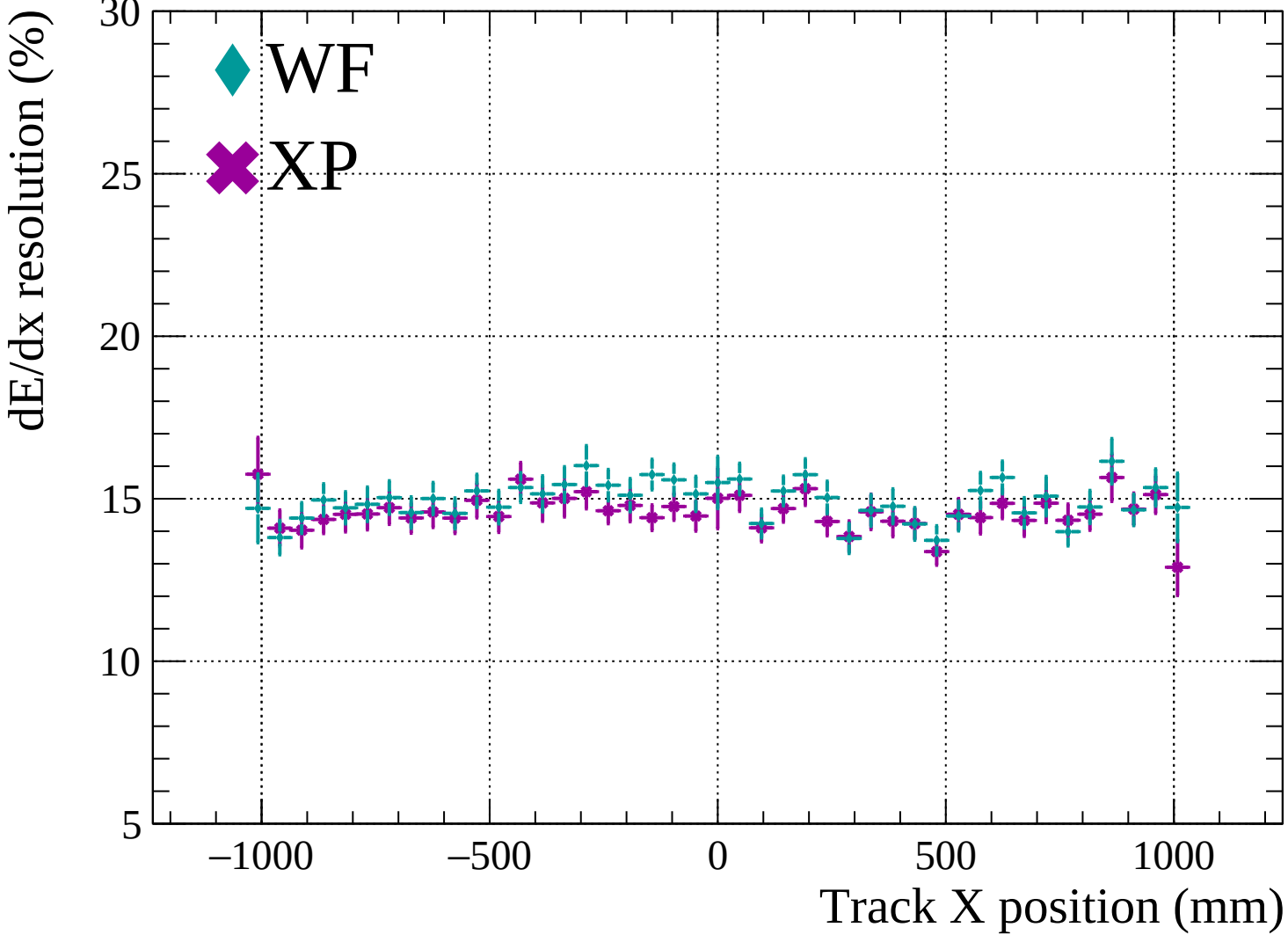


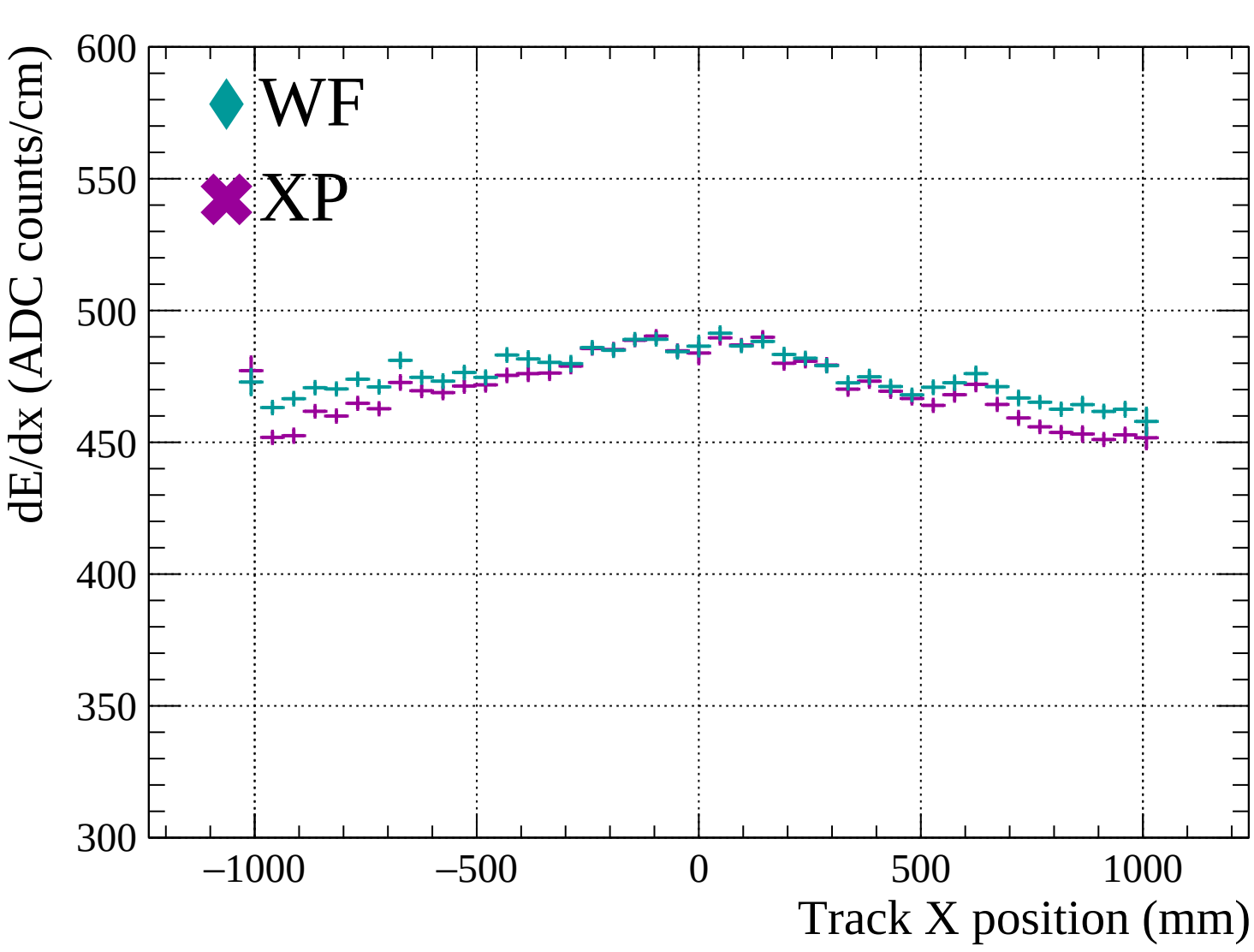


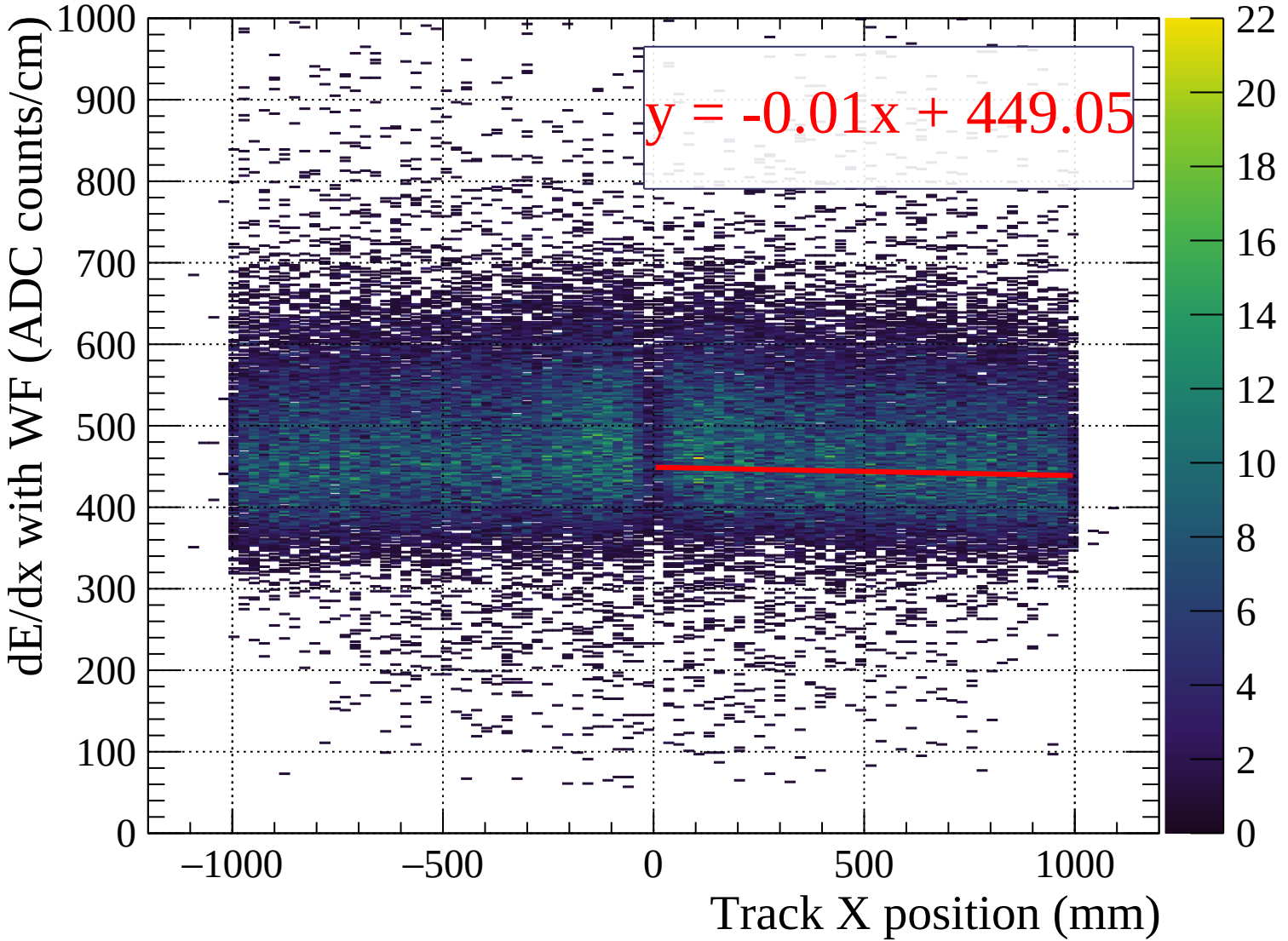


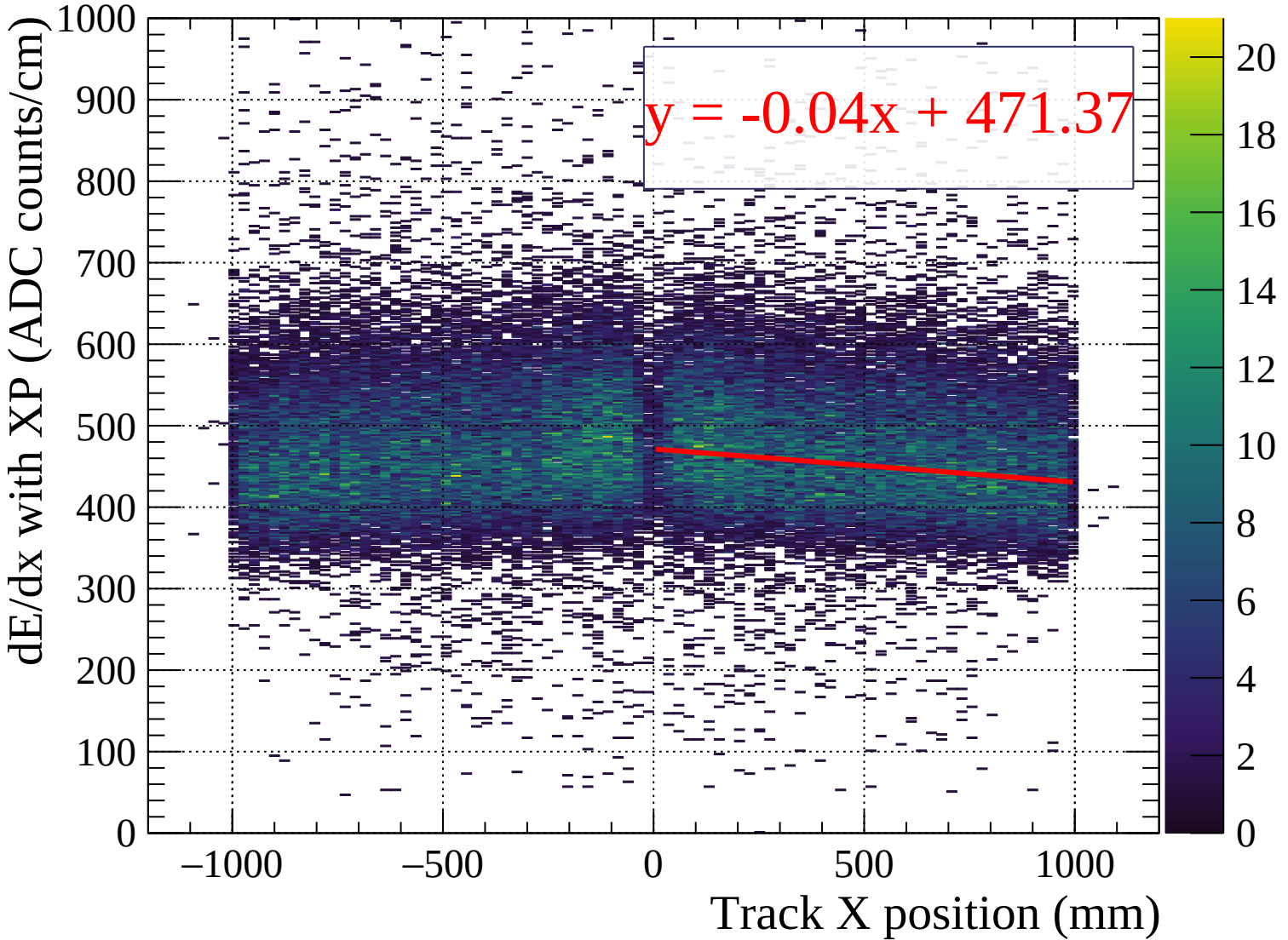


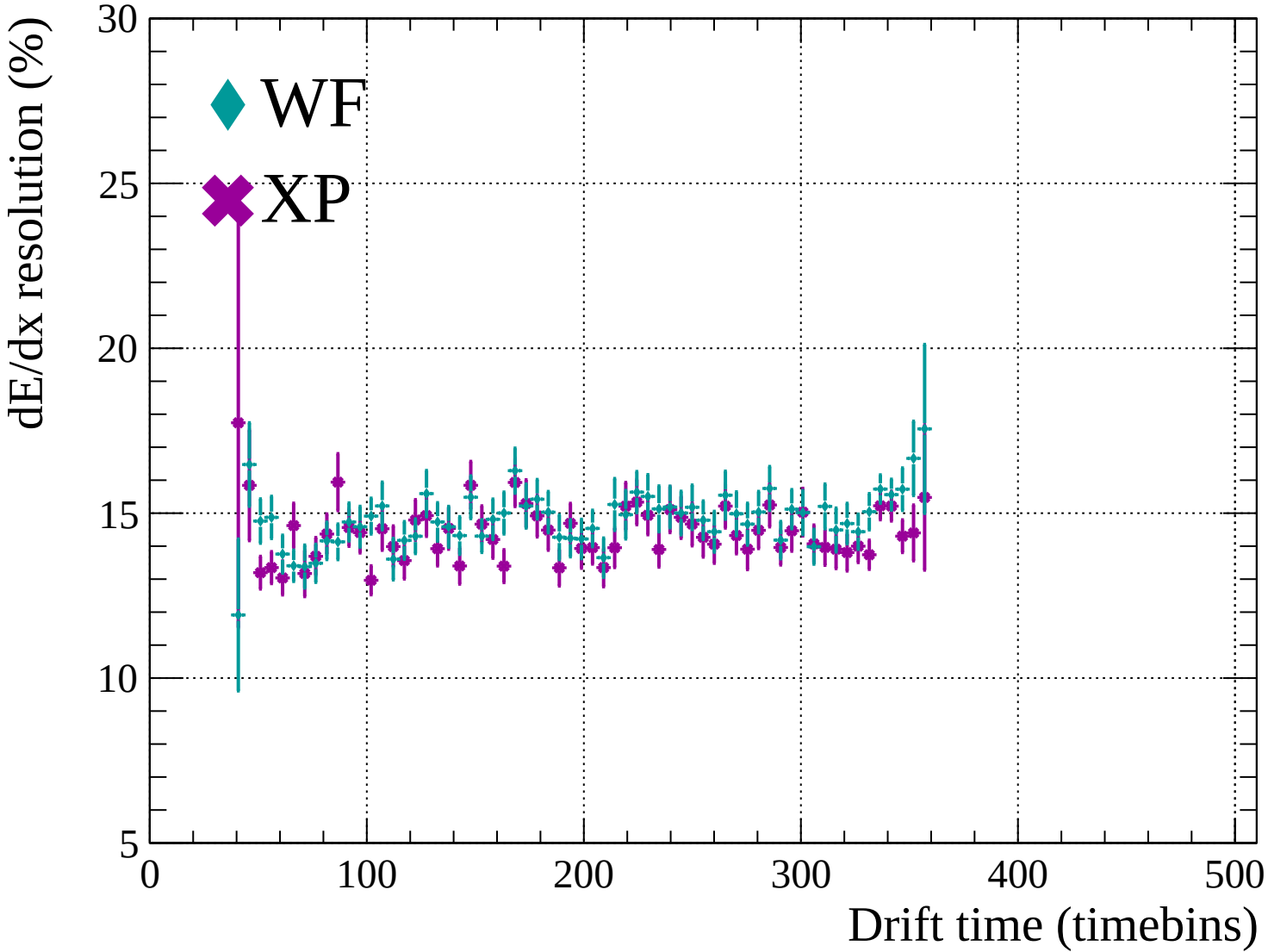


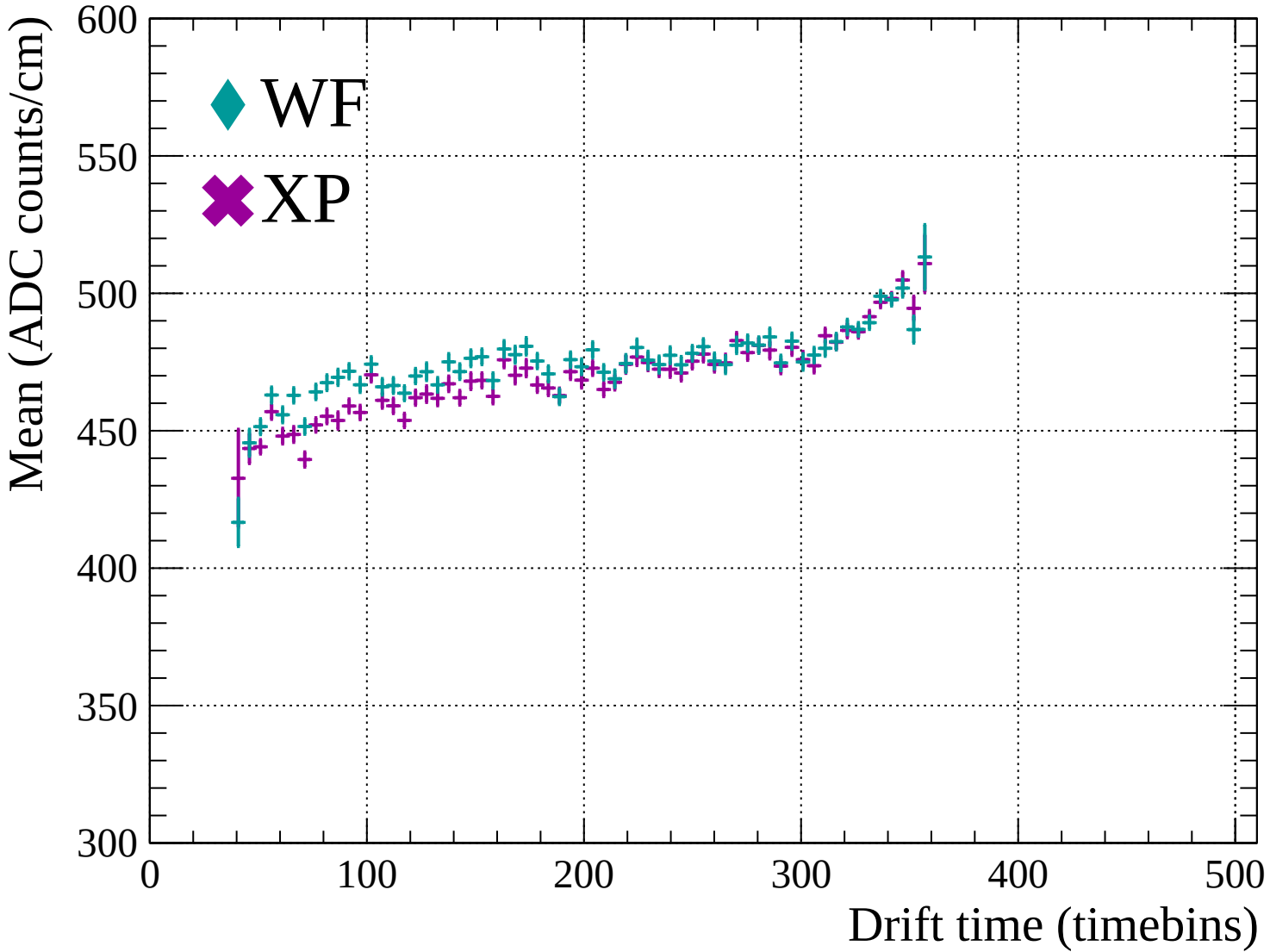


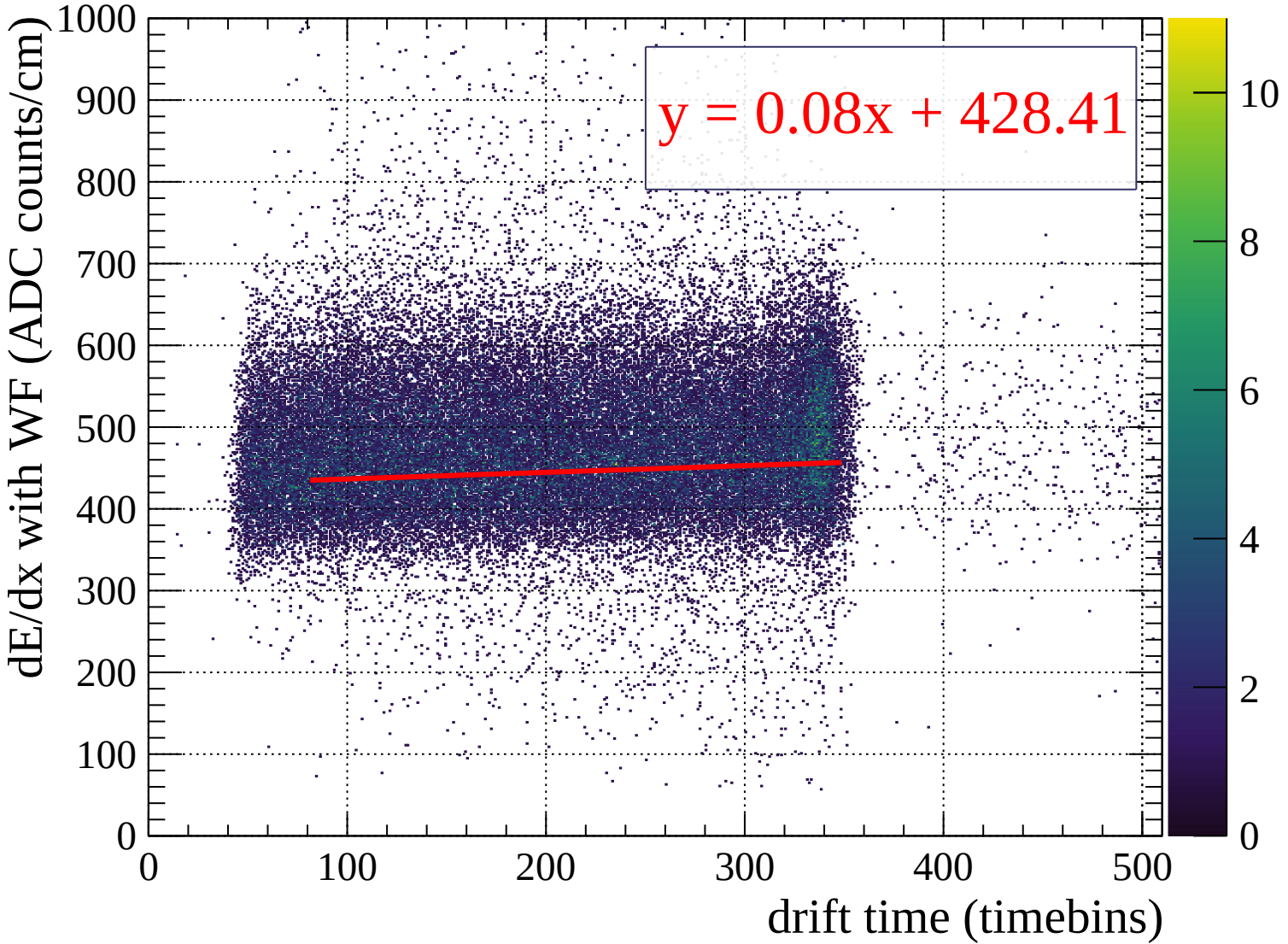


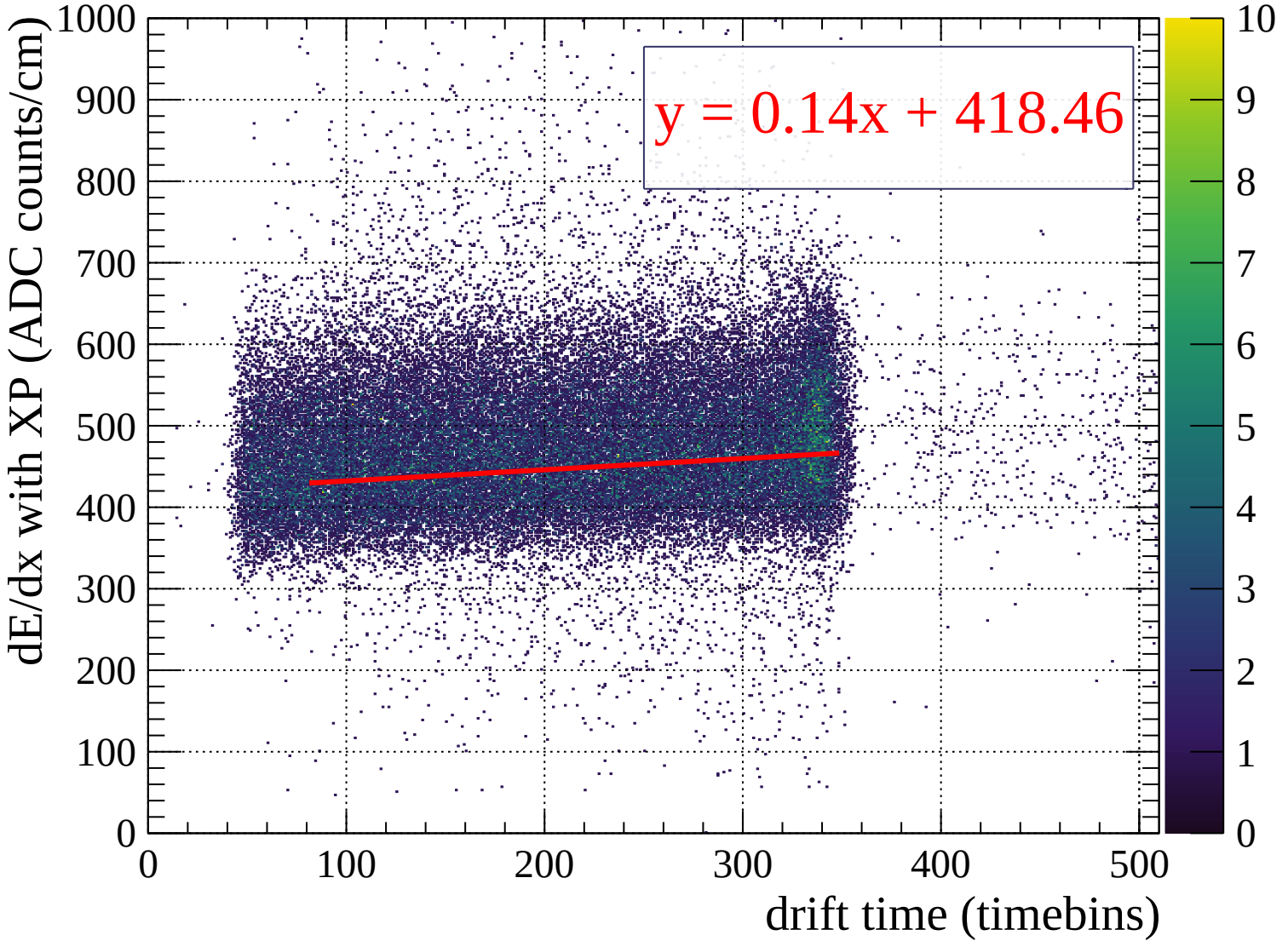


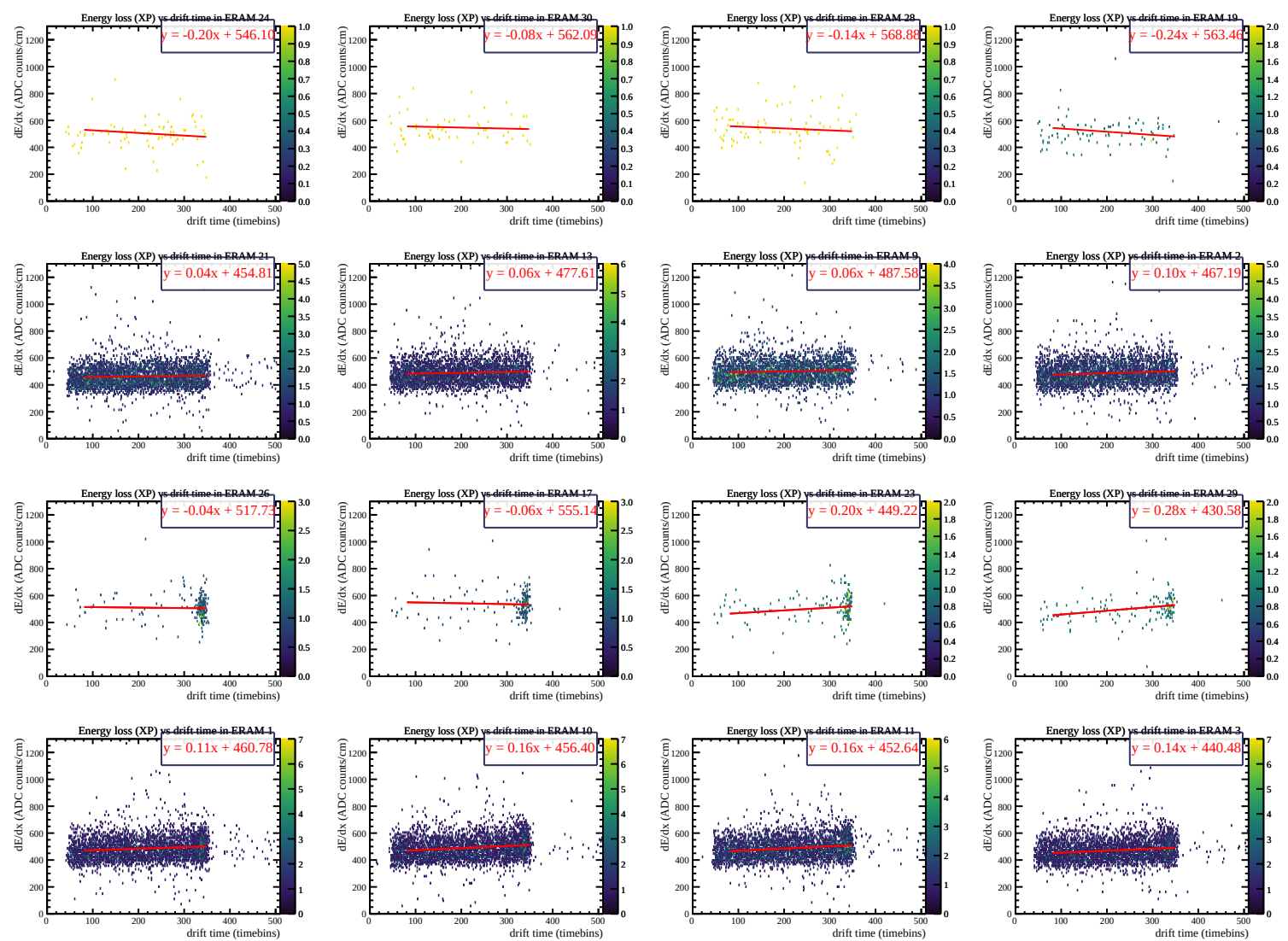


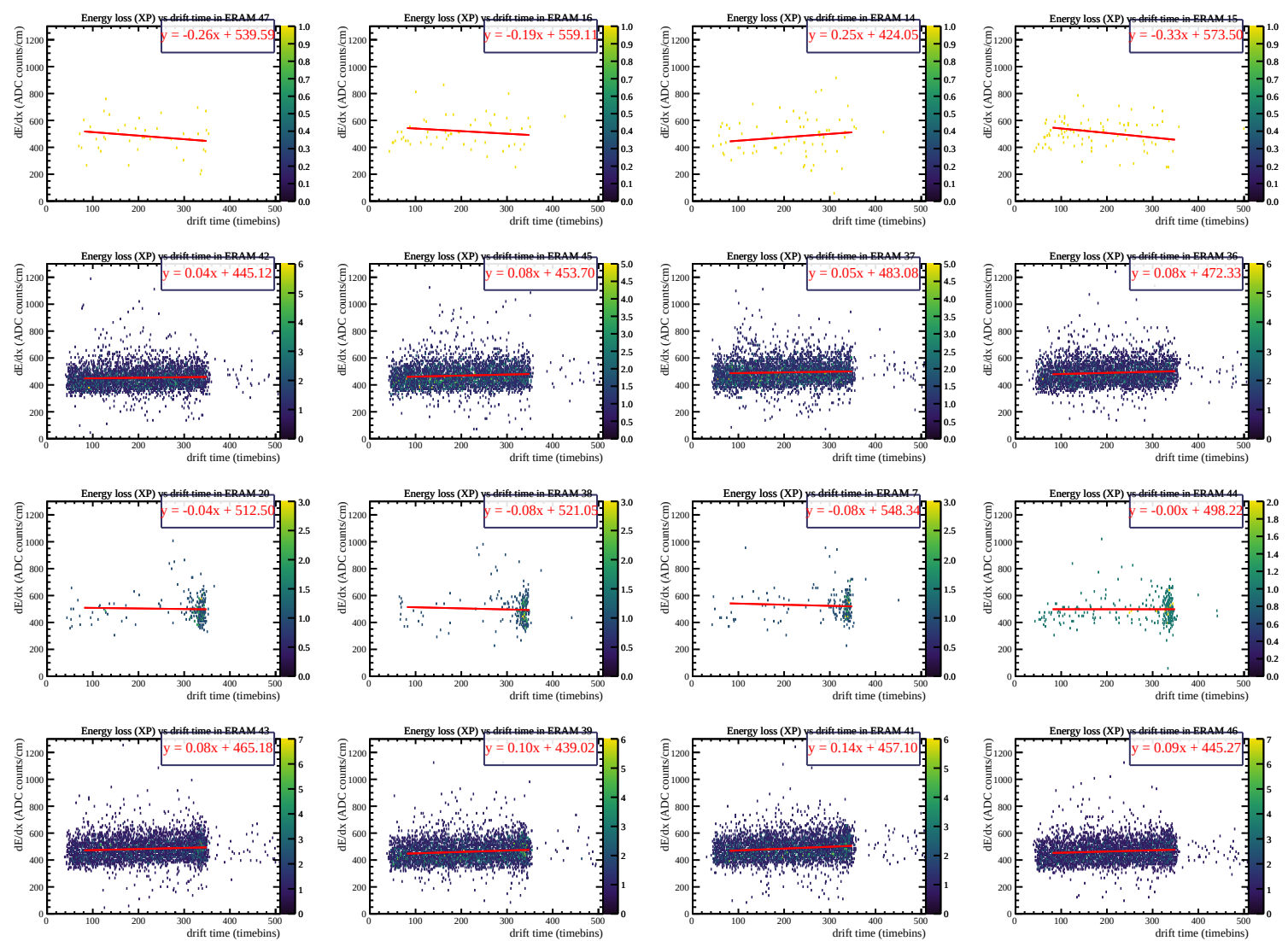


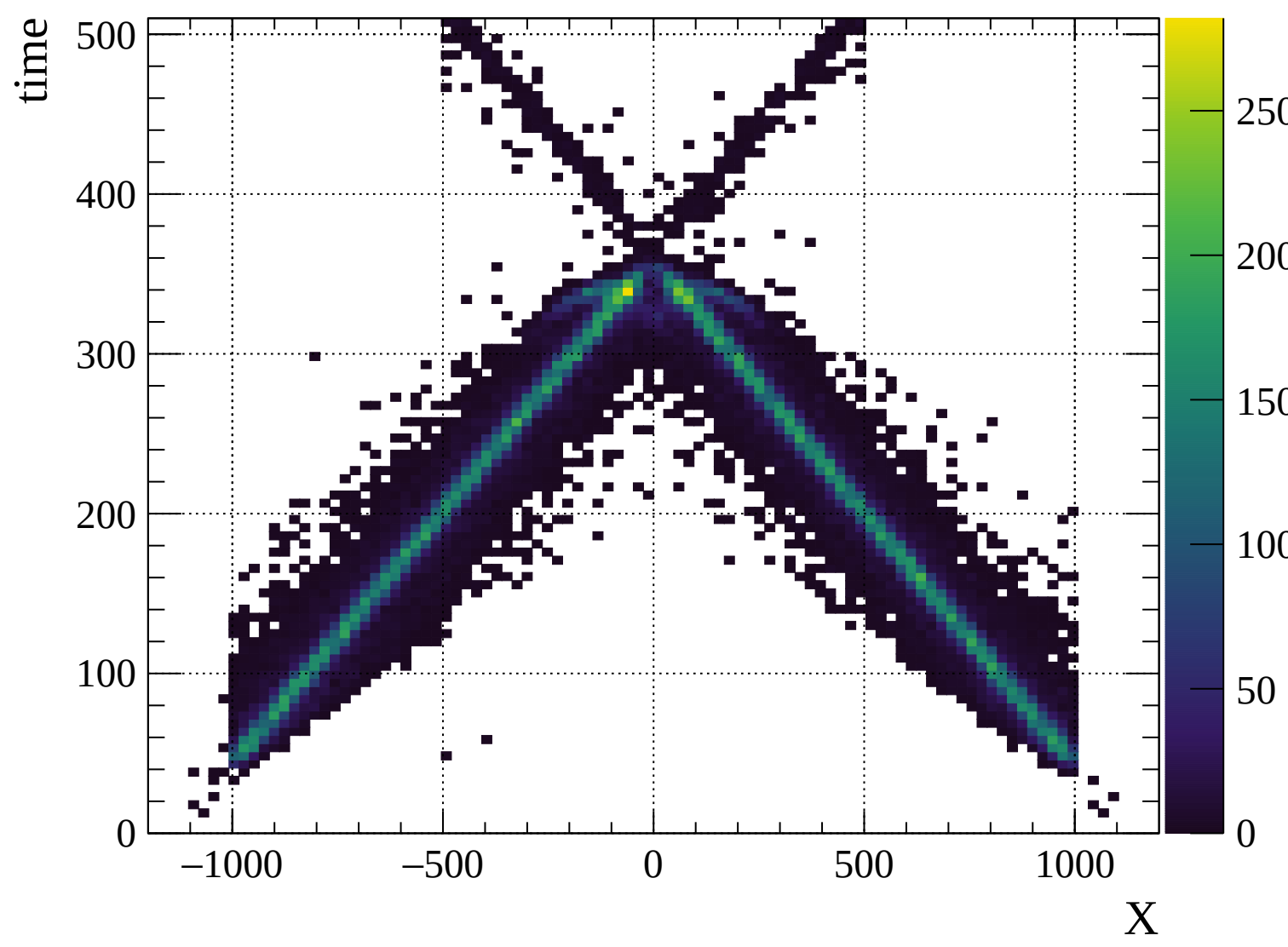


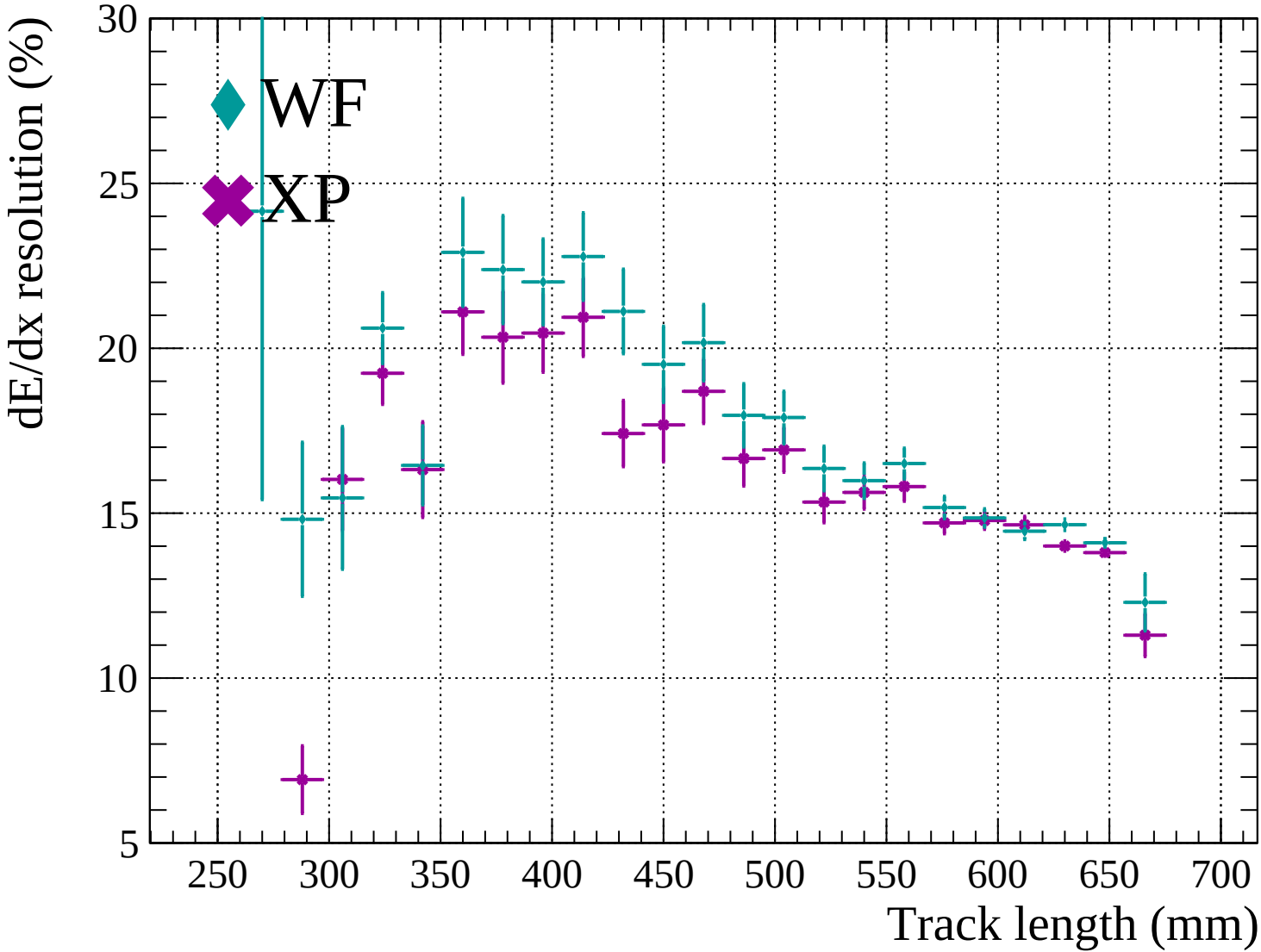


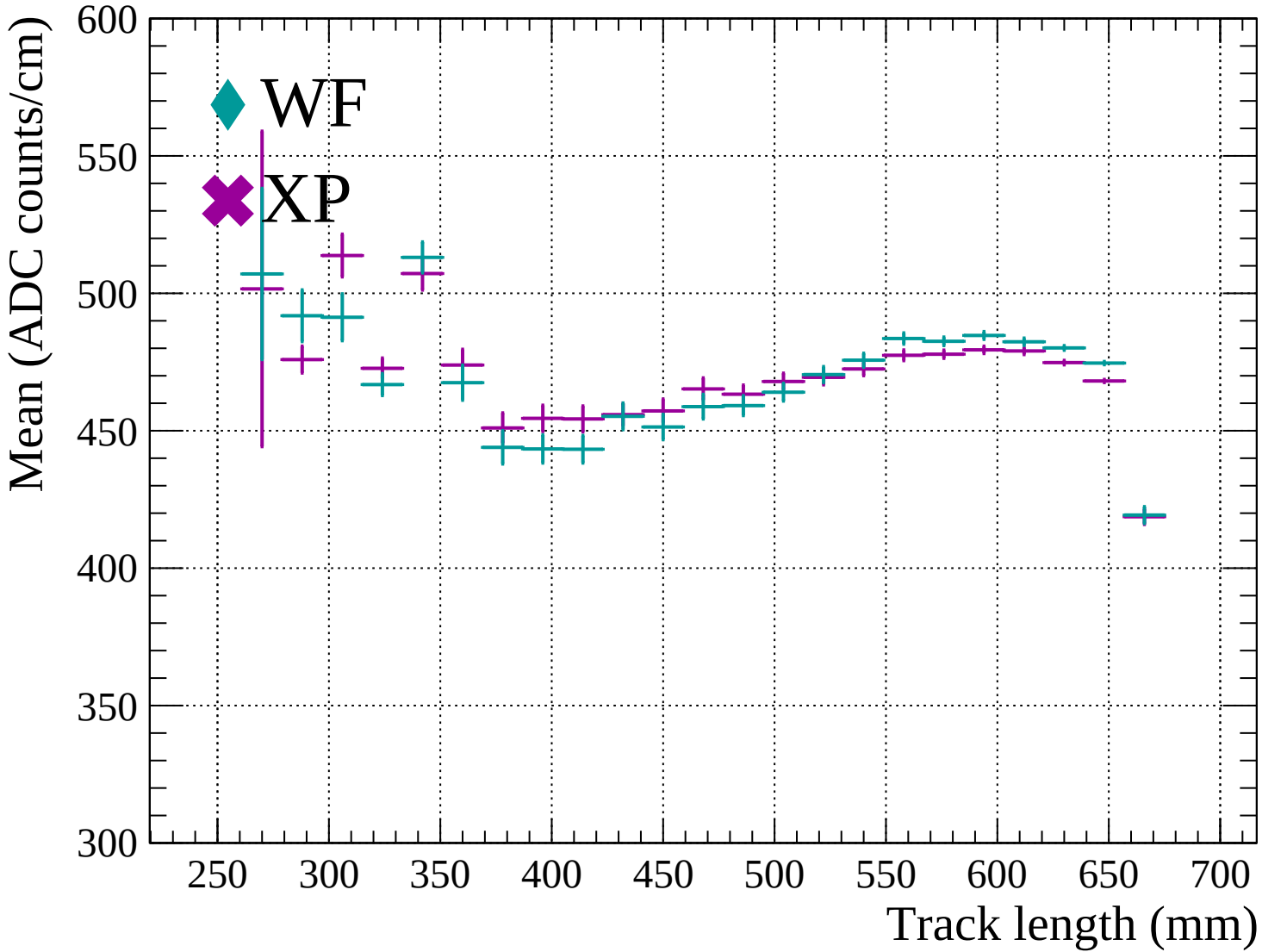


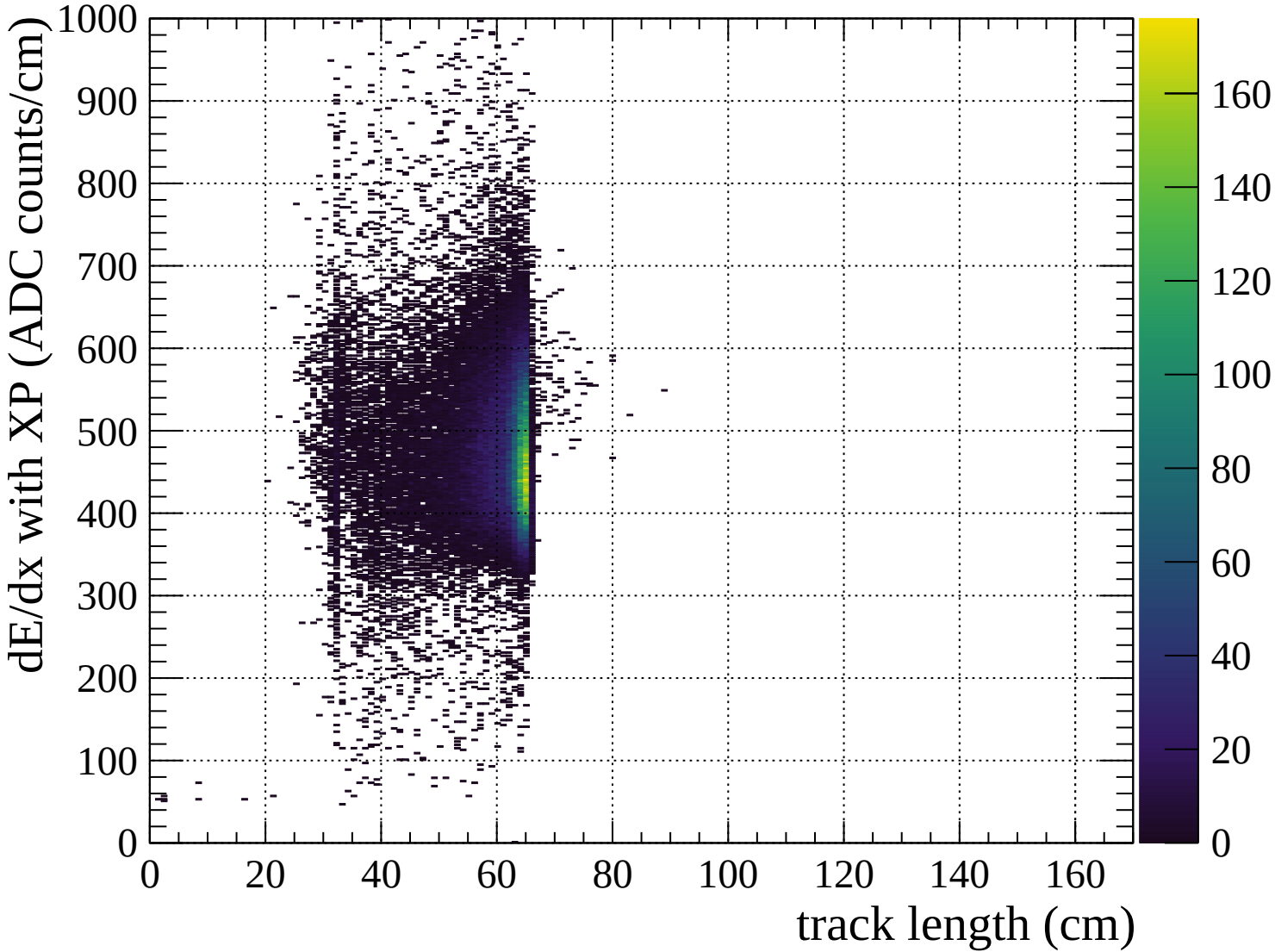


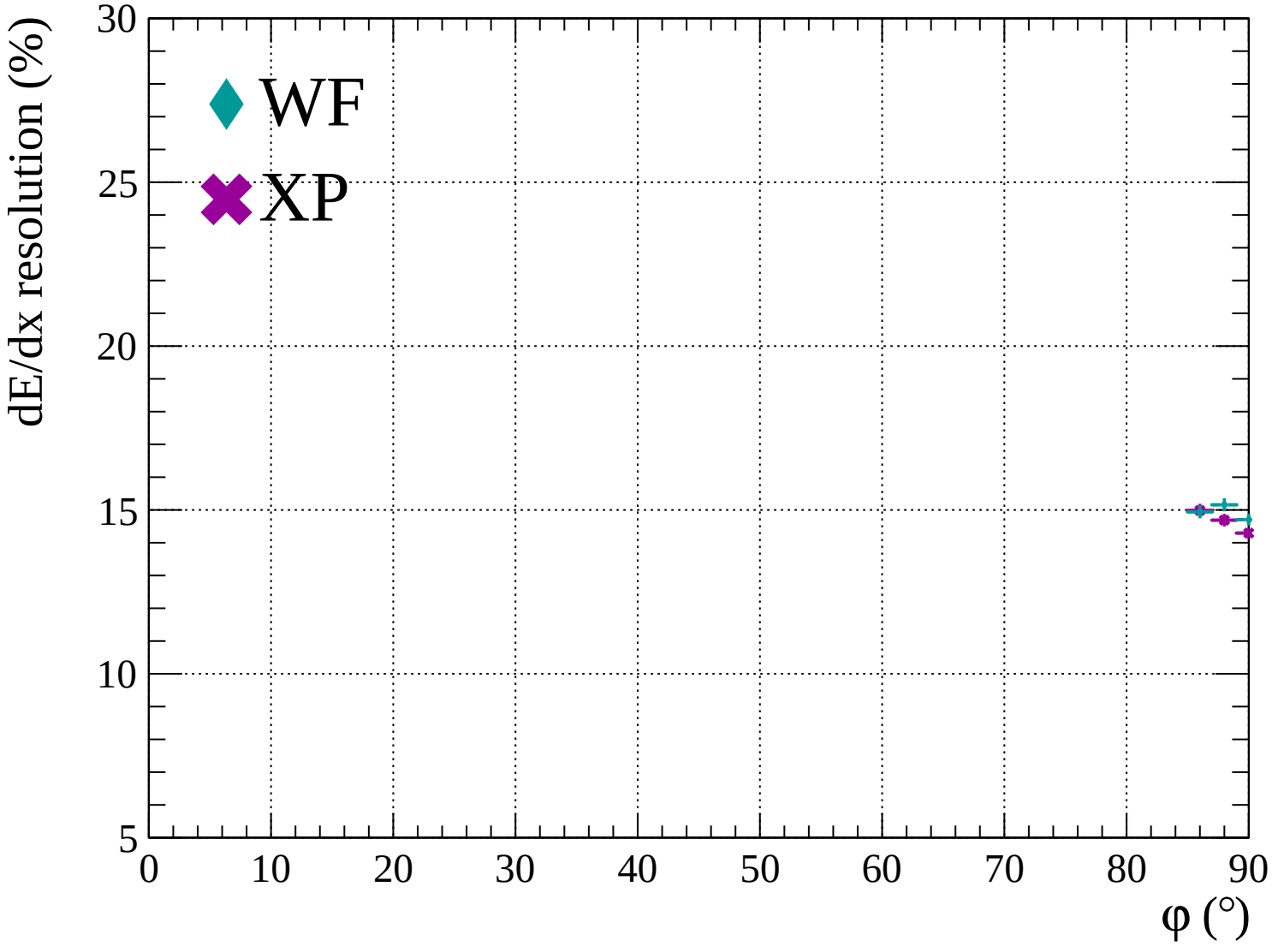


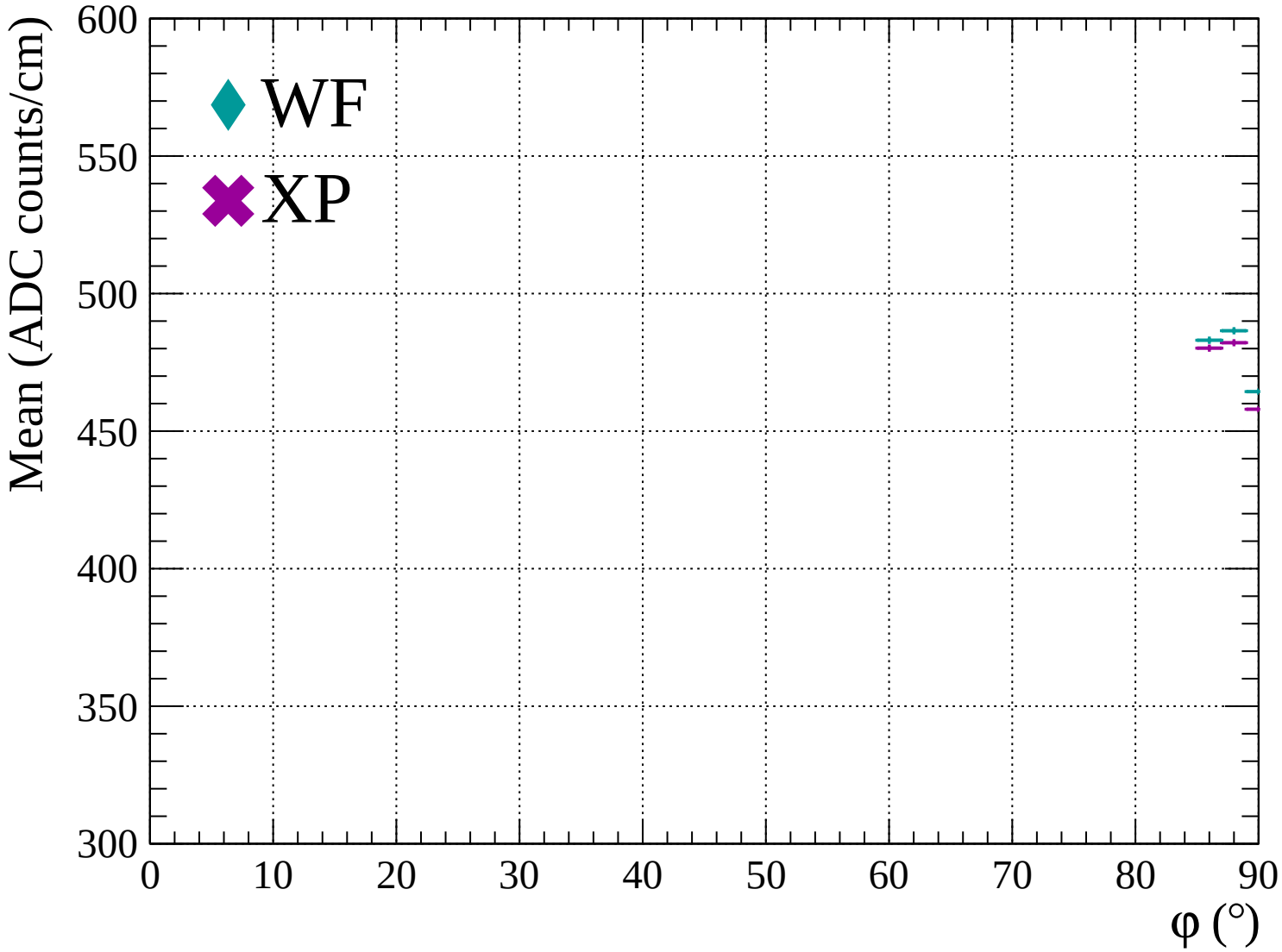


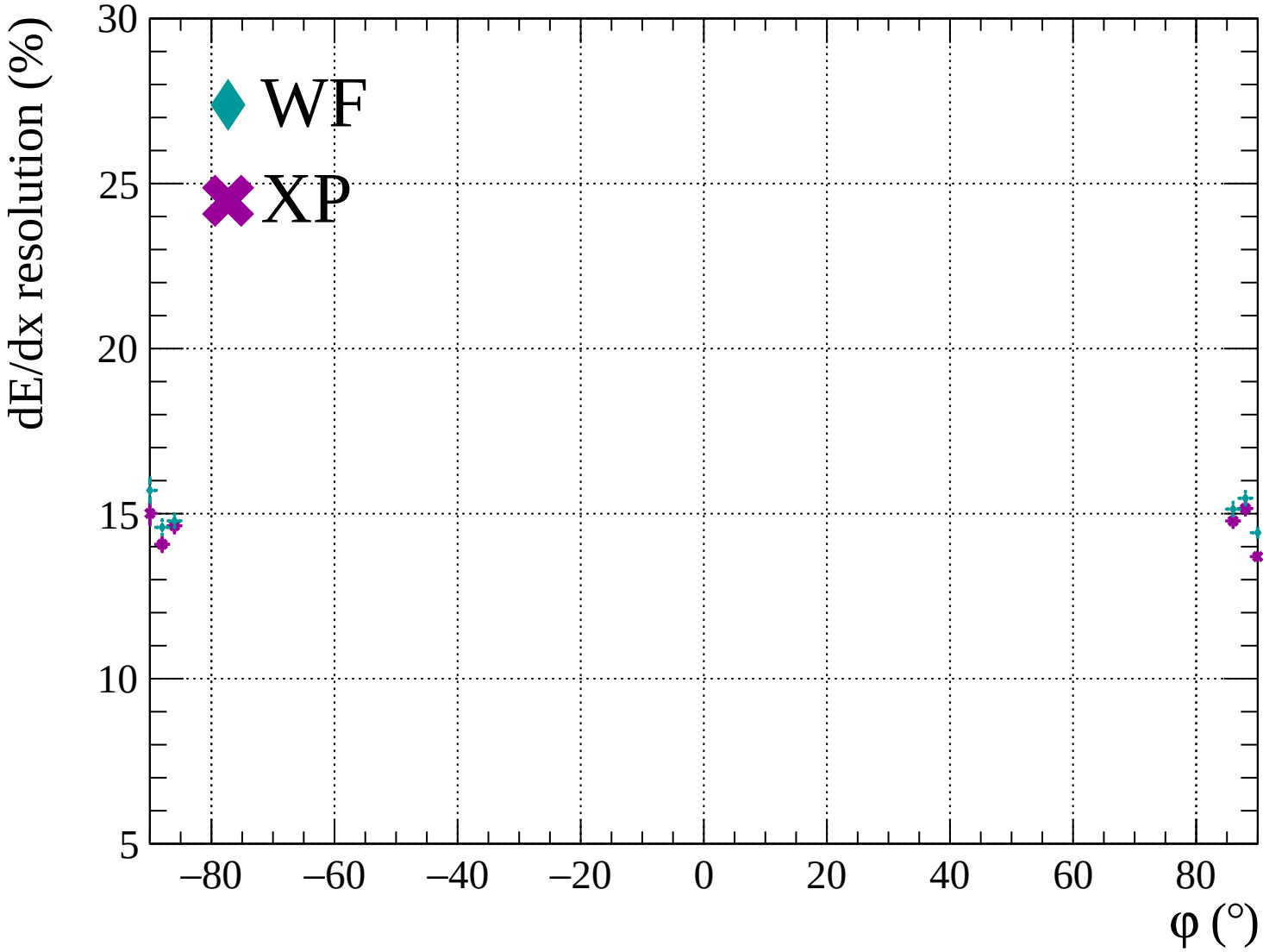


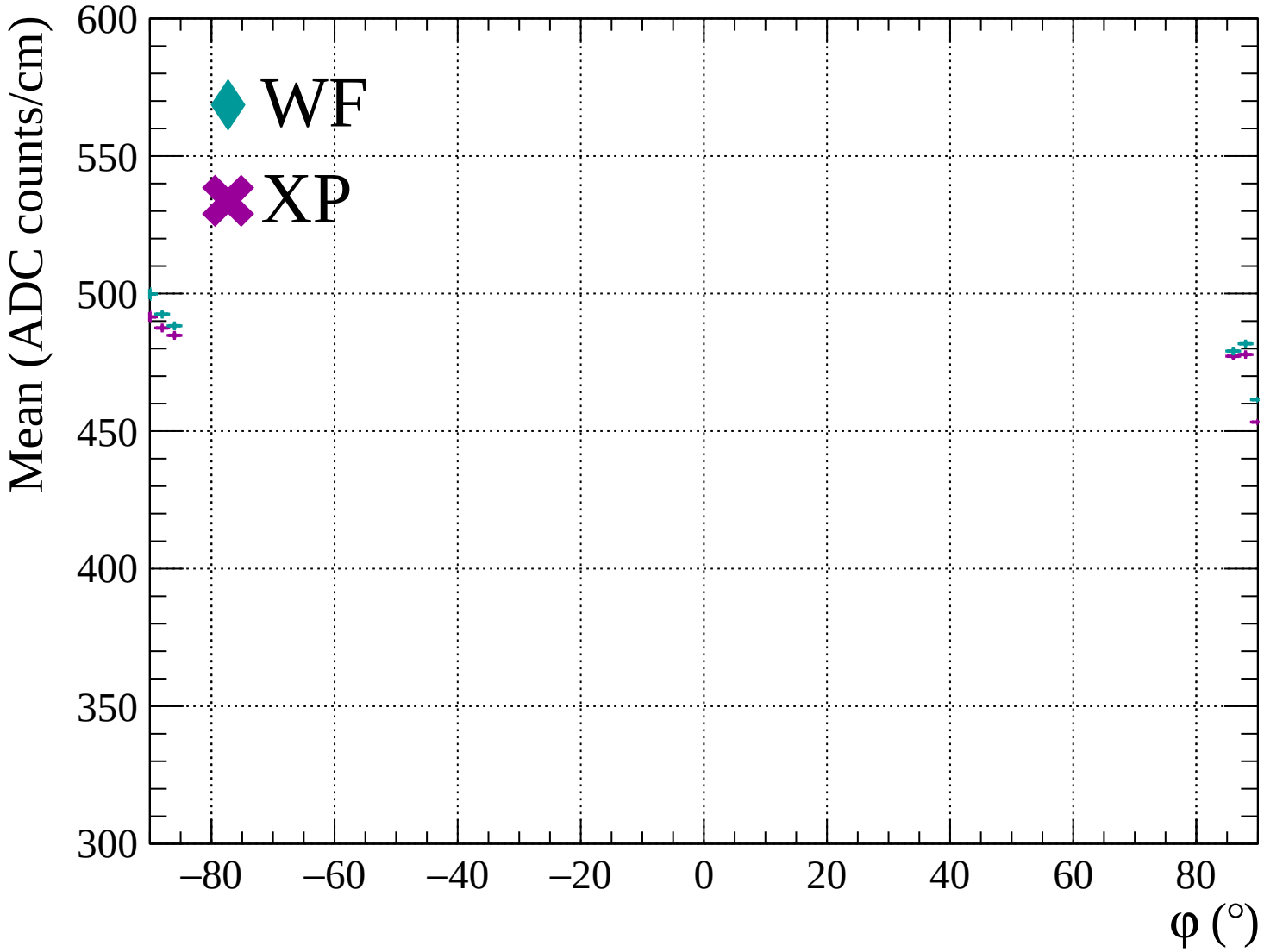


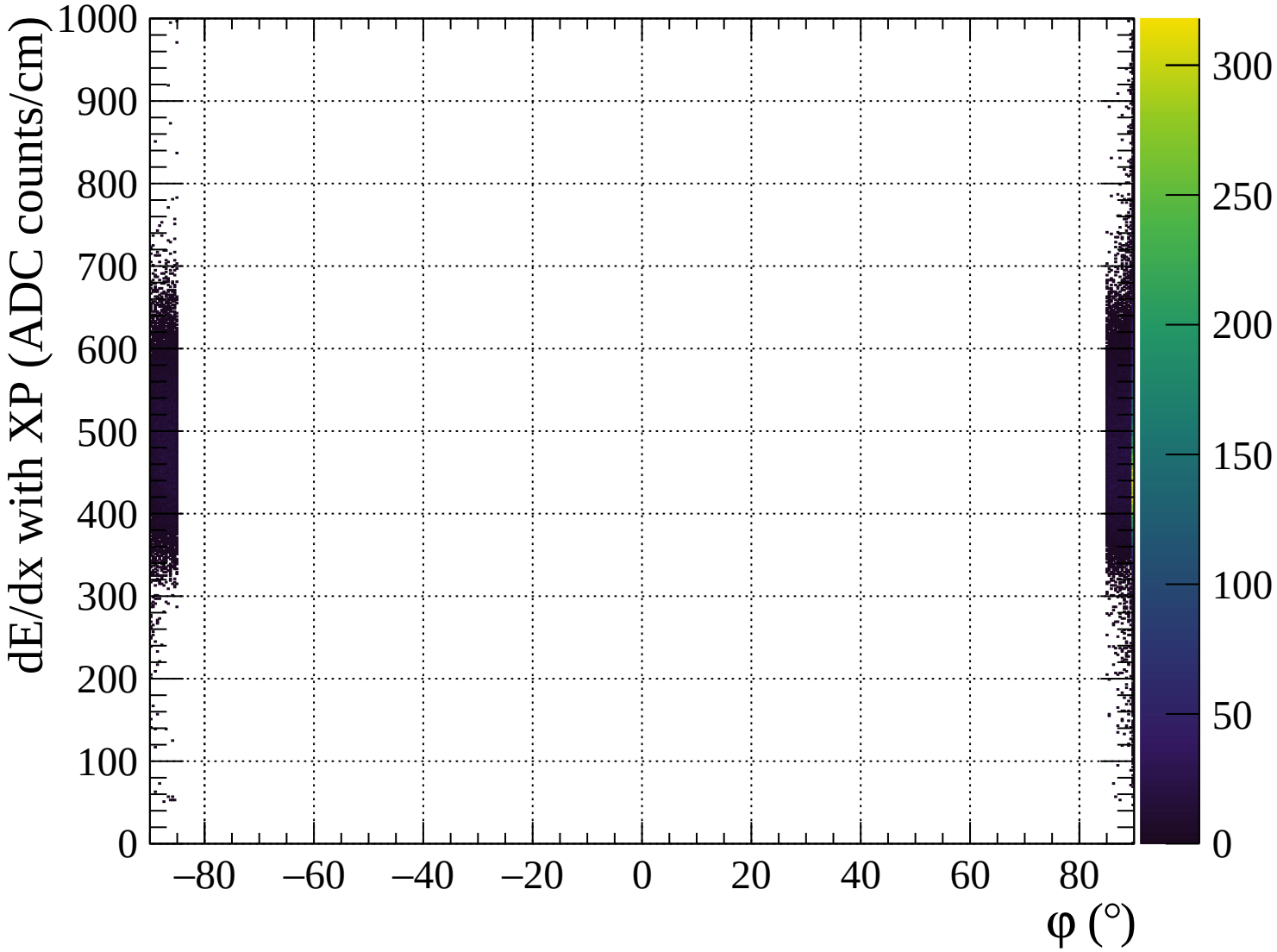


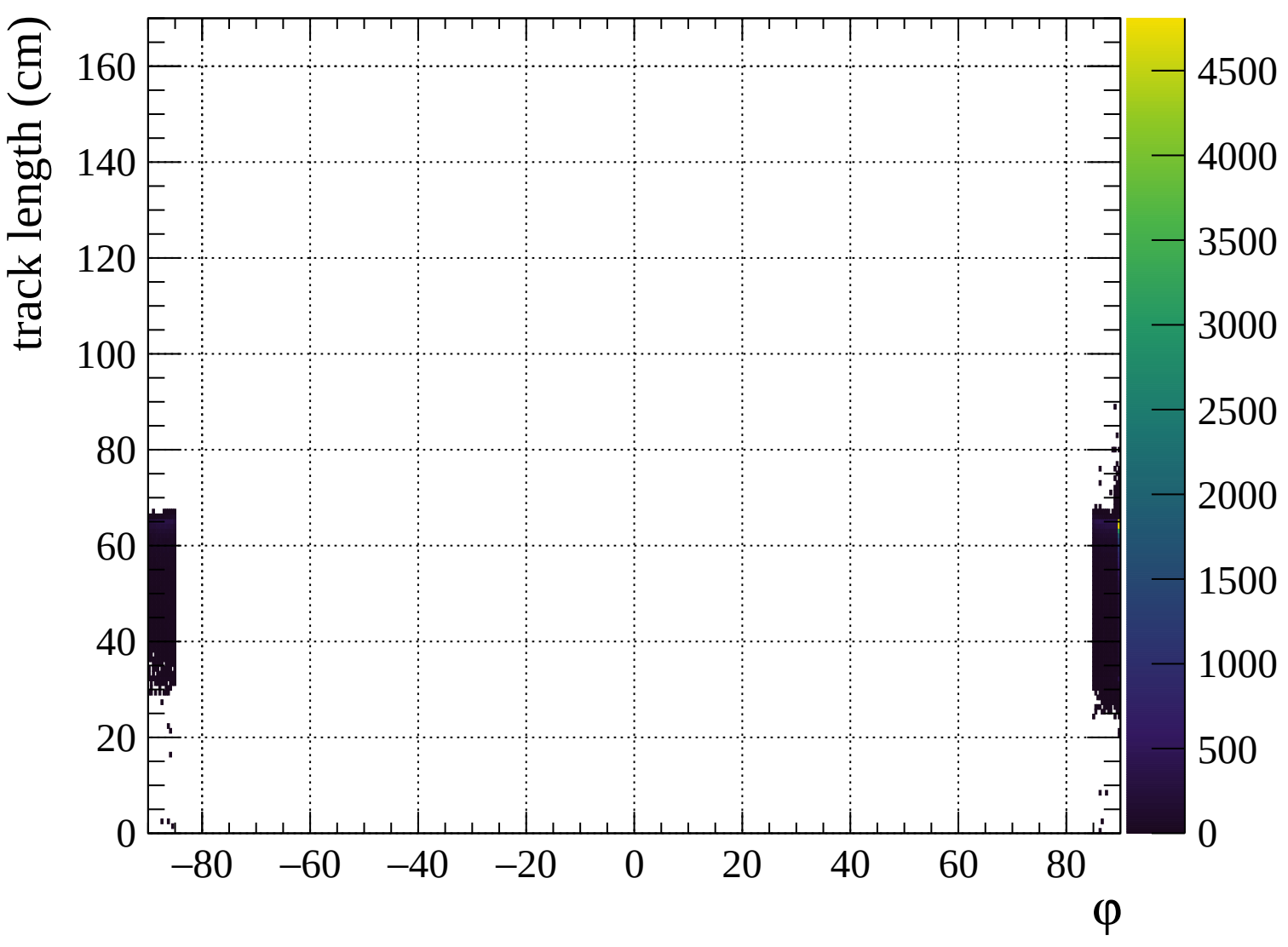


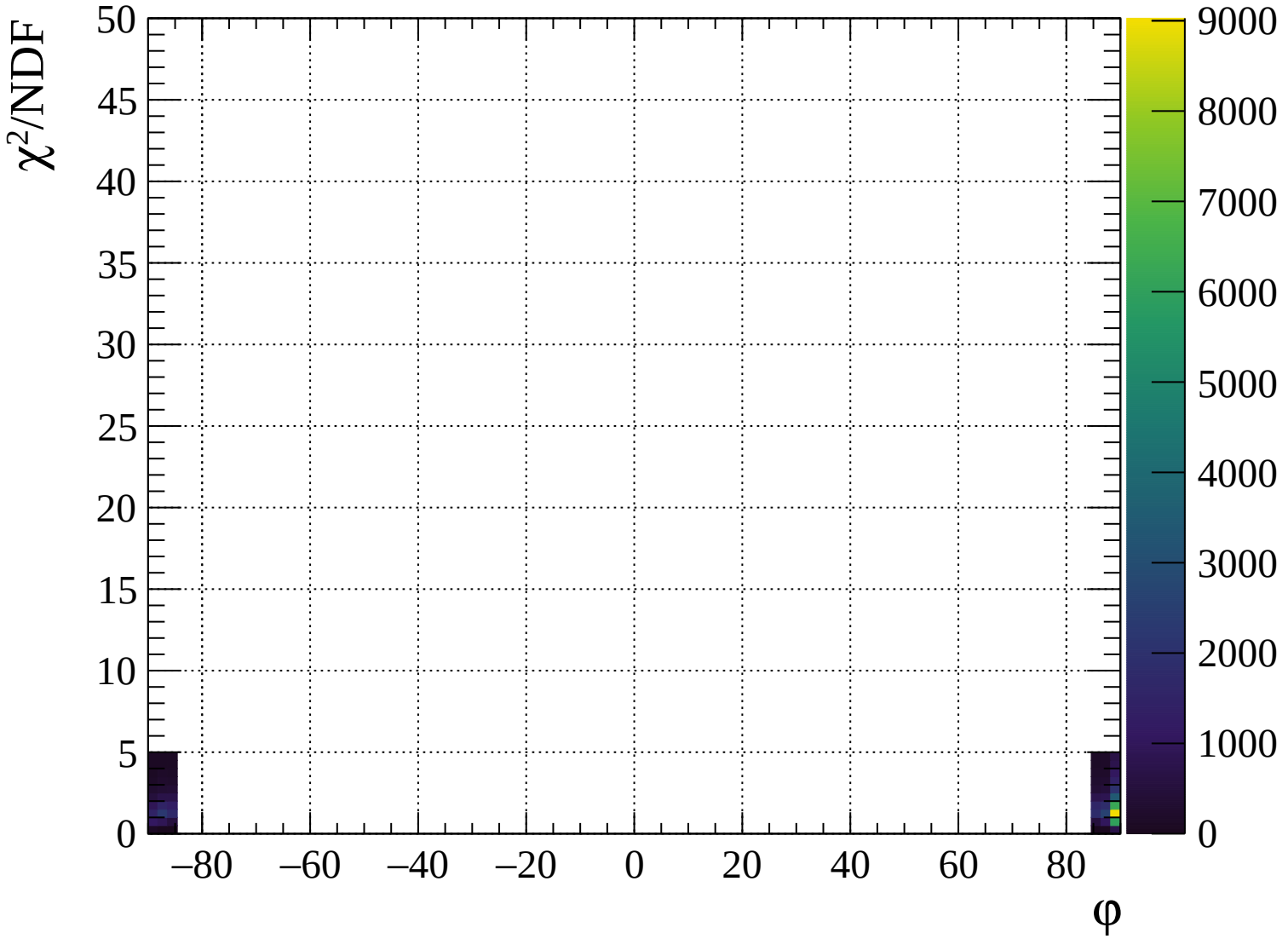


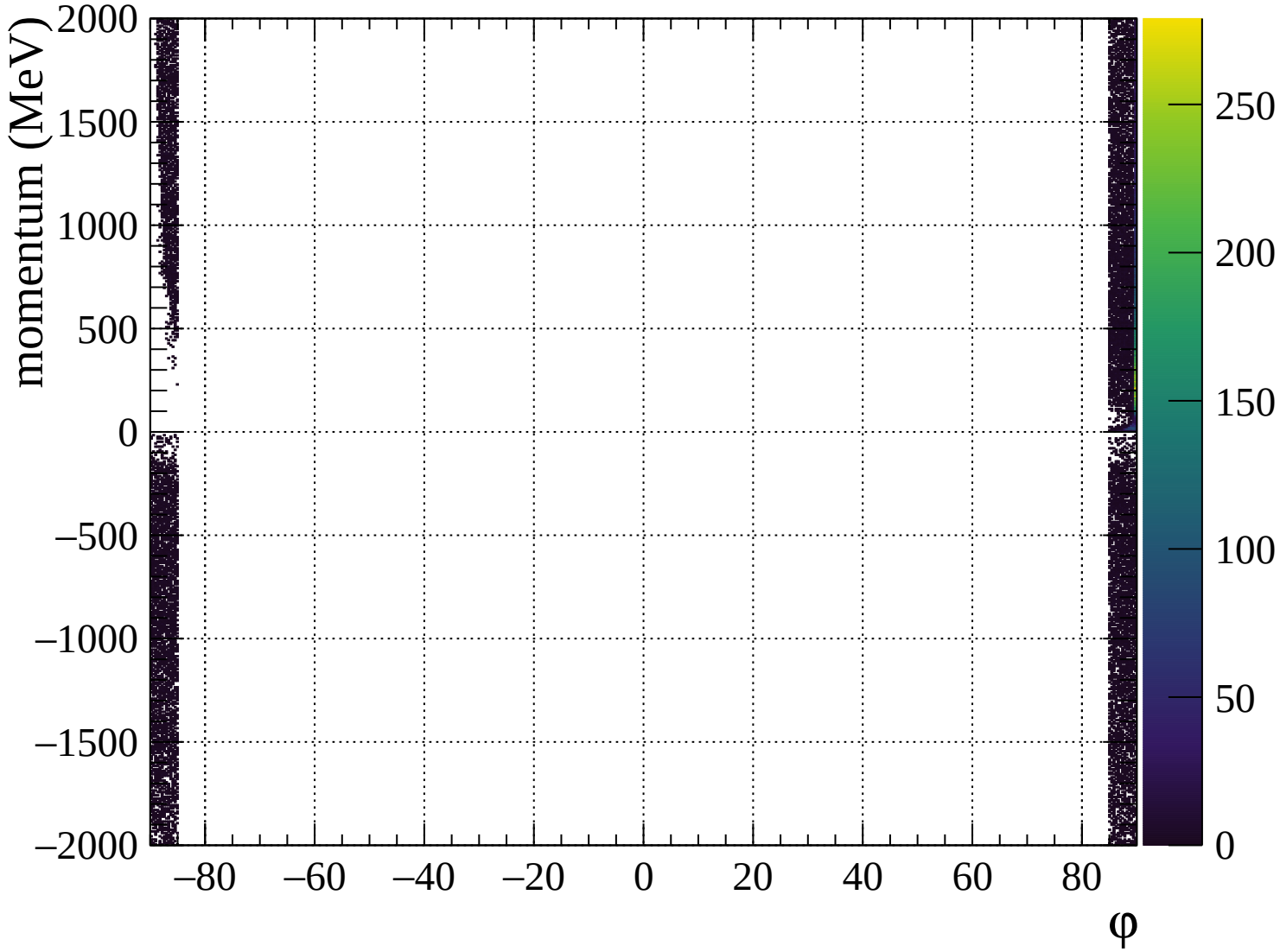


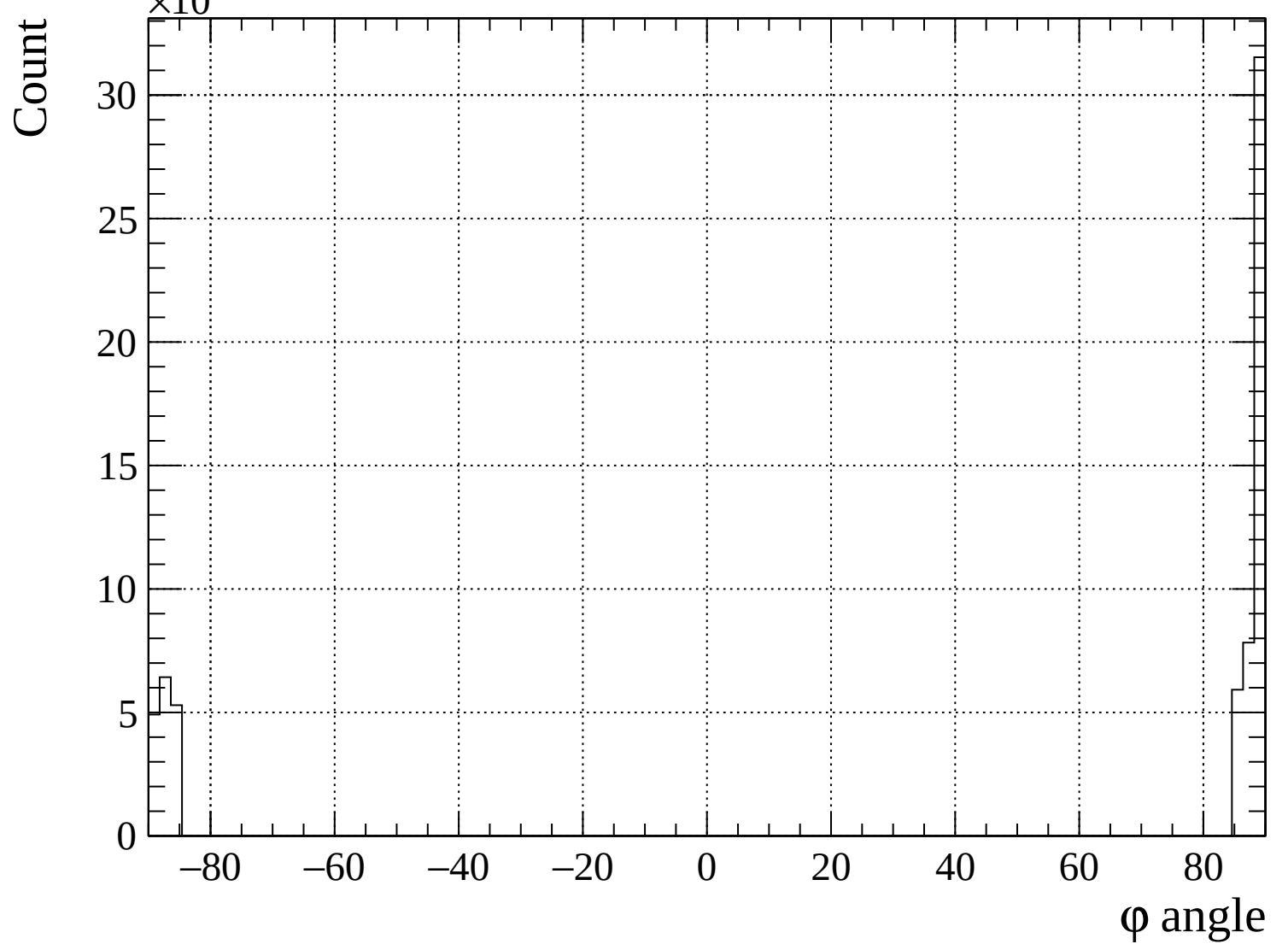


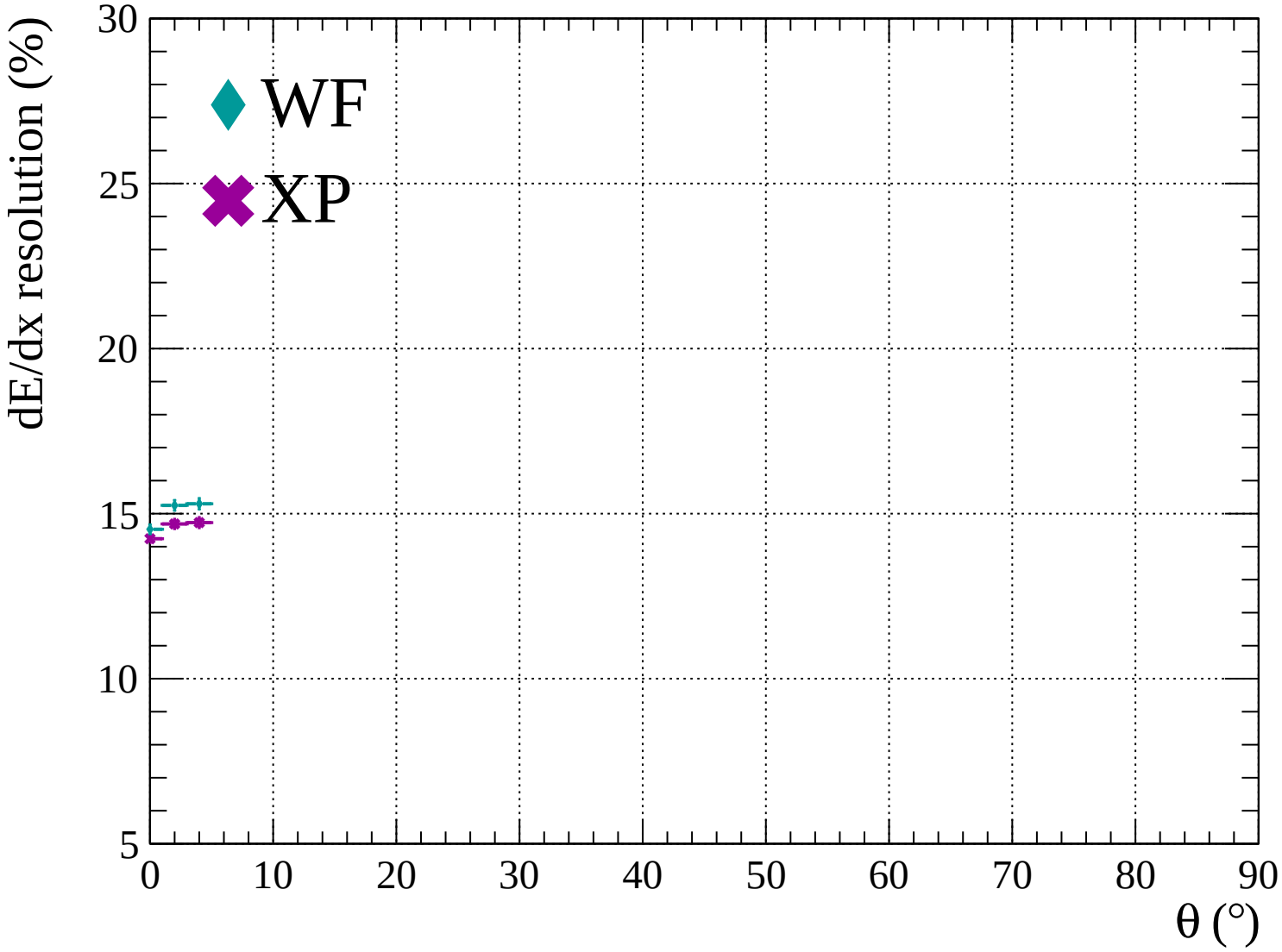


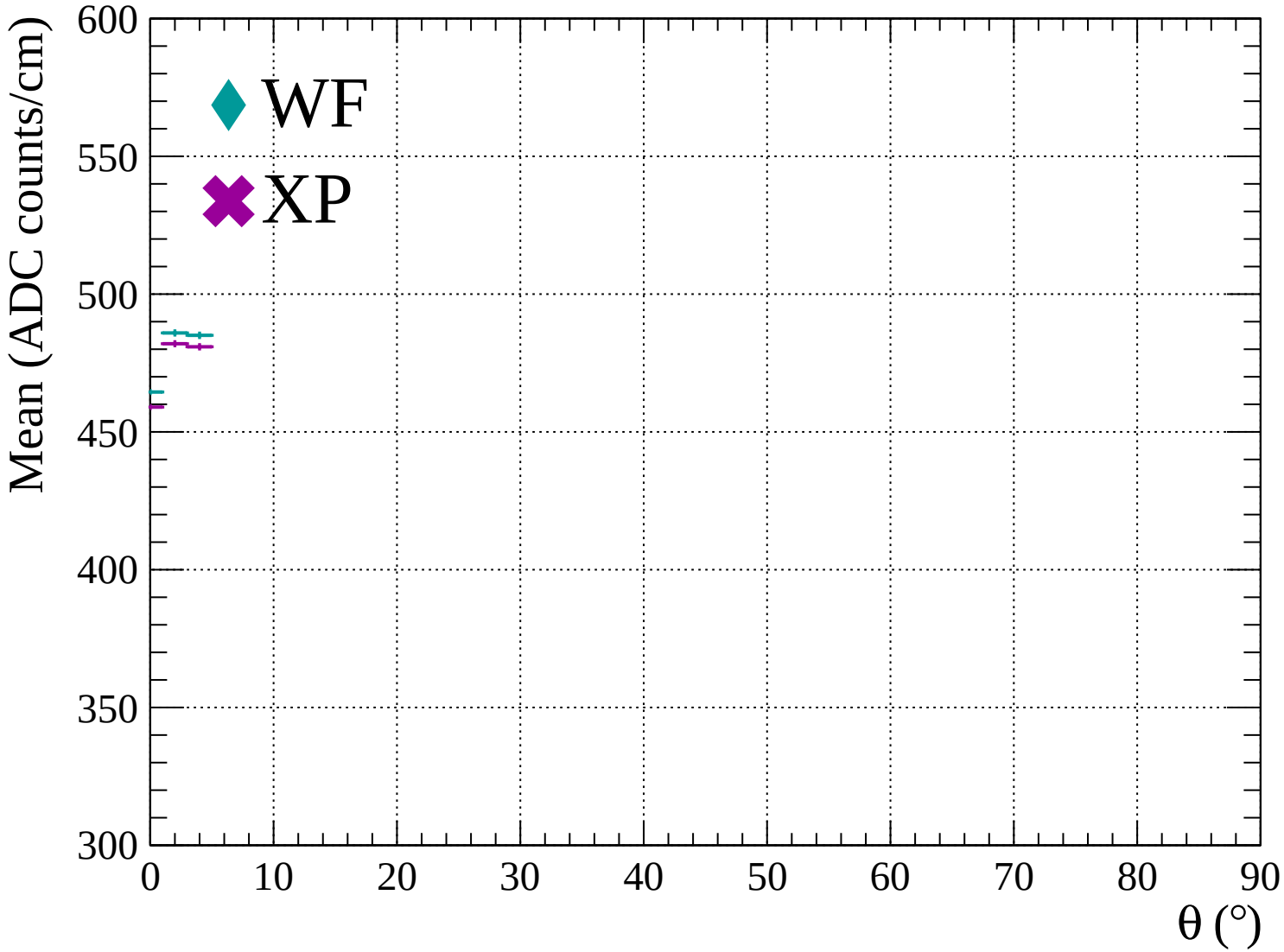


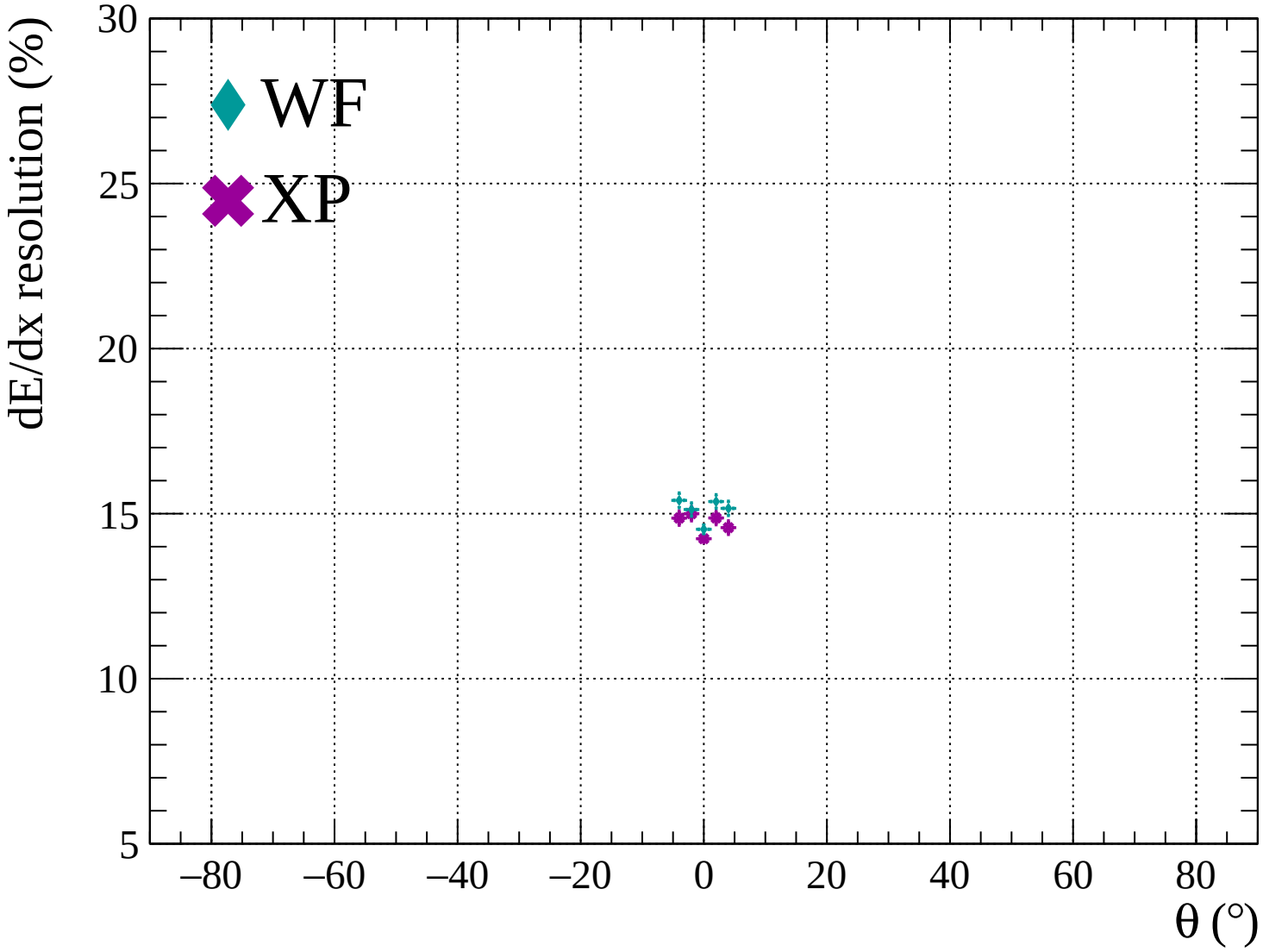


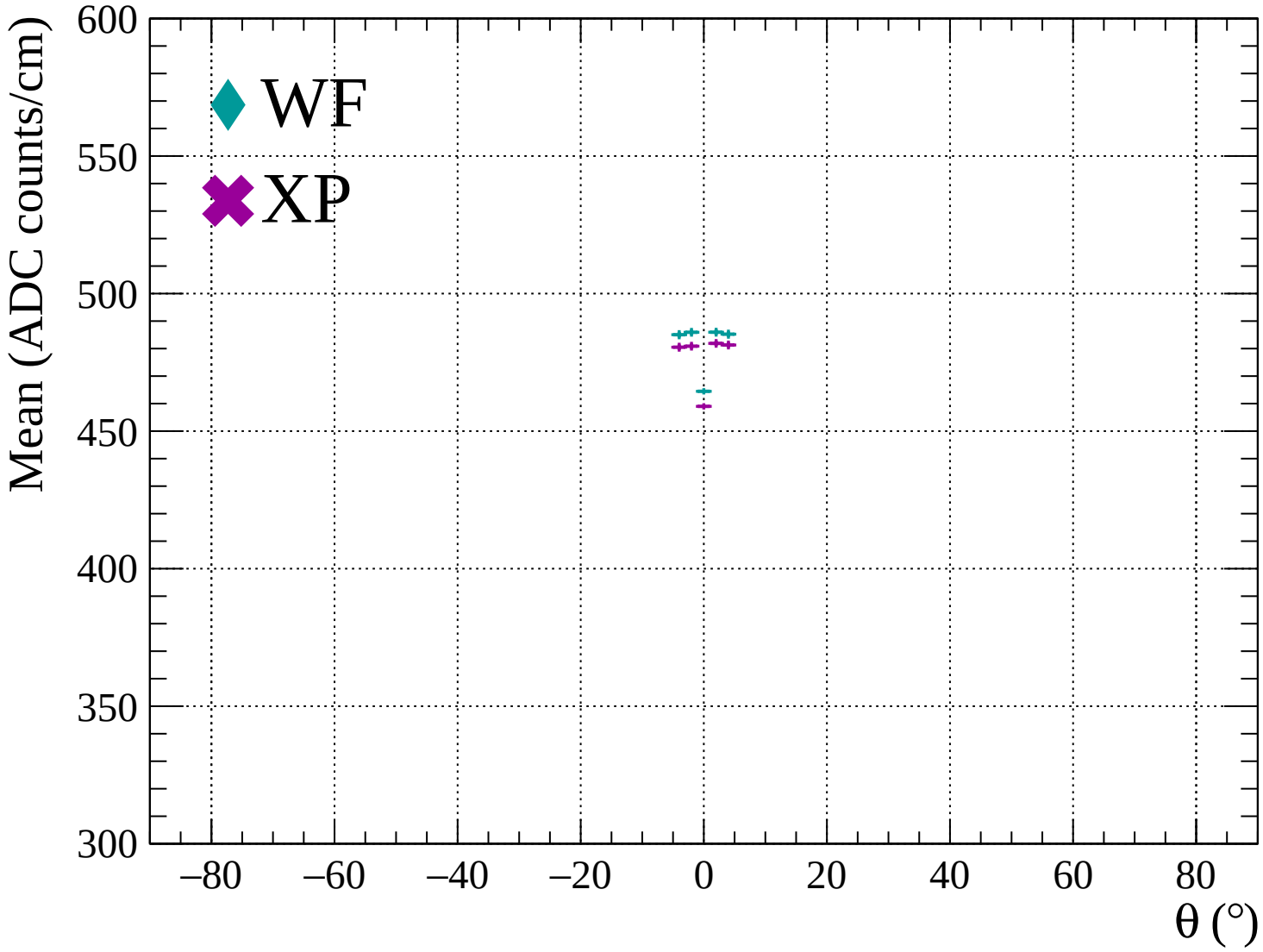


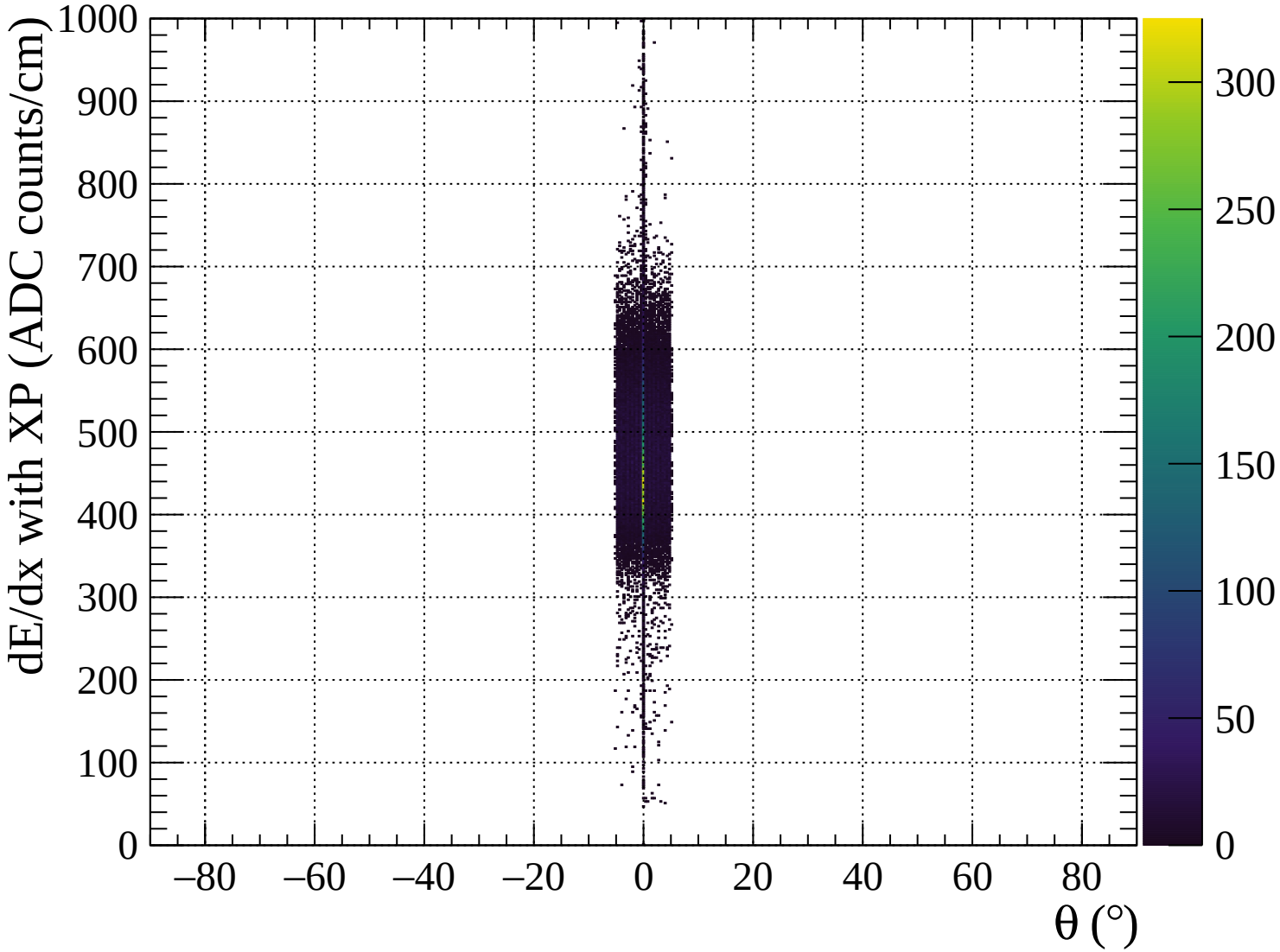


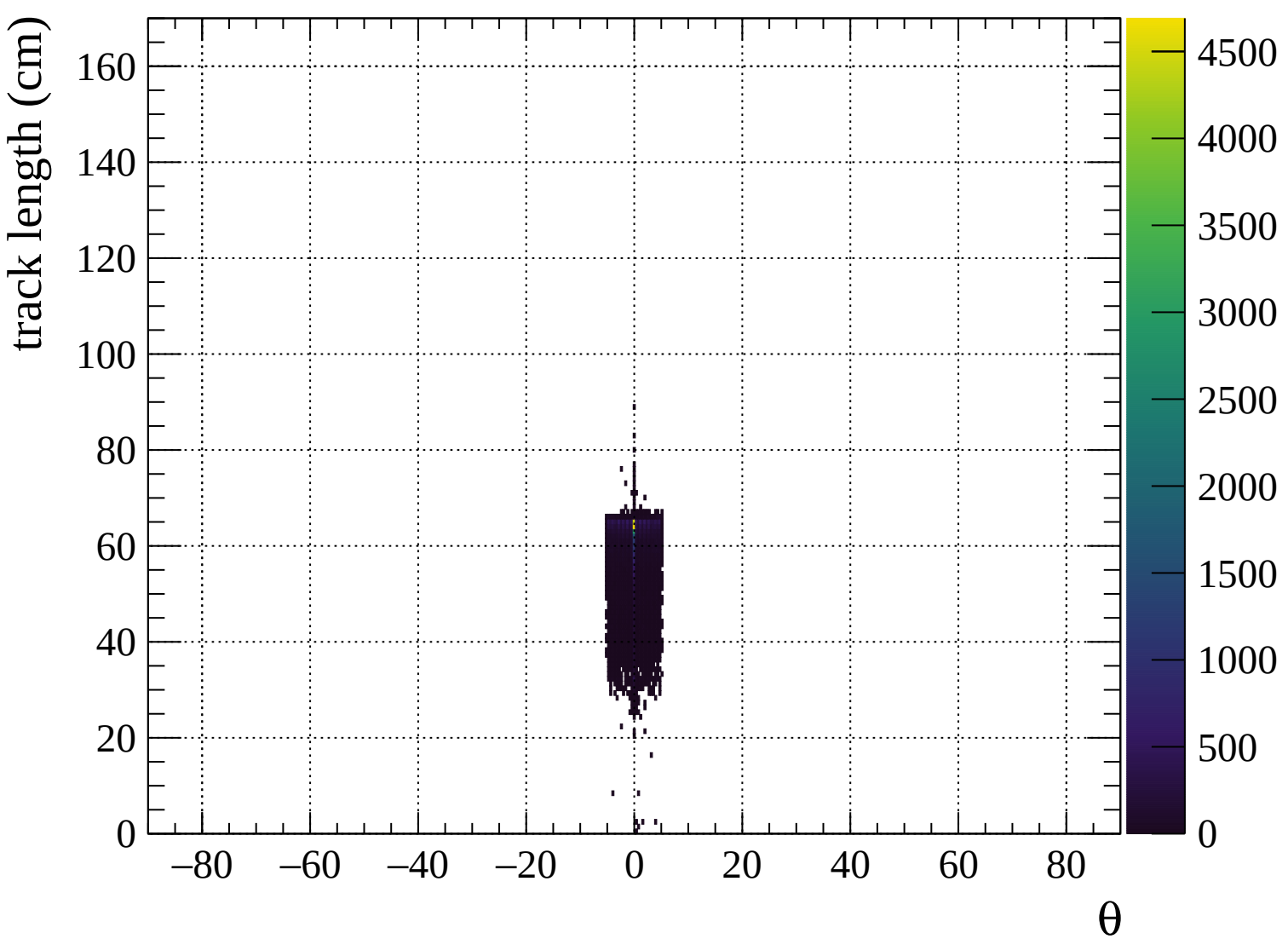


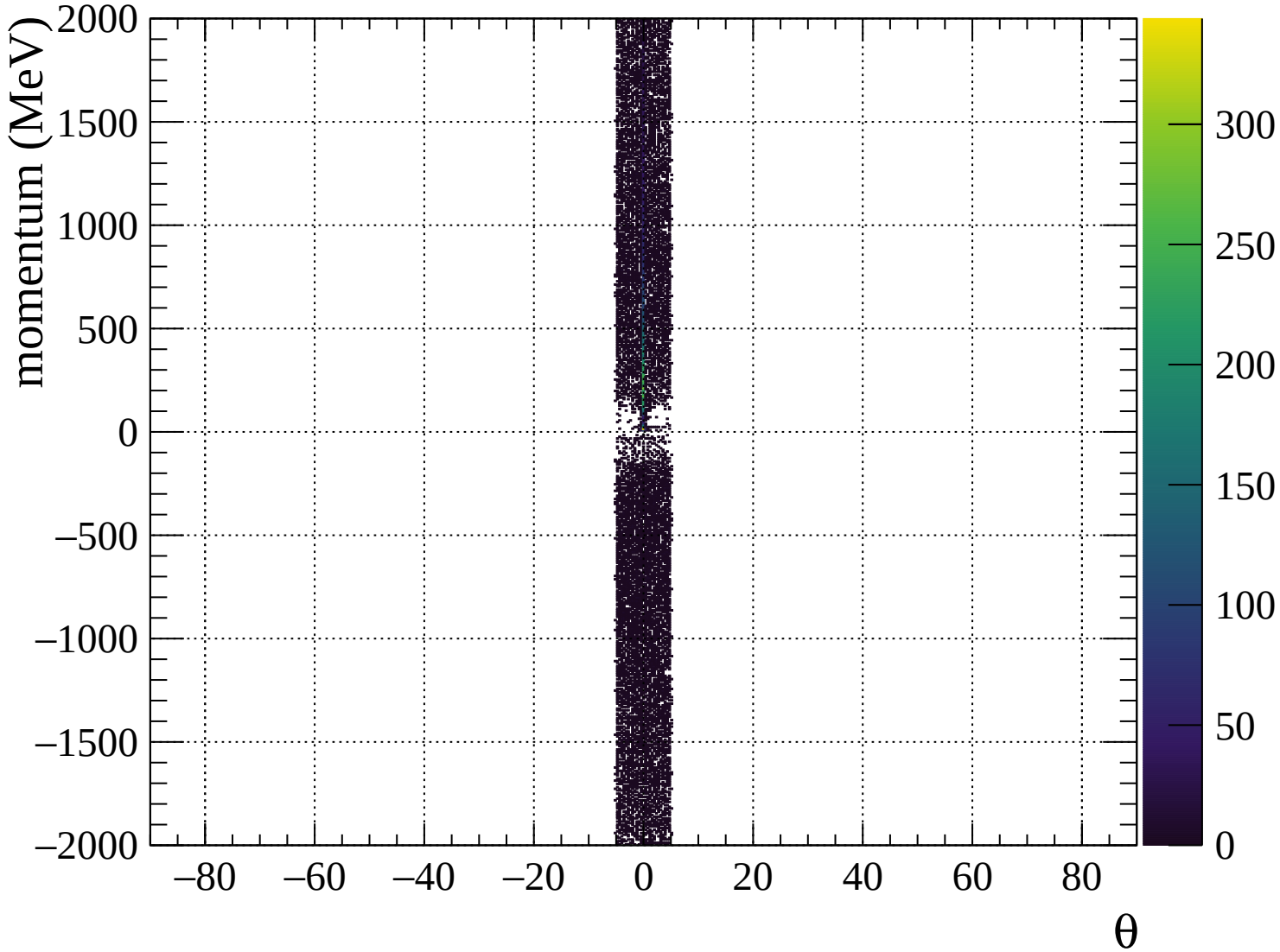


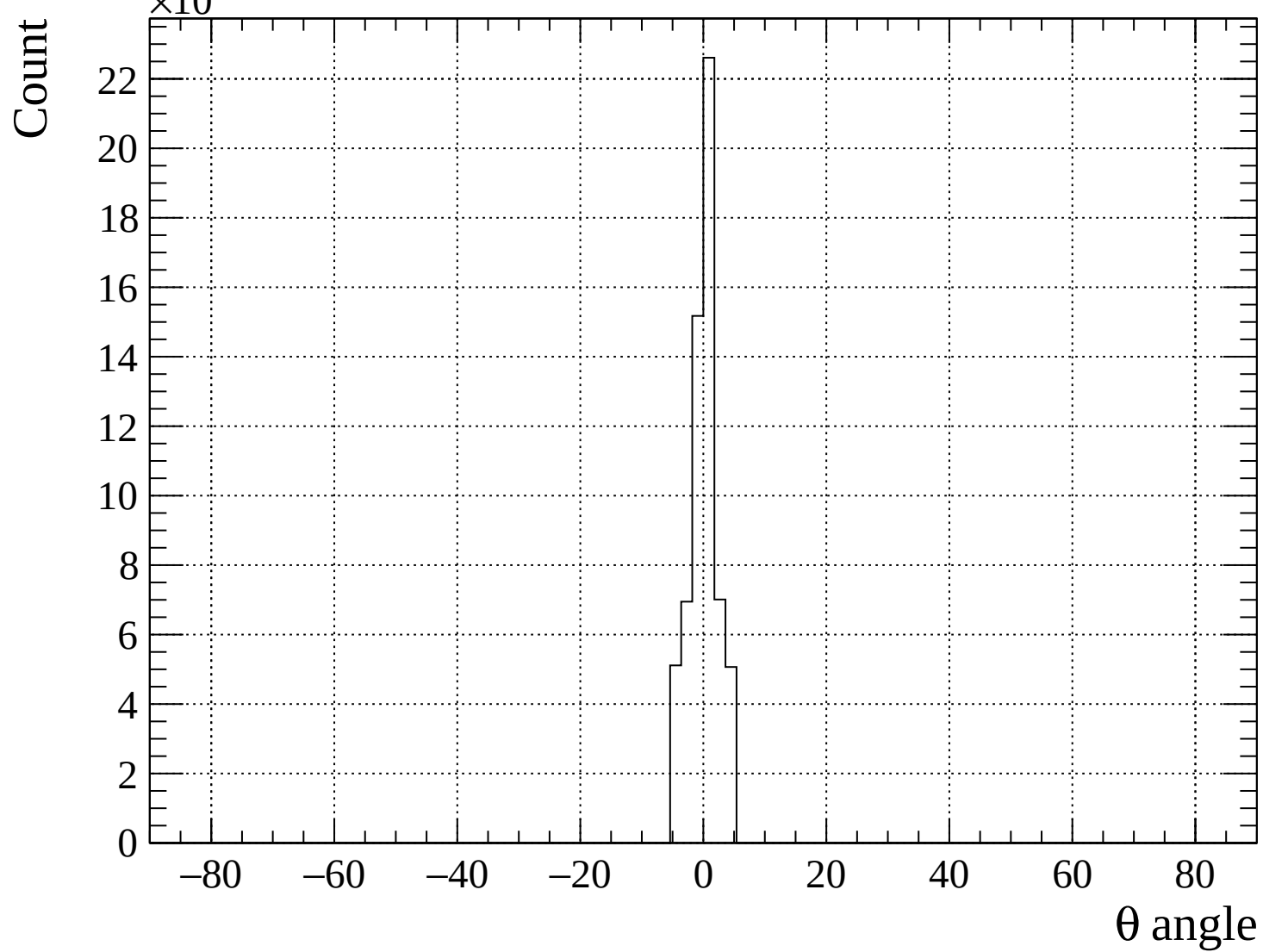


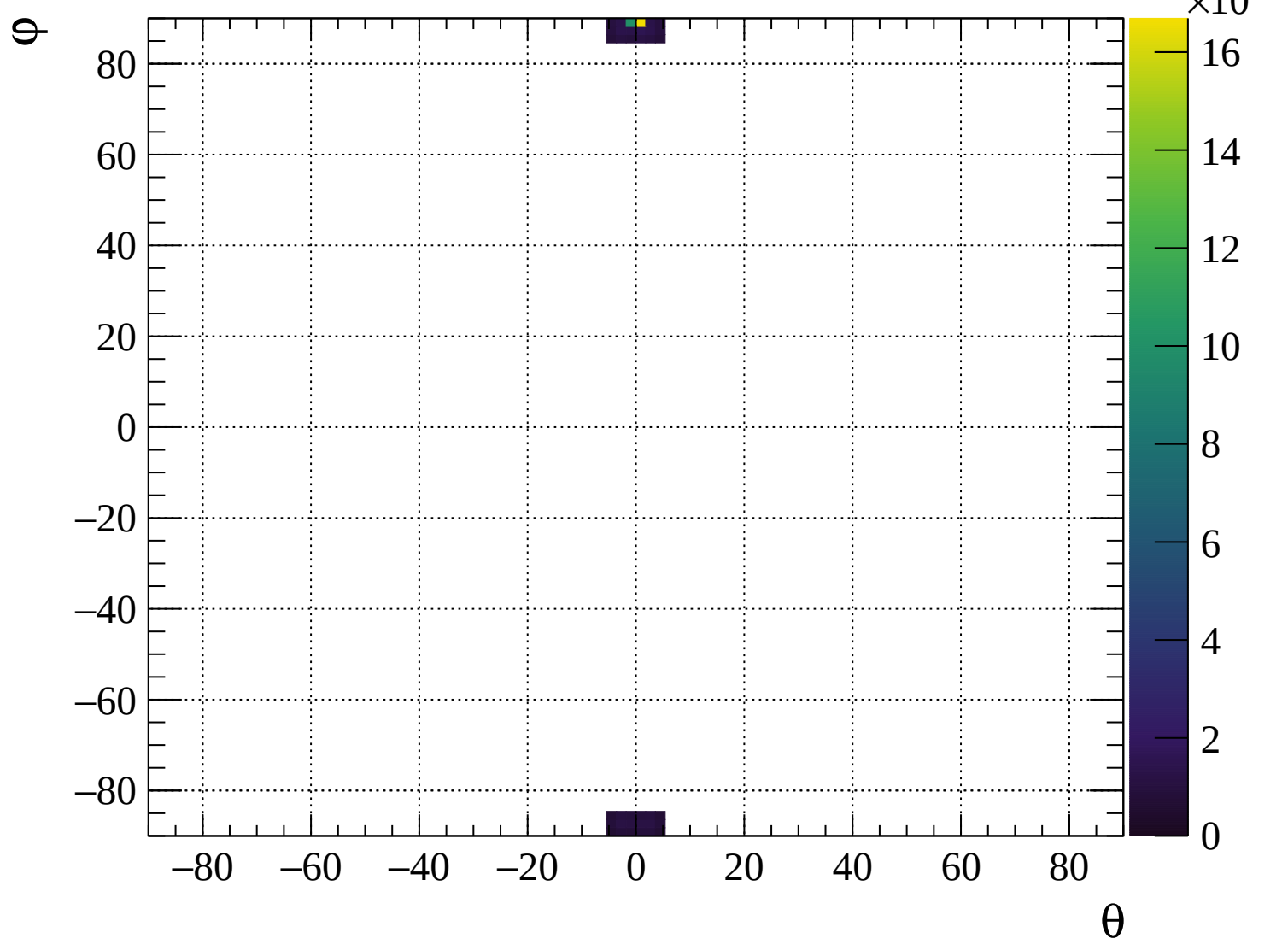


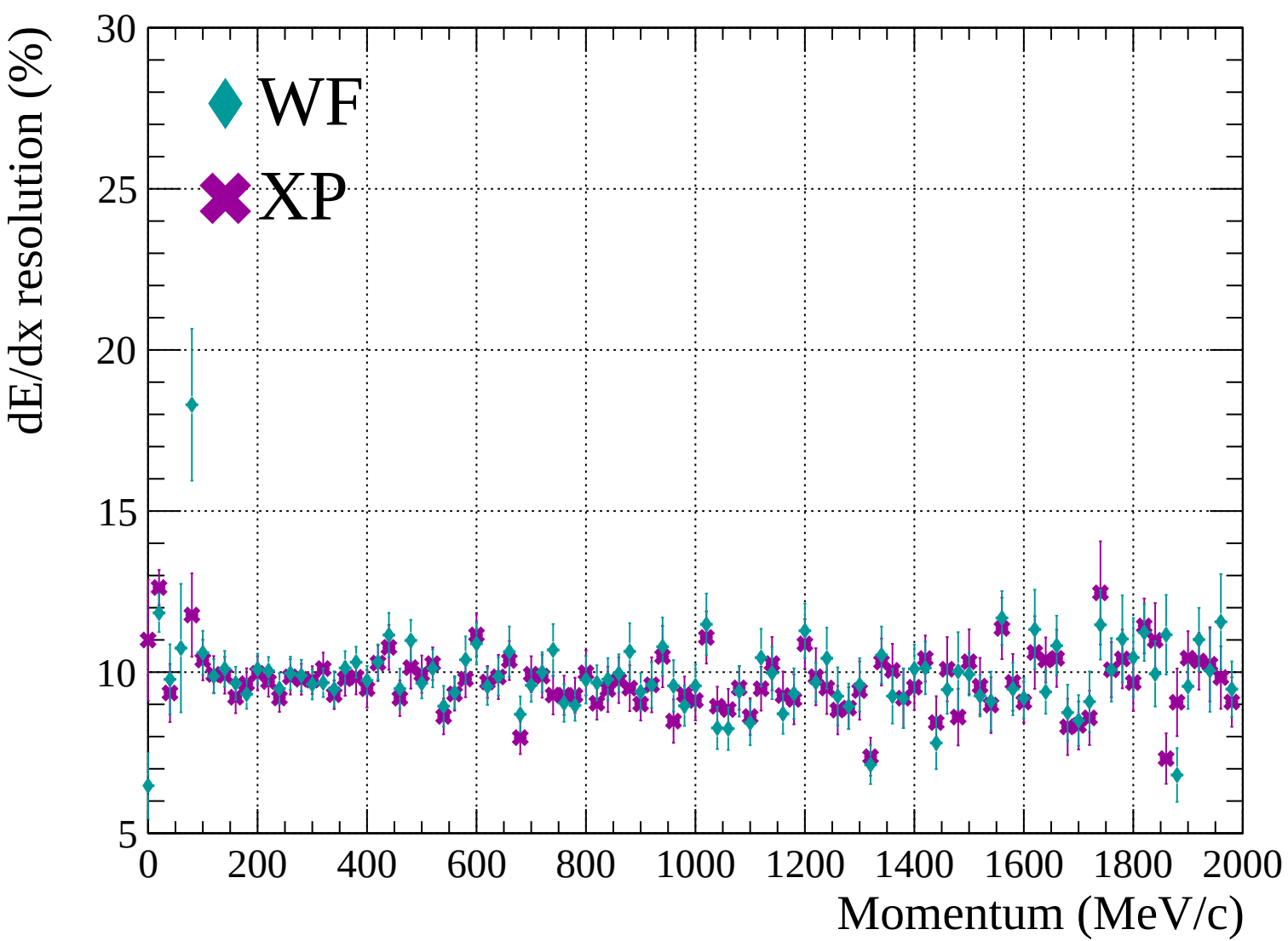


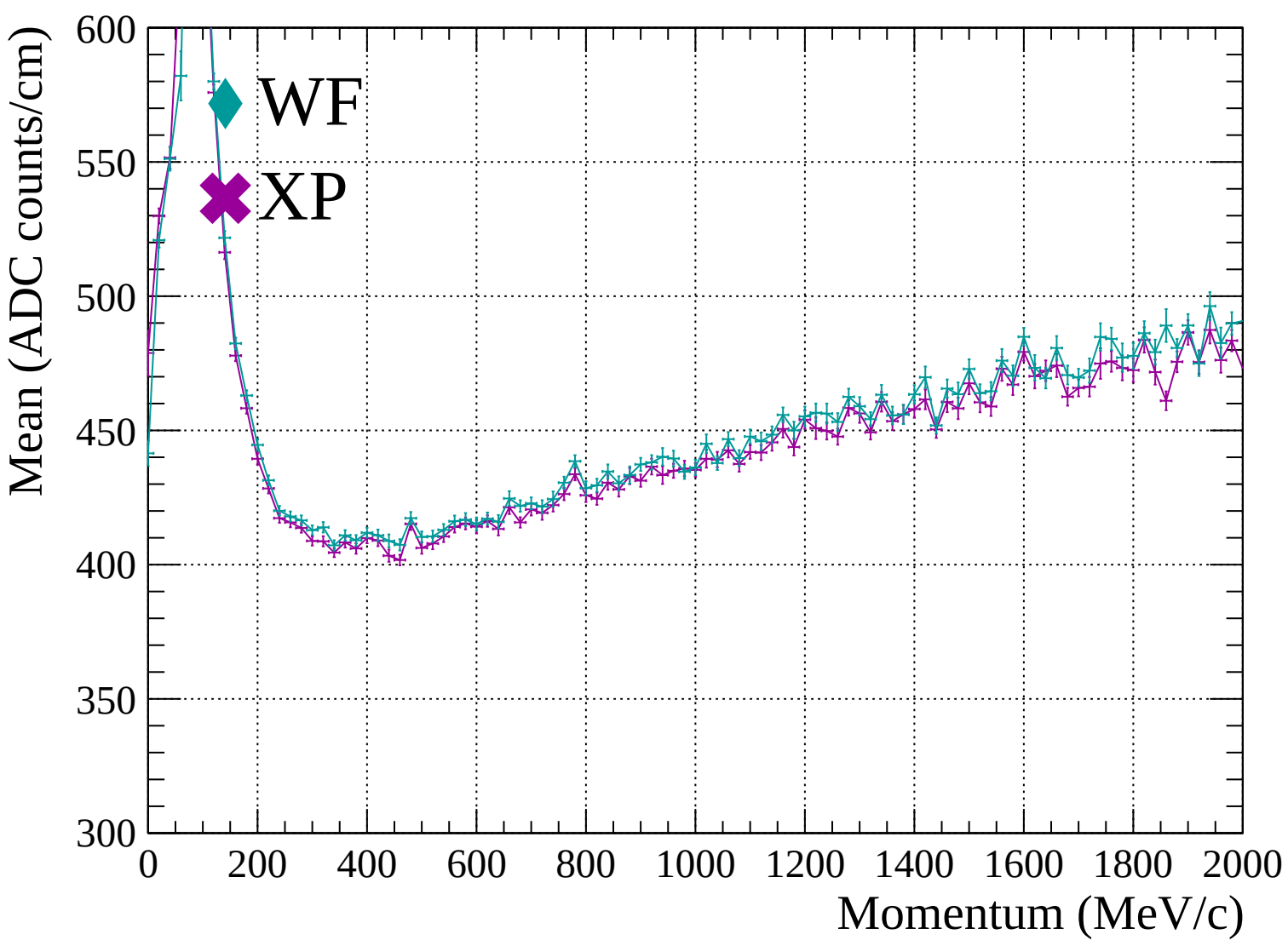


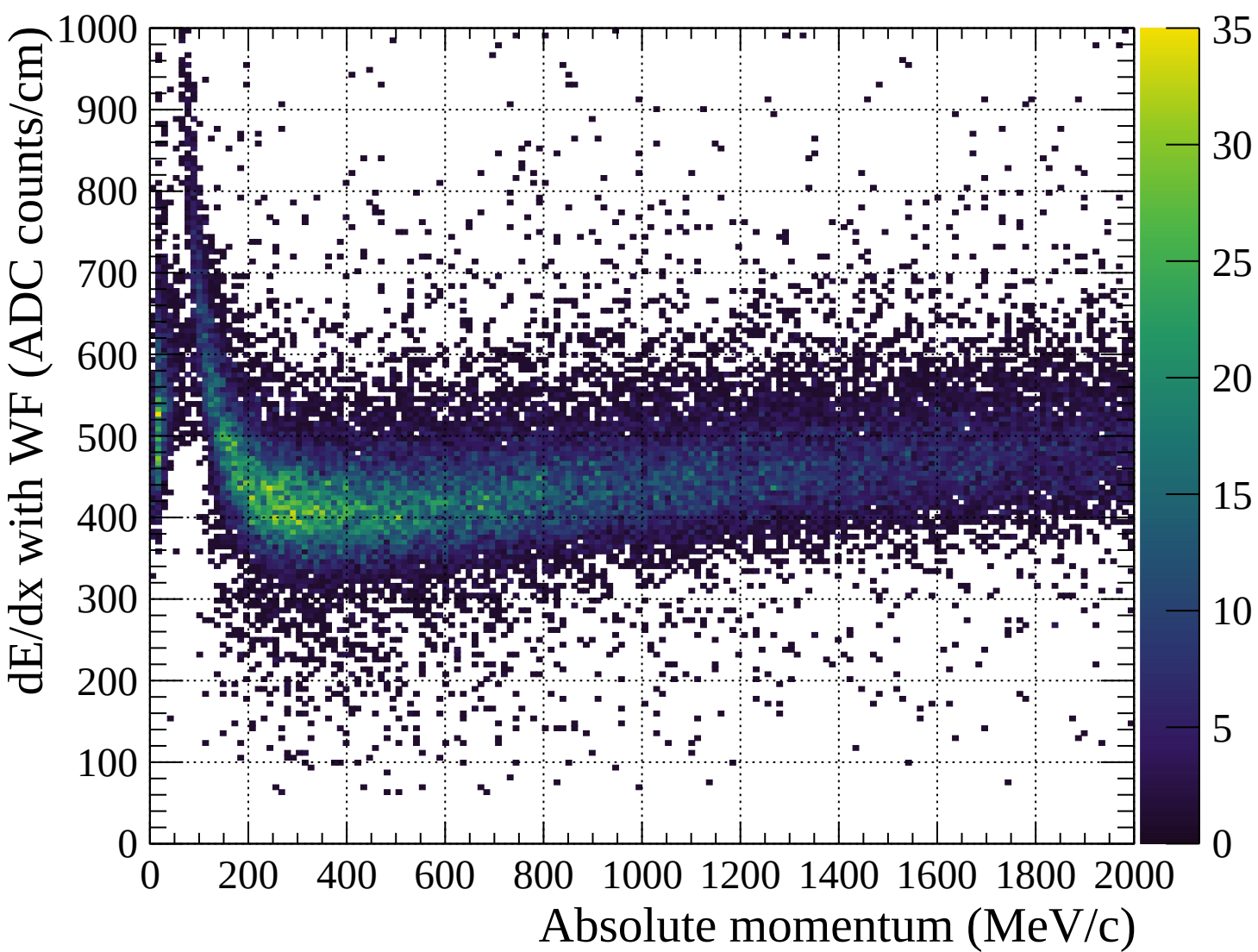


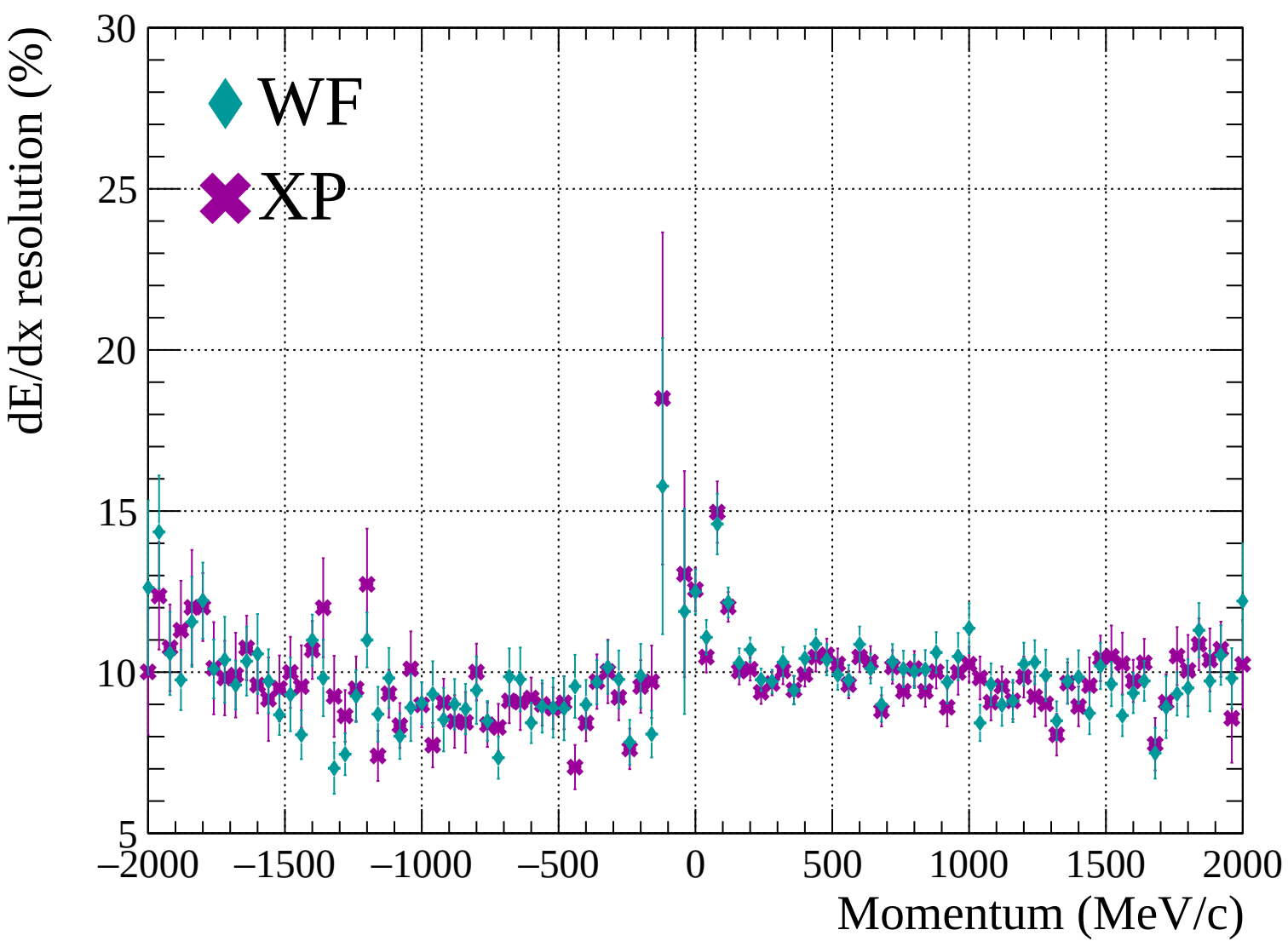


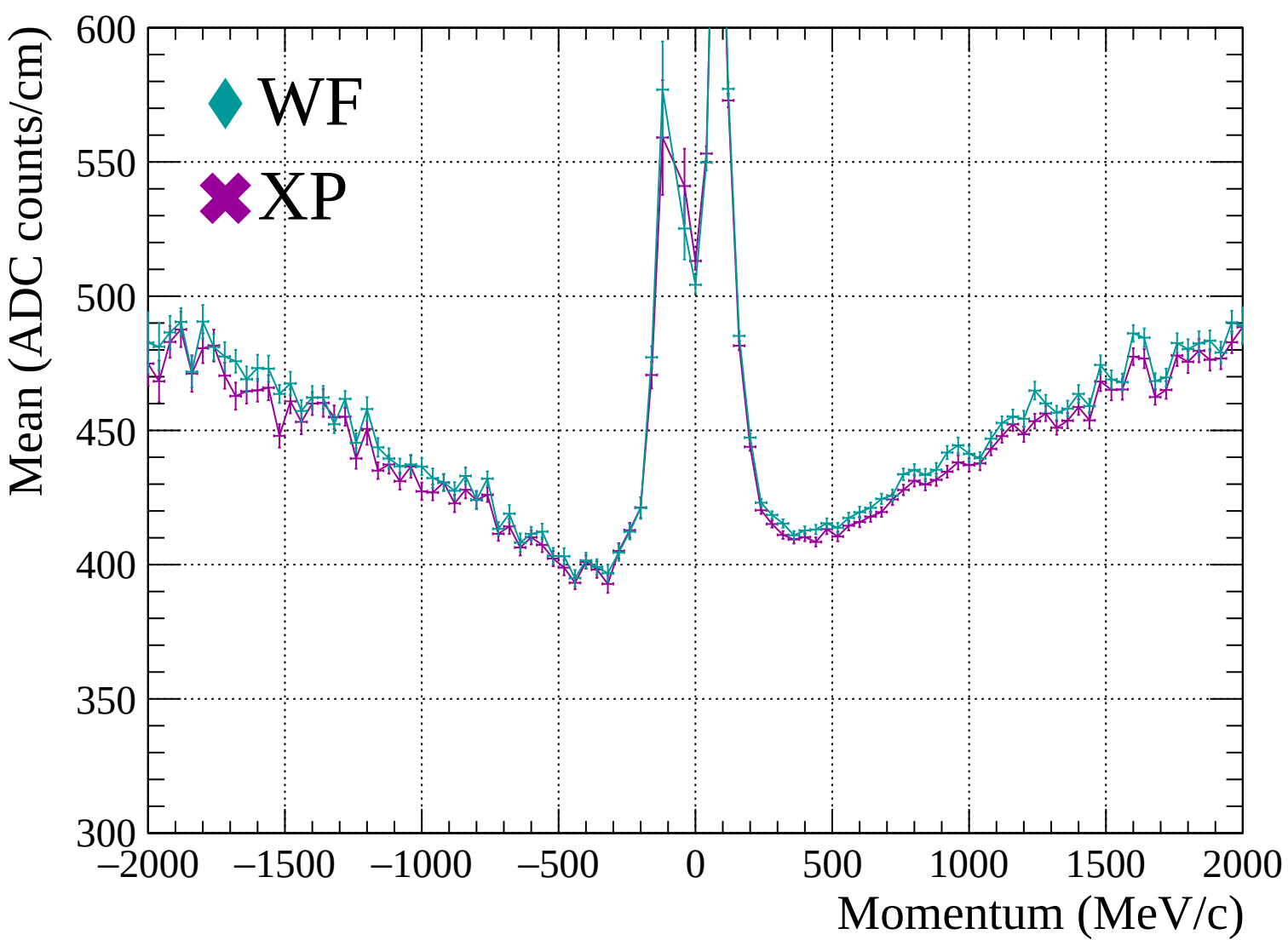


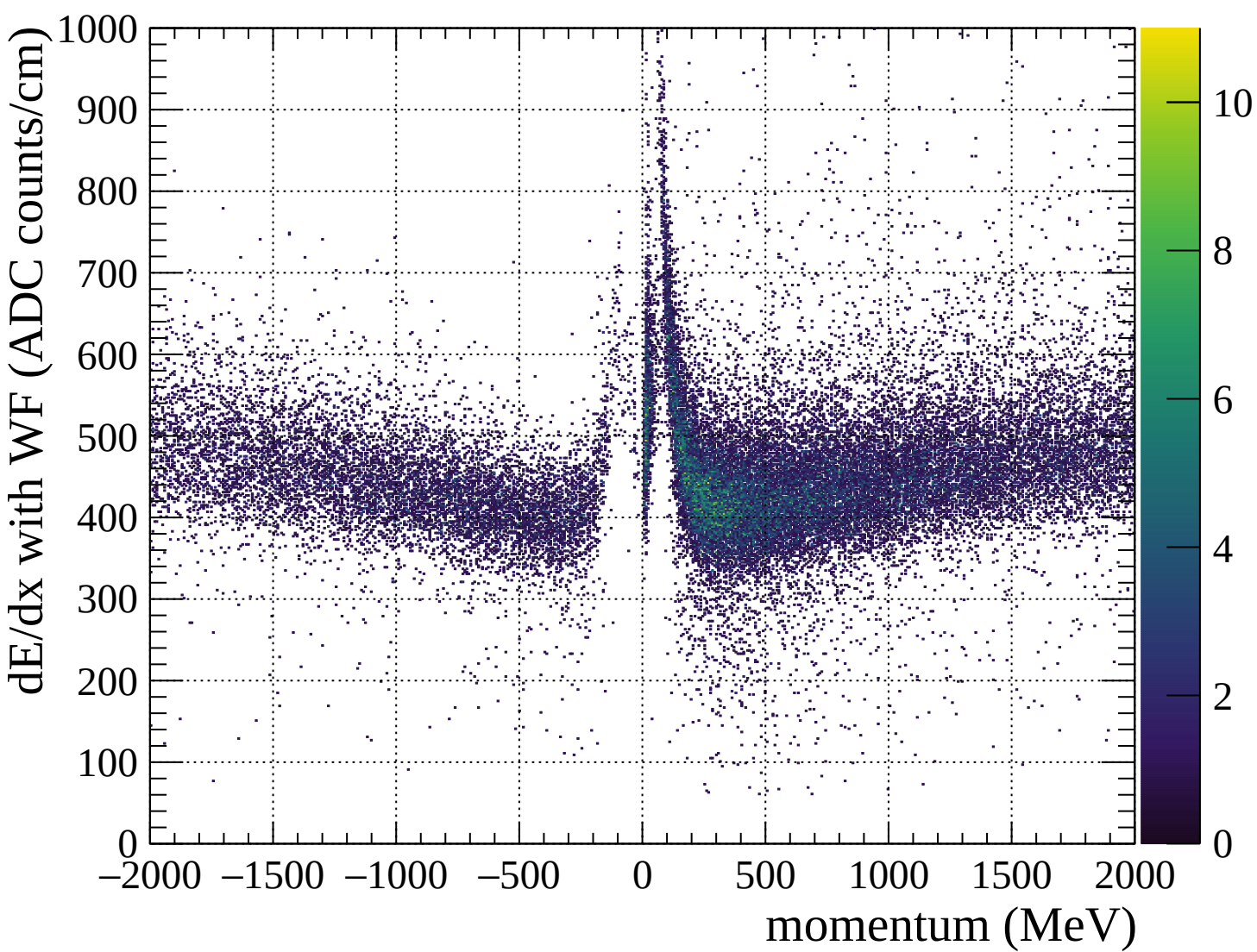


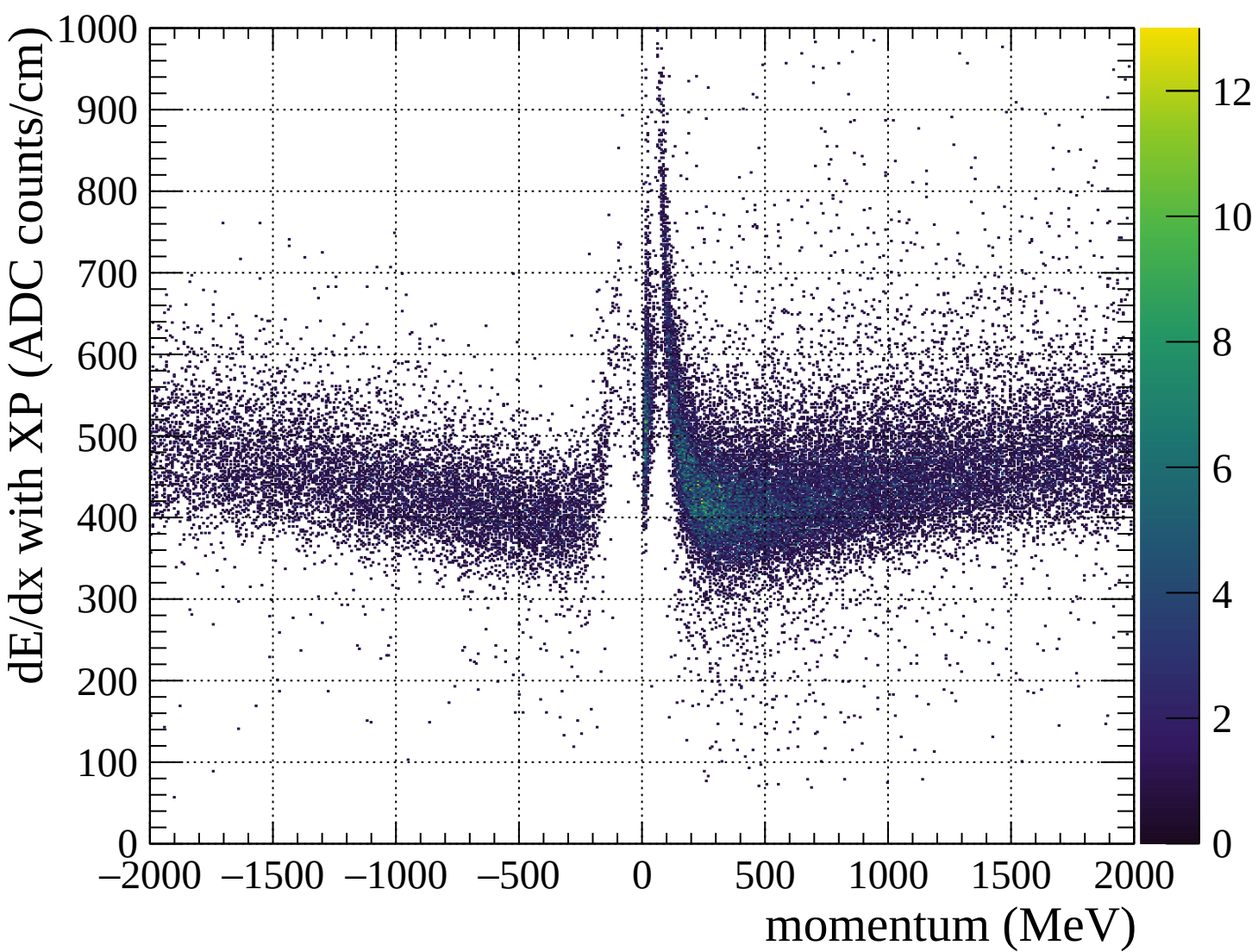


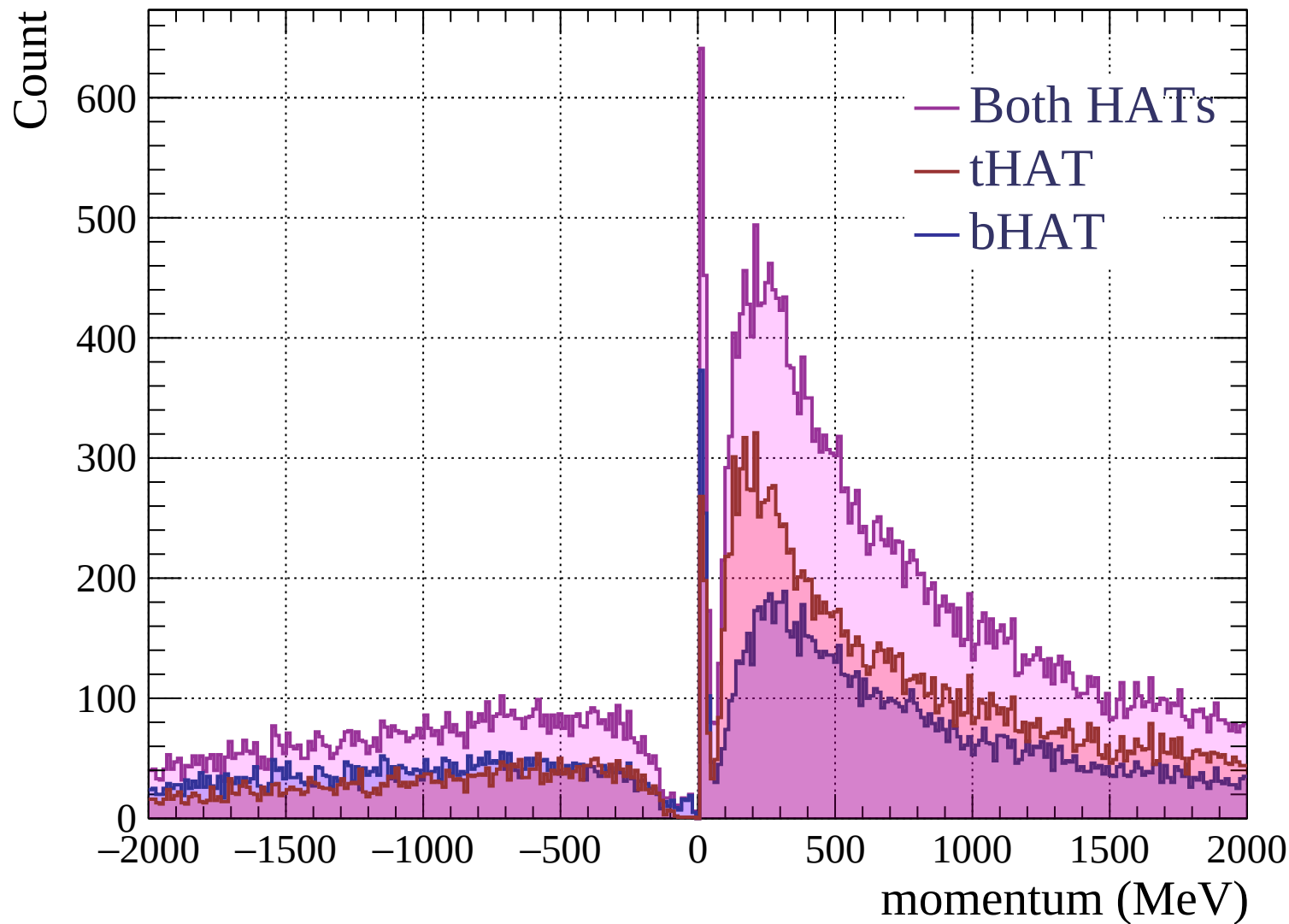


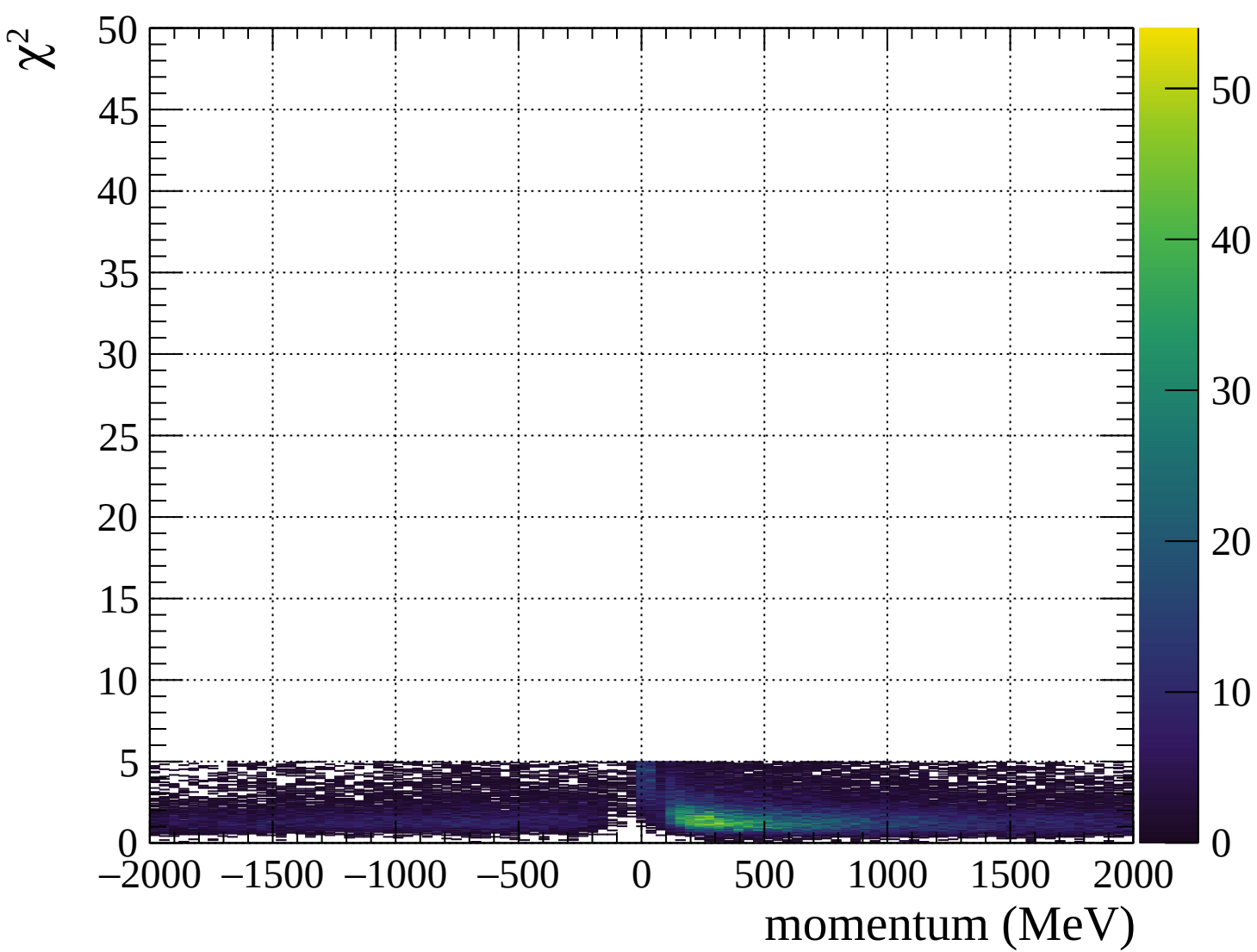




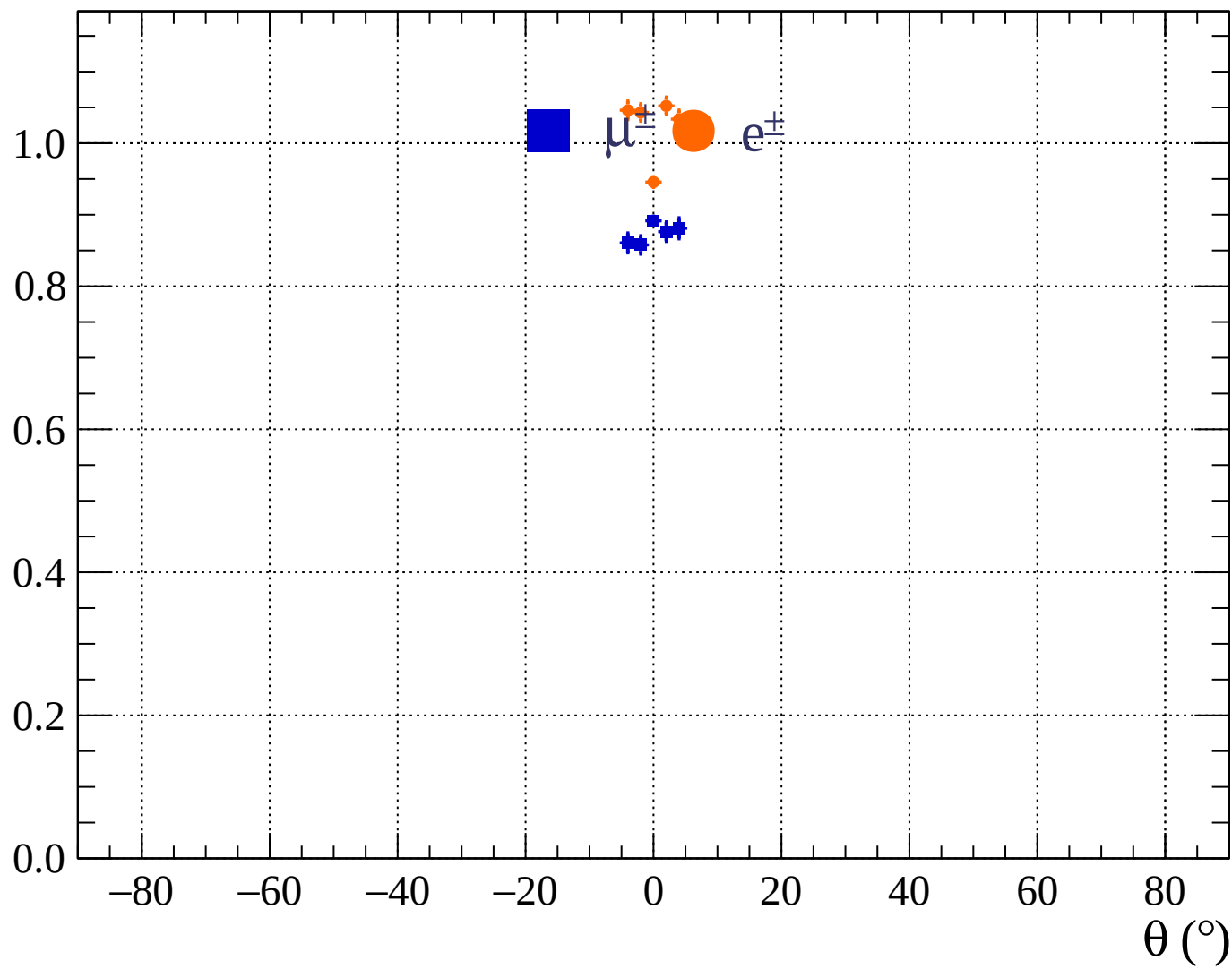


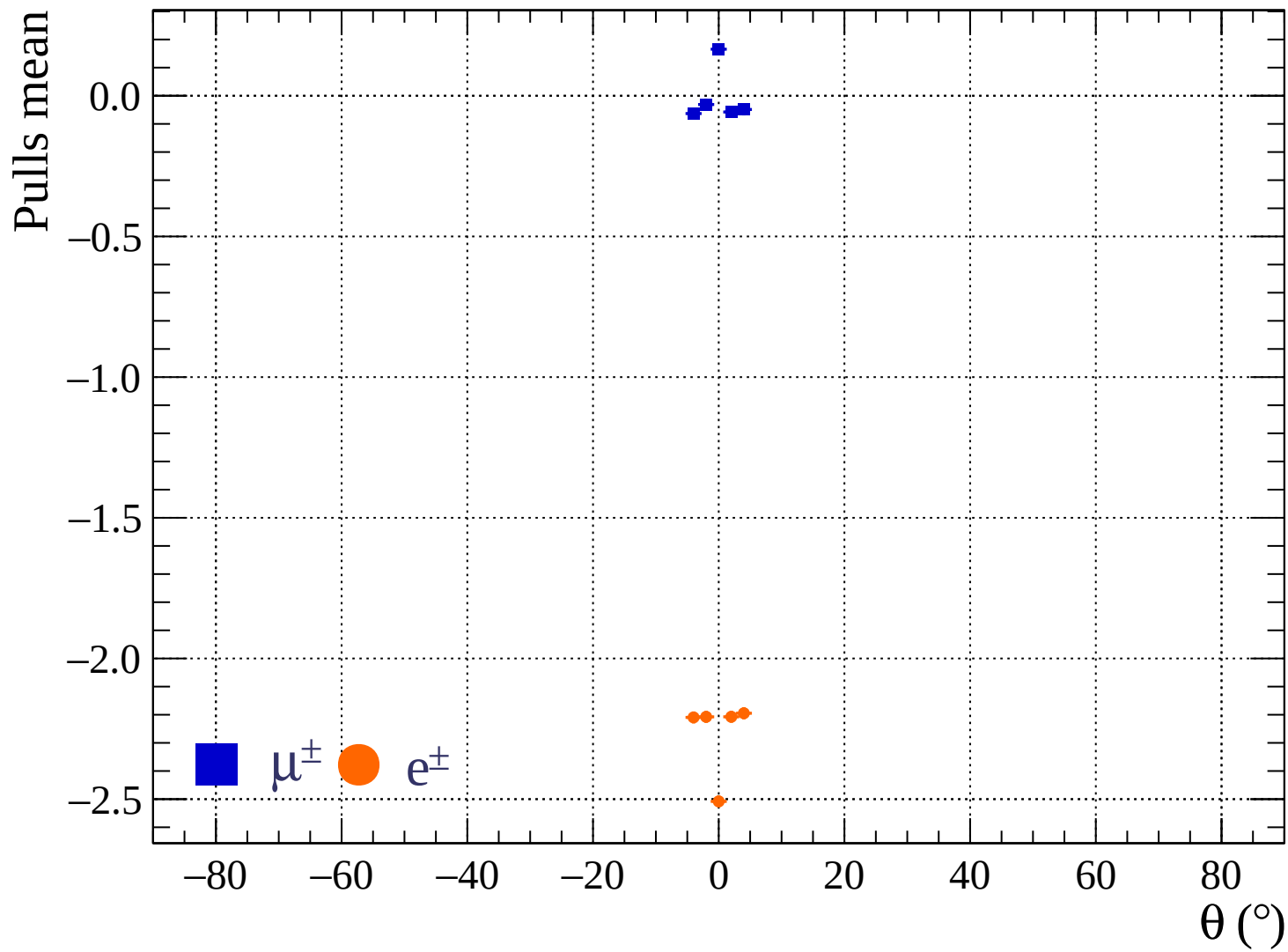




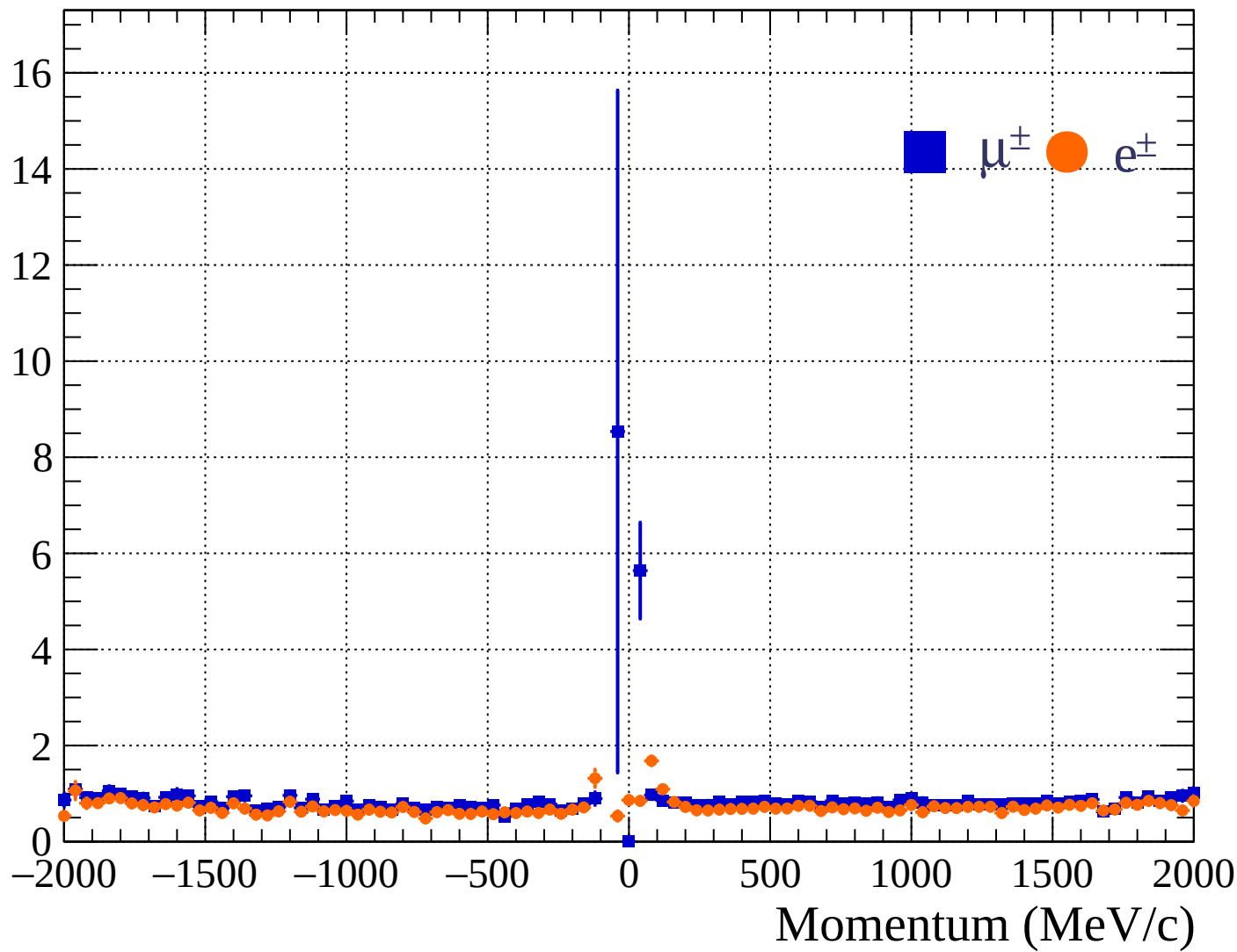


Pulls standard deviation

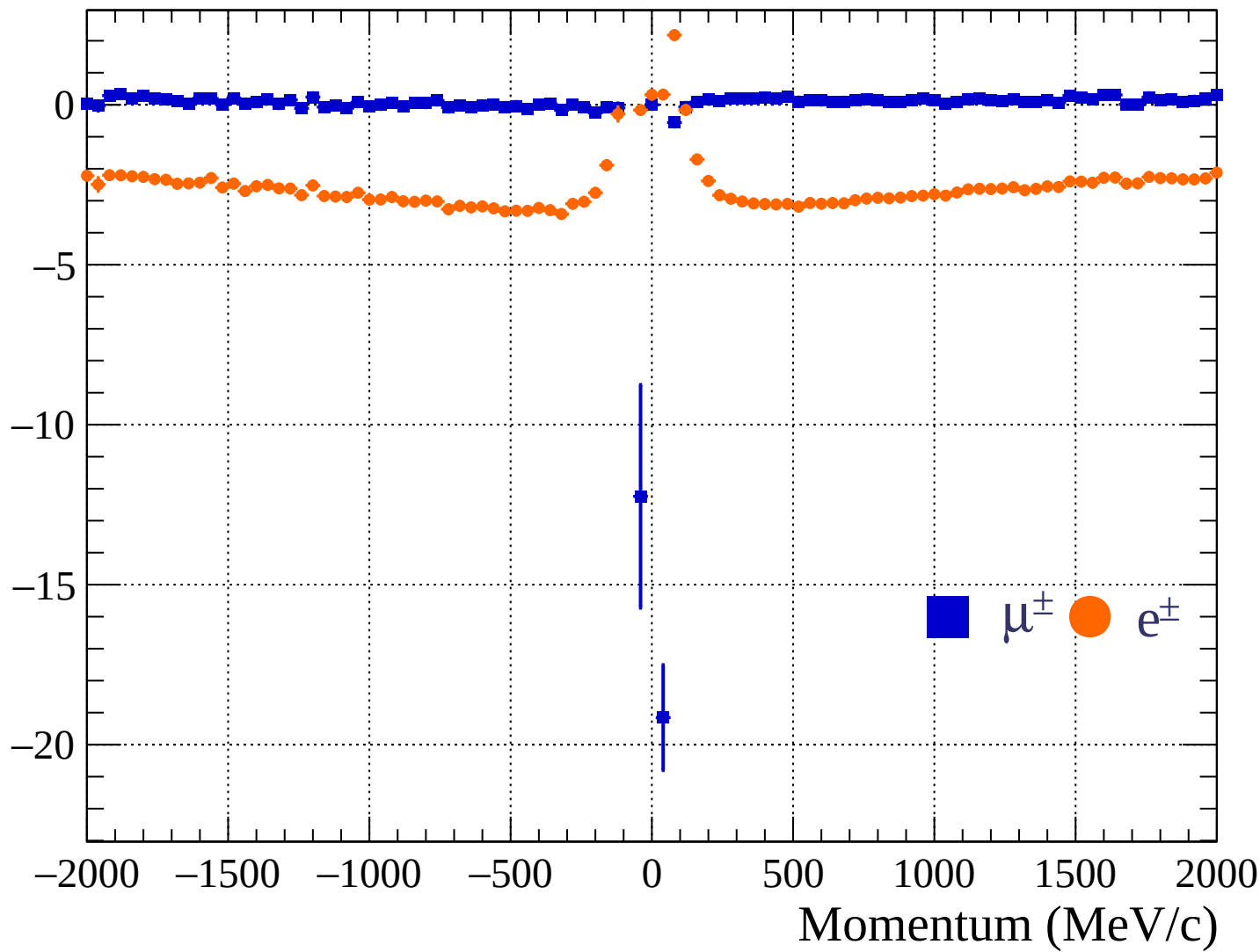


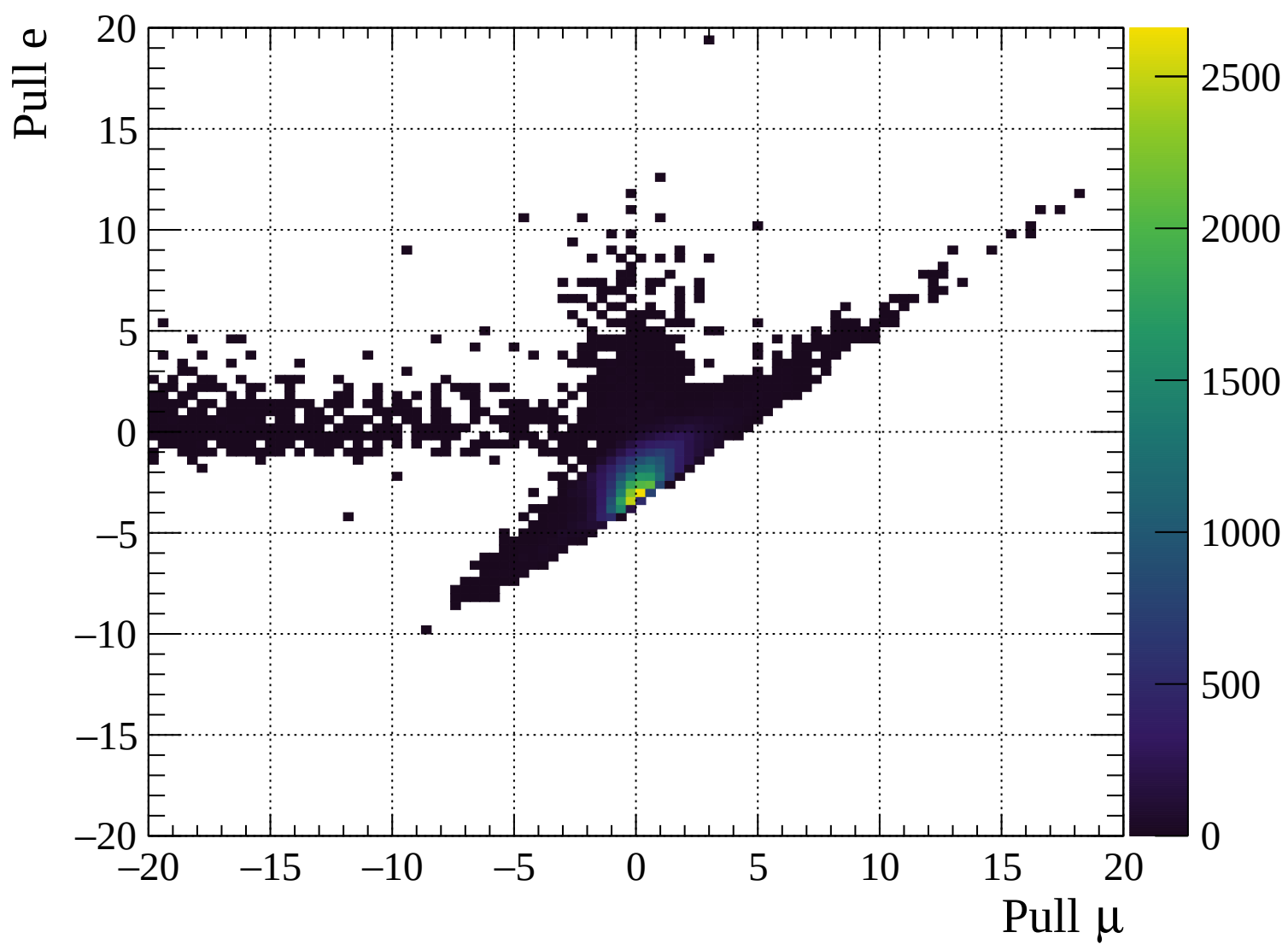


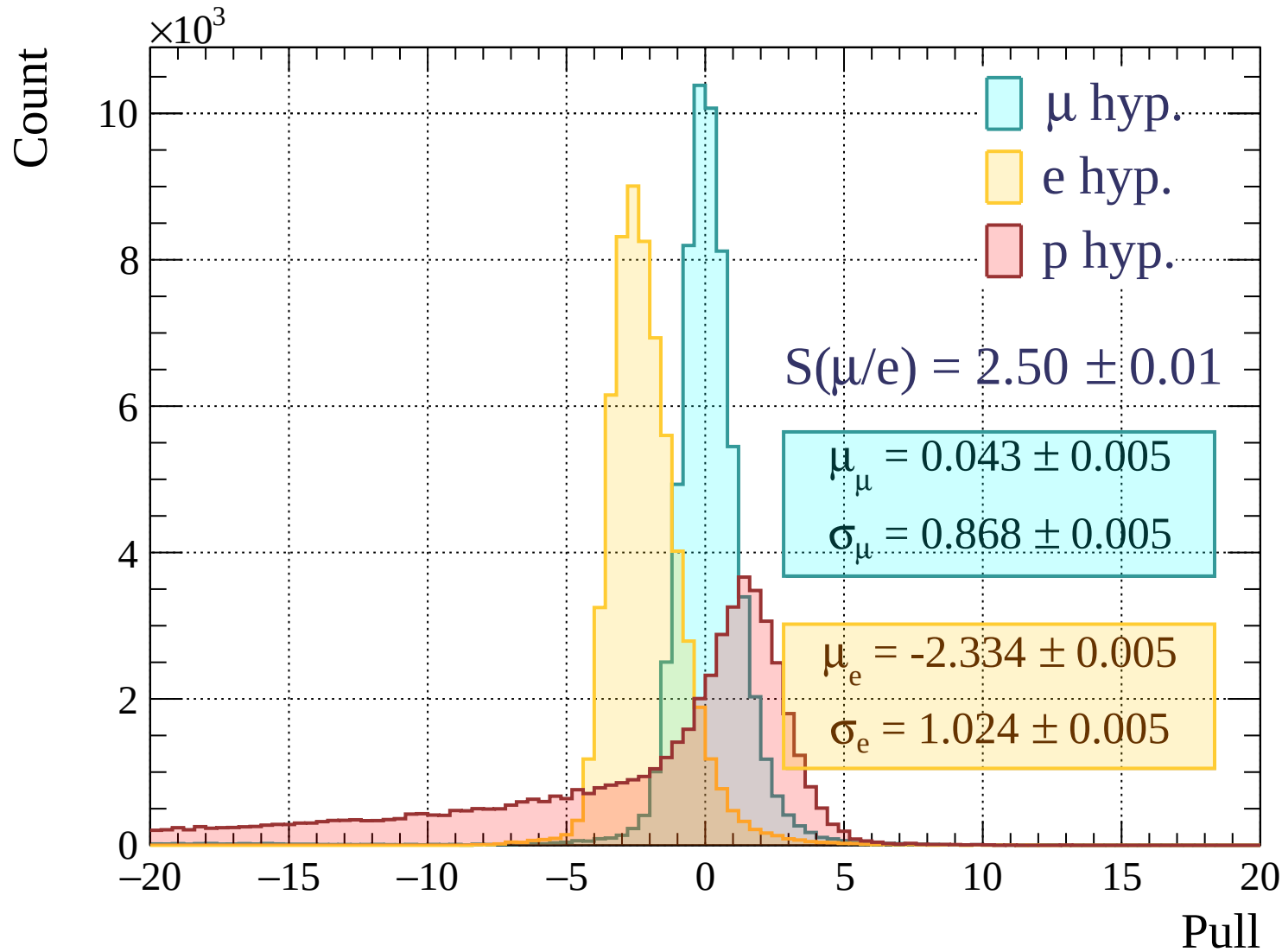
Pulls standard deviation

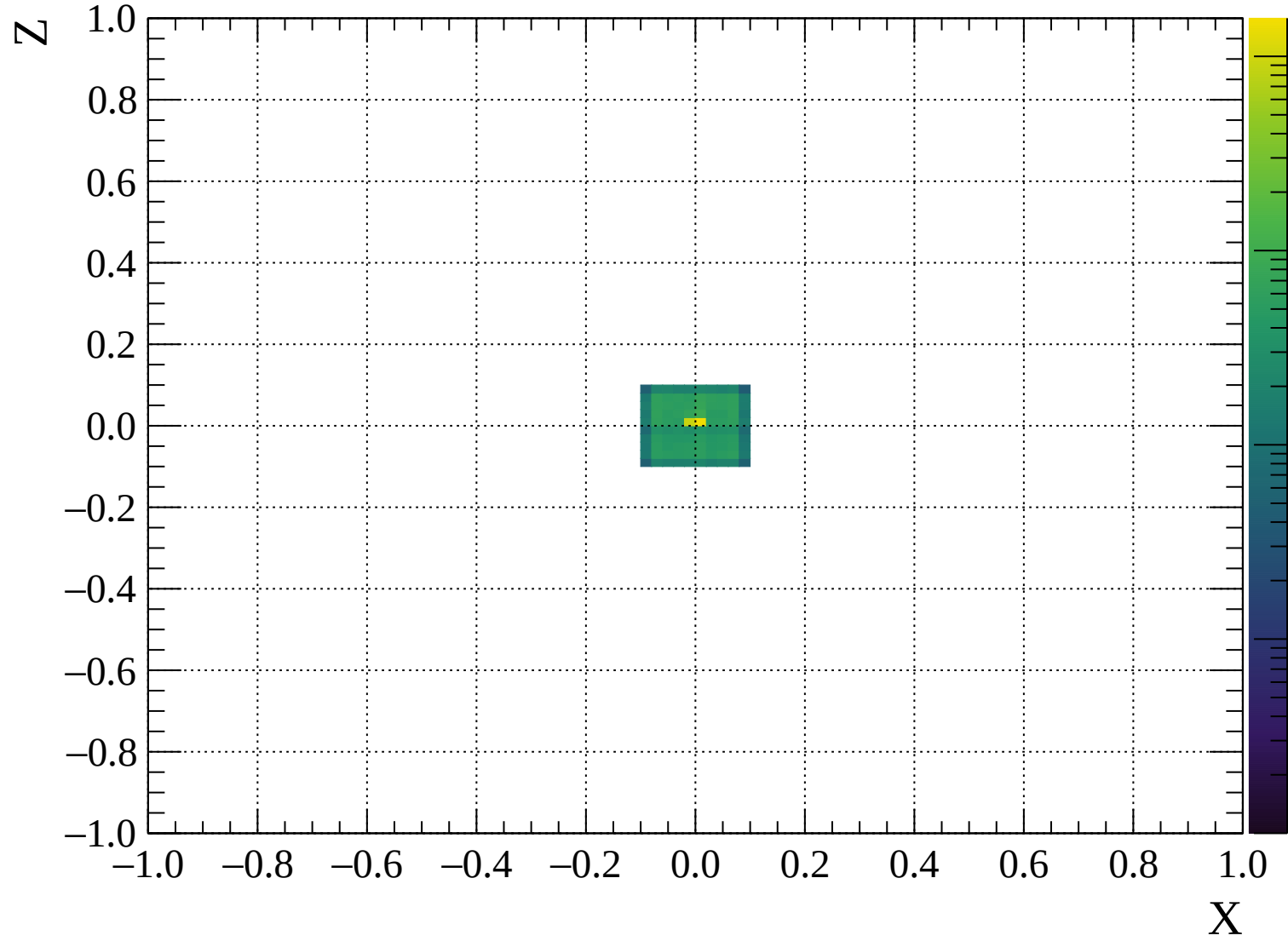


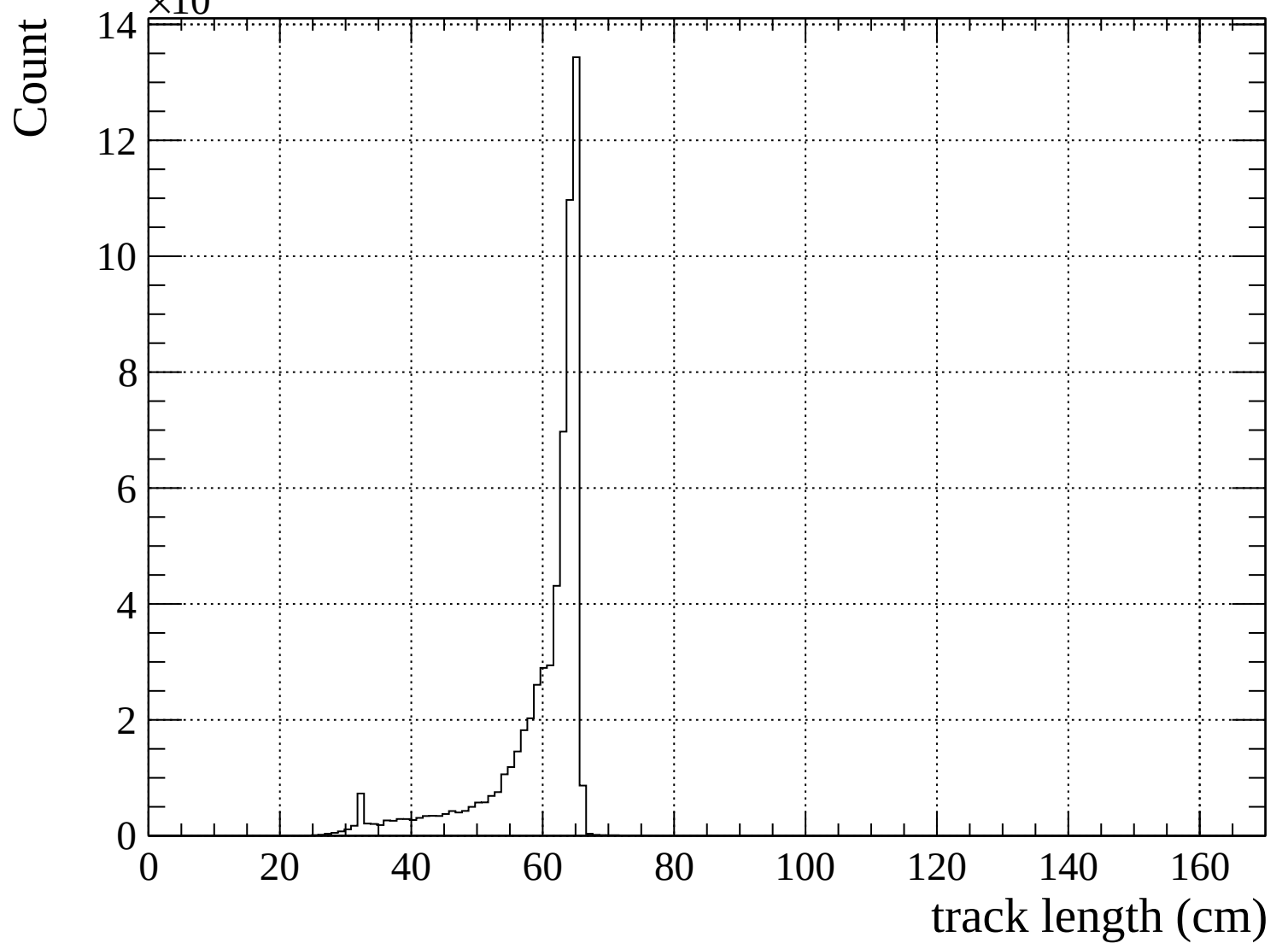
Pulls mean

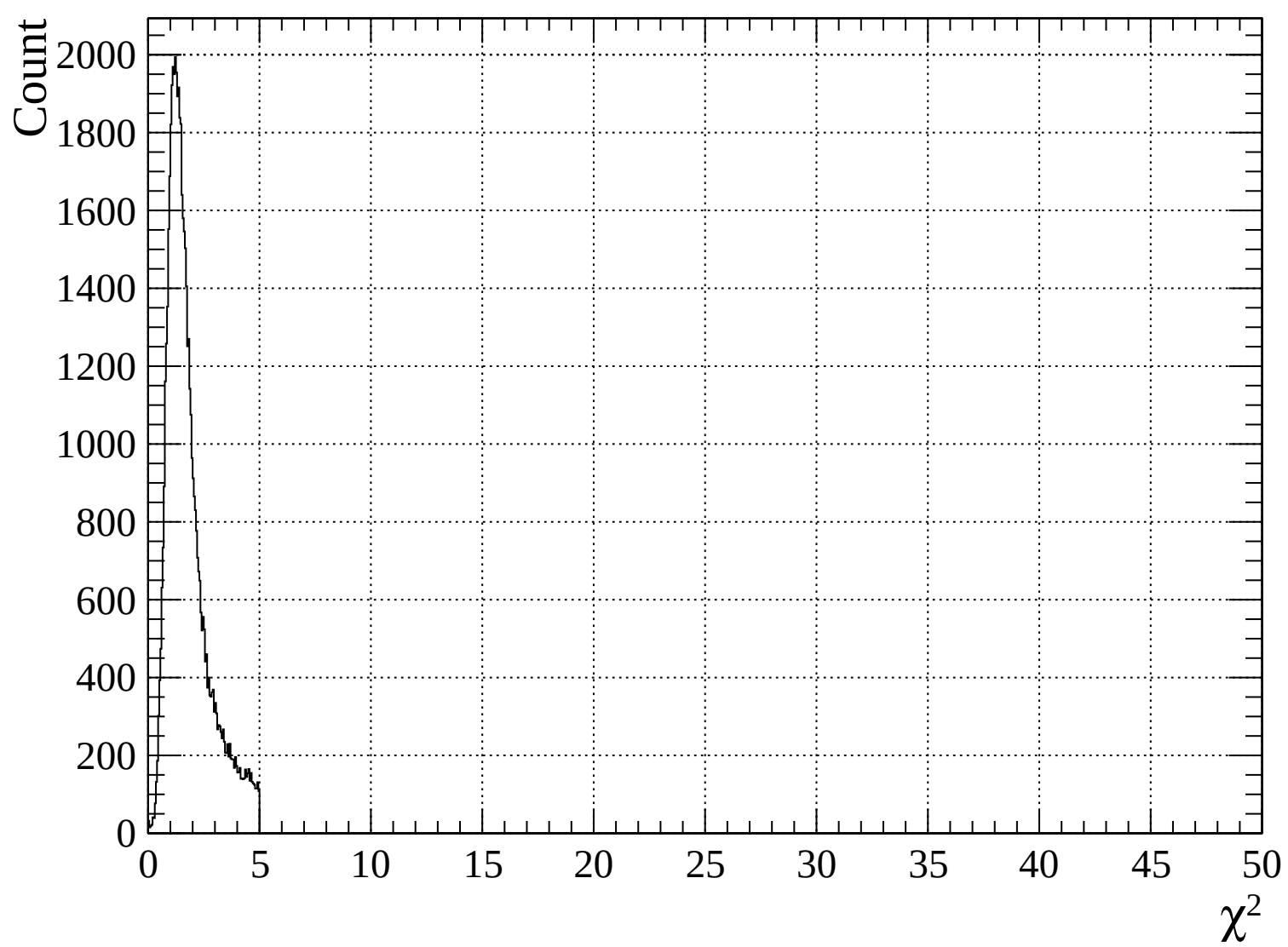


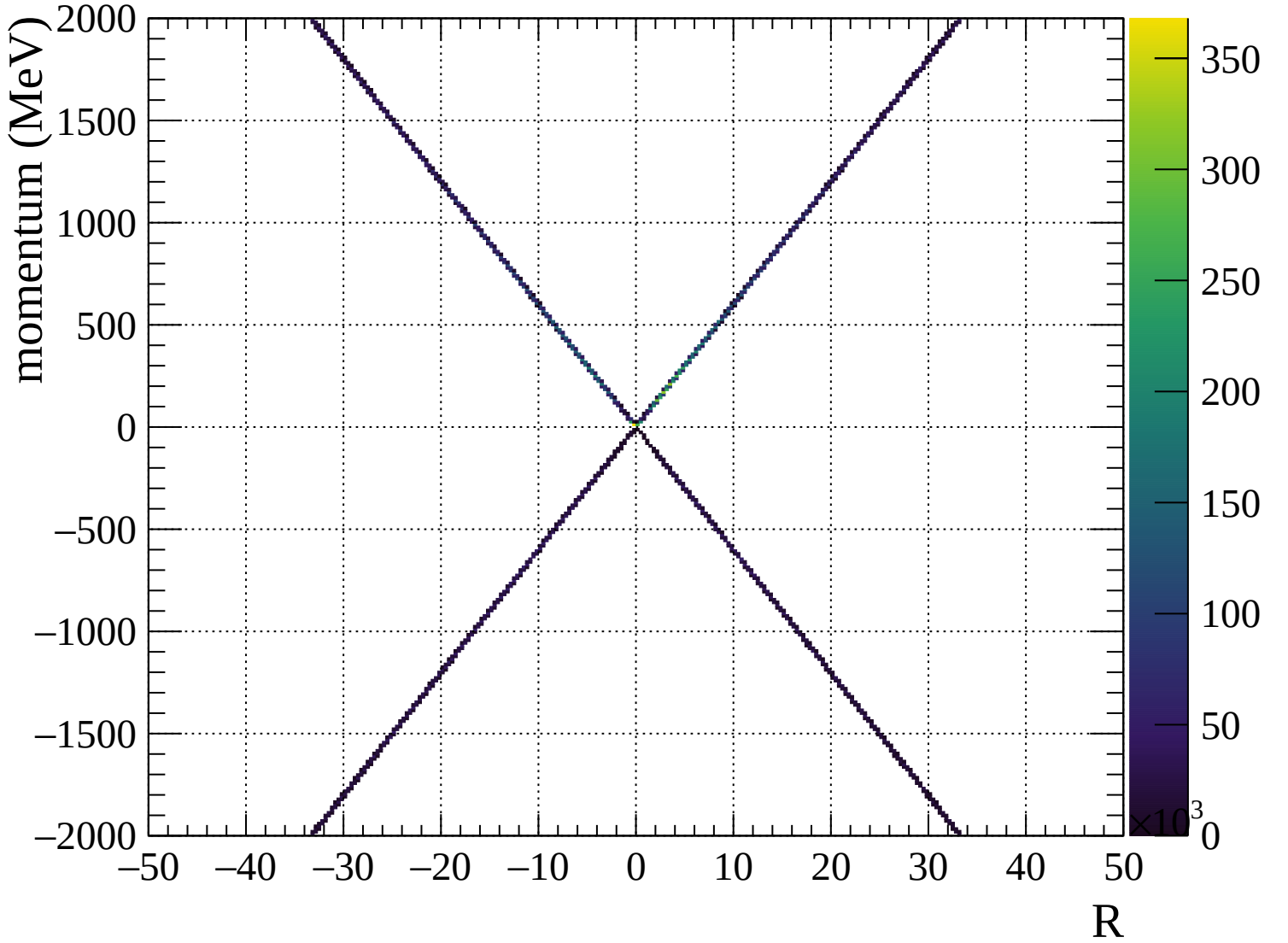


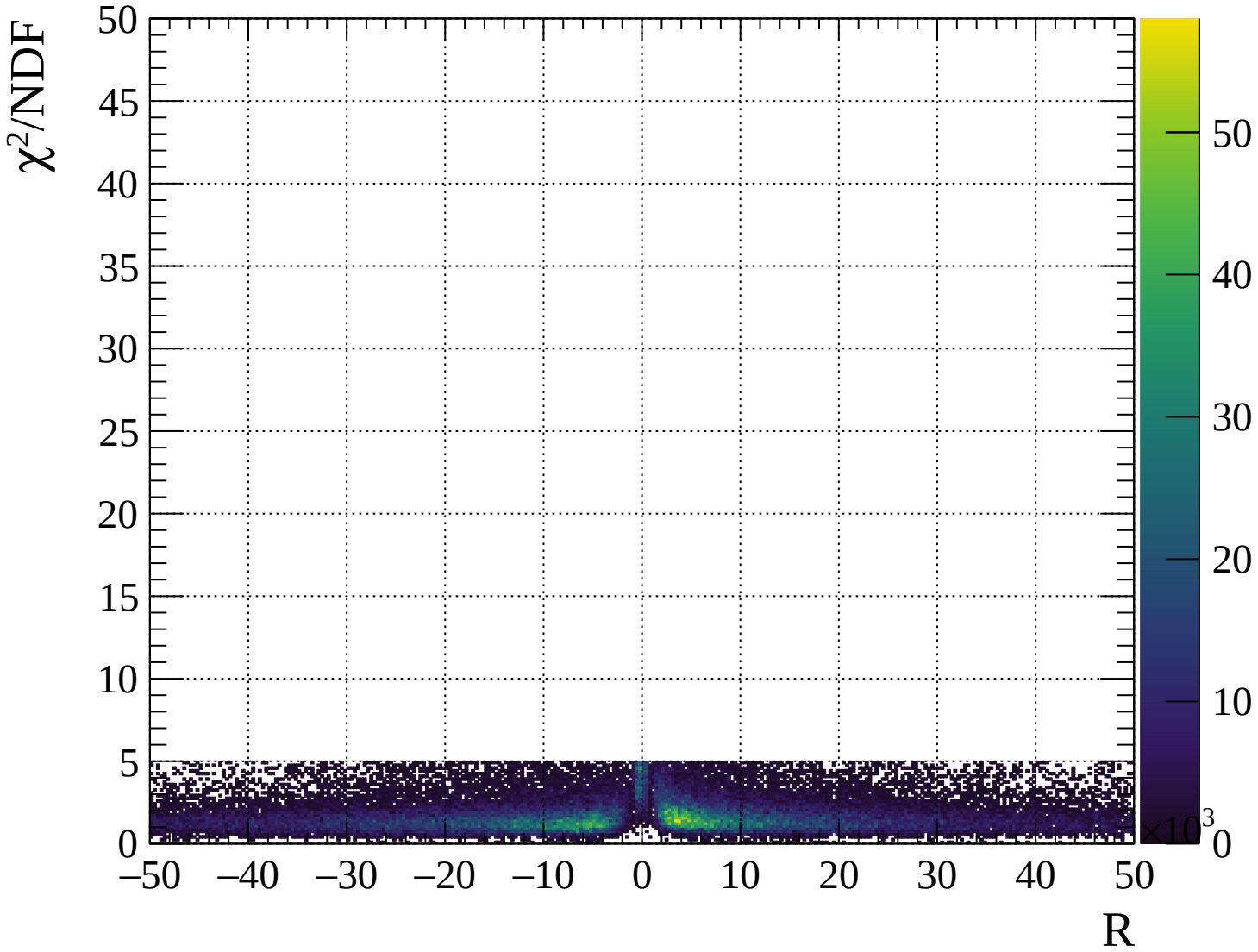






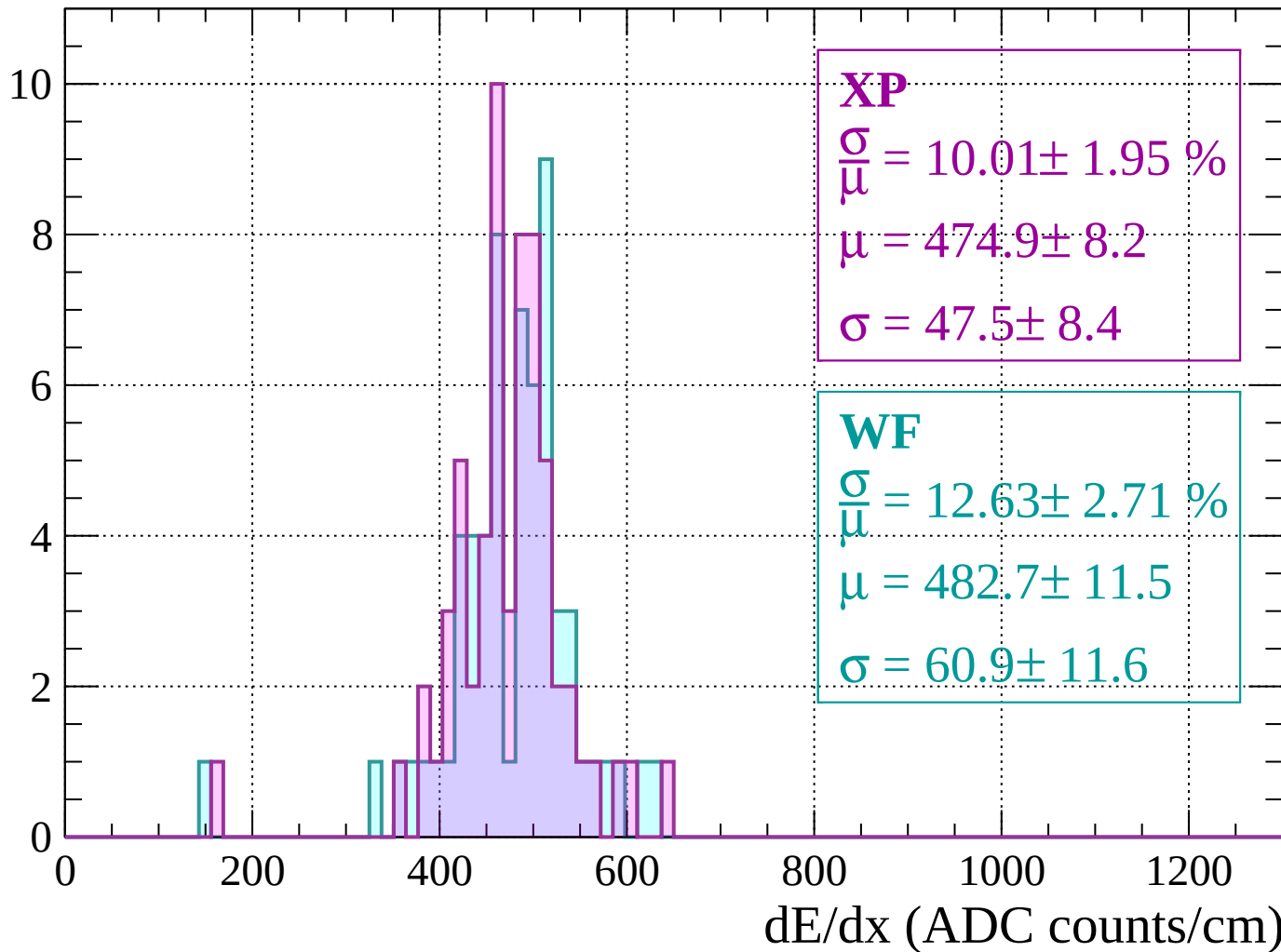






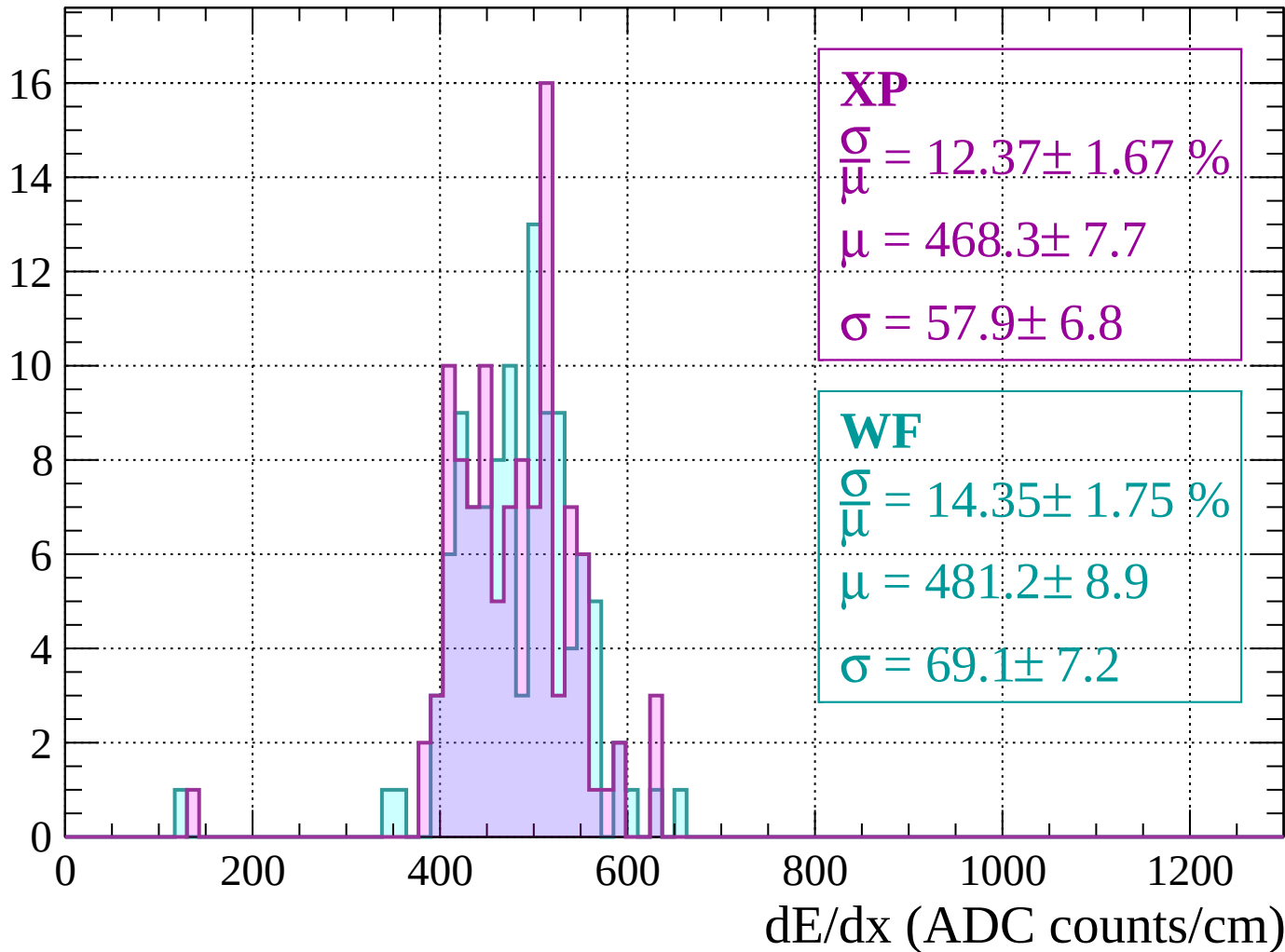
Energy loss | -2000 < p < -1960

Count



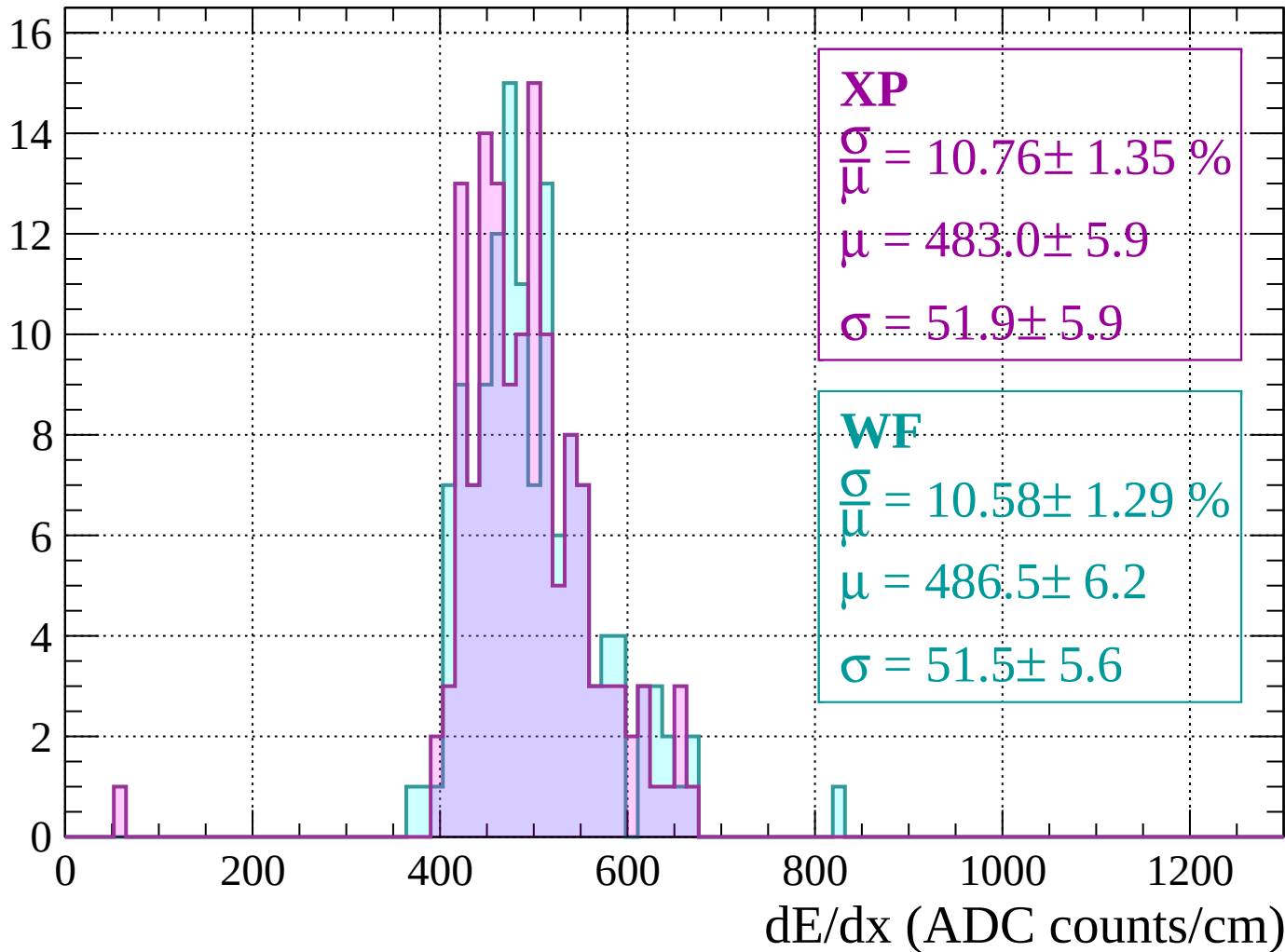
Energy loss | -1960 < p < -1920

Count



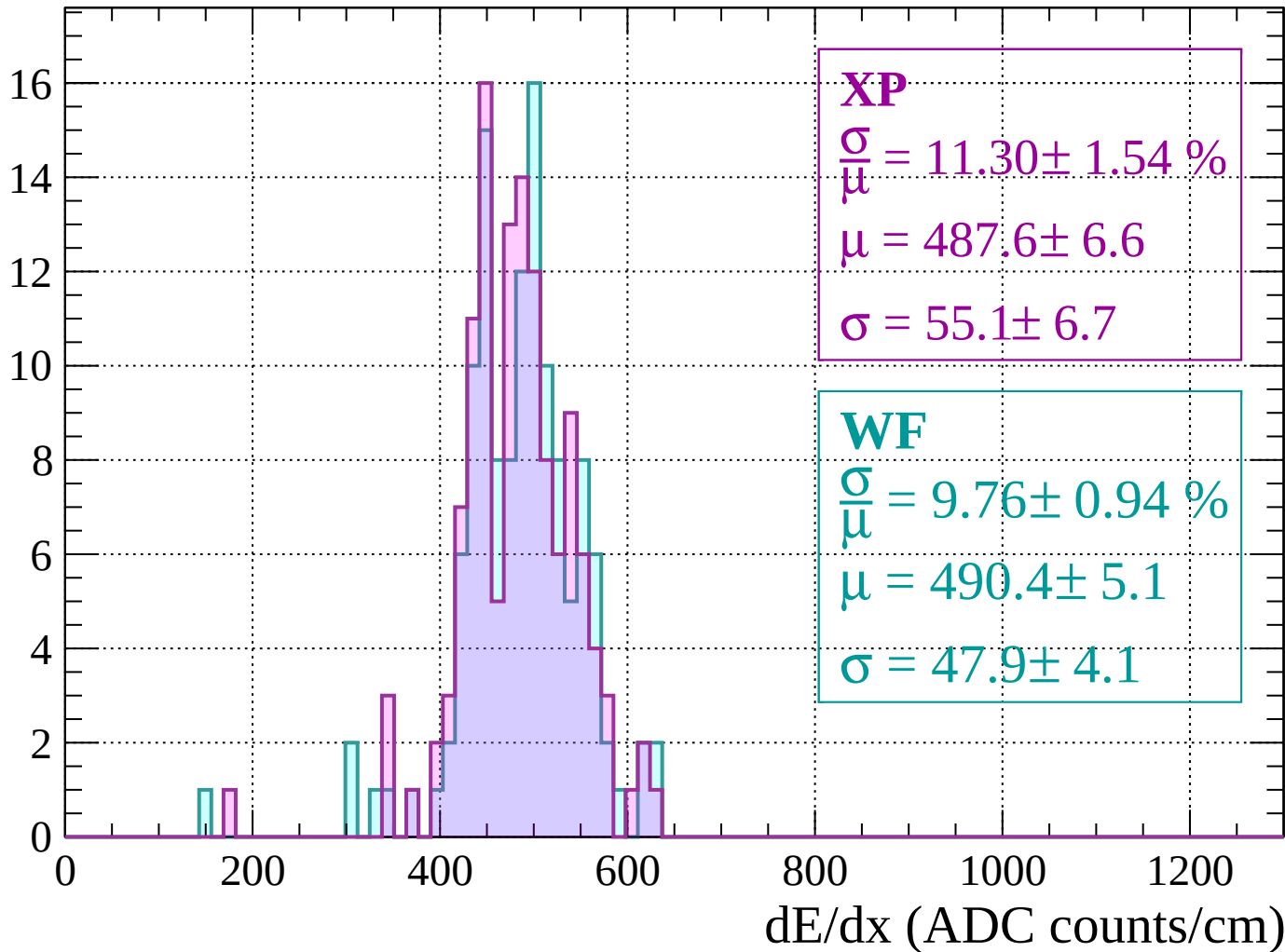
Energy loss | -1920 < p < -1880

Count



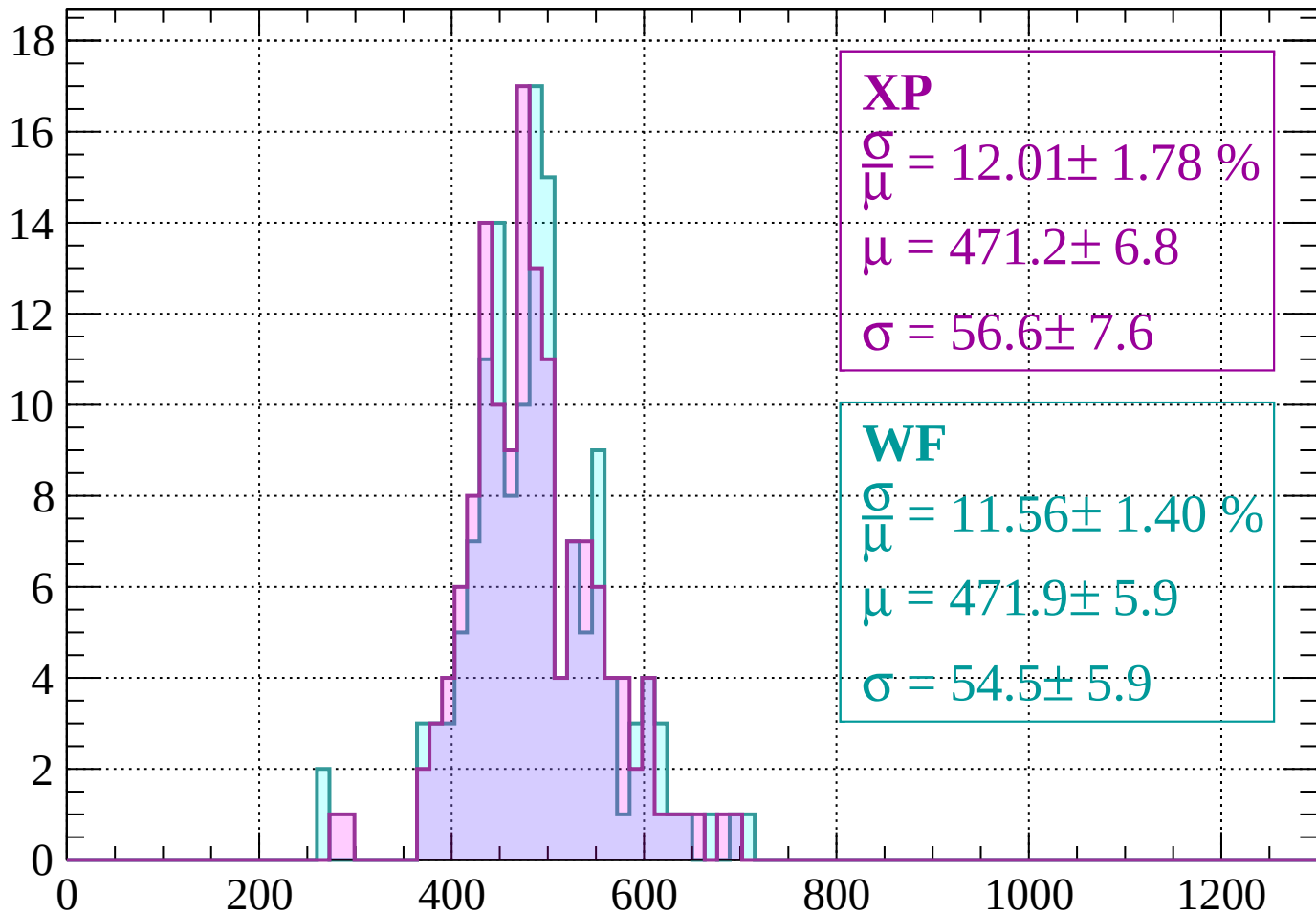
Energy loss | $-1880 < p < -1840$

Count



Energy loss | -1840 < p < -1800

Count



XP

$$\frac{\sigma}{\mu} = 12.01 \pm 1.78 \%$$

$$\mu = 471.2 \pm 6.8$$

$$\sigma = 56.6 \pm 7.6$$

WF

$$\frac{\sigma}{\mu} = 11.56 \pm 1.40 \%$$

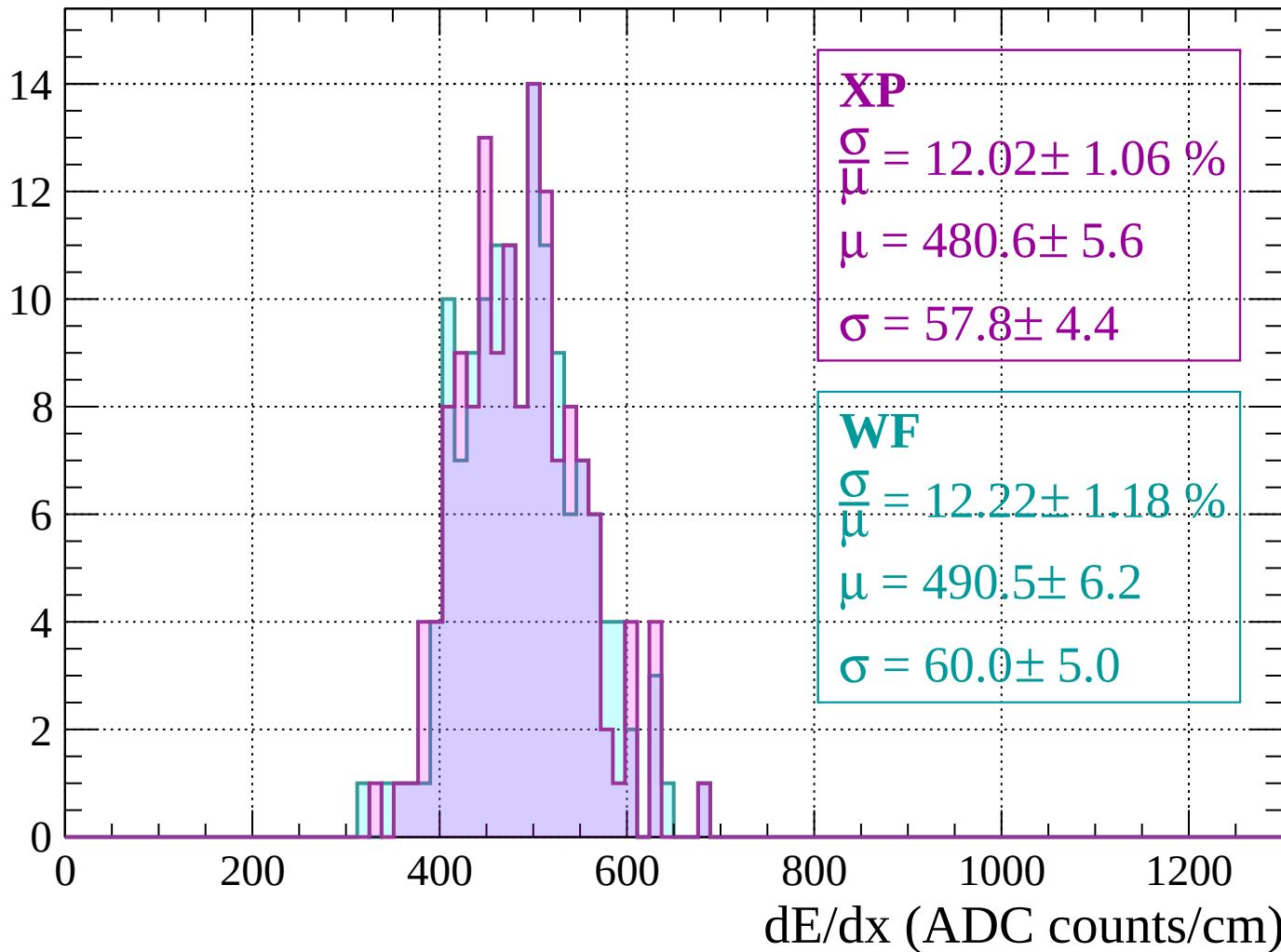
$$\mu = 471.9 \pm 5.9$$

$$\sigma = 54.5 \pm 5.9$$

dE/dx (ADC counts/cm)

Energy loss | -1800 < p < -1760

Count



Energy loss | $-1760 < p < -1720$

Count

XP

$$\frac{\sigma}{\mu} = 10.12 \pm 1.43 \%$$

$$\mu = 481.6 \pm 6.0$$

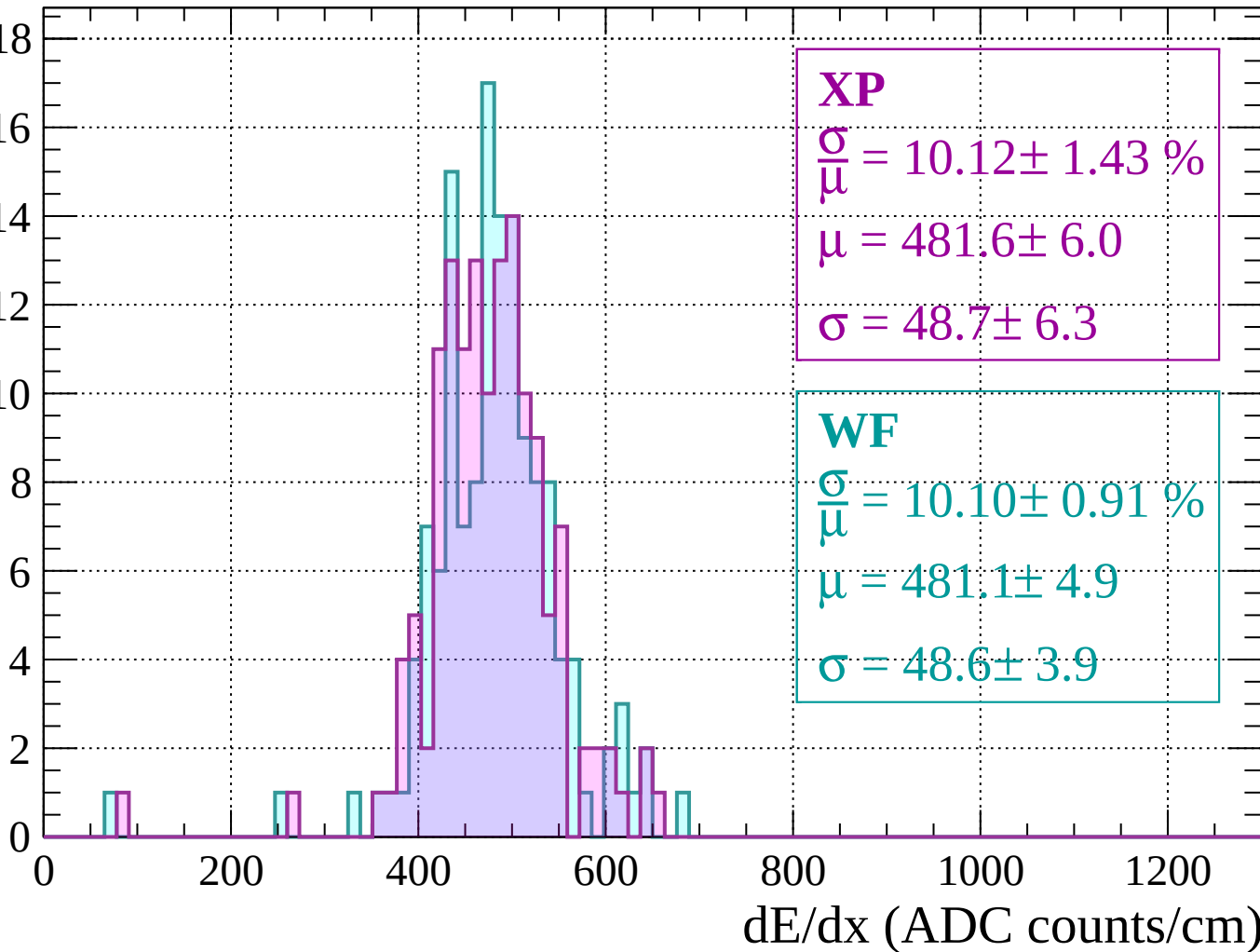
$$\sigma = 48.7 \pm 6.3$$

WF

$$\frac{\sigma}{\mu} = 10.10 \pm 0.91 \%$$

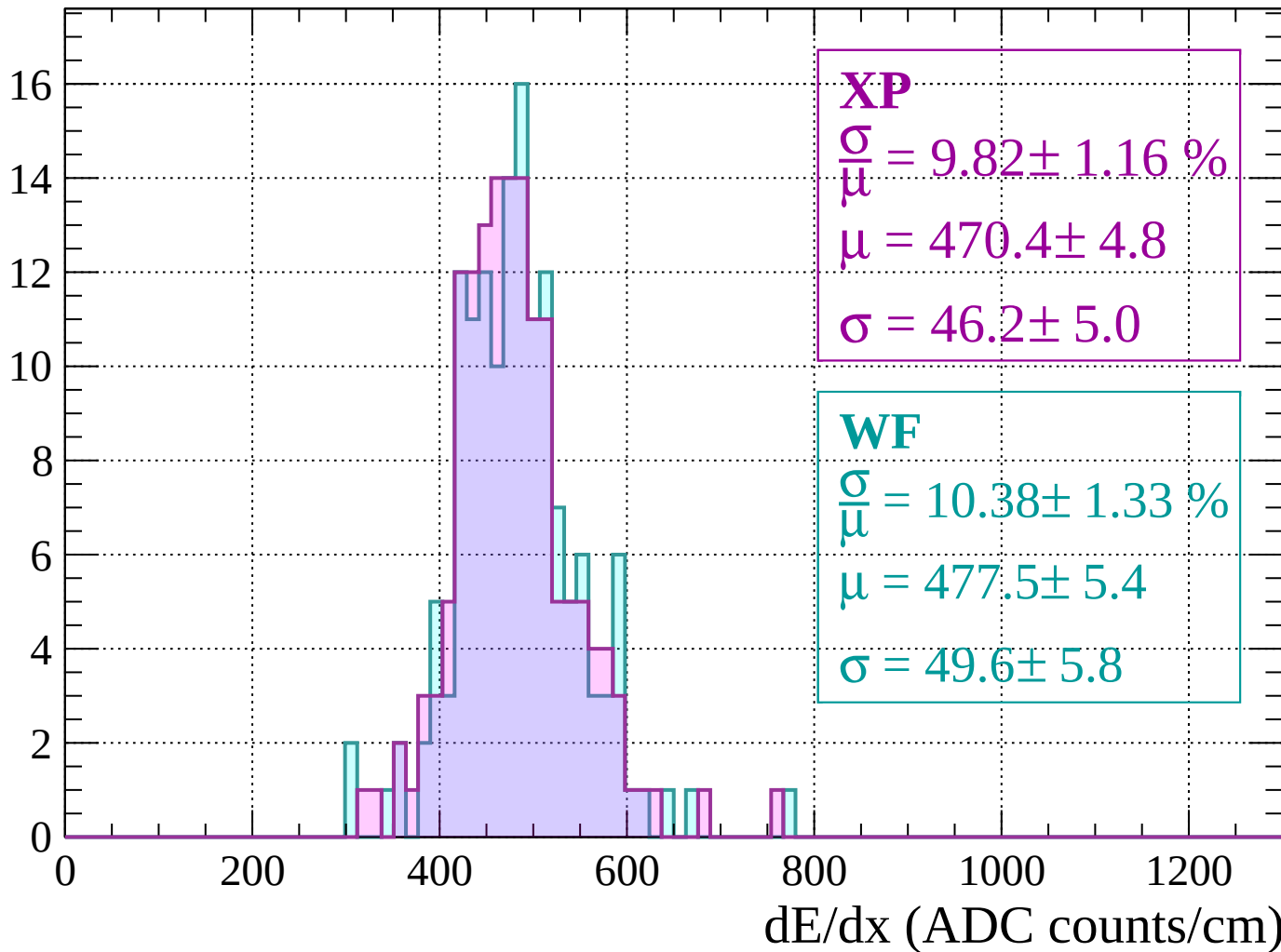
$$\mu = 481.1 \pm 4.9$$

$$\sigma = 48.6 \pm 3.9$$



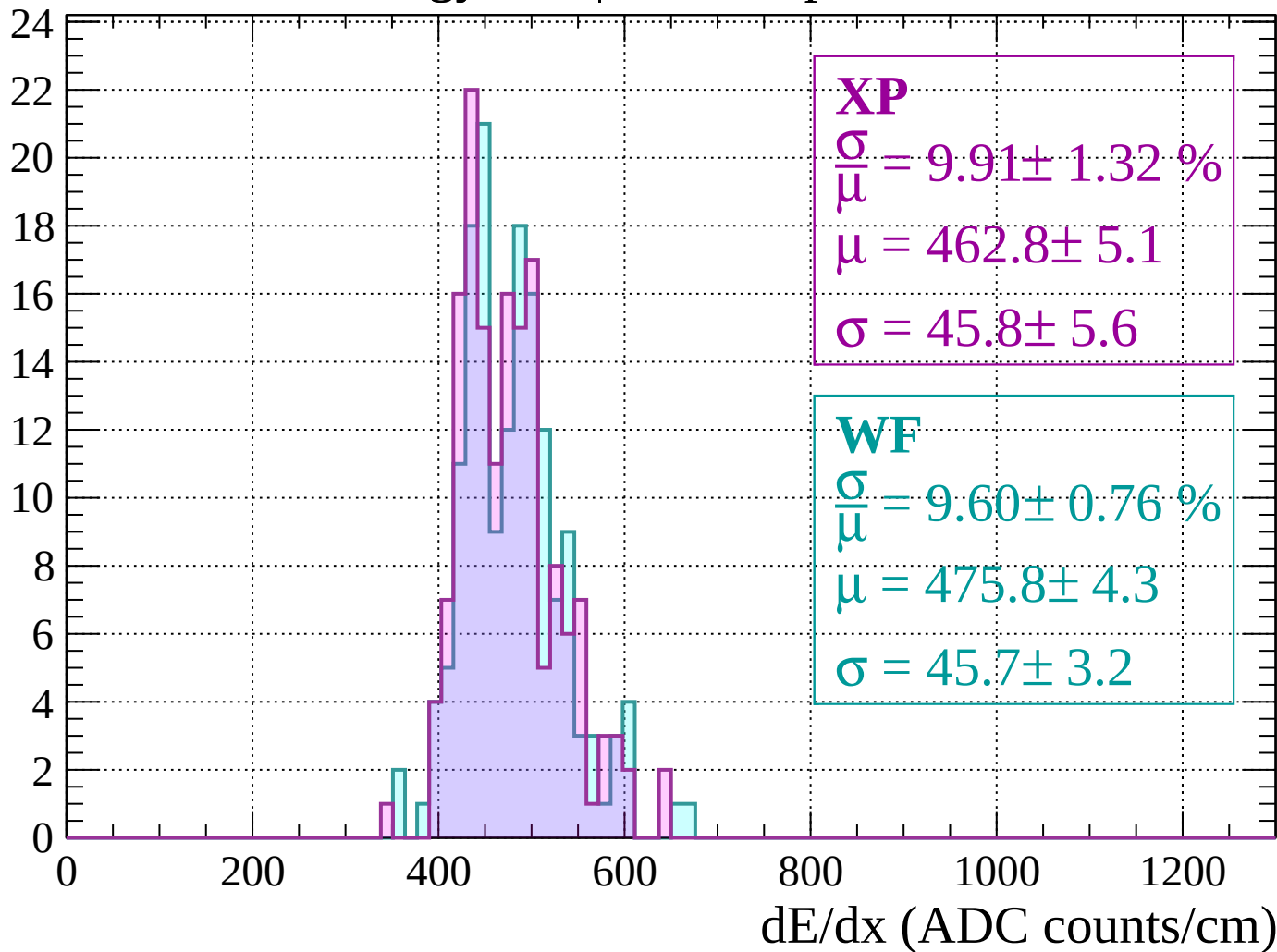
Energy loss | $-1720 < p < -1680$

Count



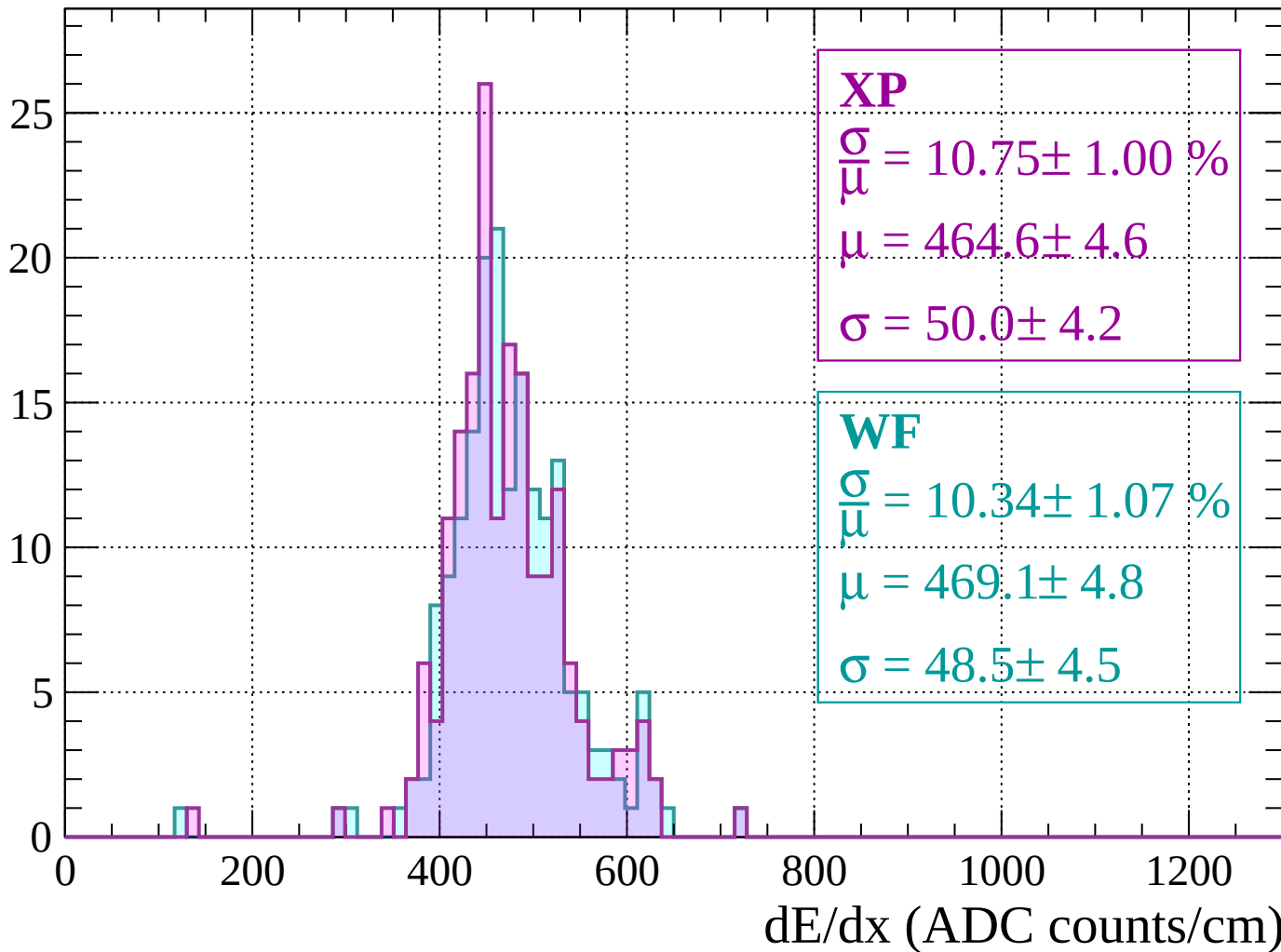
Energy loss | -1680 < p < -1640

Count



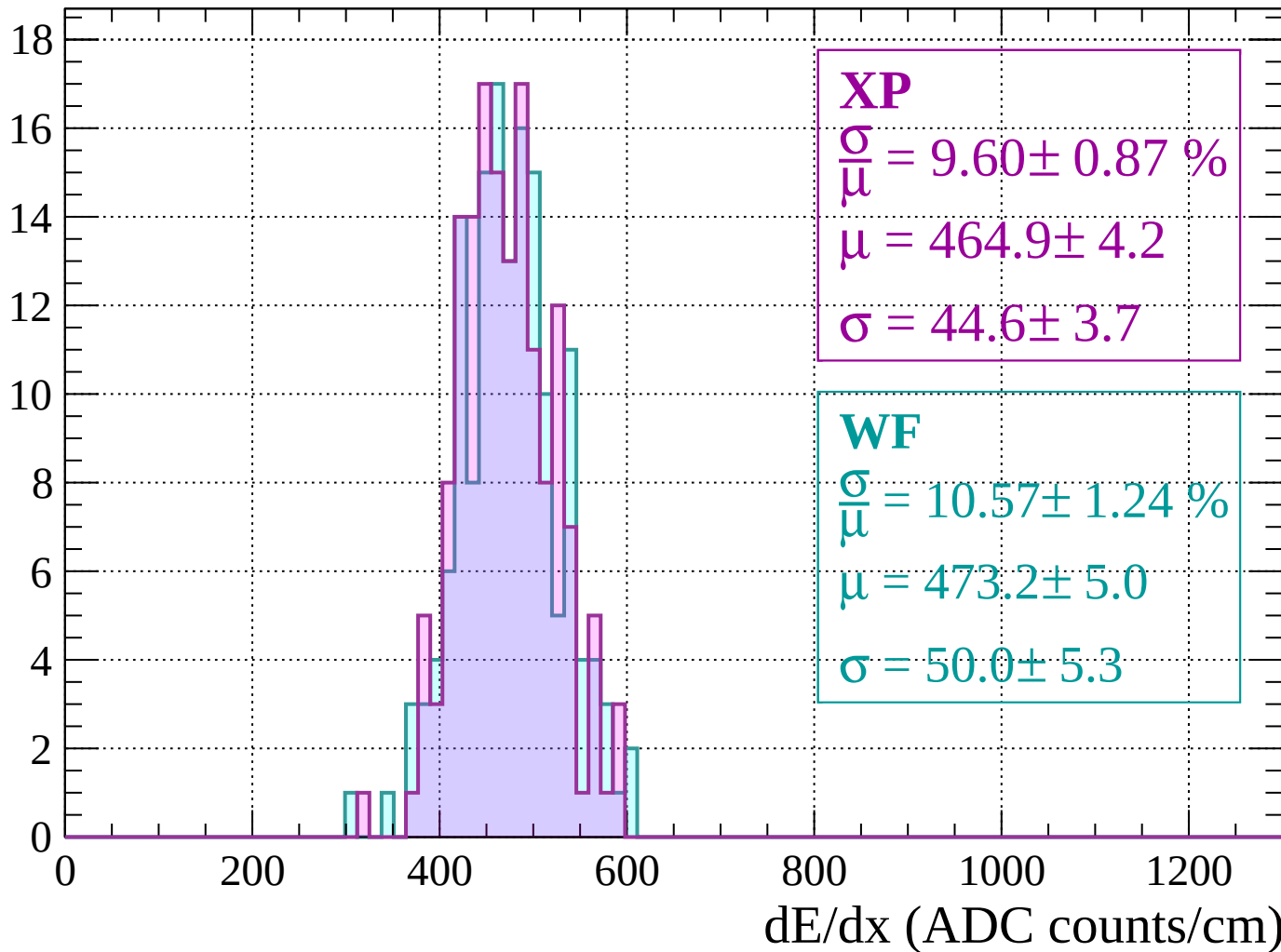
Energy loss | $-1640 < p < -1600$

Count



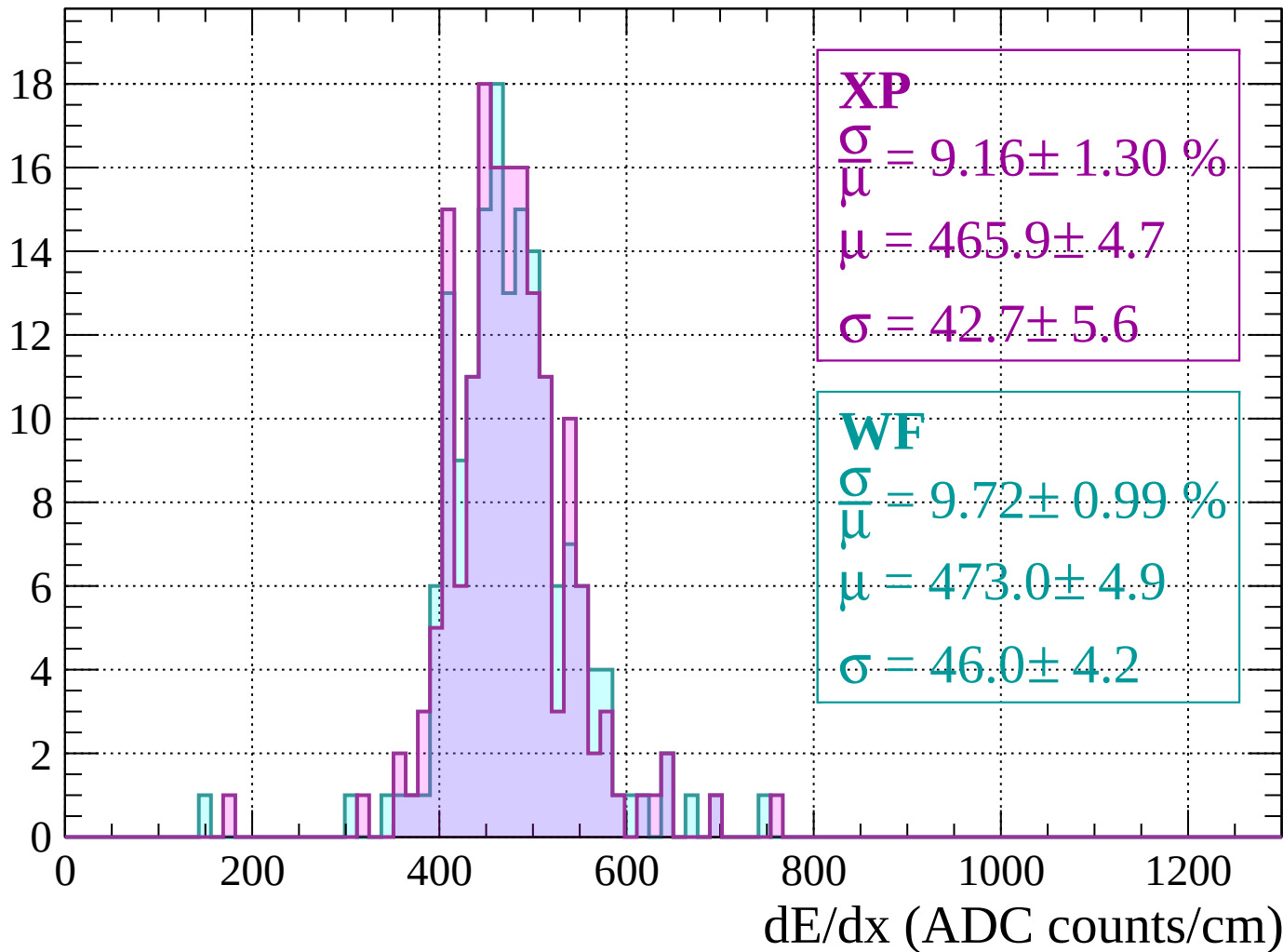
Energy loss | $-1600 < p < -1560$

Count



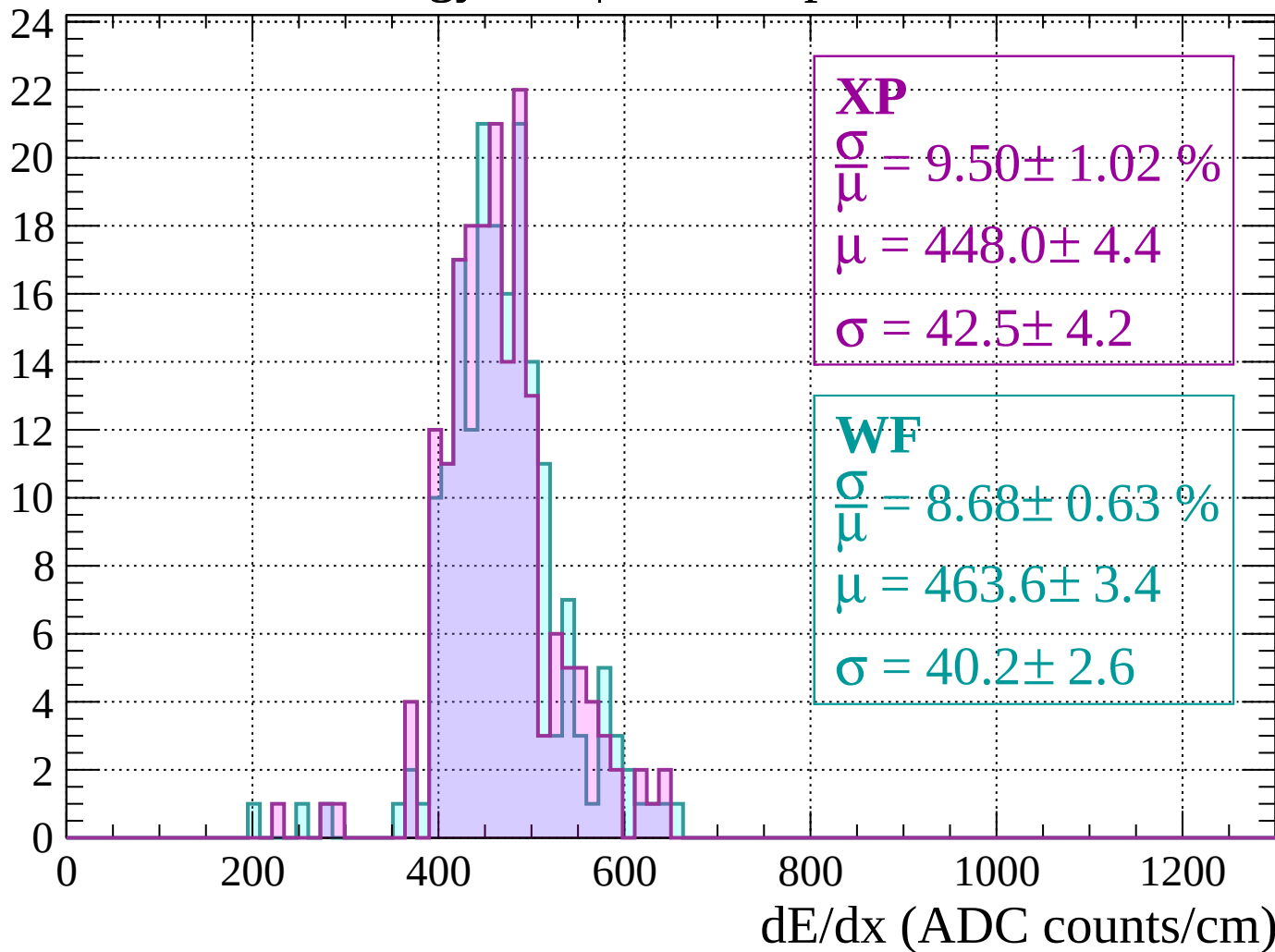
Energy loss | $-1560 < p < -1520$

Count



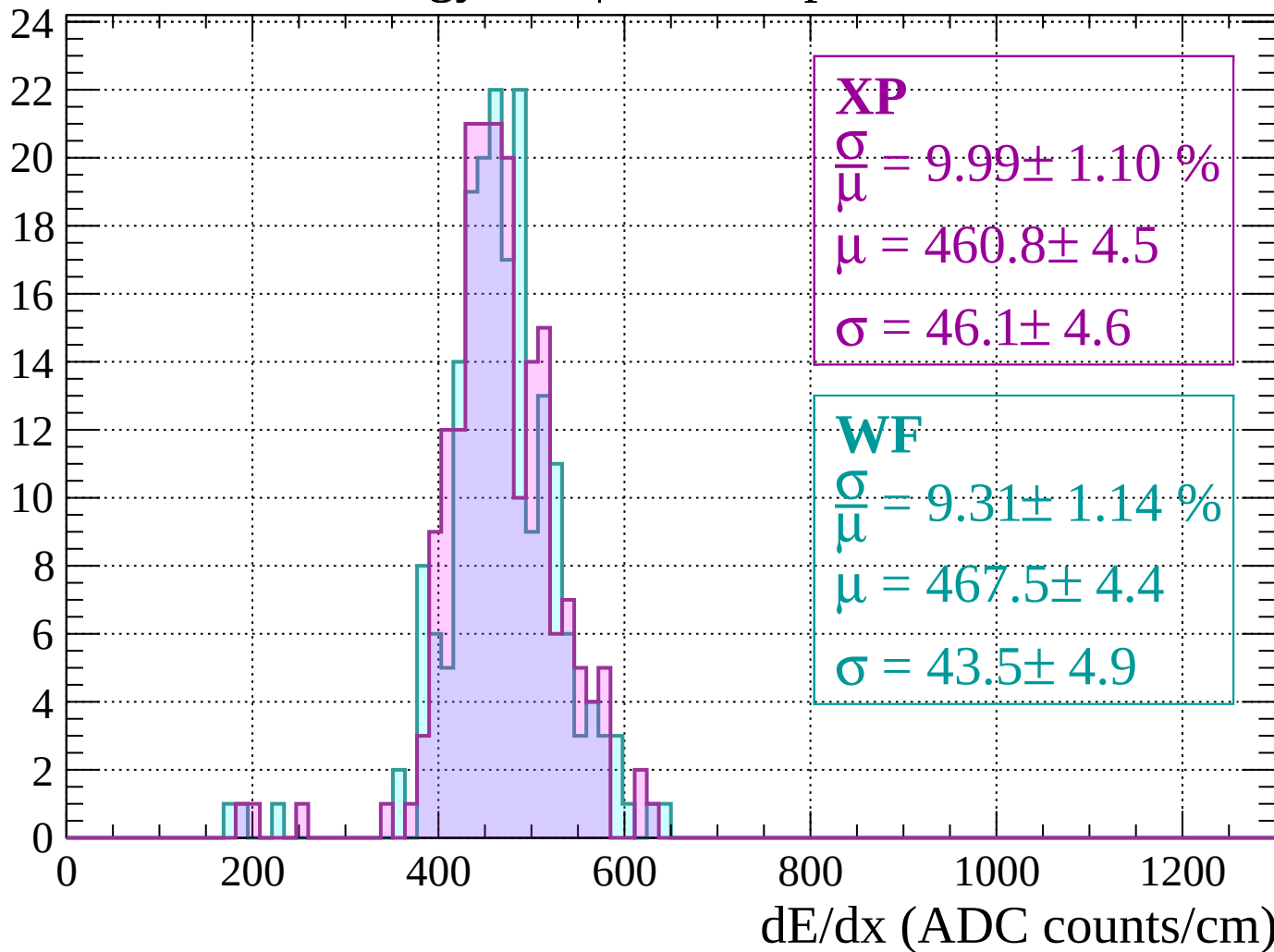
Energy loss | -1520 < p < -1480

Count



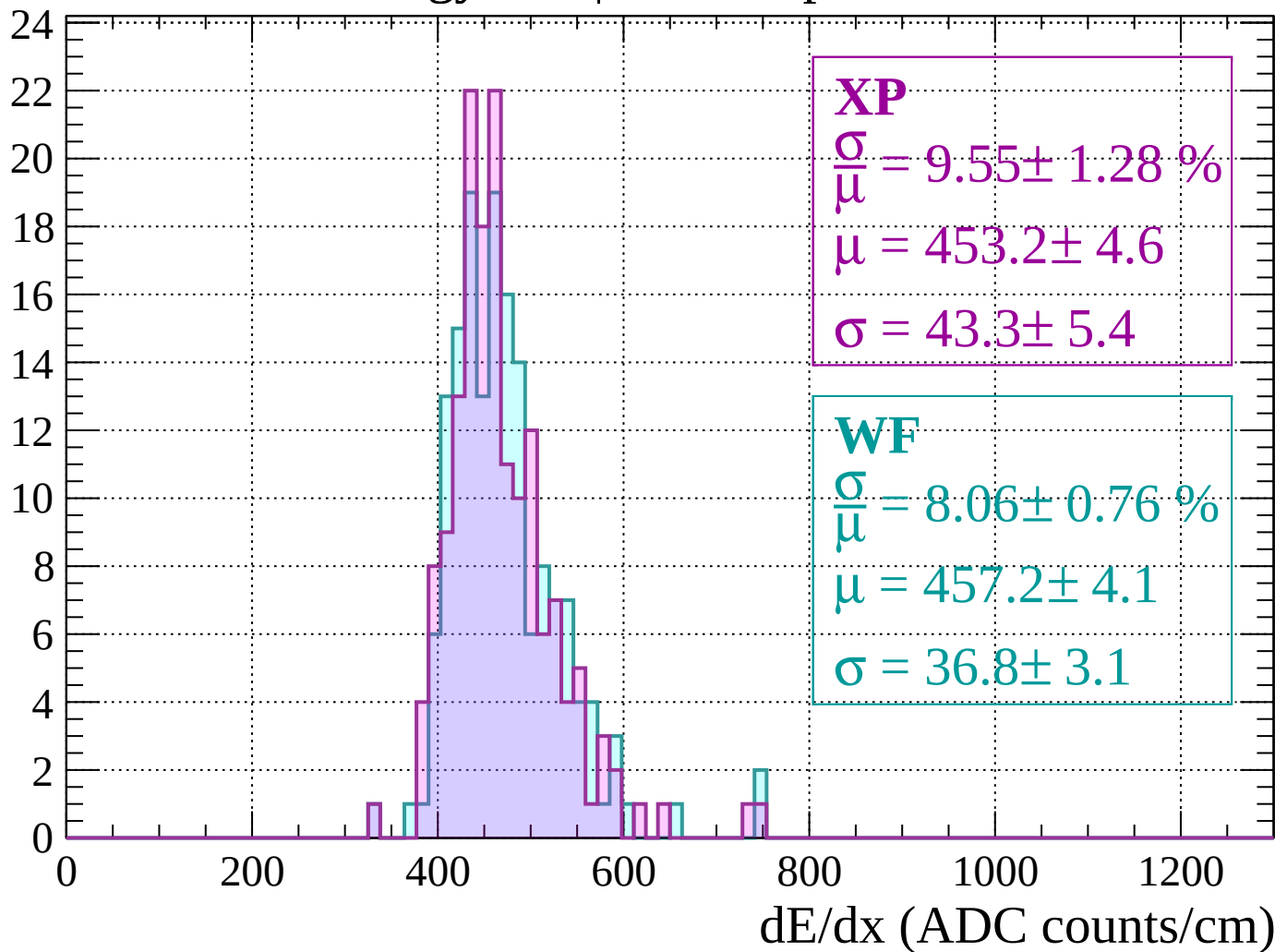
Energy loss | -1480 < p < -1440

Count



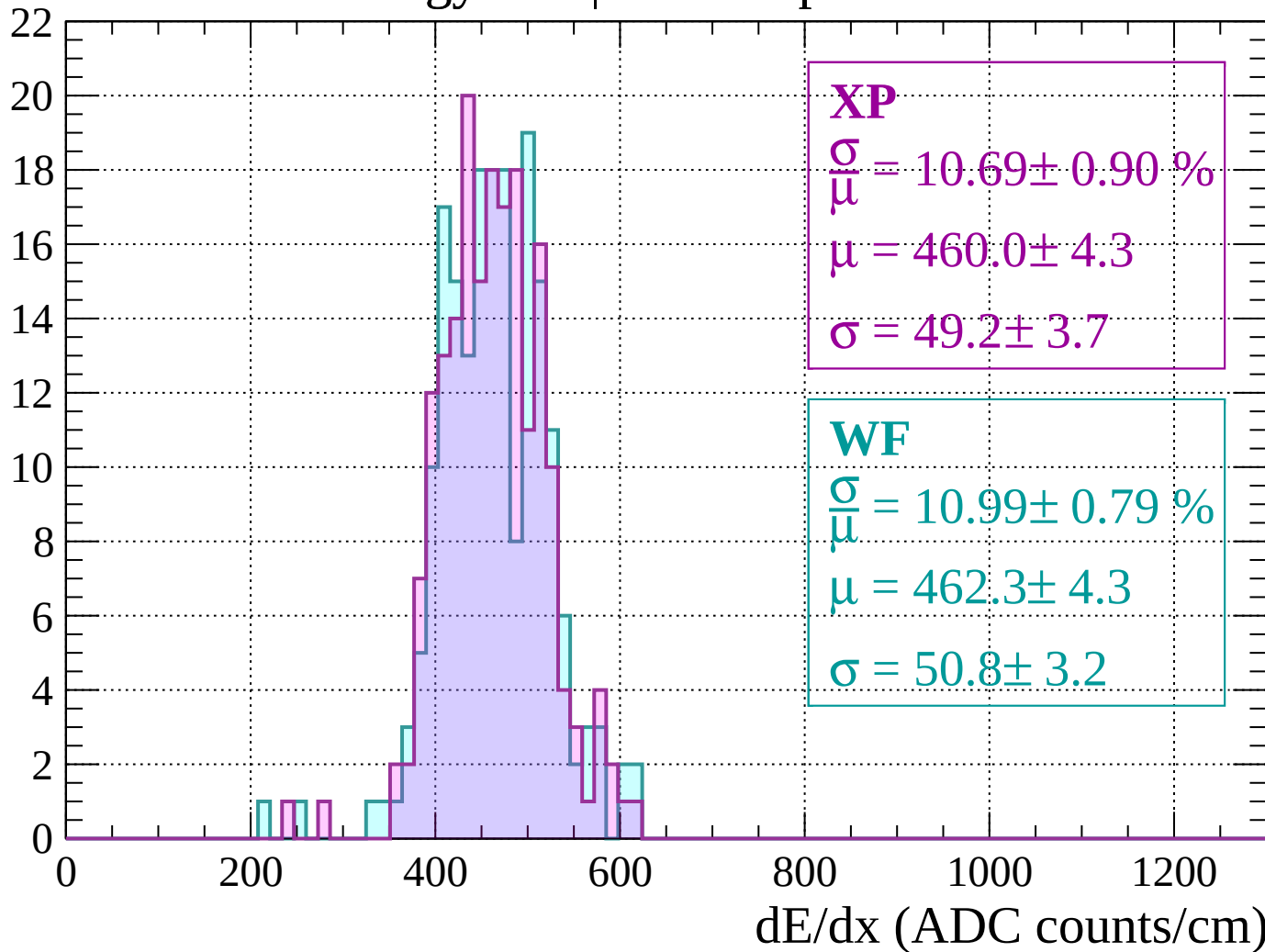
Energy loss | $-1440 < p < -1400$

Count



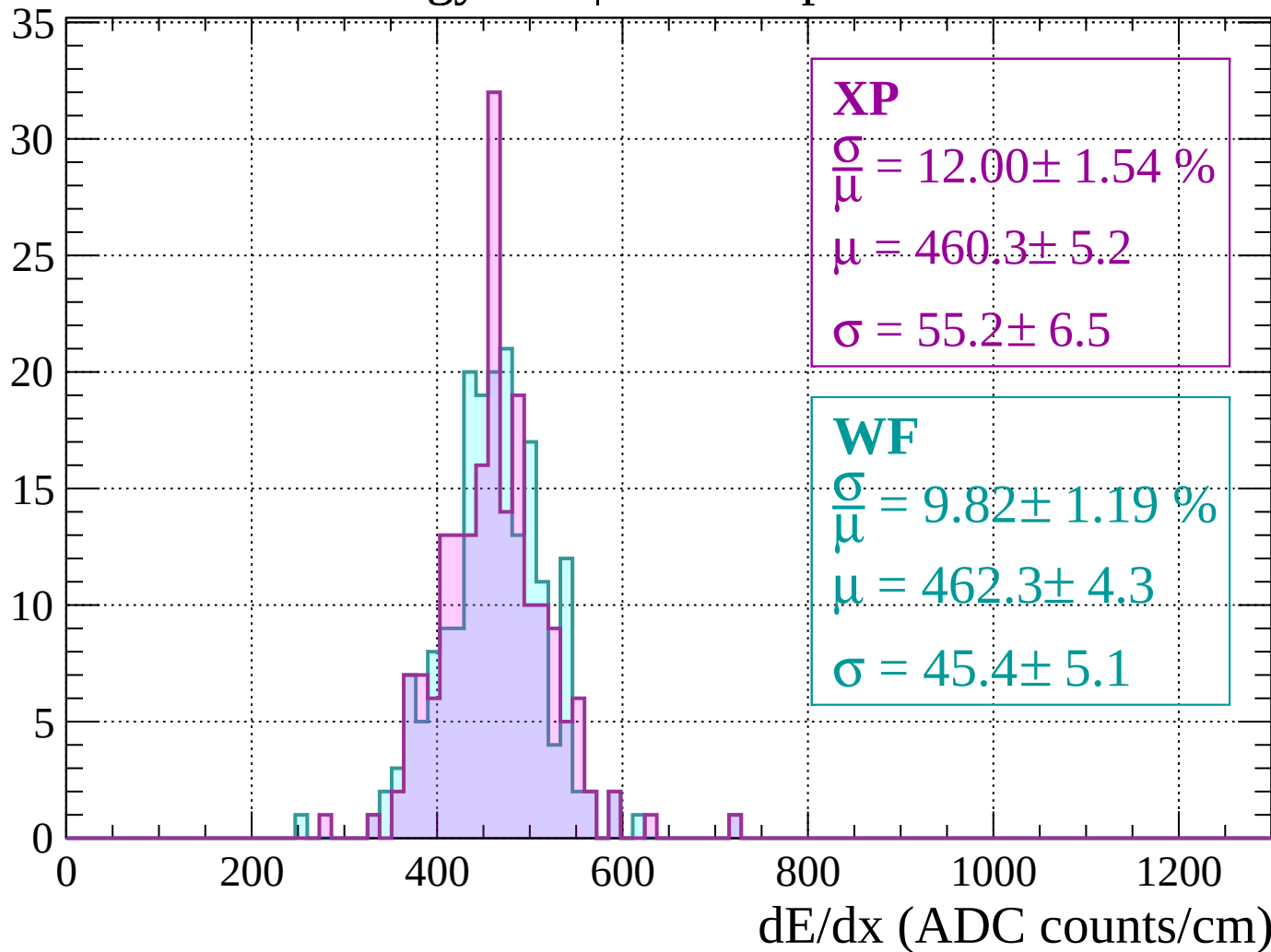
Energy loss | $-1400 < p < -1360$

Count



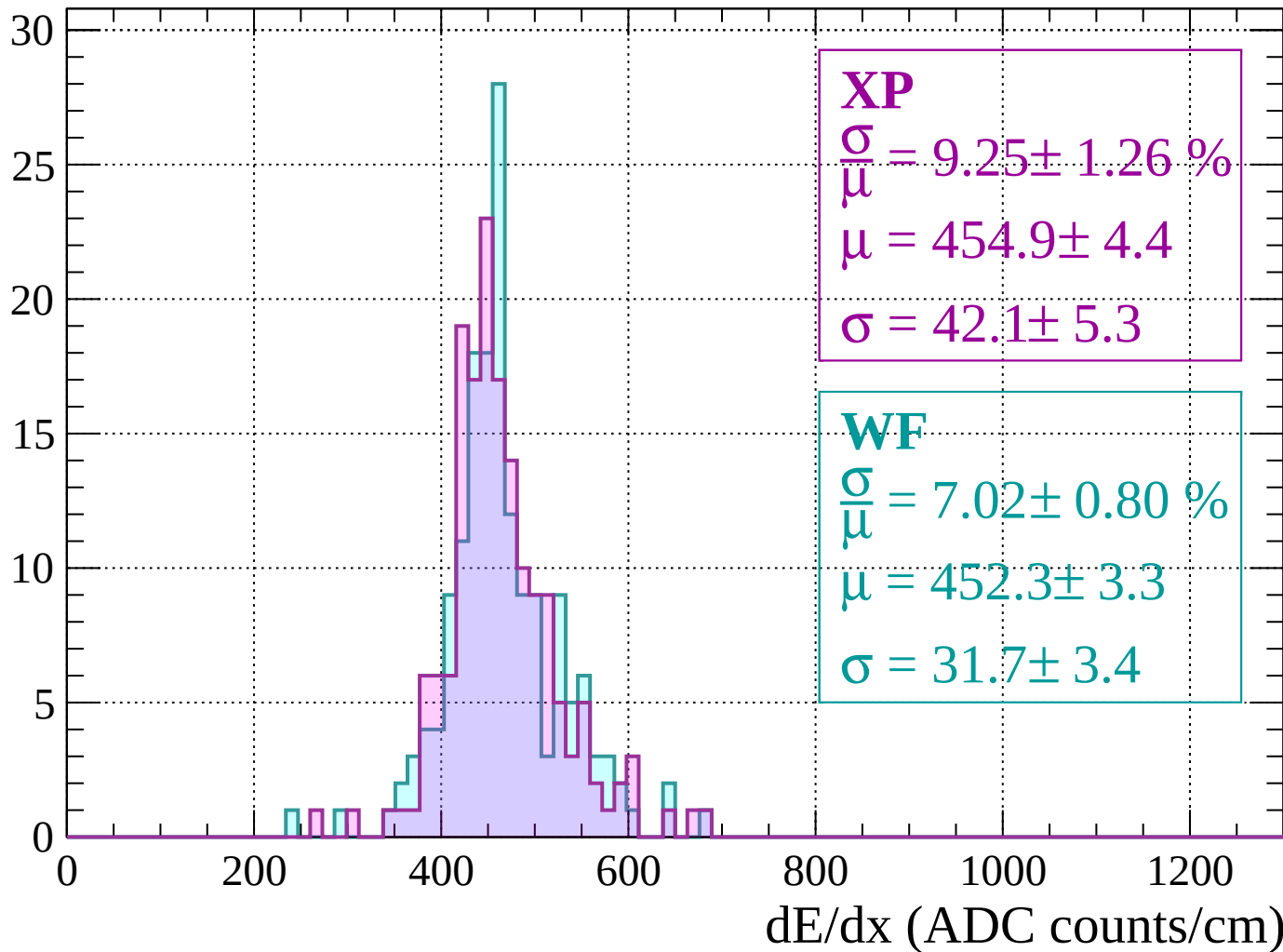
Energy loss | $-1360 < p < -1320$

Count



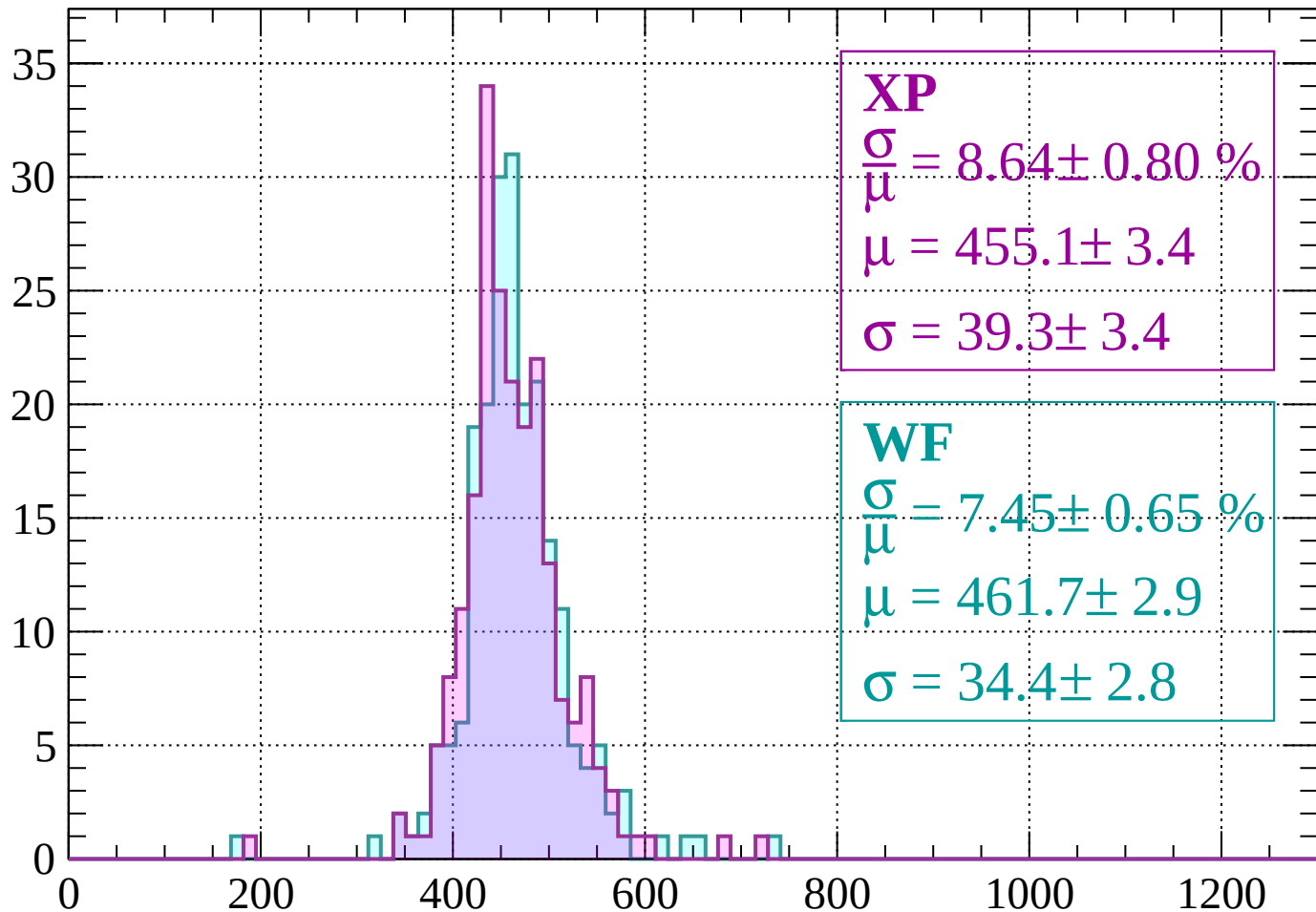
Energy loss | $-1320 < p < -1280$

Count



Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 8.64 \pm 0.80 \%$$

$$\mu = 455.1 \pm 3.4$$

$$\sigma = 39.3 \pm 3.4$$

WF

$$\frac{\sigma}{\mu} = 7.45 \pm 0.65 \%$$

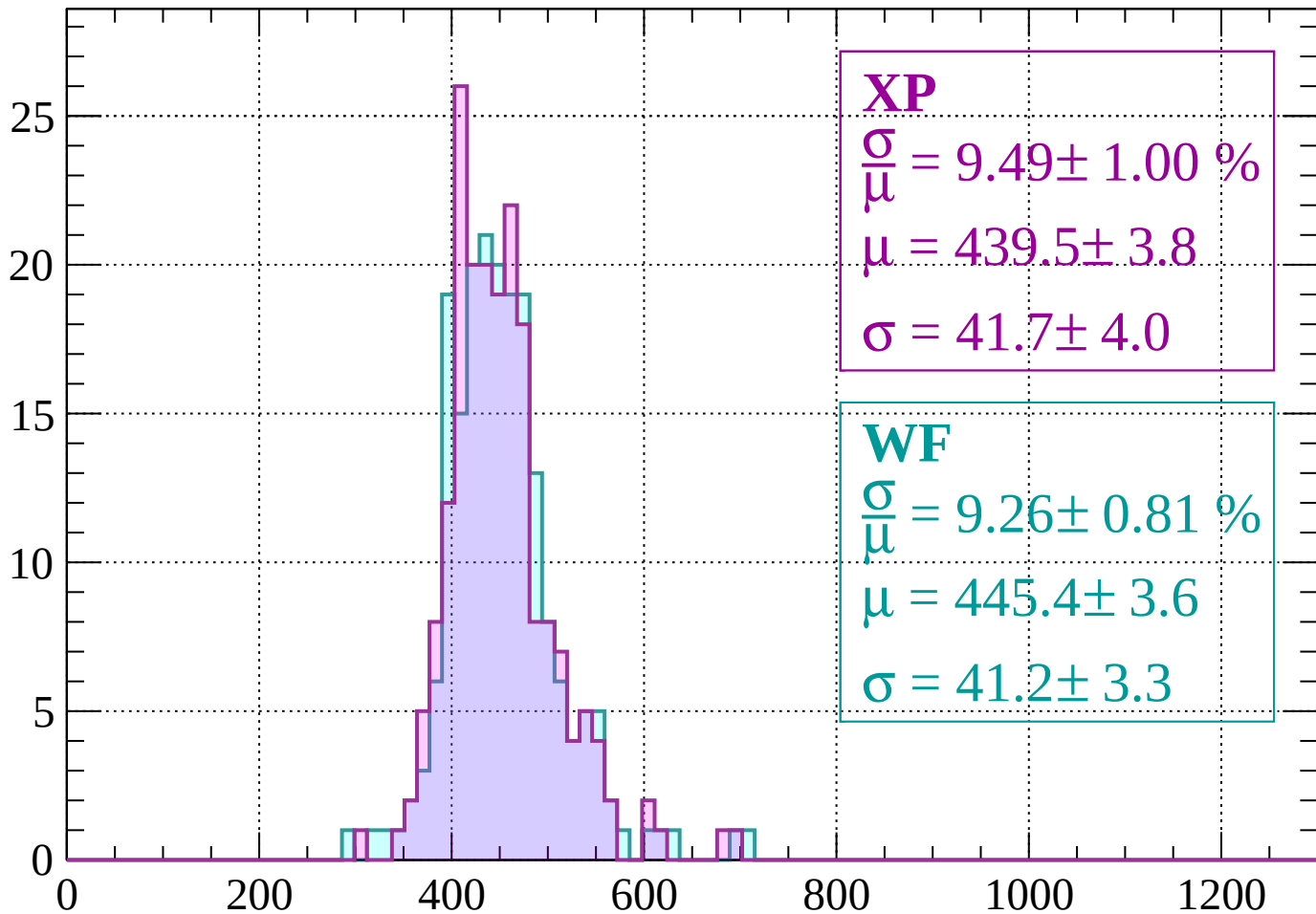
$$\mu = 461.7 \pm 2.9$$

$$\sigma = 34.4 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 9.49 \pm 1.00 \%$$

$$\mu = 439.5 \pm 3.8$$

$$\sigma = 41.7 \pm 4.0$$

WF

$$\frac{\sigma}{\mu} = 9.26 \pm 0.81 \%$$

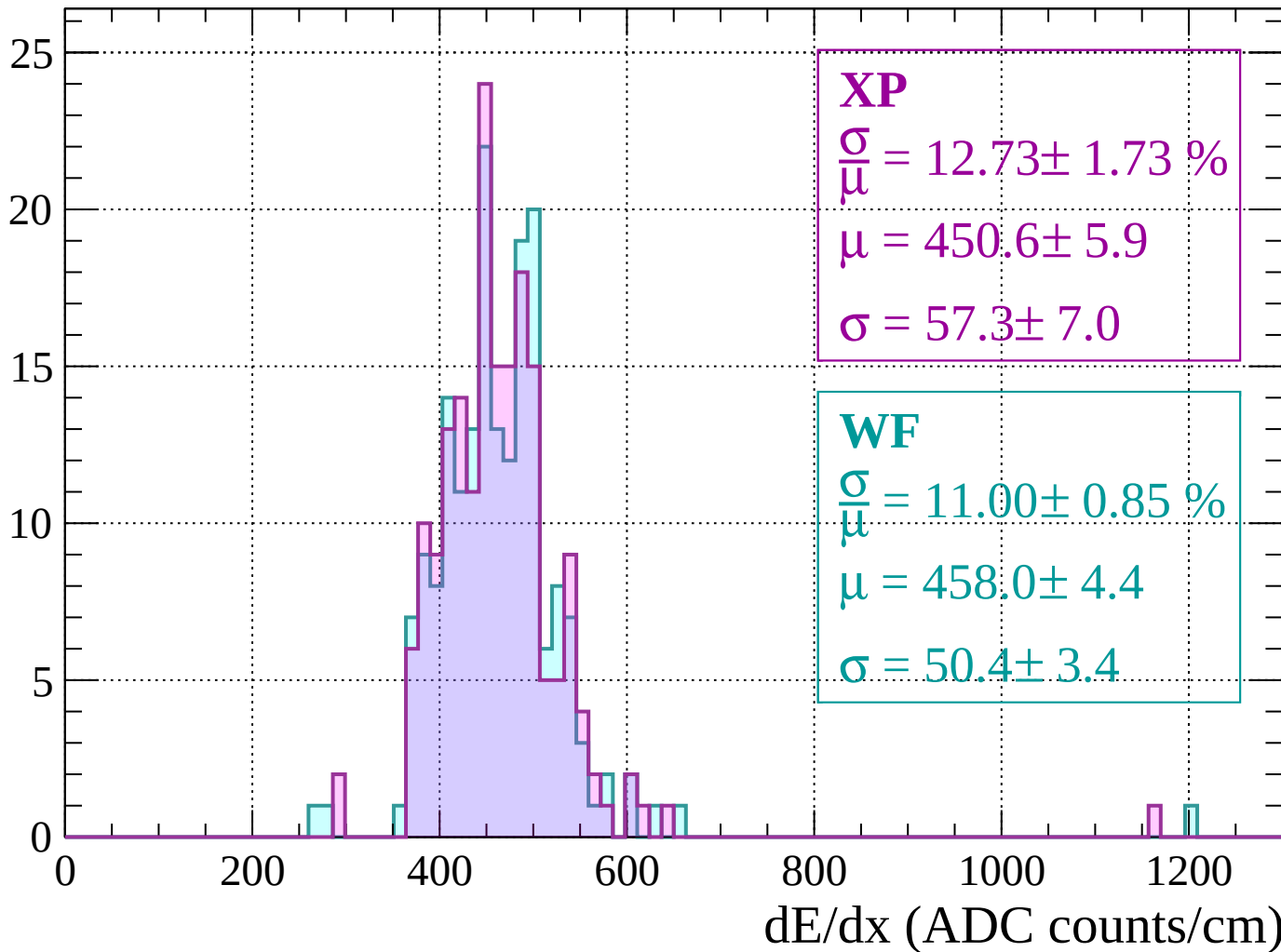
$$\mu = 445.4 \pm 3.6$$

$$\sigma = 41.2 \pm 3.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1160$

Count



Energy loss | $-1160 < p < -1120$

Count

XP

$$\frac{\sigma}{\bar{\mu}} = 7.40 \pm 0.78 \%$$

$$\mu = 435.0 \pm 3.1$$

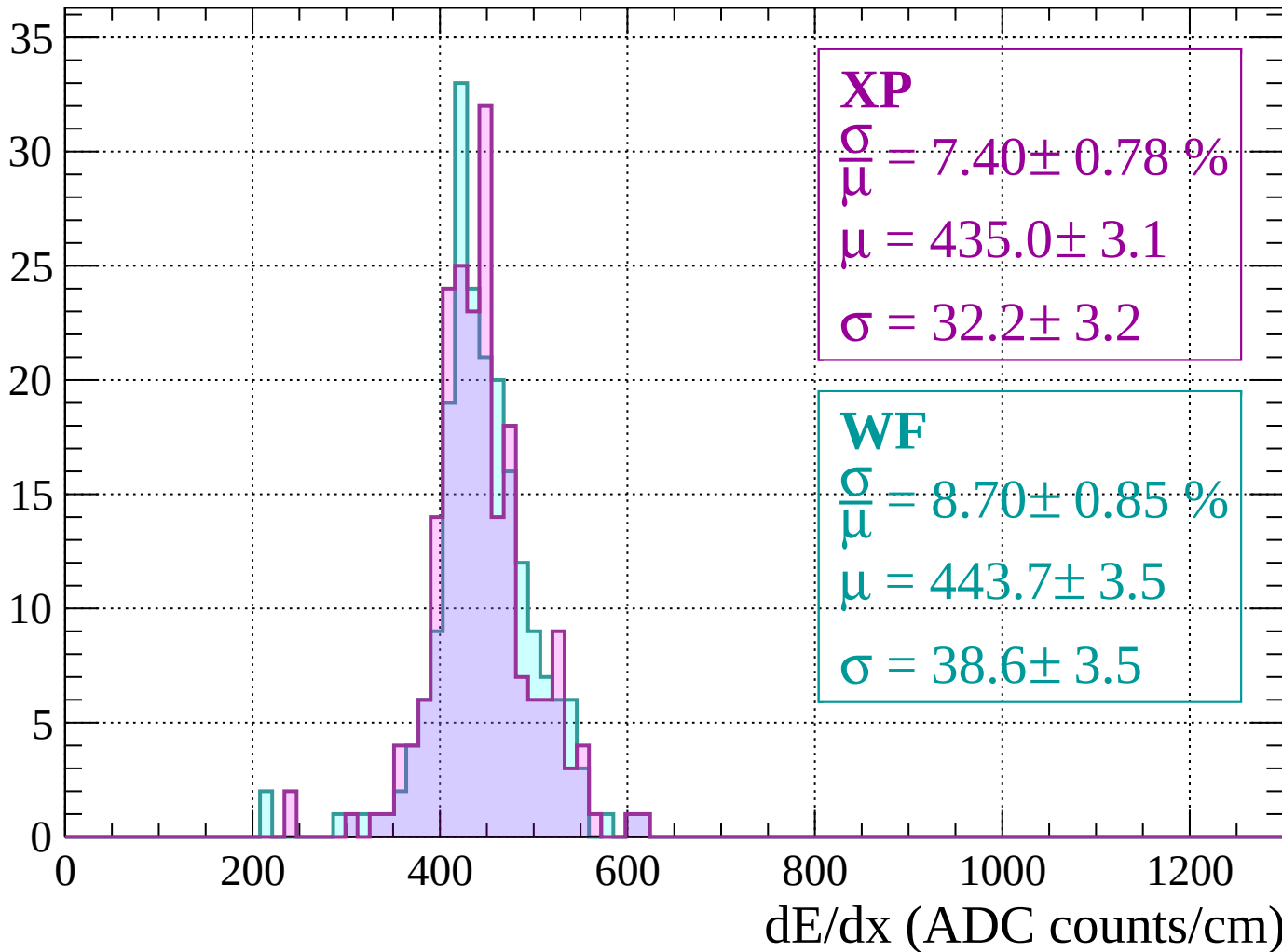
$$\sigma = 32.2 \pm 3.2$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.70 \pm 0.85 \%$$

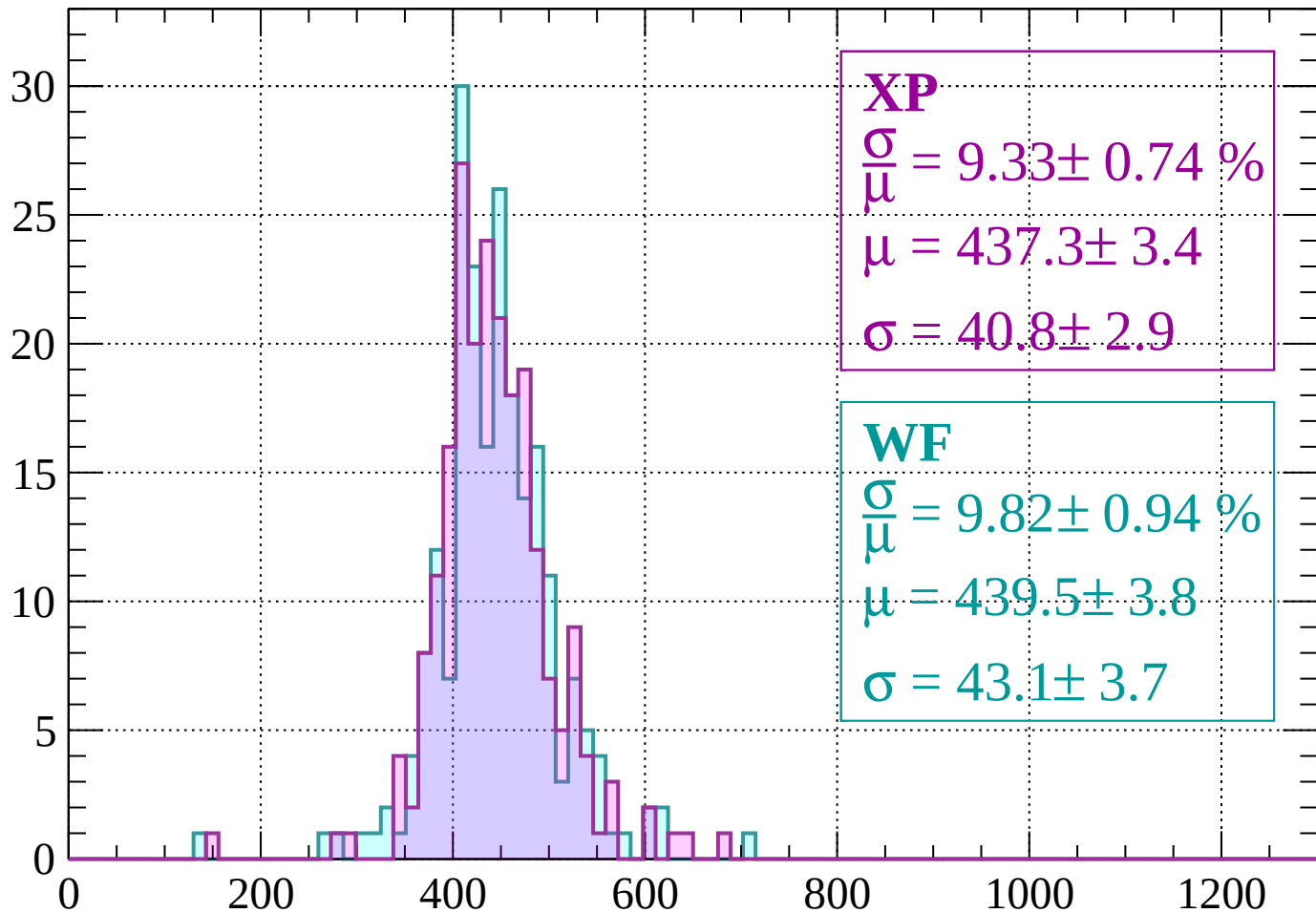
$$\mu = 443.7 \pm 3.5$$

$$\sigma = 38.6 \pm 3.5$$



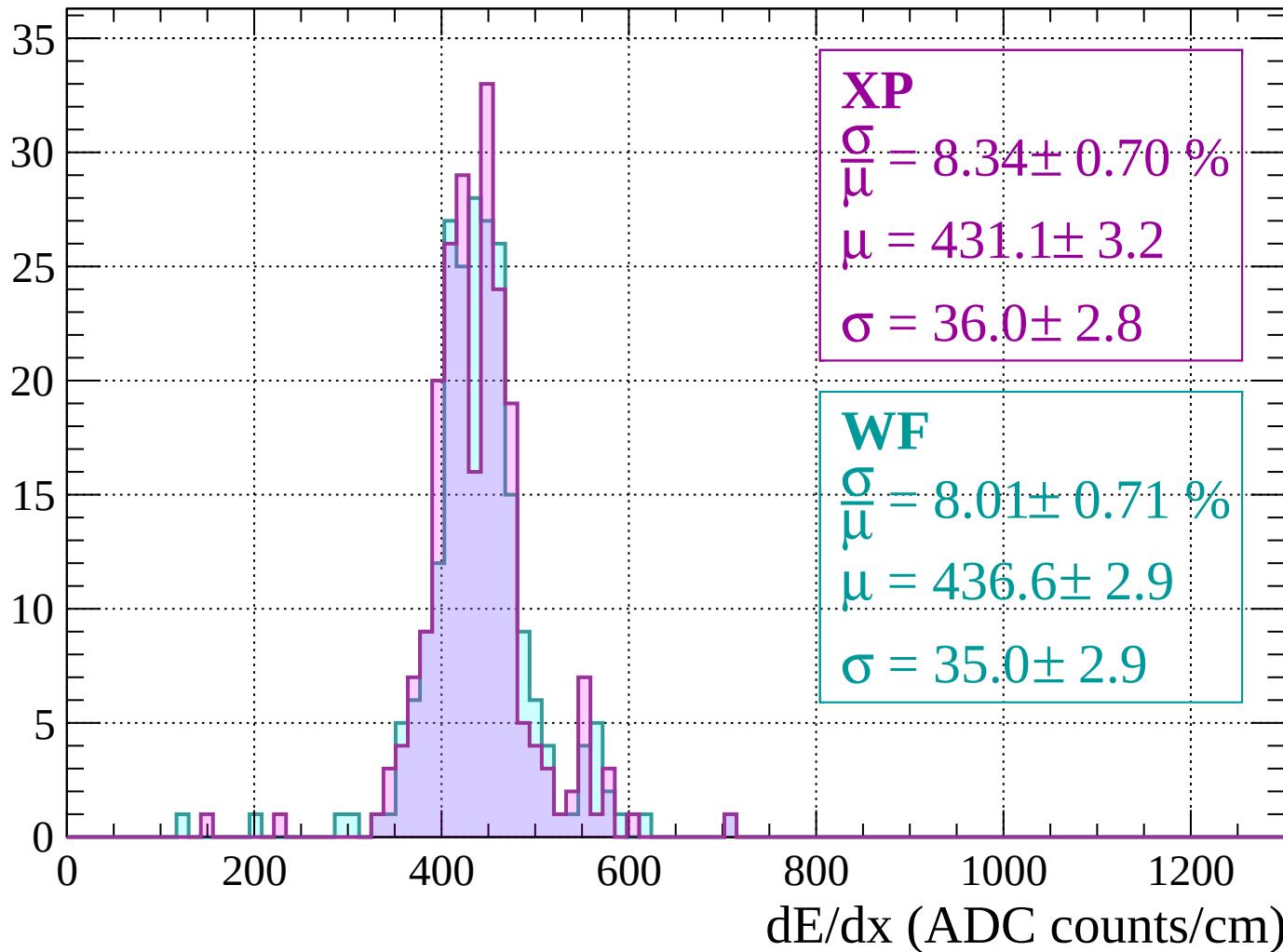
Energy loss | $-1120 < p < -1080$

Count



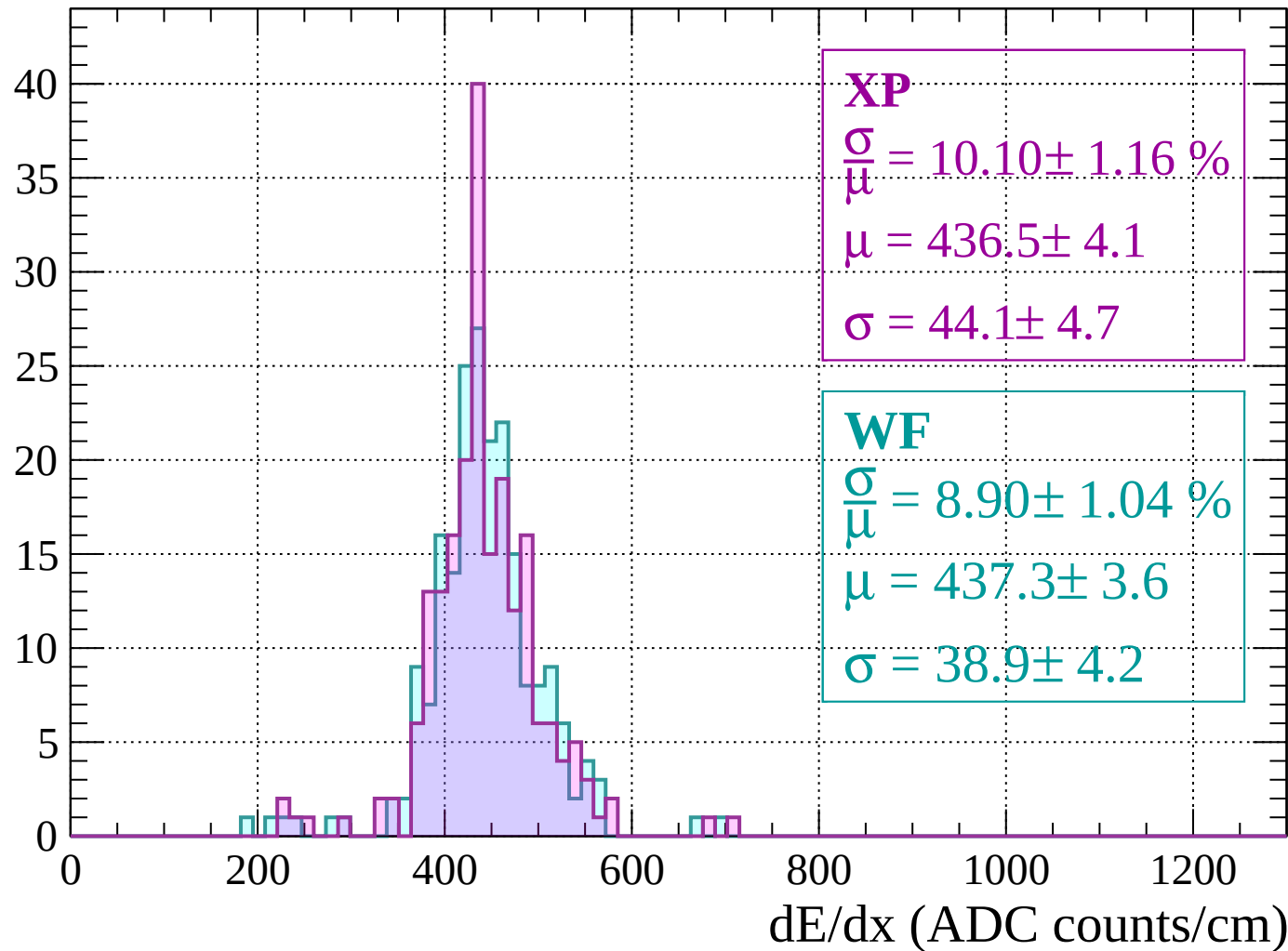
Energy loss | -1080 < p < -1040

Count



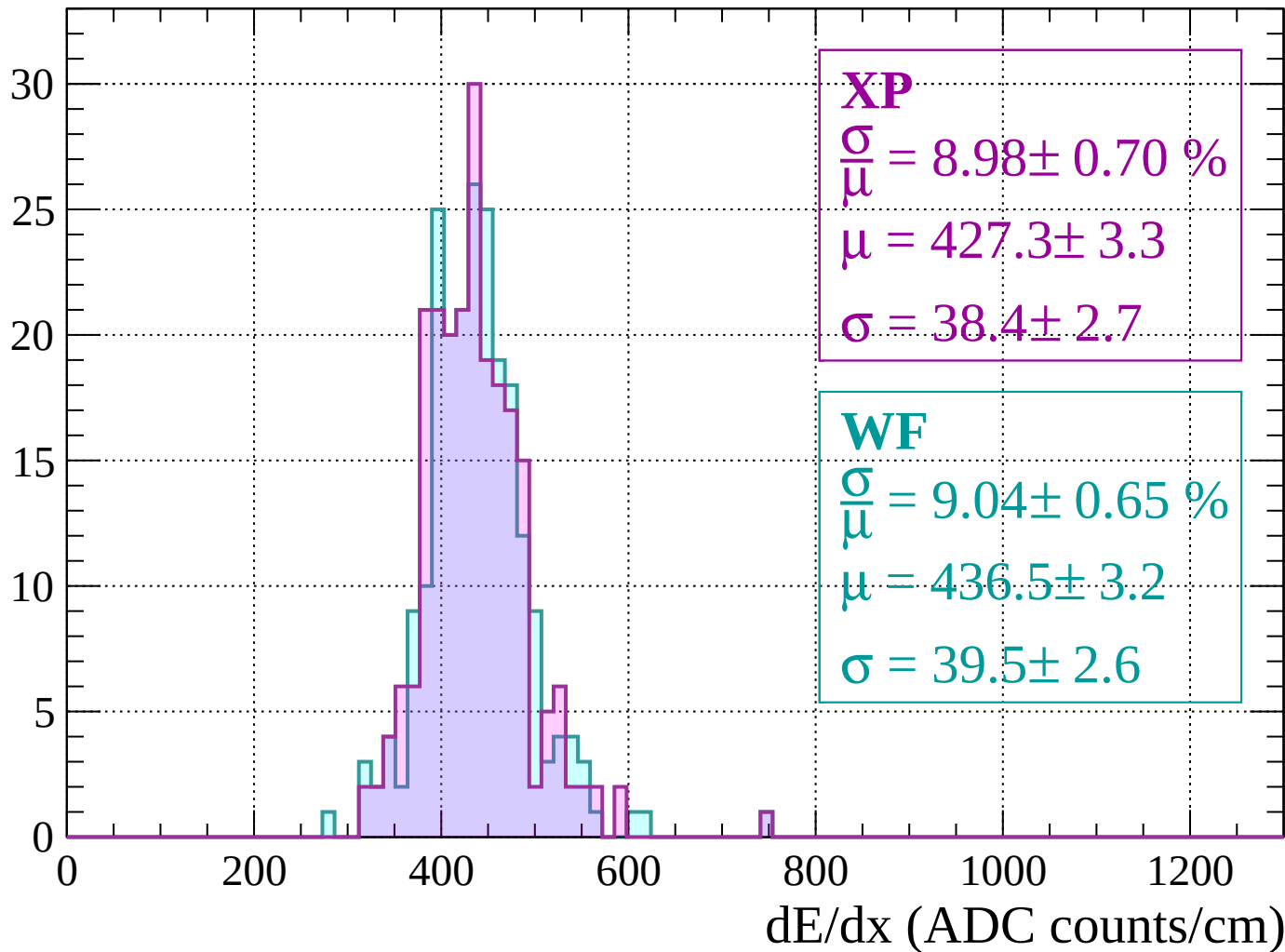
Energy loss | $-1040 < p < -1000$

Count



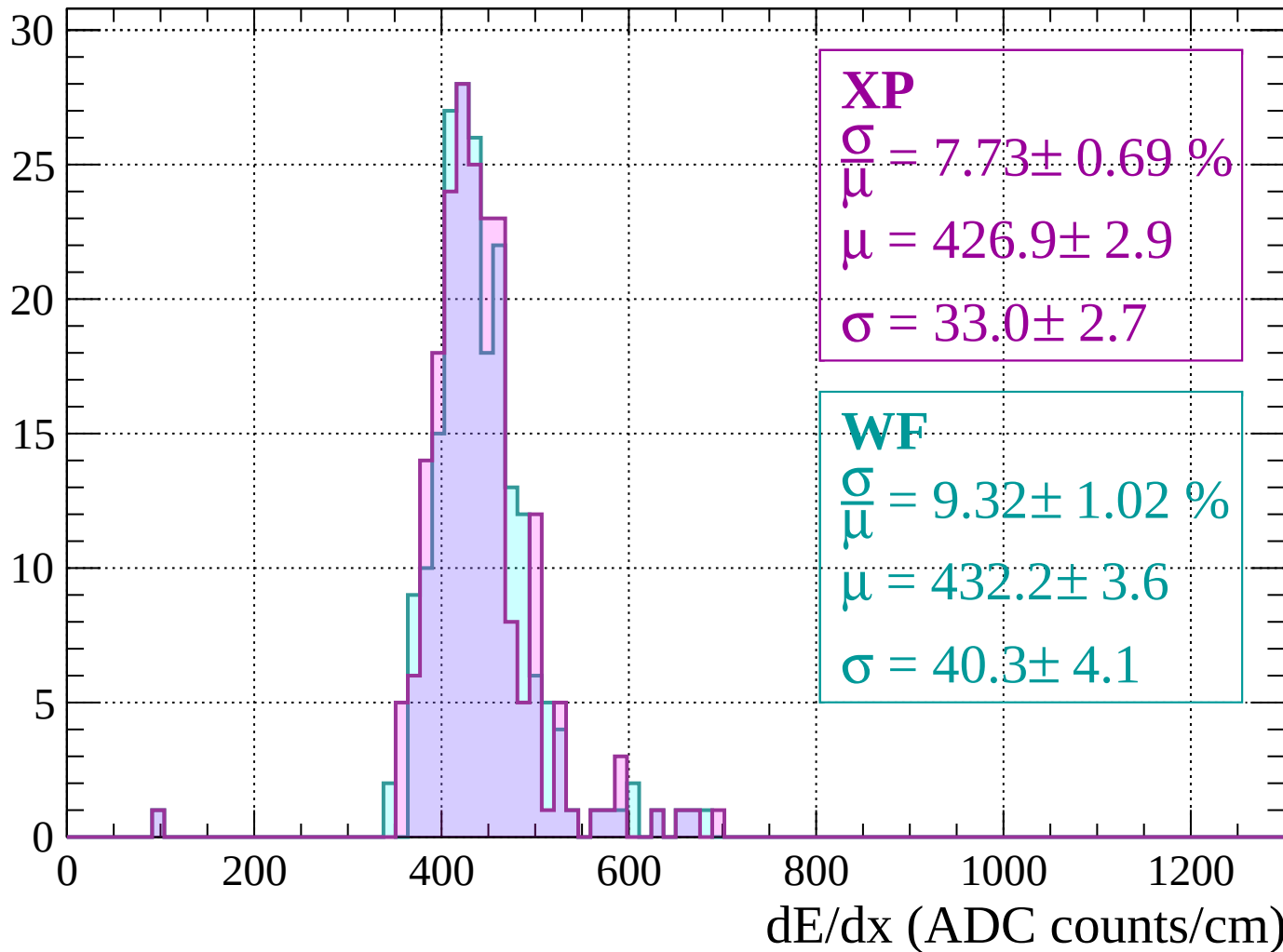
Energy loss | $-1000 < p < -960$

Count



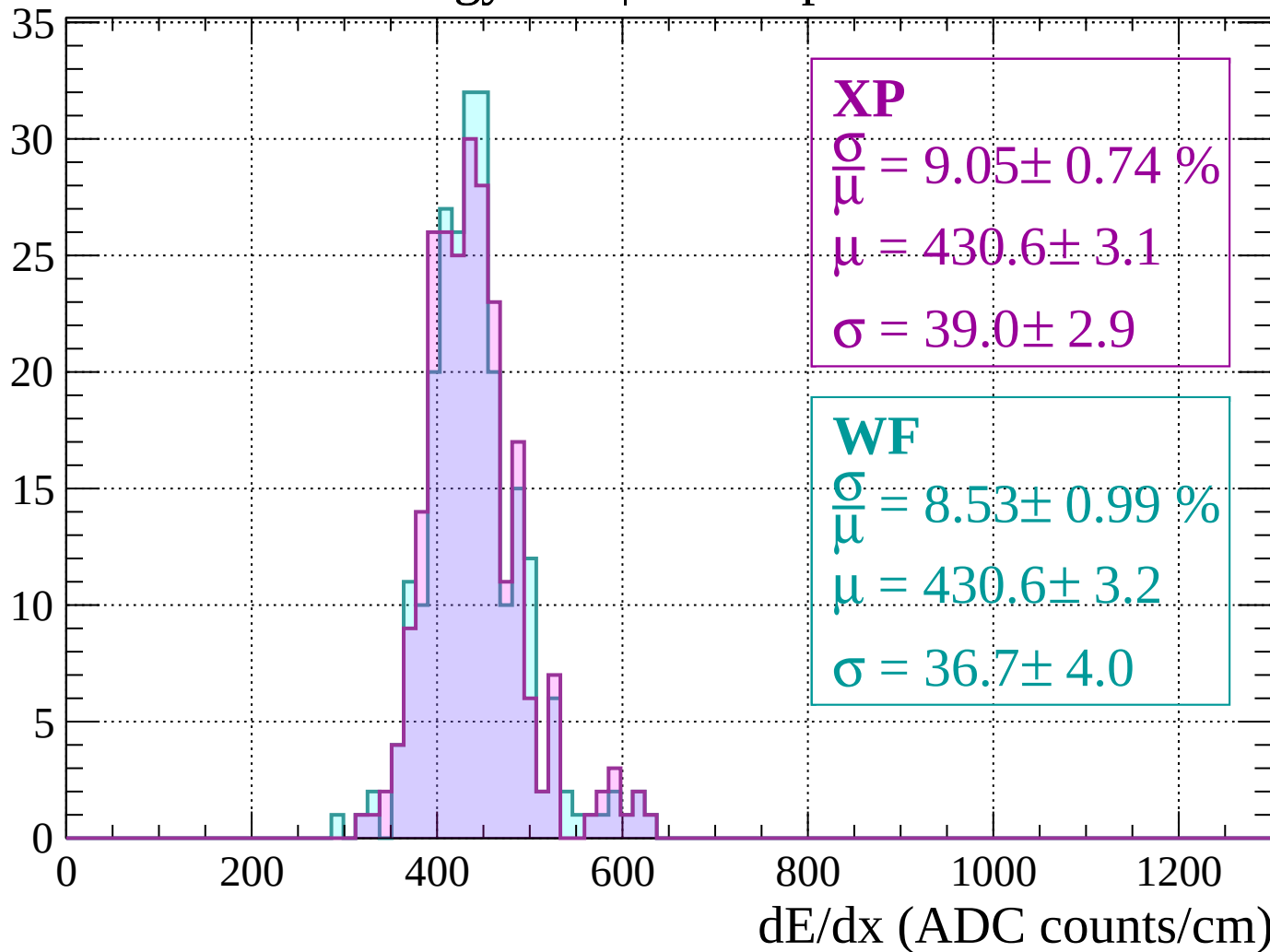
Energy loss | $-960 < p < -920$

Count



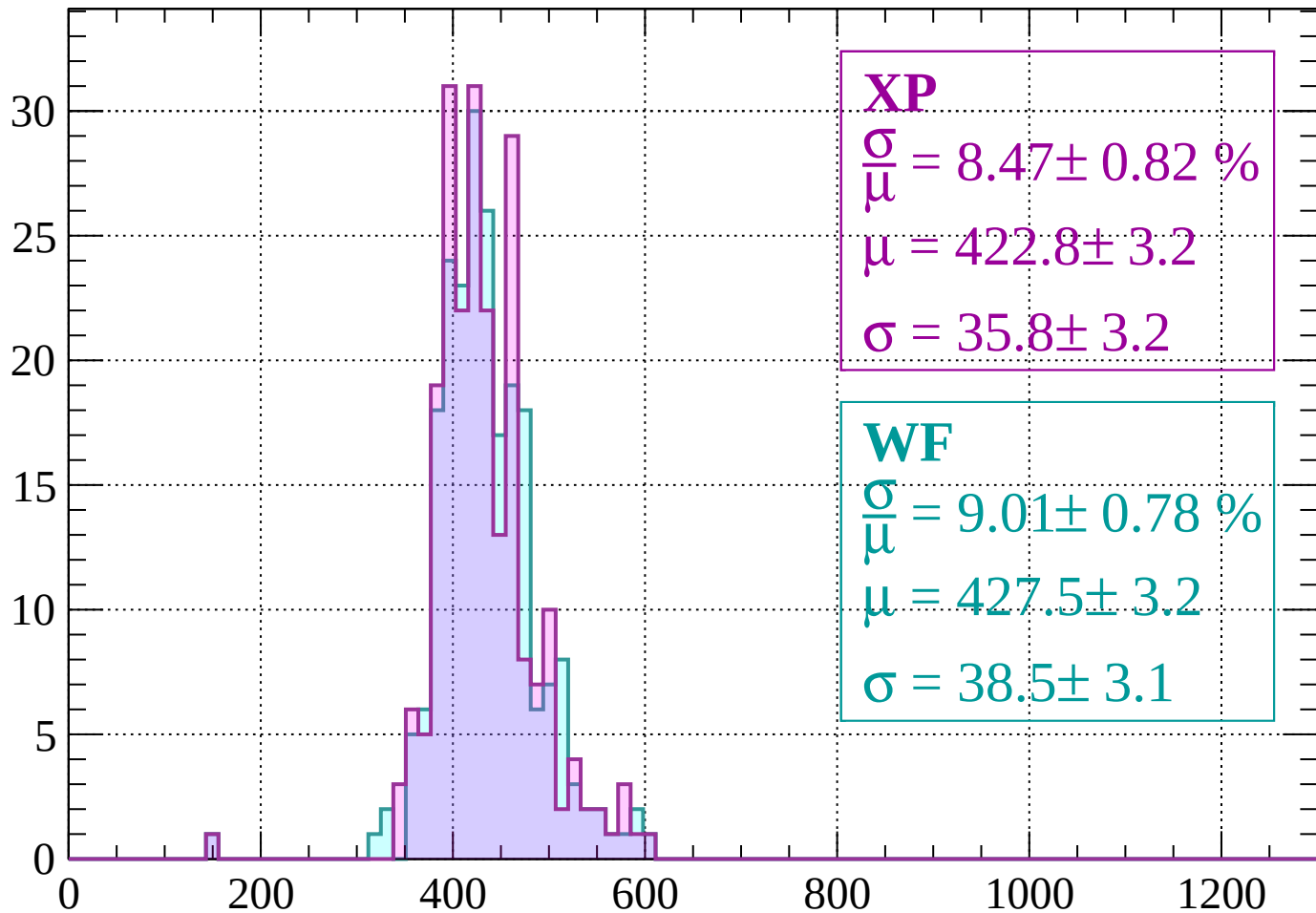
Energy loss | $-920 < p < -880$

Count



Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 8.47 \pm 0.82 \%$$

$$\mu = 422.8 \pm 3.2$$

$$\sigma = 35.8 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 9.01 \pm 0.78 \%$$

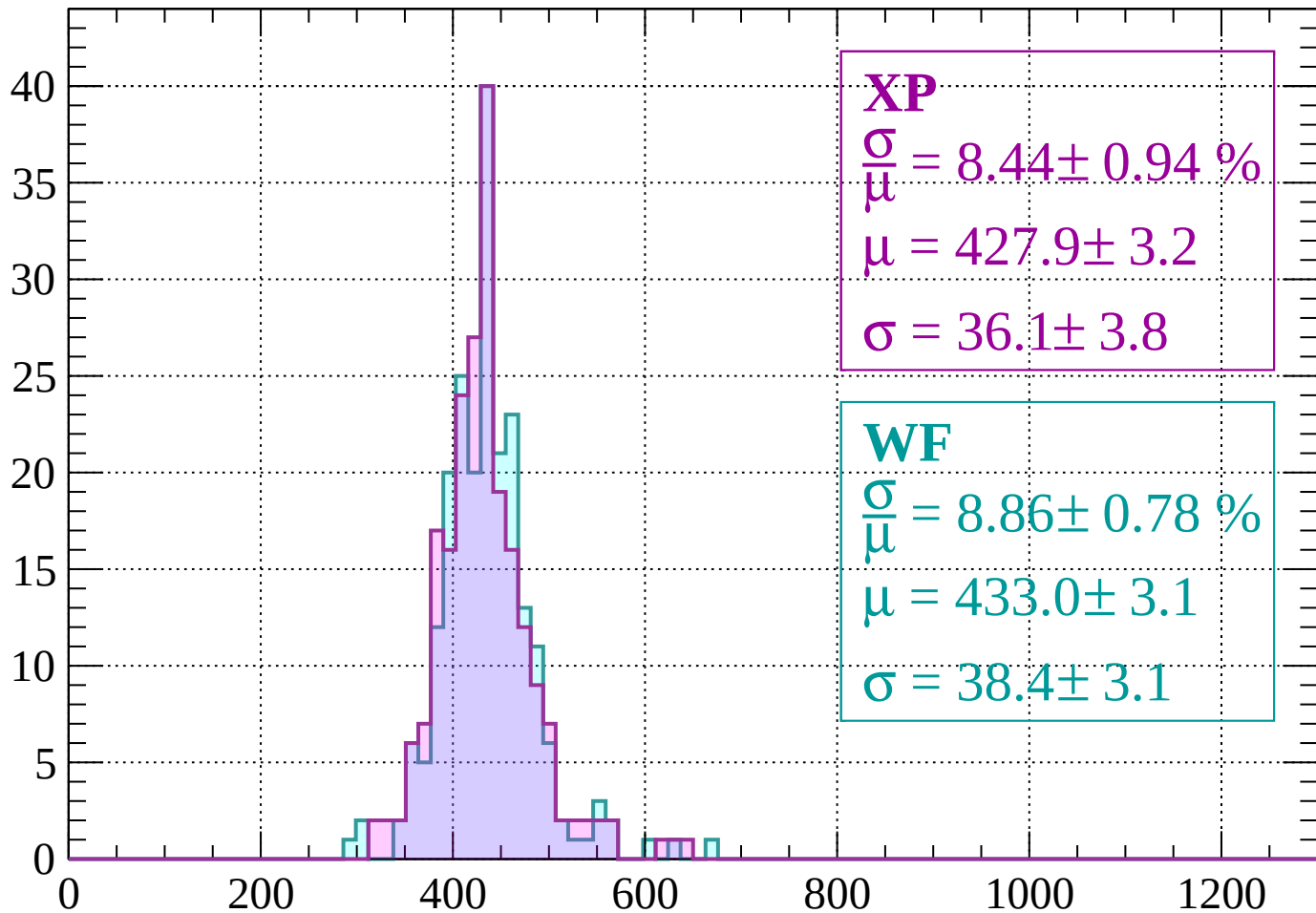
$$\mu = 427.5 \pm 3.2$$

$$\sigma = 38.5 \pm 3.1$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 8.44 \pm 0.94 \%$$

$$\mu = 427.9 \pm 3.2$$

$$\sigma = 36.1 \pm 3.8$$

WF

$$\frac{\sigma}{\mu} = 8.86 \pm 0.78 \%$$

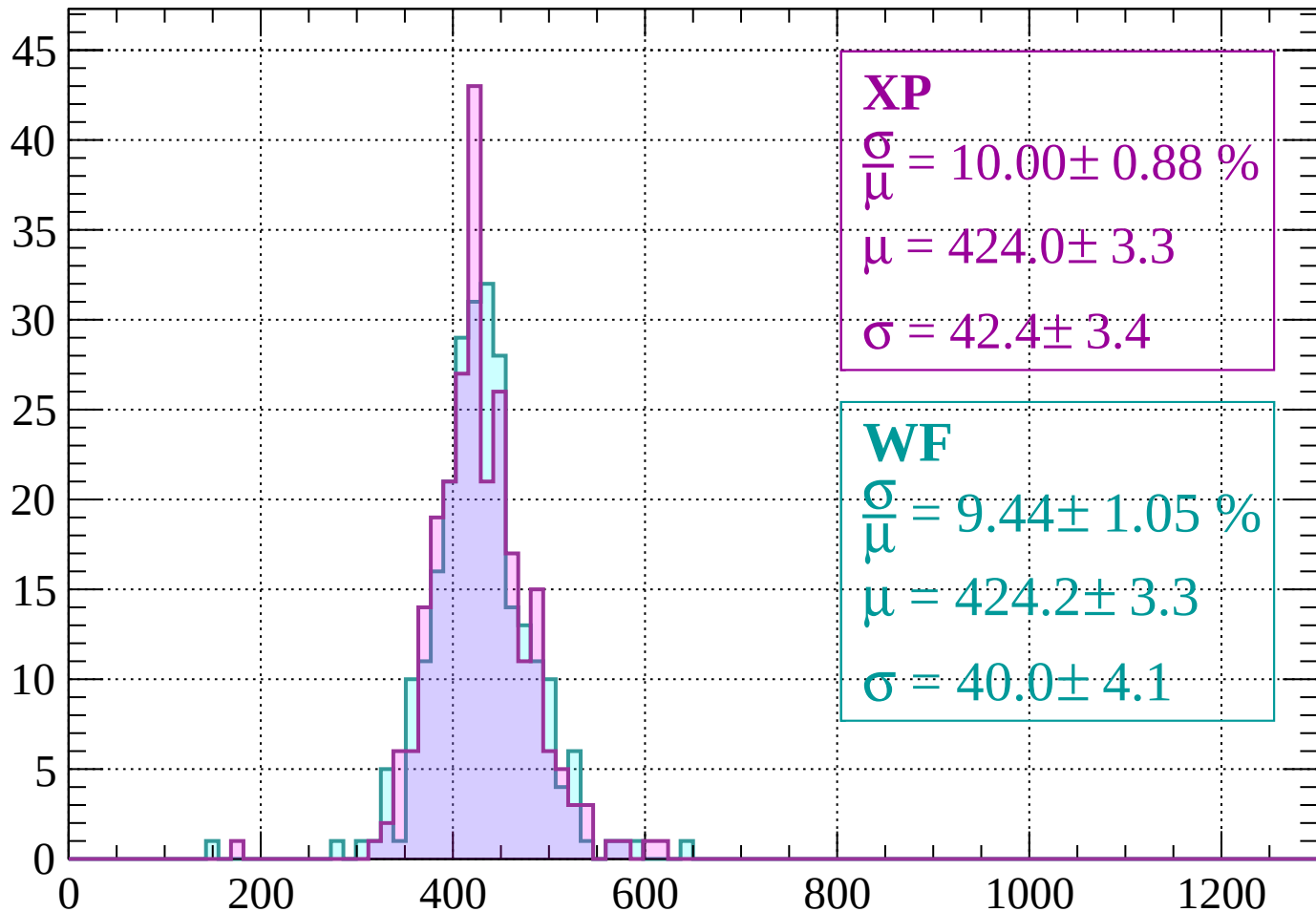
$$\mu = 433.0 \pm 3.1$$

$$\sigma = 38.4 \pm 3.1$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 10.00 \pm 0.88 \%$$

$$\mu = 424.0 \pm 3.3$$

$$\sigma = 42.4 \pm 3.4$$

WF

$$\frac{\sigma}{\mu} = 9.44 \pm 1.05 \%$$

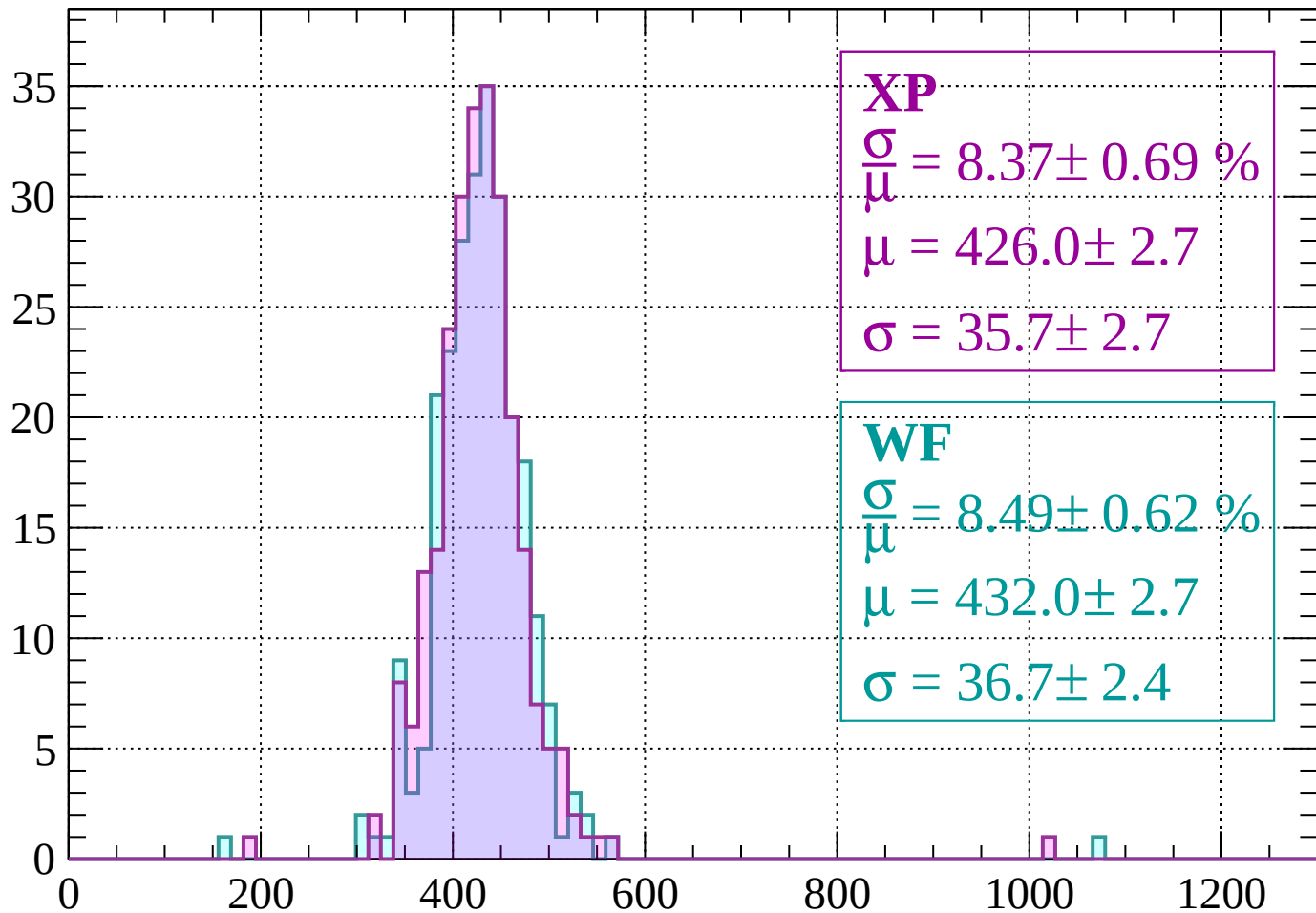
$$\mu = 424.2 \pm 3.3$$

$$\sigma = 40.0 \pm 4.1$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 8.37 \pm 0.69 \%$$

$$\mu = 426.0 \pm 2.7$$

$$\sigma = 35.7 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 8.49 \pm 0.62 \%$$

$$\mu = 432.0 \pm 2.7$$

$$\sigma = 36.7 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count

XP

$$\frac{\sigma}{\mu} = 8.29 \pm 0.73 \%$$

$$\mu = 411.5 \pm 2.6$$

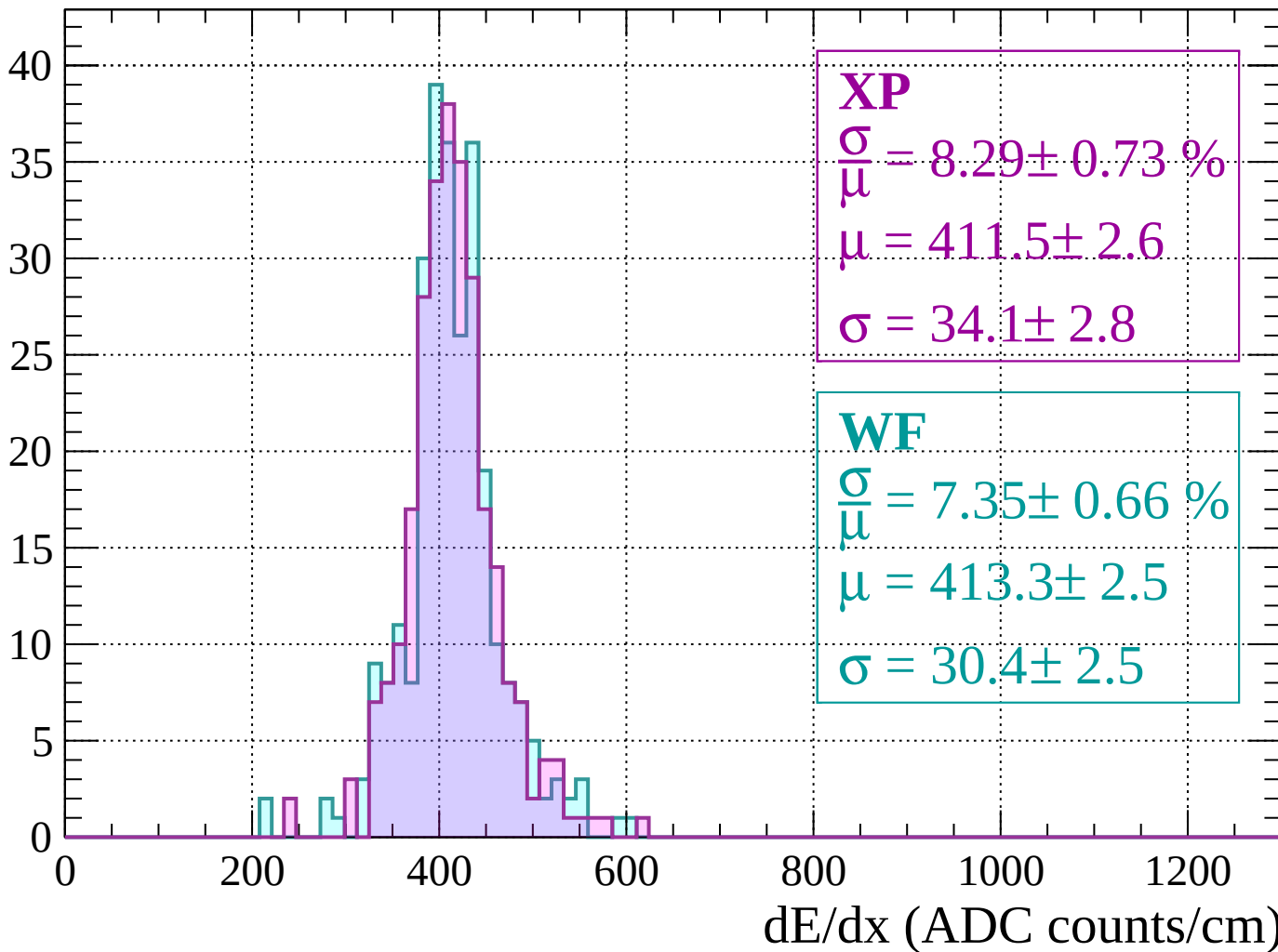
$$\sigma = 34.1 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 7.35 \pm 0.66 \%$$

$$\mu = 413.3 \pm 2.5$$

$$\sigma = 30.4 \pm 2.5$$



Energy loss | $-680 < p < -640$

Count

XP

$$\frac{\sigma}{\mu} = 9.11 \pm 0.70 \%$$

$$\mu = 414.2 \pm 2.8$$

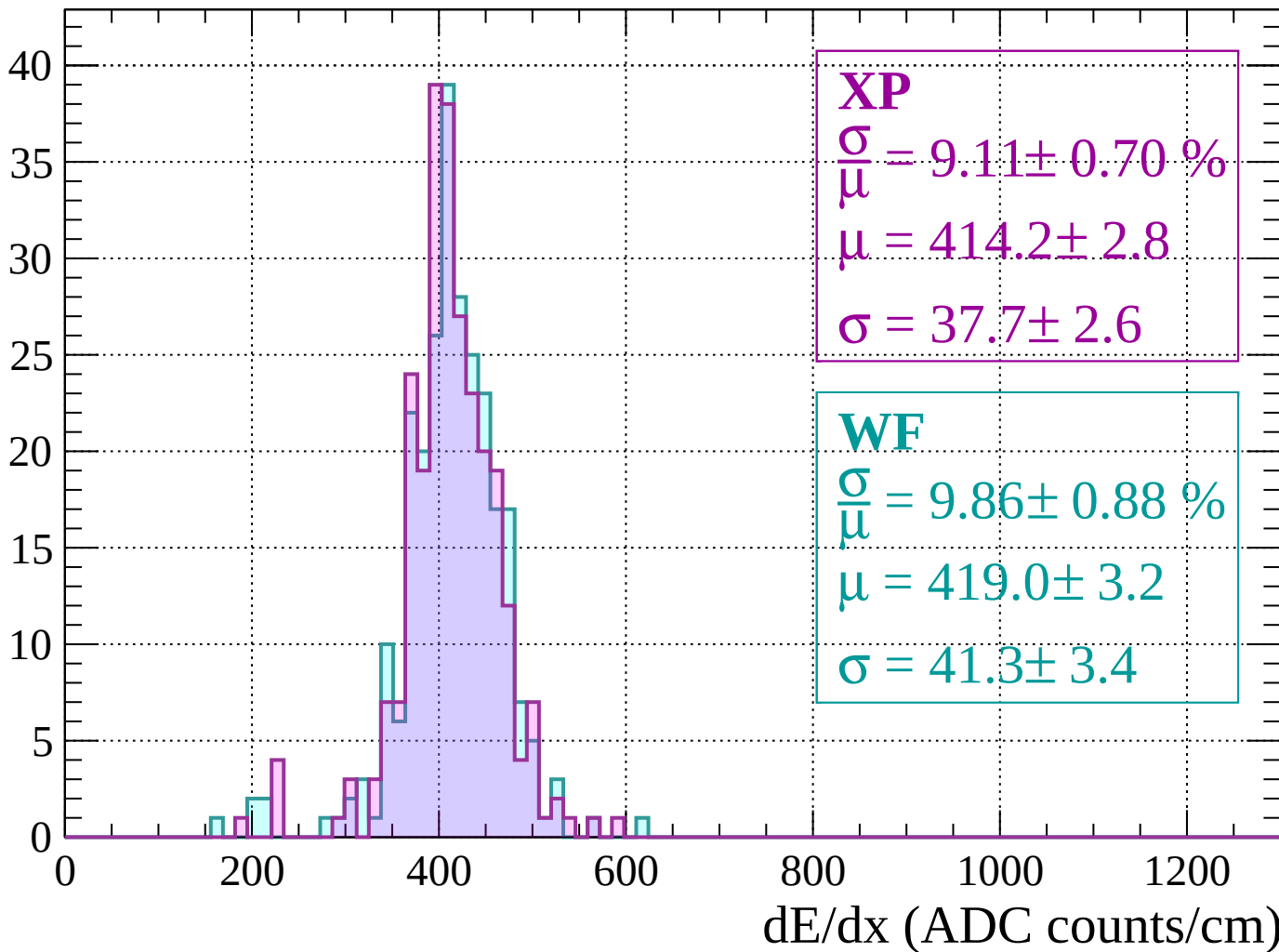
$$\sigma = 37.7 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 9.86 \pm 0.88 \%$$

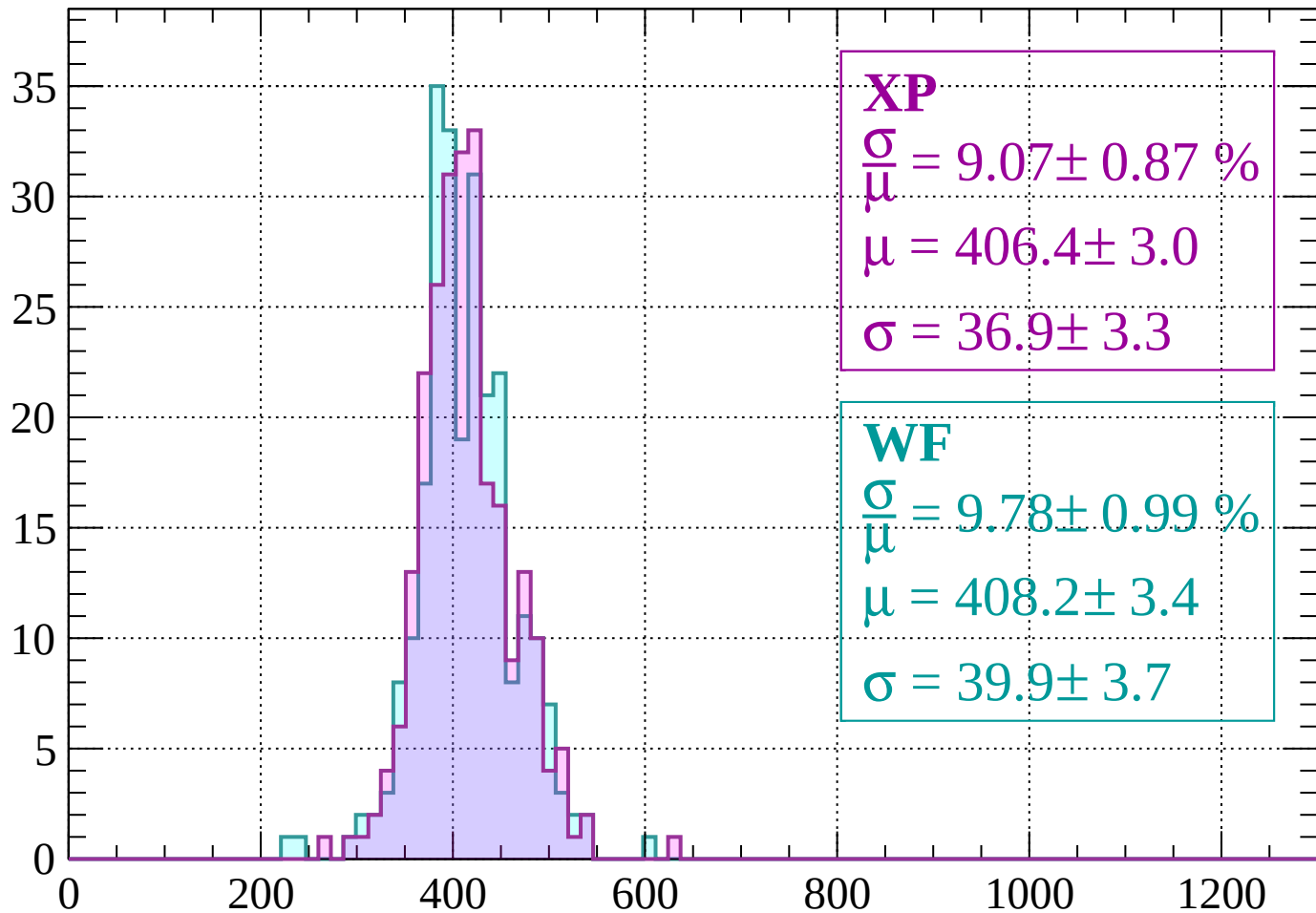
$$\mu = 419.0 \pm 3.2$$

$$\sigma = 41.3 \pm 3.4$$



Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 9.07 \pm 0.87 \%$$

$$\mu = 406.4 \pm 3.0$$

$$\sigma = 36.9 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 9.78 \pm 0.99 \%$$

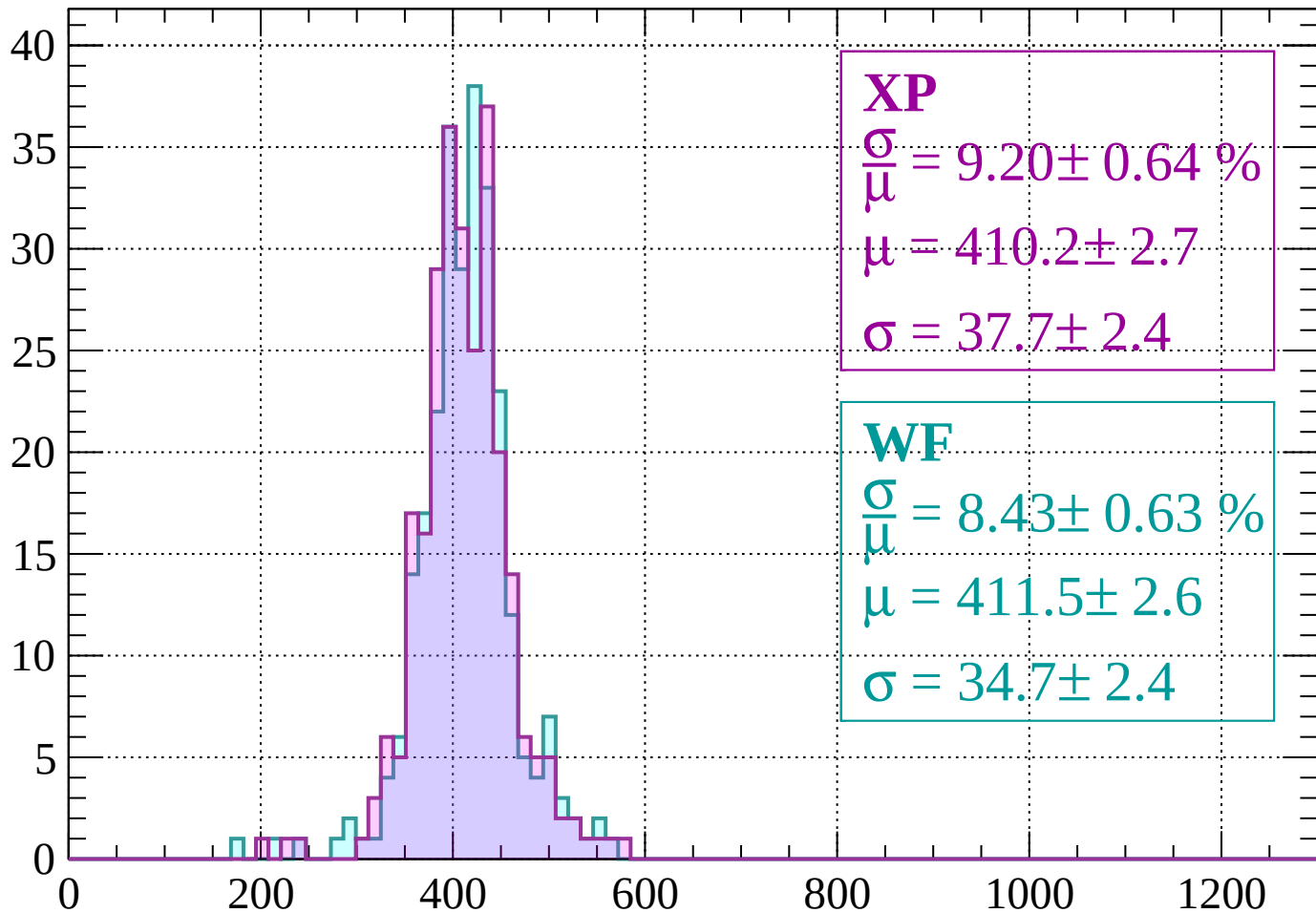
$$\mu = 408.2 \pm 3.4$$

$$\sigma = 39.9 \pm 3.7$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 9.20 \pm 0.64 \%$$

$$\mu = 410.2 \pm 2.7$$

$$\sigma = 37.7 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 8.43 \pm 0.63 \%$$

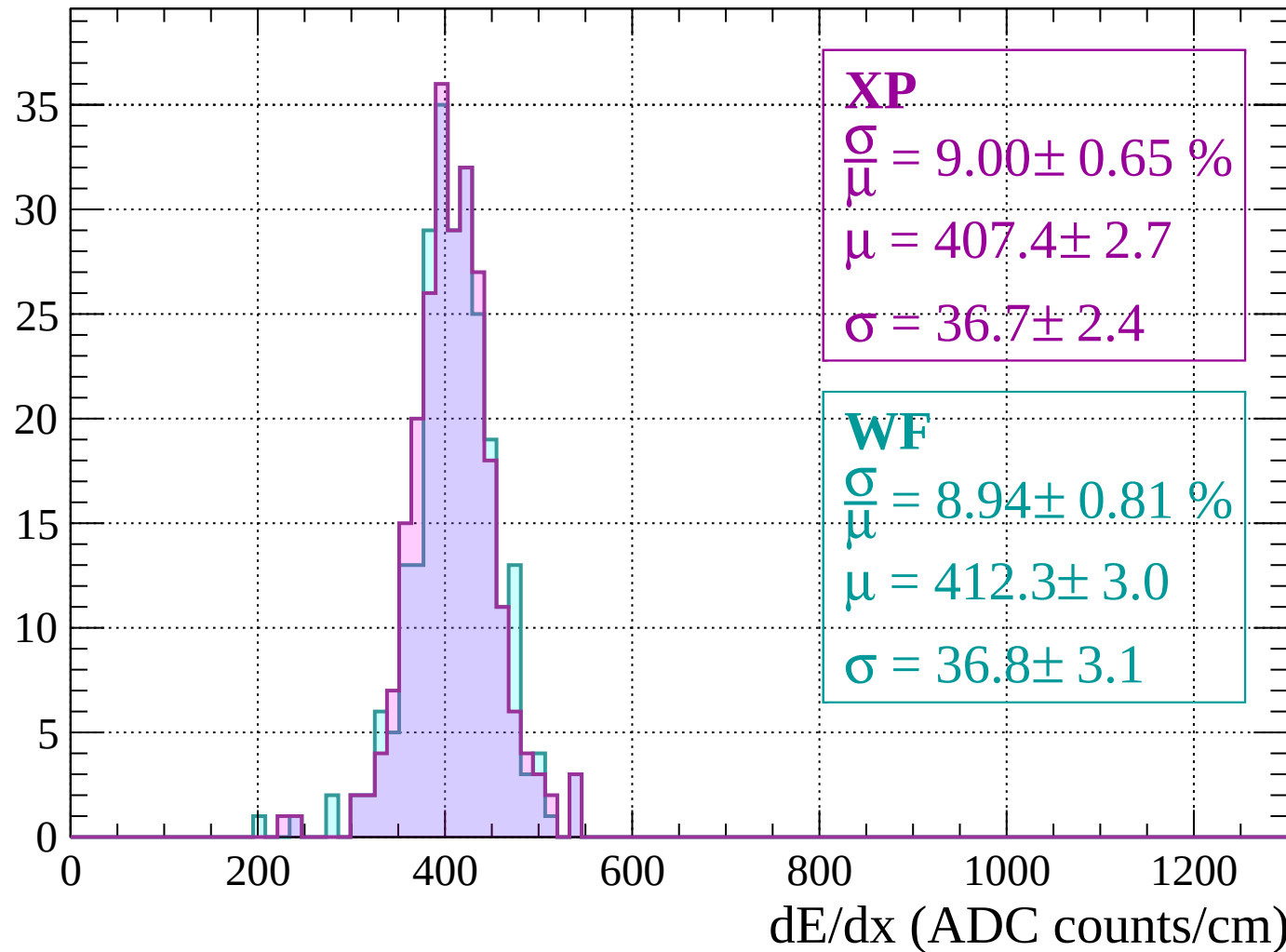
$$\mu = 411.5 \pm 2.6$$

$$\sigma = 34.7 \pm 2.4$$

dE/dx (ADC counts/cm)

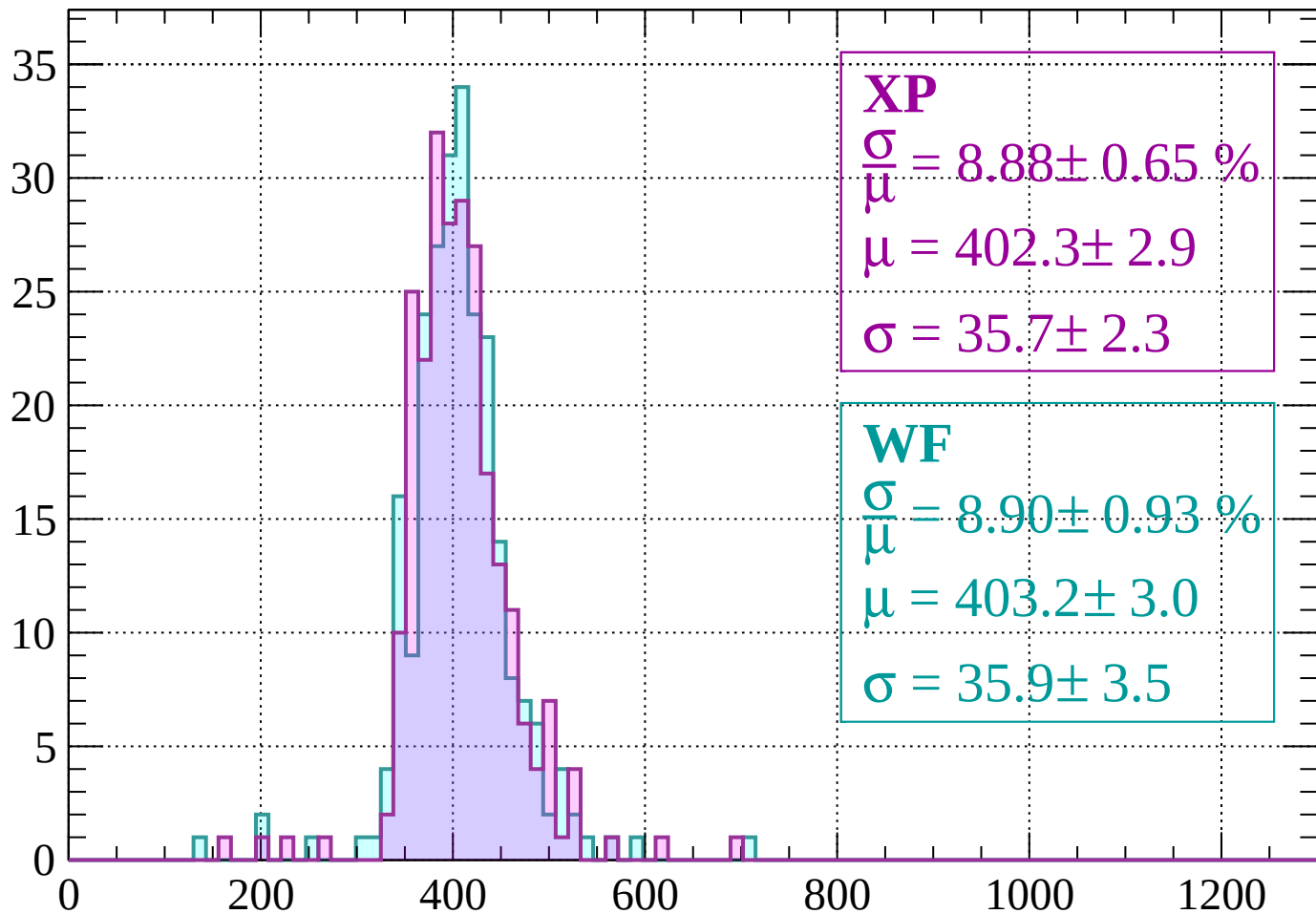
Energy loss | $-560 < p < -520$

Count



Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 8.88 \pm 0.65 \%$$

$$\mu = 402.3 \pm 2.9$$

$$\sigma = 35.7 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 8.90 \pm 0.93 \%$$

$$\mu = 403.2 \pm 3.0$$

$$\sigma = 35.9 \pm 3.5$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count

XP

$$\frac{\sigma}{\mu} = 9.05 \pm 0.82 \%$$

$$\mu = 398.9 \pm 2.9$$

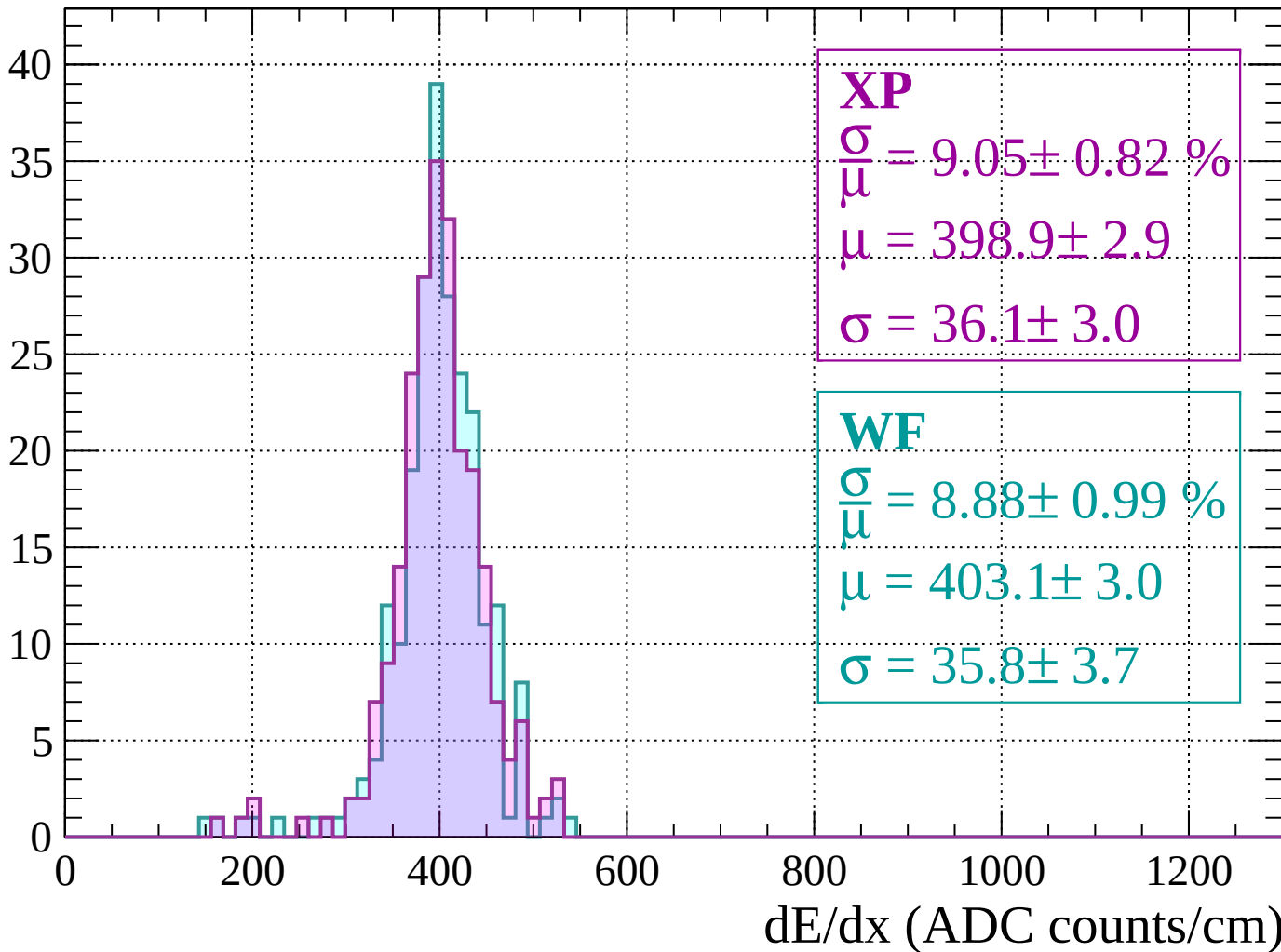
$$\sigma = 36.1 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 8.88 \pm 0.99 \%$$

$$\mu = 403.1 \pm 3.0$$

$$\sigma = 35.8 \pm 3.7$$



Energy loss | $-440 < p < -400$

Count

XP

$$\frac{\sigma}{\mu} = 7.05 \pm 0.69 \%$$

$$\mu = 393.2 \pm 2.4$$

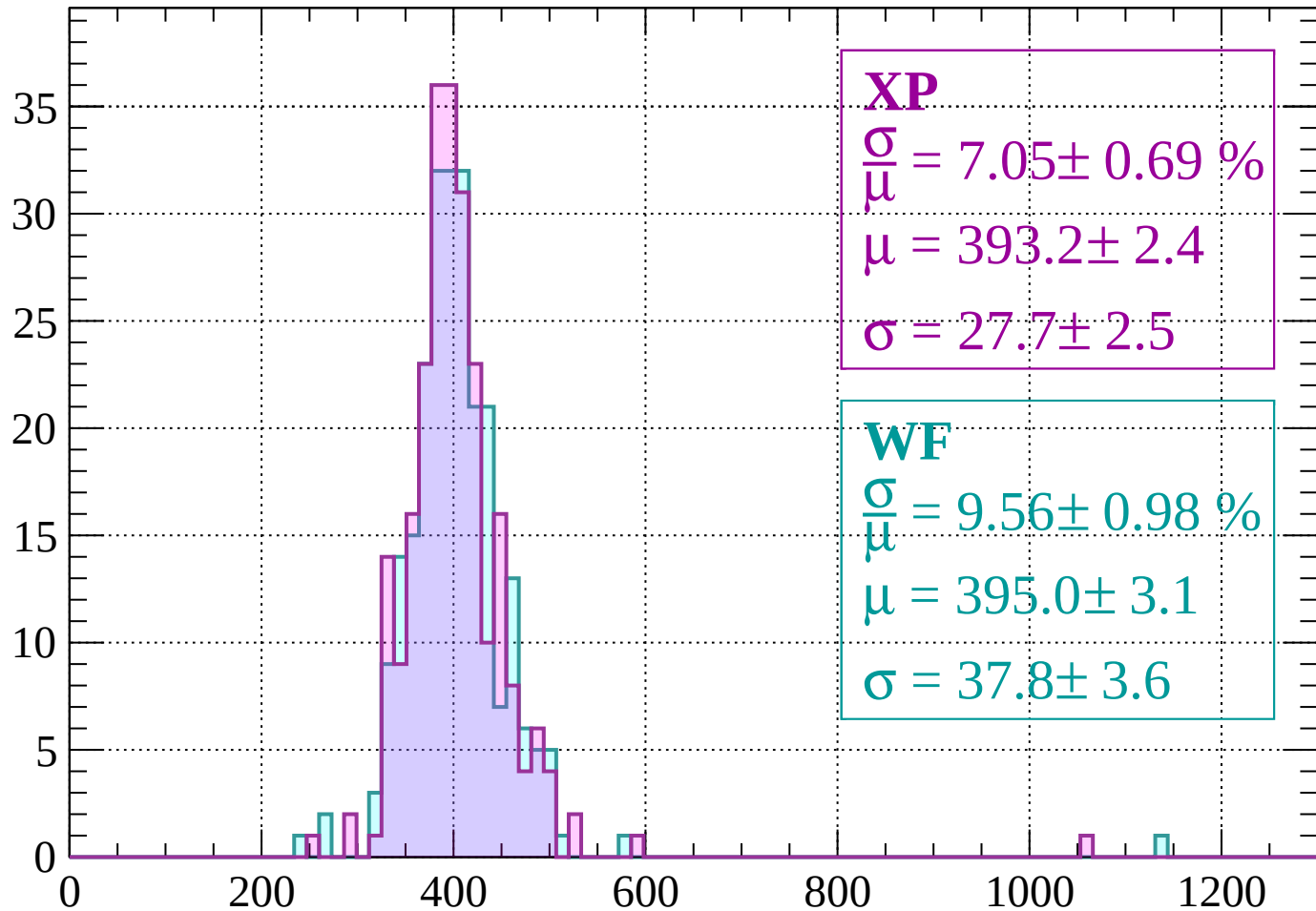
$$\sigma = 27.7 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.98 \%$$

$$\mu = 395.0 \pm 3.1$$

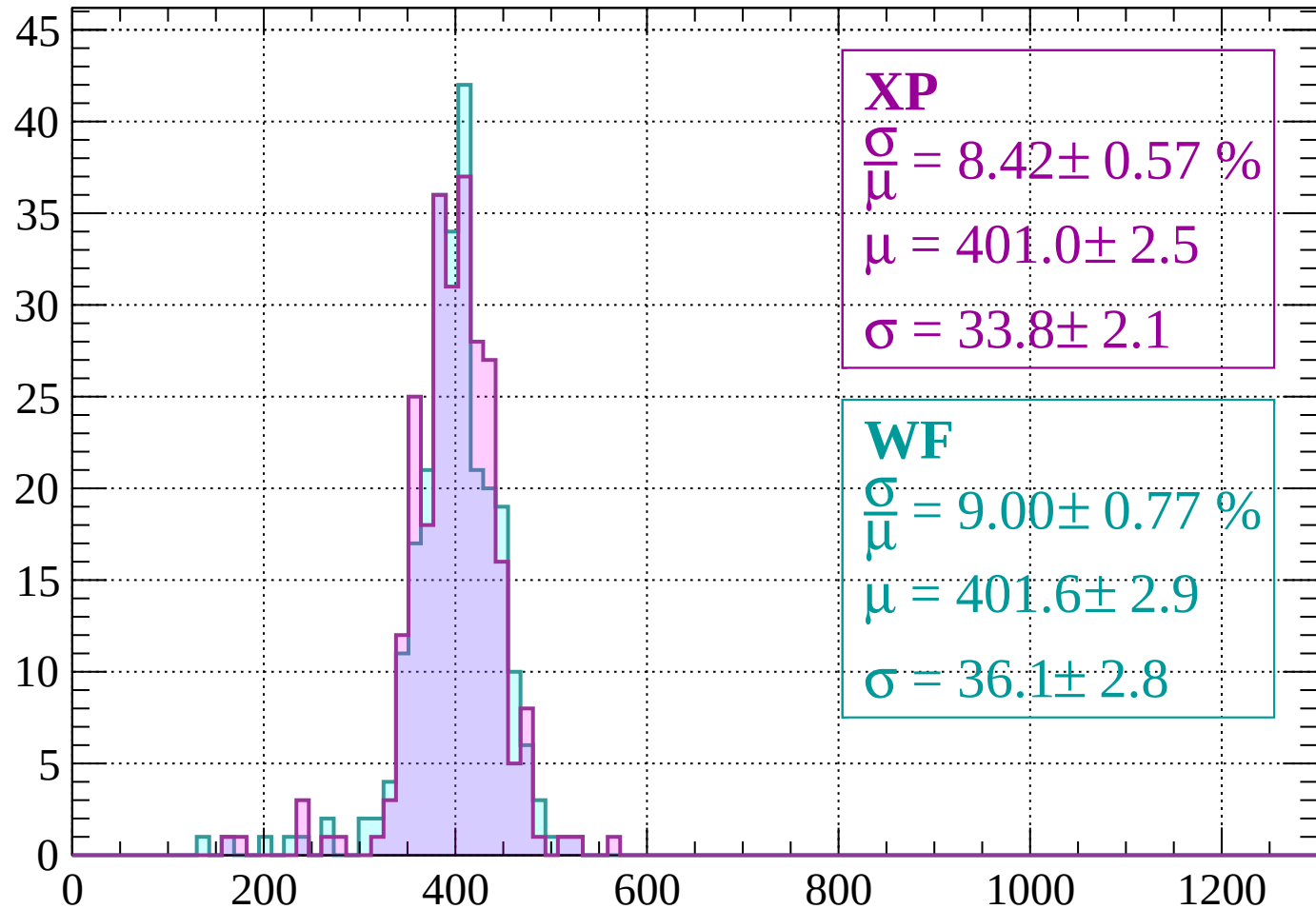
$$\sigma = 37.8 \pm 3.6$$



dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.42 \pm 0.57 \%$$

$$\mu = 401.0 \pm 2.5$$

$$\sigma = 33.8 \pm 2.1$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.00 \pm 0.77 \%$$

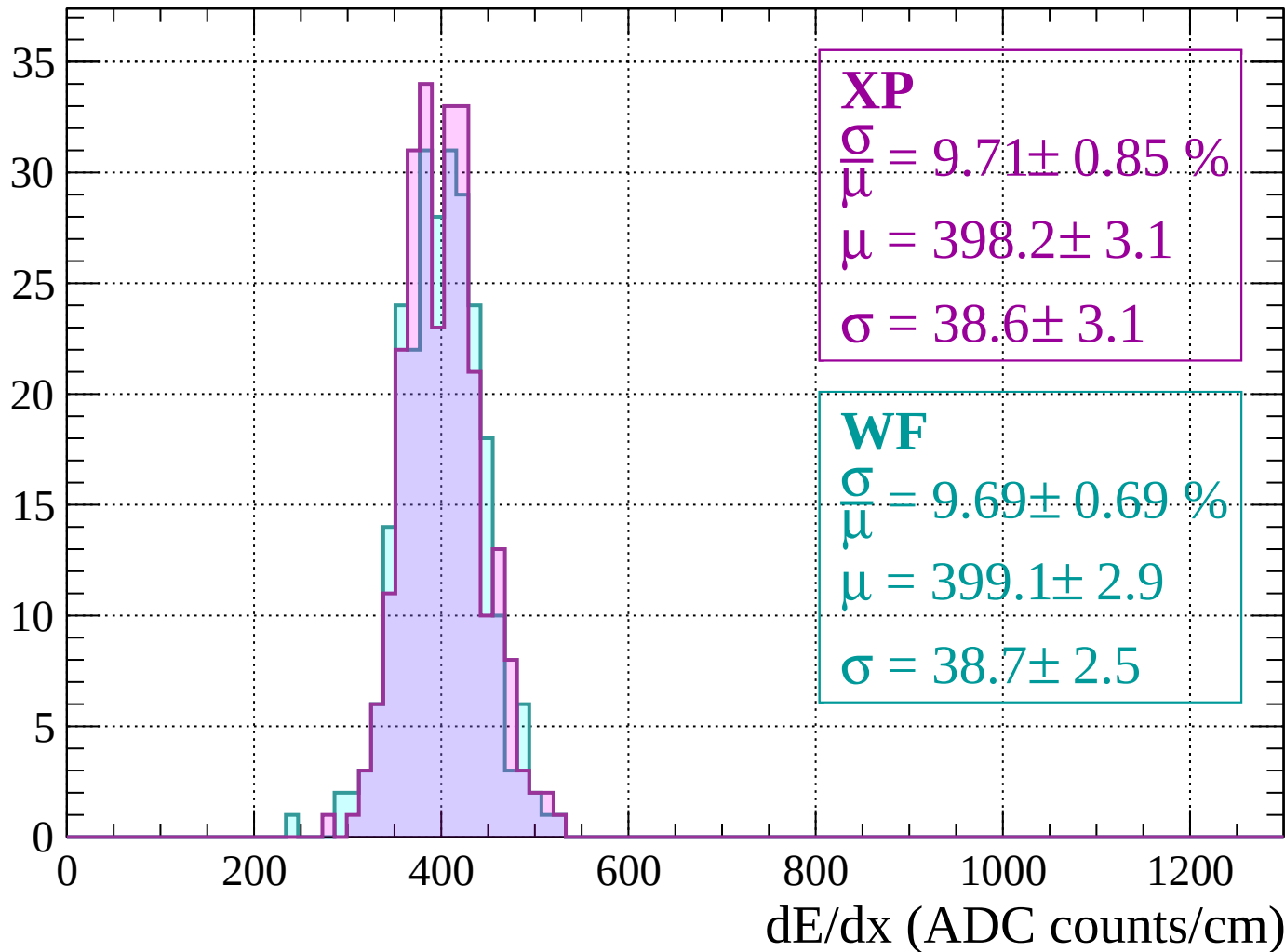
$$\mu = 401.6 \pm 2.9$$

$$\sigma = 36.1 \pm 2.8$$

dE/dx (ADC counts/cm)

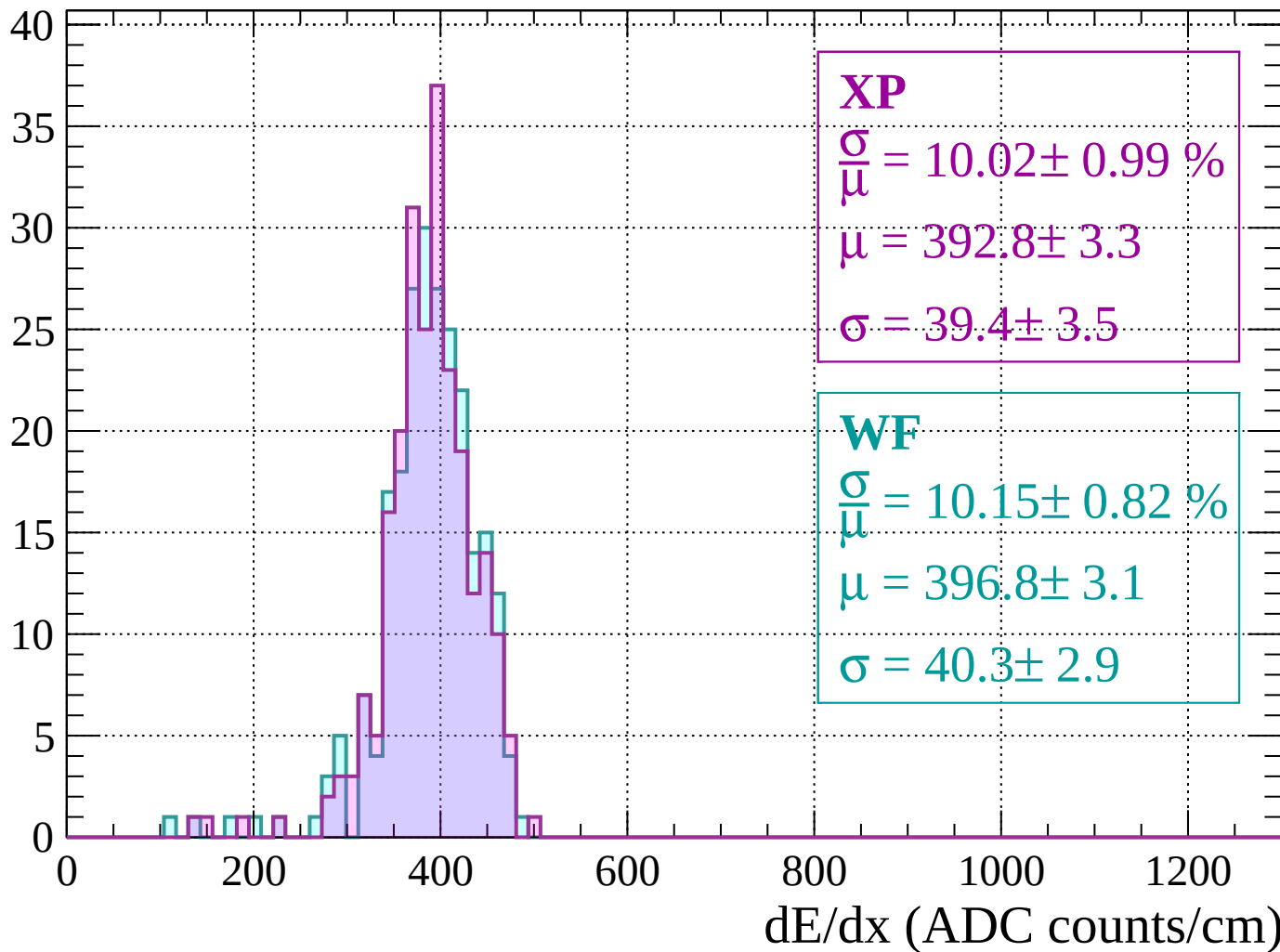
Energy loss | $-360 < p < -320$

Count



Energy loss | $-320 < p < -280$

Count



Energy loss | $-280 < p < -240$

Count

XP

$$\frac{\sigma}{\mu} = 9.22 \pm 0.71 \%$$

$$\mu = 405.1 \pm 2.9$$

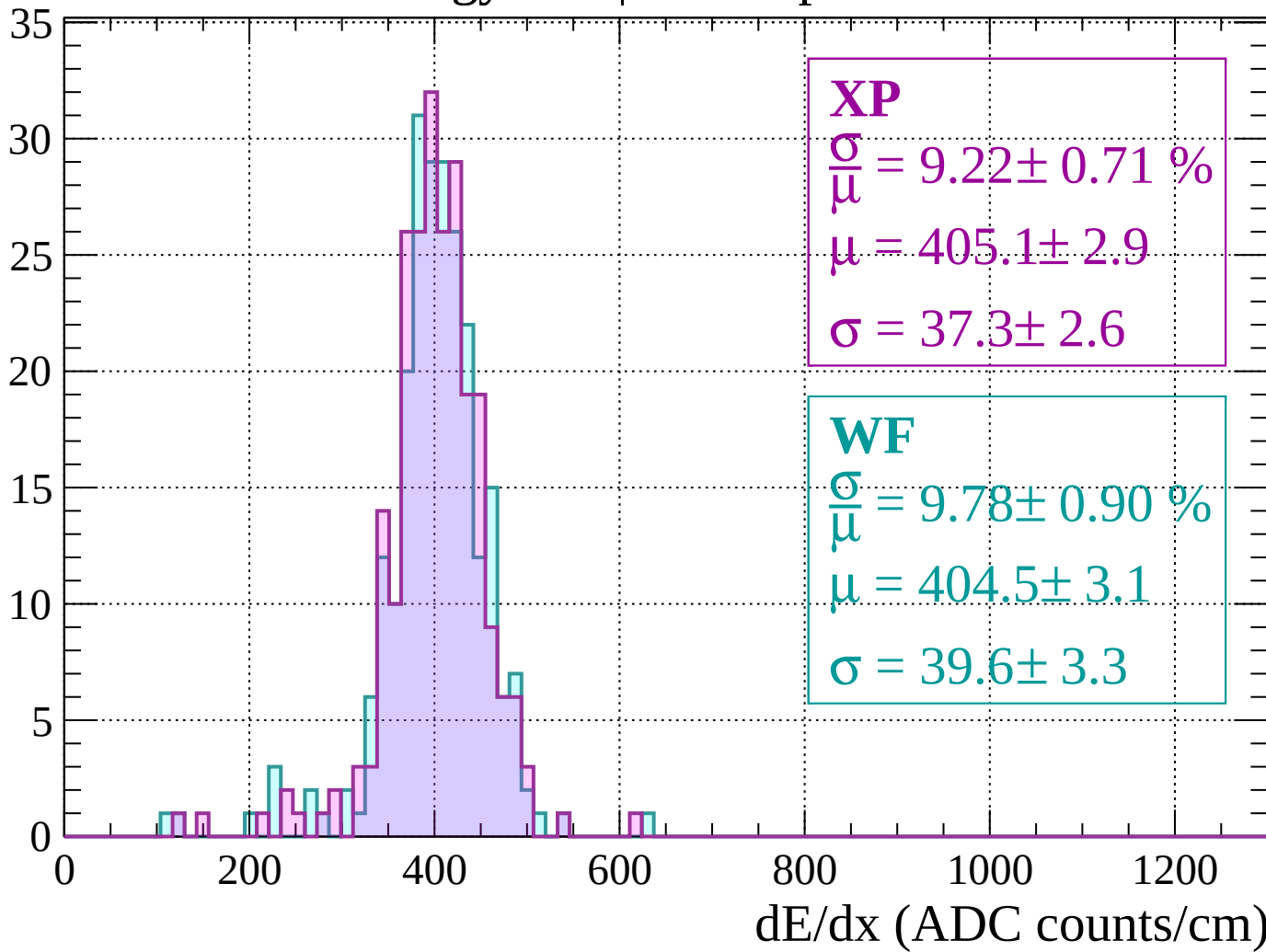
$$\sigma = 37.3 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 9.78 \pm 0.90 \%$$

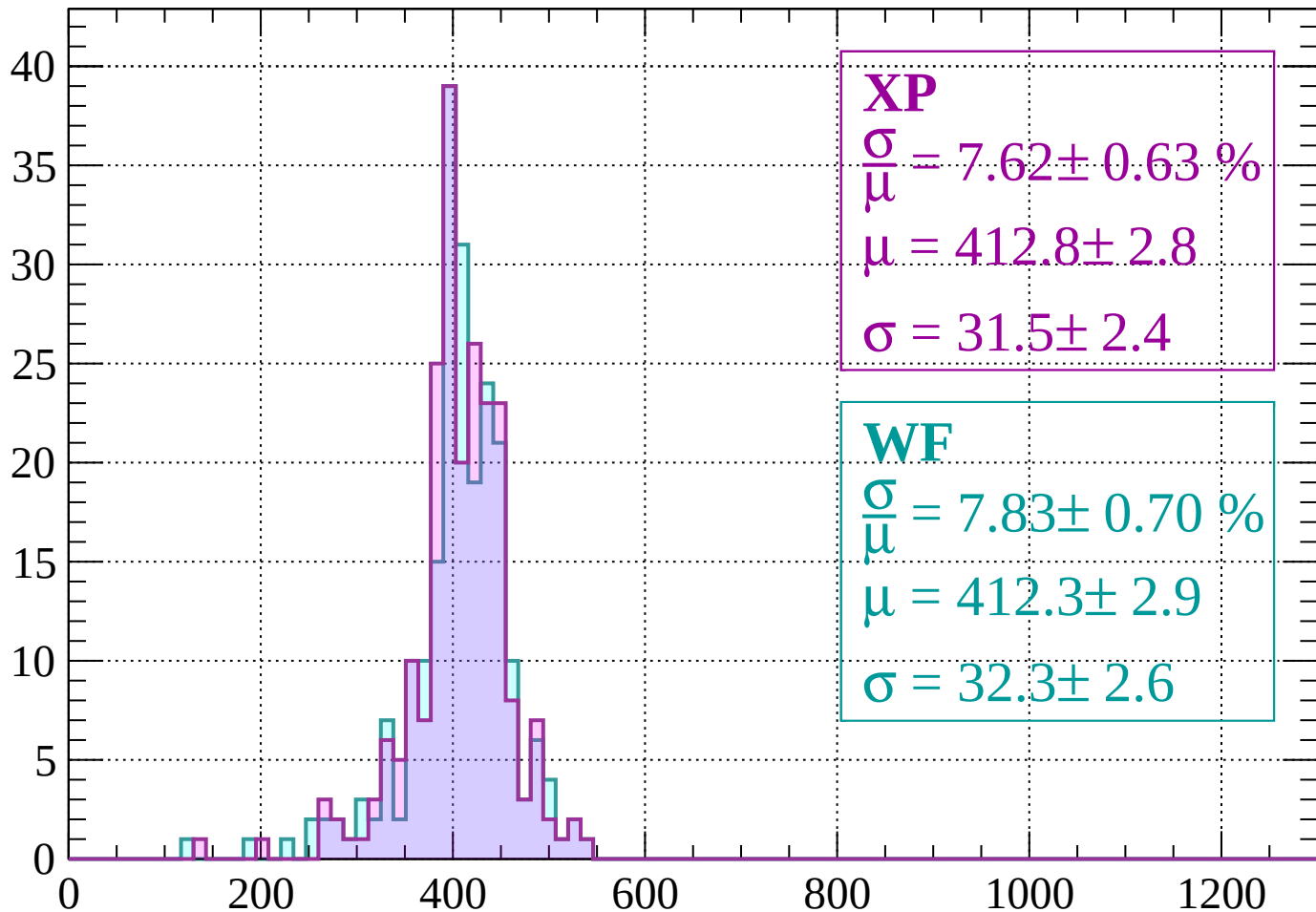
$$\mu = 404.5 \pm 3.1$$

$$\sigma = 39.6 \pm 3.3$$



Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 7.62 \pm 0.63 \%$$

$$\mu = 412.8 \pm 2.8$$

$$\sigma = 31.5 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 7.83 \pm 0.70 \%$$

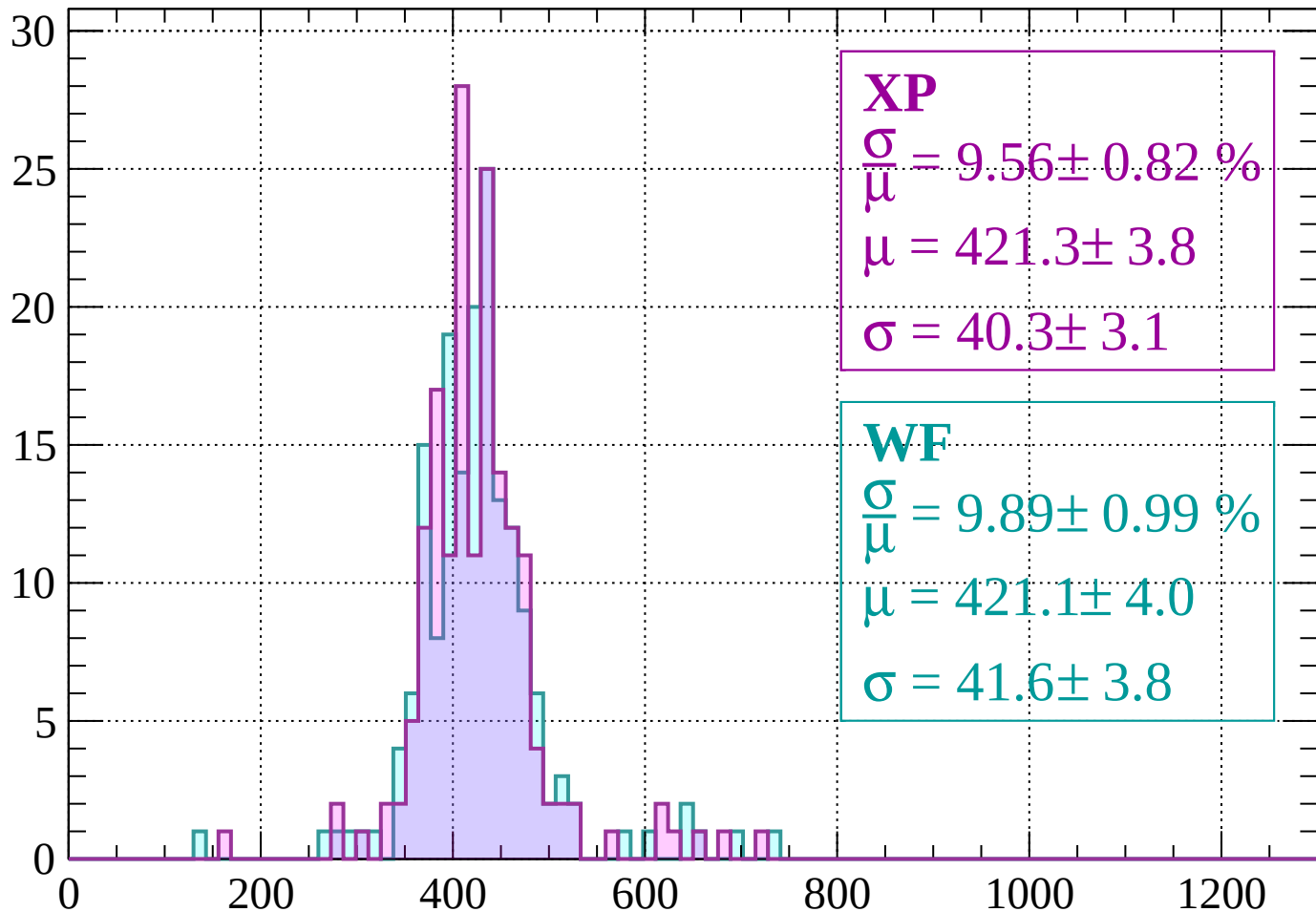
$$\mu = 412.3 \pm 2.9$$

$$\sigma = 32.3 \pm 2.6$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 9.56 \pm 0.82 \%$$

$$\mu = 421.3 \pm 3.8$$

$$\sigma = 40.3 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = 9.89 \pm 0.99 \%$$

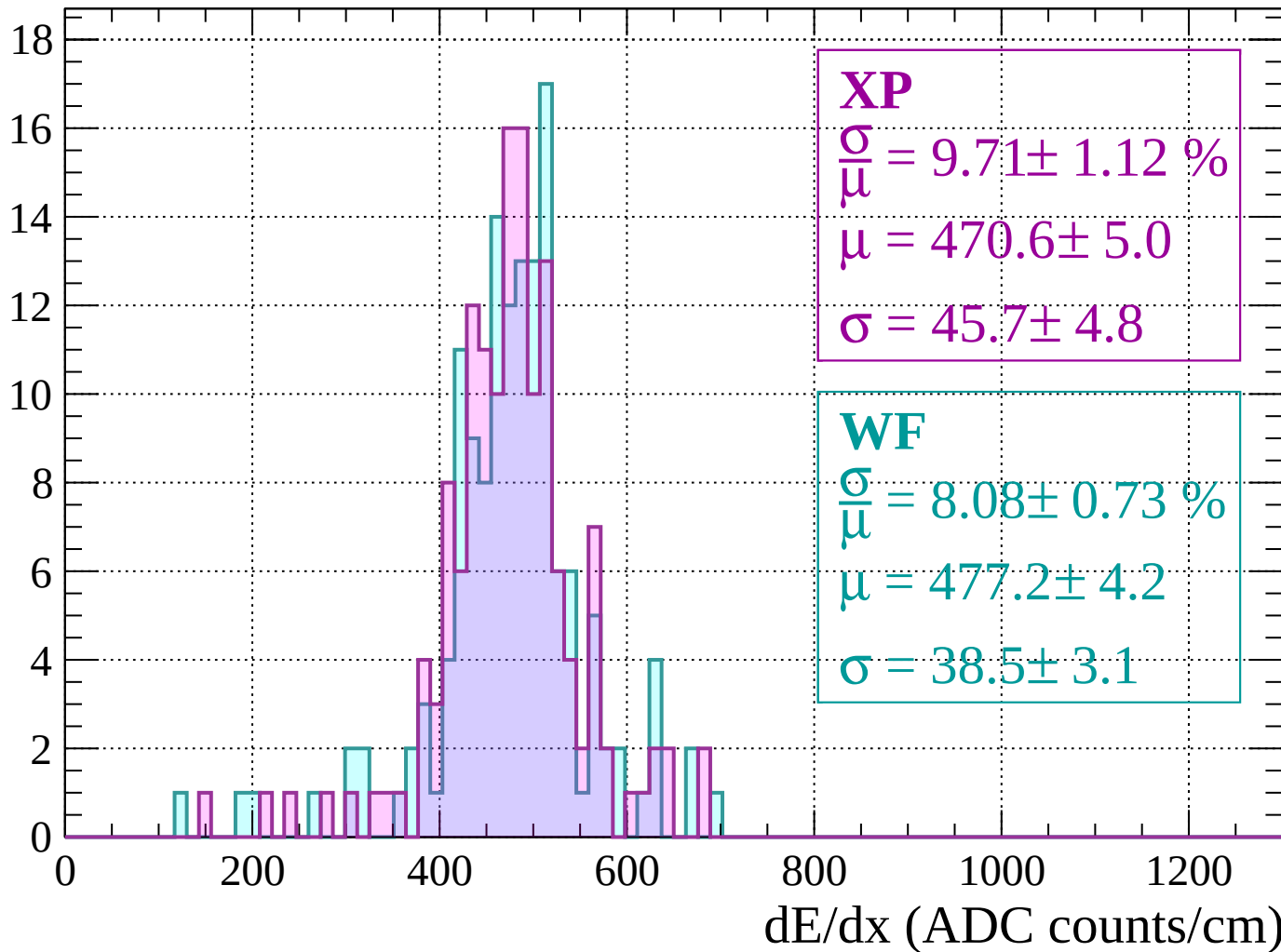
$$\mu = 421.1 \pm 4.0$$

$$\sigma = 41.6 \pm 3.8$$

dE/dx (ADC counts/cm)

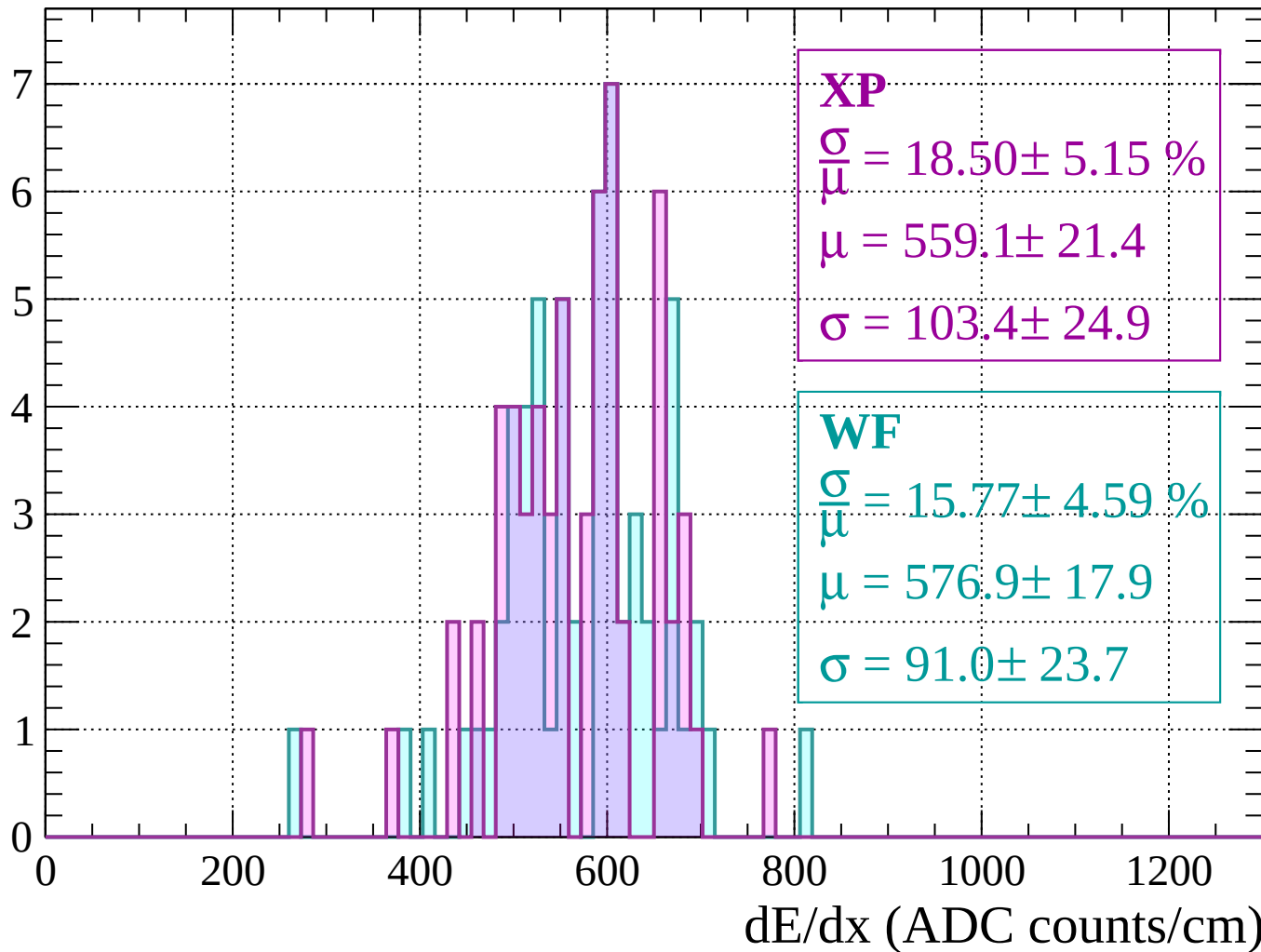
Energy loss | $-160 < p < -120$

Count



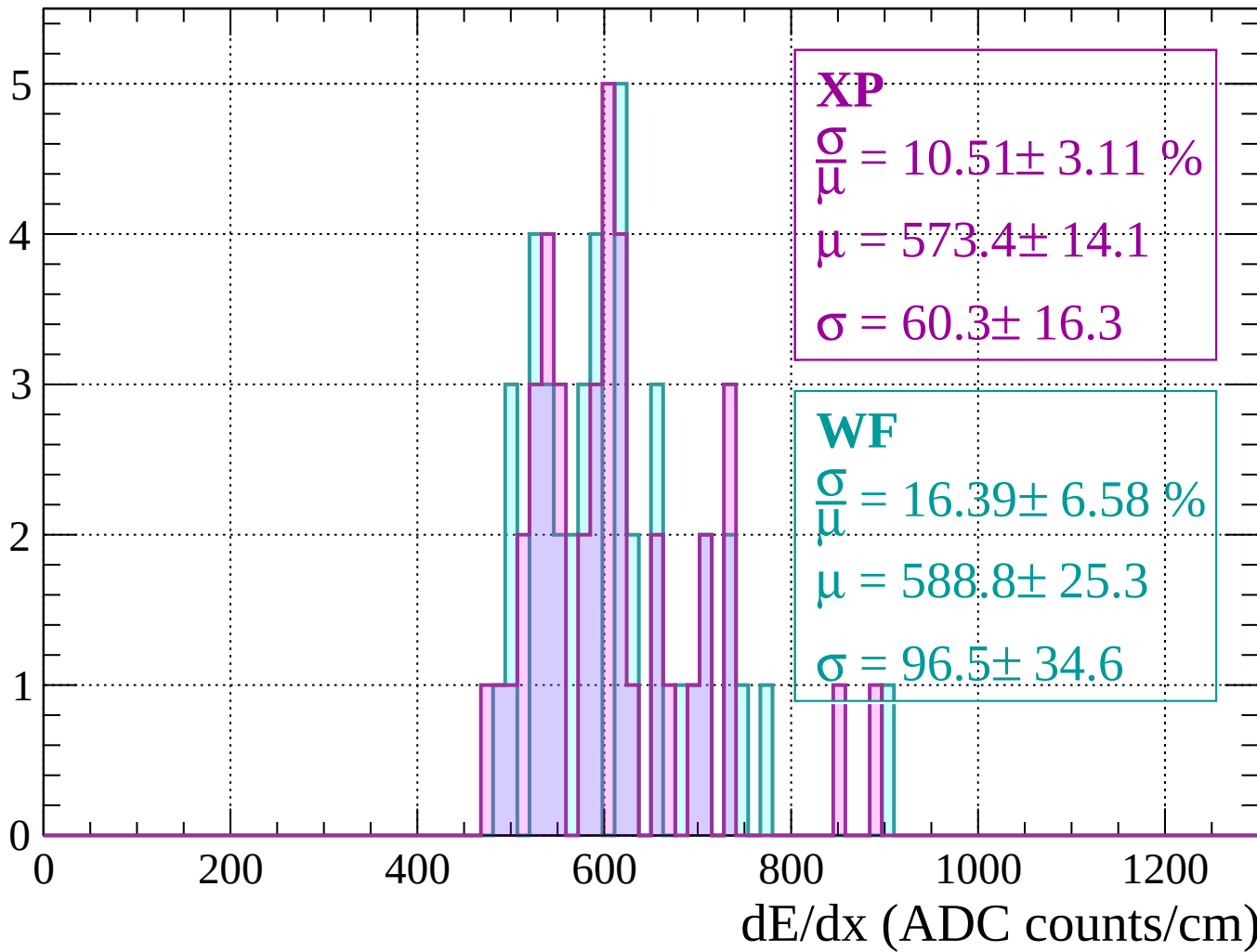
Energy loss | $-120 < p < -80$

Count



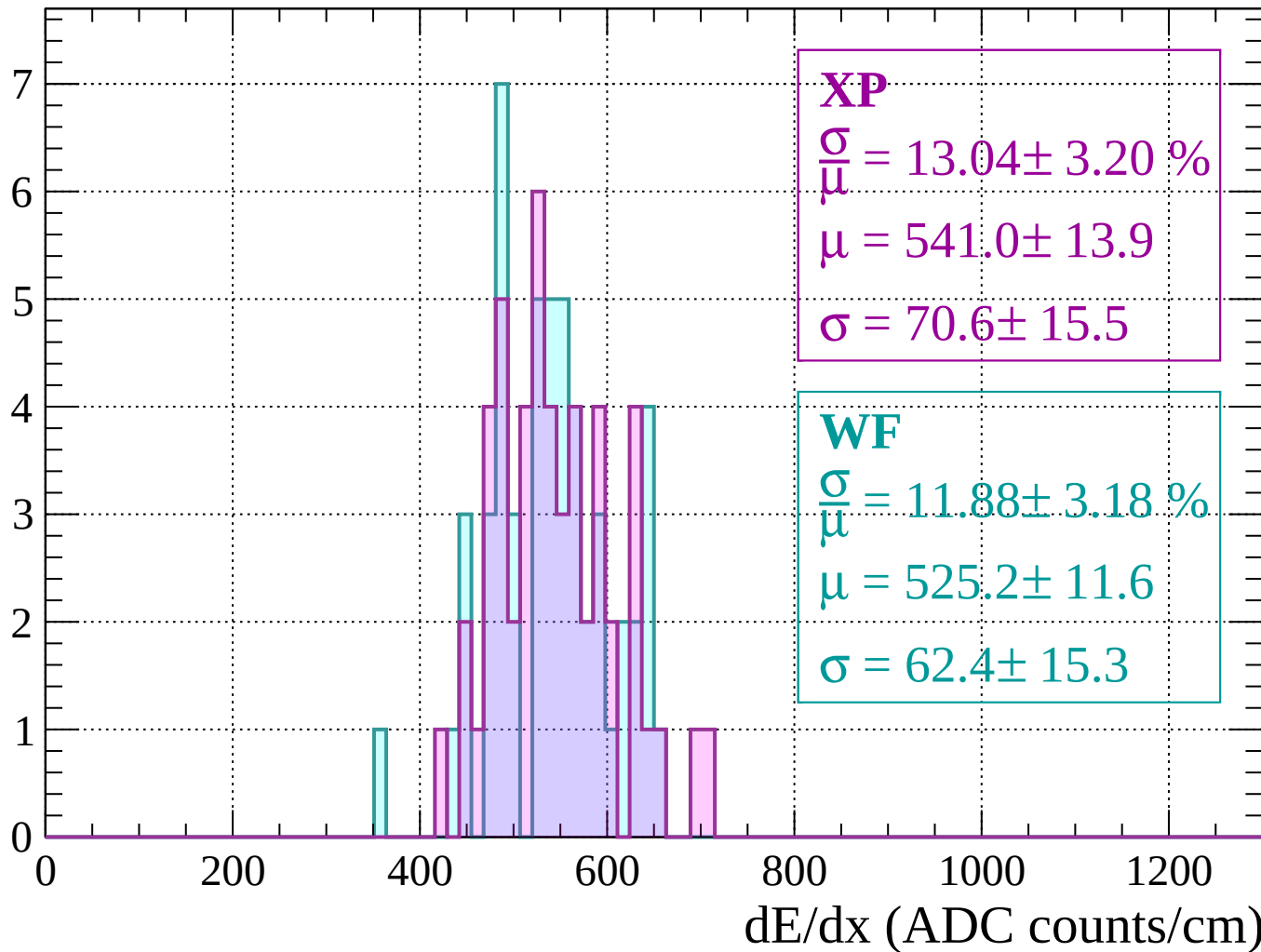
Energy loss | $-80 < p < -40$

Count



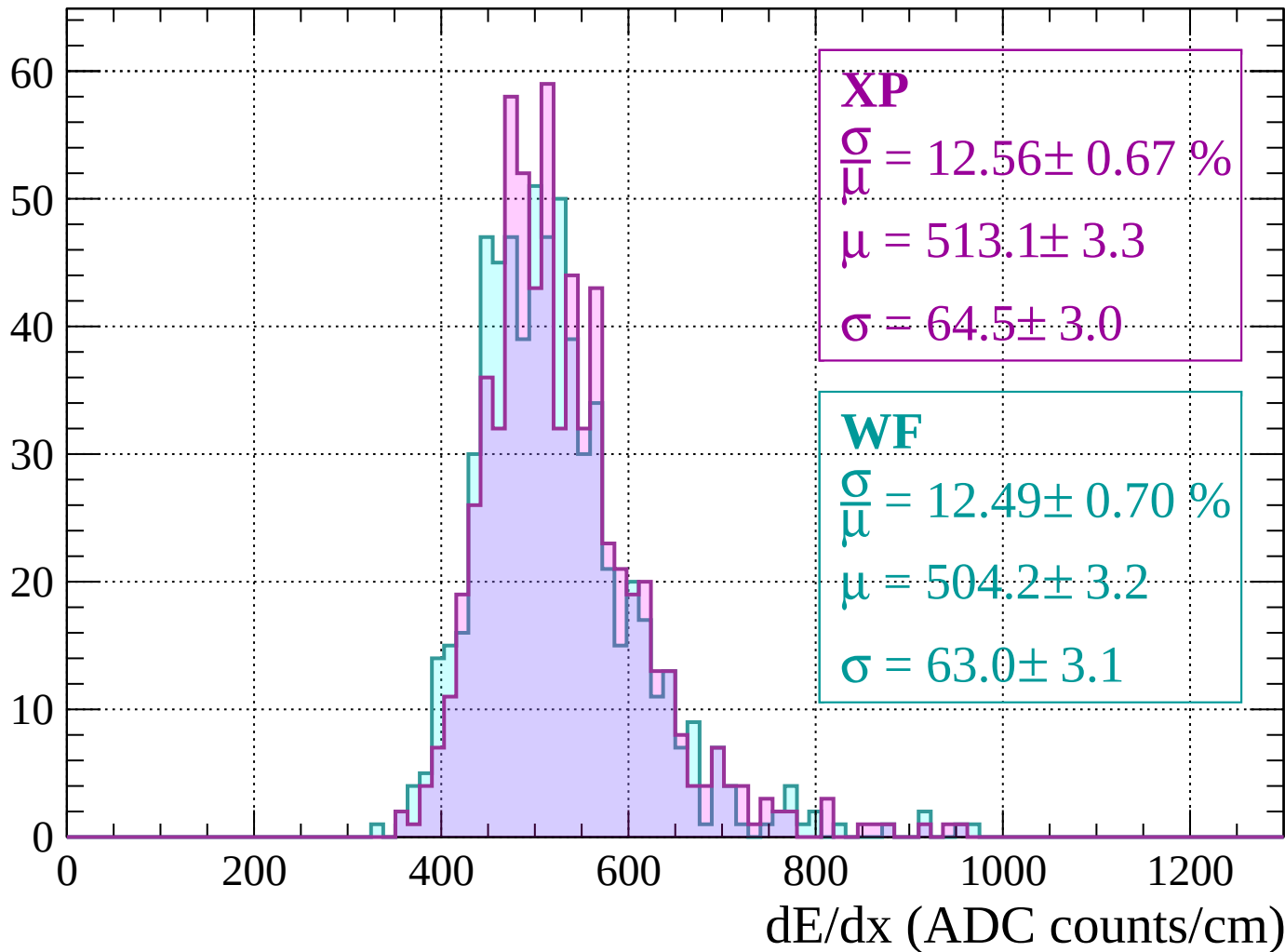
Energy loss | $-40 < p < 0$

Count



Energy loss | $0 < p < 40$

Count



Energy loss | $40 < p < 80$

Count

XP

$$\frac{\sigma}{\mu} = 10.47 \pm 0.48 \%$$

$$\mu = 553.1 \pm 2.8$$

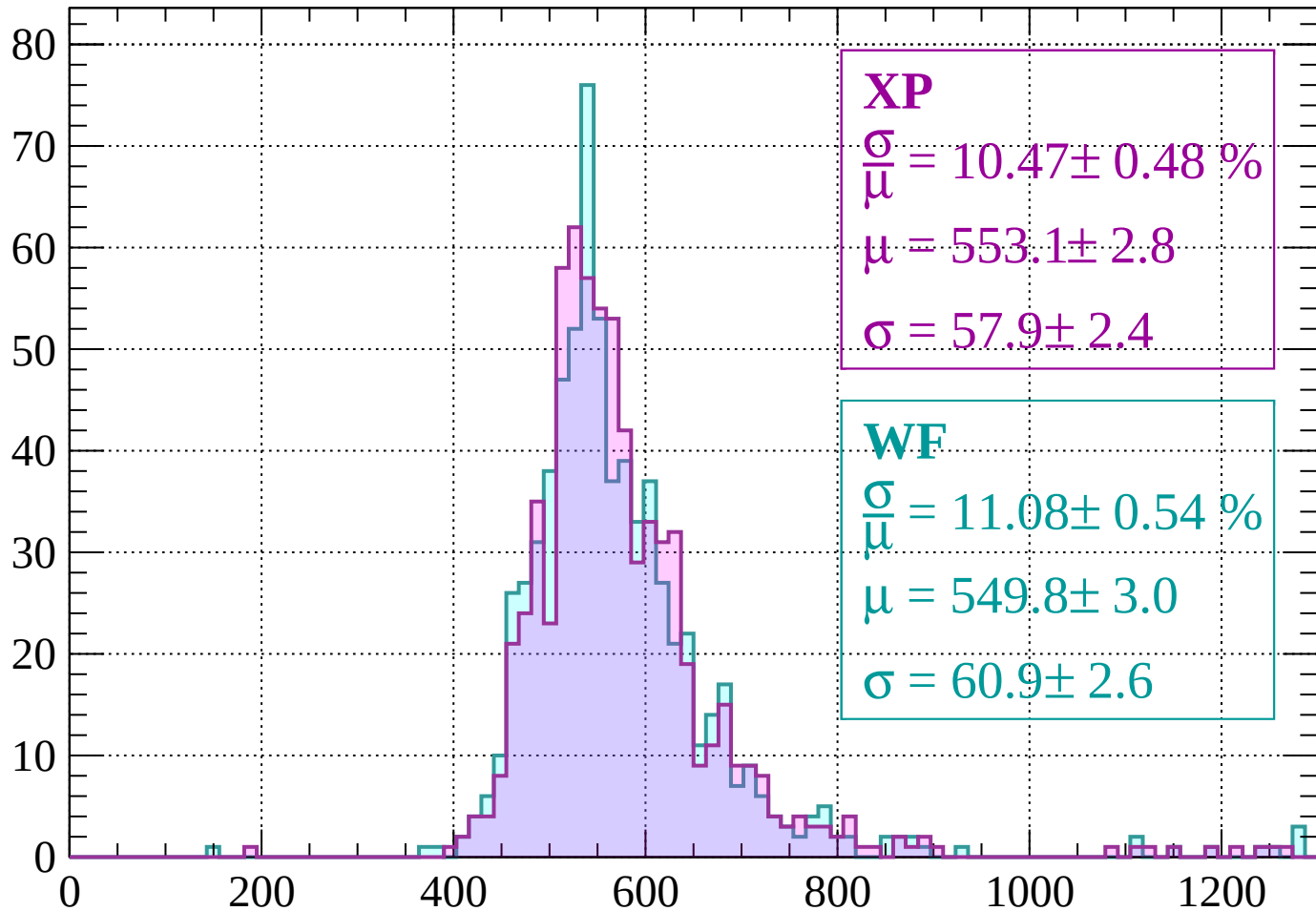
$$\sigma = 57.9 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 11.08 \pm 0.54 \%$$

$$\mu = 549.8 \pm 3.0$$

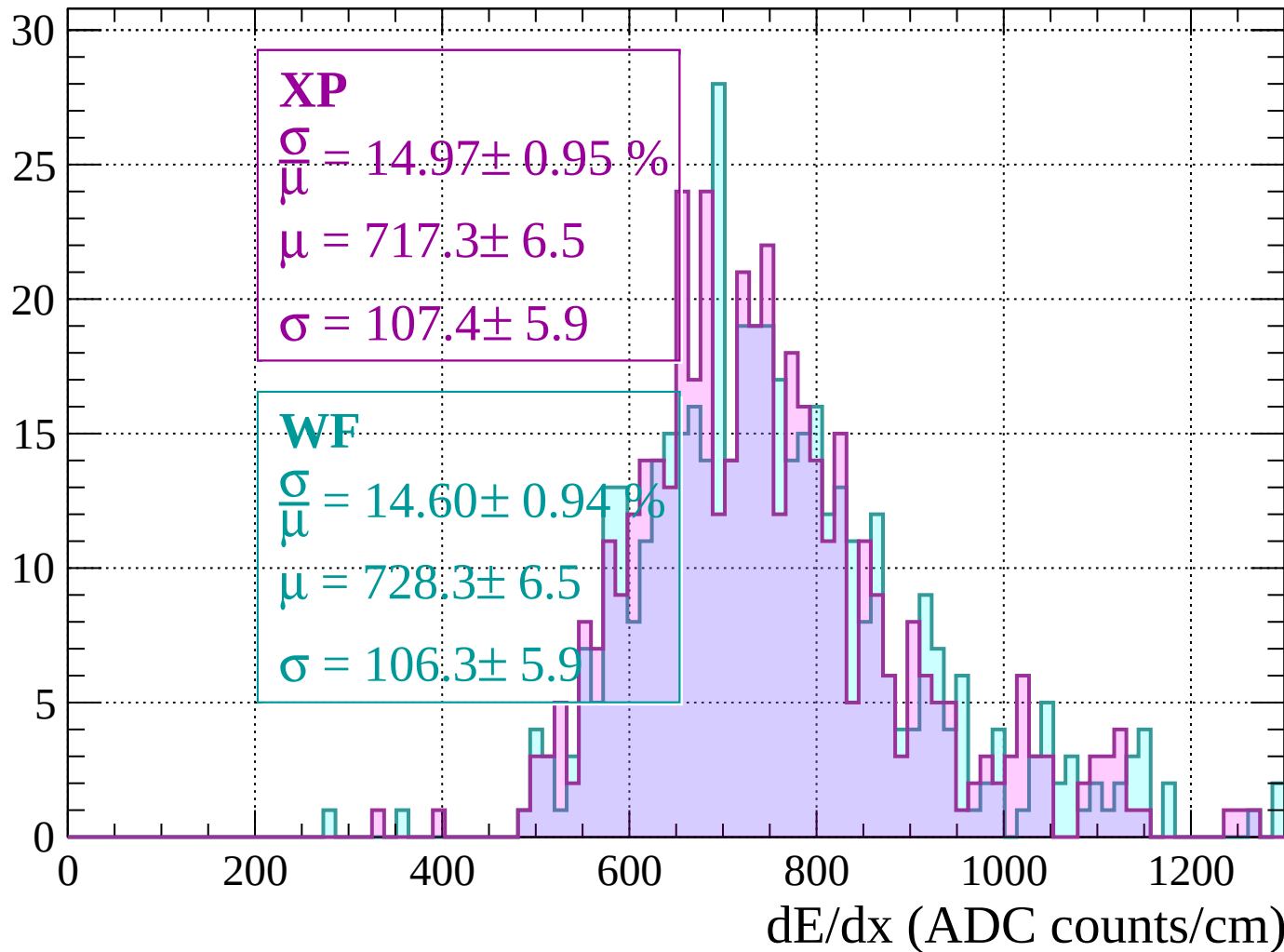
$$\sigma = 60.9 \pm 2.6$$



dE/dx (ADC counts/cm)

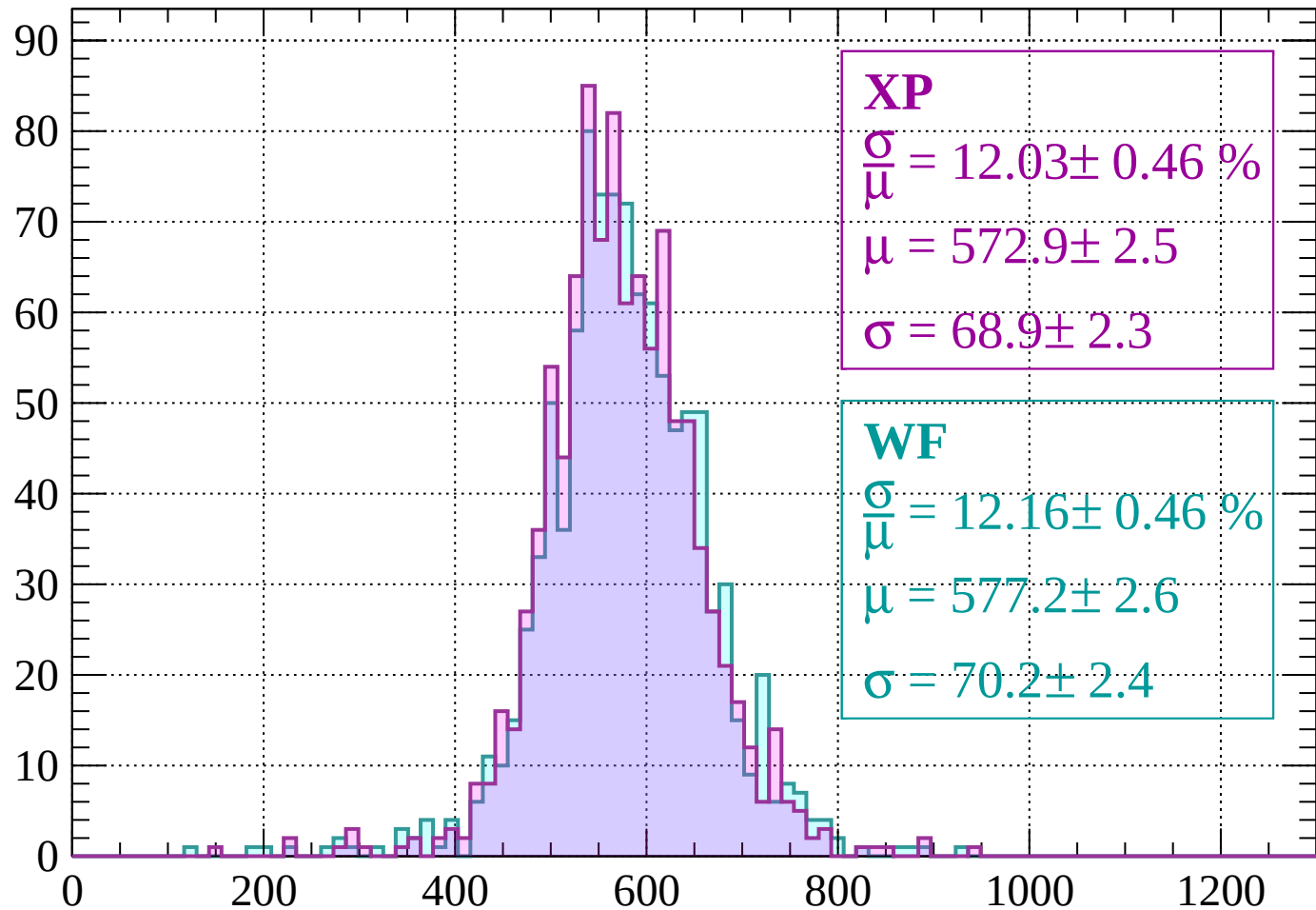
Energy loss | $80 < p < 120$

Count



Energy loss | $120 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 12.03 \pm 0.46 \%$$

$$\mu = 572.9 \pm 2.5$$

$$\sigma = 68.9 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 12.16 \pm 0.46 \%$$

$$\mu = 577.2 \pm 2.6$$

$$\sigma = 70.2 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $160 < p < 200$

Count

140

120

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 10.03 \pm 0.41 \%$$

$$\mu = 481.5 \pm 1.7$$

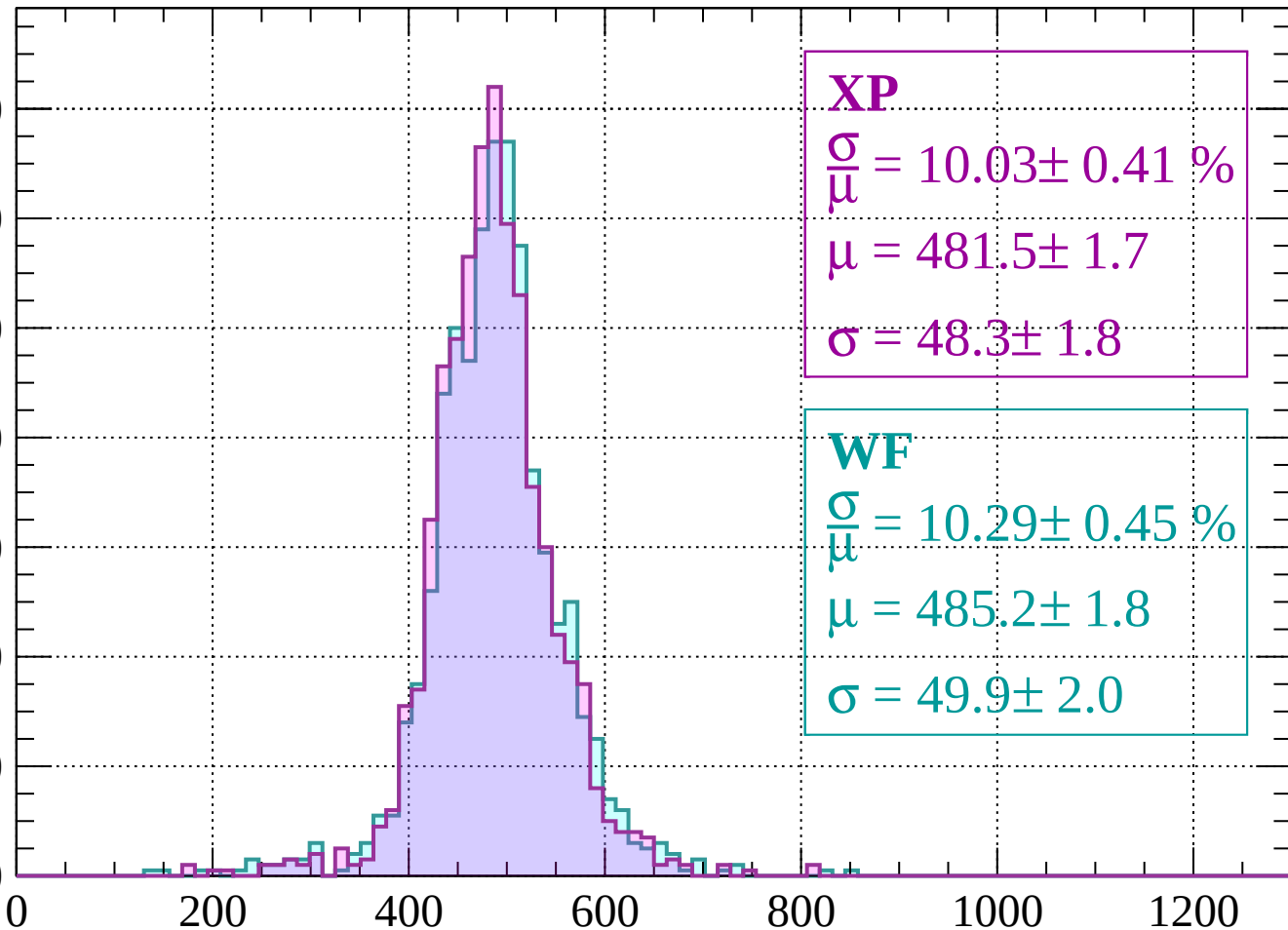
$$\sigma = 48.3 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 10.29 \pm 0.45 \%$$

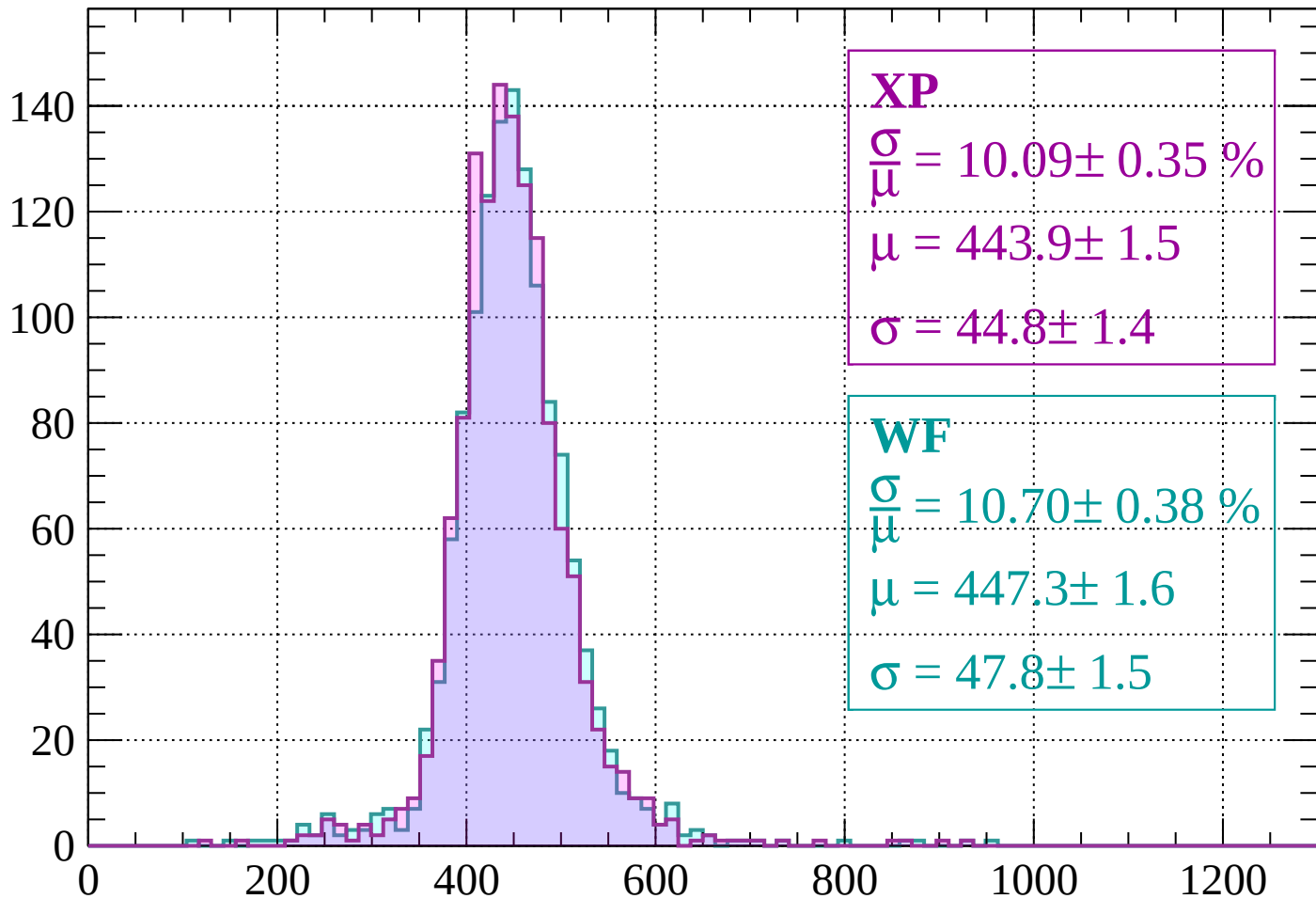
$$\mu = 485.2 \pm 1.8$$

$$\sigma = 49.9 \pm 2.0$$



Energy loss | $200 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 10.09 \pm 0.35 \%$$

$$\mu = 443.9 \pm 1.5$$

$$\sigma = 44.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.70 \pm 0.38 \%$$

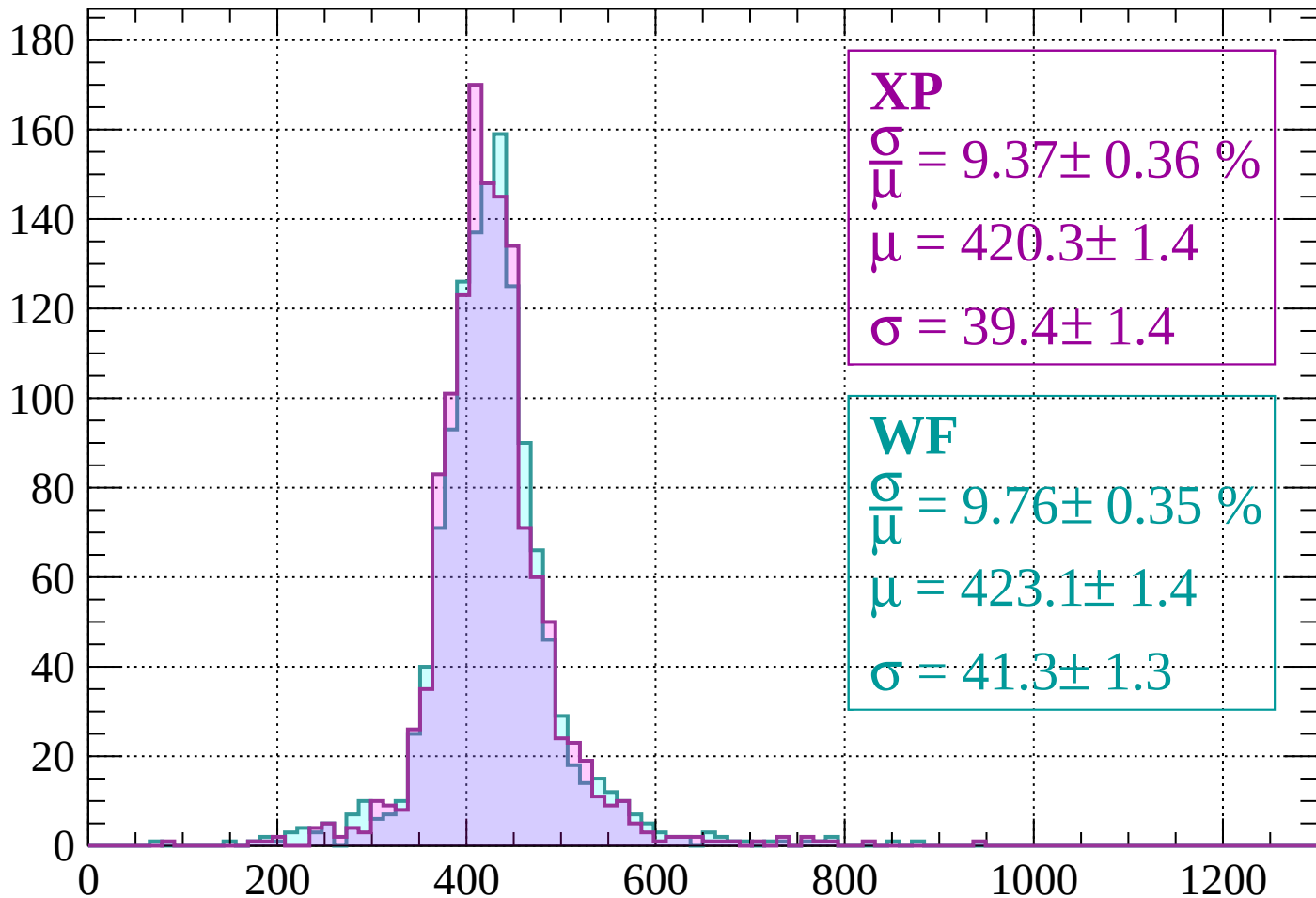
$$\mu = 447.3 \pm 1.6$$

$$\sigma = 47.8 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 9.37 \pm 0.36 \%$$

$$\mu = 420.3 \pm 1.4$$

$$\sigma = 39.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.76 \pm 0.35 \%$$

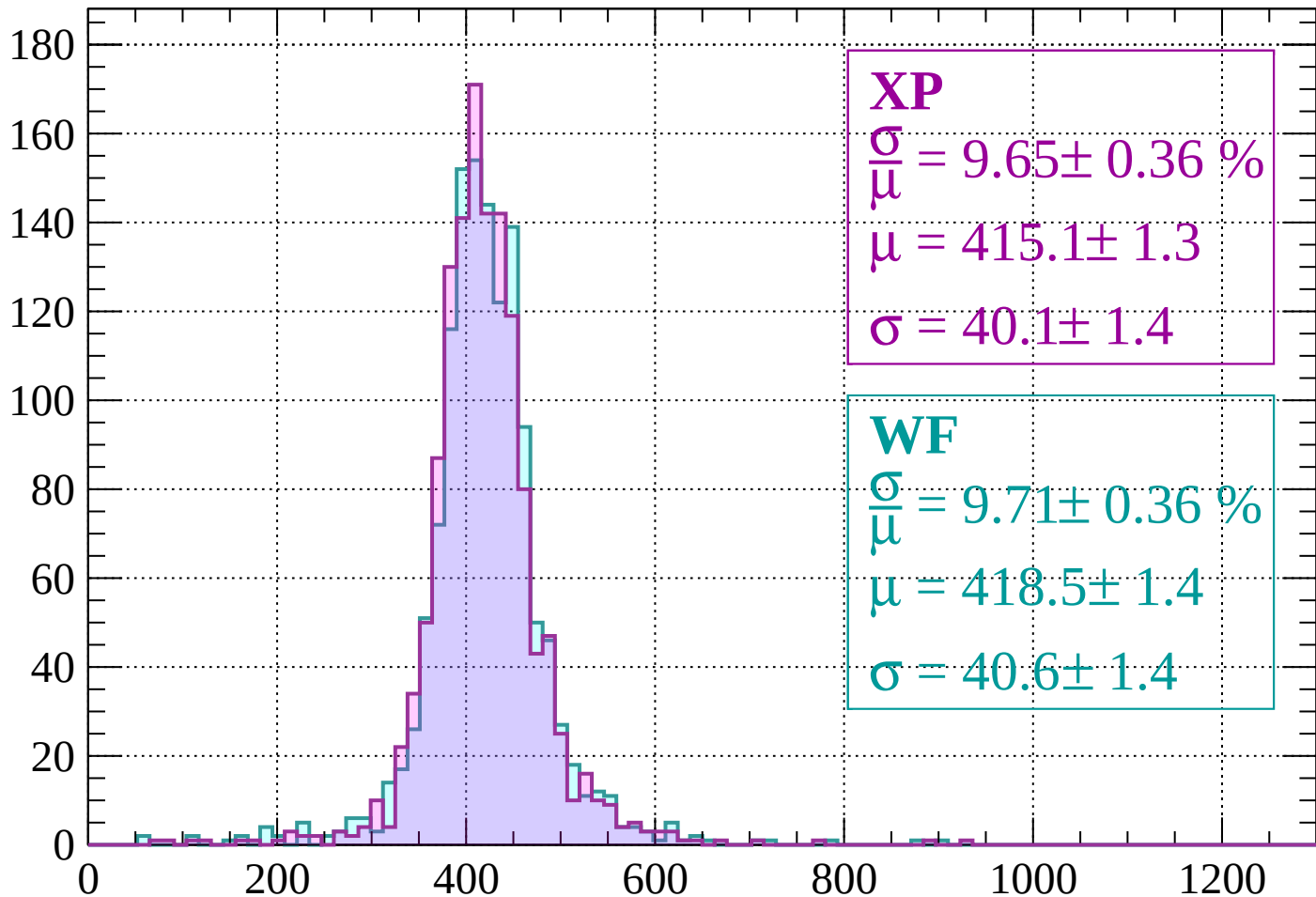
$$\mu = 423.1 \pm 1.4$$

$$\sigma = 41.3 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 9.65 \pm 0.36 \%$$

$$\mu = 415.1 \pm 1.3$$

$$\sigma = 40.1 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.71 \pm 0.36 \%$$

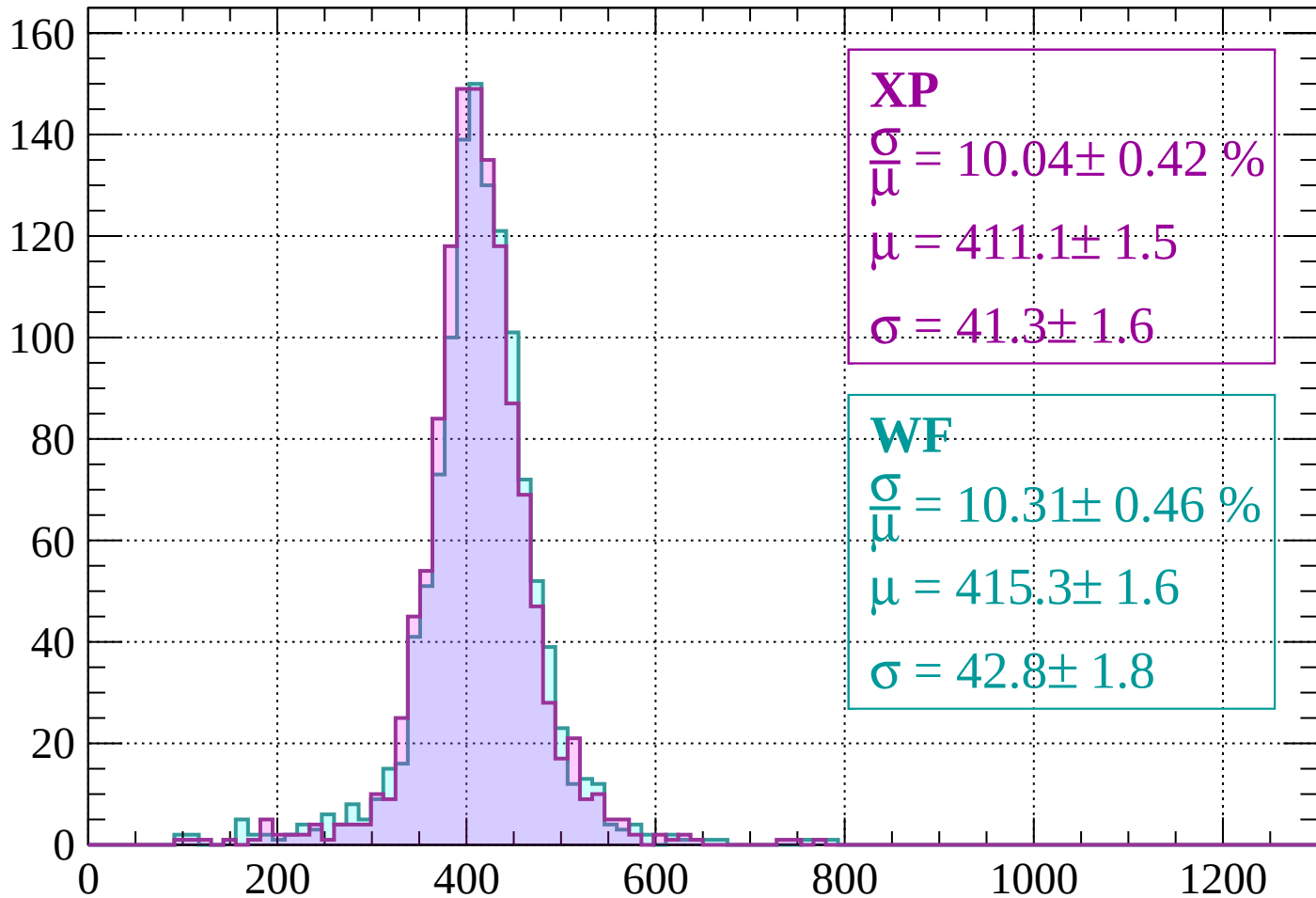
$$\mu = 418.5 \pm 1.4$$

$$\sigma = 40.6 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 10.04 \pm 0.42 \%$$

$$\mu = 411.1 \pm 1.5$$

$$\sigma = 41.3 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 10.31 \pm 0.46 \%$$

$$\mu = 415.3 \pm 1.6$$

$$\sigma = 42.8 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 400$

Count

140

120

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.45 \pm 0.43 \%$$

$$\mu = 409.4 \pm 1.5$$

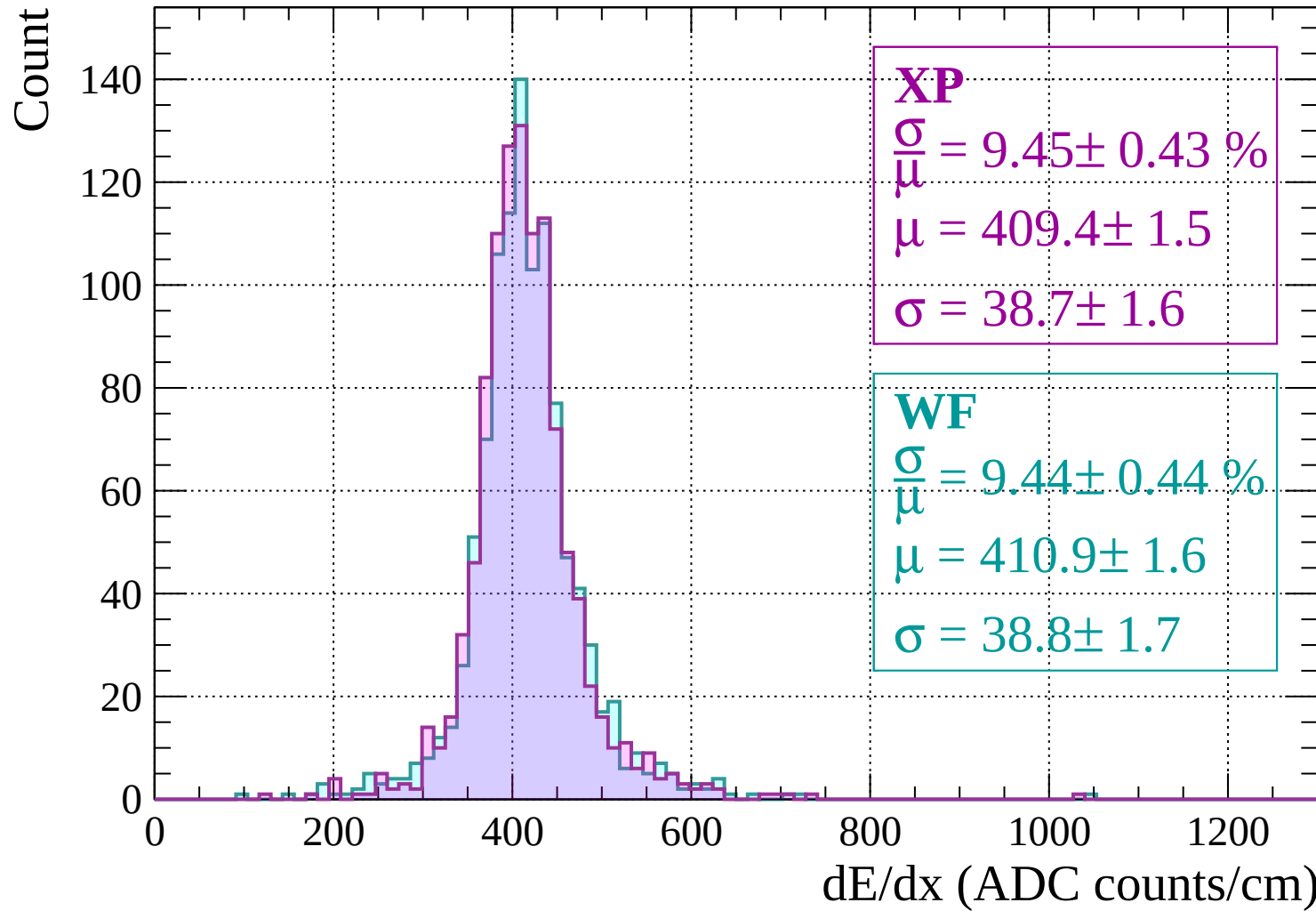
$$\sigma = 38.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 9.44 \pm 0.44 \%$$

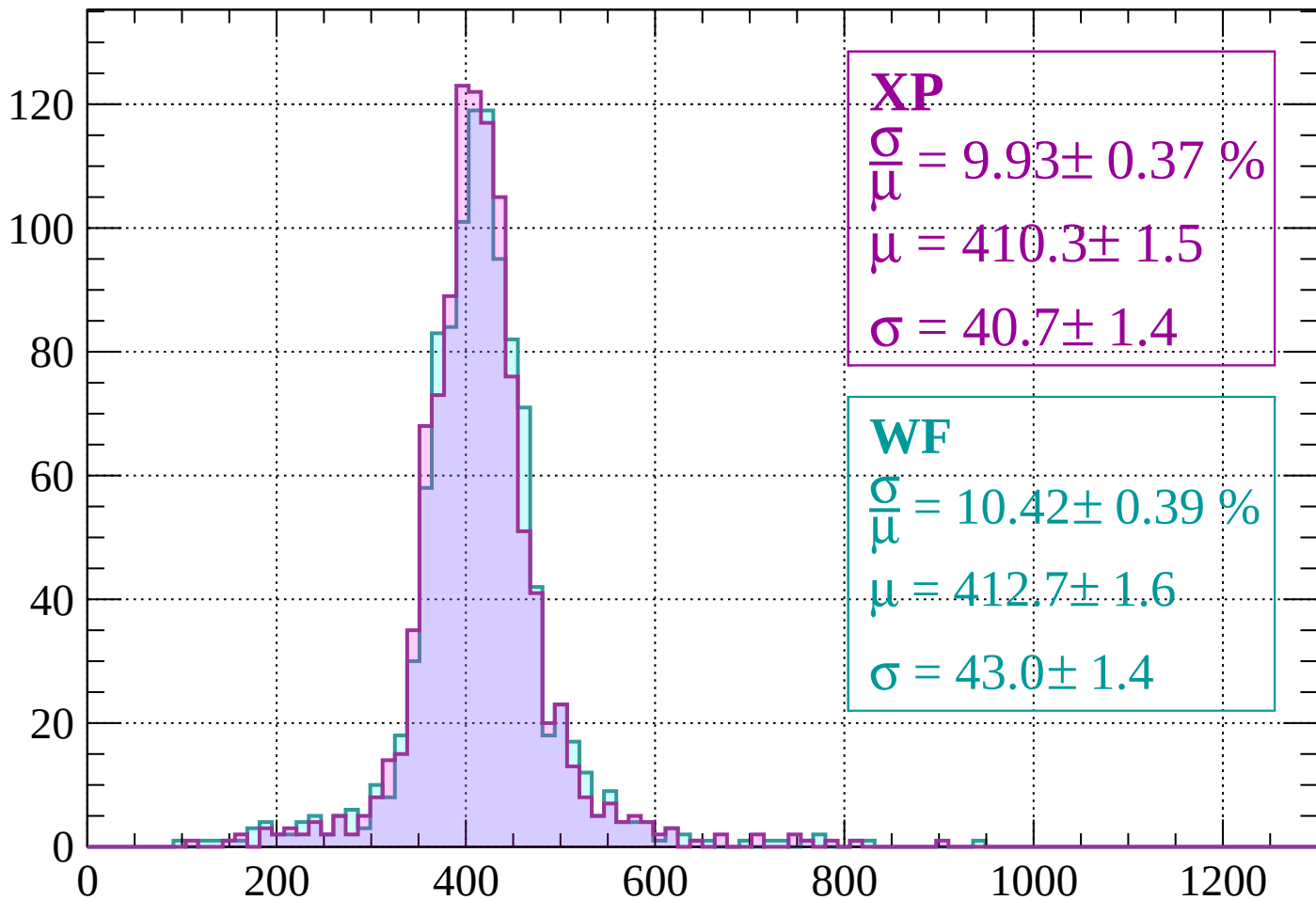
$$\mu = 410.9 \pm 1.6$$

$$\sigma = 38.8 \pm 1.7$$



Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 9.93 \pm 0.37 \%$$

$$\mu = 410.3 \pm 1.5$$

$$\sigma = 40.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.42 \pm 0.39 \%$$

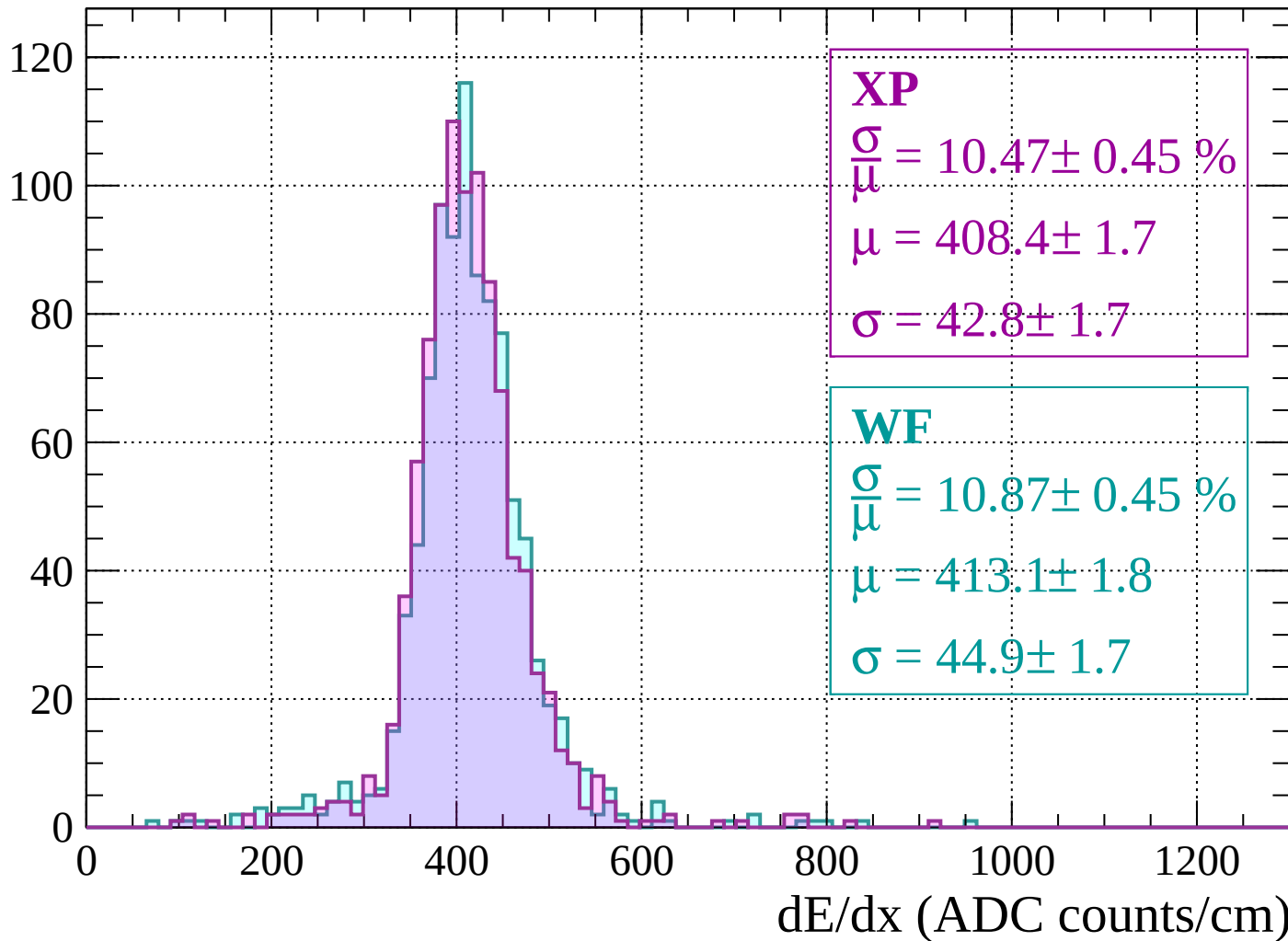
$$\mu = 412.7 \pm 1.6$$

$$\sigma = 43.0 \pm 1.4$$

dE/dx (ADC counts/cm)

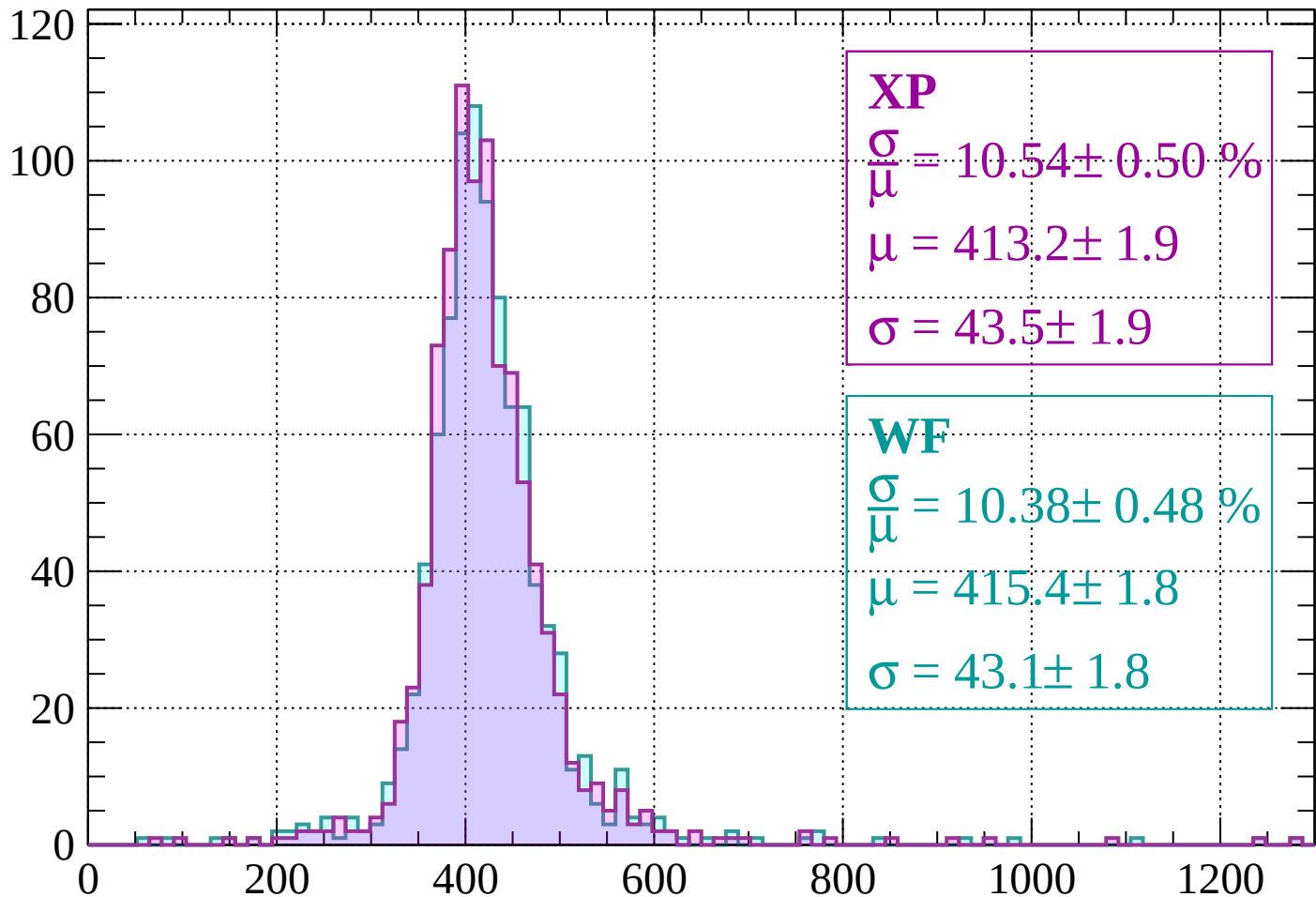
Energy loss | $440 < p < 480$

Count



Energy loss | $480 < p < 520$

Count



XP

$$\frac{\sigma}{\mu} = 10.54 \pm 0.50 \%$$

$$\mu = 413.2 \pm 1.9$$

$$\sigma = 43.5 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 10.38 \pm 0.48 \%$$

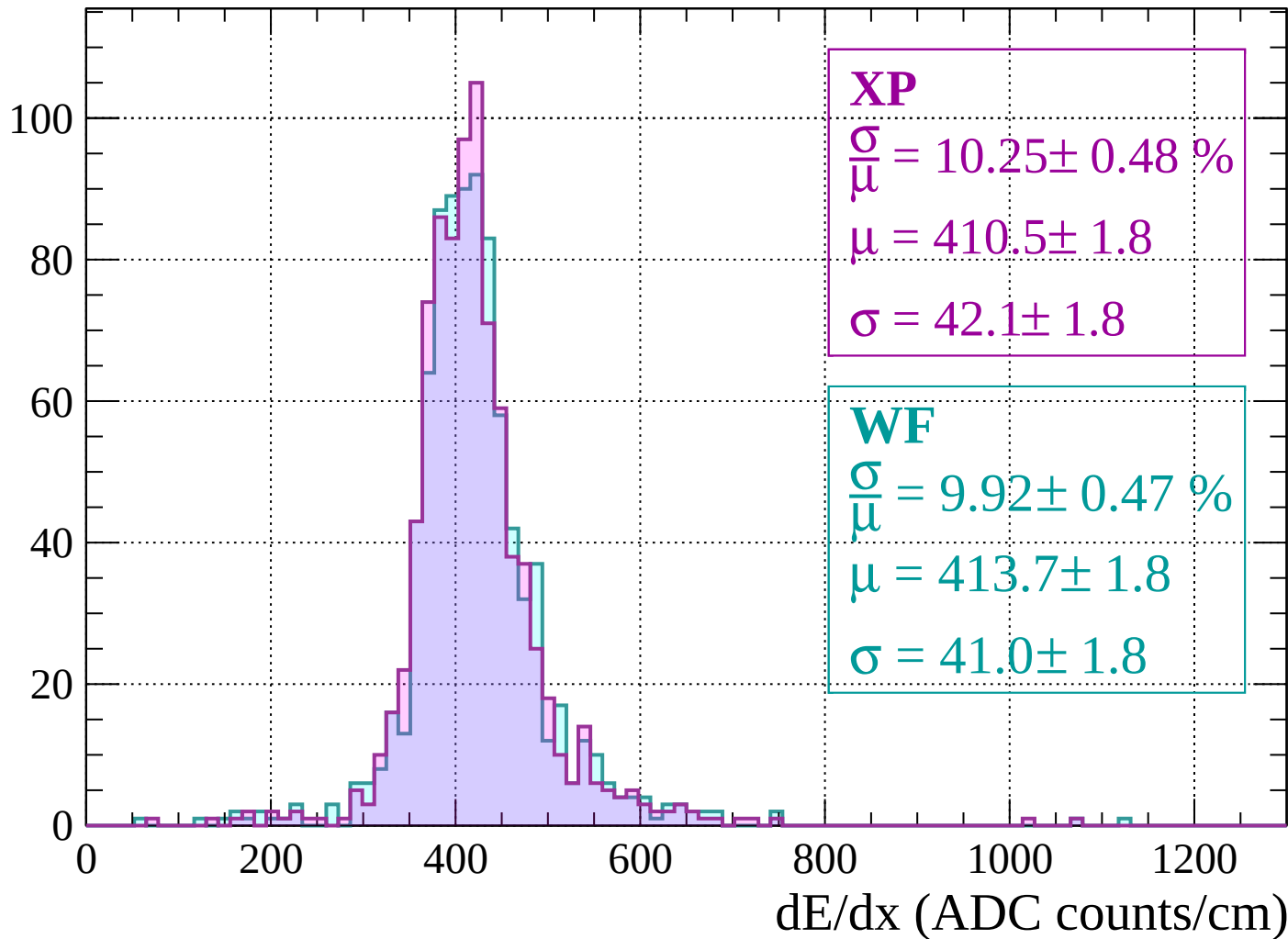
$$\mu = 415.4 \pm 1.8$$

$$\sigma = 43.1 \pm 1.8$$

dE/dx (ADC counts/cm)

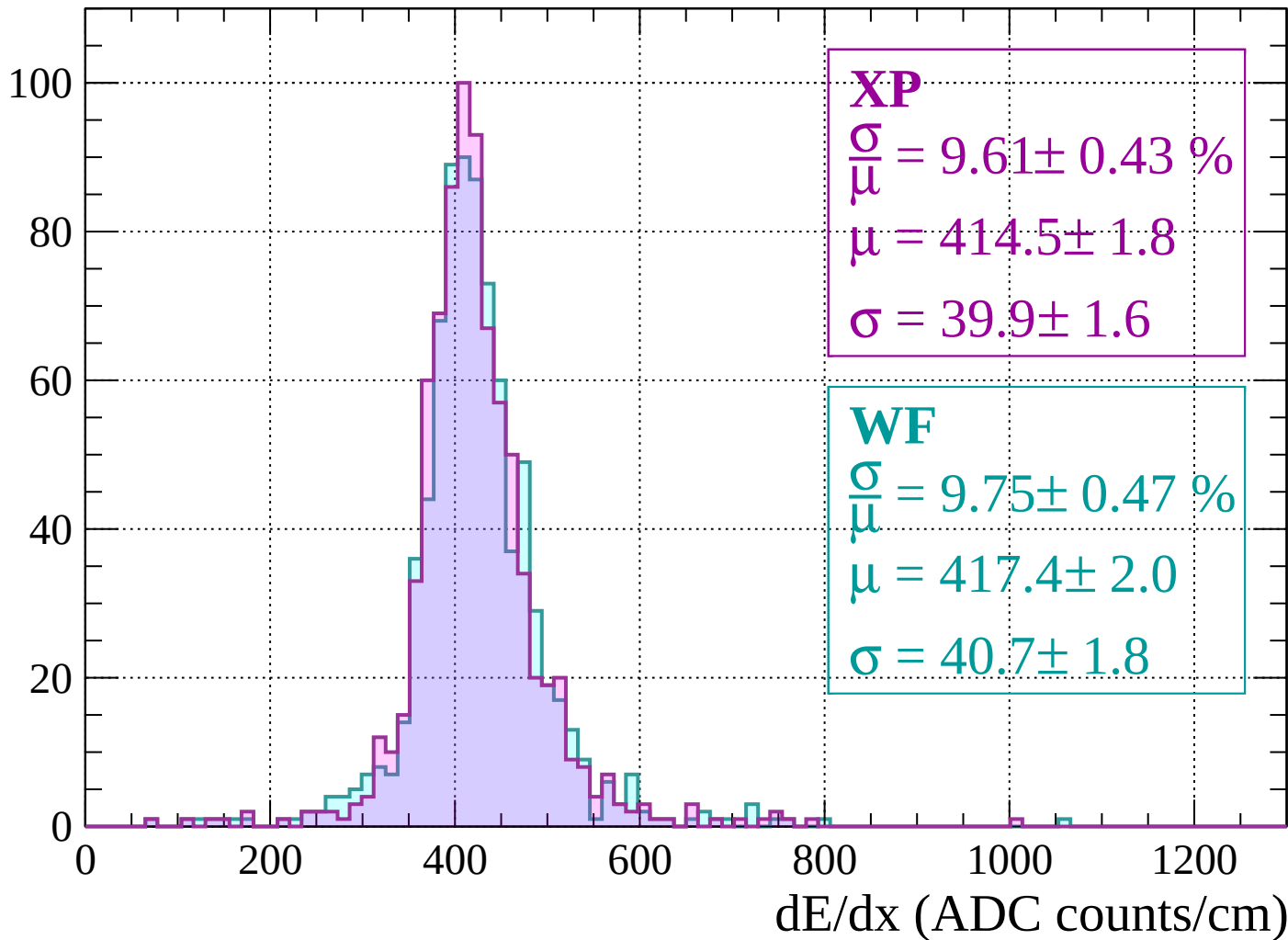
Energy loss | $520 < p < 560$

Count



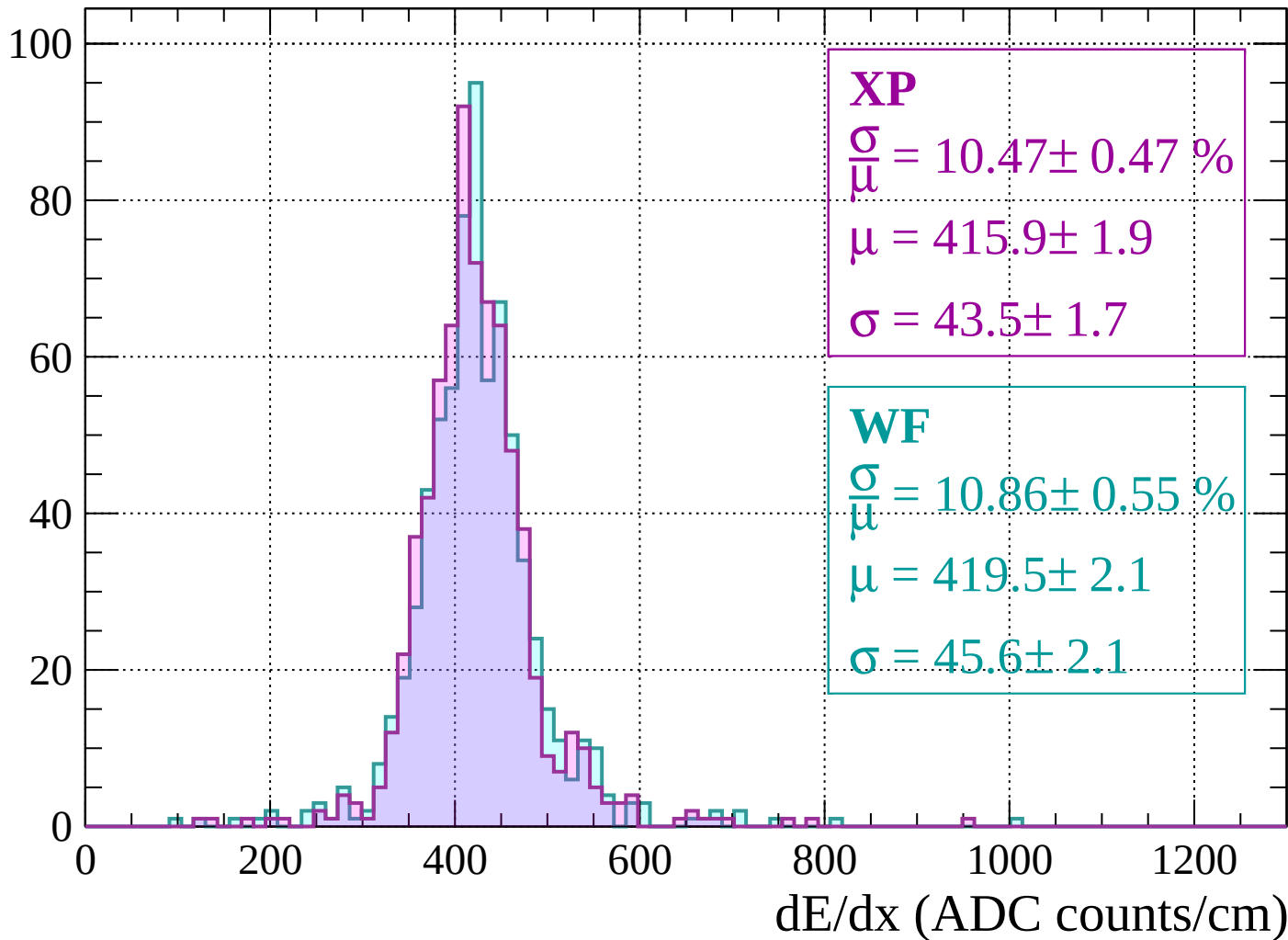
Energy loss | $560 < p < 600$

Count



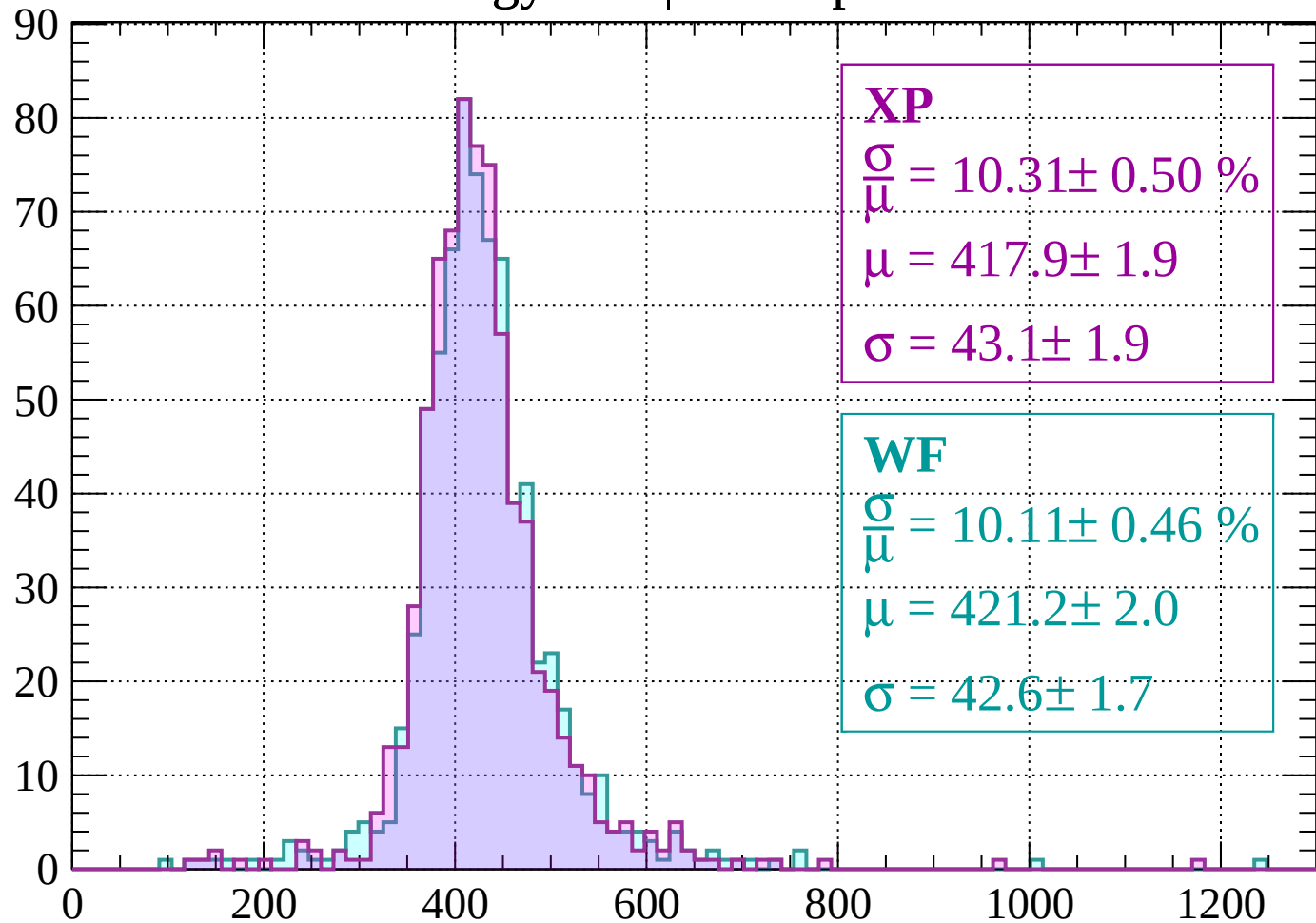
Energy loss | $600 < p < 640$

Count



Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\mu} = 10.31 \pm 0.50 \%$$

$$\mu = 417.9 \pm 1.9$$

$$\sigma = 43.1 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 10.11 \pm 0.46 \%$$

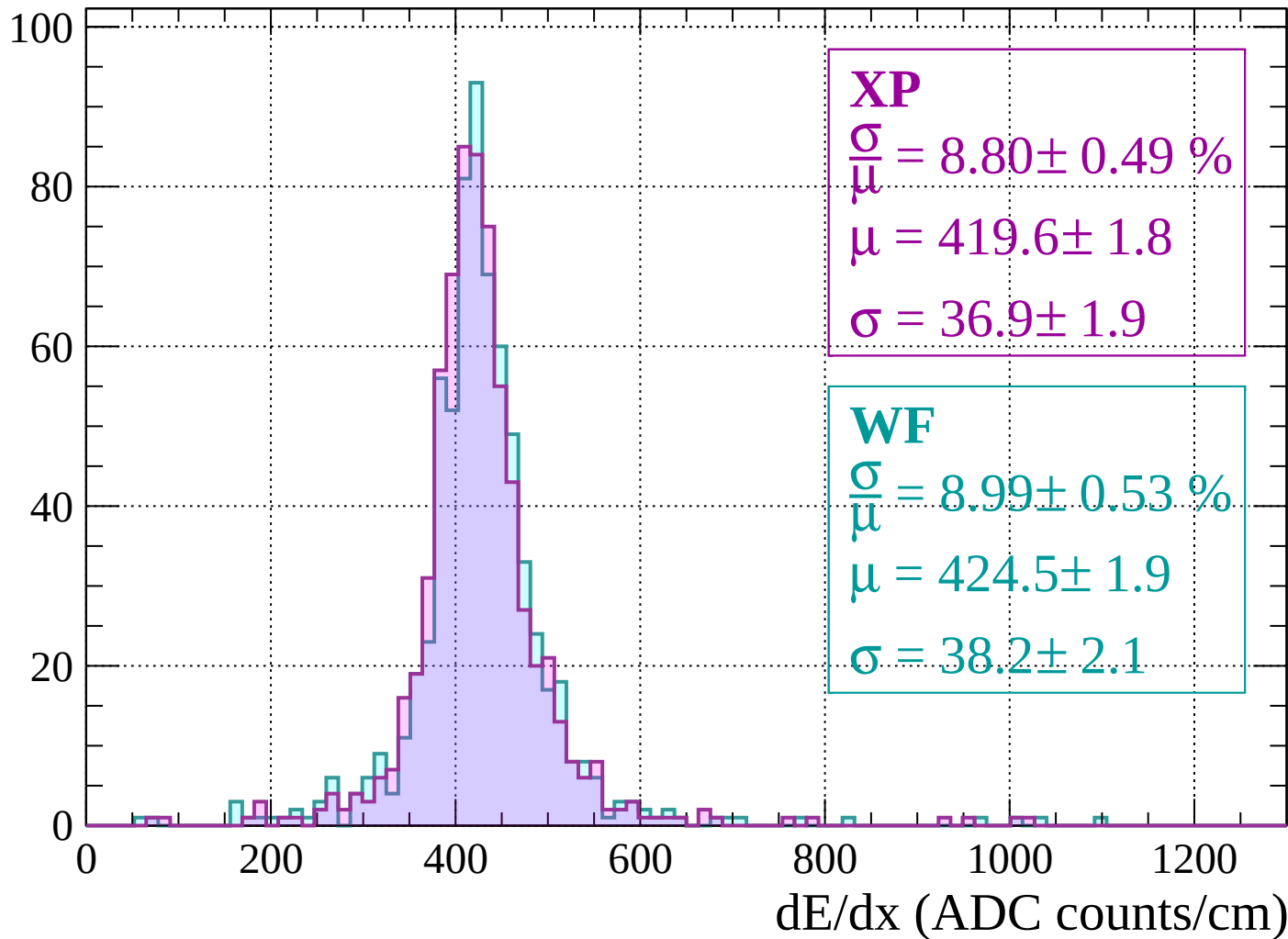
$$\mu = 421.2 \pm 2.0$$

$$\sigma = 42.6 \pm 1.7$$

dE/dx (ADC counts/cm)

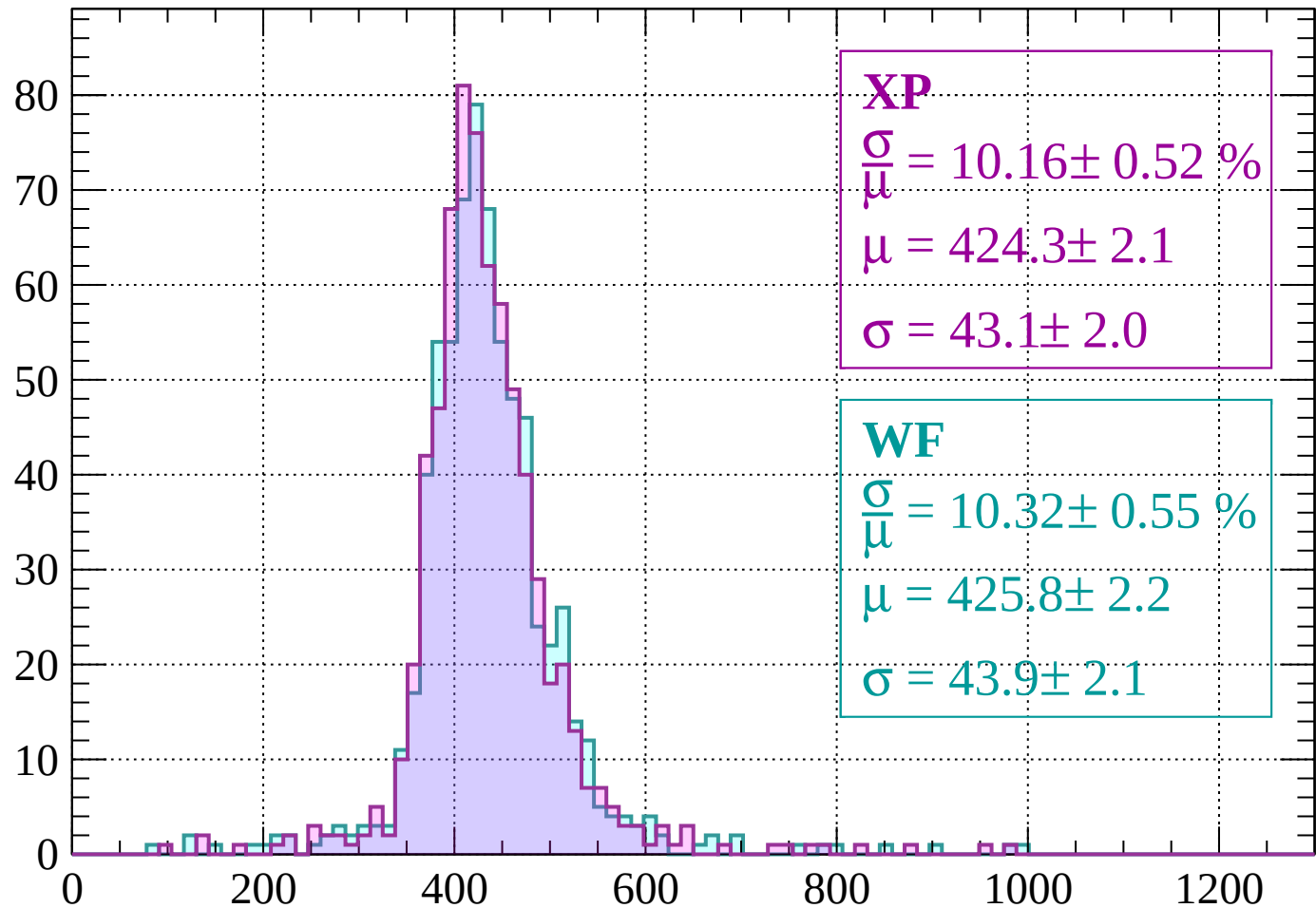
Energy loss | $680 < p < 720$

Count



Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 10.16 \pm 0.52 \%$$

$$\mu = 424.3 \pm 2.1$$

$$\sigma = 43.1 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 10.32 \pm 0.55 \%$$

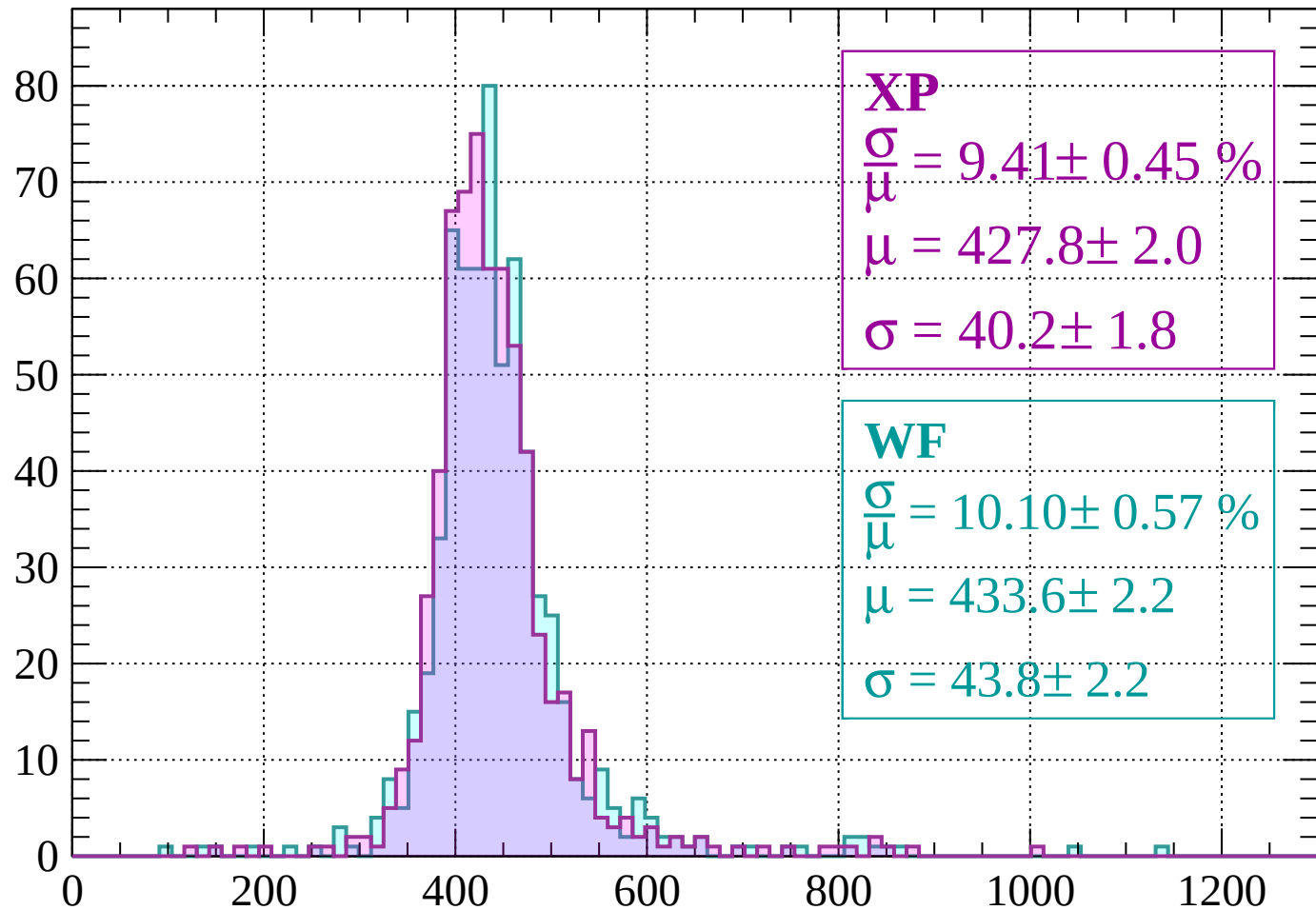
$$\mu = 425.8 \pm 2.2$$

$$\sigma = 43.9 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 9.41 \pm 0.45 \%$$

$$\mu = 427.8 \pm 2.0$$

$$\sigma = 40.2 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 10.10 \pm 0.57 \%$$

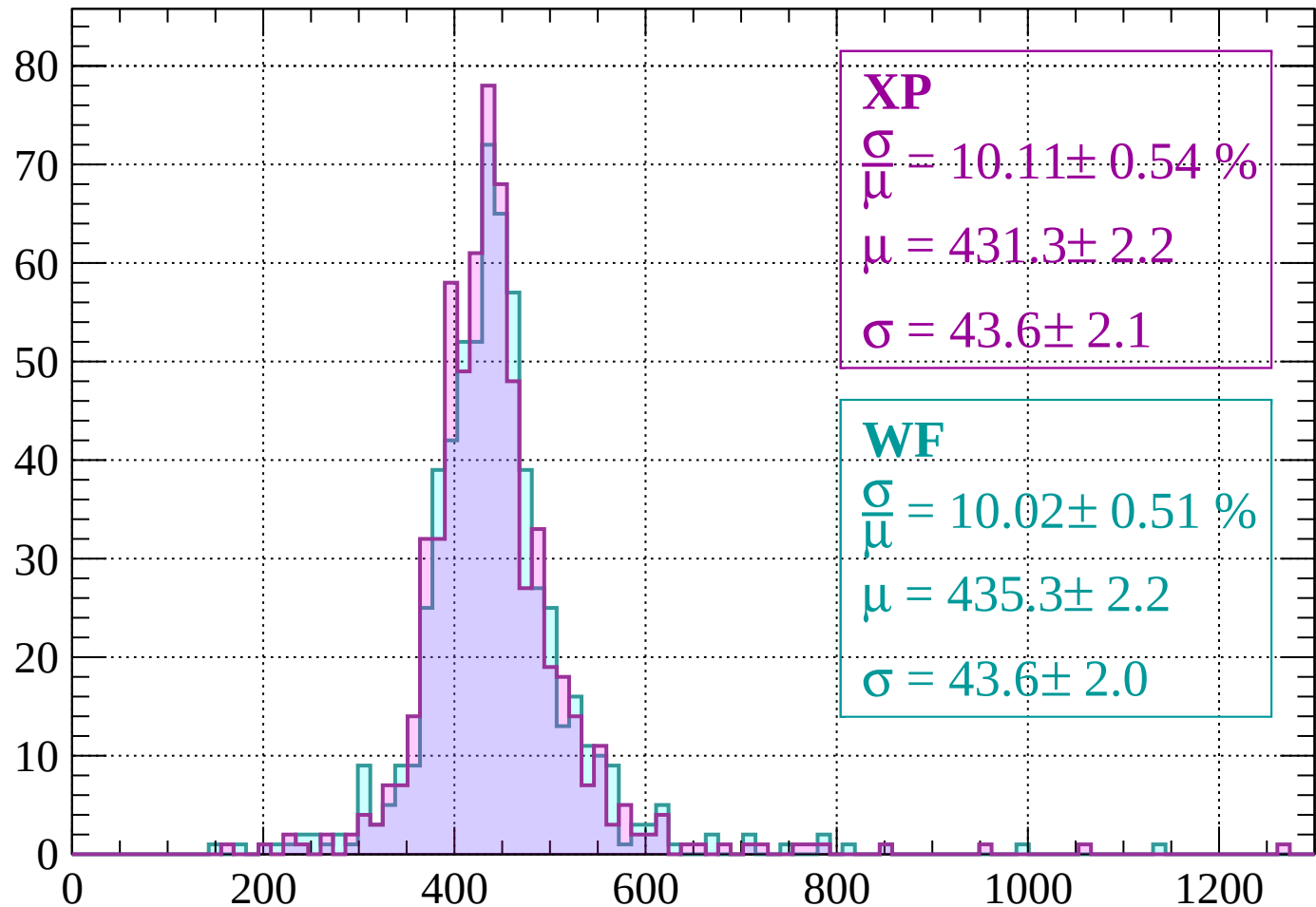
$$\mu = 433.6 \pm 2.2$$

$$\sigma = 43.8 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 10.11 \pm 0.54 \%$$

$$\mu = 431.3 \pm 2.2$$

$$\sigma = 43.6 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 10.02 \pm 0.51 \%$$

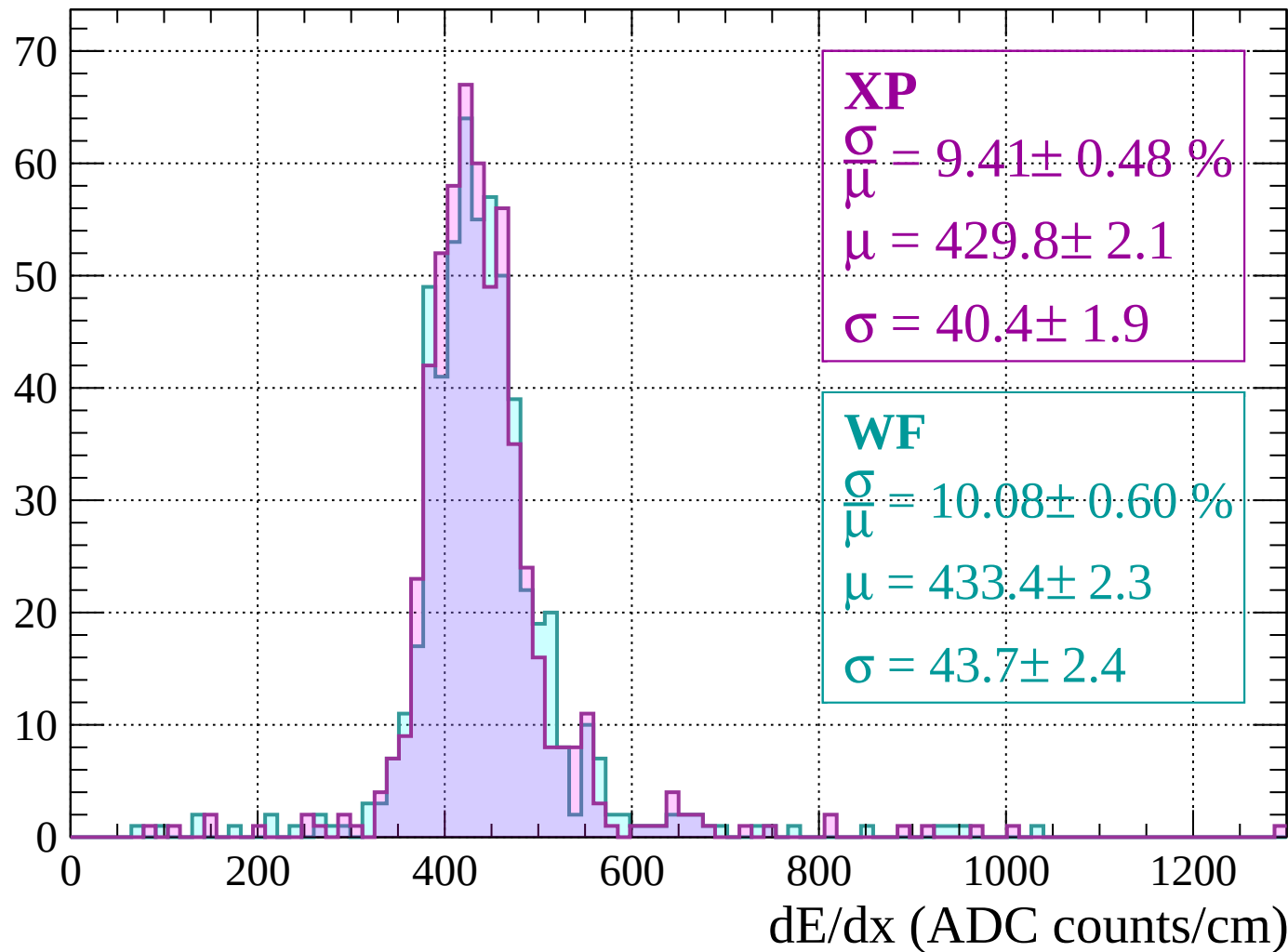
$$\mu = 435.3 \pm 2.2$$

$$\sigma = 43.6 \pm 2.0$$

dE/dx (ADC counts/cm)

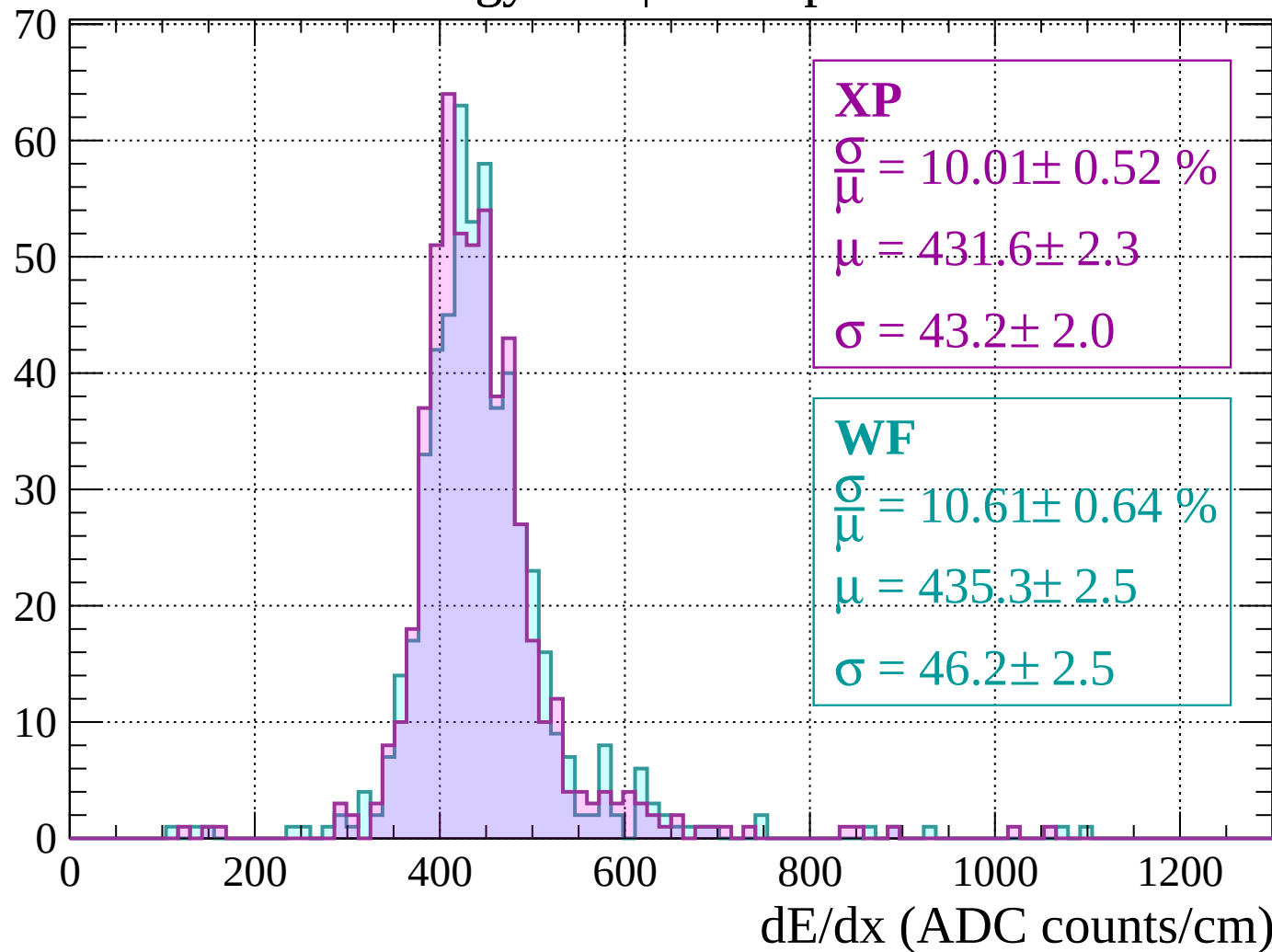
Energy loss | $840 < p < 880$

Count



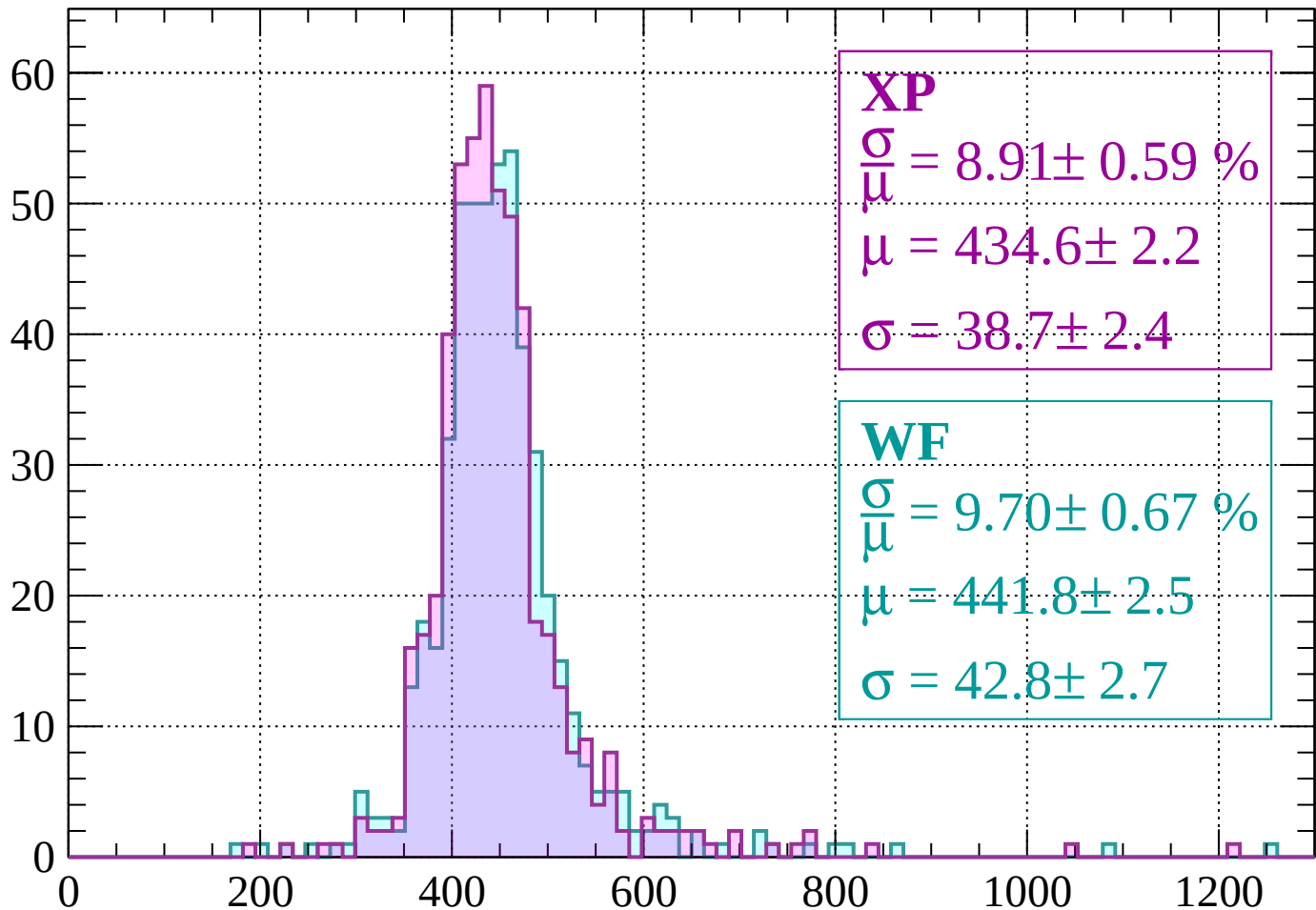
Energy loss | $880 < p < 920$

Count



Energy loss | $920 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.91 \pm 0.59 \%$$

$$\mu = 434.6 \pm 2.2$$

$$\sigma = 38.7 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 9.70 \pm 0.67 \%$$

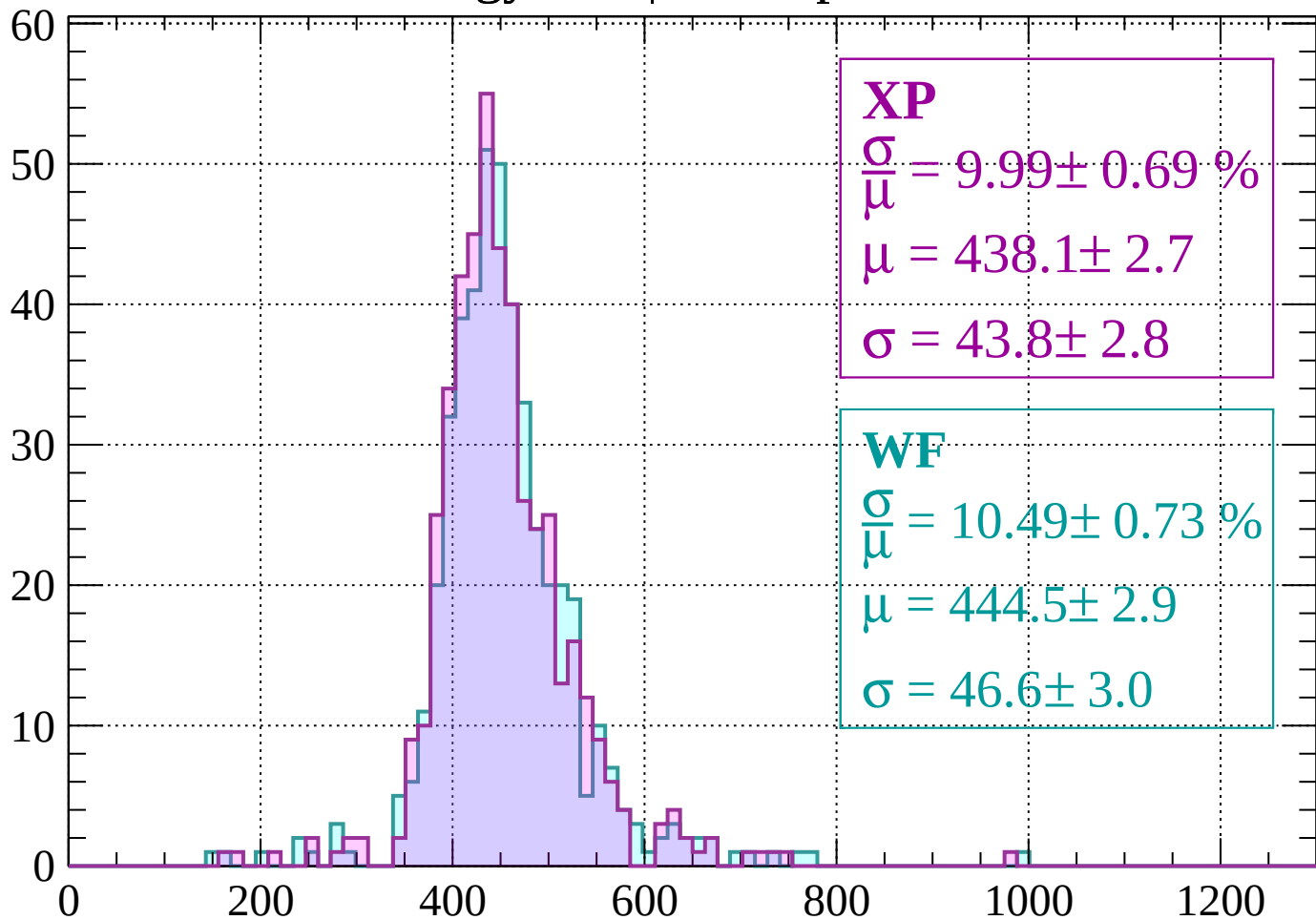
$$\mu = 441.8 \pm 2.5$$

$$\sigma = 42.8 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.99 \pm 0.69 \%$$

$$\mu = 438.1 \pm 2.7$$

$$\sigma = 43.8 \pm 2.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 10.49 \pm 0.73 \%$$

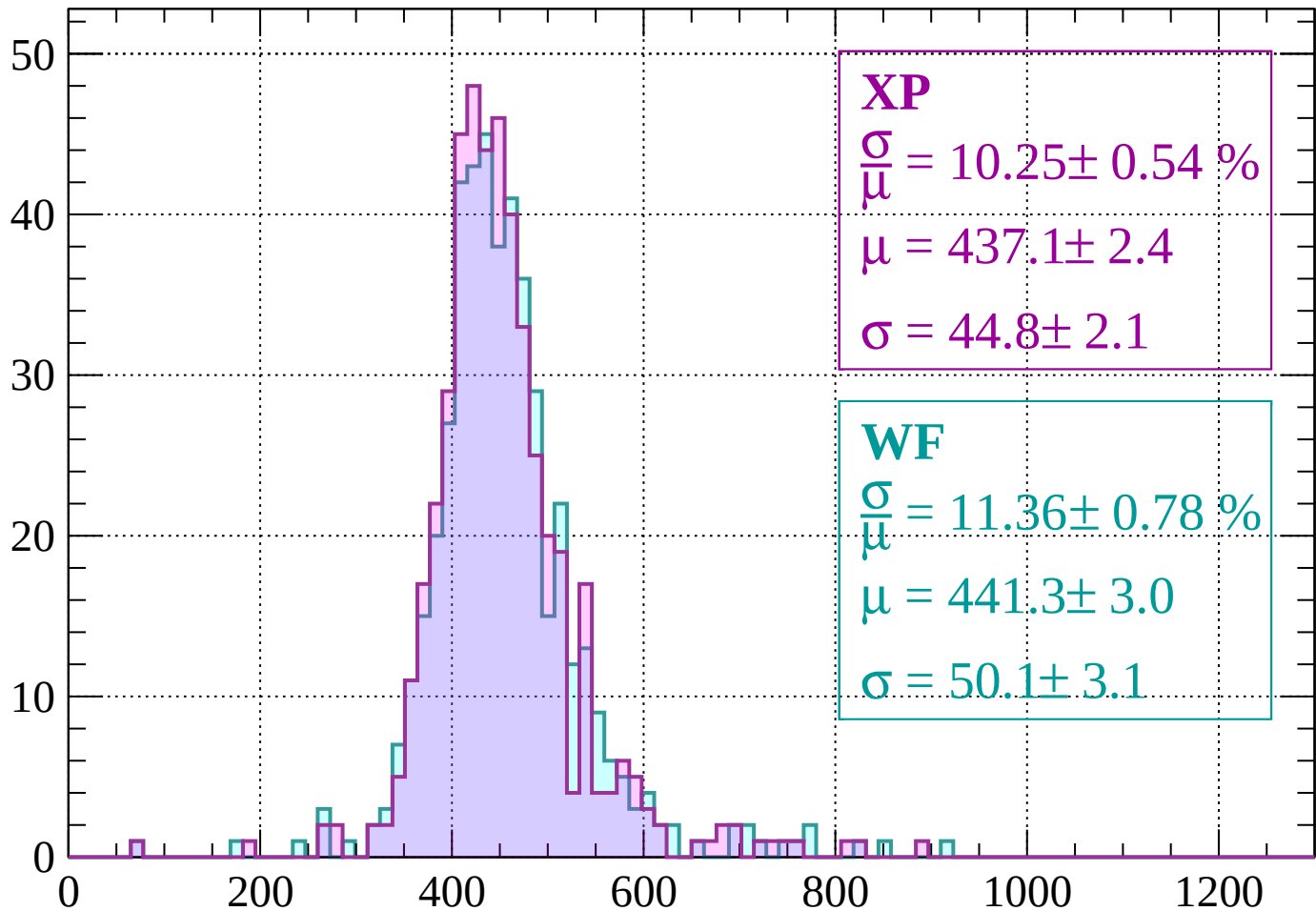
$$\mu = 444.5 \pm 2.9$$

$$\sigma = 46.6 \pm 3.0$$

dE/dx (ADC counts/cm)

Energy loss | $1000 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 10.25 \pm 0.54 \%$$

$$\mu = 437.1 \pm 2.4$$

$$\sigma = 44.8 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 11.36 \pm 0.78 \%$$

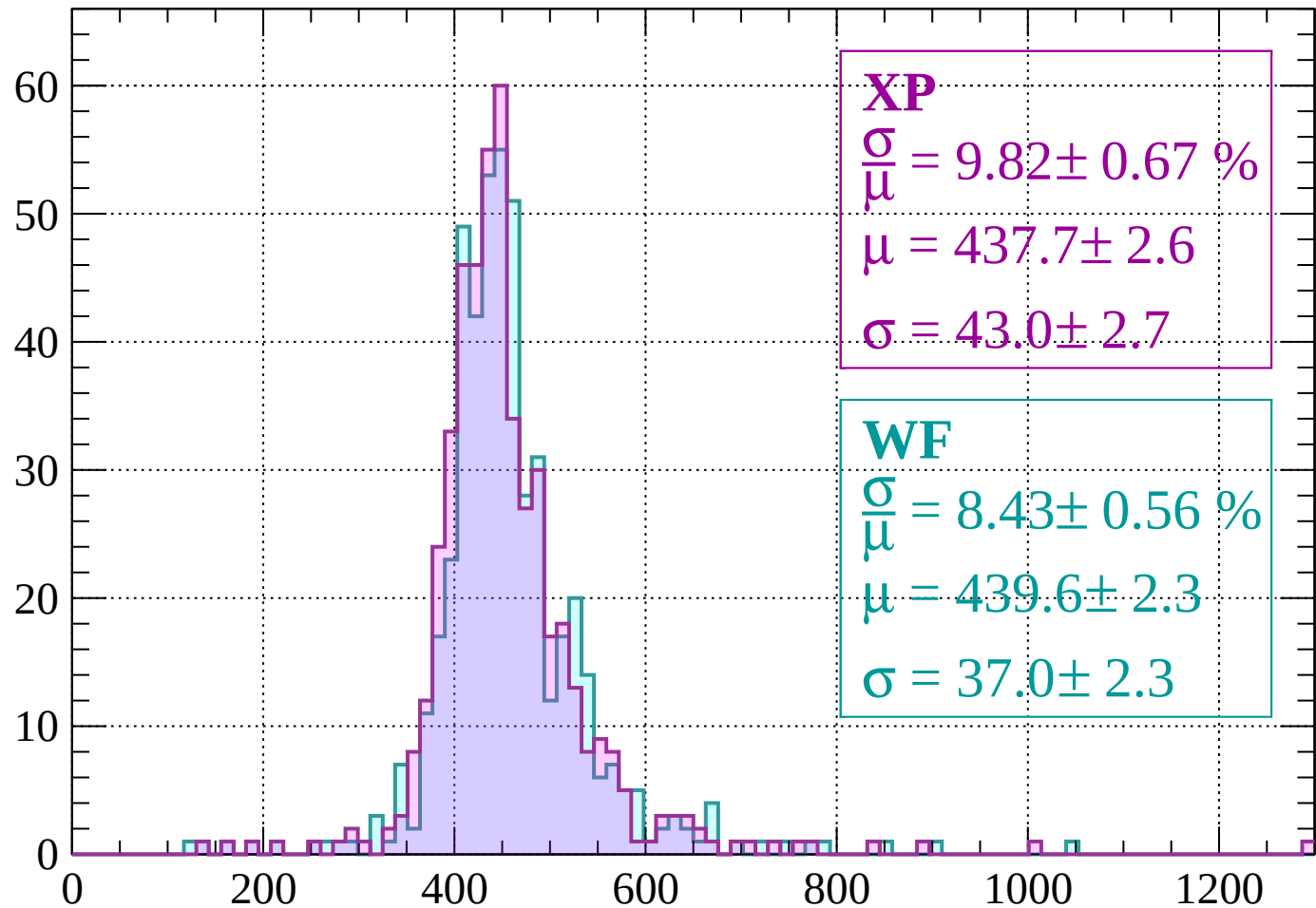
$$\mu = 441.3 \pm 3.0$$

$$\sigma = 50.1 \pm 3.1$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1080$

Count



XP

$$\frac{\sigma}{\mu} = 9.82 \pm 0.67 \%$$

$$\mu = 437.7 \pm 2.6$$

$$\sigma = 43.0 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 8.43 \pm 0.56 \%$$

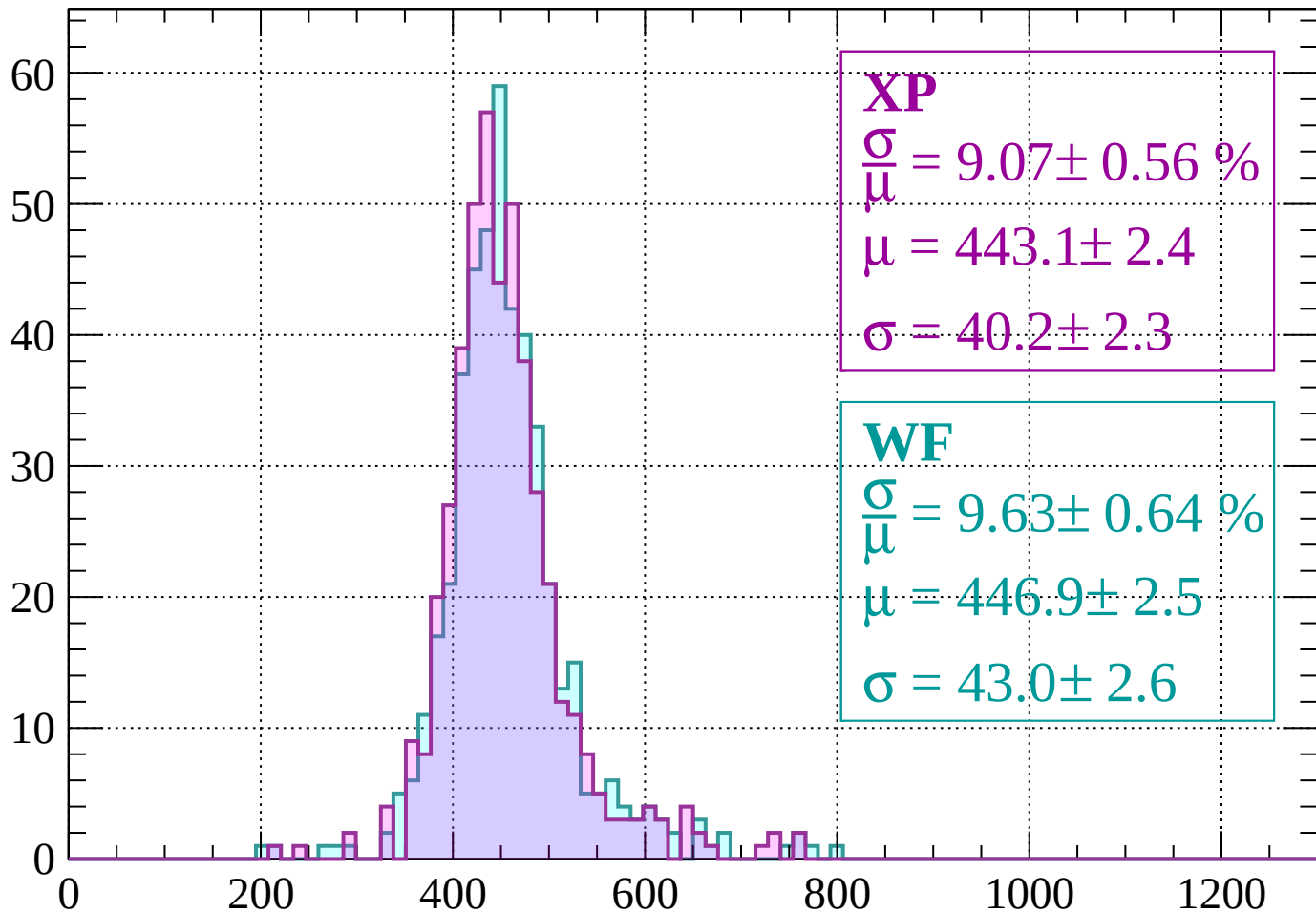
$$\mu = 439.6 \pm 2.3$$

$$\sigma = 37.0 \pm 2.3$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1120$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.07 \pm 0.56 \%$$

$$\mu = 443.1 \pm 2.4$$

$$\sigma = 40.2 \pm 2.3$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.63 \pm 0.64 \%$$

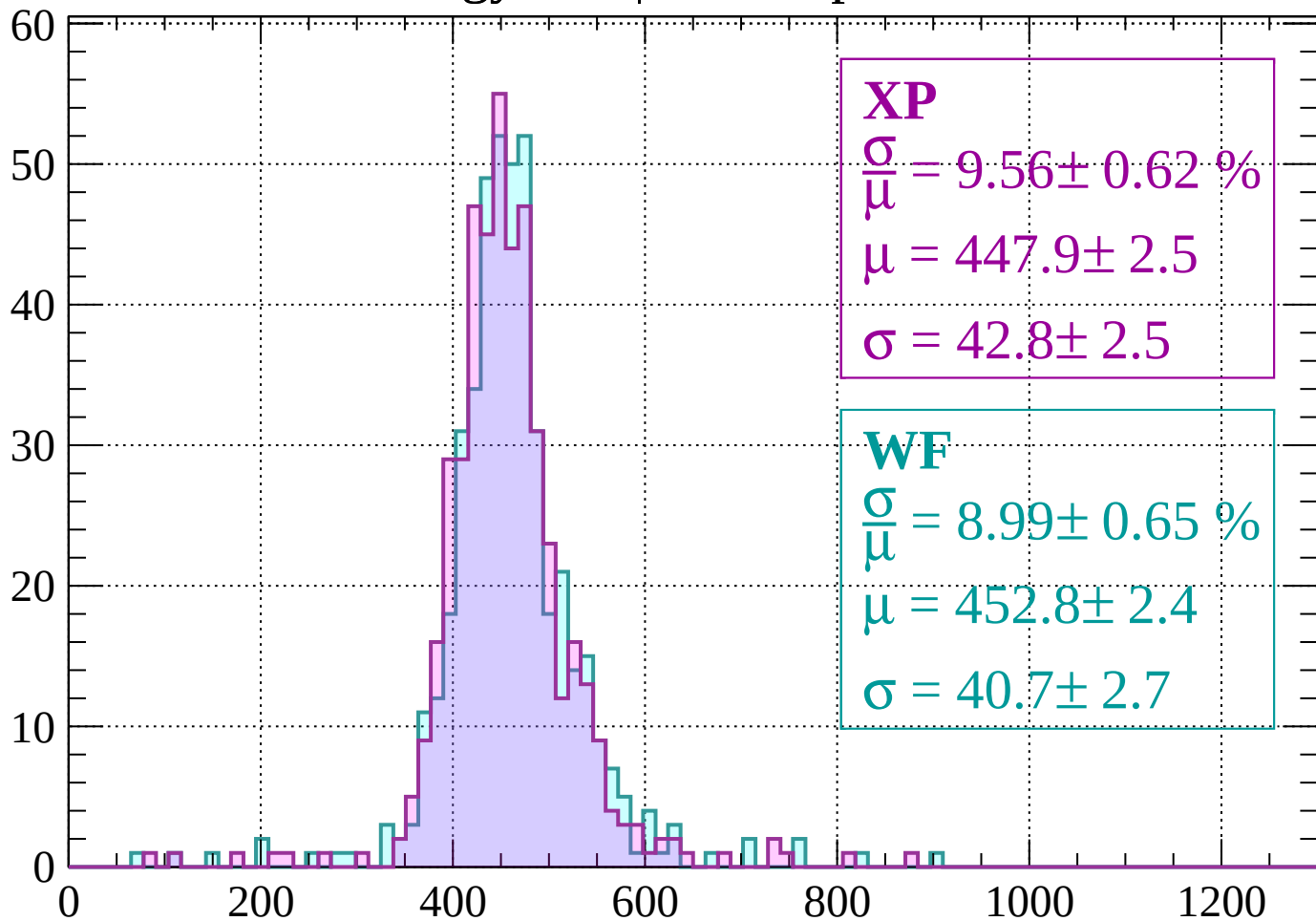
$$\mu = 446.9 \pm 2.5$$

$$\sigma = 43.0 \pm 2.6$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1160$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.56 \pm 0.62 \%$$

$$\mu = 447.9 \pm 2.5$$

$$\sigma = 42.8 \pm 2.5$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.99 \pm 0.65 \%$$

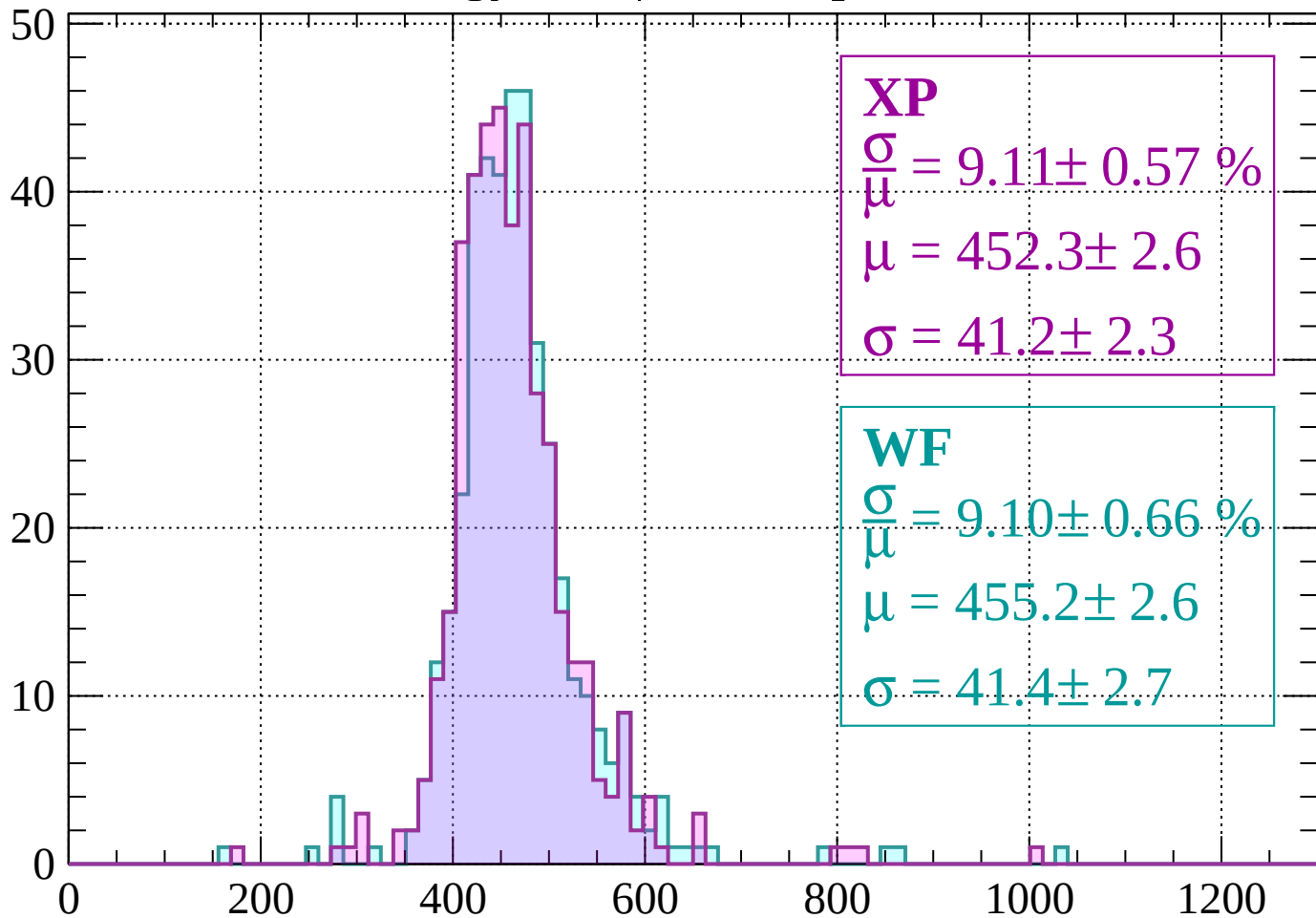
$$\mu = 452.8 \pm 2.4$$

$$\sigma = 40.7 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $1160 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 9.11 \pm 0.57 \%$$

$$\mu = 452.3 \pm 2.6$$

$$\sigma = 41.2 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 9.10 \pm 0.66 \%$$

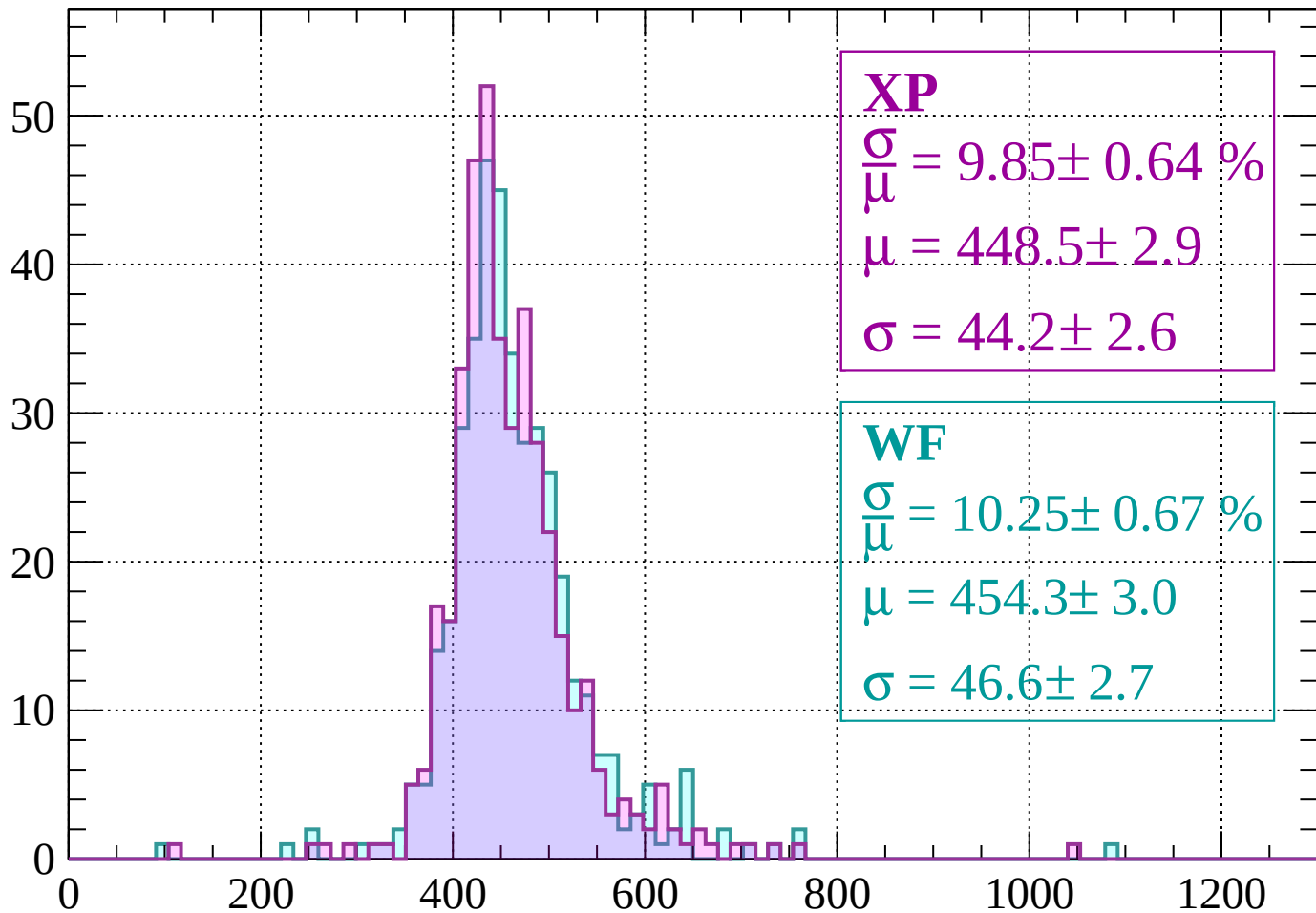
$$\mu = 455.2 \pm 2.6$$

$$\sigma = 41.4 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

Count



XP

$$\frac{\sigma}{\mu} = 9.85 \pm 0.64 \%$$

$$\mu = 448.5 \pm 2.9$$

$$\sigma = 44.2 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 10.25 \pm 0.67 \%$$

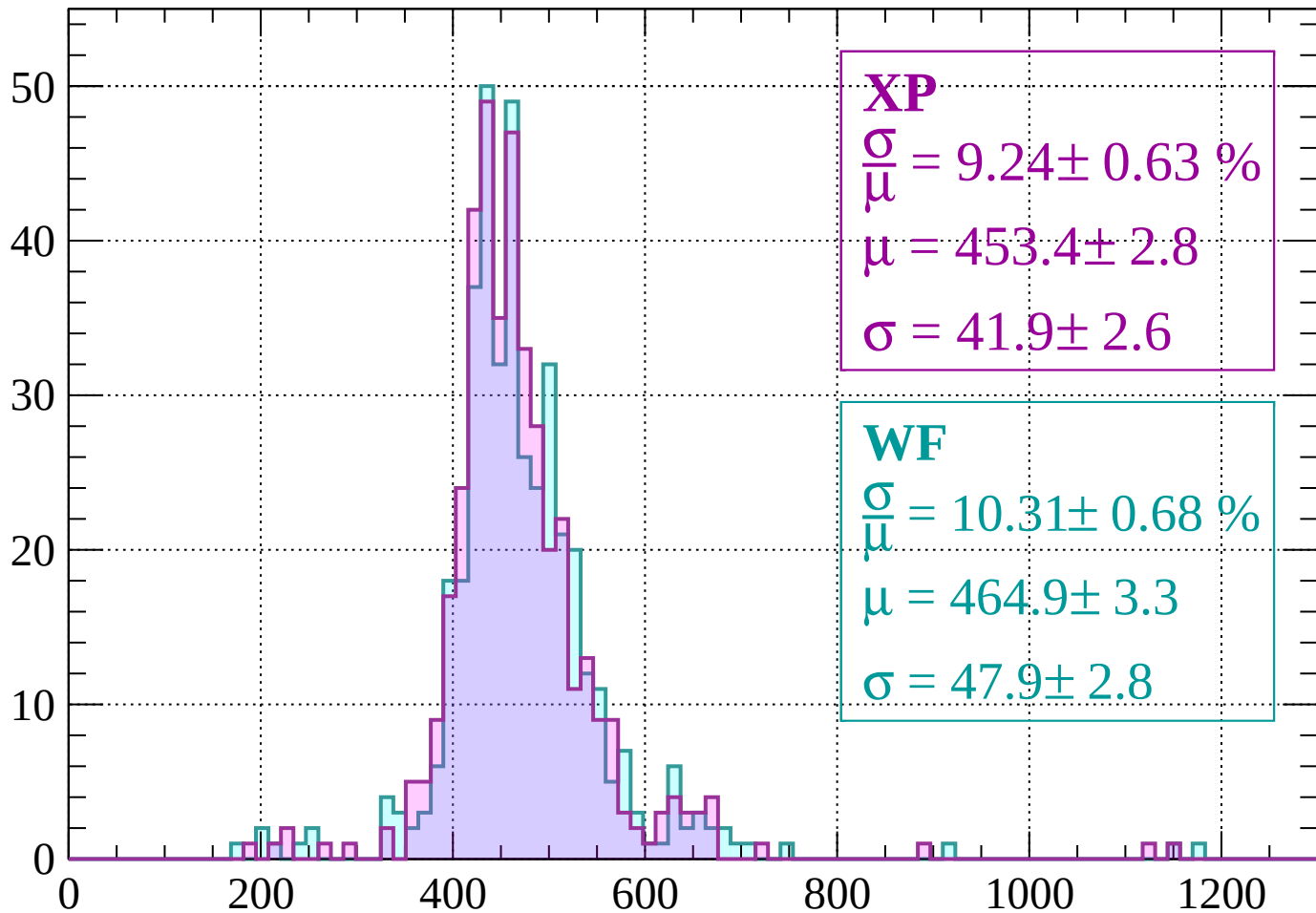
$$\mu = 454.3 \pm 3.0$$

$$\sigma = 46.6 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.24 \pm 0.63 \%$$

$$\mu = 453.4 \pm 2.8$$

$$\sigma = 41.9 \pm 2.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 10.31 \pm 0.68 \%$$

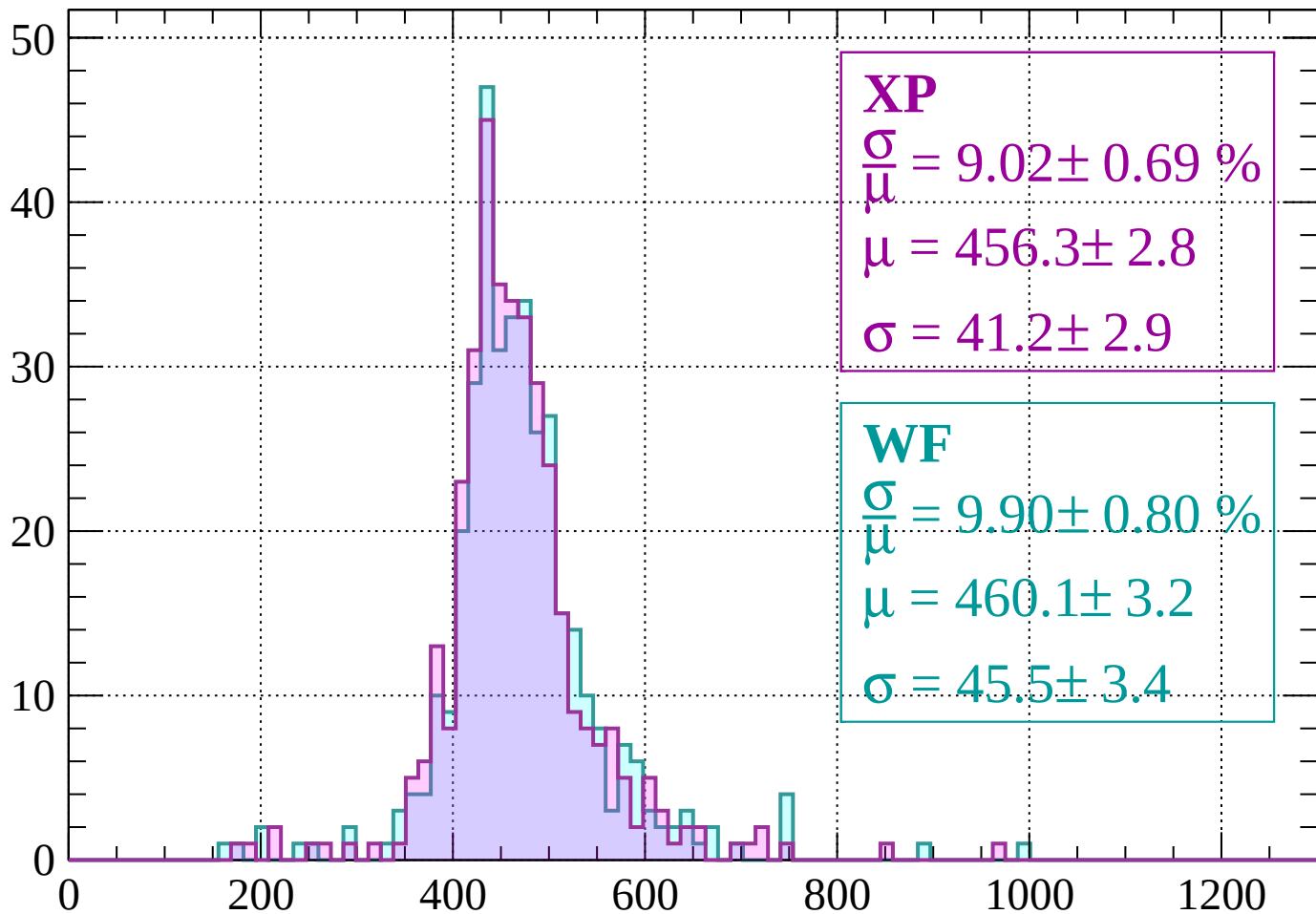
$$\mu = 464.9 \pm 3.3$$

$$\sigma = 47.9 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$

Count



XP

$$\frac{\sigma}{\mu} = 9.02 \pm 0.69 \%$$

$$\mu = 456.3 \pm 2.8$$

$$\sigma = 41.2 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 9.90 \pm 0.80 \%$$

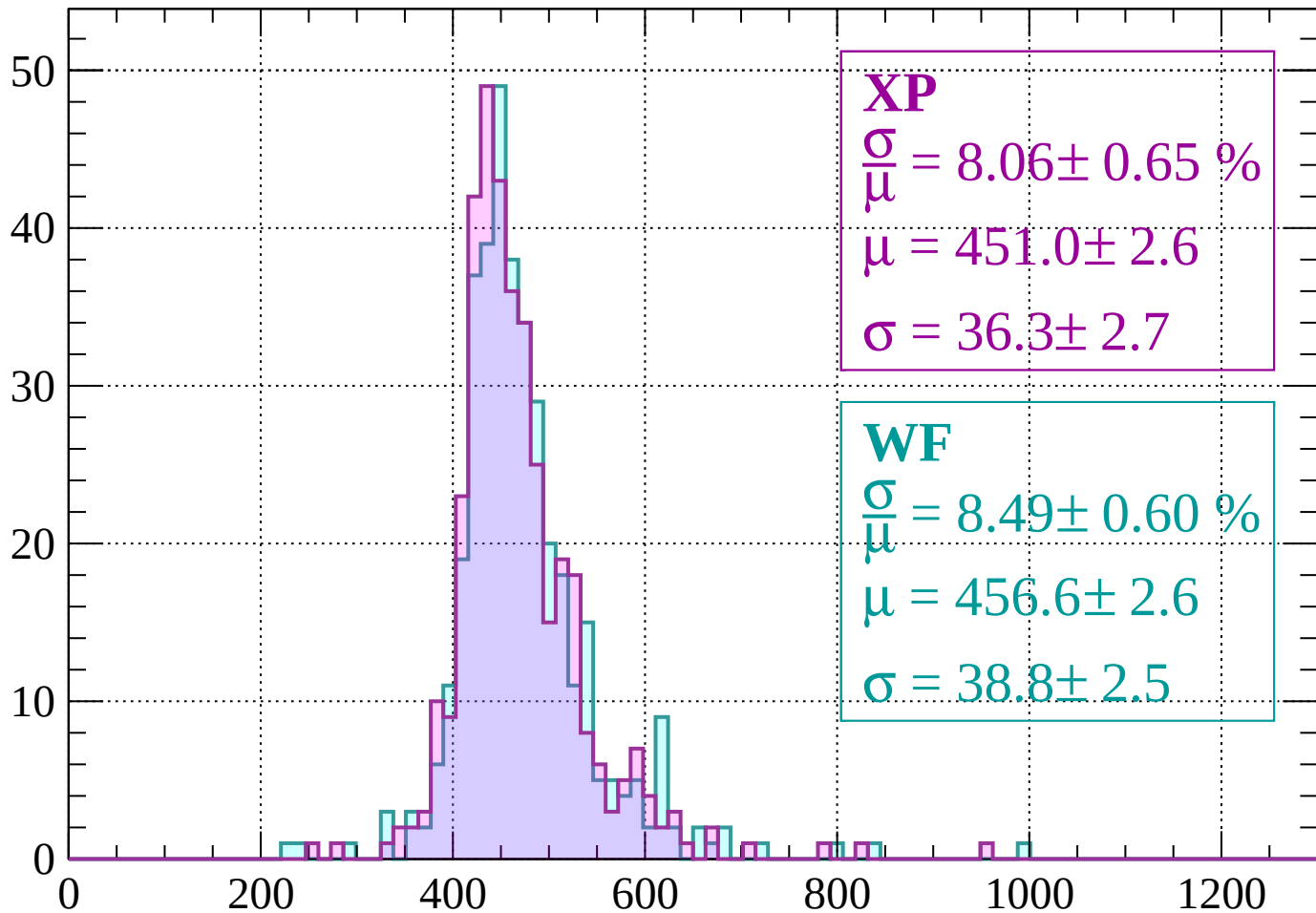
$$\mu = 460.1 \pm 3.2$$

$$\sigma = 45.5 \pm 3.4$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.06 \pm 0.65 \%$$

$$\mu = 451.0 \pm 2.6$$

$$\sigma = 36.3 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 8.49 \pm 0.60 \%$$

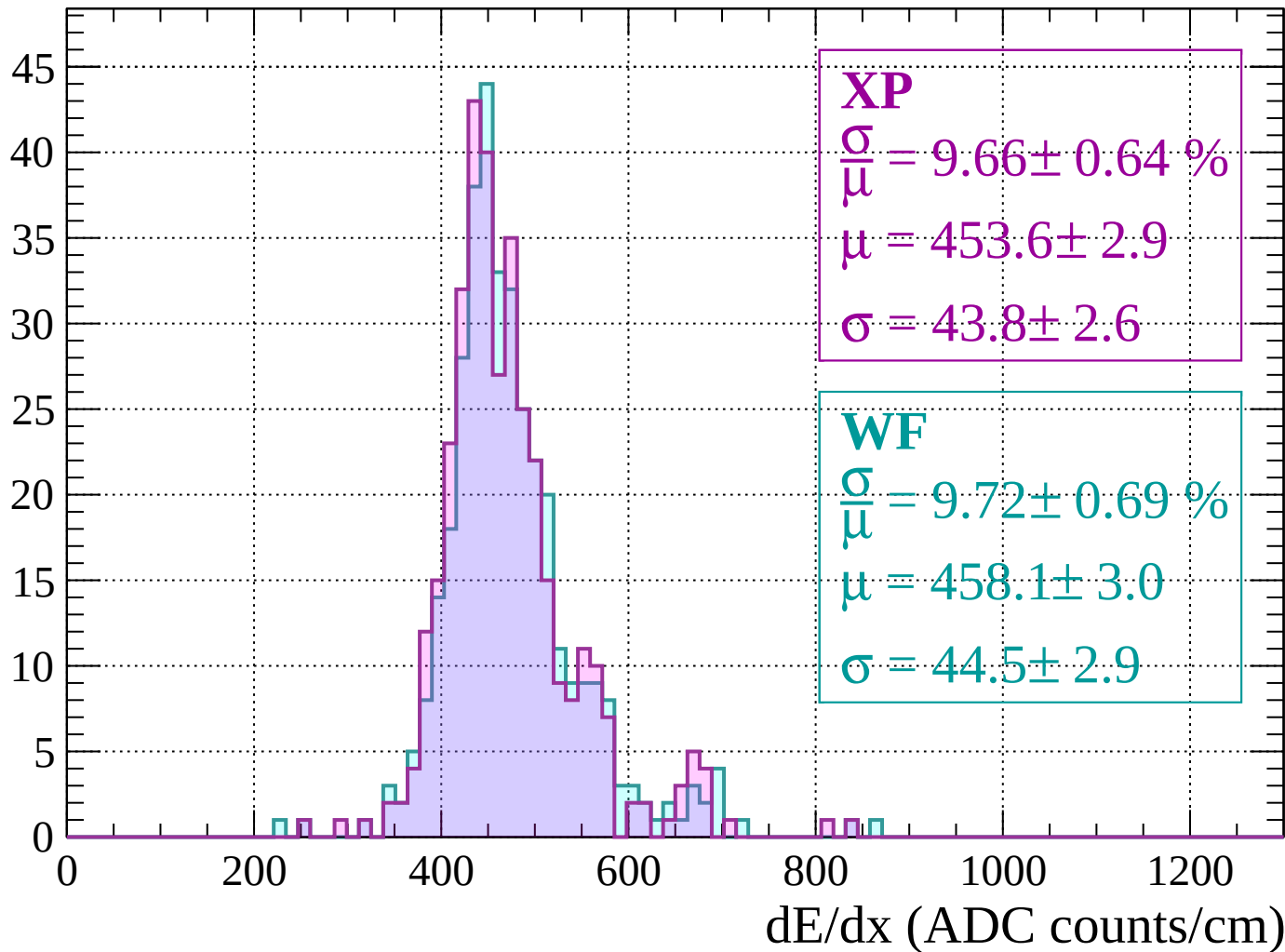
$$\mu = 456.6 \pm 2.6$$

$$\sigma = 38.8 \pm 2.5$$

dE/dx (ADC counts/cm)

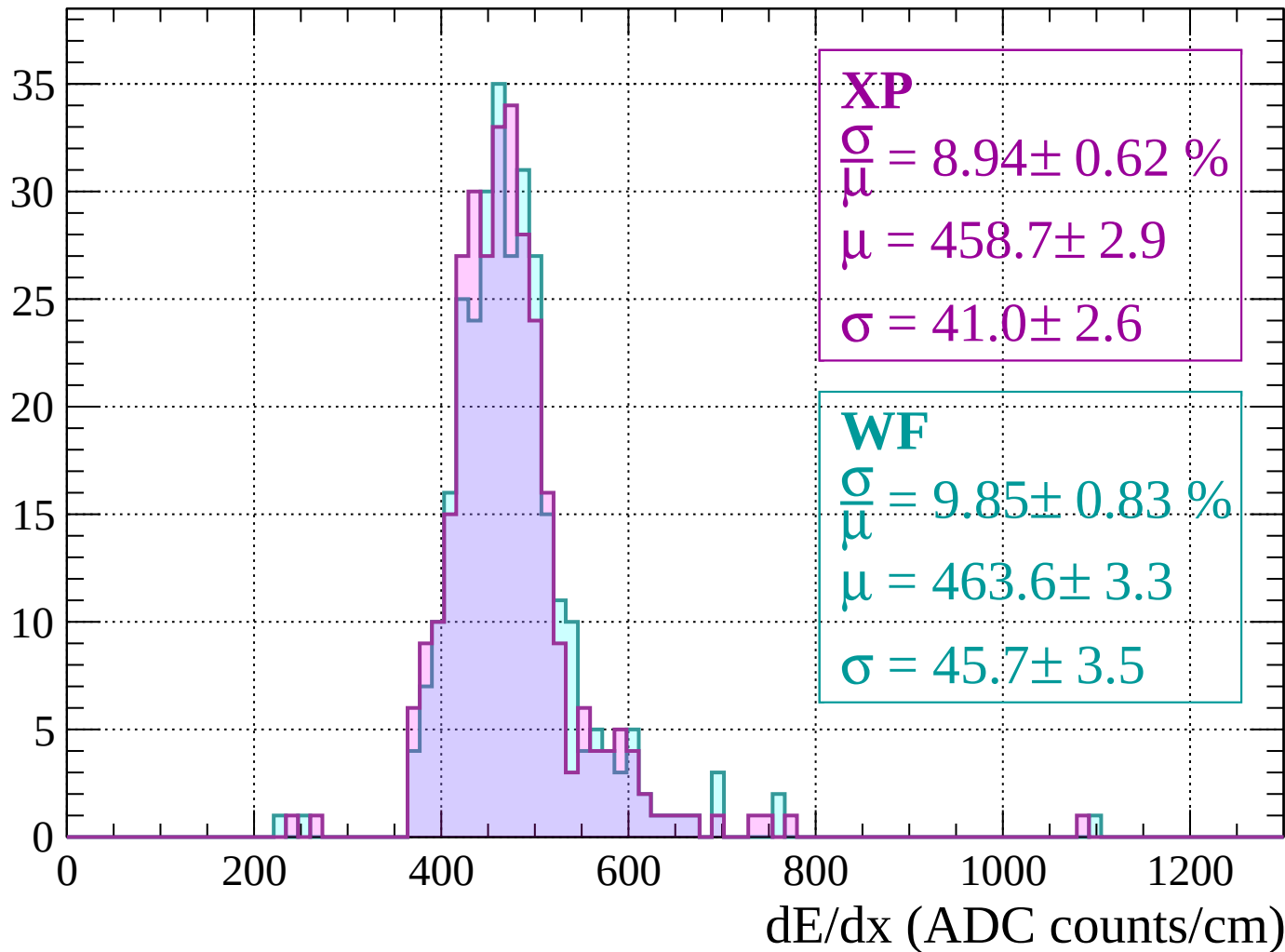
Energy loss | $1360 < p < 1400$

Count



Energy loss | $1400 < p < 1440$

Count



Energy loss | $1440 < p < 1480$

Count

XP

$$\frac{\sigma}{\mu} = 9.58 \pm 0.87 \%$$

$$\mu = 453.8 \pm 3.1$$

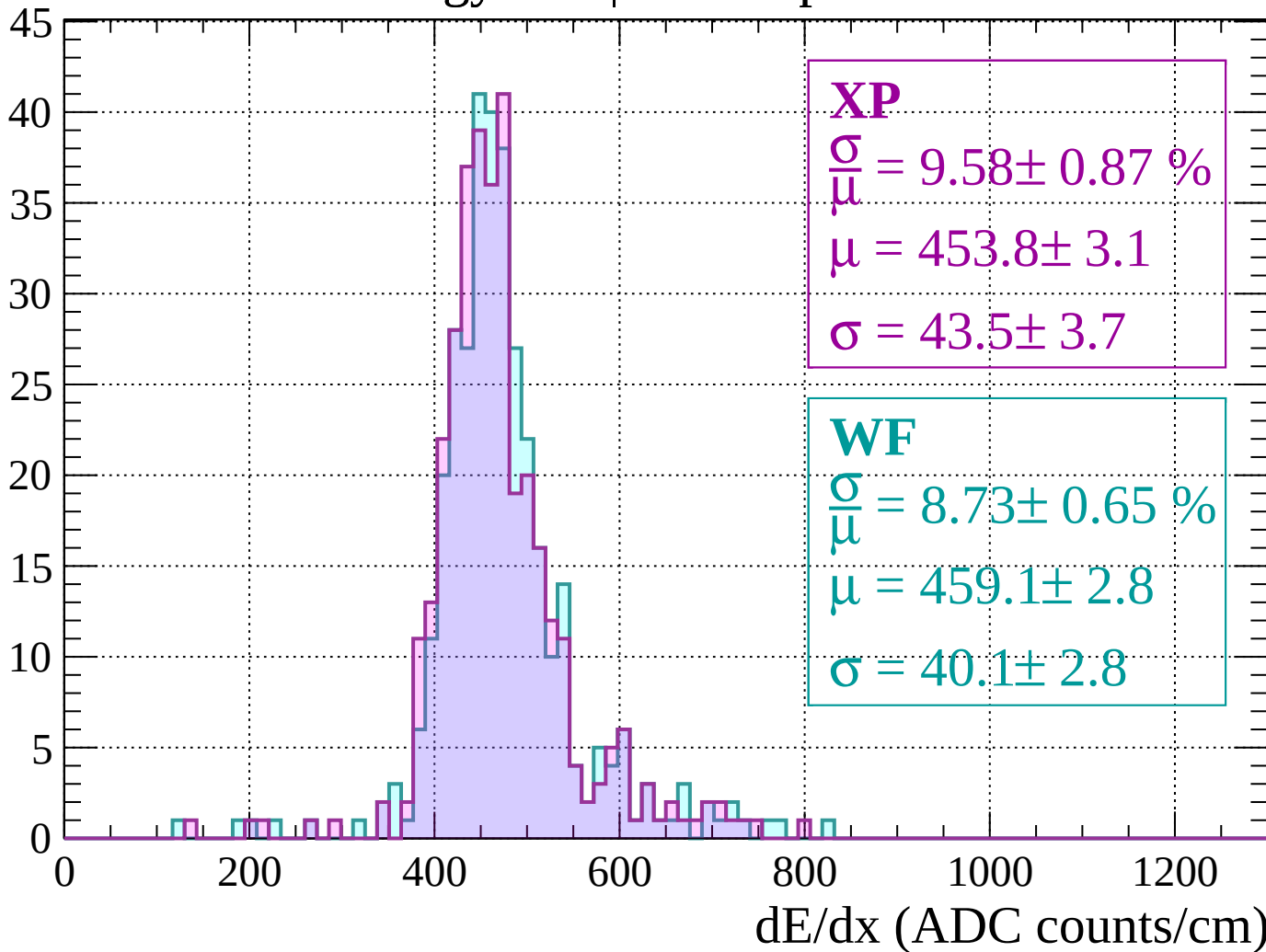
$$\sigma = 43.5 \pm 3.7$$

WF

$$\frac{\sigma}{\mu} = 8.73 \pm 0.65 \%$$

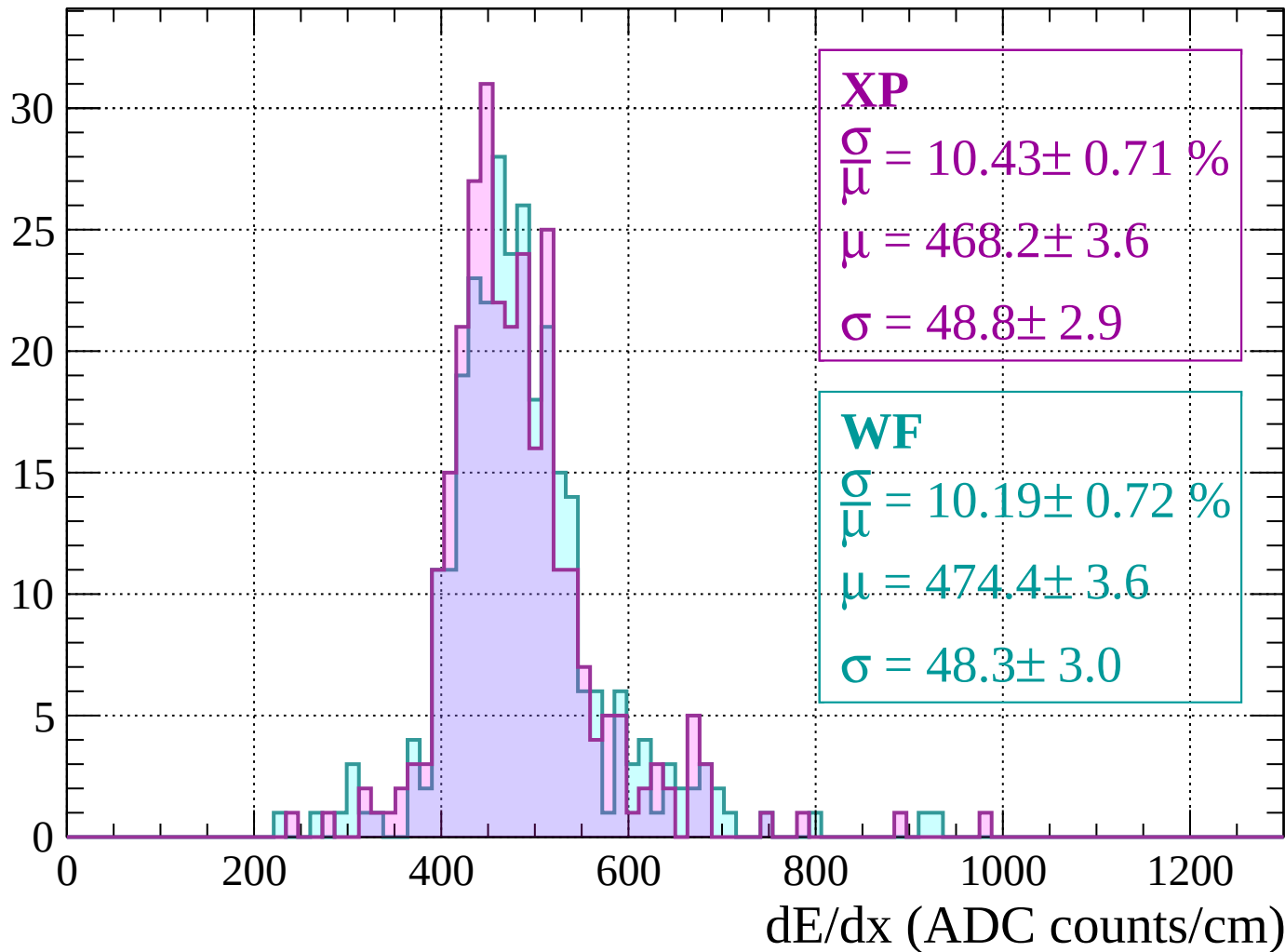
$$\mu = 459.1 \pm 2.8$$

$$\sigma = 40.1 \pm 2.8$$



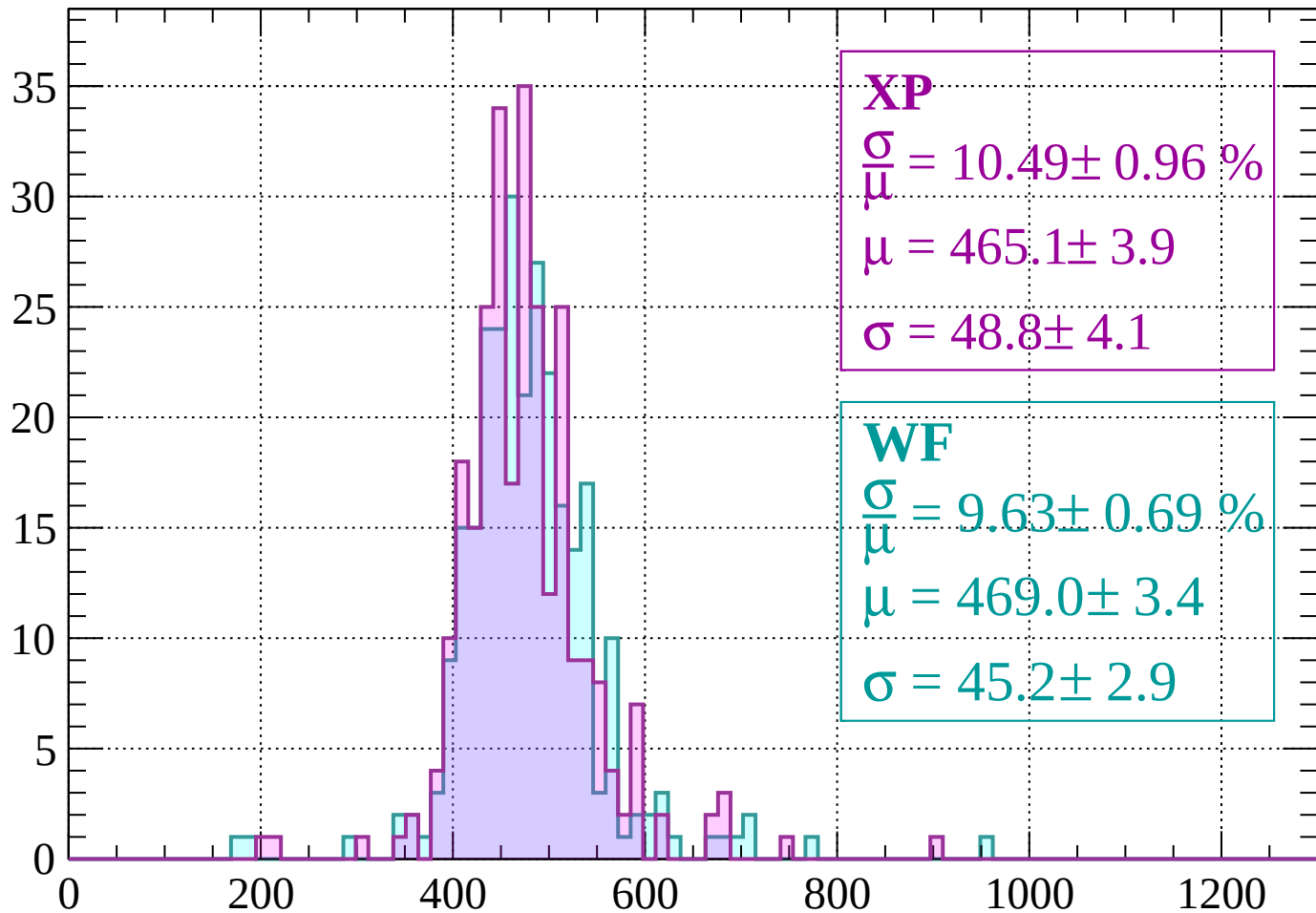
Energy loss | $1480 < p < 1520$

Count



Energy loss | $1520 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 10.49 \pm 0.96 \%$$

$$\mu = 465.1 \pm 3.9$$

$$\sigma = 48.8 \pm 4.1$$

WF

$$\frac{\sigma}{\mu} = 9.63 \pm 0.69 \%$$

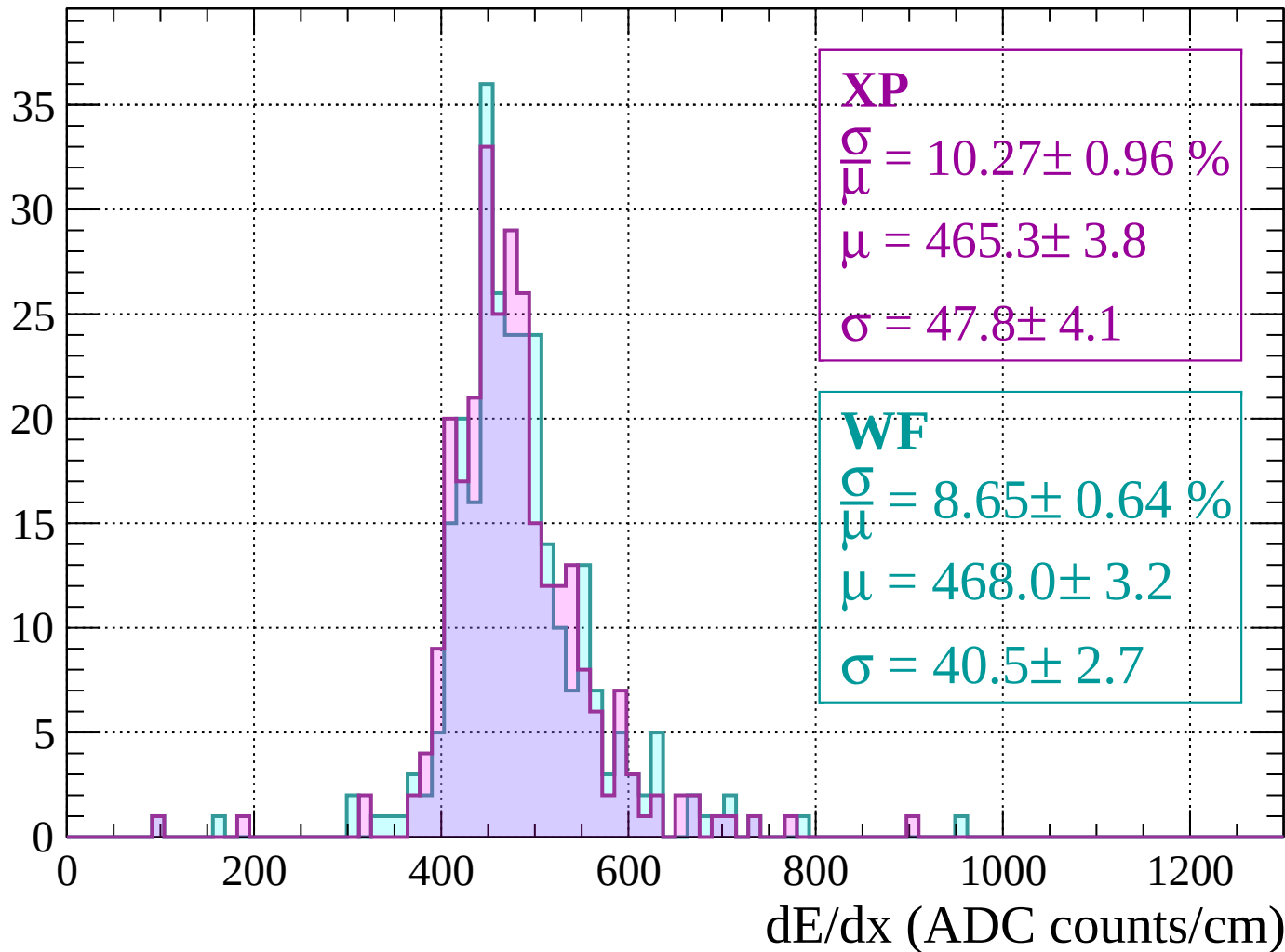
$$\mu = 469.0 \pm 3.4$$

$$\sigma = 45.2 \pm 2.9$$

dE/dx (ADC counts/cm)

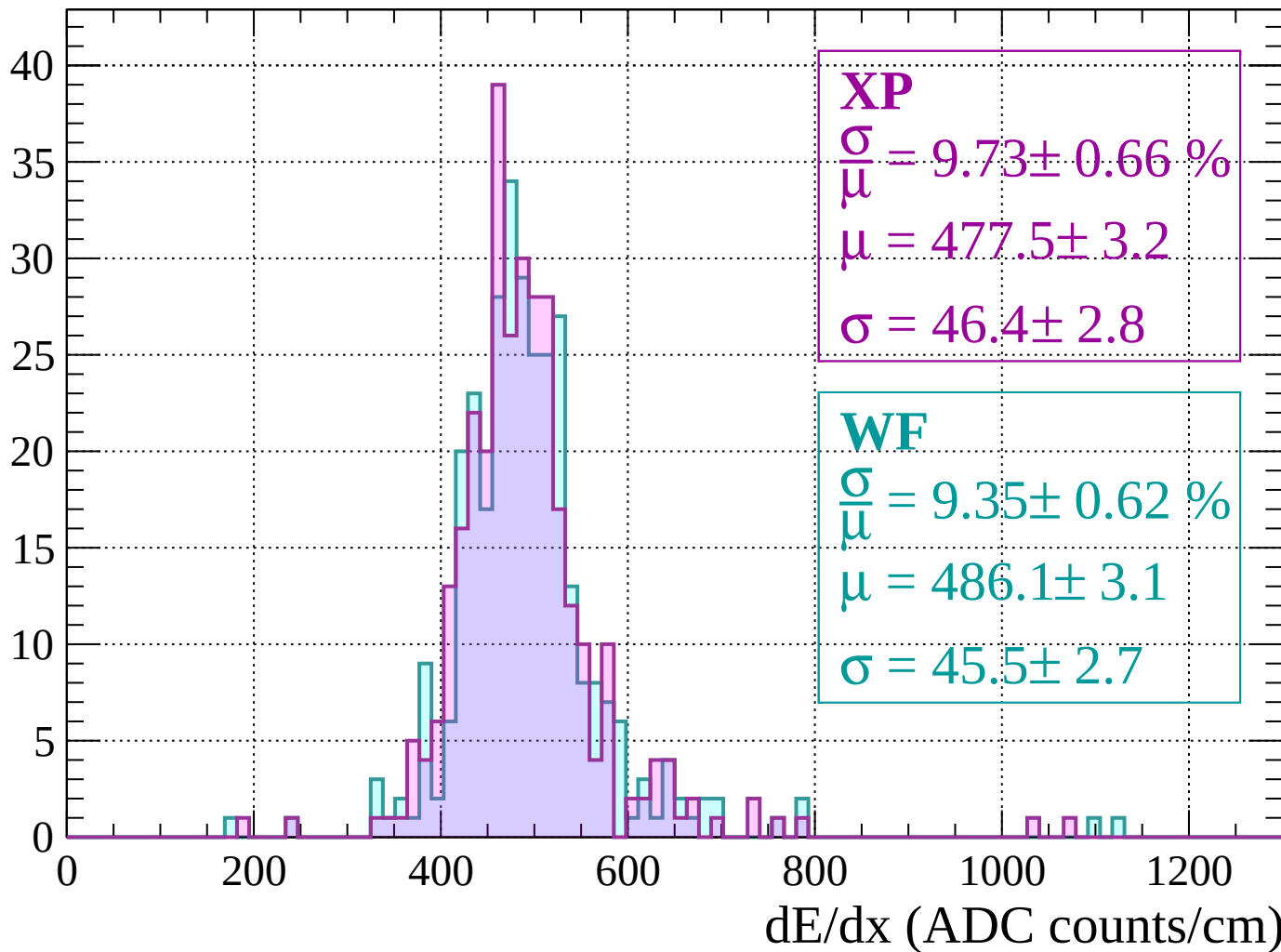
Energy loss | $1560 < p < 1600$

Count



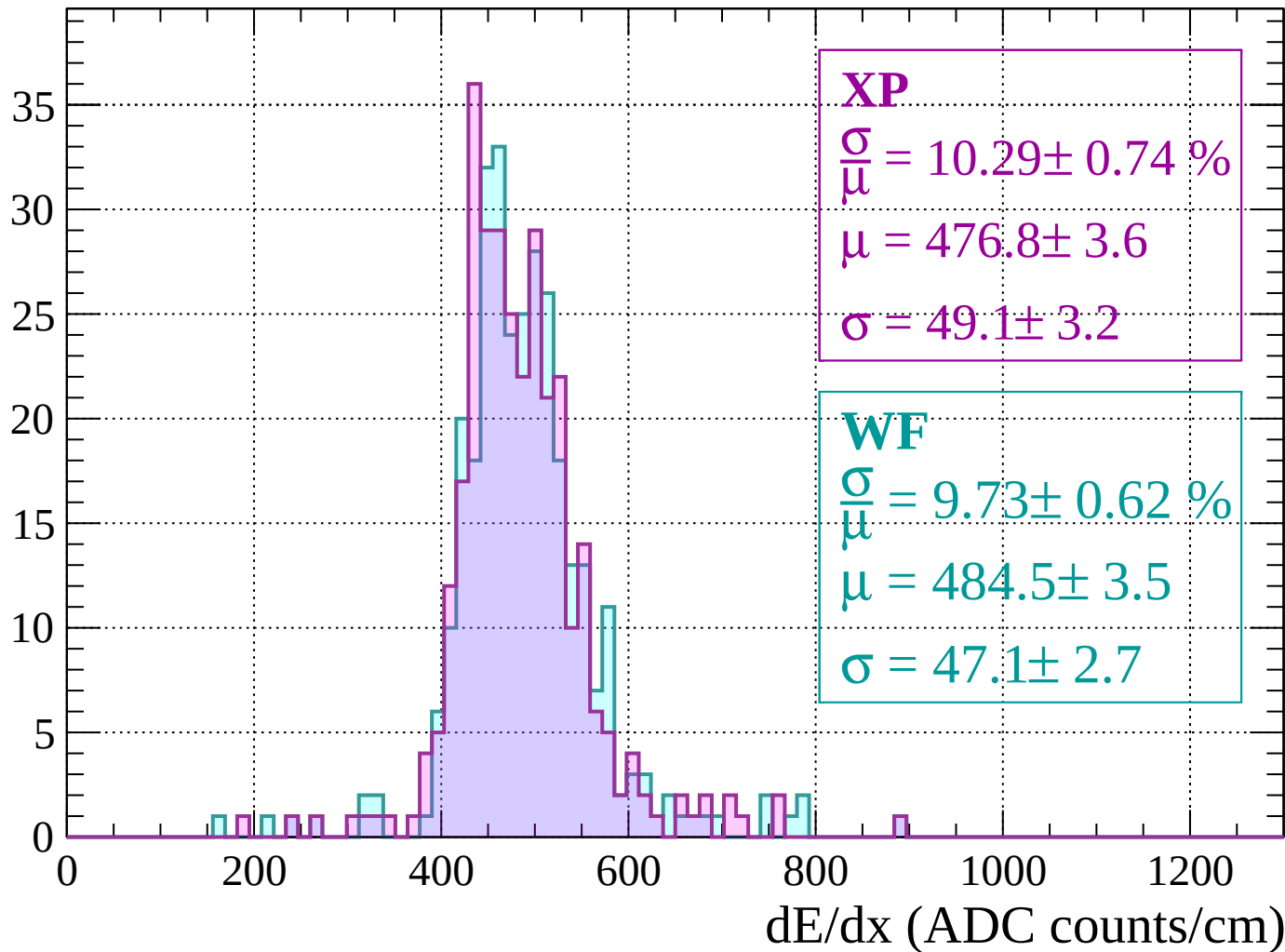
Energy loss | $1600 < p < 1640$

Count



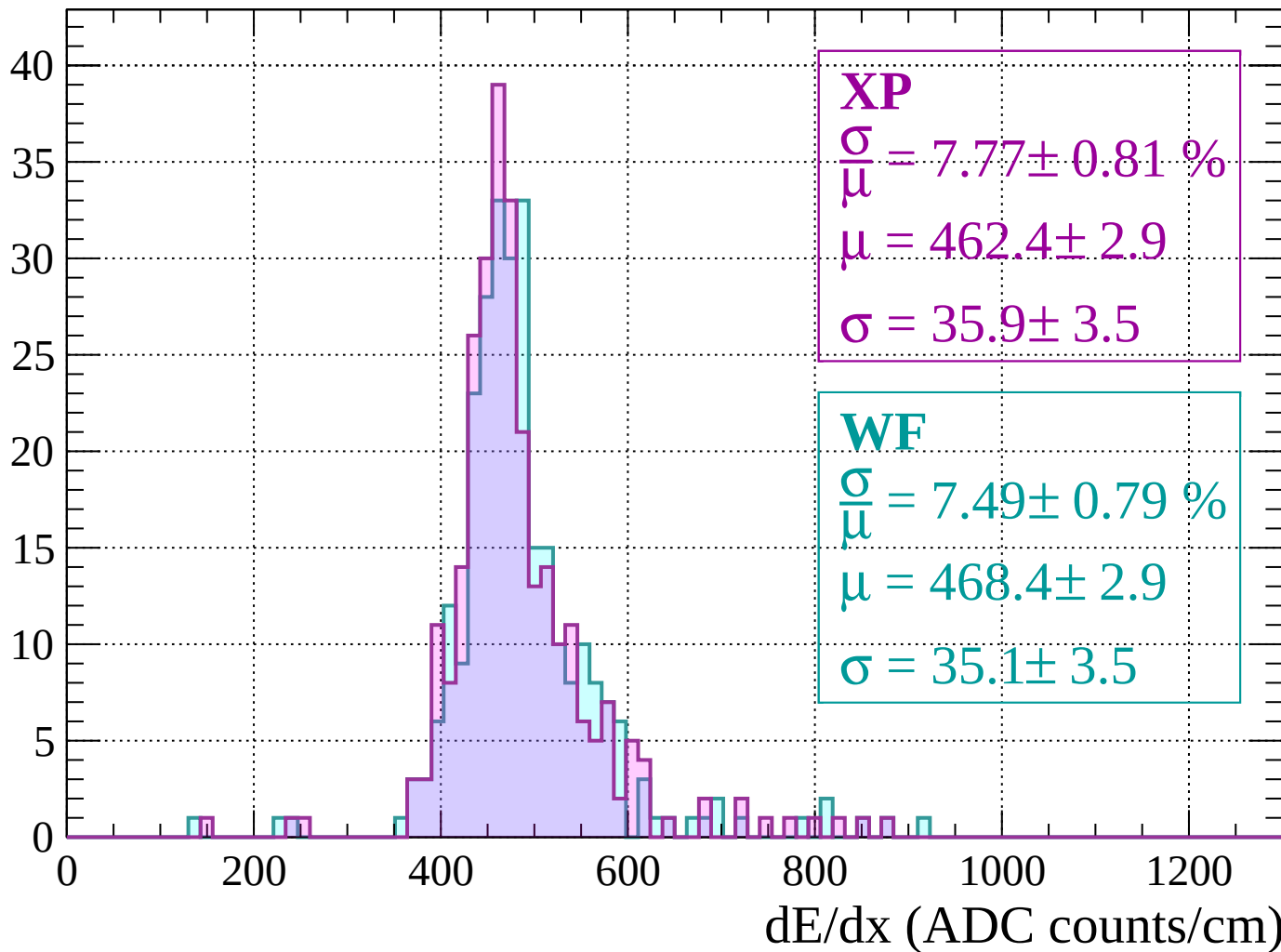
Energy loss | $1640 < p < 1680$

Count



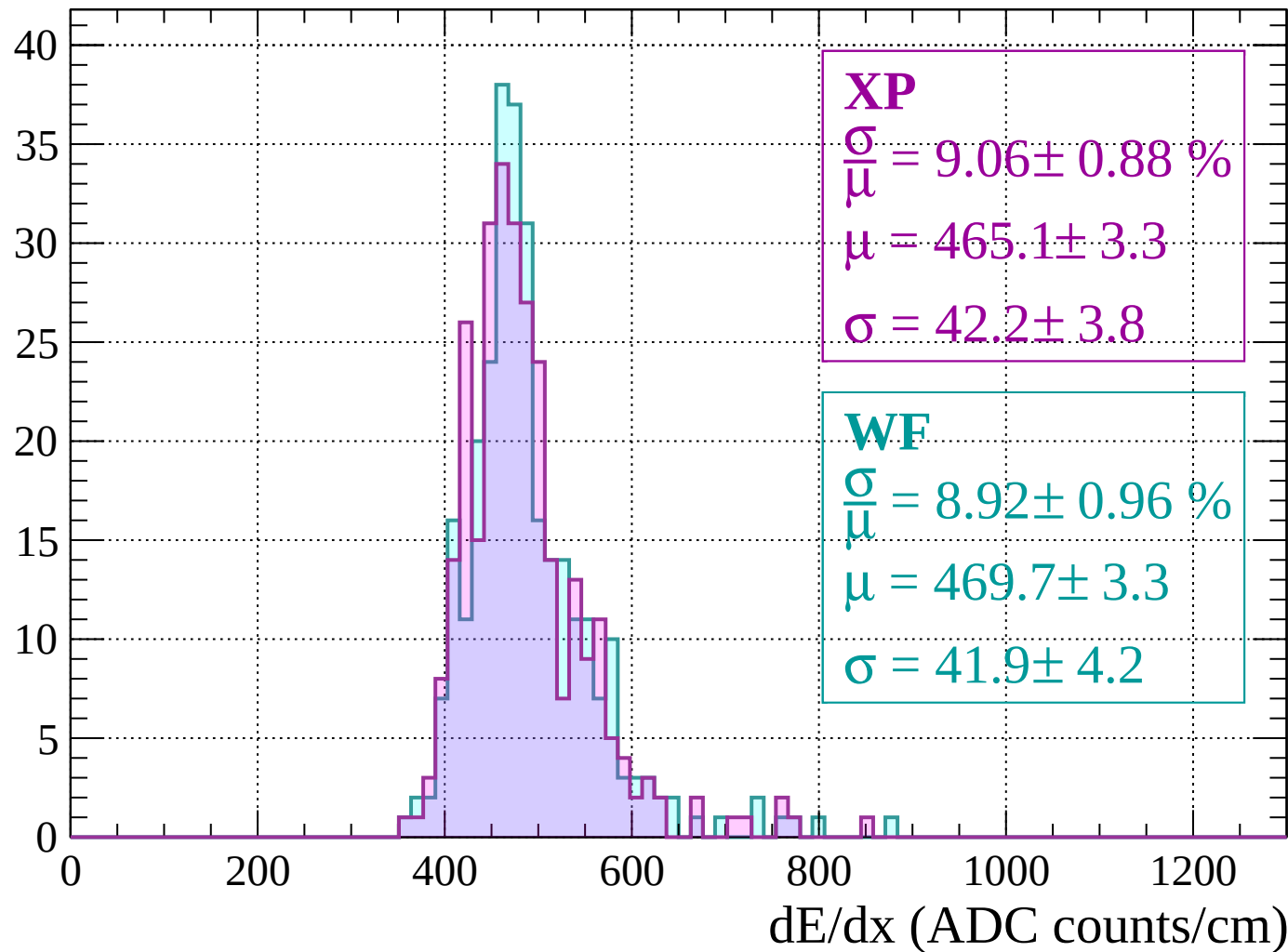
Energy loss | $1680 < p < 1720$

Count



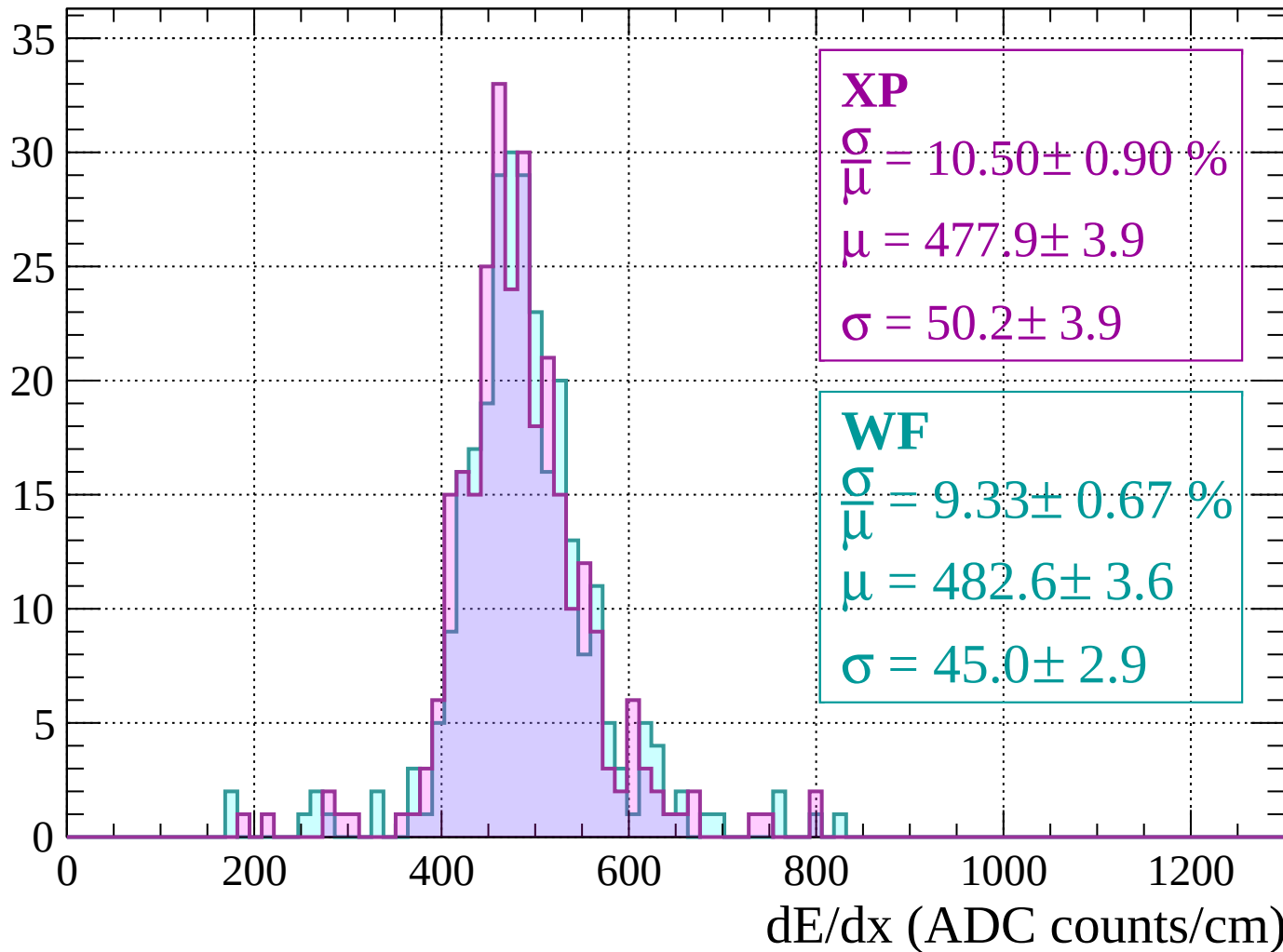
Energy loss | $1720 < p < 1760$

Count



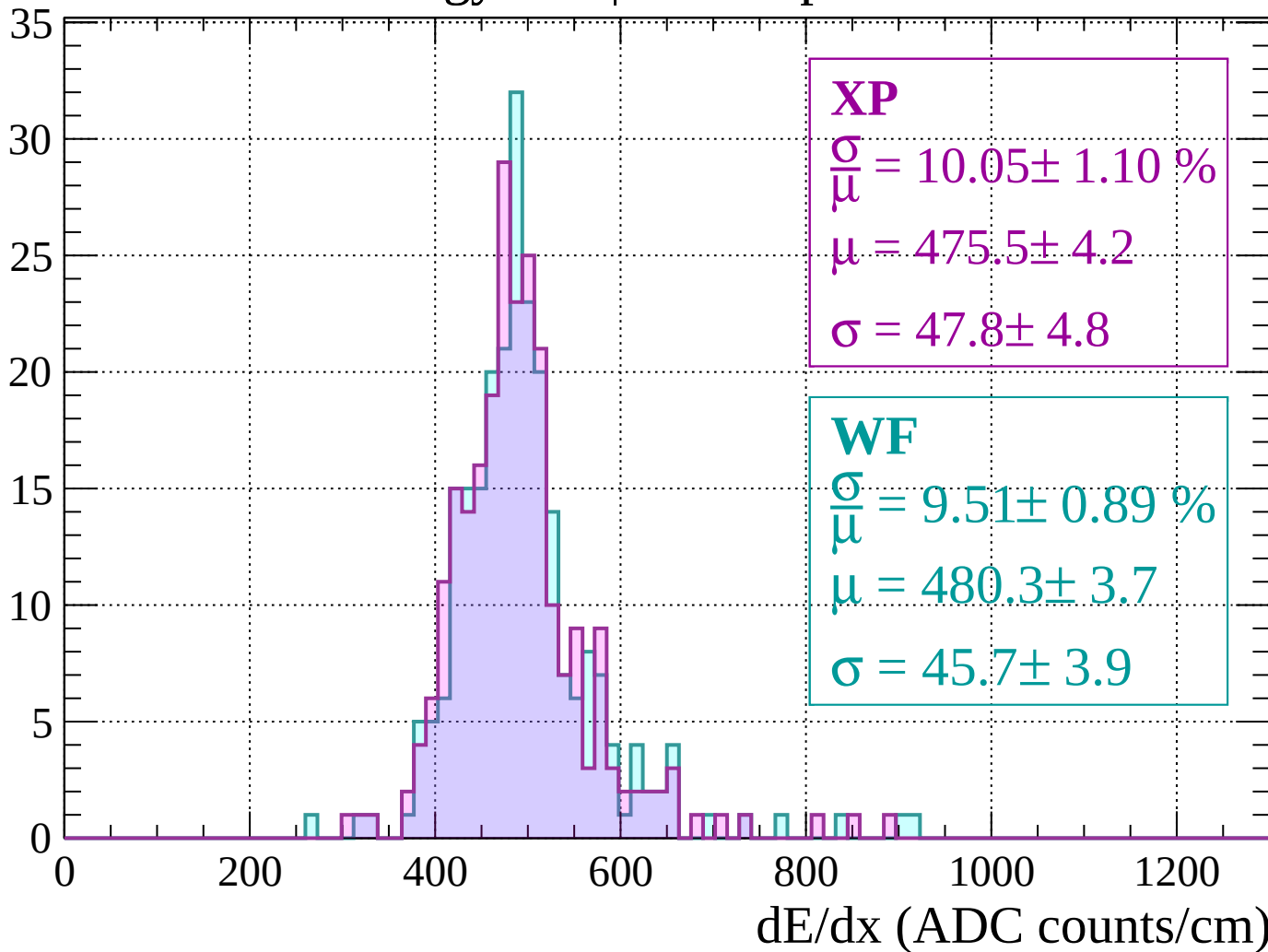
Energy loss | $1760 < p < 1800$

Count



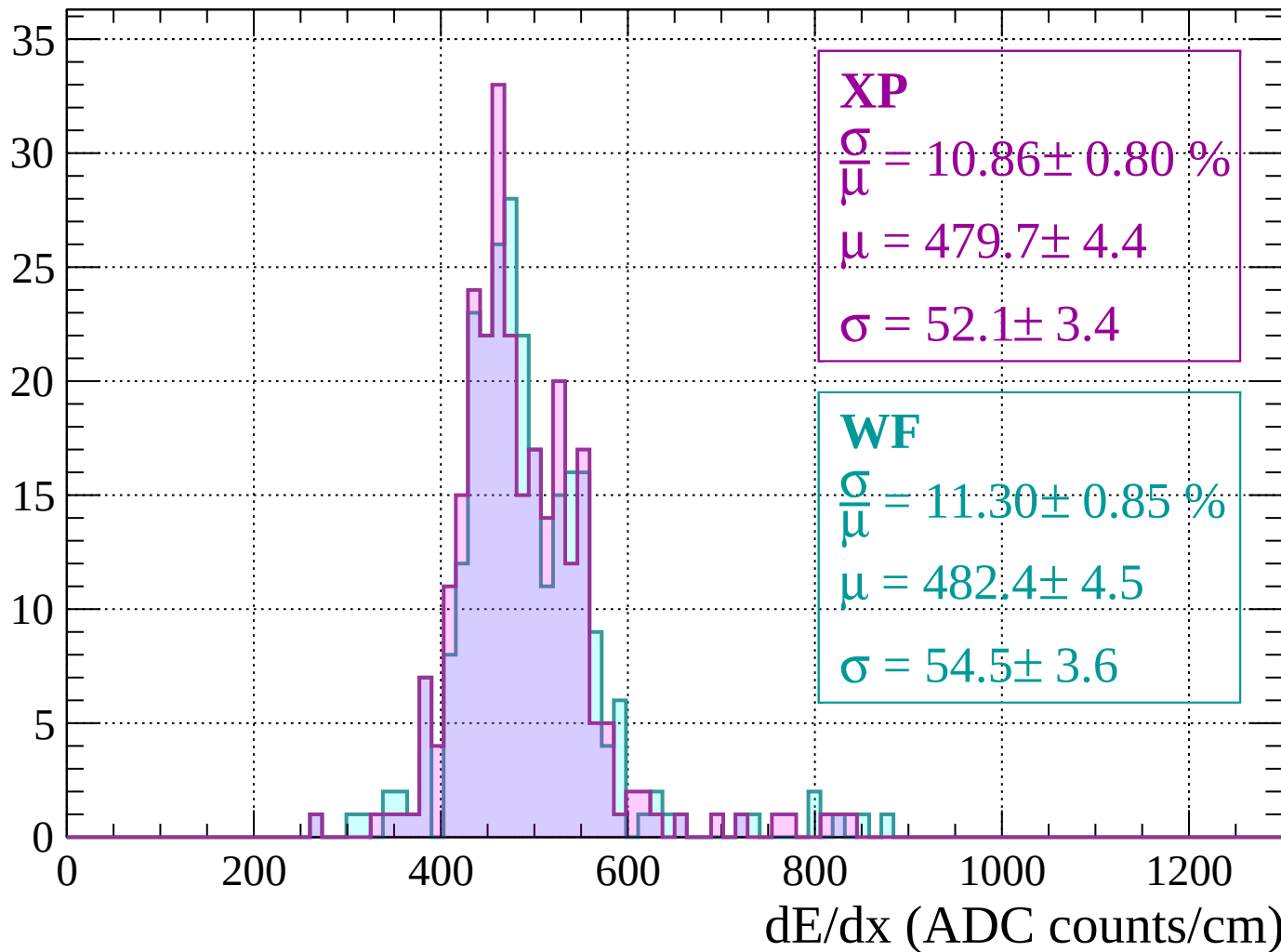
Energy loss | $1800 < p < 1840$

Count



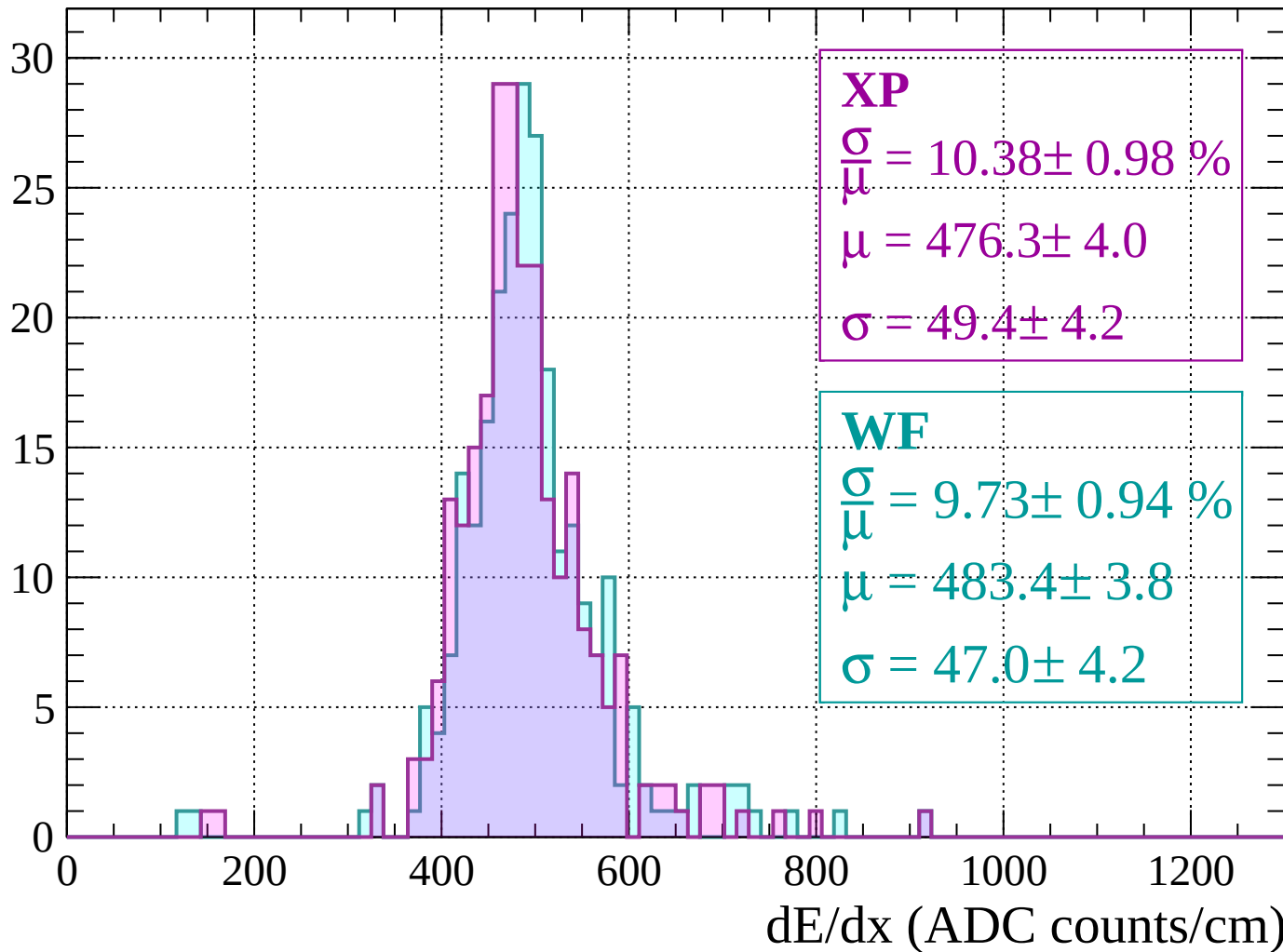
Energy loss | $1840 < p < 1880$

Count



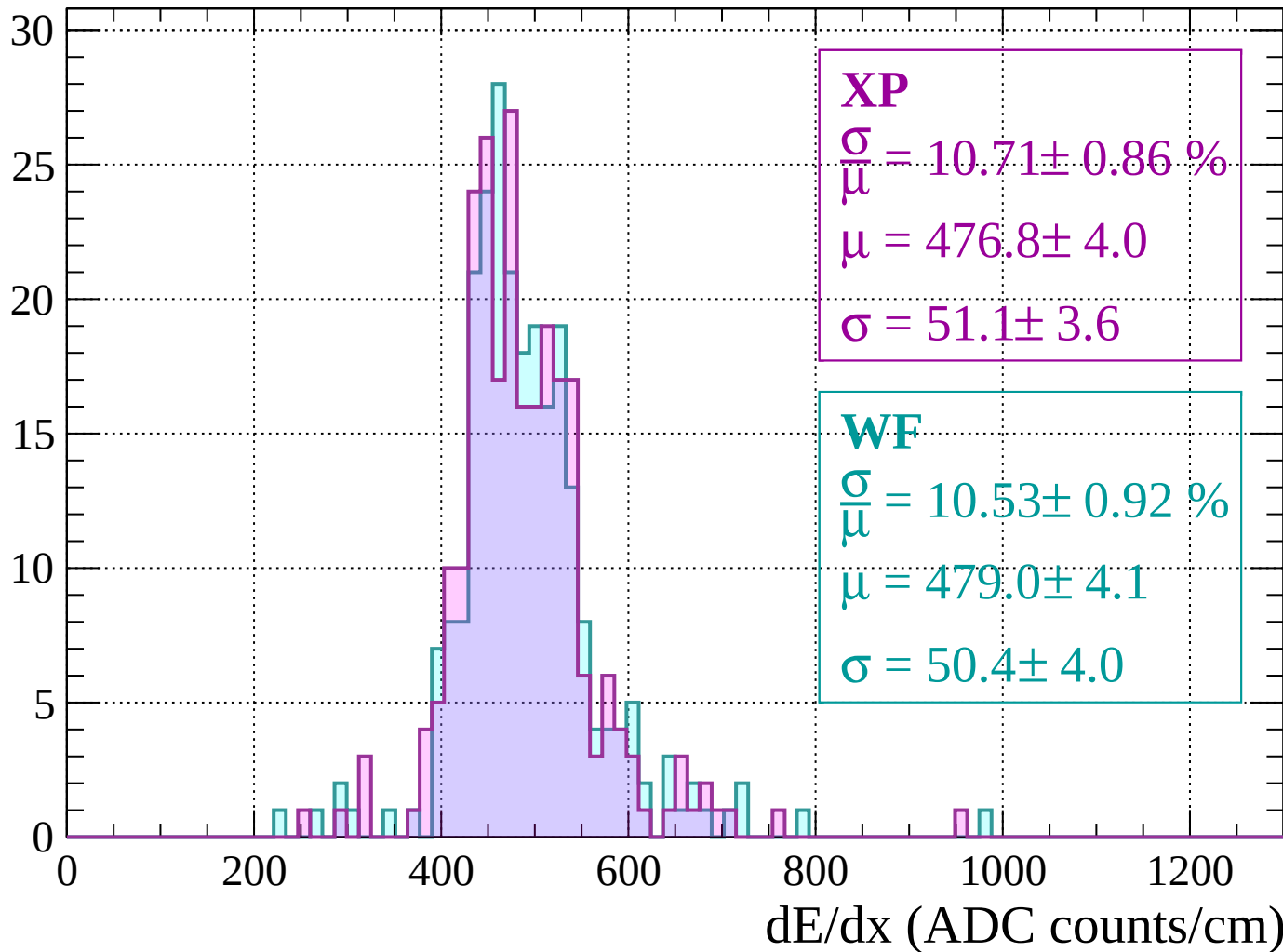
Energy loss | $1880 < p < 1920$

Count



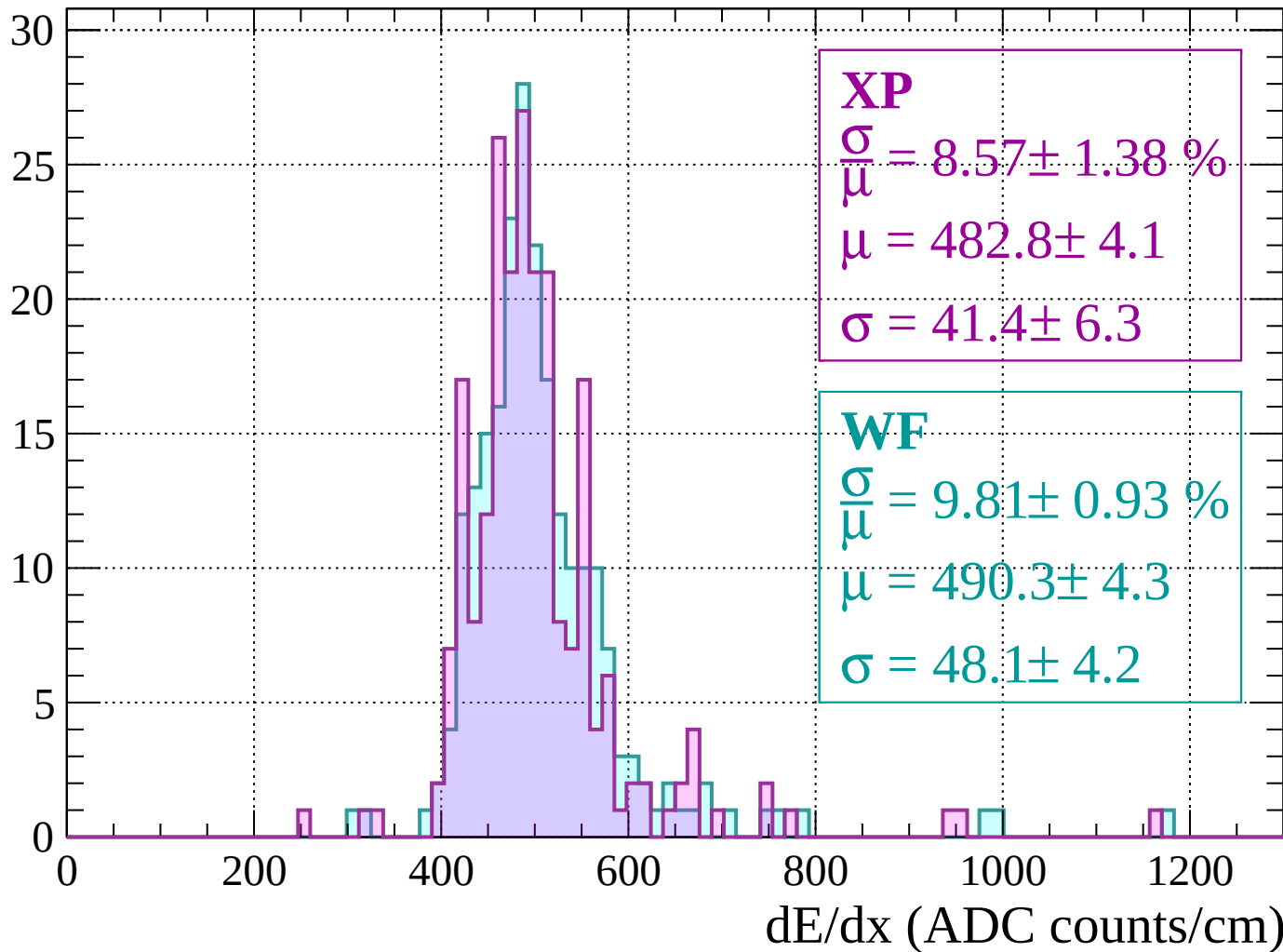
Energy loss | $1920 < p < 1960$

Count



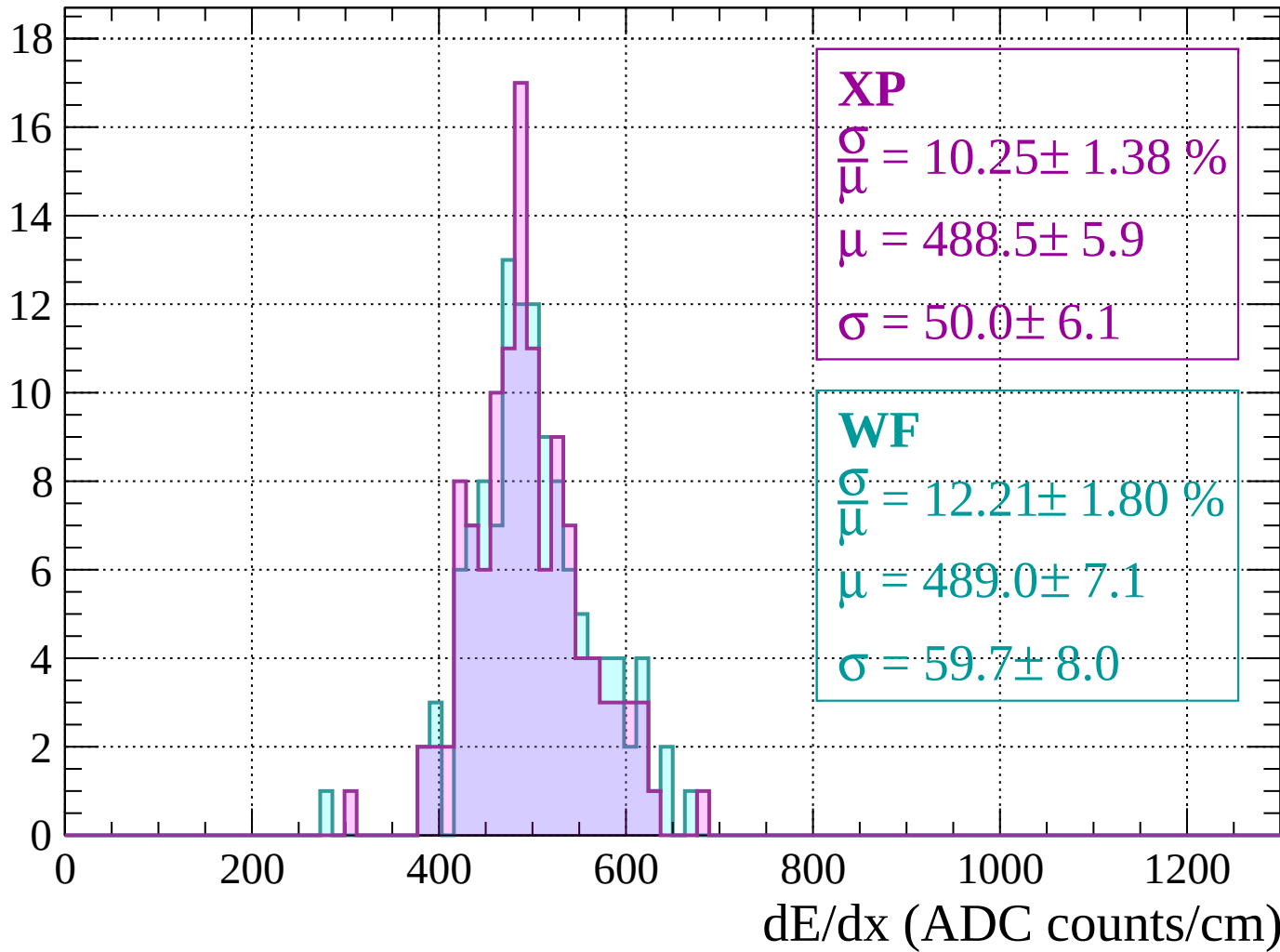
Energy loss | 1960 < p < 2000

Count



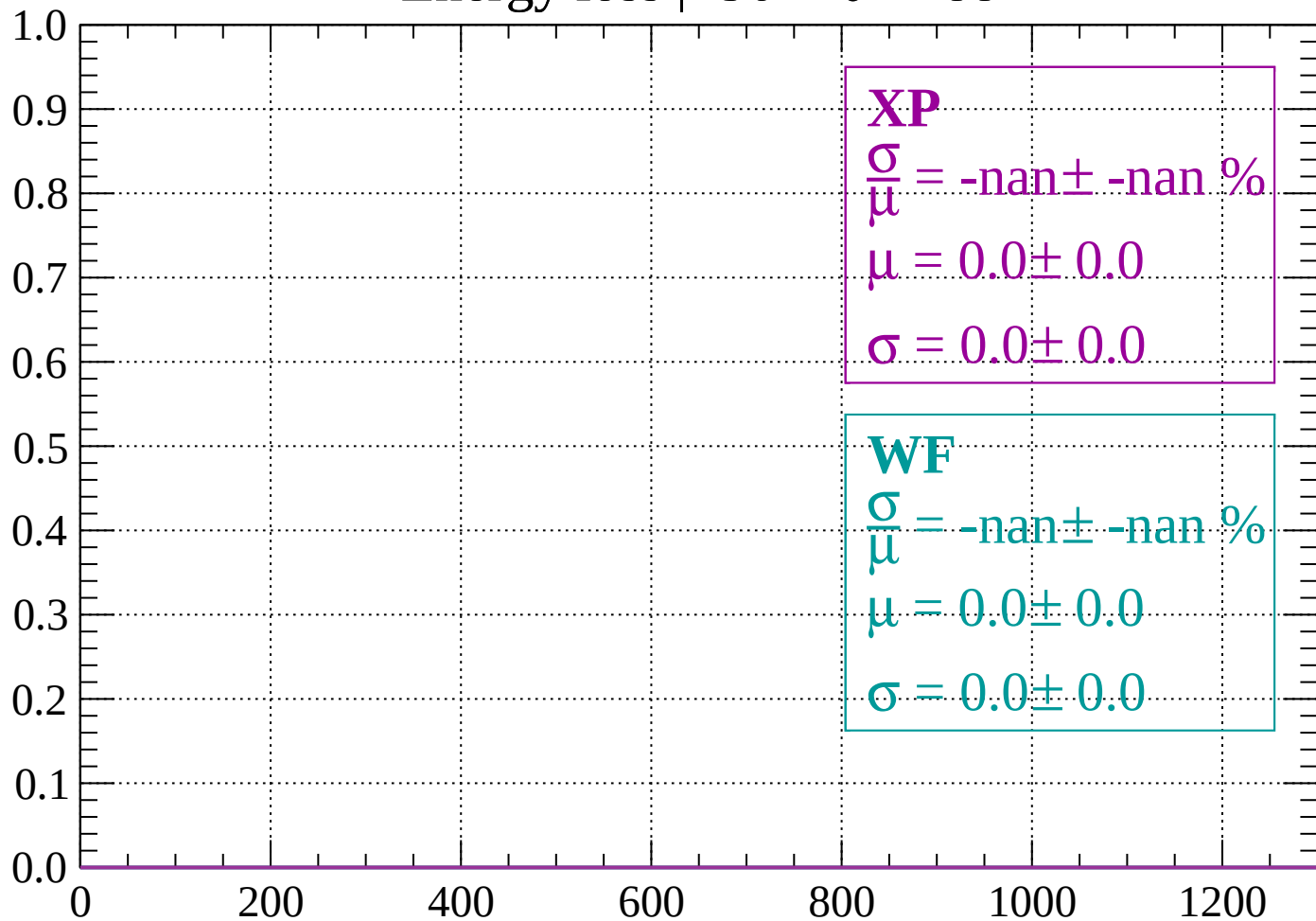
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

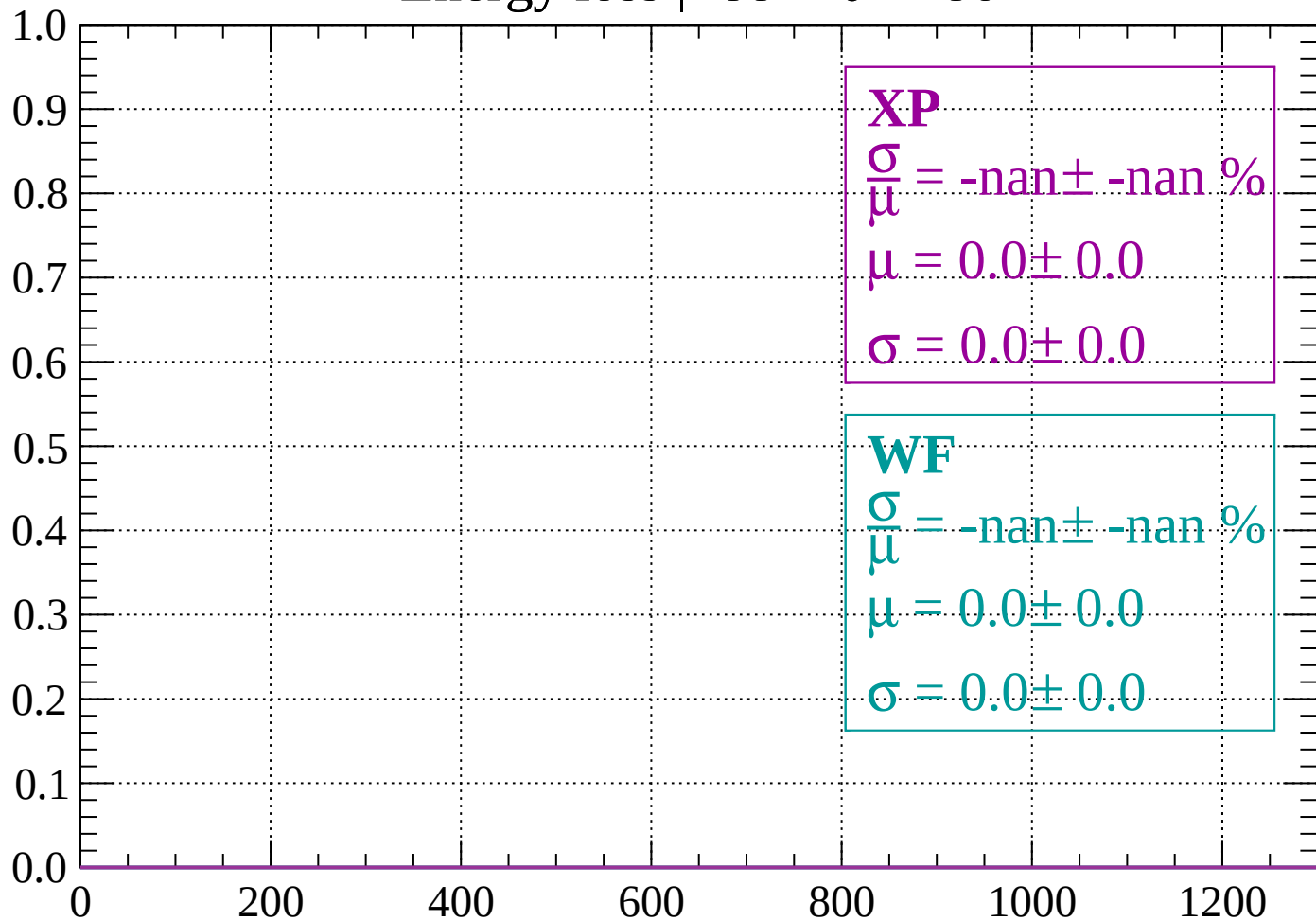
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

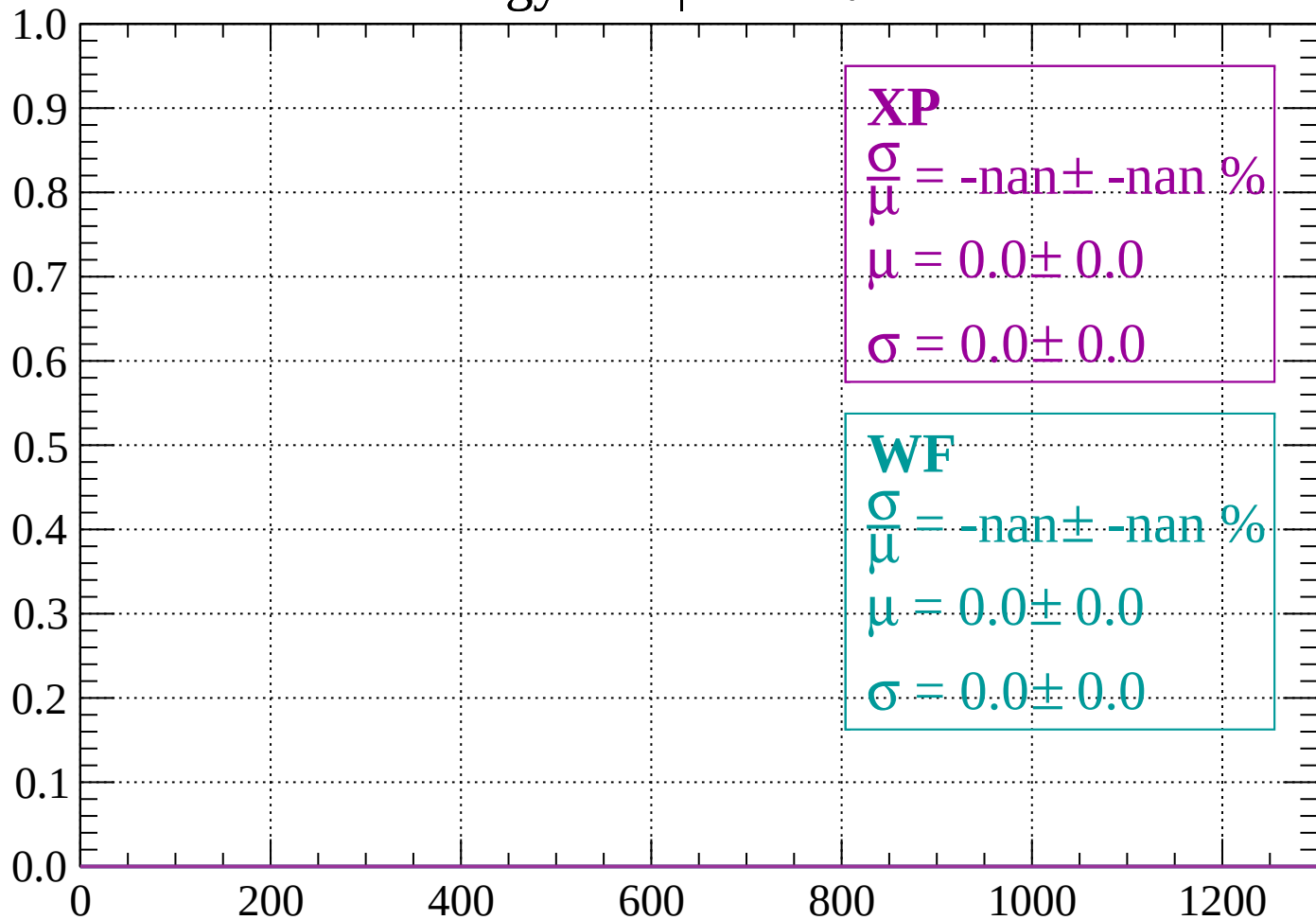
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

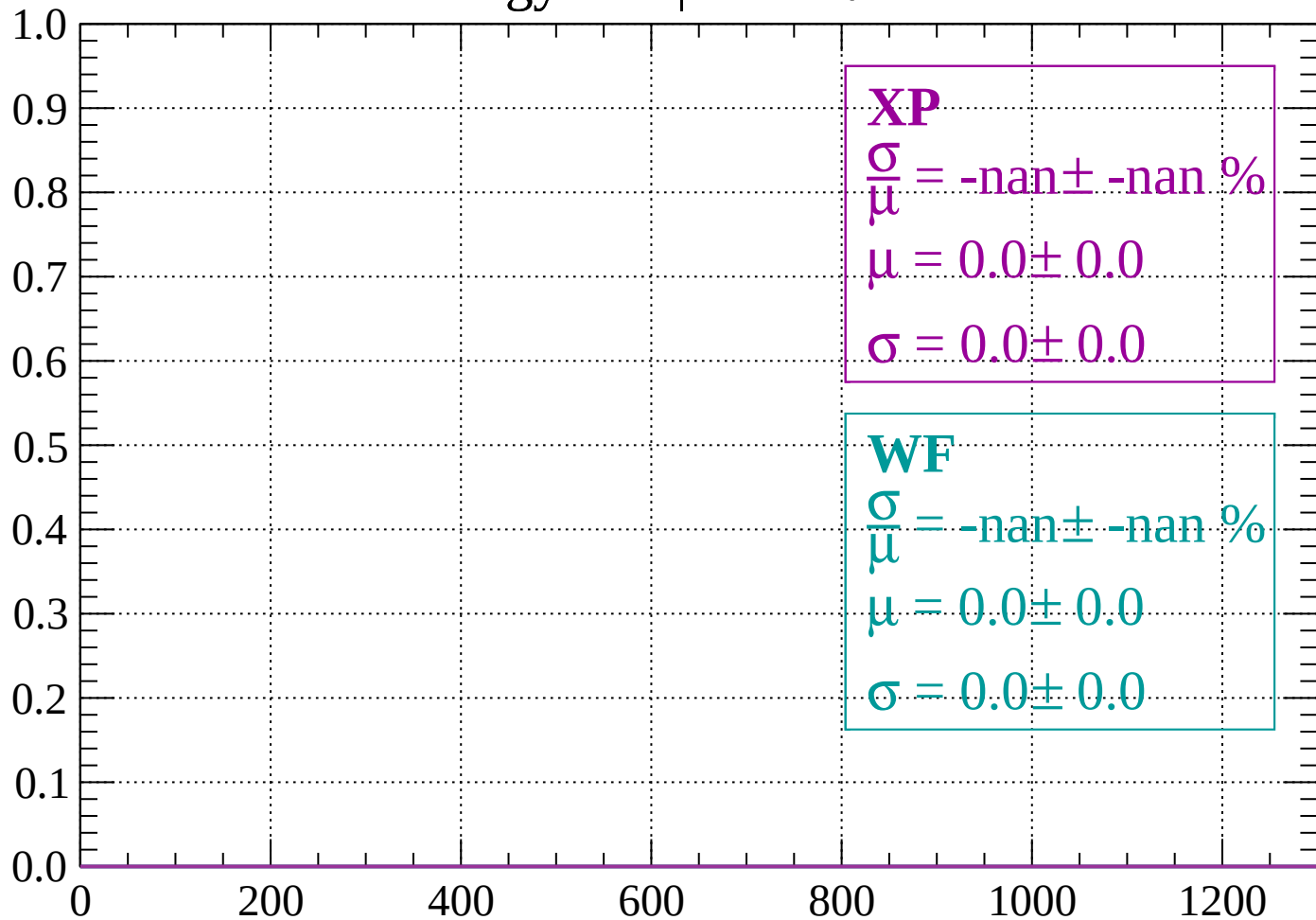
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

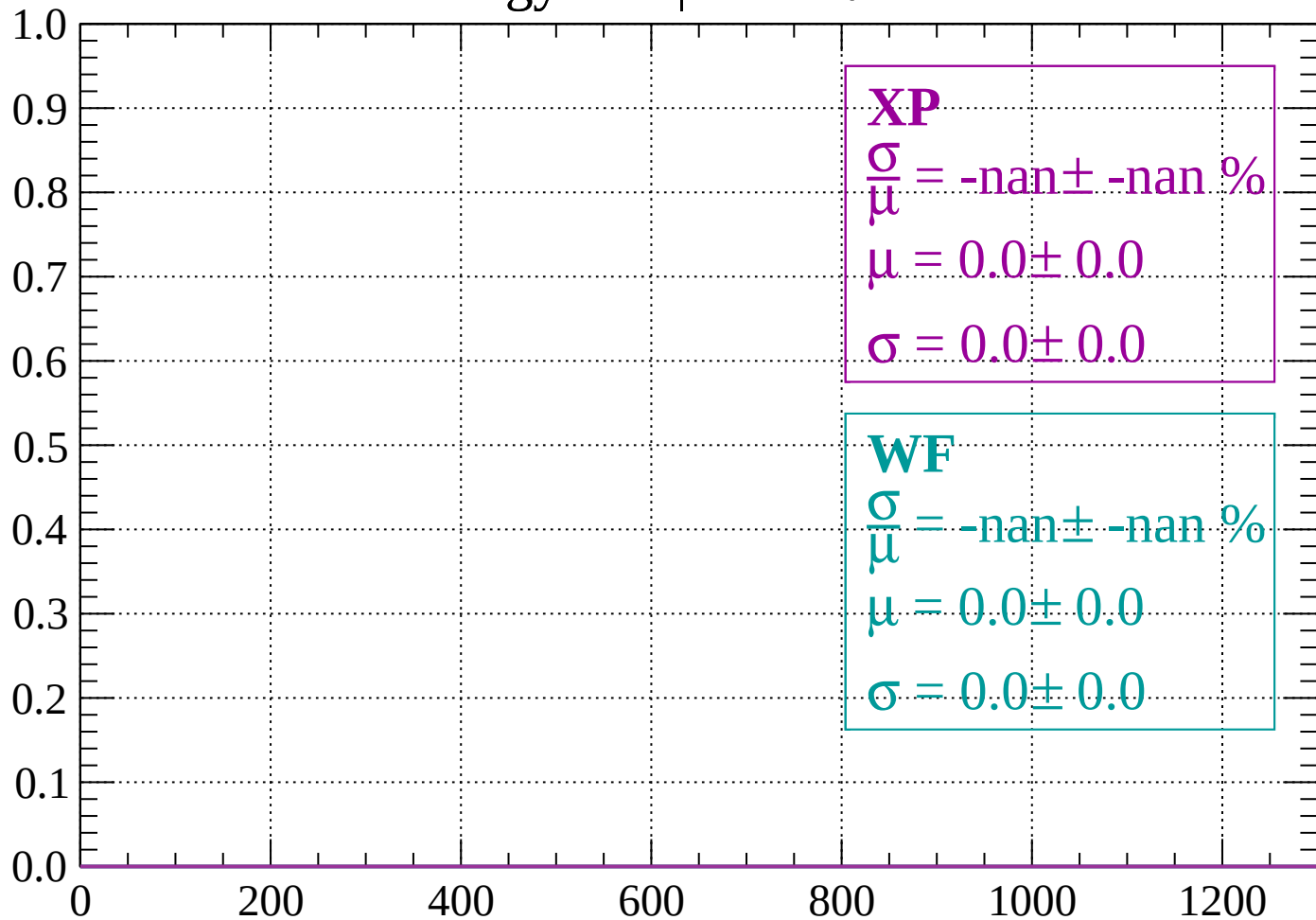
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

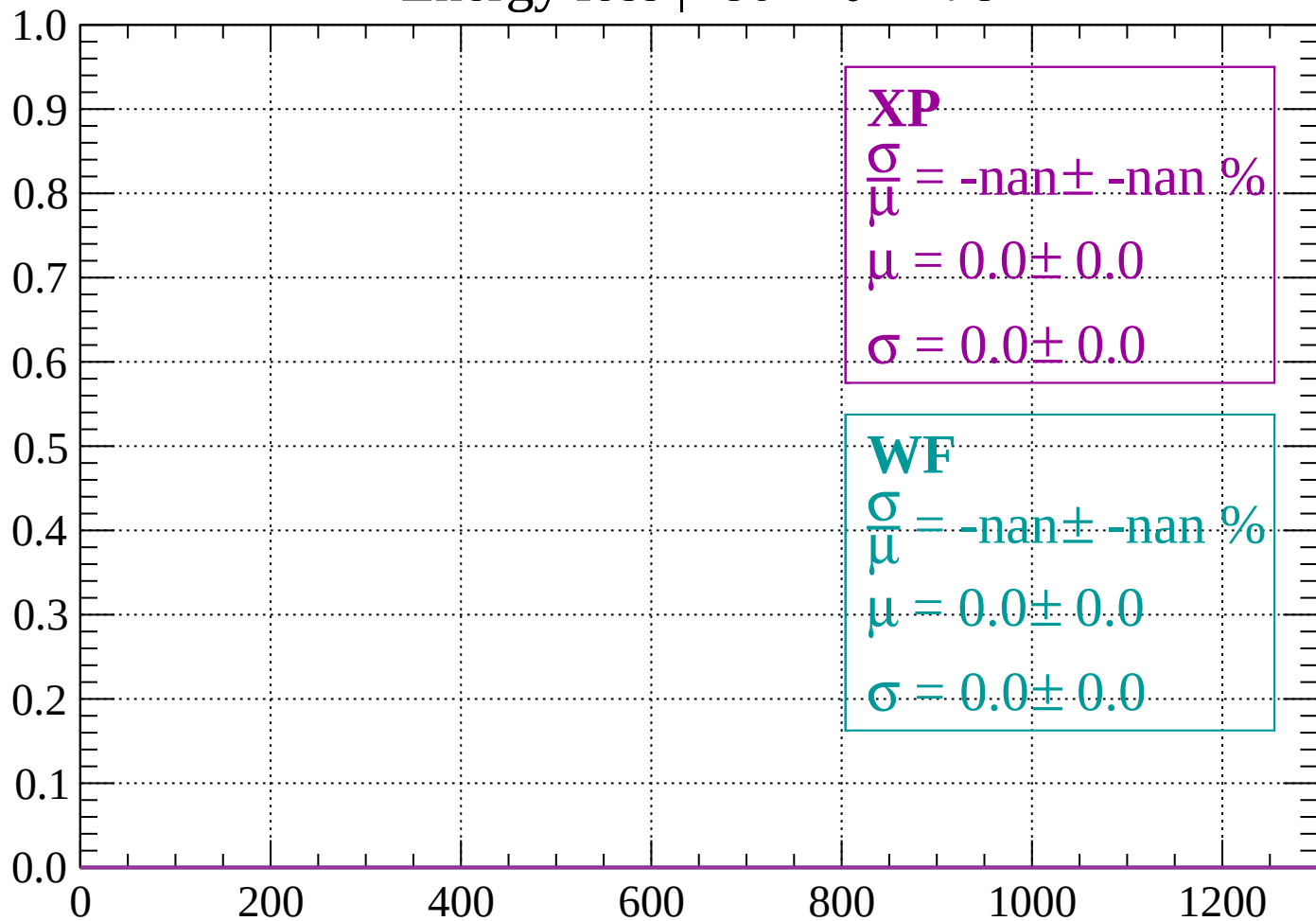
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

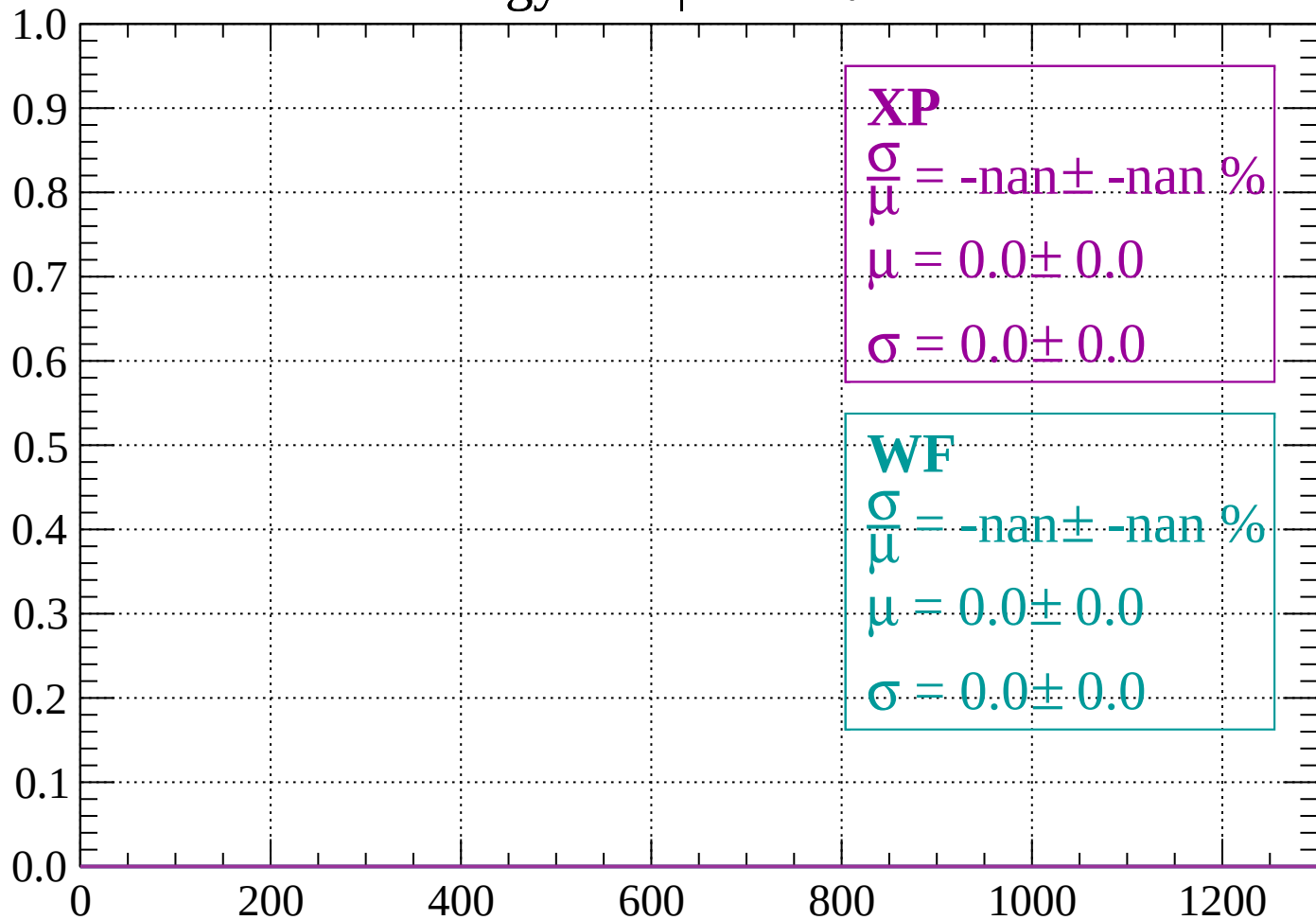
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

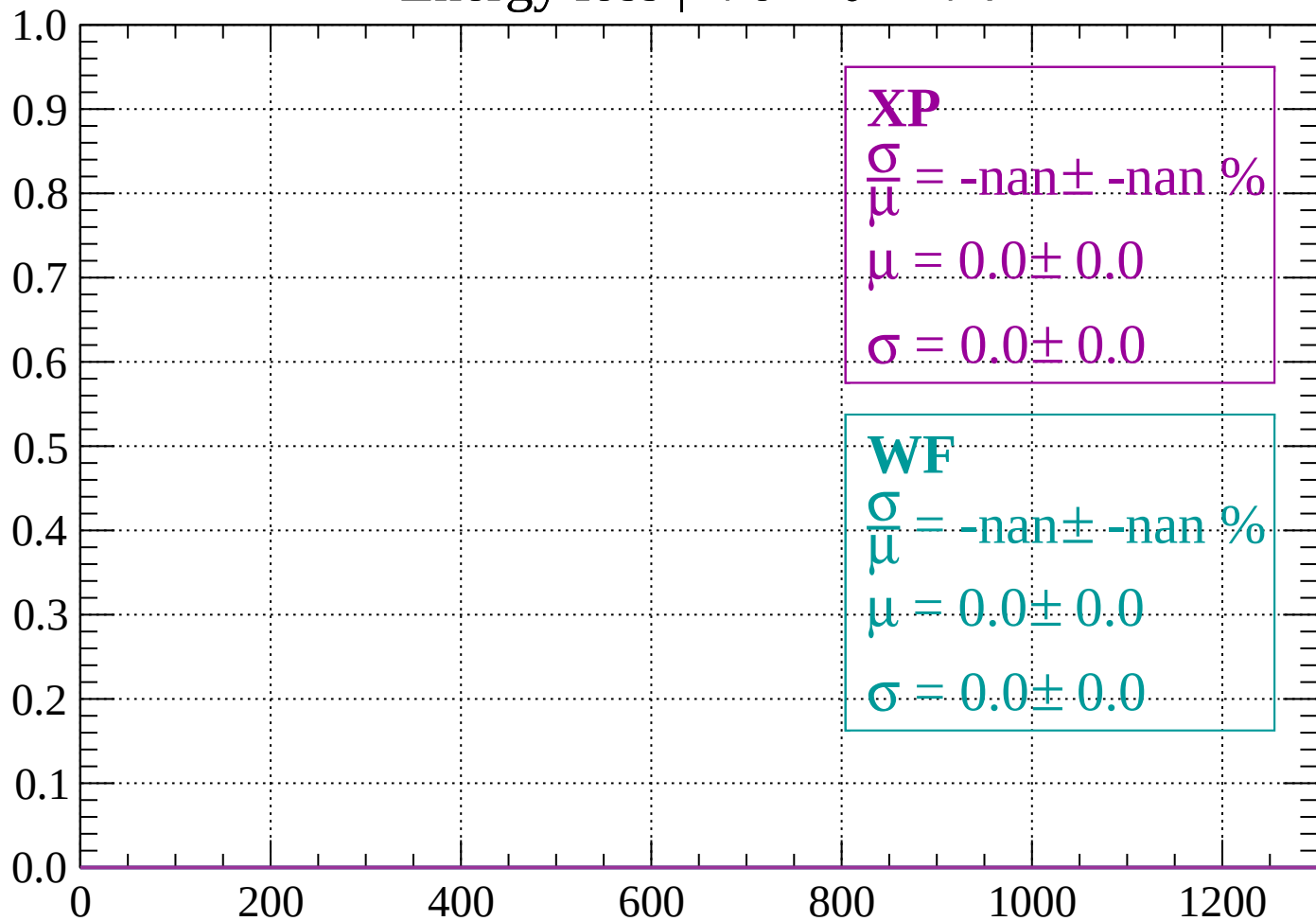
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

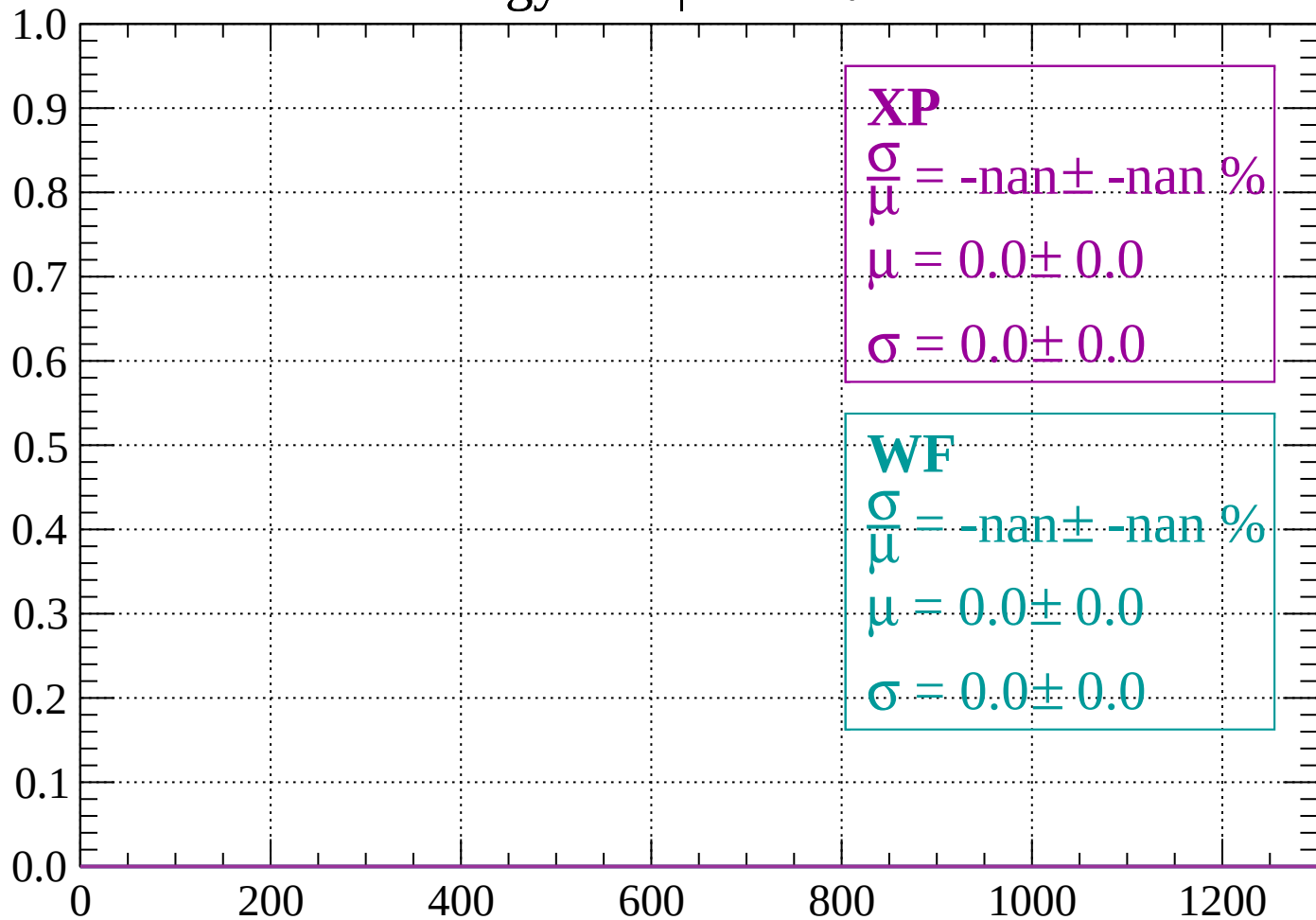
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

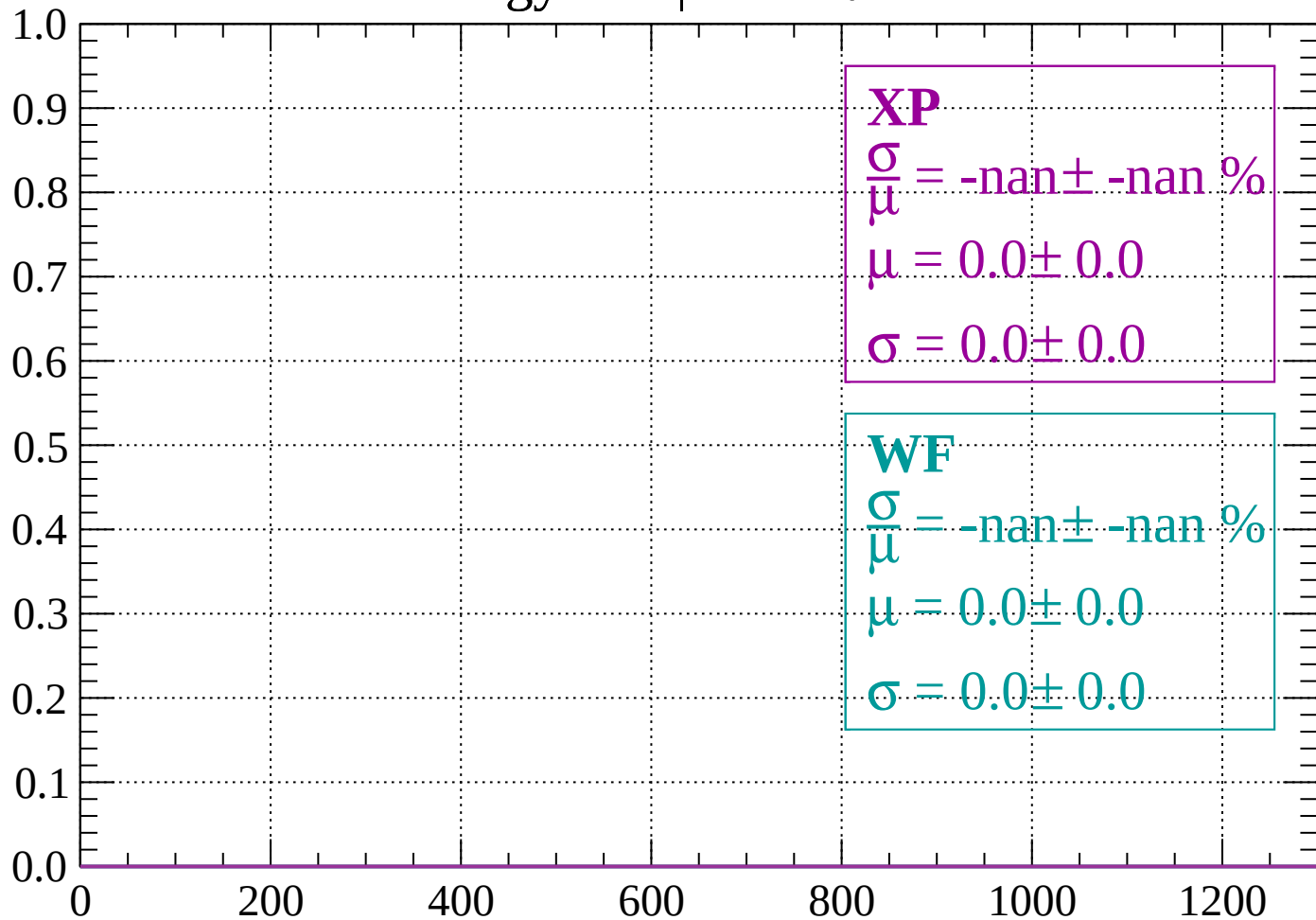
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

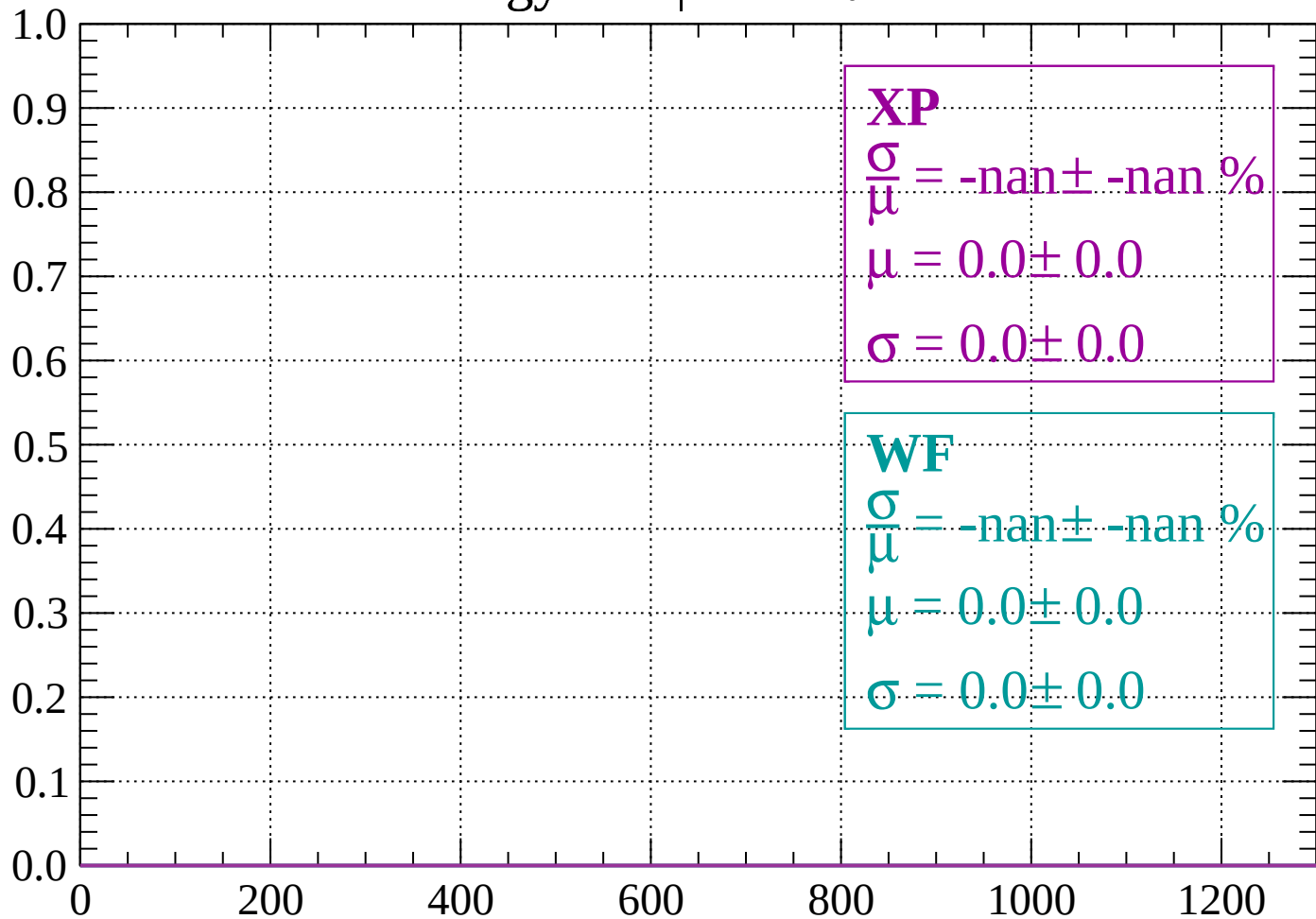
Count



dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$

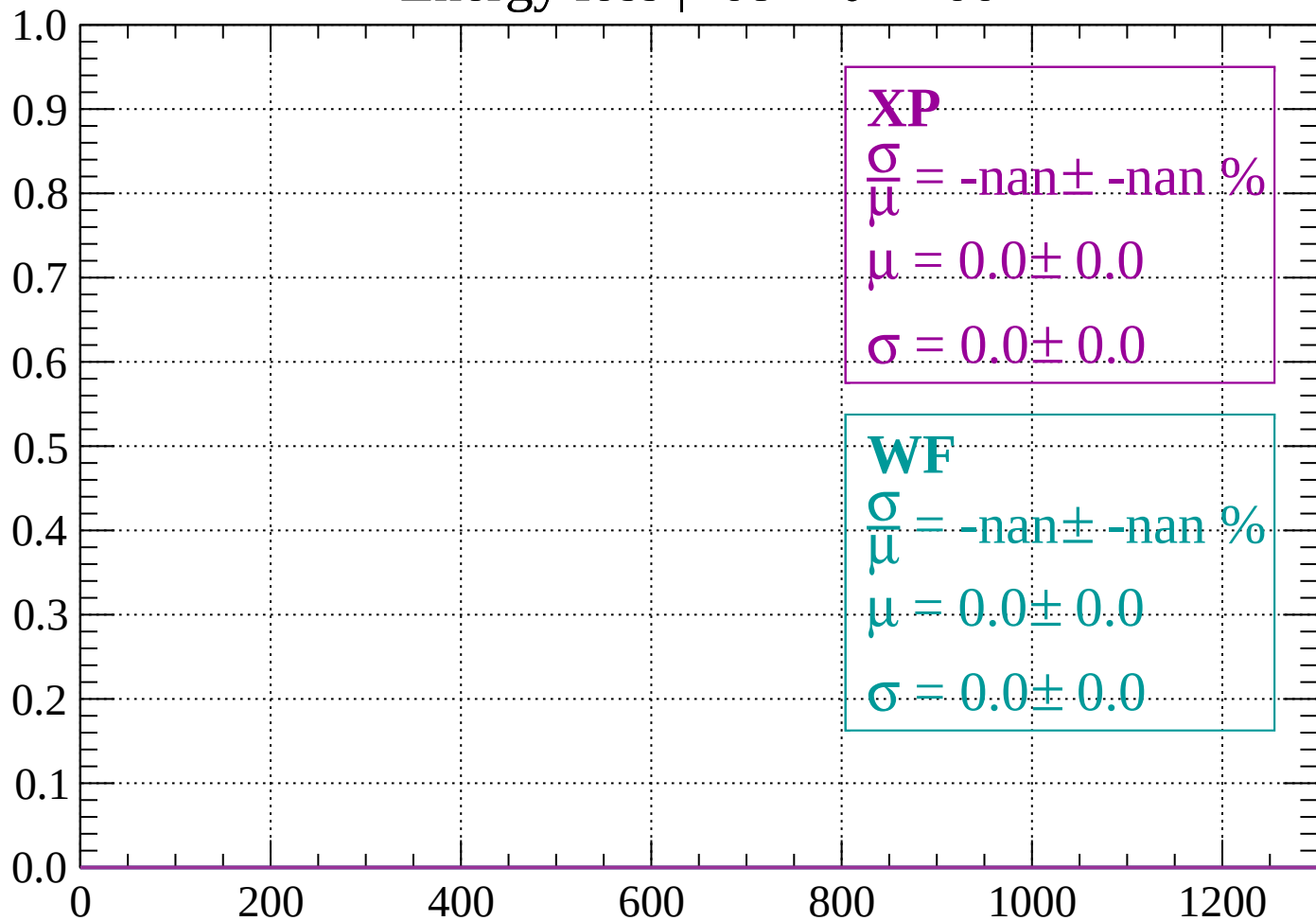
Count



dE/dx (ADC counts/cm)

Energy loss | $-68 < \theta < -66$

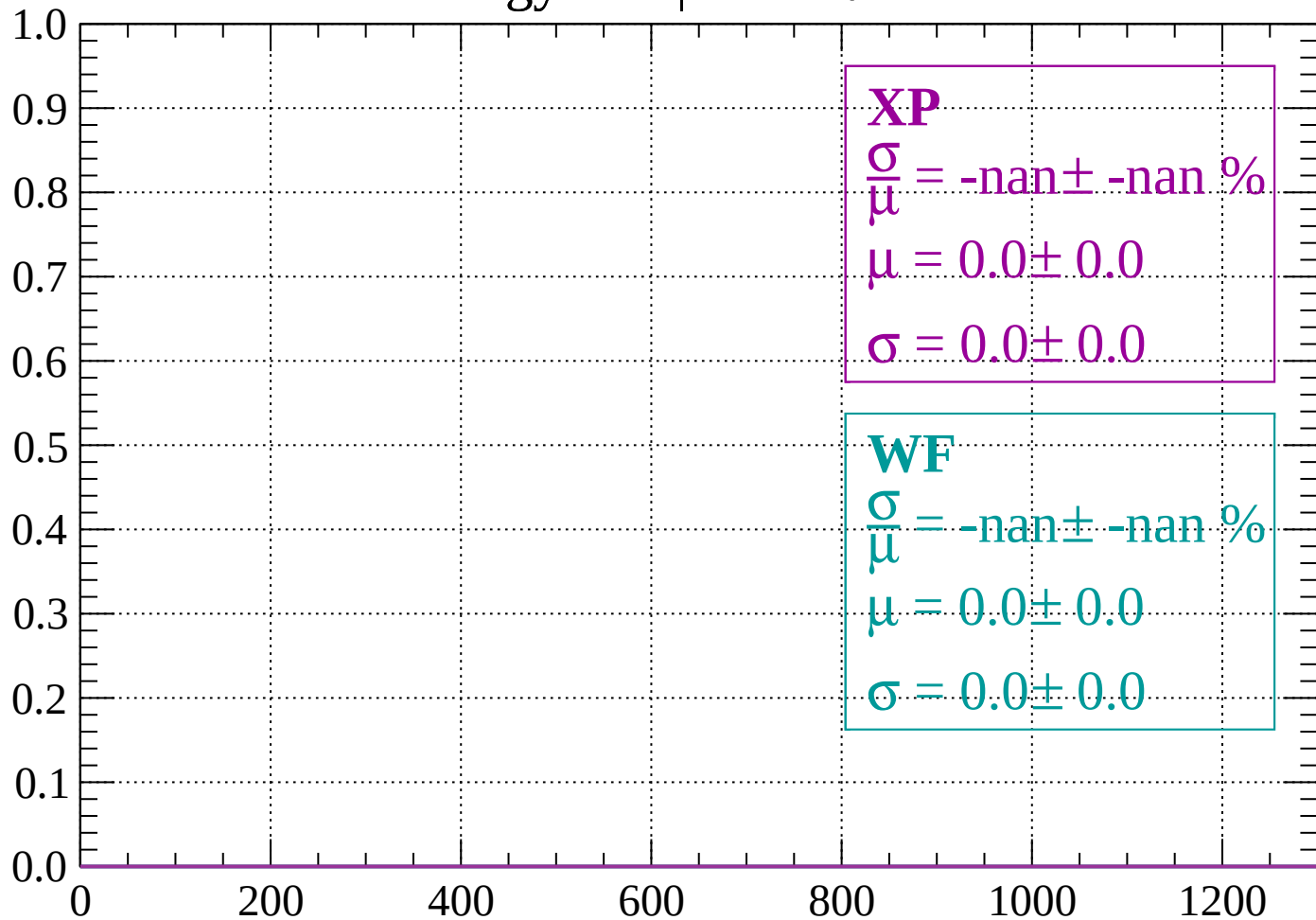
Count



dE/dx (ADC counts/cm)

Energy loss | $-66 < \theta < -64$

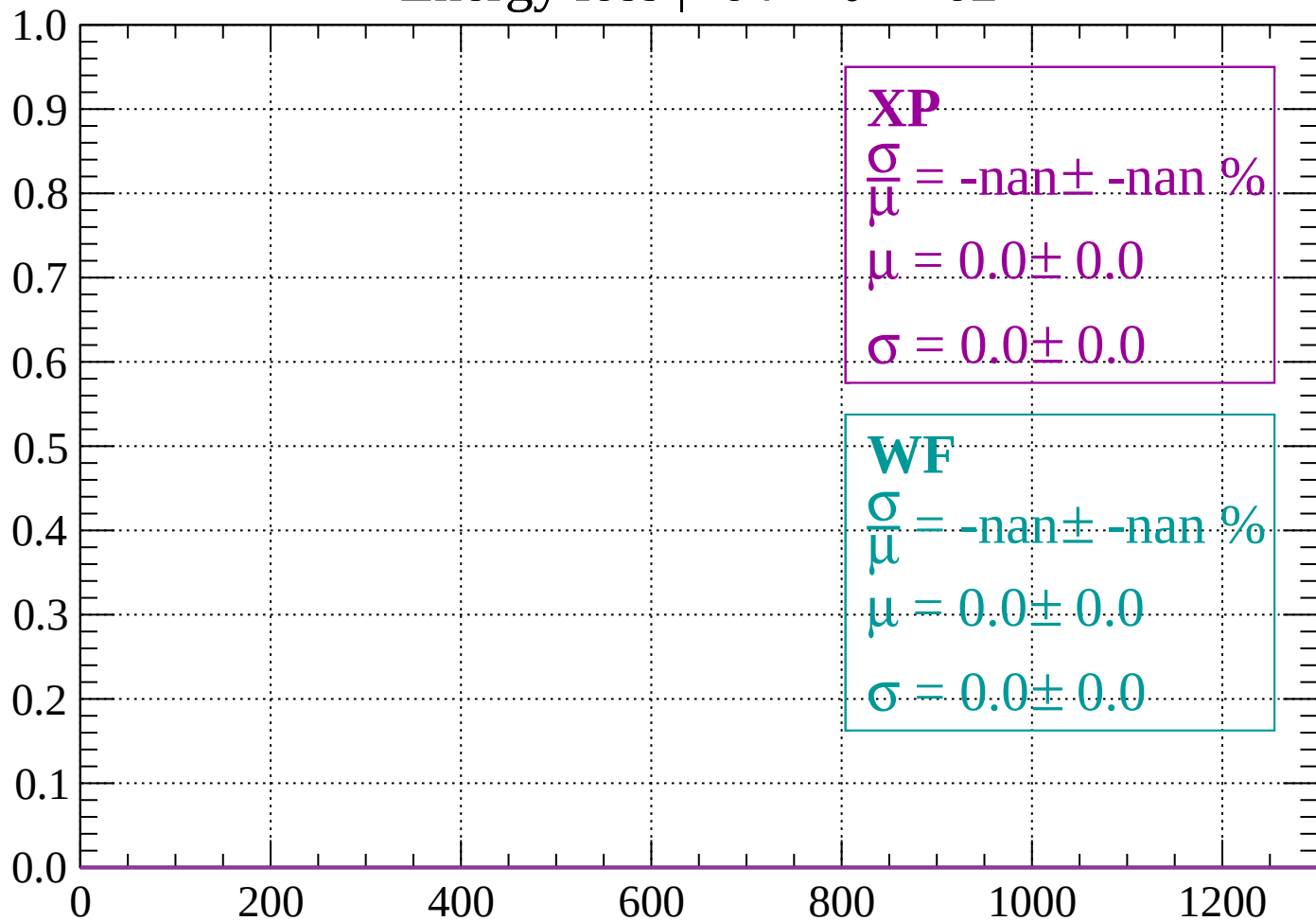
Count



dE/dx (ADC counts/cm)

Energy loss | $-64 < \theta < -62$

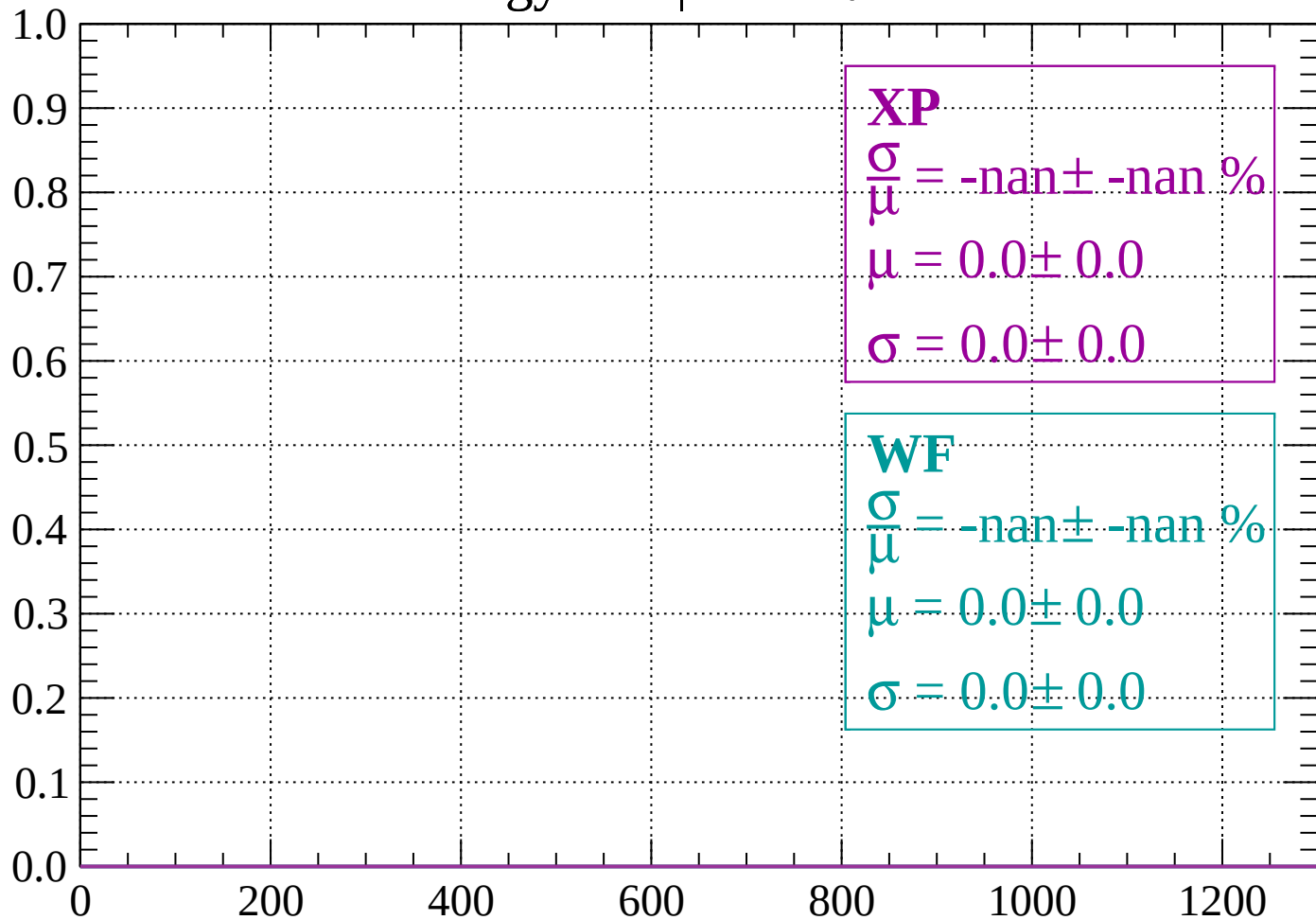
Count



dE/dx (ADC counts/cm)

Energy loss | $-62 < \theta < -60$

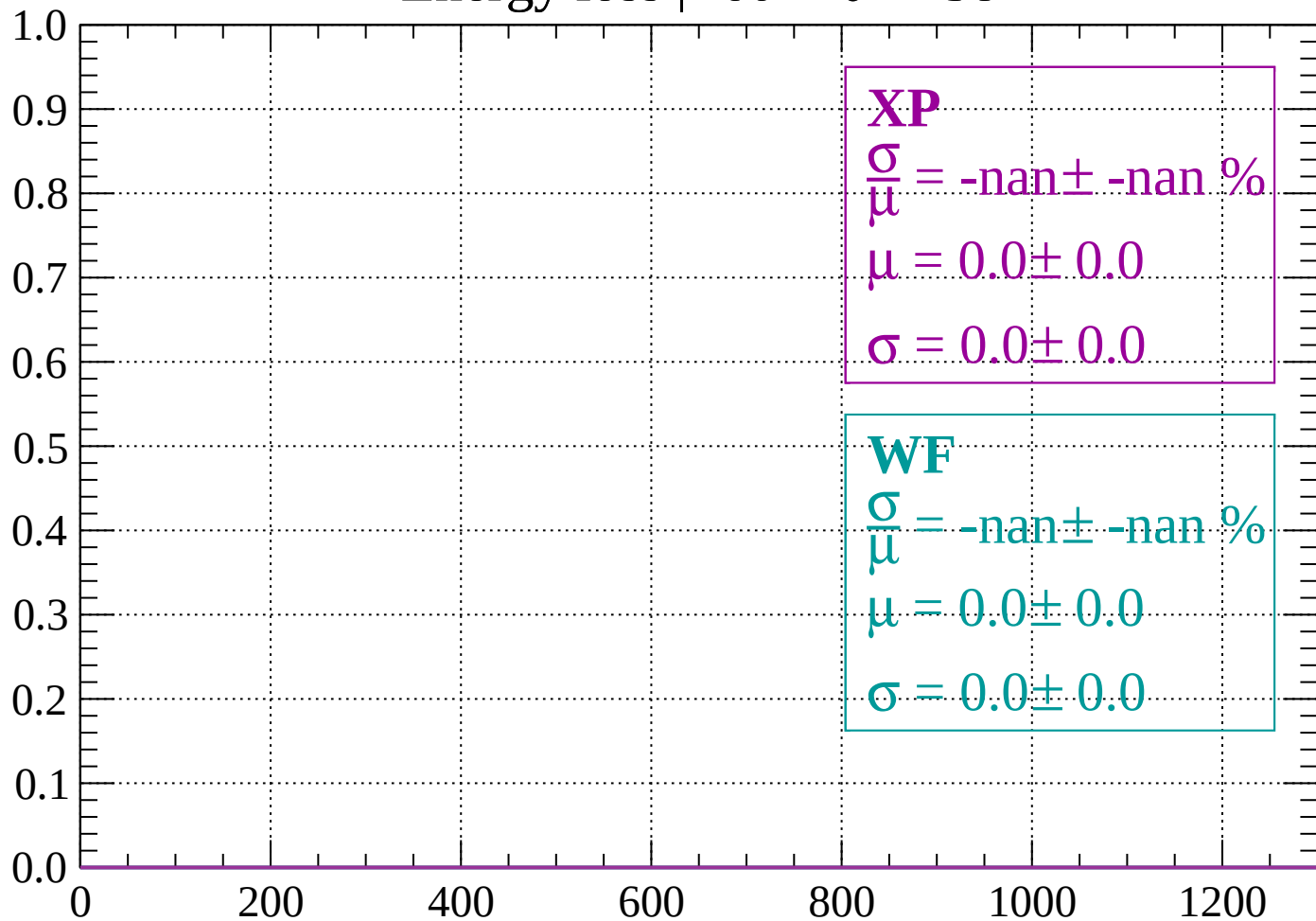
Count



dE/dx (ADC counts/cm)

Energy loss | $-60 < \theta < -58$

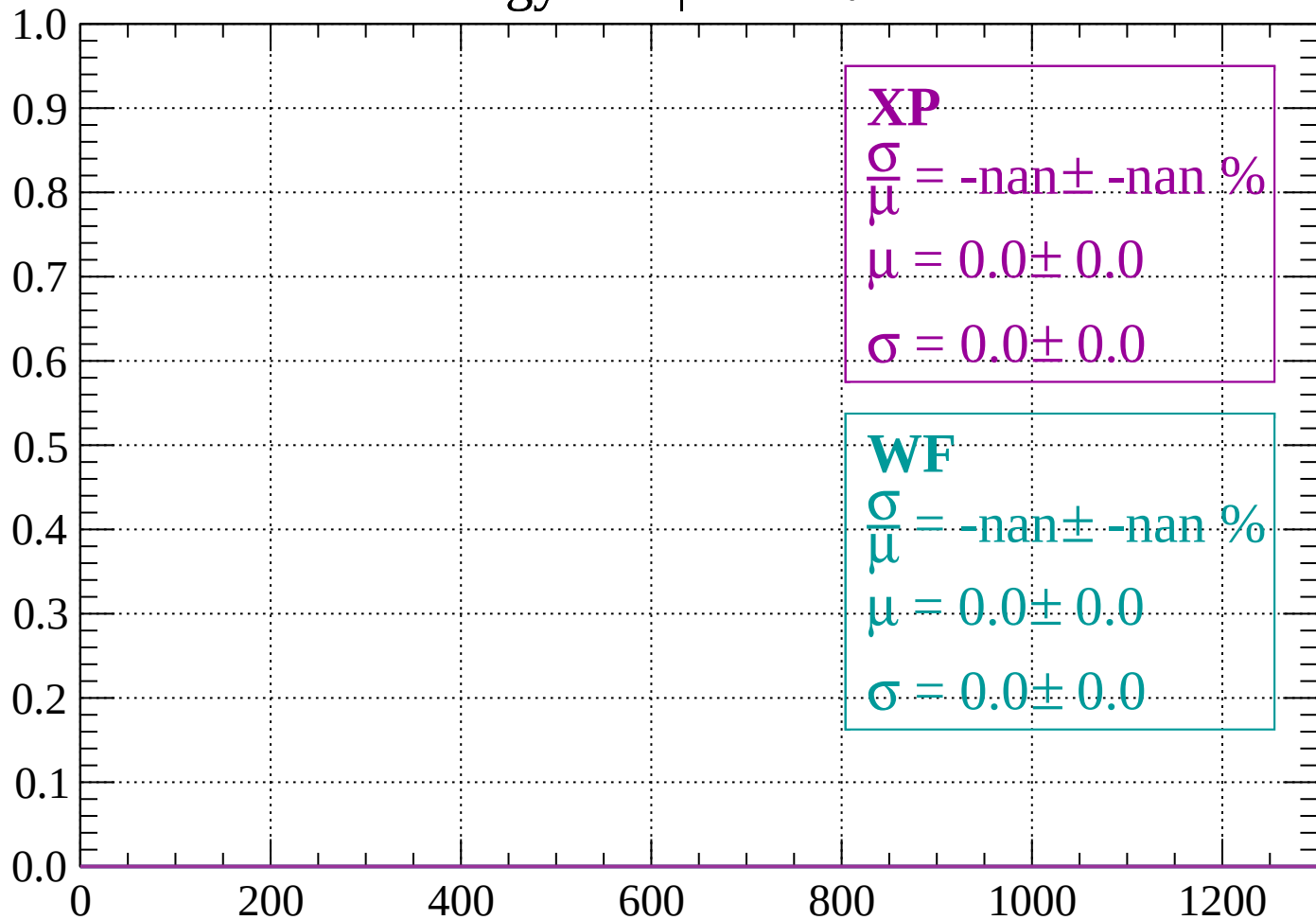
Count



dE/dx (ADC counts/cm)

Energy loss | $-58 < \theta < -56$

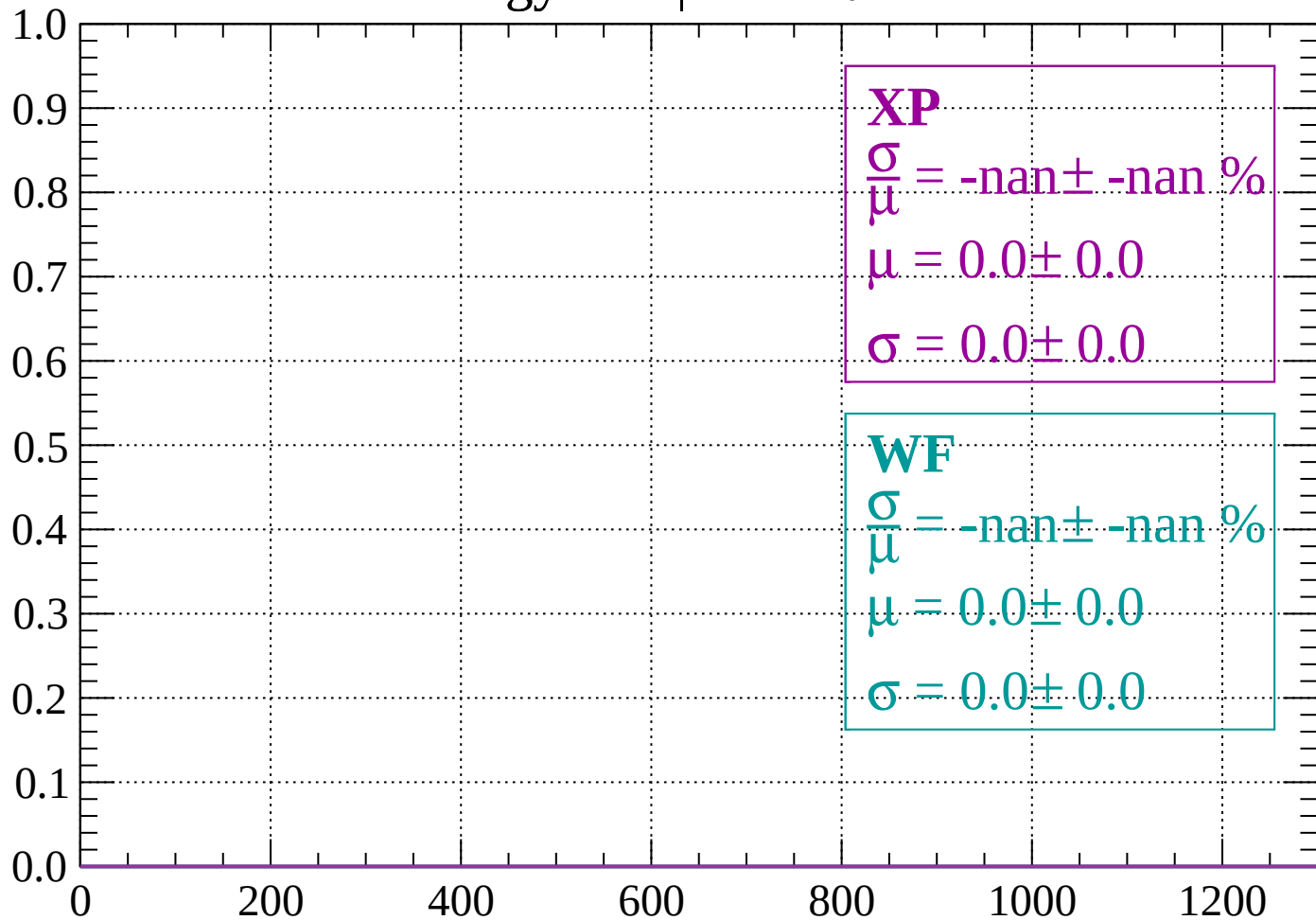
Count



dE/dx (ADC counts/cm)

Energy loss | $-56 < \theta < -54$

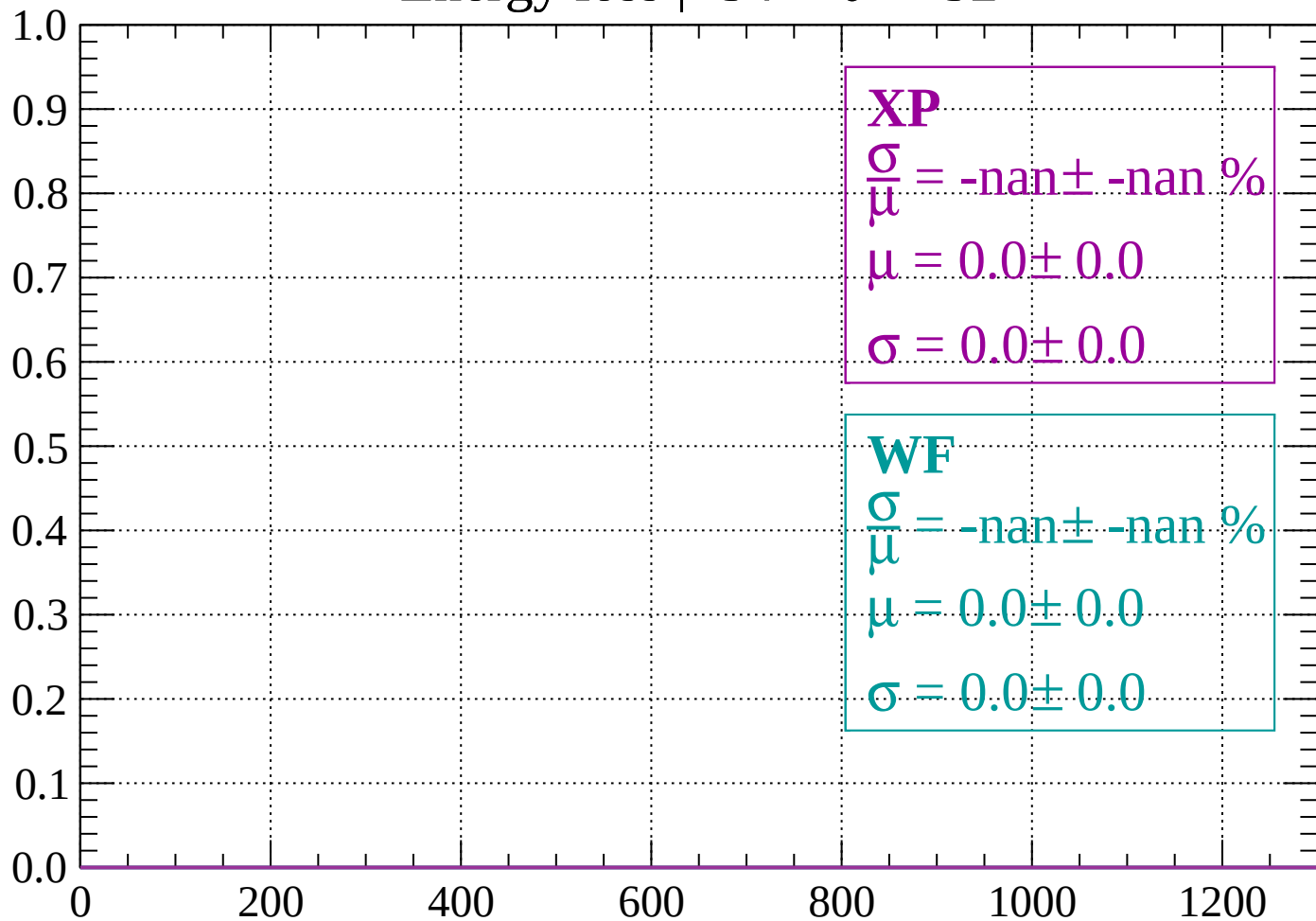
Count



dE/dx (ADC counts/cm)

Energy loss | $-54 < \theta < -52$

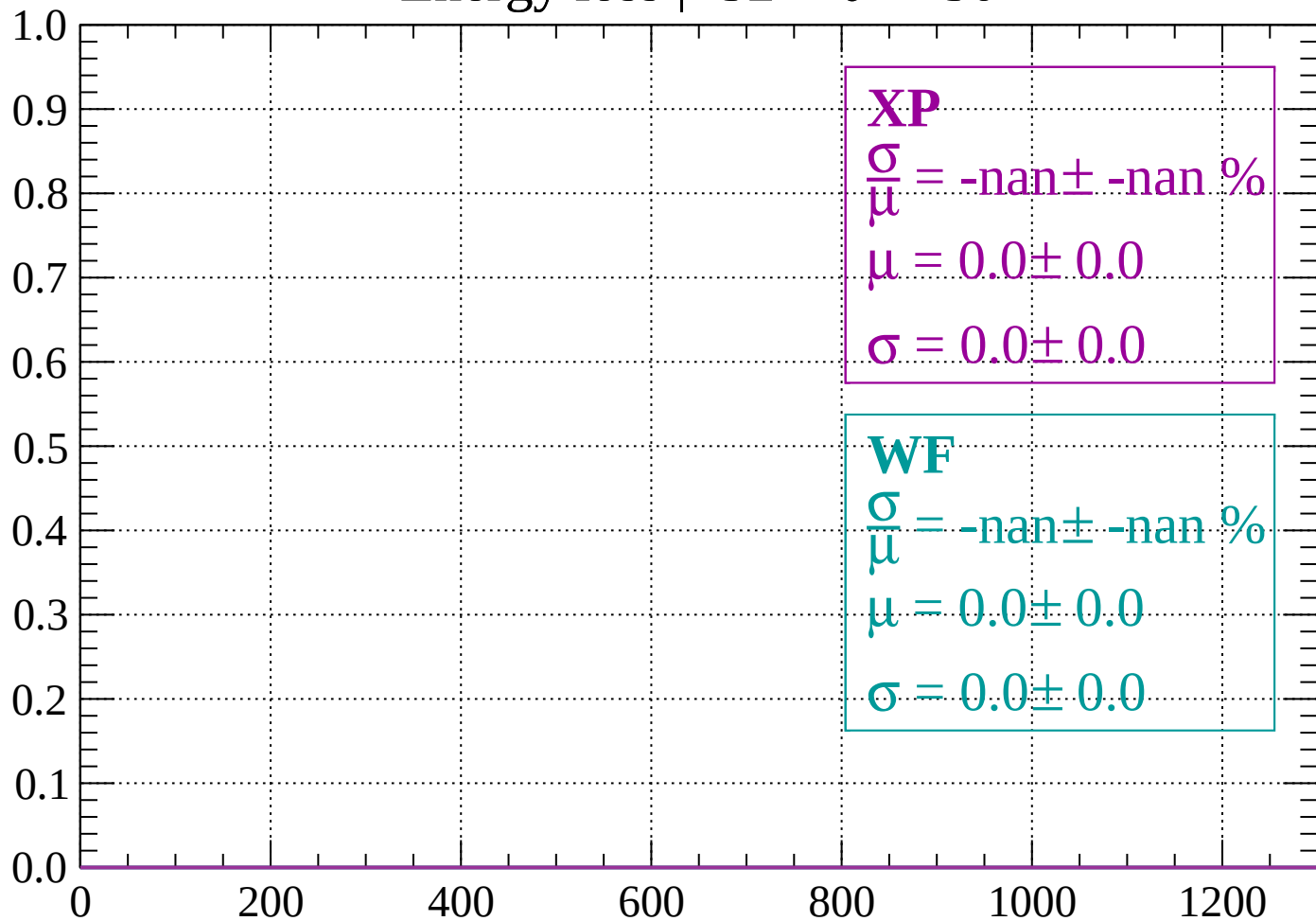
Count



dE/dx (ADC counts/cm)

Energy loss | $-52 < \theta < -50$

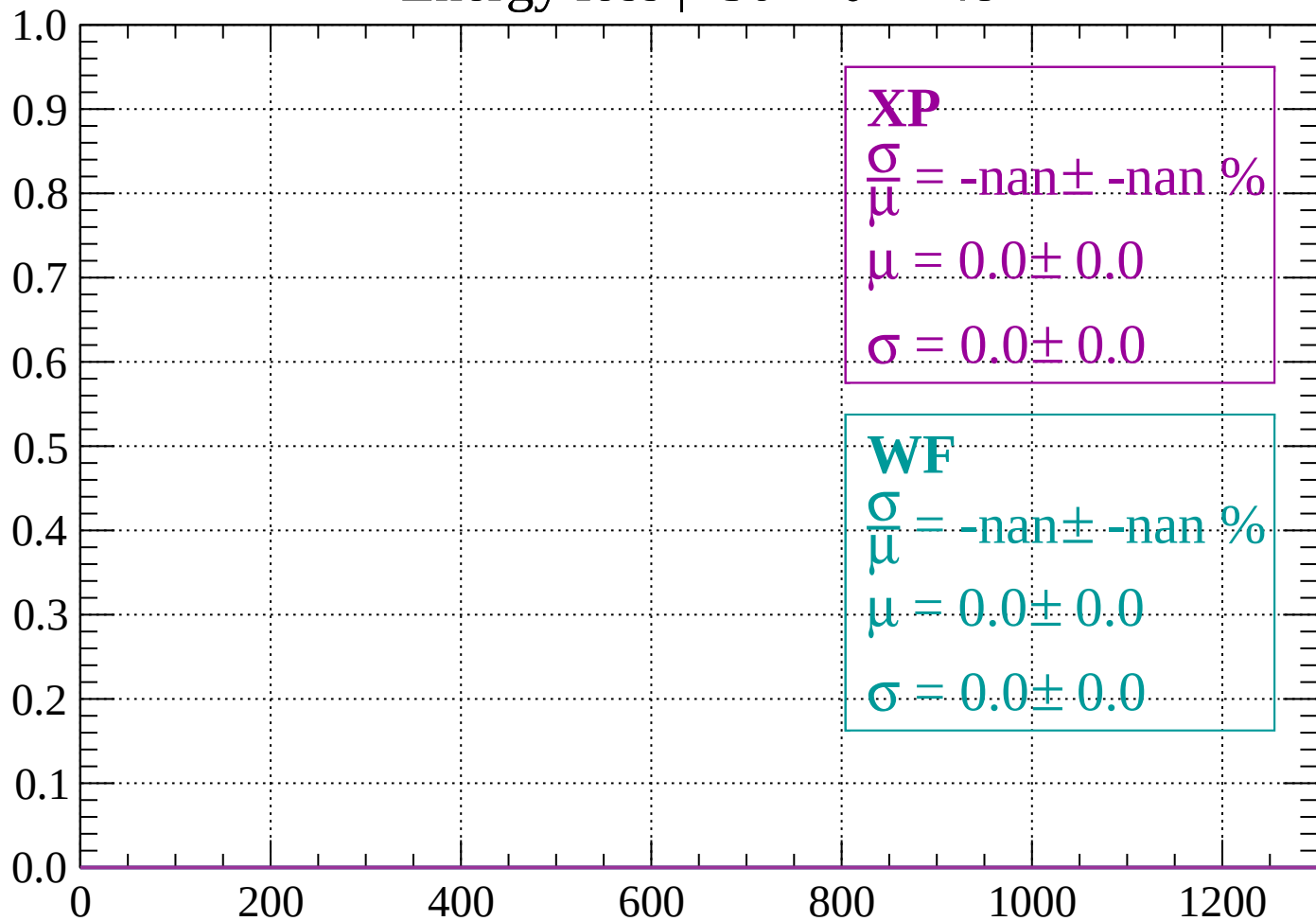
Count



dE/dx (ADC counts/cm)

Energy loss | $-50 < \theta < -48$

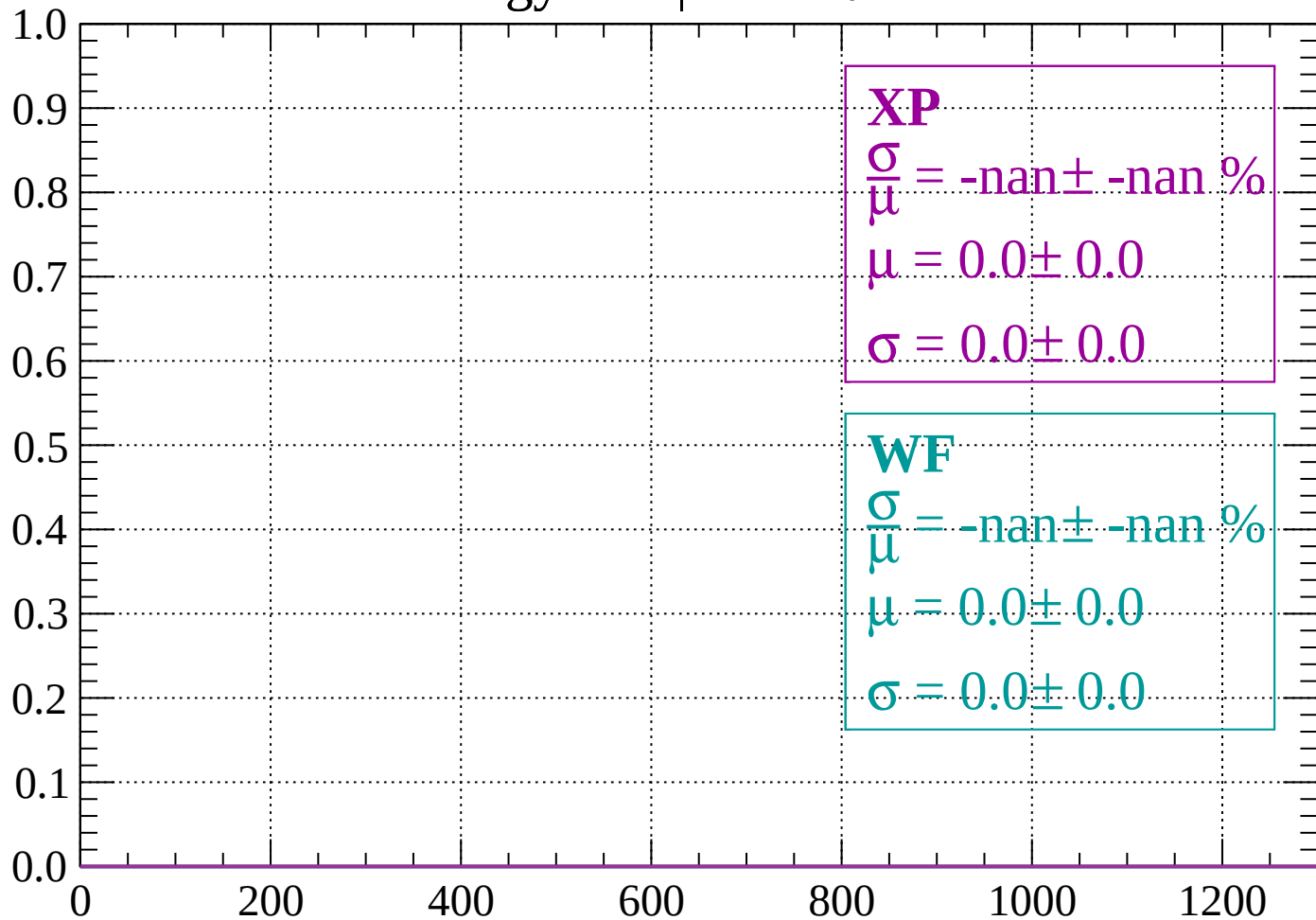
Count



dE/dx (ADC counts/cm)

Energy loss | $-48 < \theta < -46$

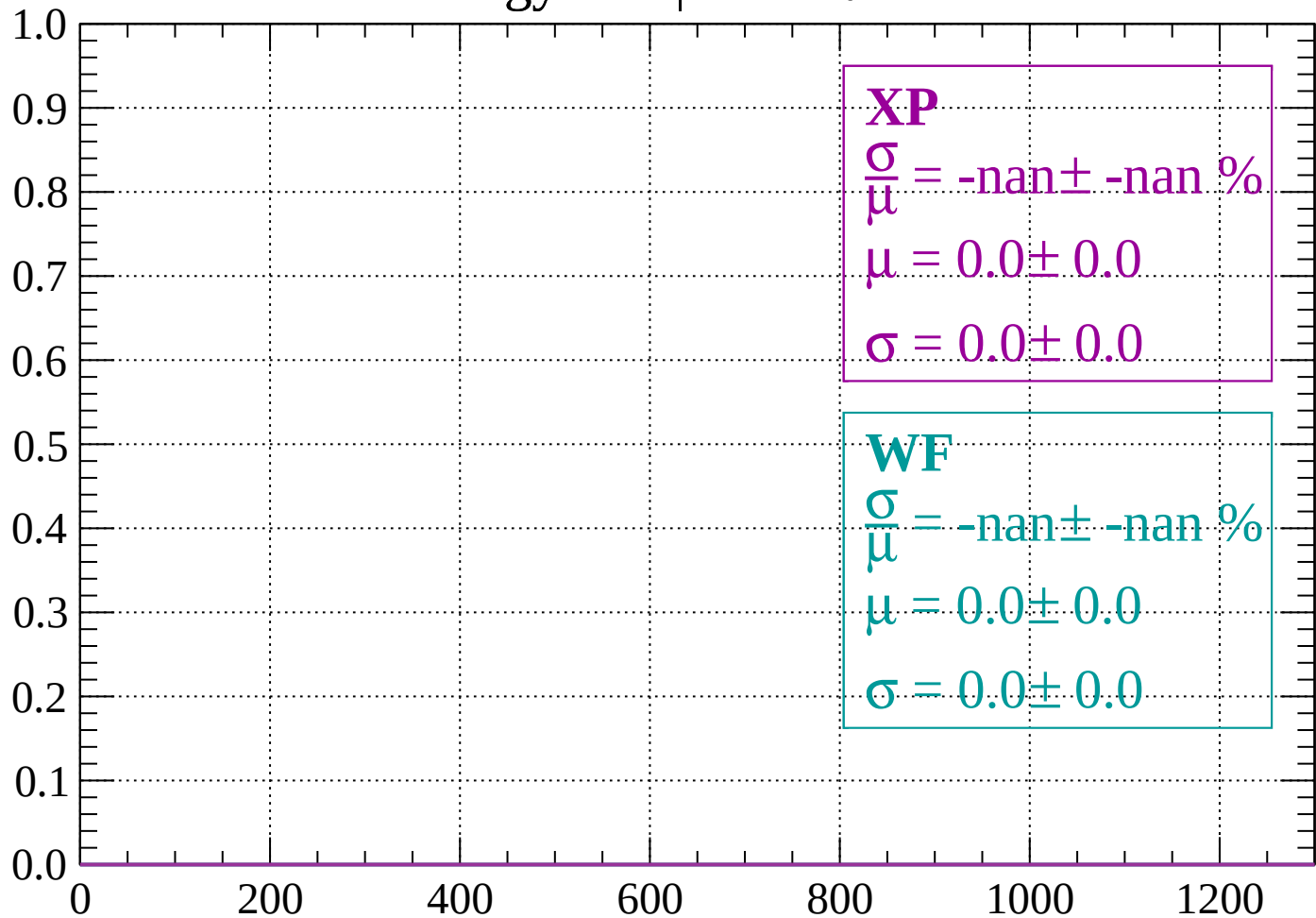
Count



dE/dx (ADC counts/cm)

Energy loss | $-46 < \theta < -44$

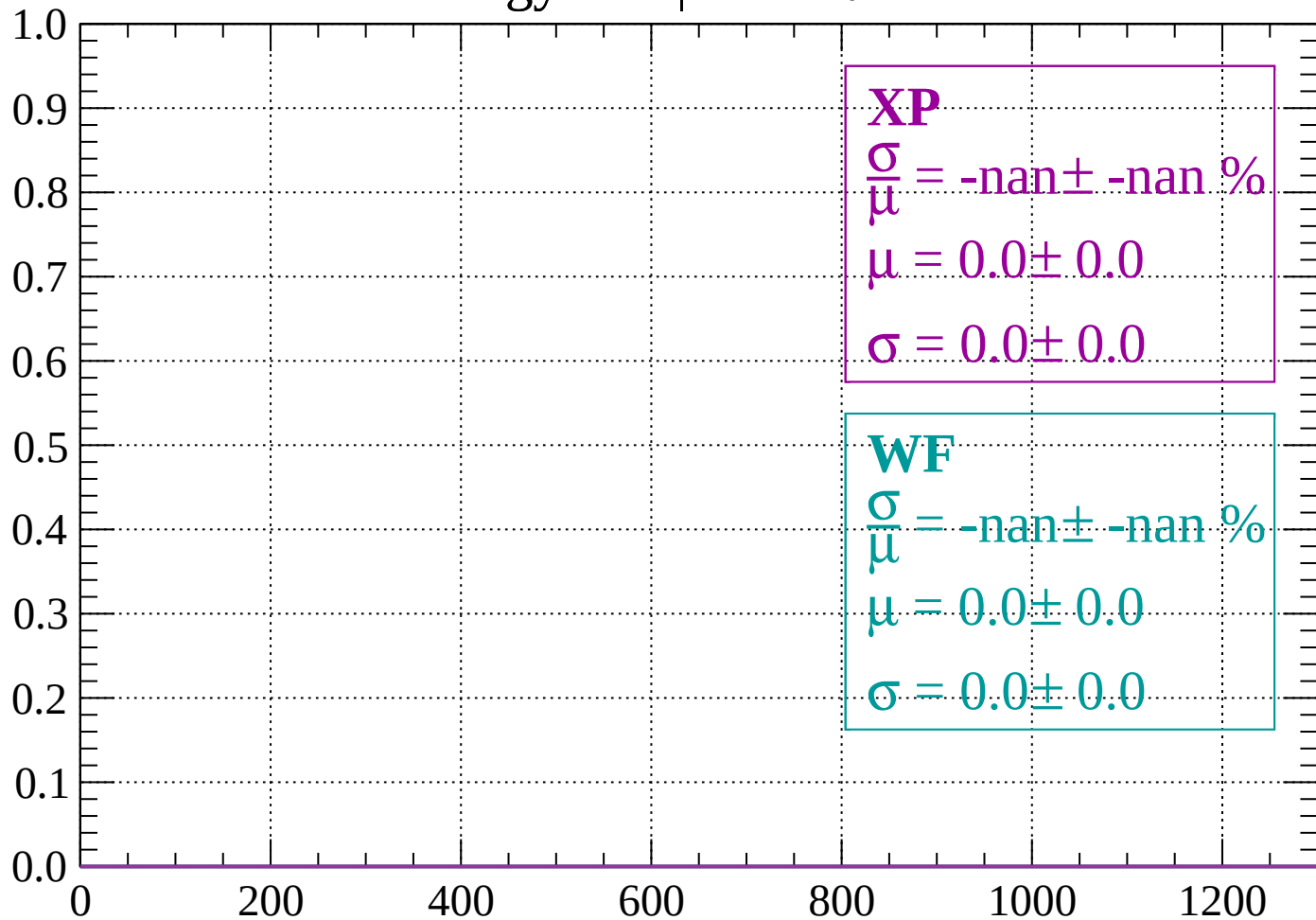
Count



dE/dx (ADC counts/cm)

Energy loss | $-44 < \theta < -42$

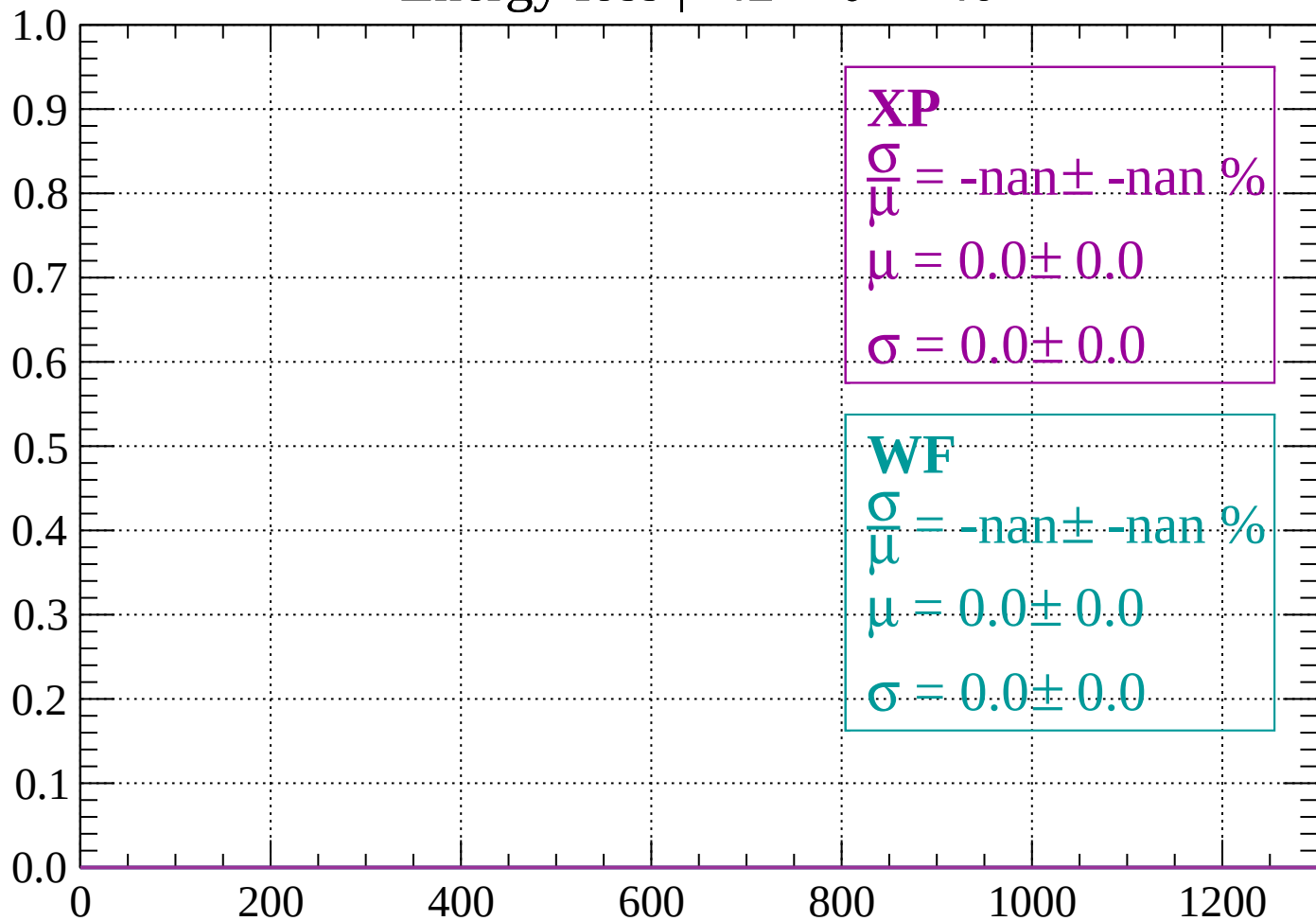
Count



dE/dx (ADC counts/cm)

Energy loss | $-42 < \theta < -40$

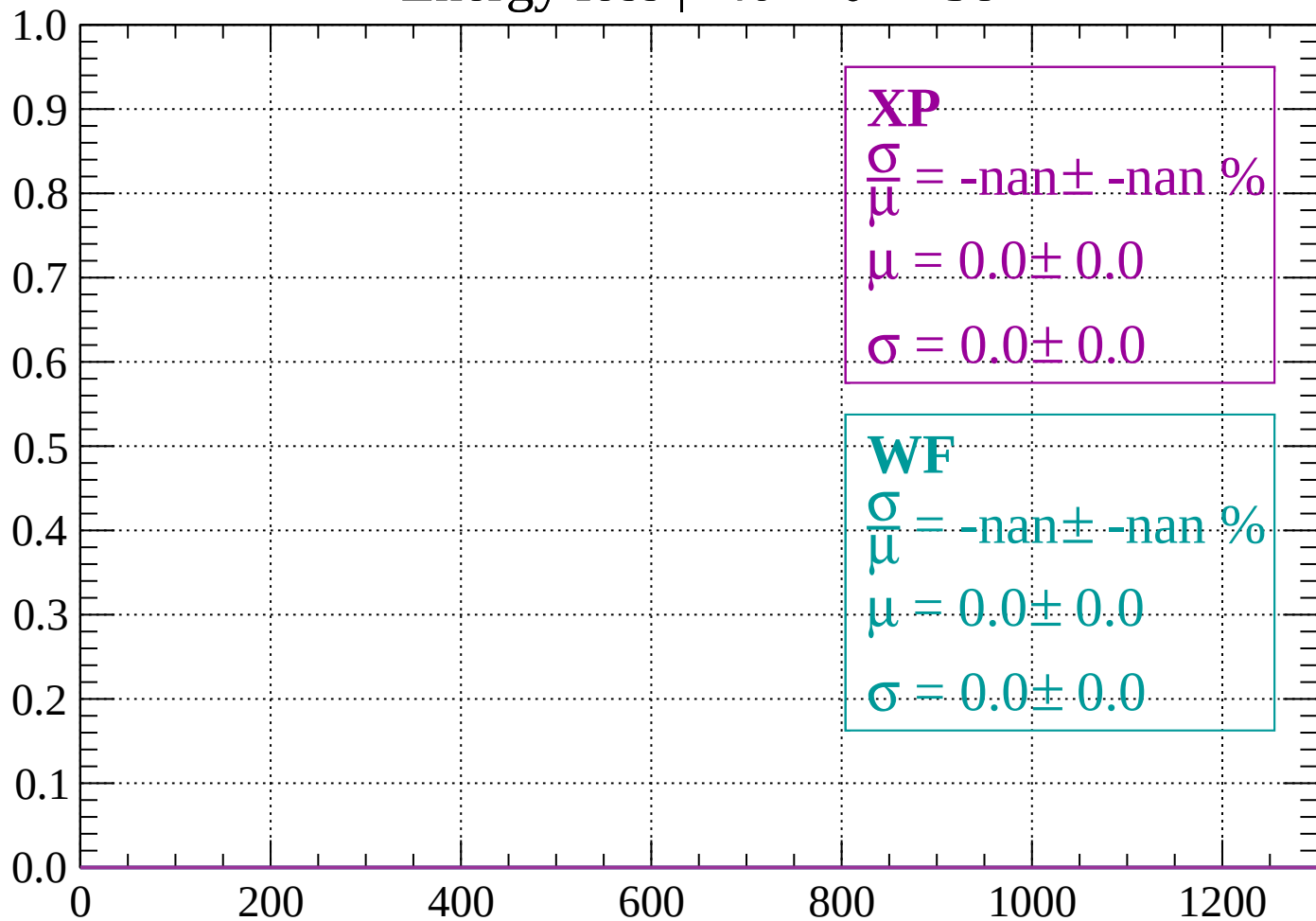
Count



dE/dx (ADC counts/cm)

Energy loss | $-40 < \theta < -38$

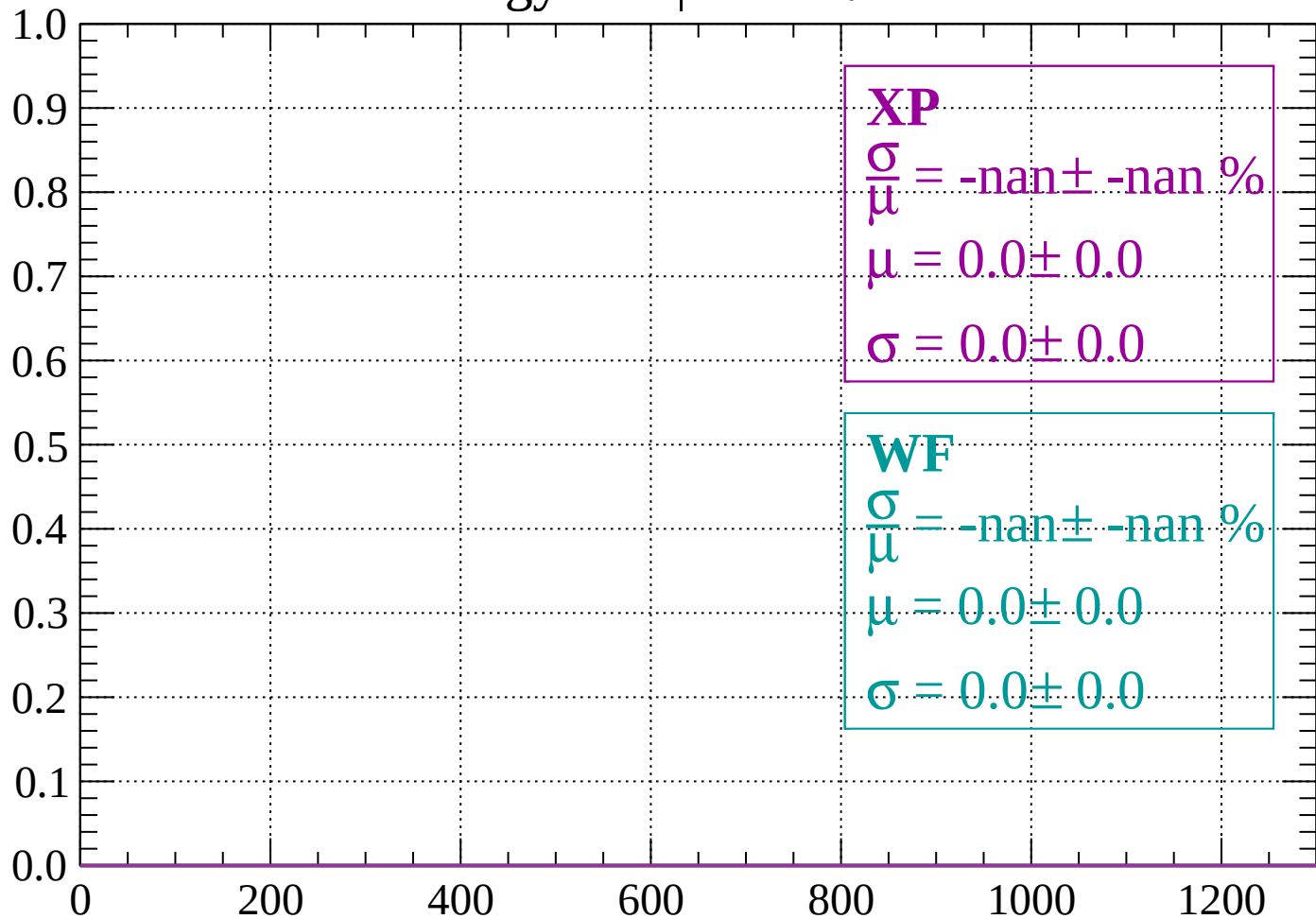
Count



dE/dx (ADC counts/cm)

Energy loss | $-38 < \theta < -36$

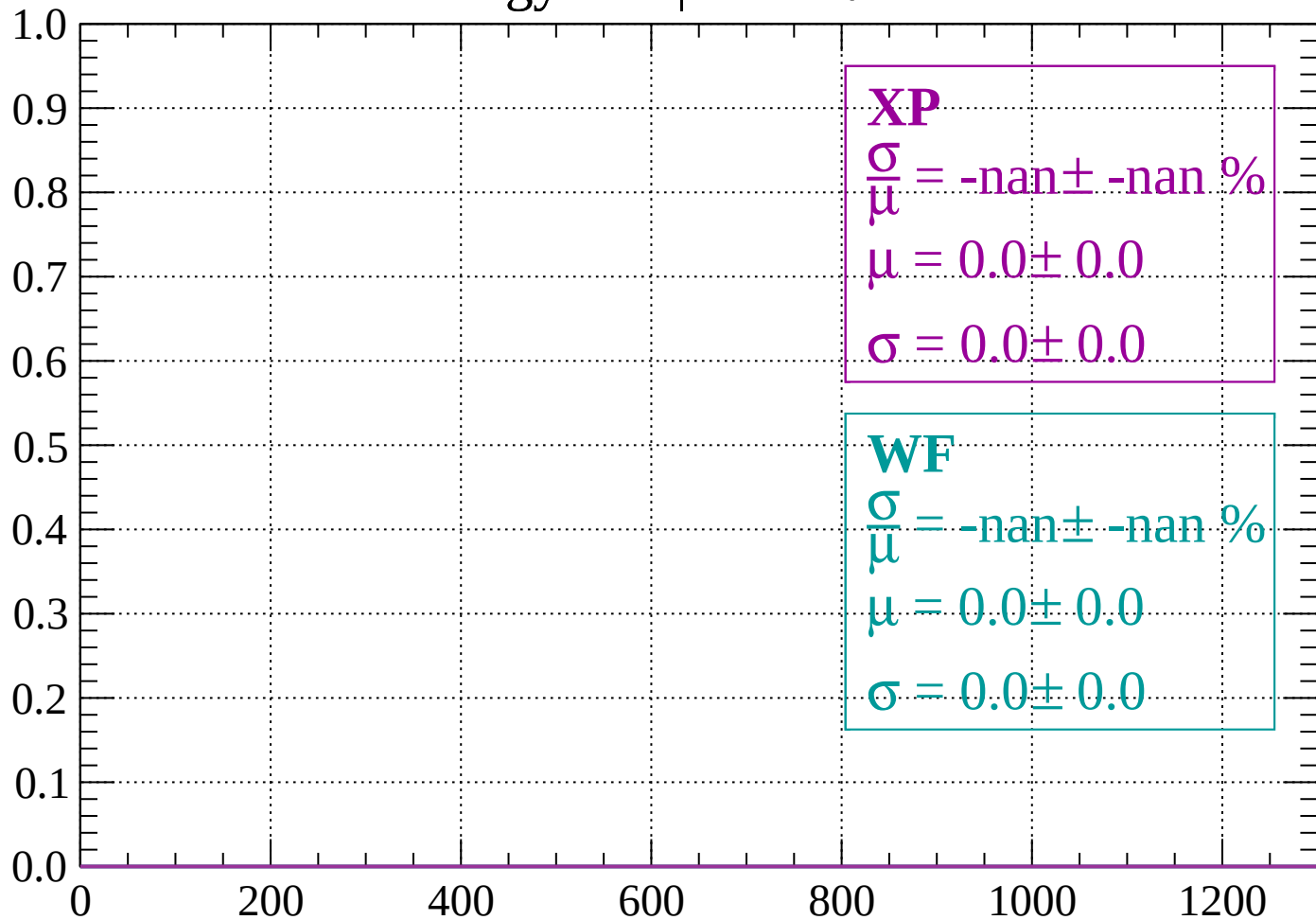
Count



dE/dx (ADC counts/cm)

Energy loss | $-36 < \theta < -34$

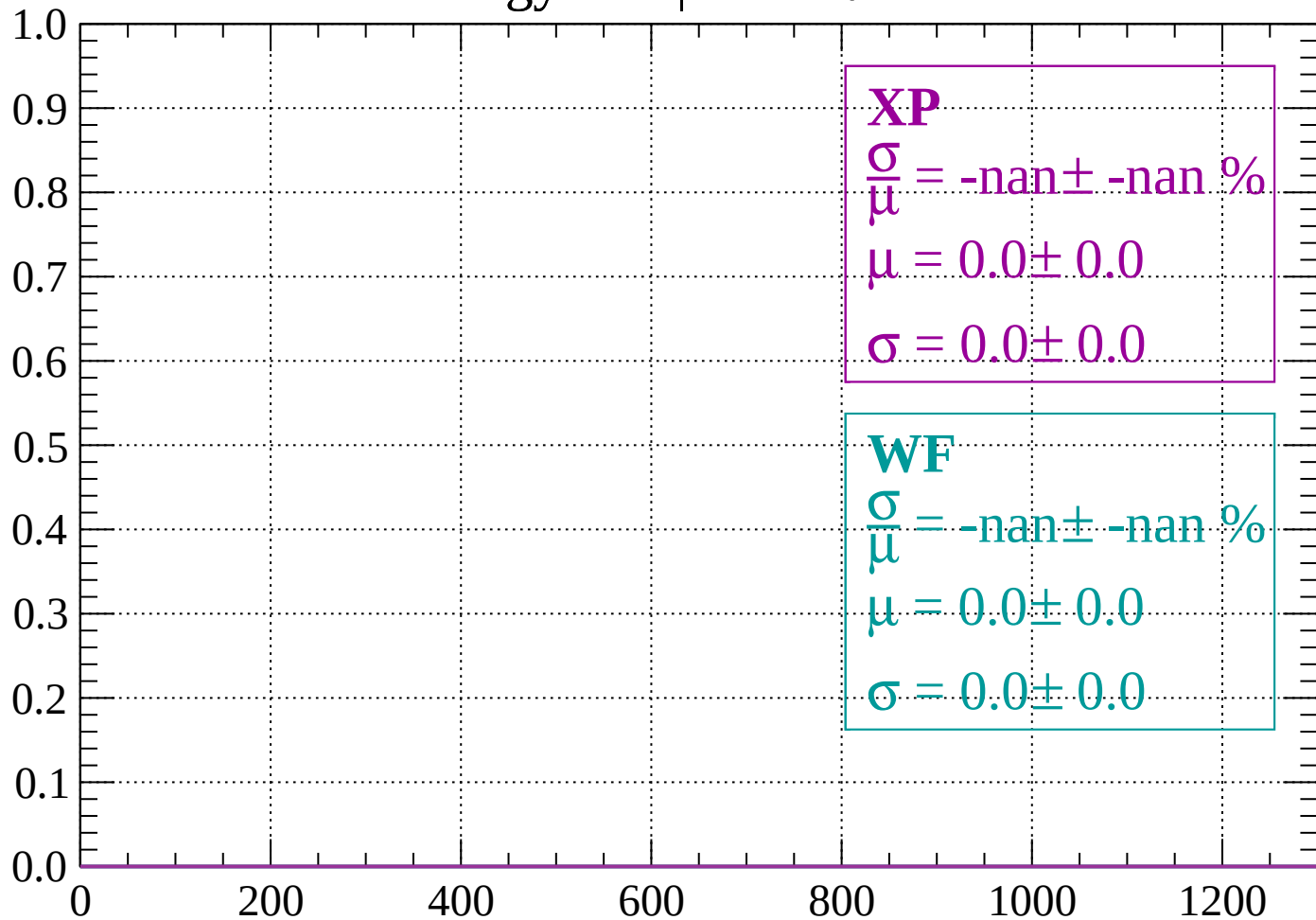
Count



dE/dx (ADC counts/cm)

Energy loss | $-34 < \theta < -32$

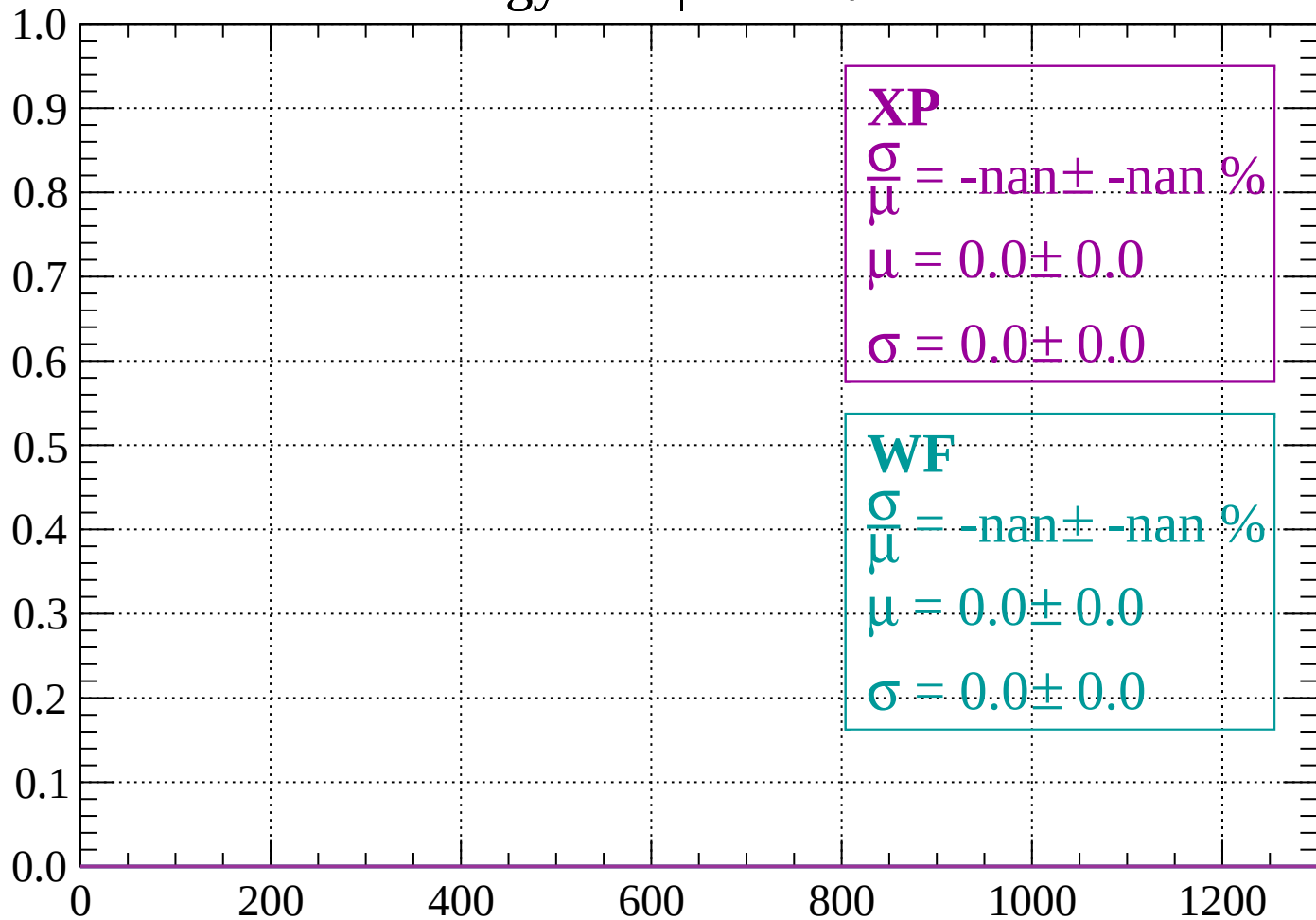
Count



dE/dx (ADC counts/cm)

Energy loss | $-32 < \theta < -30$

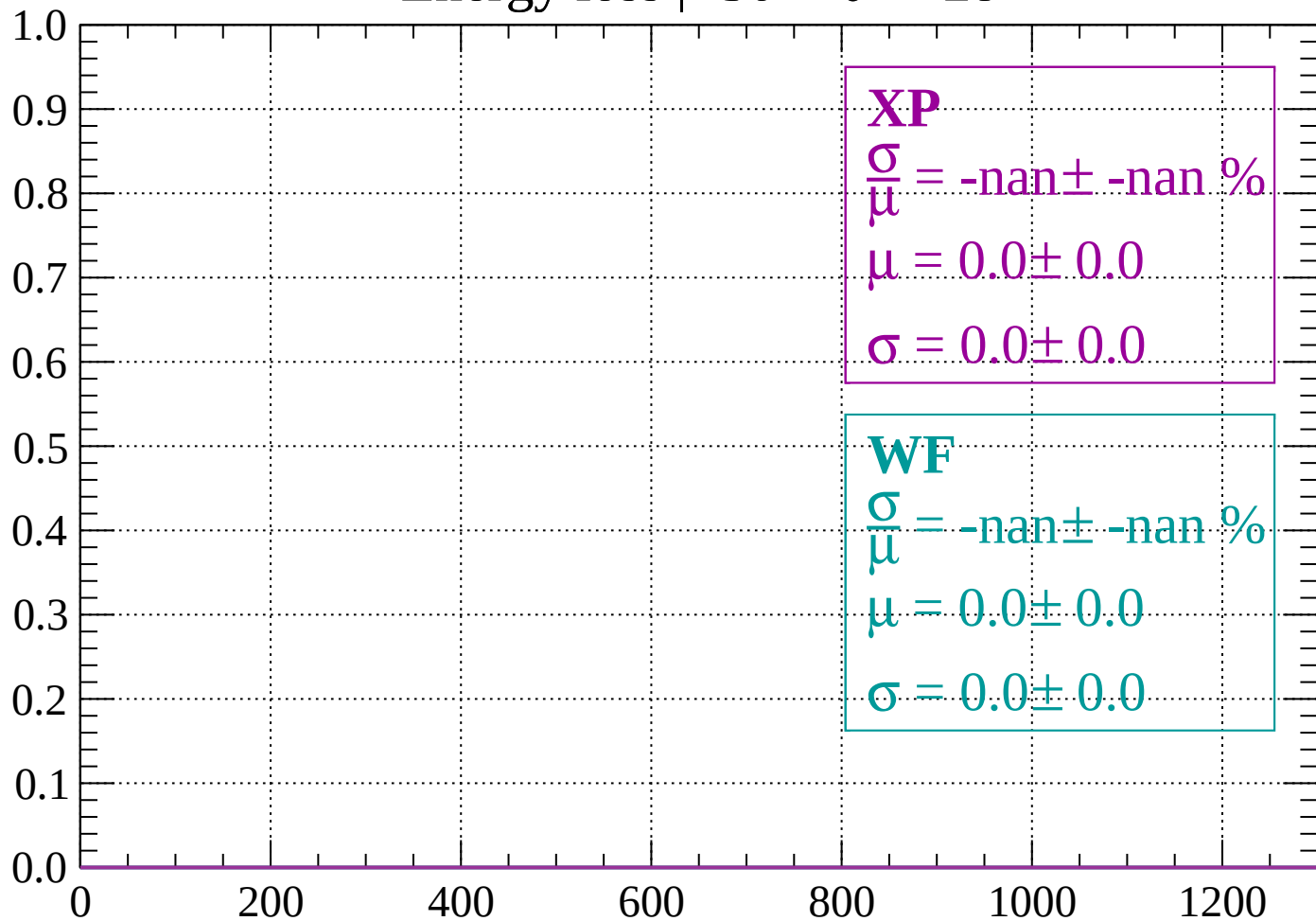
Count



dE/dx (ADC counts/cm)

Energy loss | $-30 < \theta < -28$

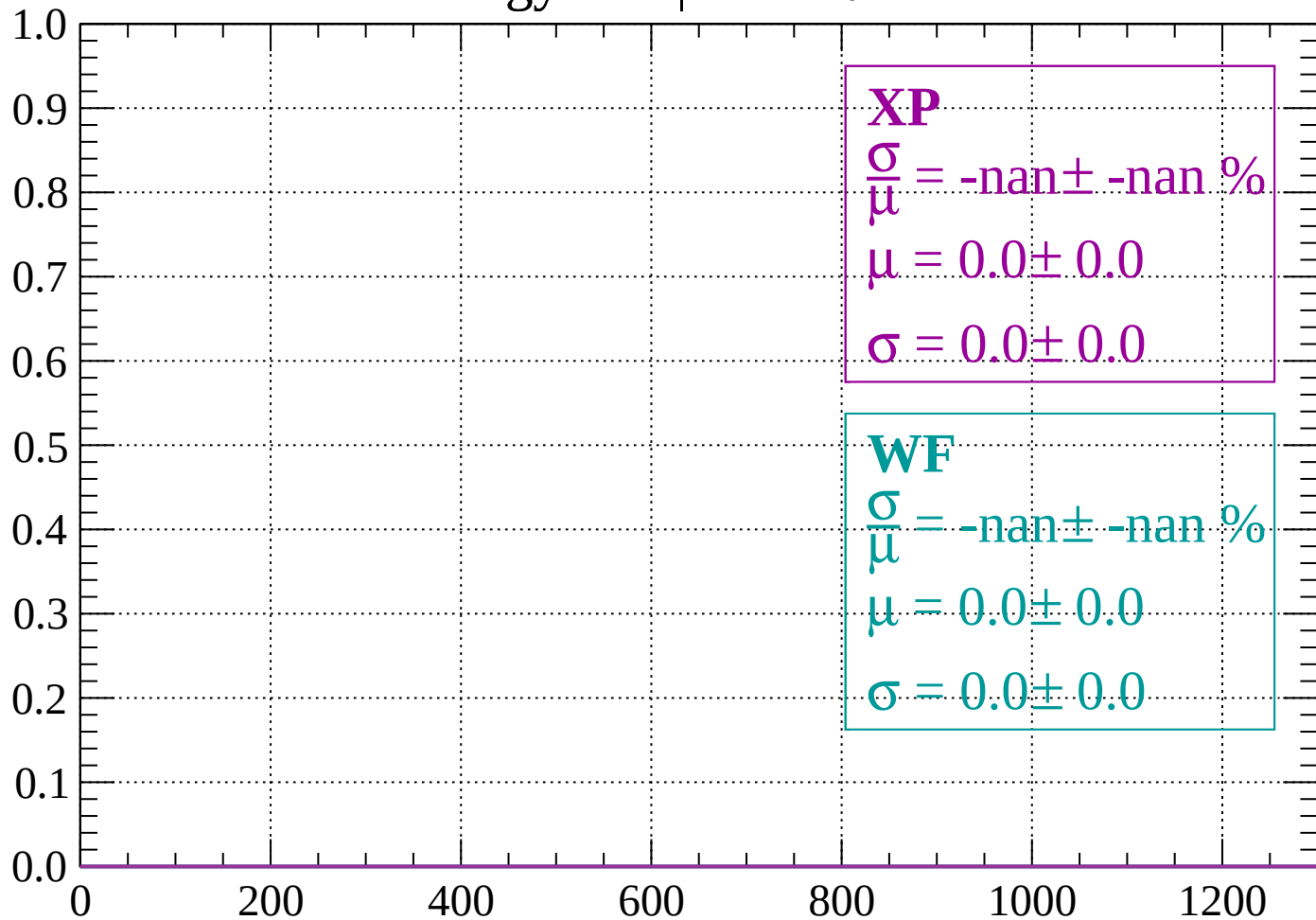
Count



dE/dx (ADC counts/cm)

Energy loss | $-28 < \theta < -26$

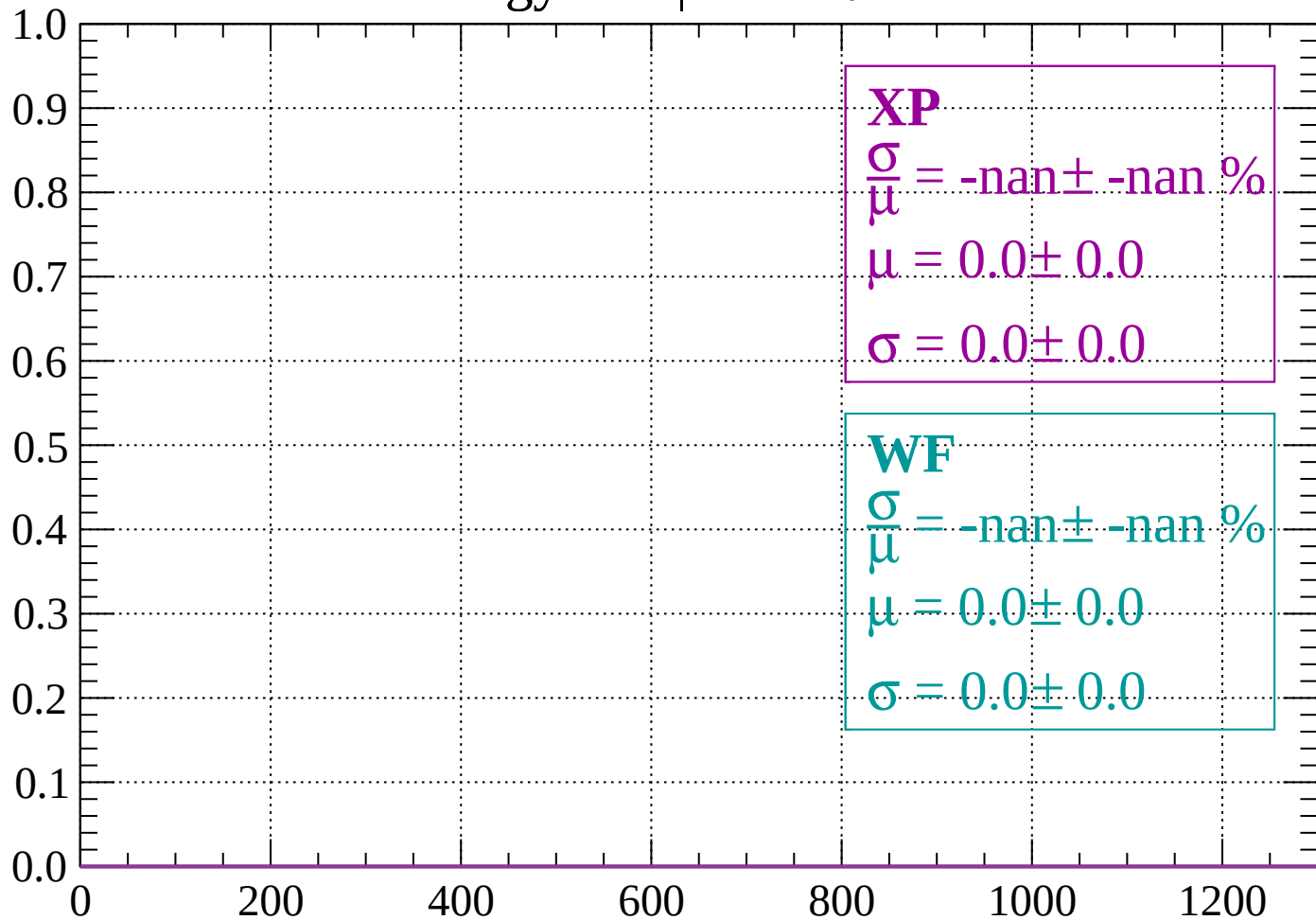
Count



dE/dx (ADC counts/cm)

Energy loss | $-26 < \theta < -24$

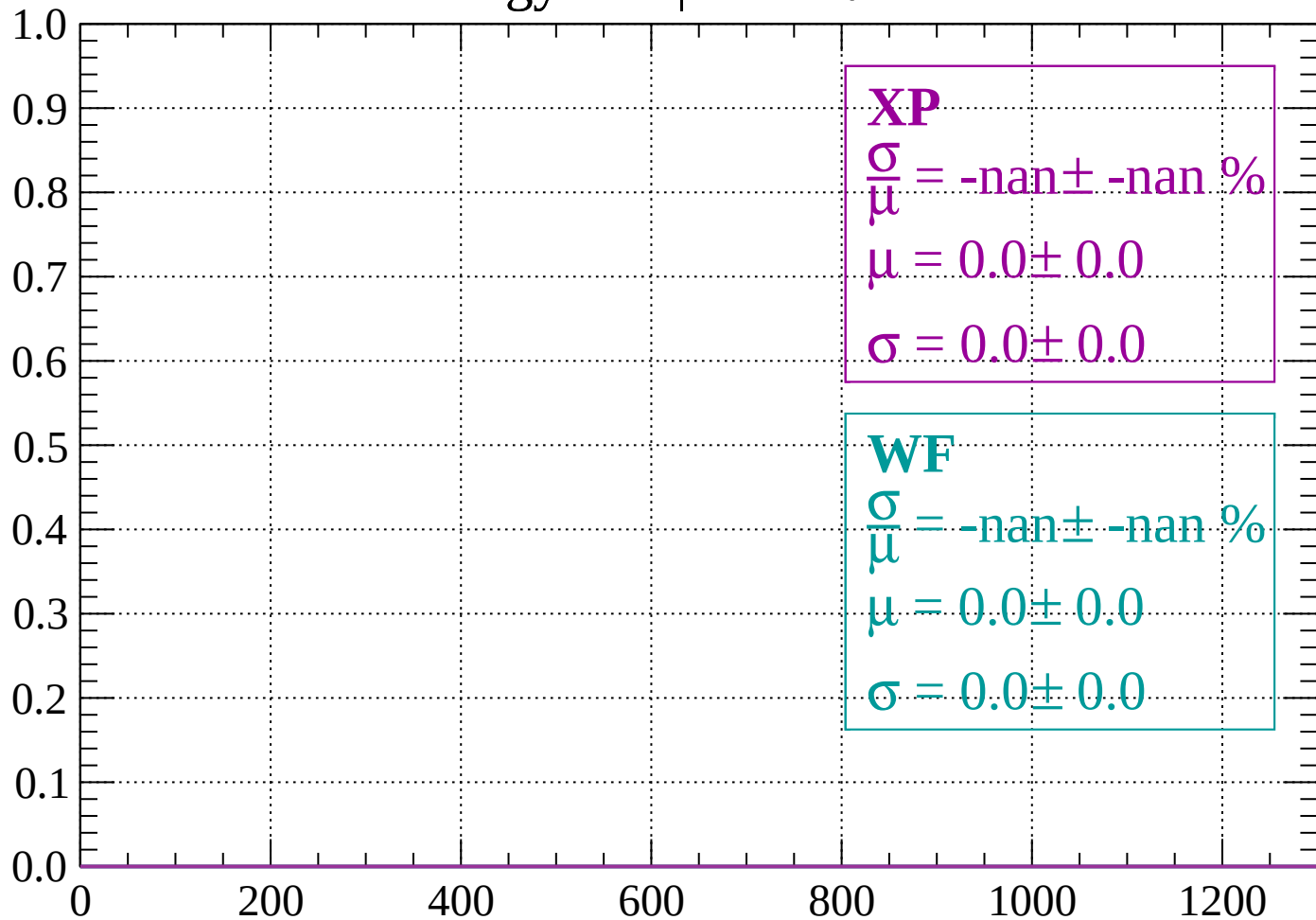
Count



dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$

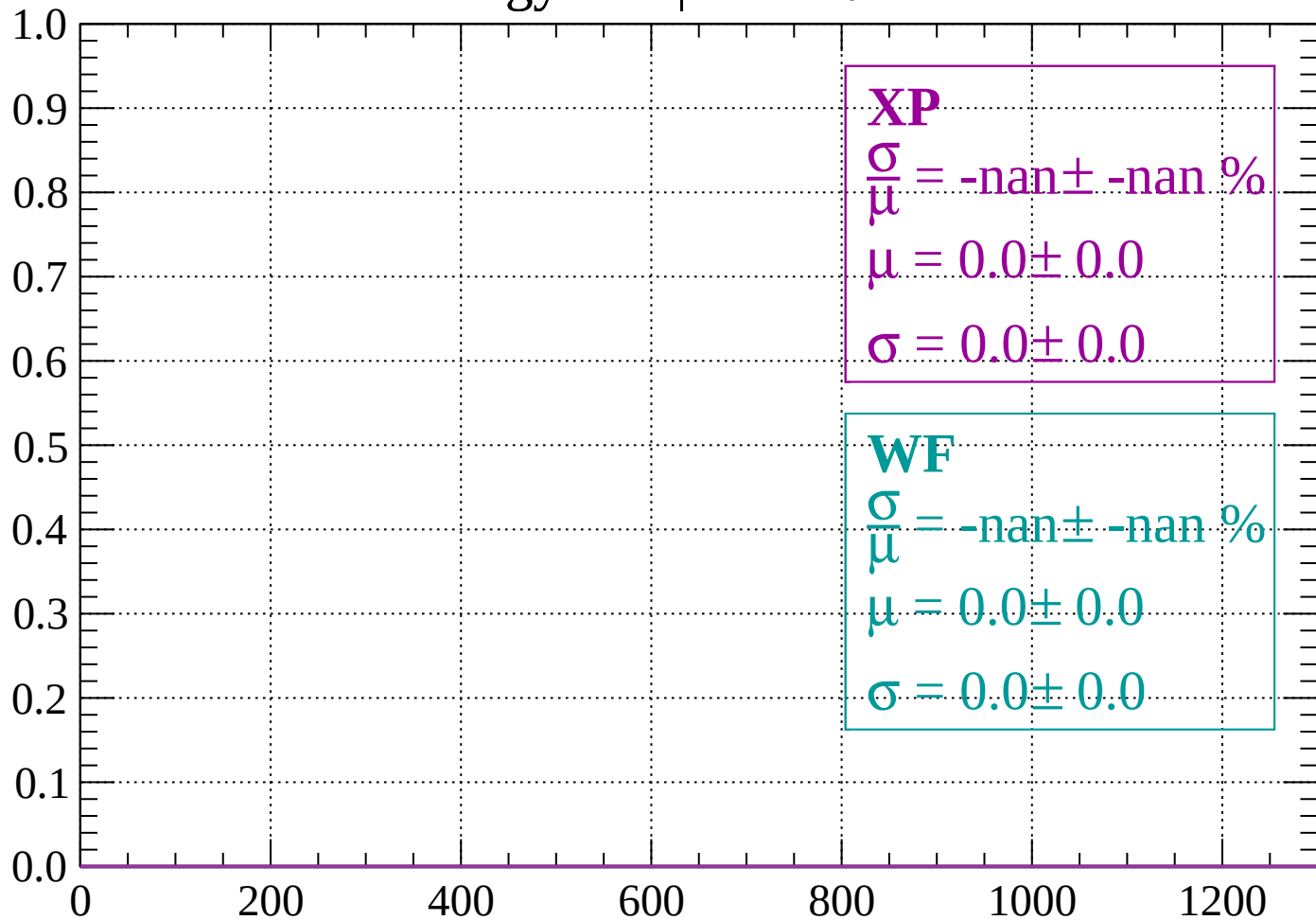
Count



dE/dx (ADC counts/cm)

Energy loss | $-22 < \theta < -20$

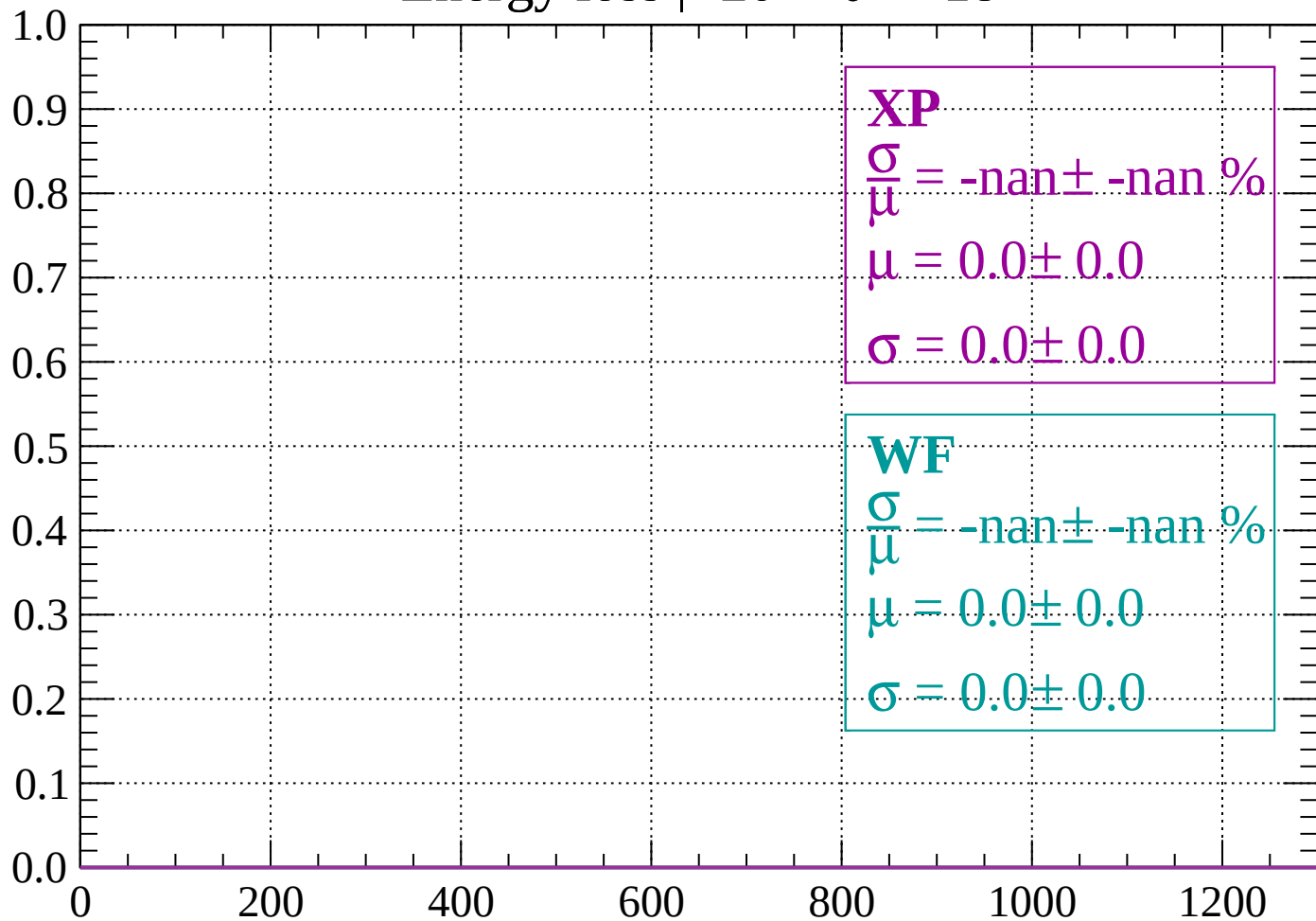
Count



dE/dx (ADC counts/cm)

Energy loss | $-20 < \theta < -18$

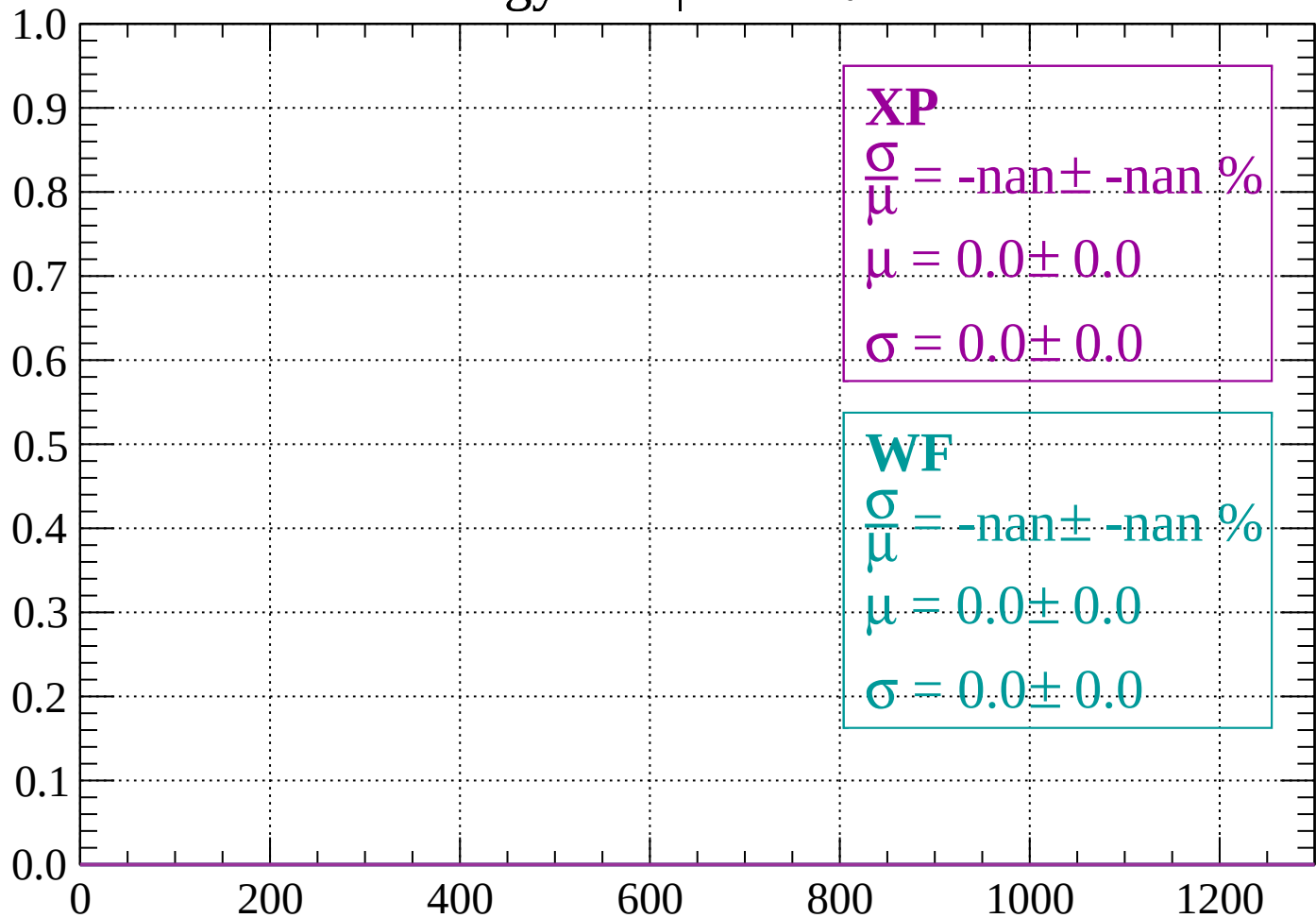
Count



dE/dx (ADC counts/cm)

Energy loss | $-18 < \theta < -16$

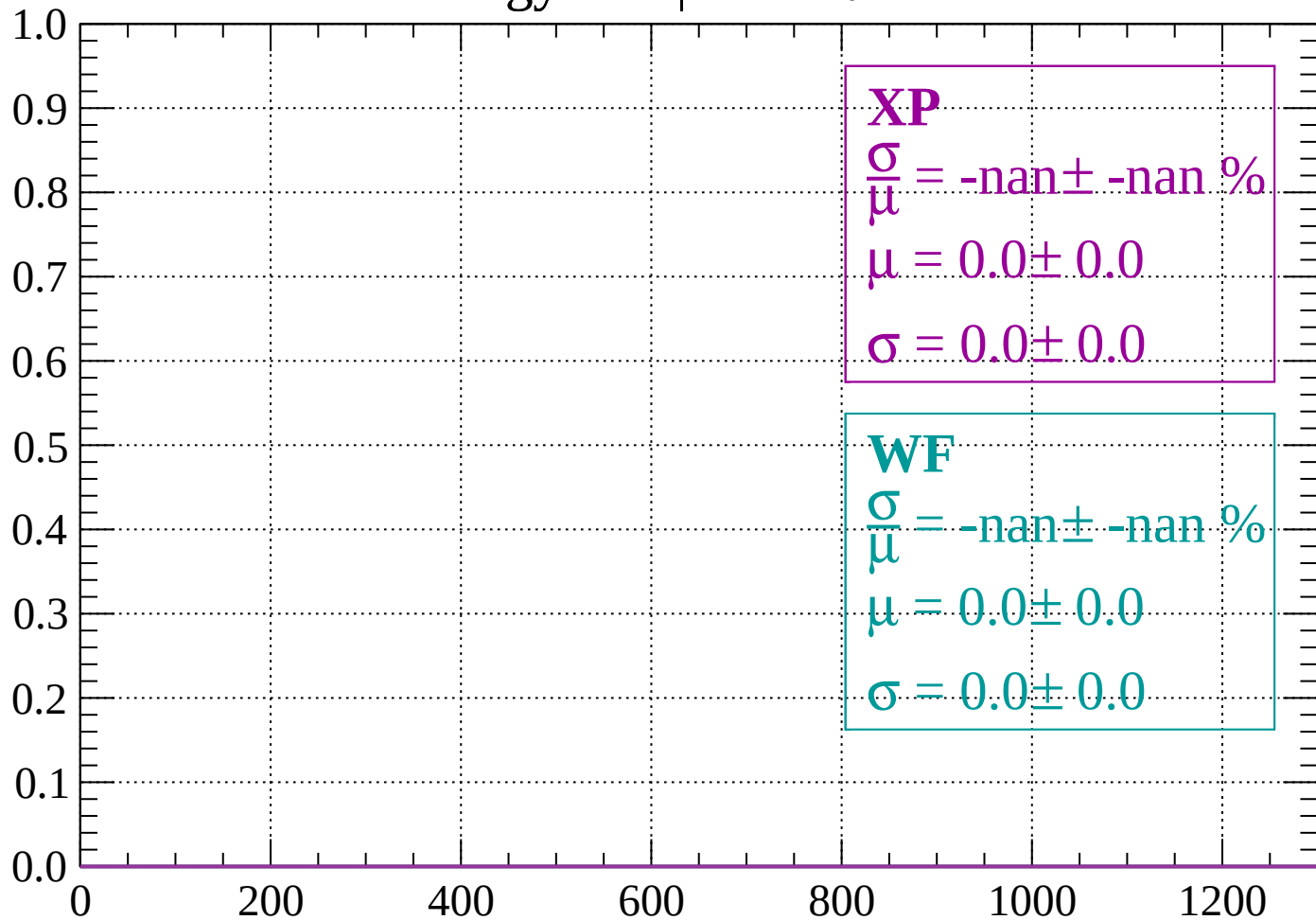
Count



dE/dx (ADC counts/cm)

Energy loss | $-16 < \theta < -14$

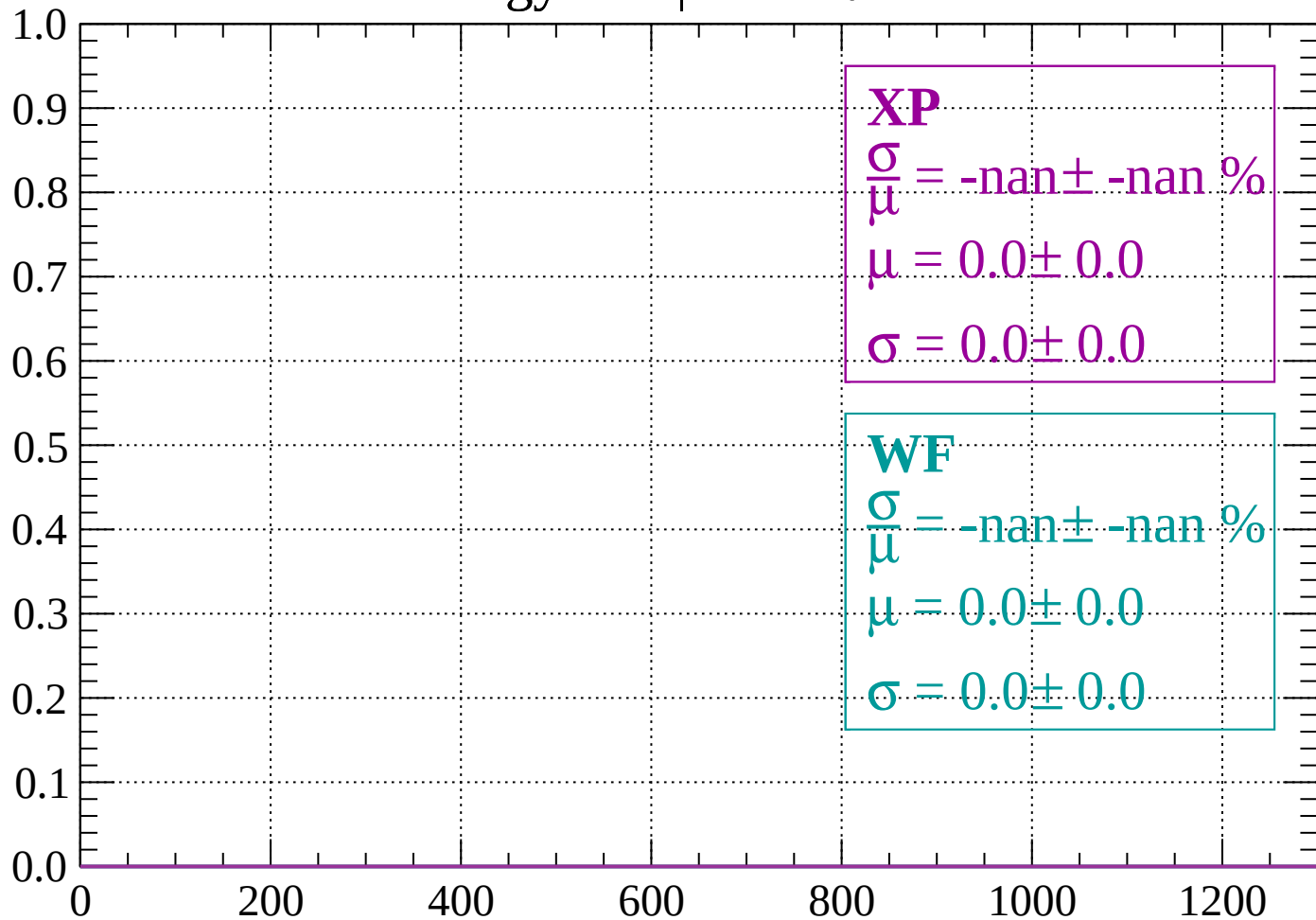
Count



dE/dx (ADC counts/cm)

Energy loss | $-14 < \theta < -12$

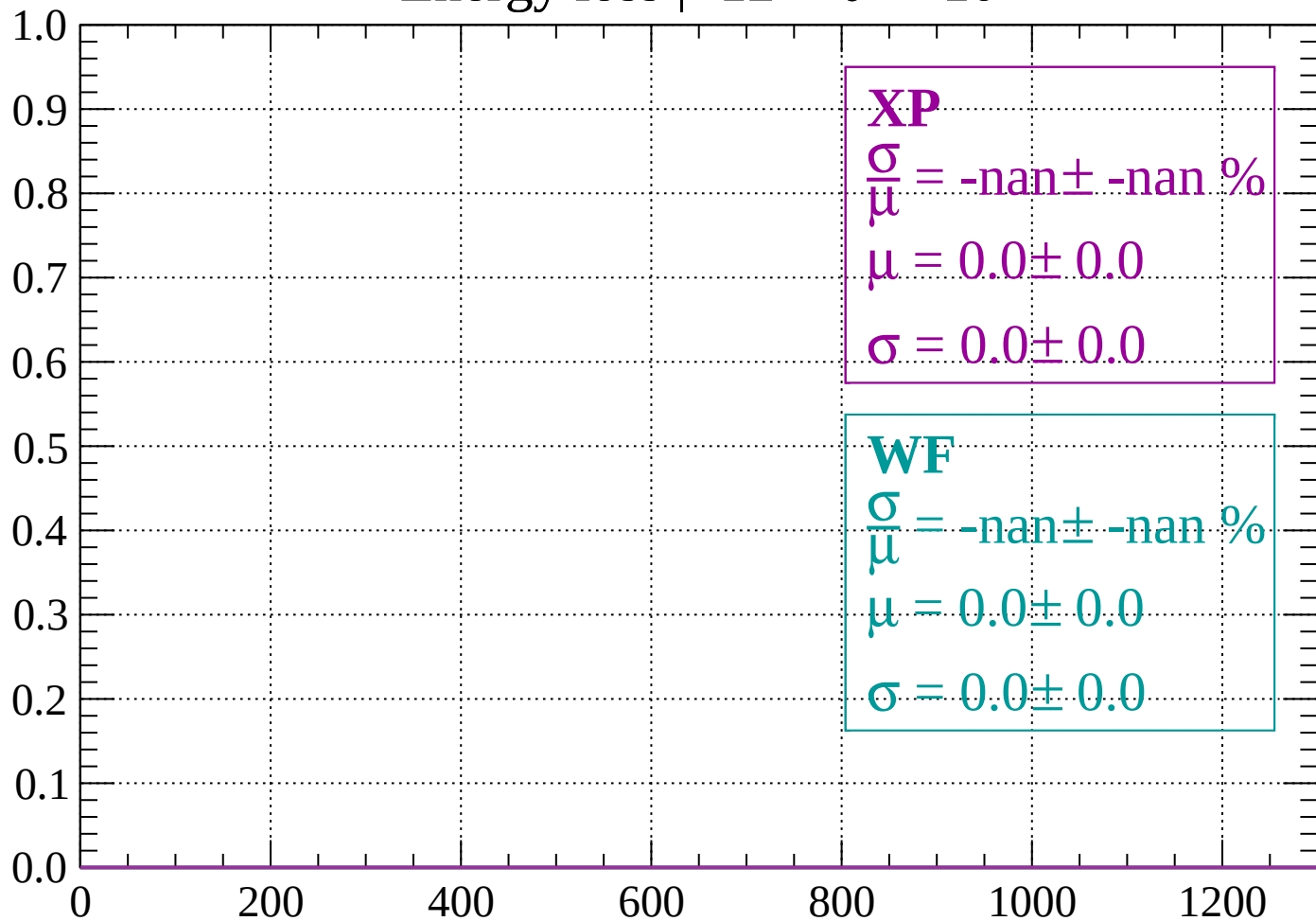
Count



dE/dx (ADC counts/cm)

Energy loss | $-12 < \theta < -10$

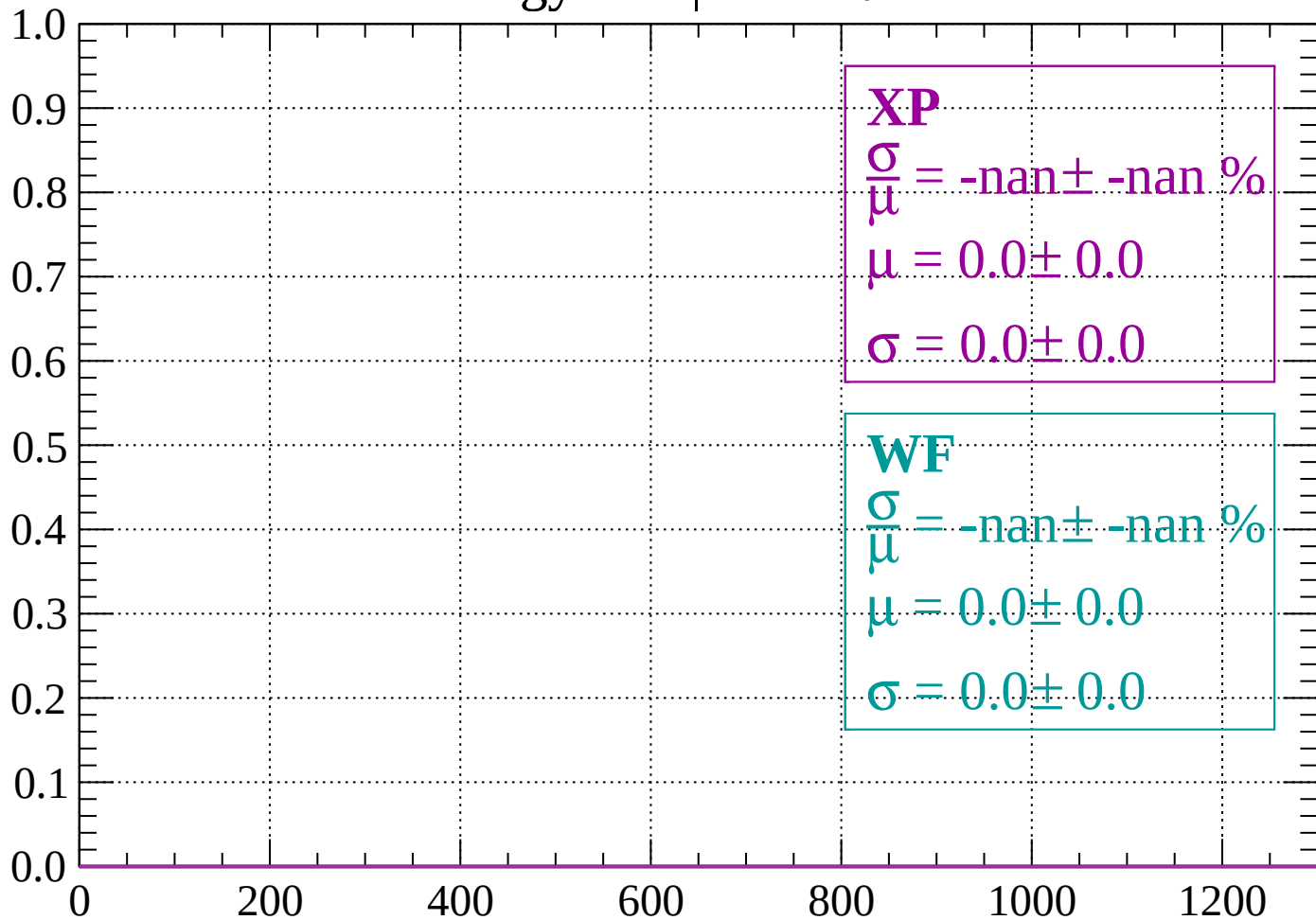
Count



dE/dx (ADC counts/cm)

Energy loss | $-10 < \theta < -8$

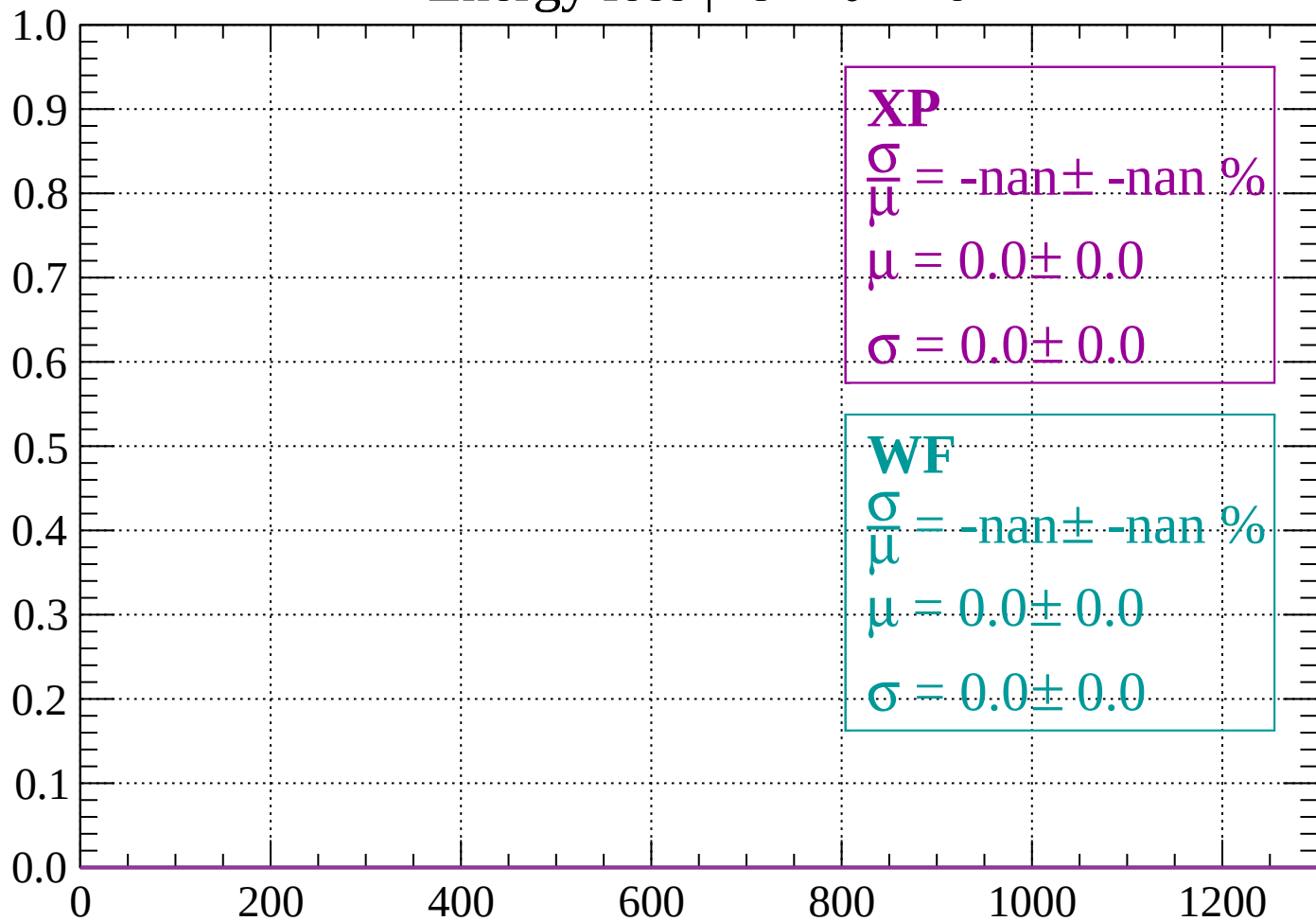
Count



dE/dx (ADC counts/cm)

Energy loss | $-8 < \theta < -6$

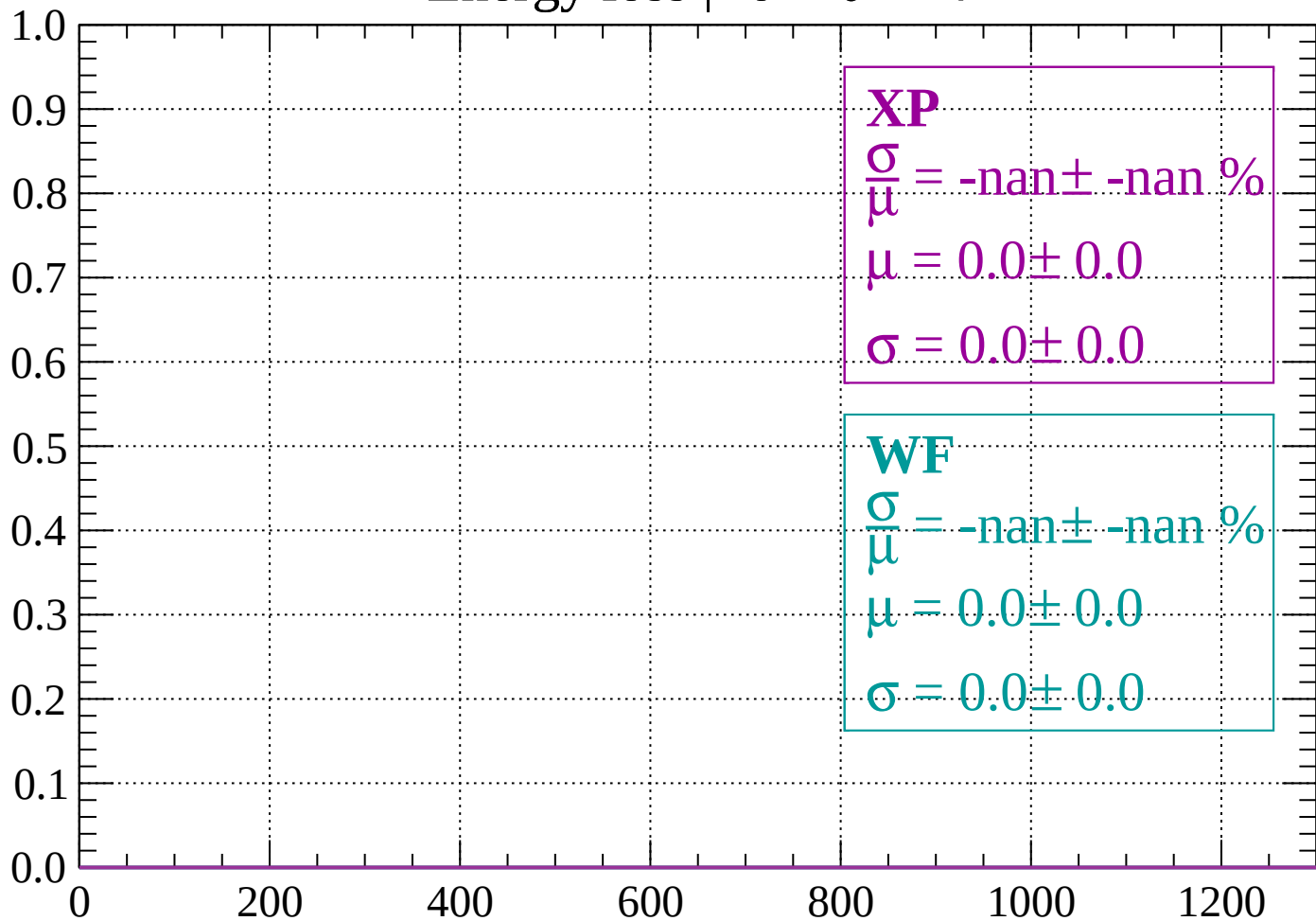
Count



dE/dx (ADC counts/cm)

Energy loss | $-6 < \theta < -4$

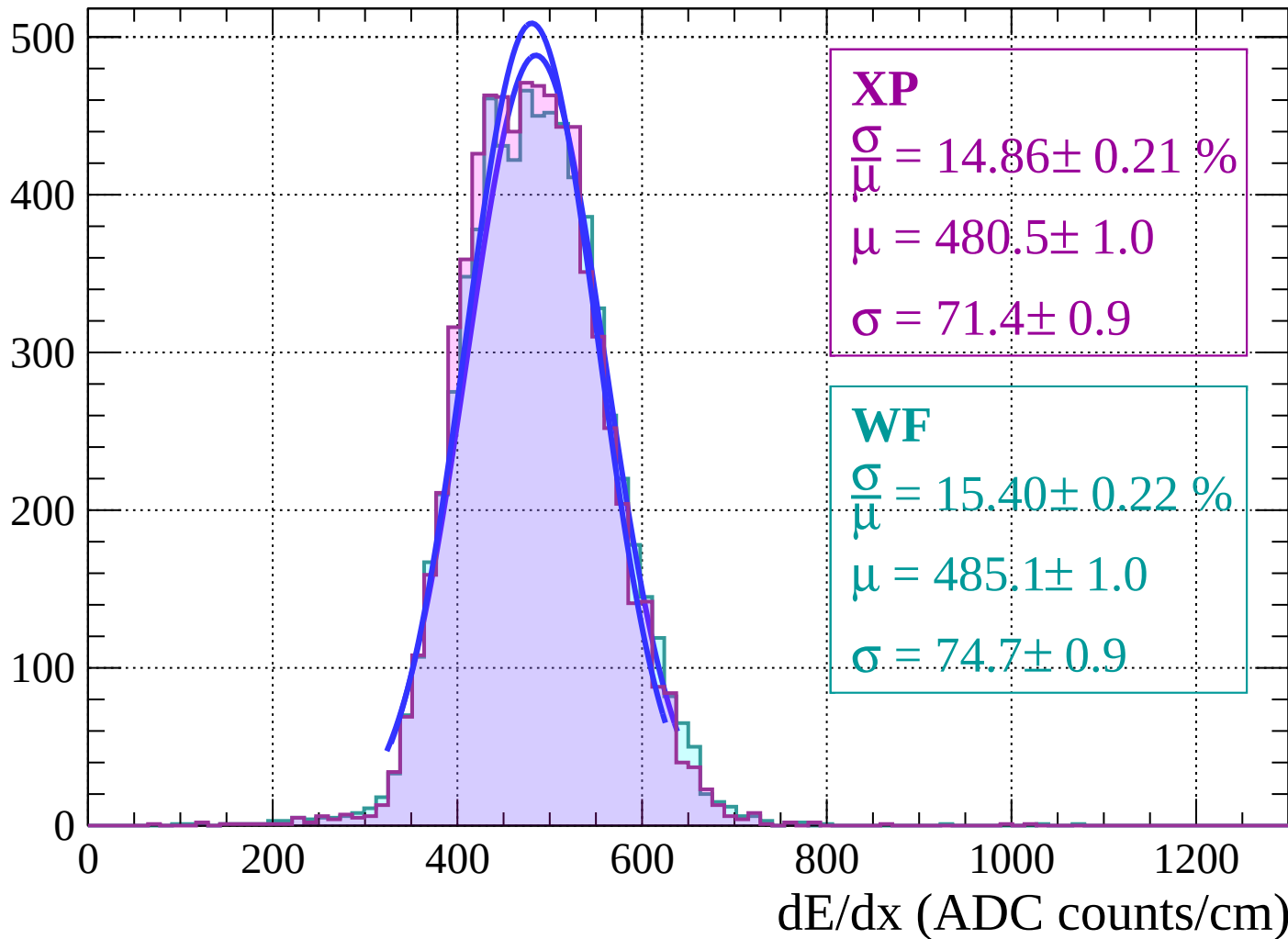
Count



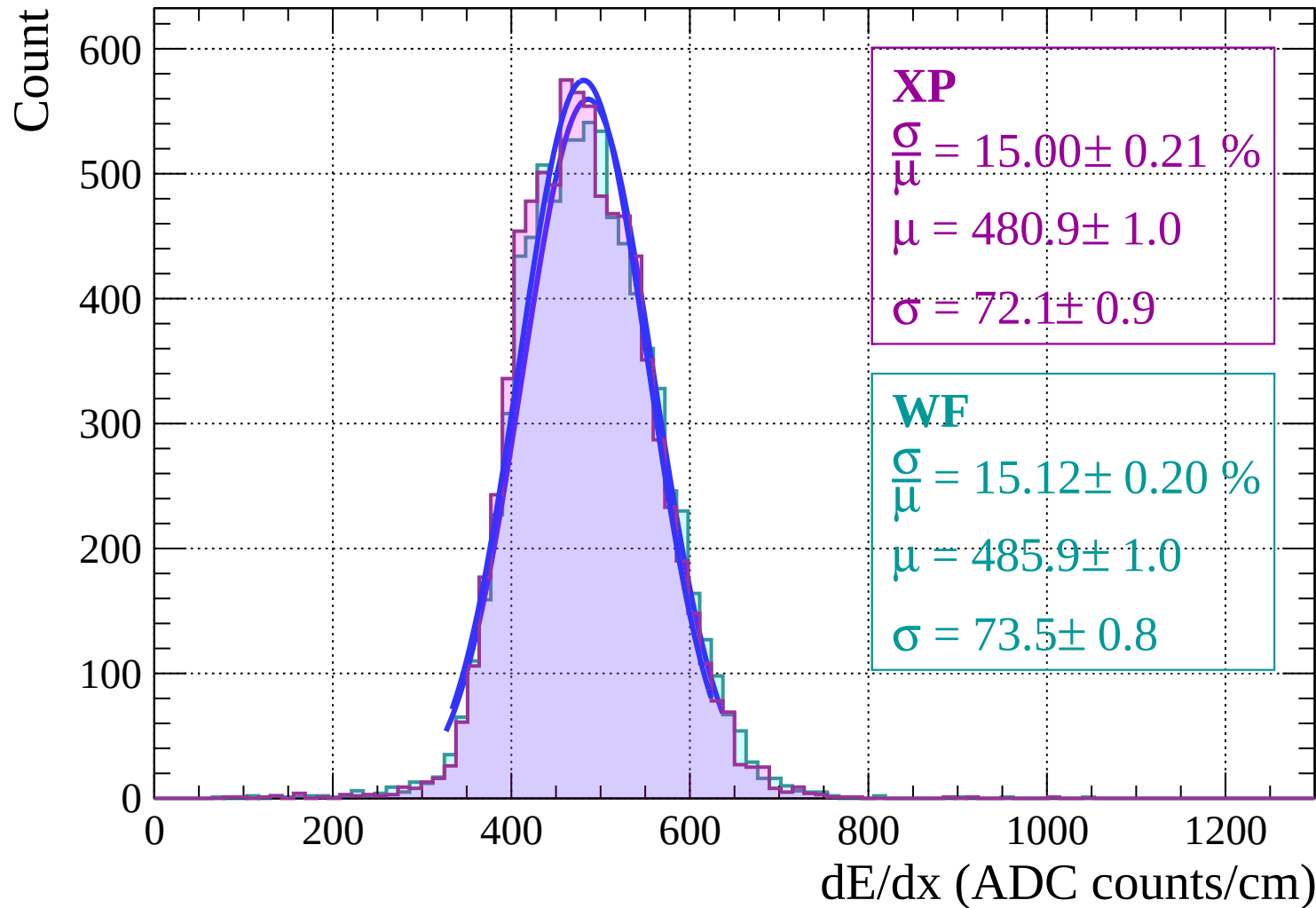
dE/dx (ADC counts/cm)

Energy loss | $-4 < \theta < -2$

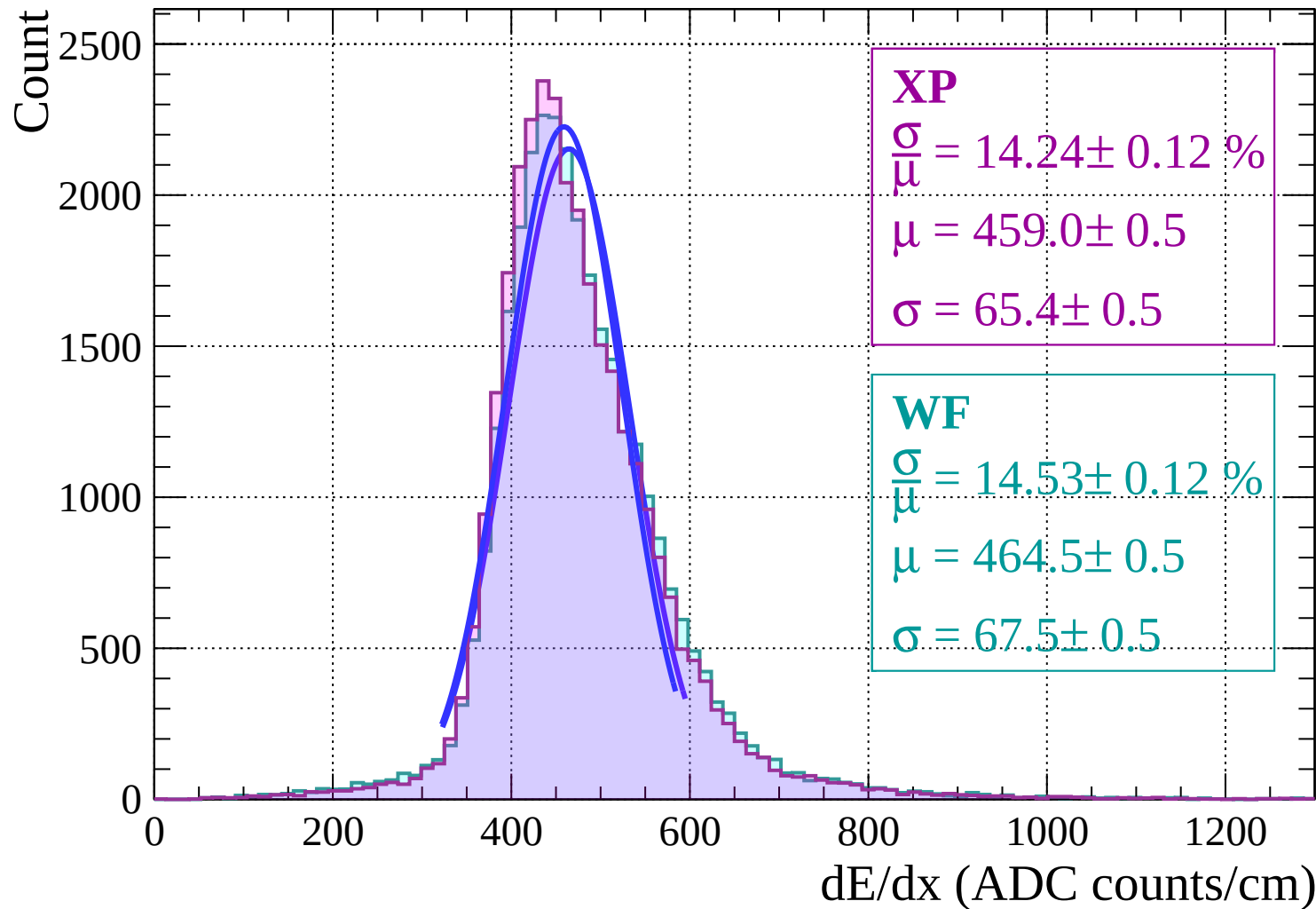
Count



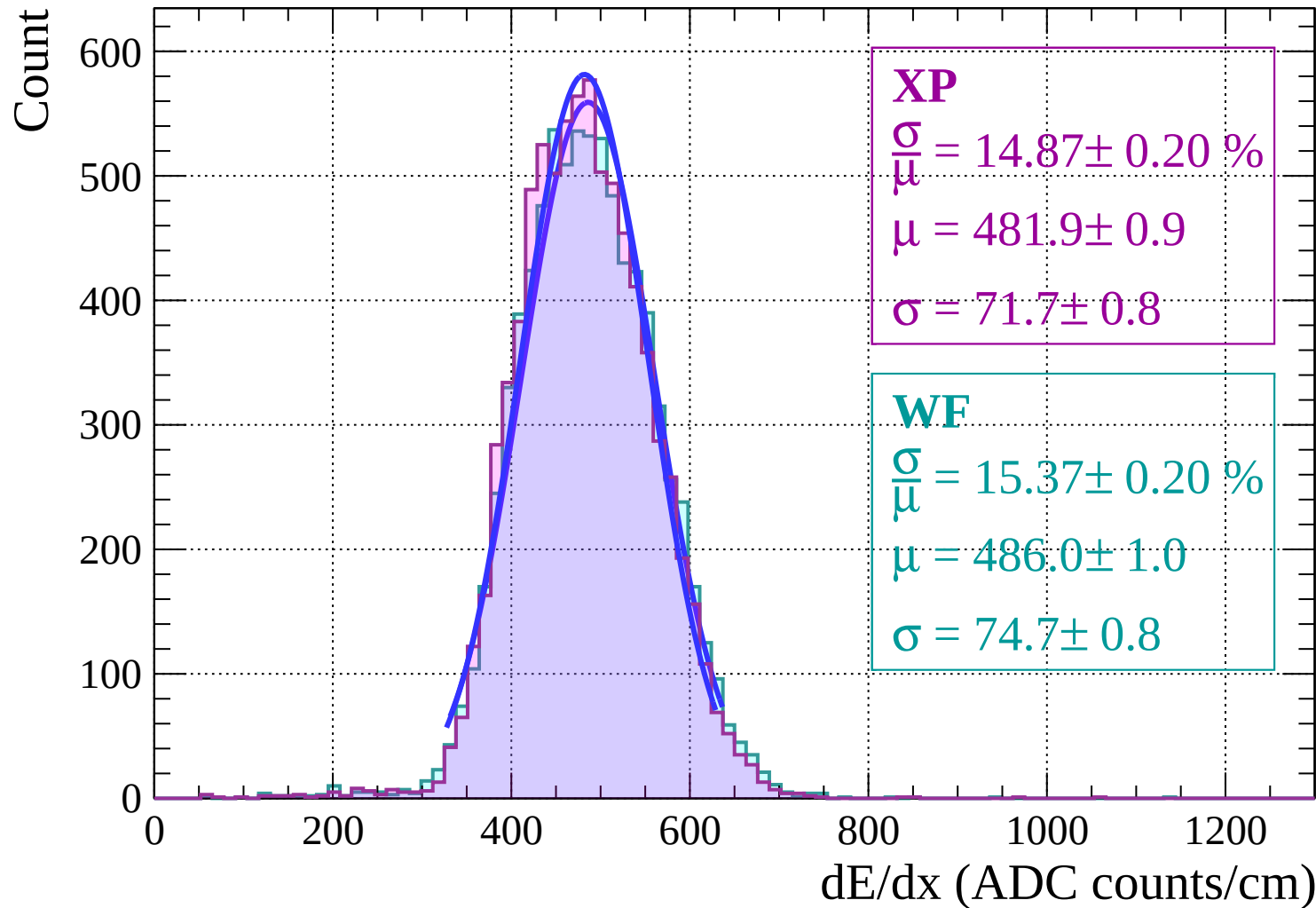
Energy loss | $-2 < \theta < 0$



Energy loss | $0 < \theta < 2$

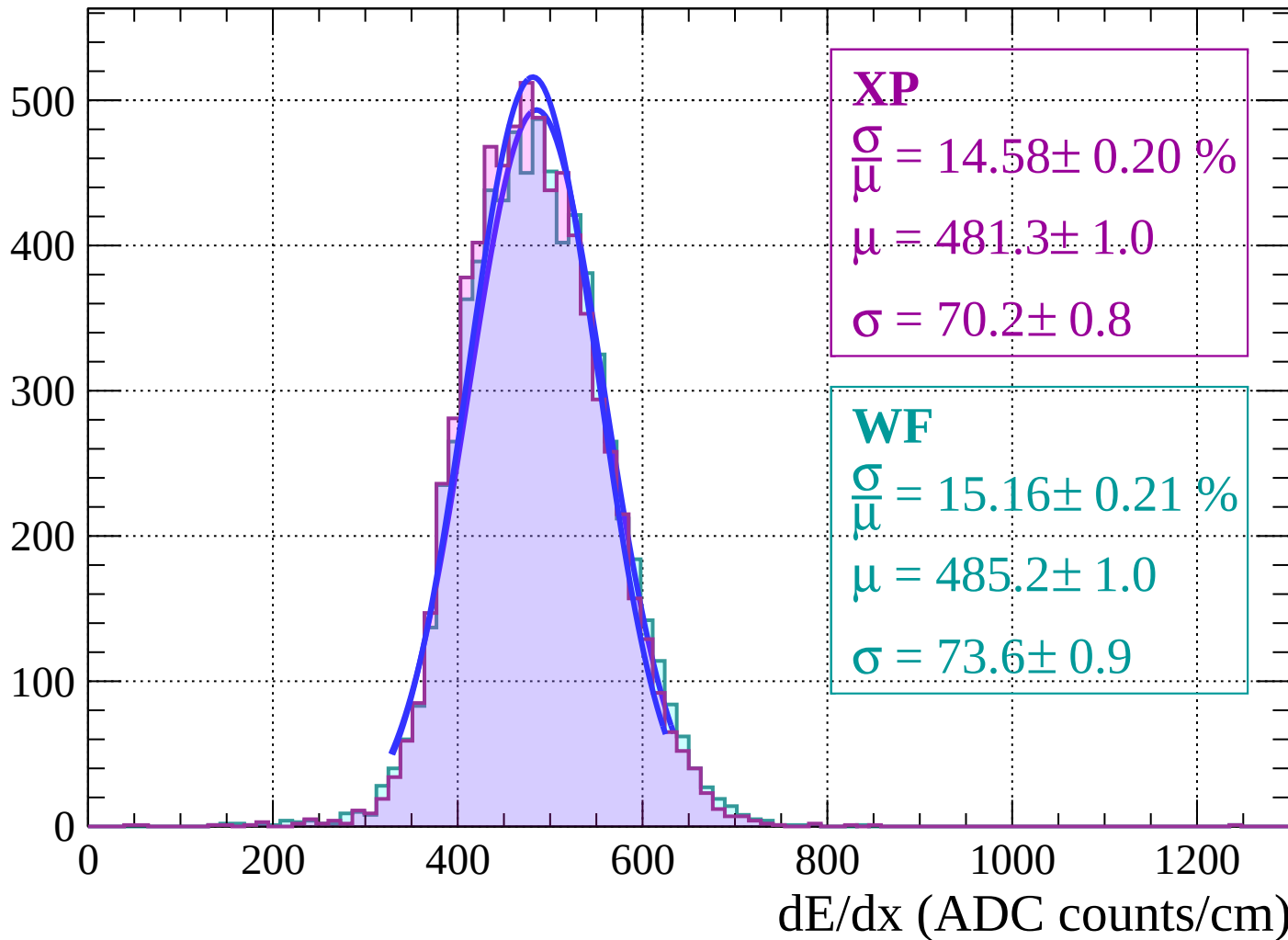


Energy loss | $2 < \theta < 4$



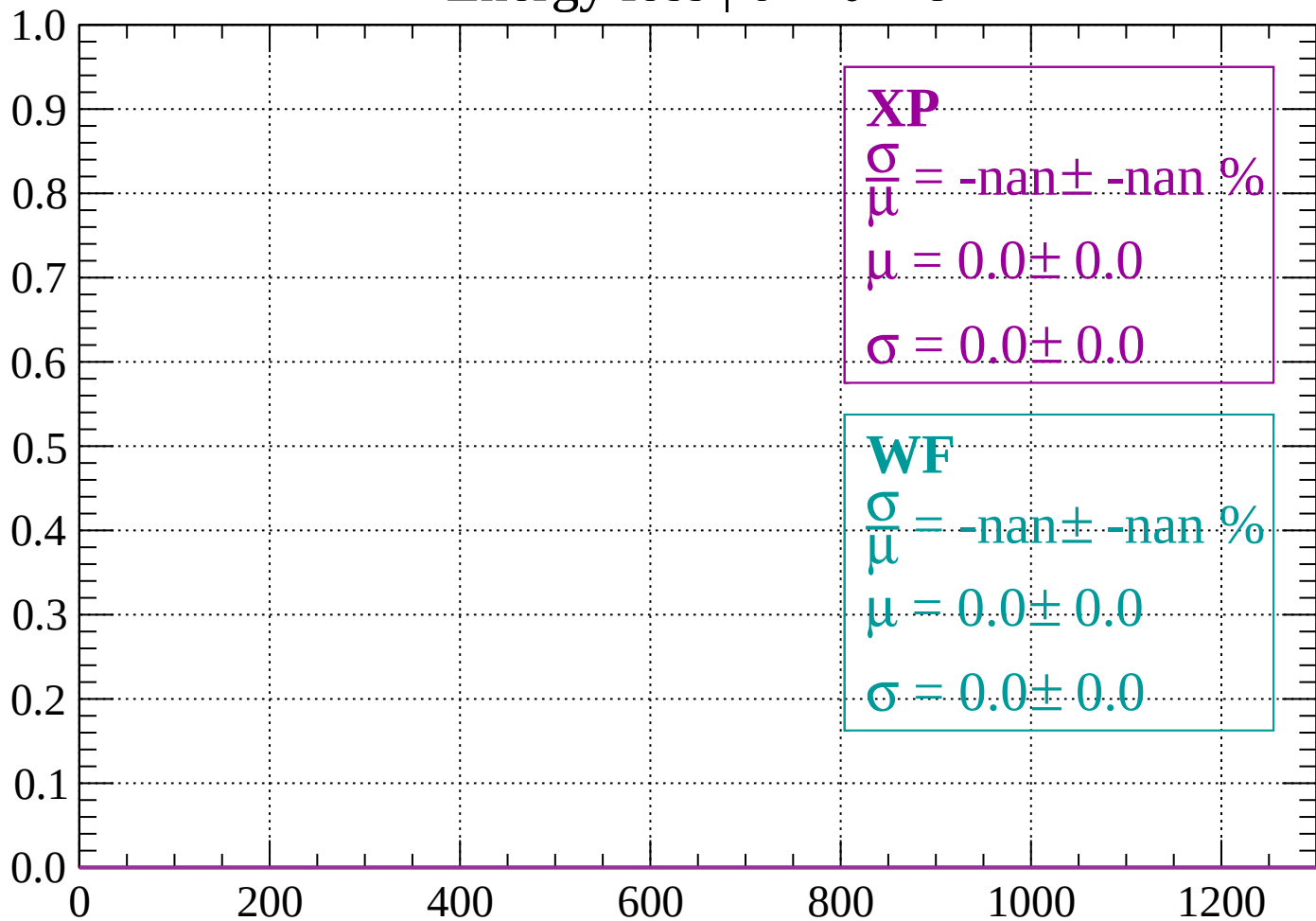
Energy loss | $4 < \theta < 6$

Count



Energy loss | $6 < \theta < 8$

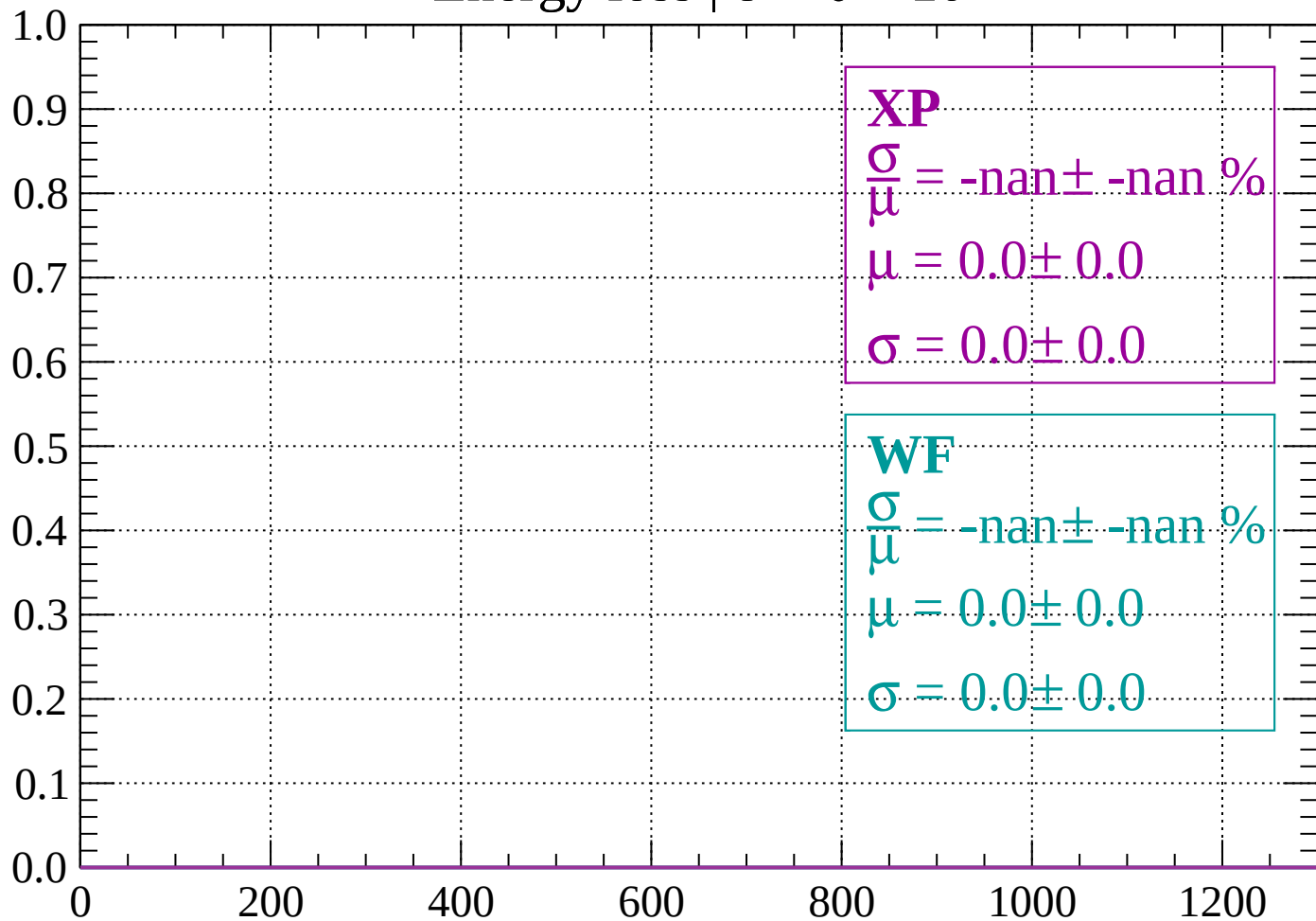
Count



dE/dx (ADC counts/cm)

Energy loss | $8 < \theta < 10$

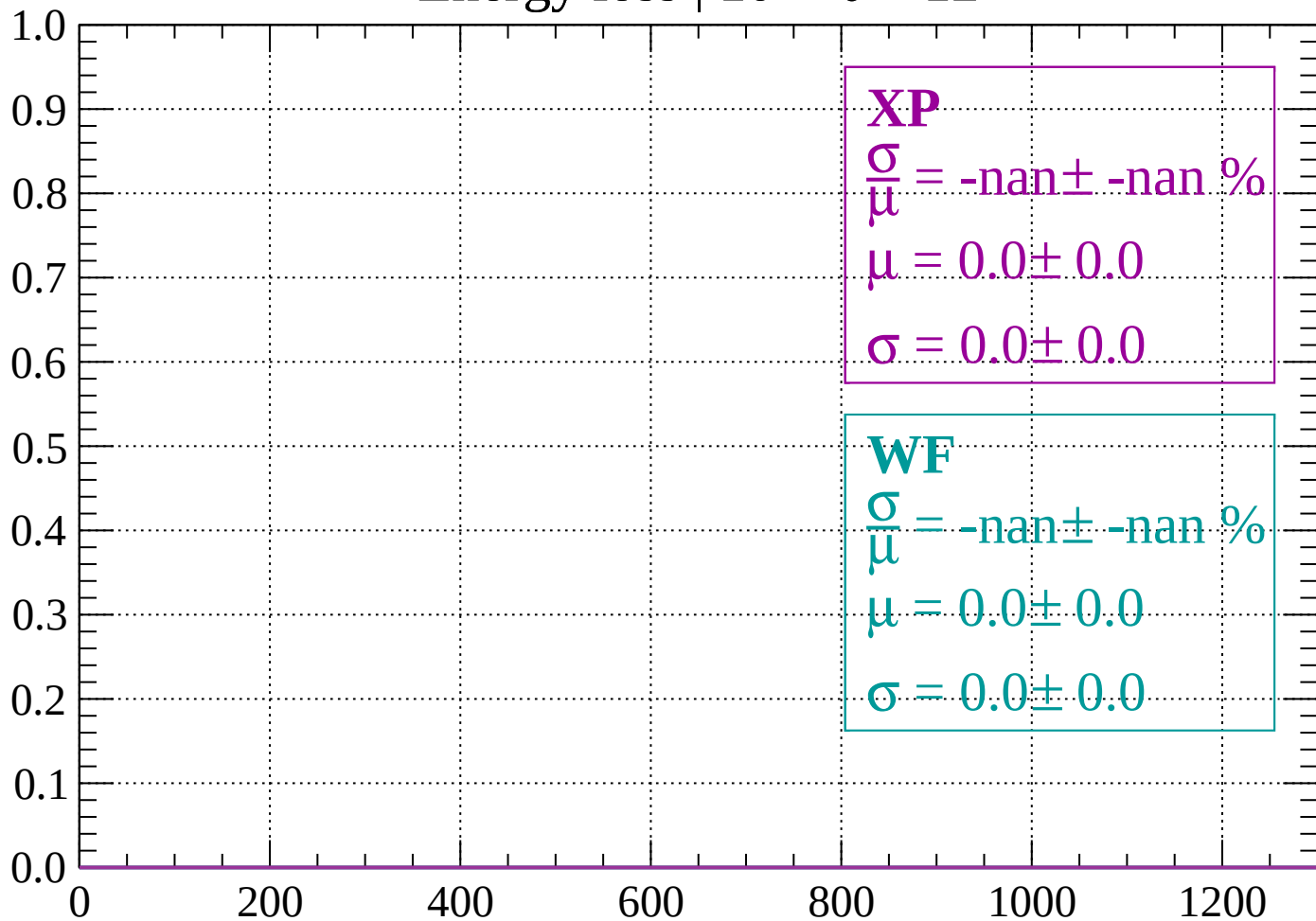
Count



dE/dx (ADC counts/cm)

Energy loss | $10 < \theta < 12$

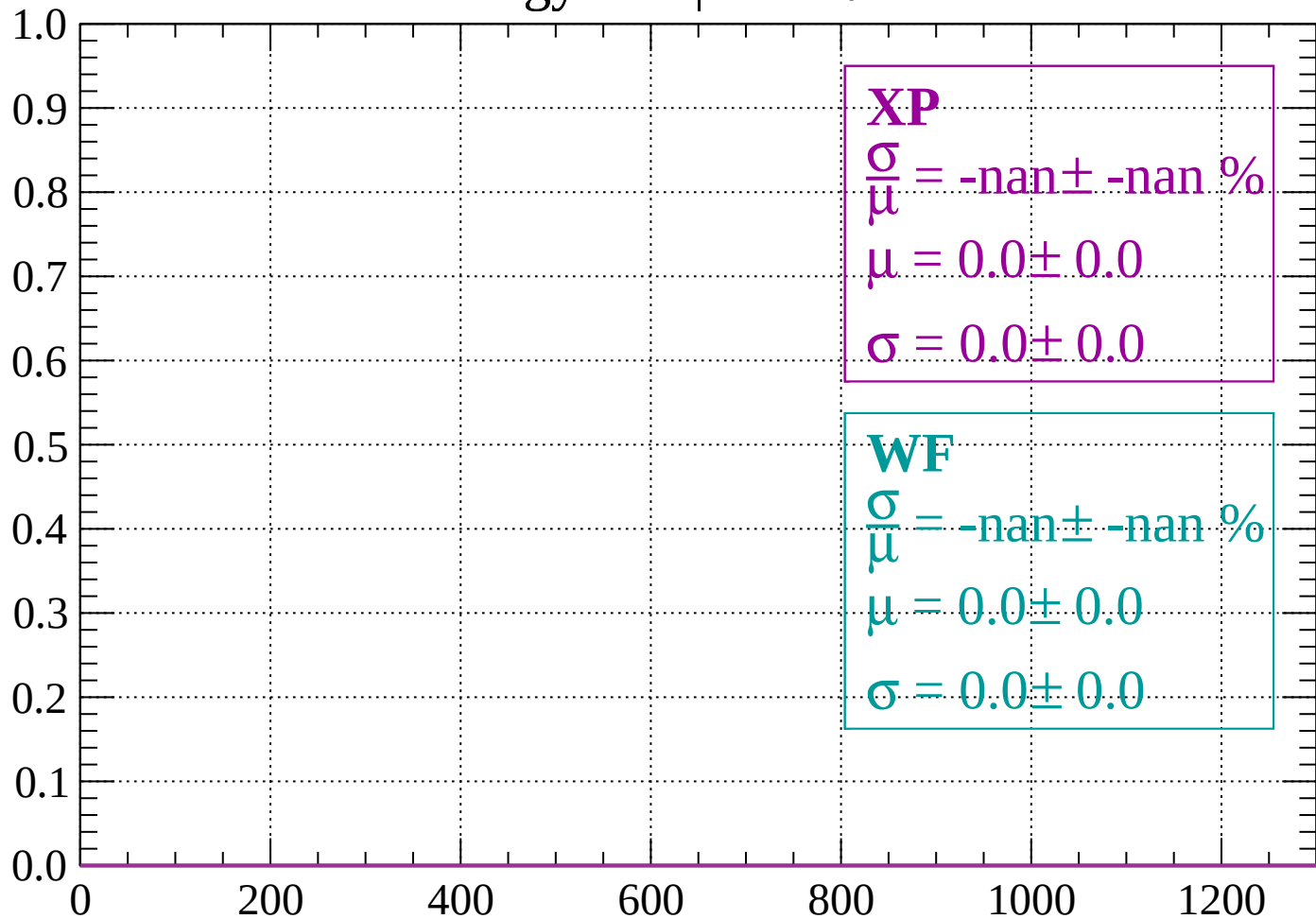
Count



dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 14$

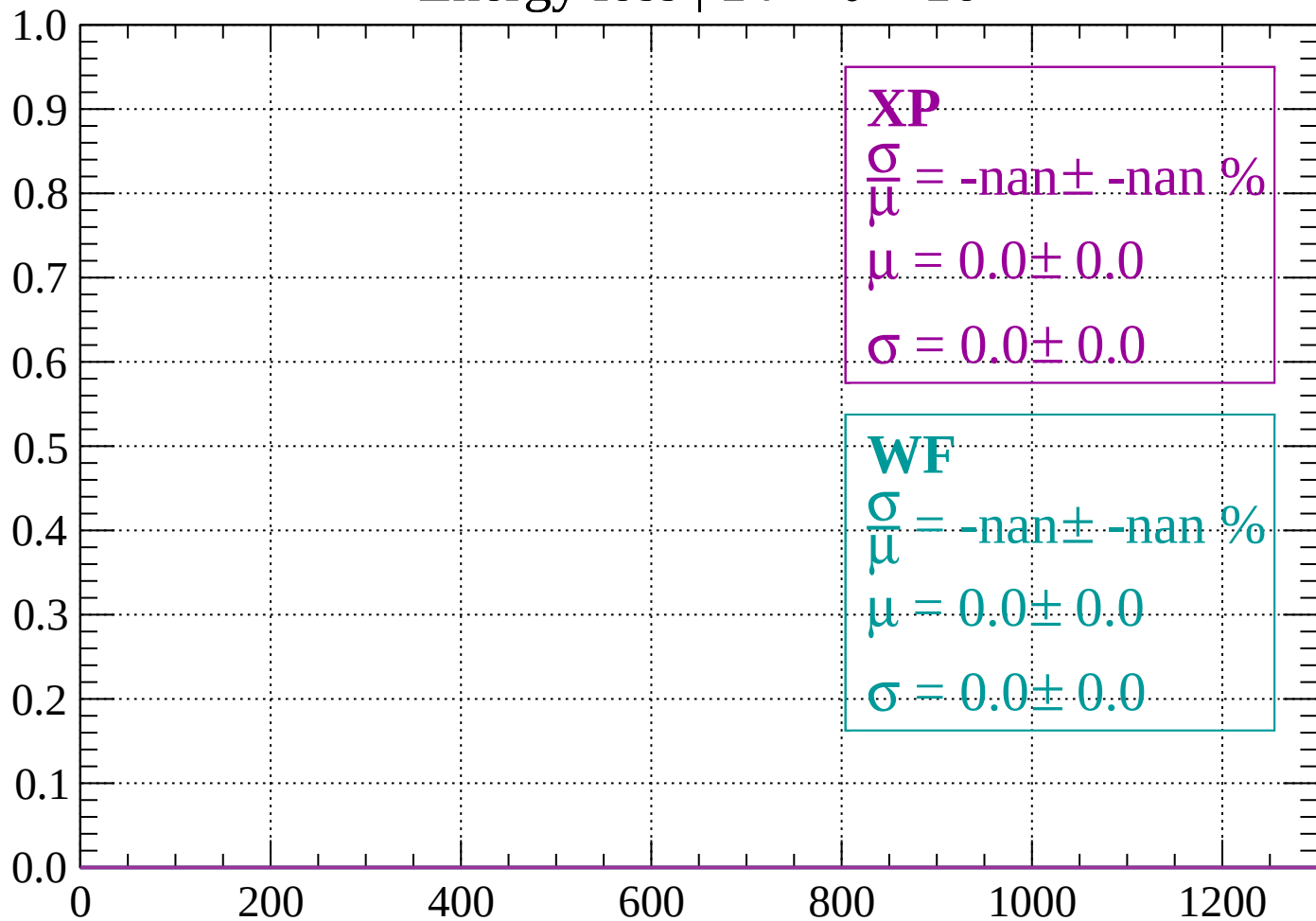
Count



dE/dx (ADC counts/cm)

Energy loss | $14 < \theta < 16$

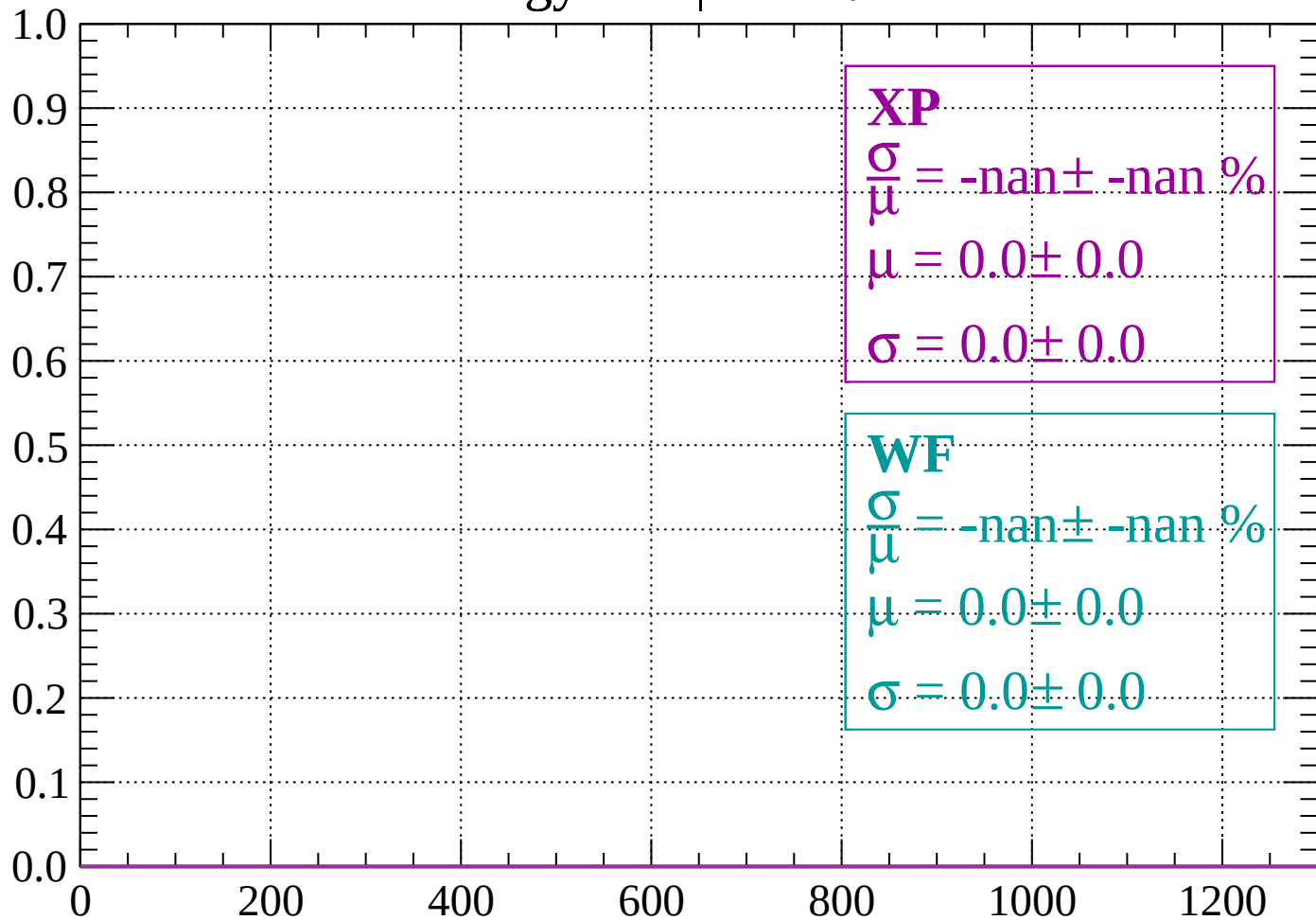
Count



dE/dx (ADC counts/cm)

Energy loss | $16 < \theta < 18$

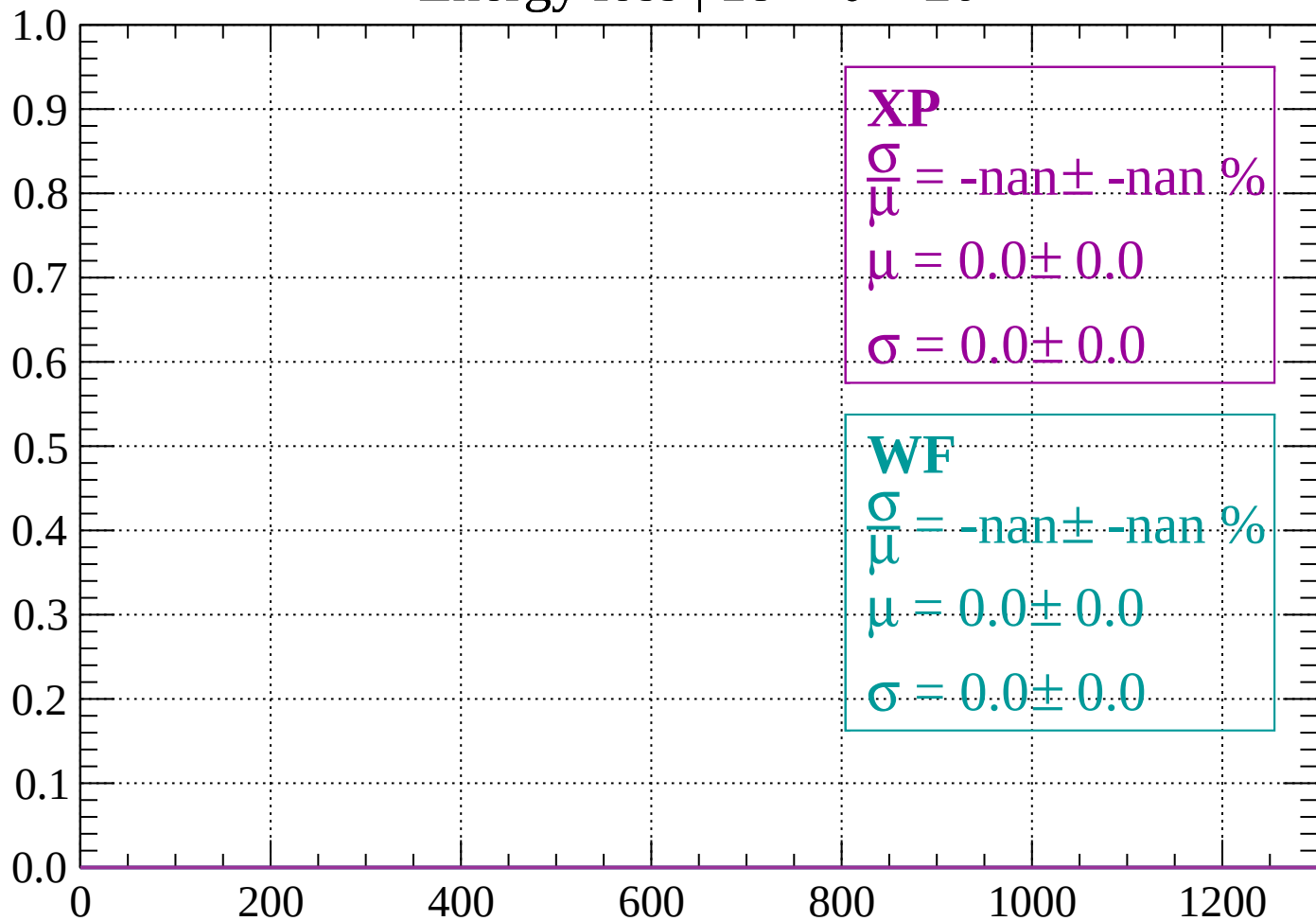
Count



dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 20$

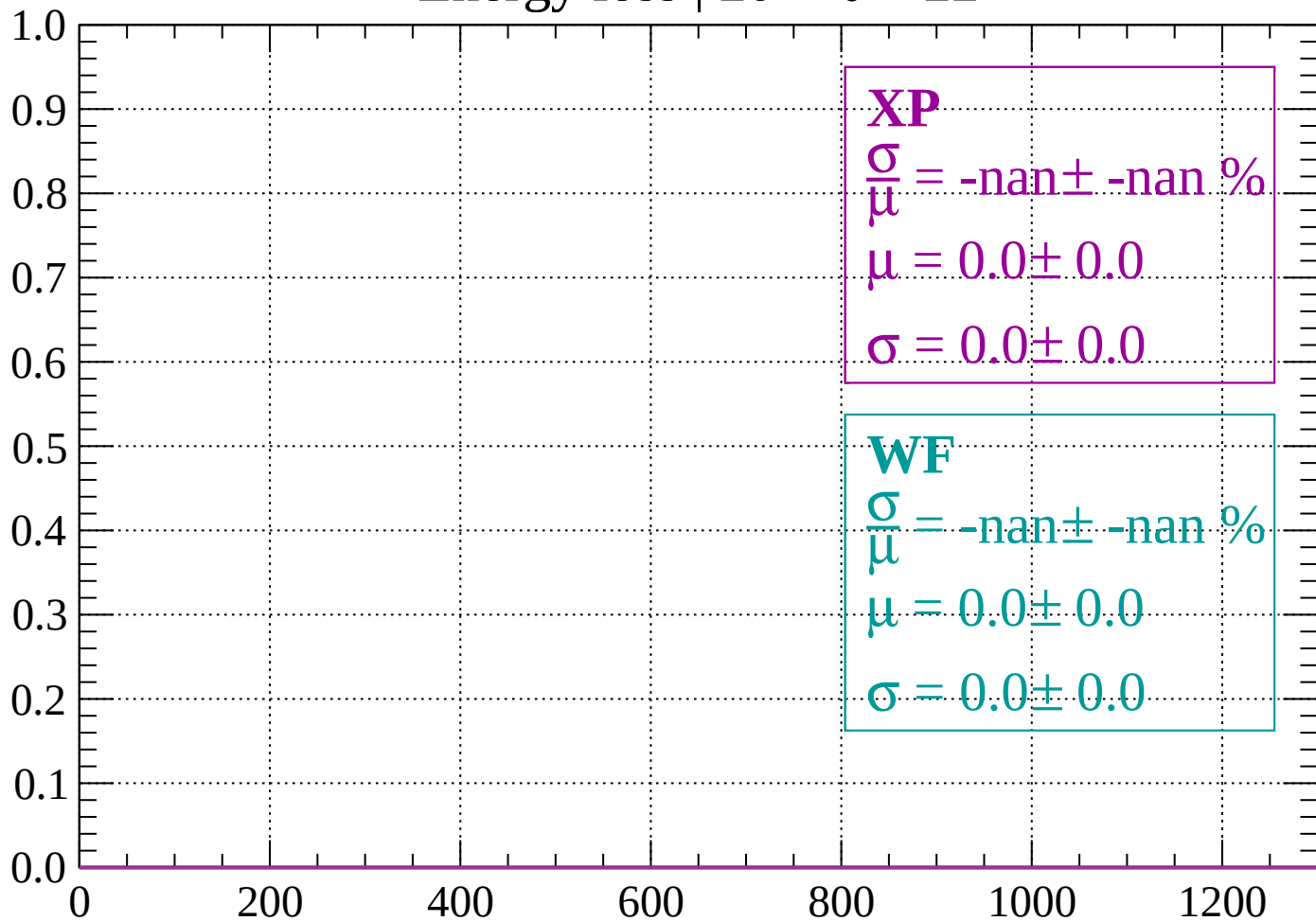
Count



dE/dx (ADC counts/cm)

Energy loss | $20 < \theta < 22$

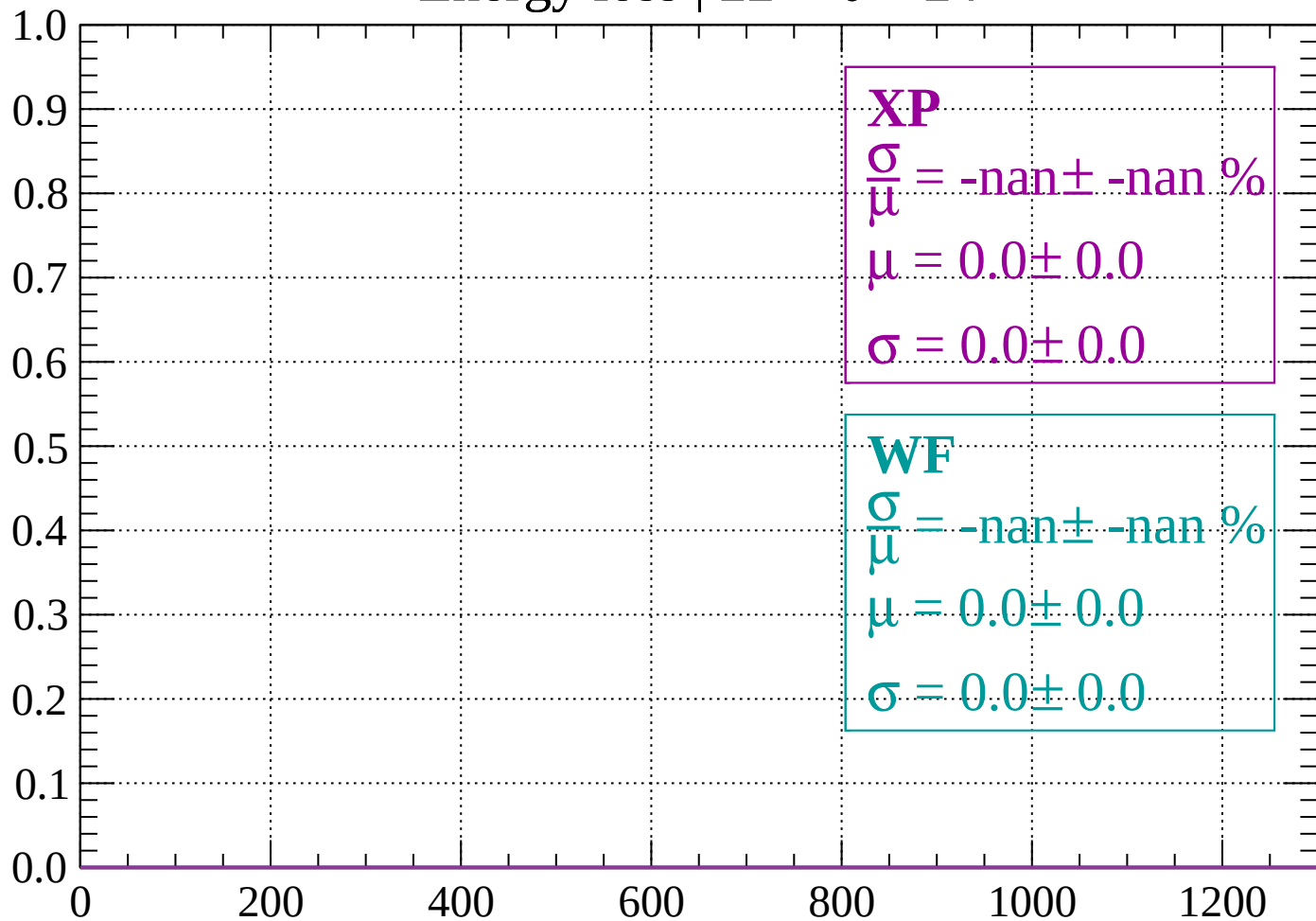
Count



dE/dx (ADC counts/cm)

Energy loss | $22 < \theta < 24$

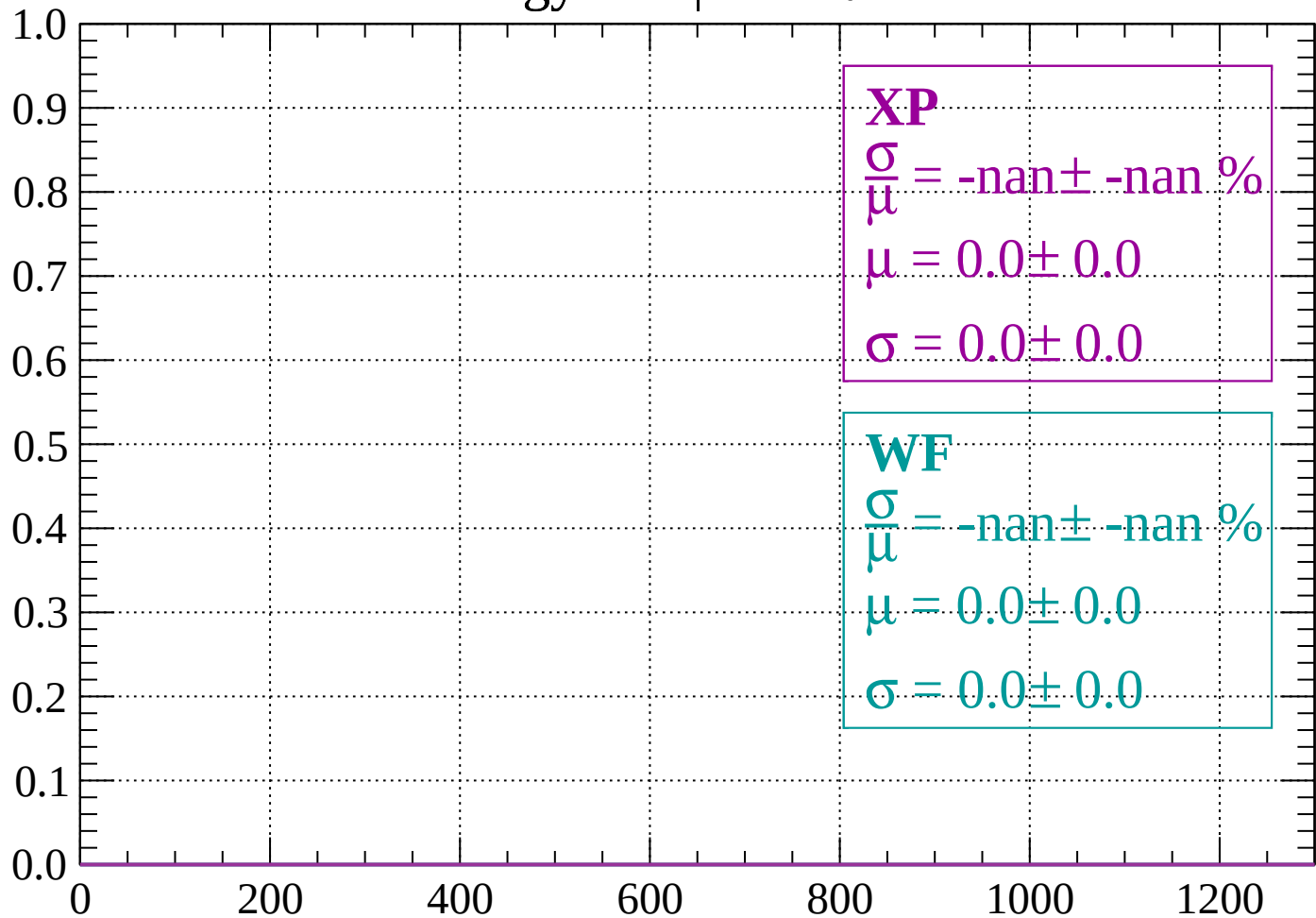
Count



dE/dx (ADC counts/cm)

Energy loss | $24 < \theta < 26$

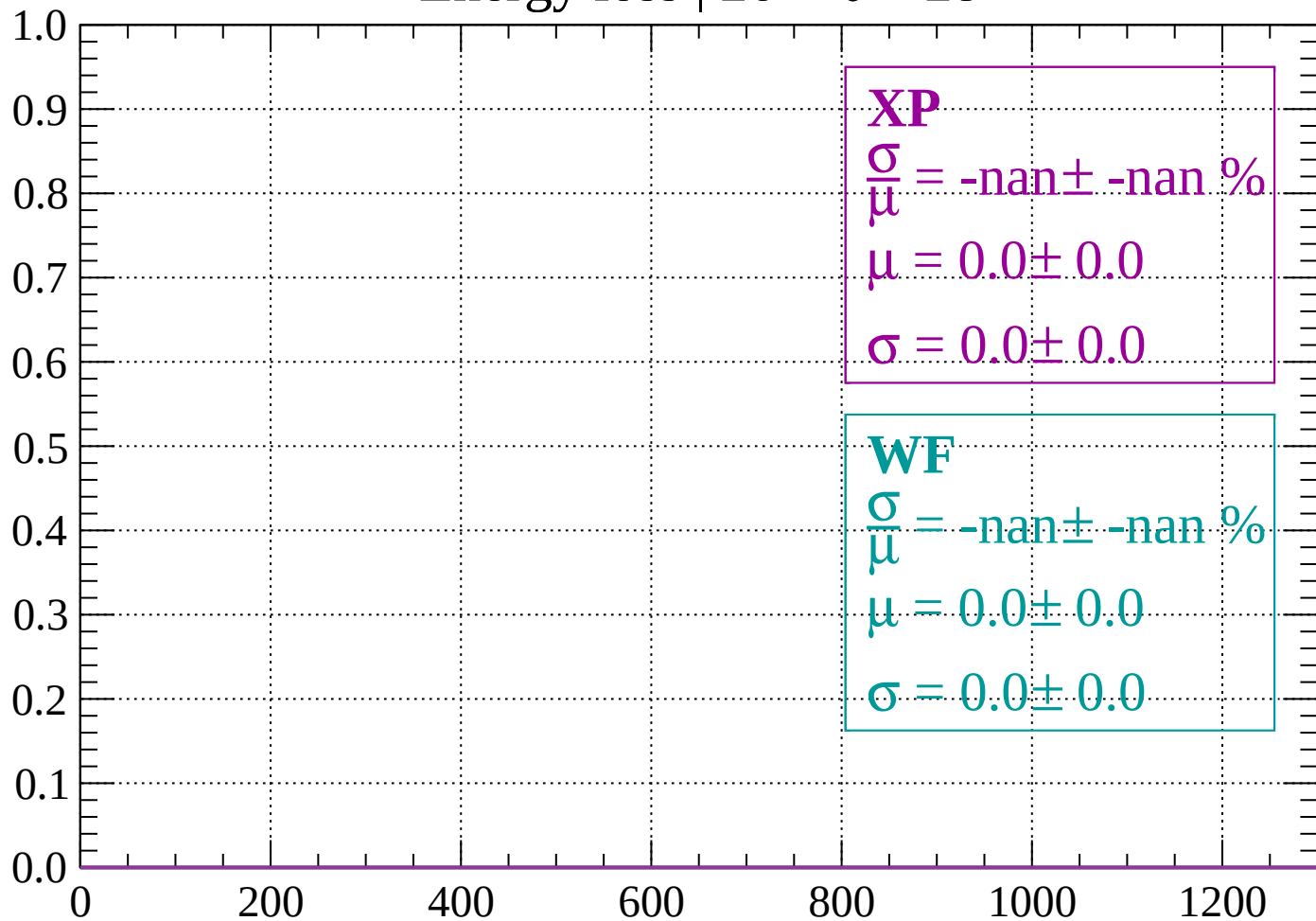
Count



dE/dx (ADC counts/cm)

Energy loss | $26 < \theta < 28$

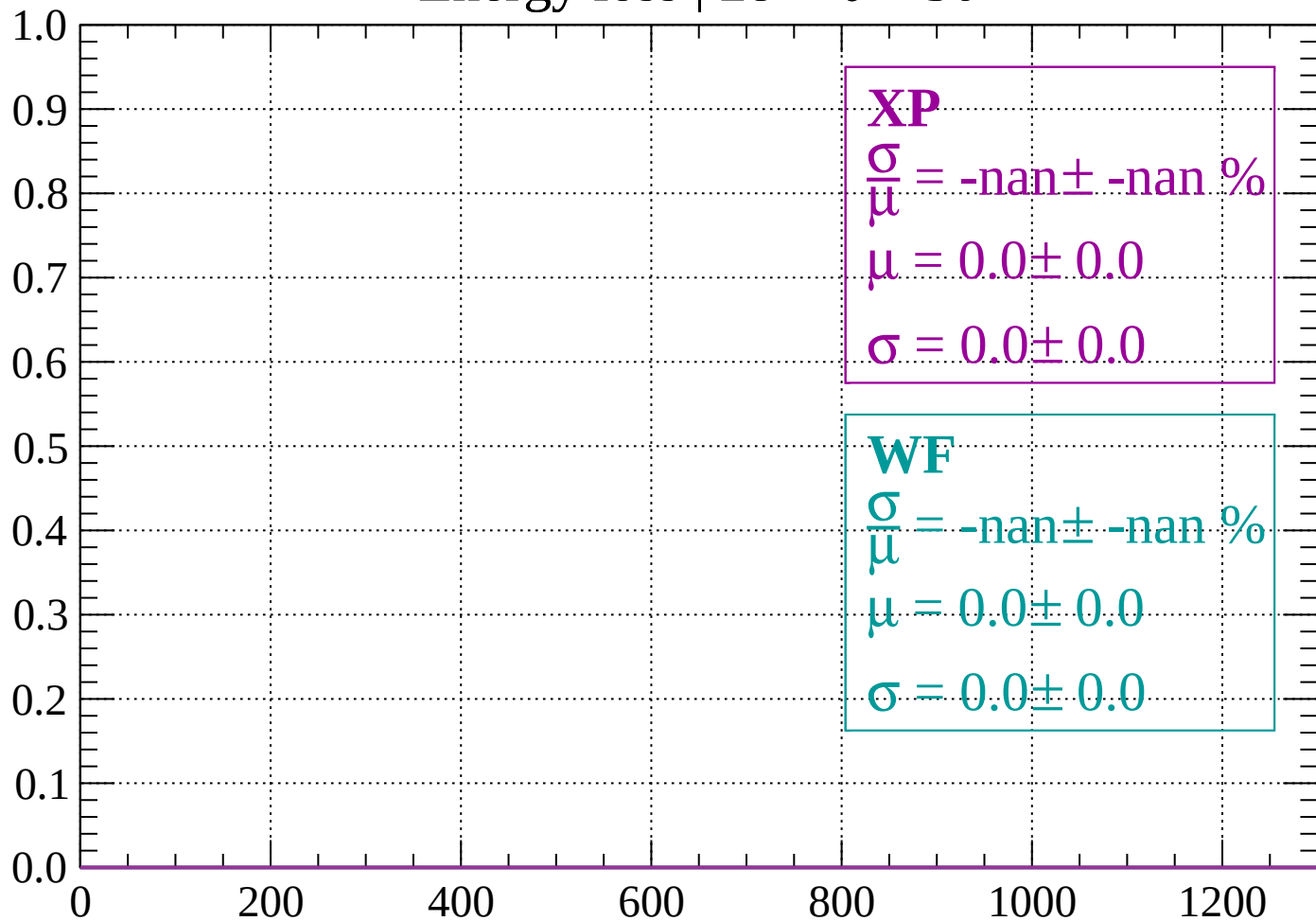
Count



dE/dx (ADC counts/cm)

Energy loss | $28 < \theta < 30$

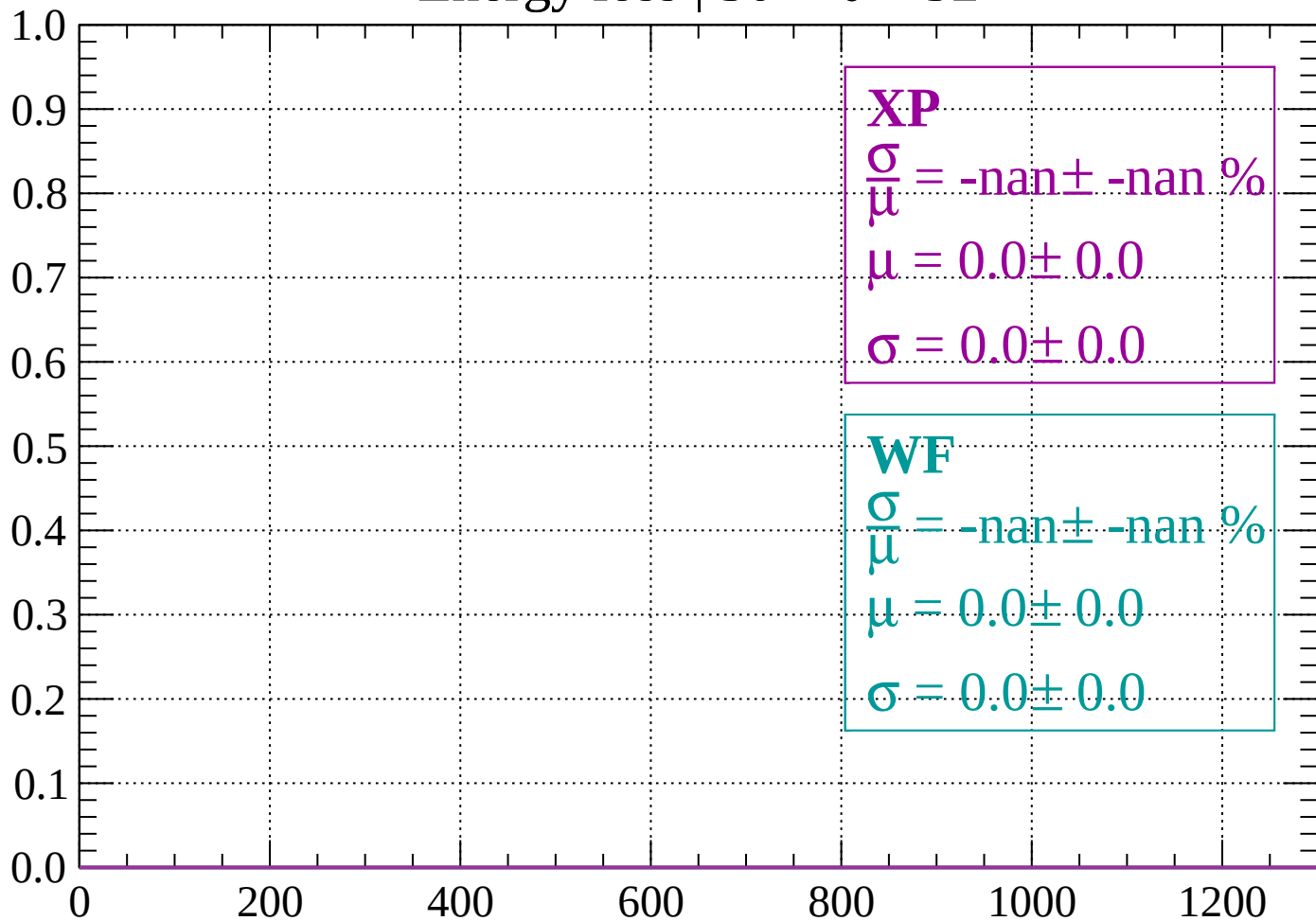
Count



dE/dx (ADC counts/cm)

Energy loss | $30 < \theta < 32$

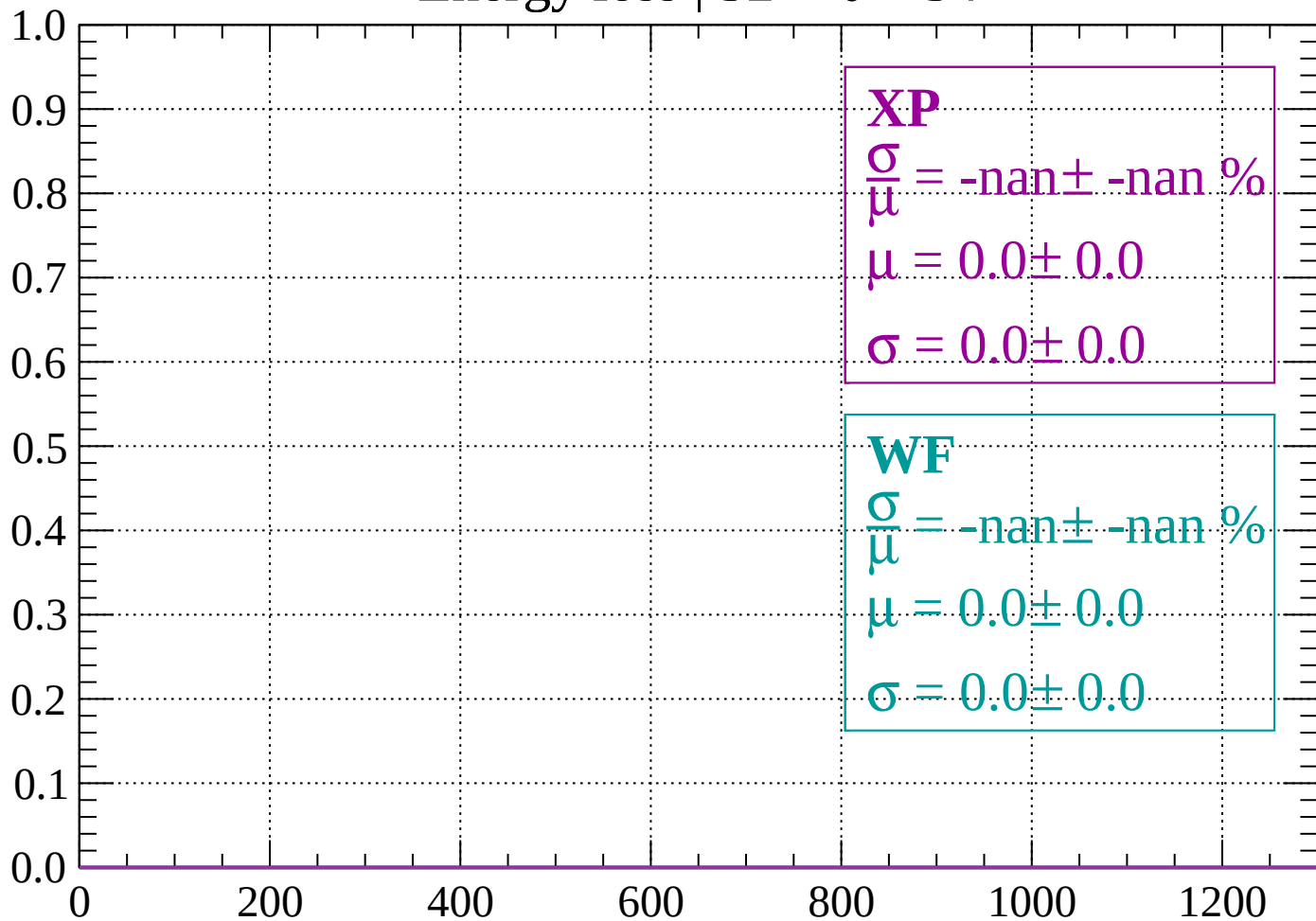
Count



dE/dx (ADC counts/cm)

Energy loss | $32 < \theta < 34$

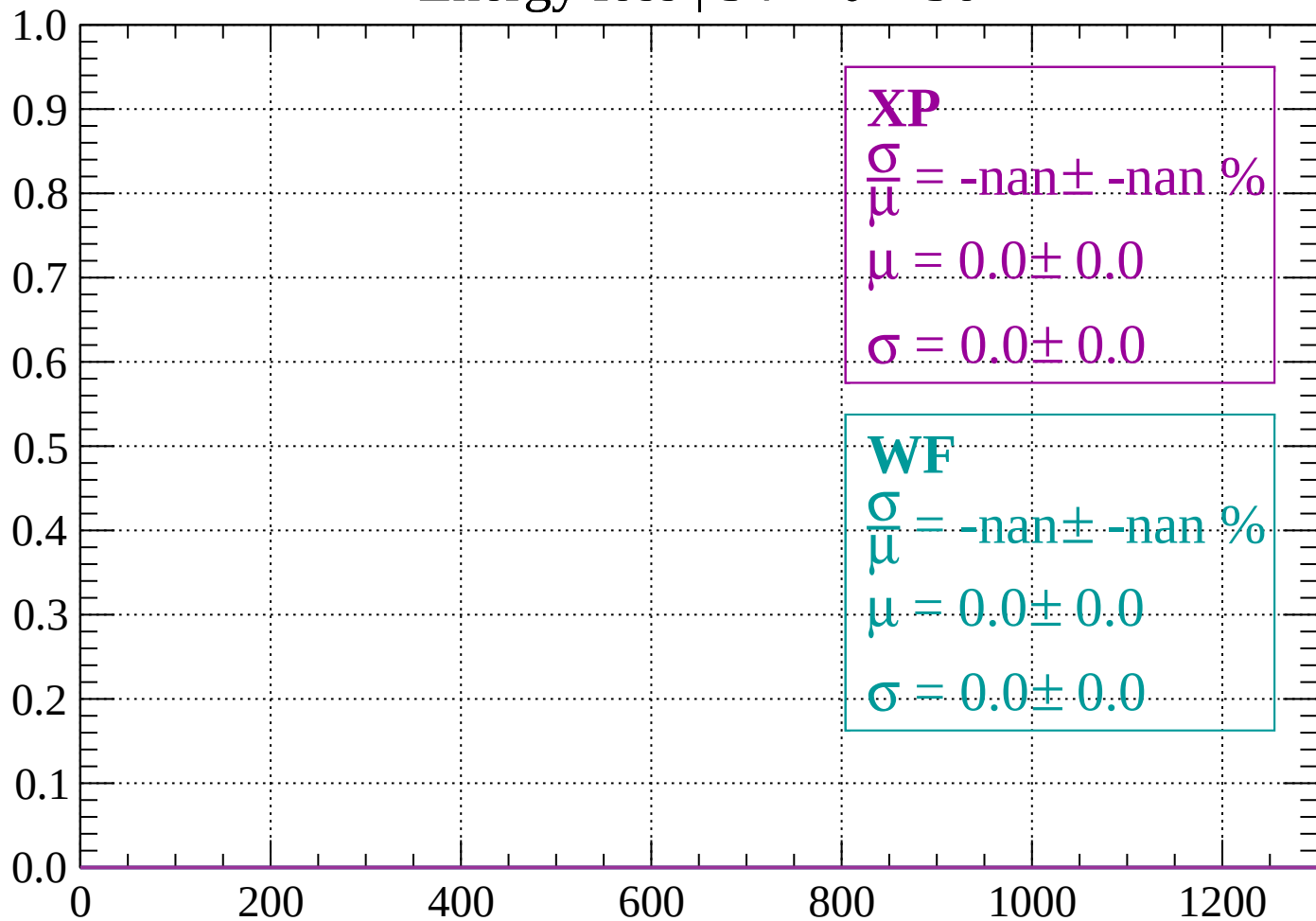
Count



dE/dx (ADC counts/cm)

Energy loss | $34 < \theta < 36$

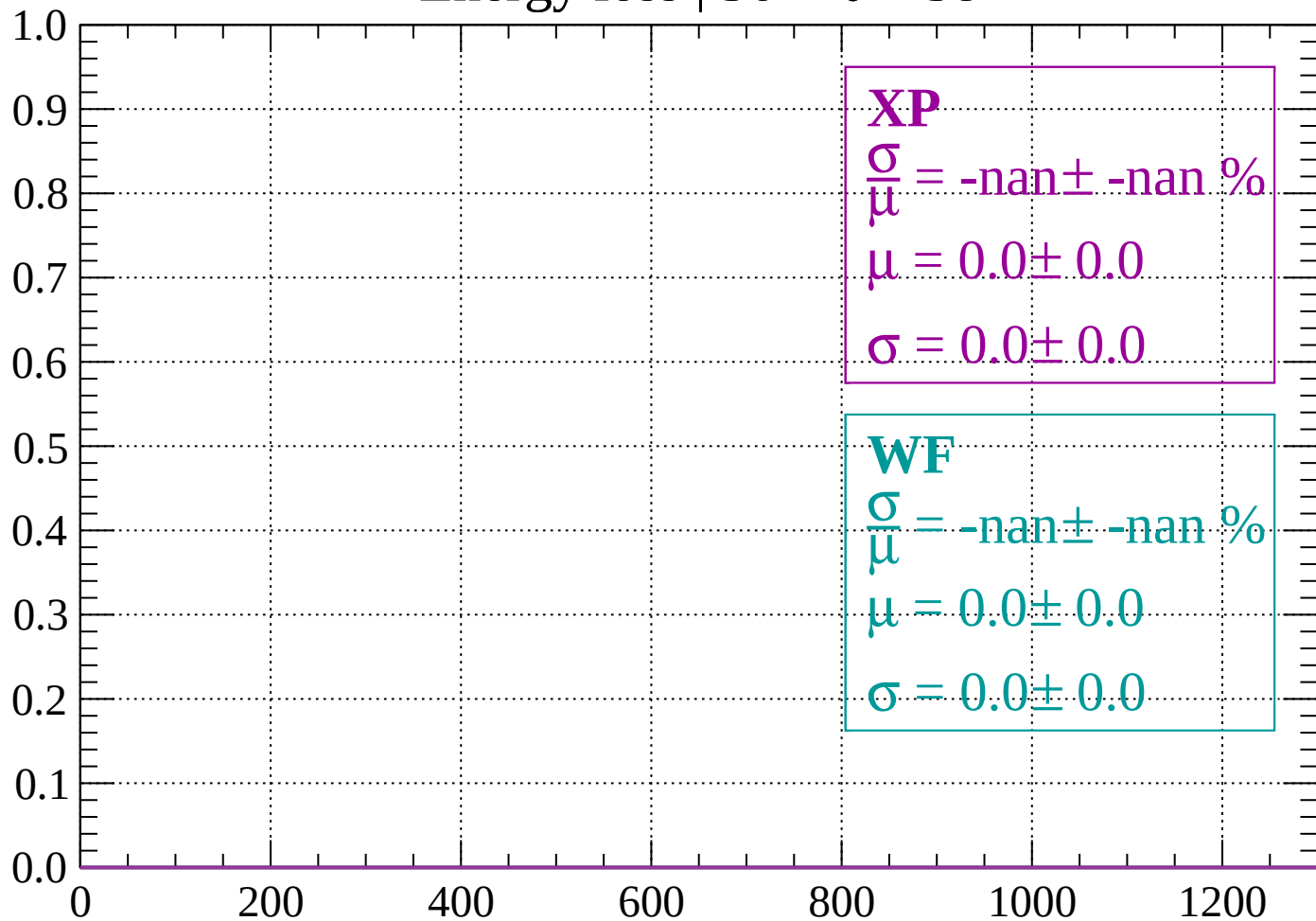
Count



dE/dx (ADC counts/cm)

Energy loss | $36 < \theta < 38$

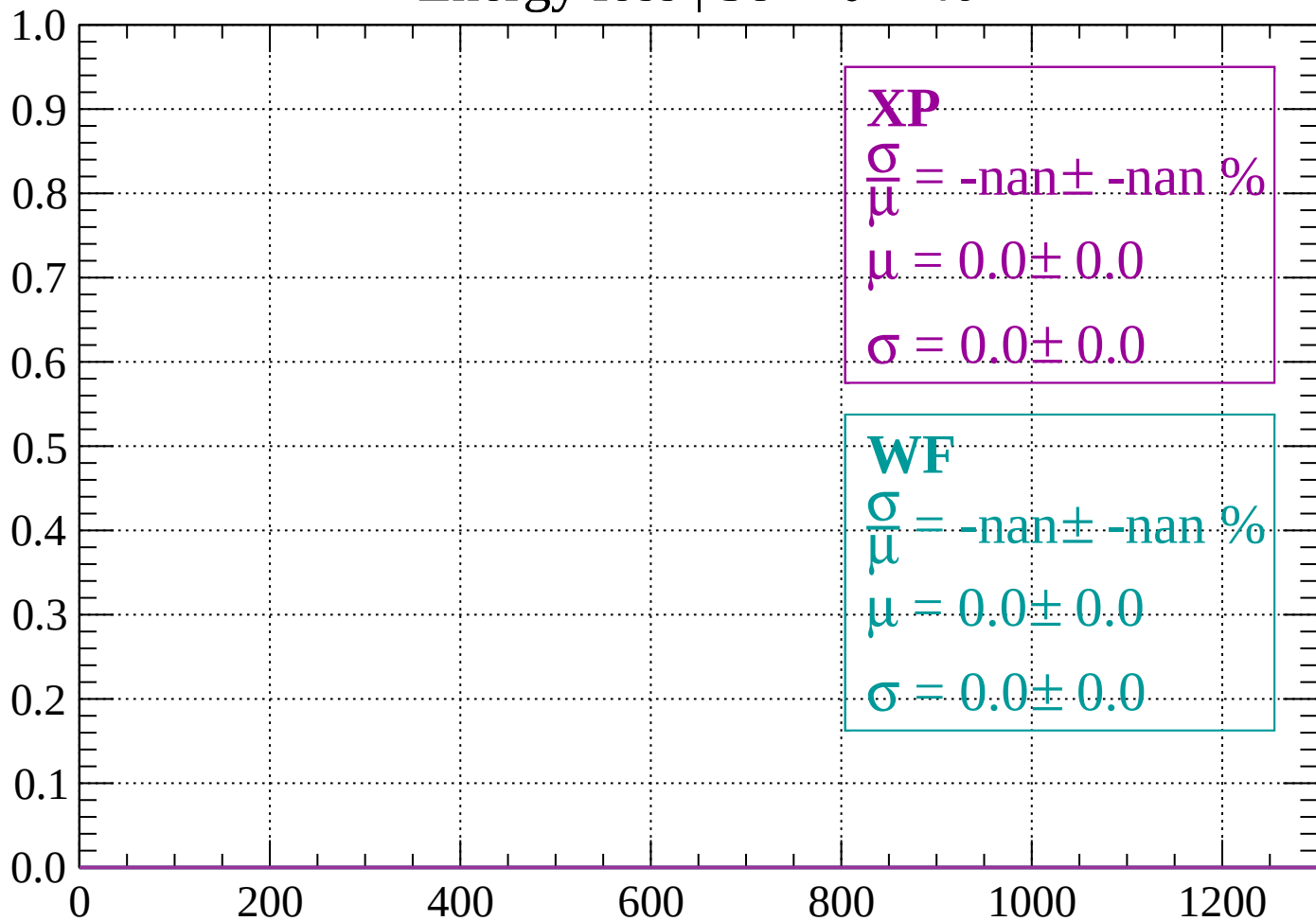
Count



dE/dx (ADC counts/cm)

Energy loss | $38 < \theta < 40$

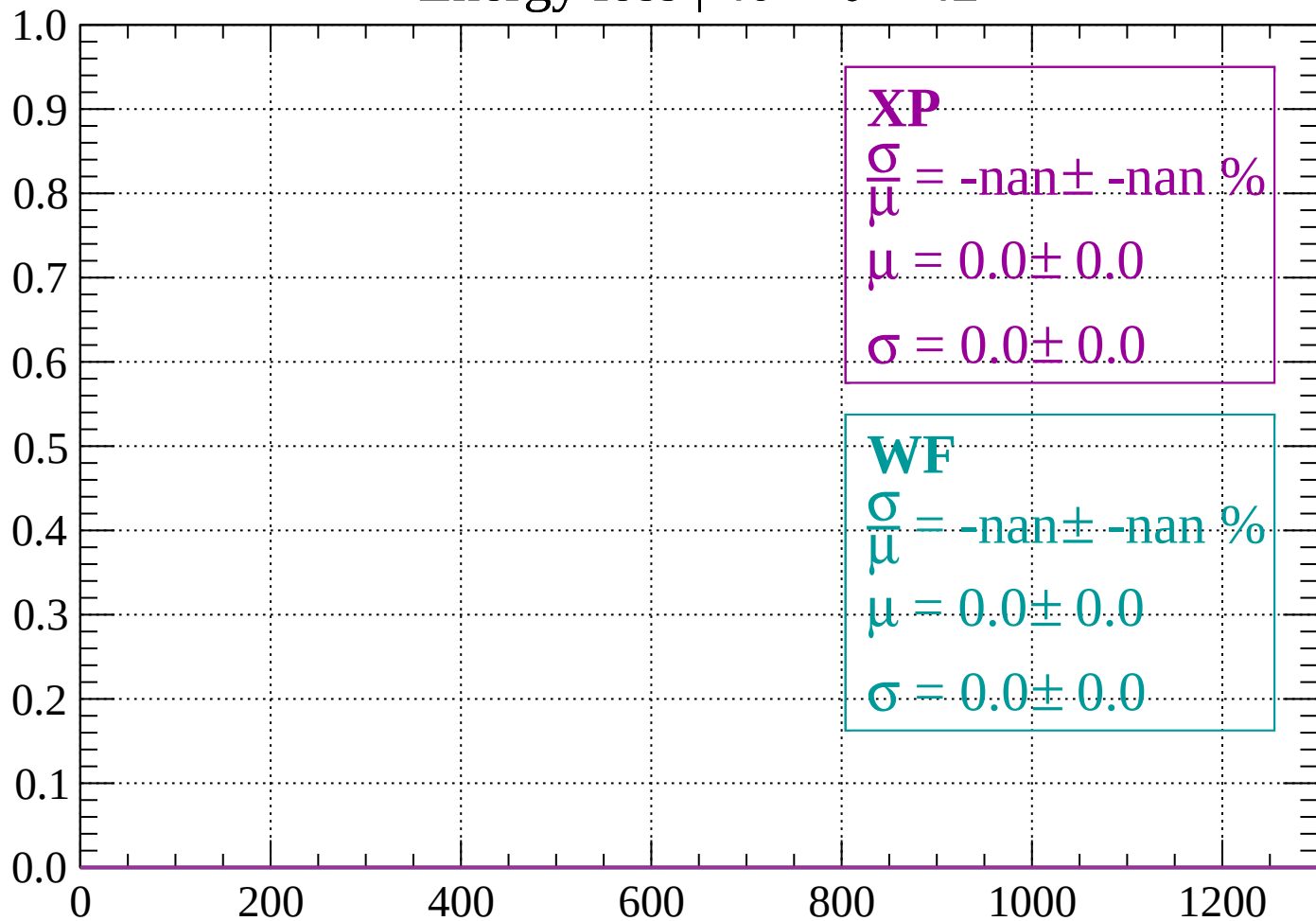
Count



dE/dx (ADC counts/cm)

Energy loss | $40 < \theta < 42$

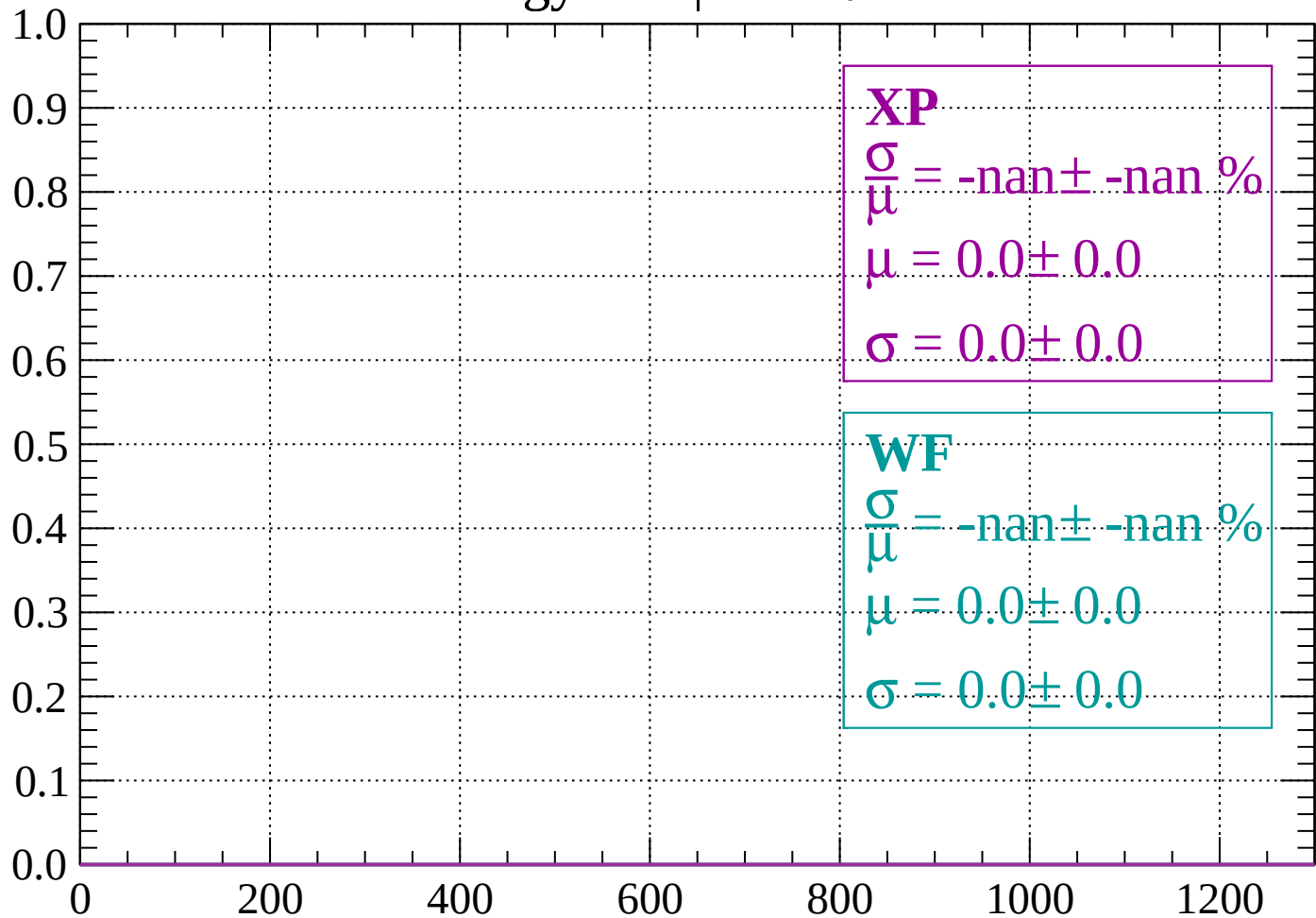
Count



dE/dx (ADC counts/cm)

Energy loss | $42 < \theta < 44$

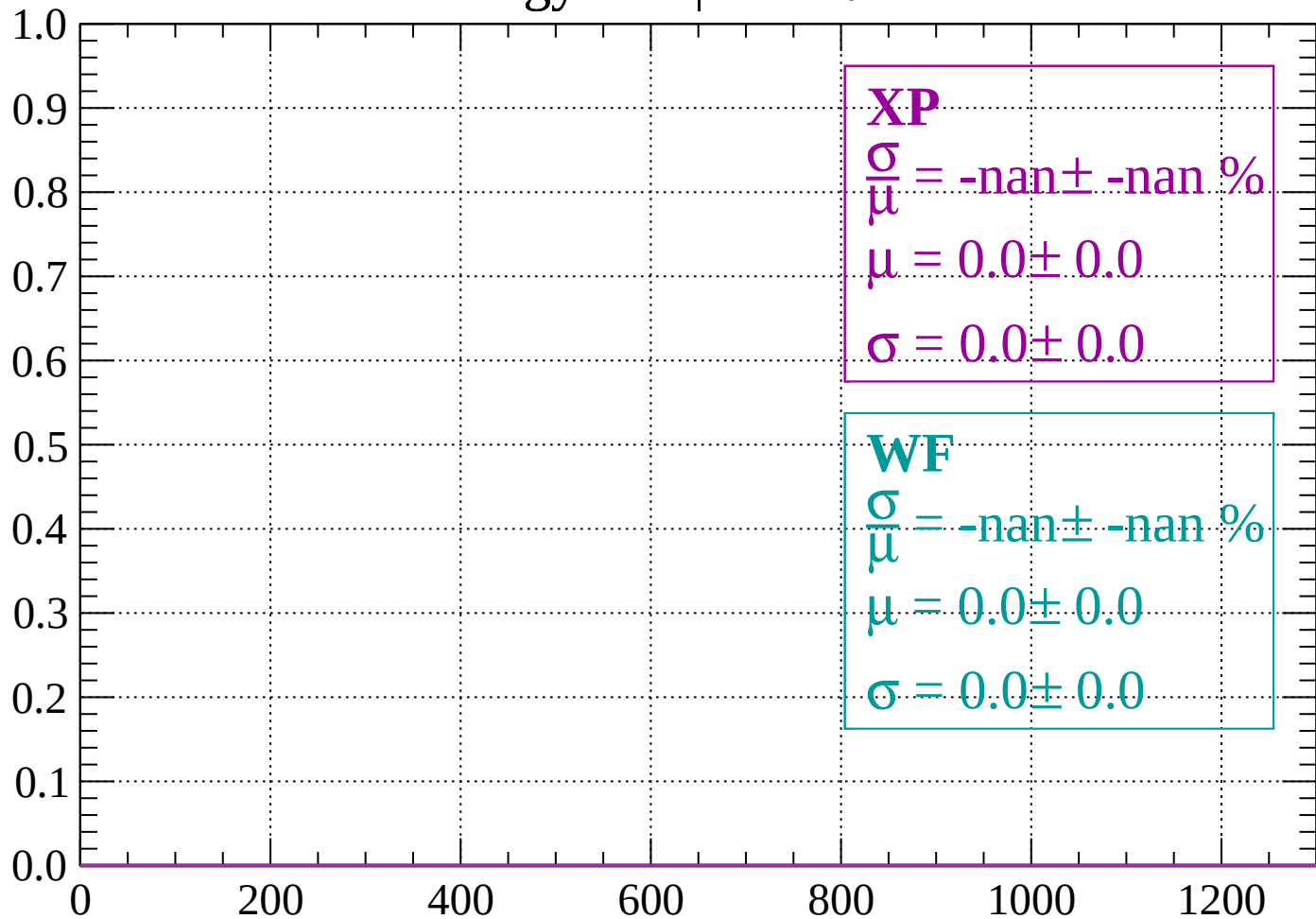
Count



dE/dx (ADC counts/cm)

Energy loss | $44 < \theta < 46$

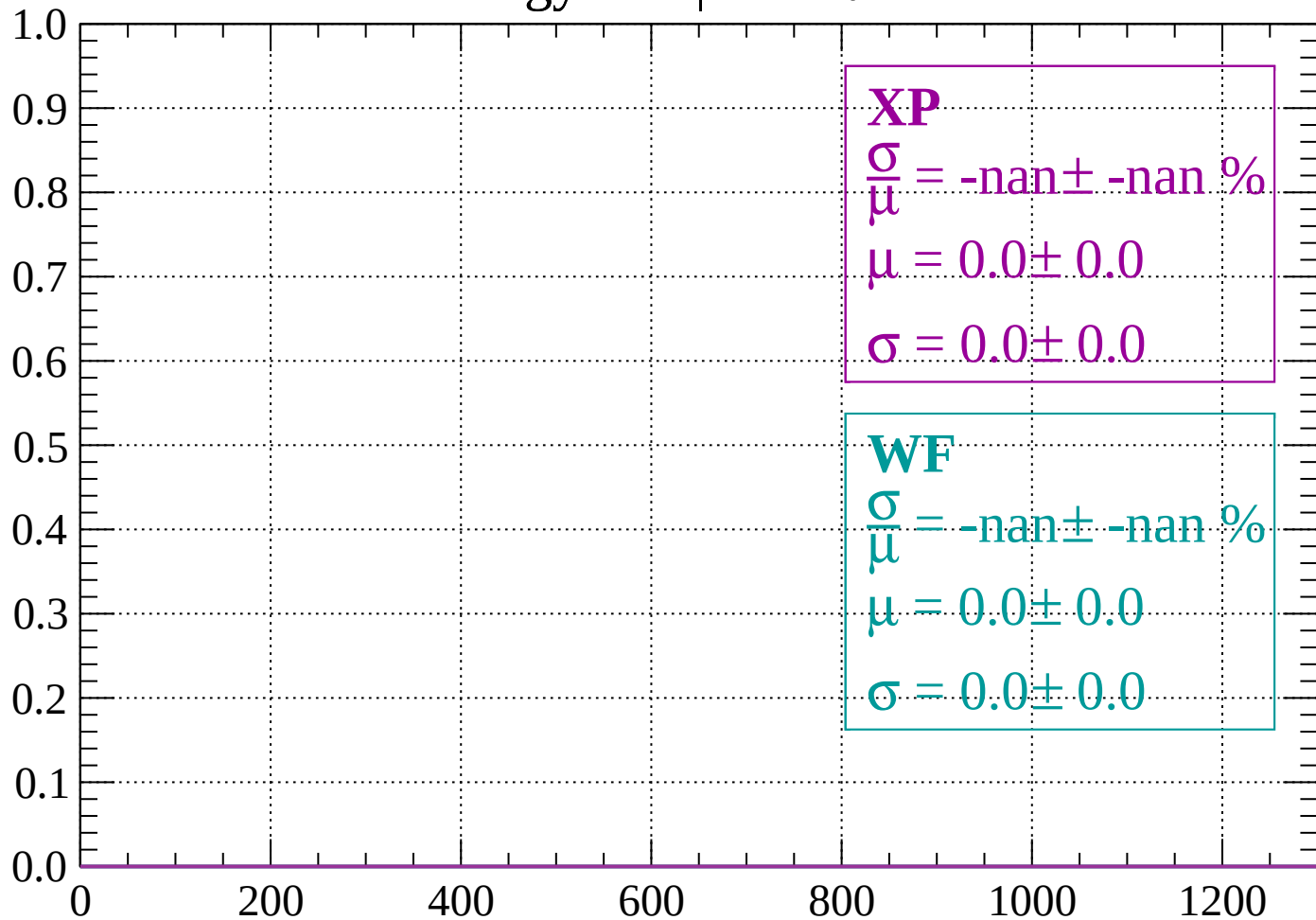
Count



dE/dx (ADC counts/cm)

Energy loss | $46 < \theta < 48$

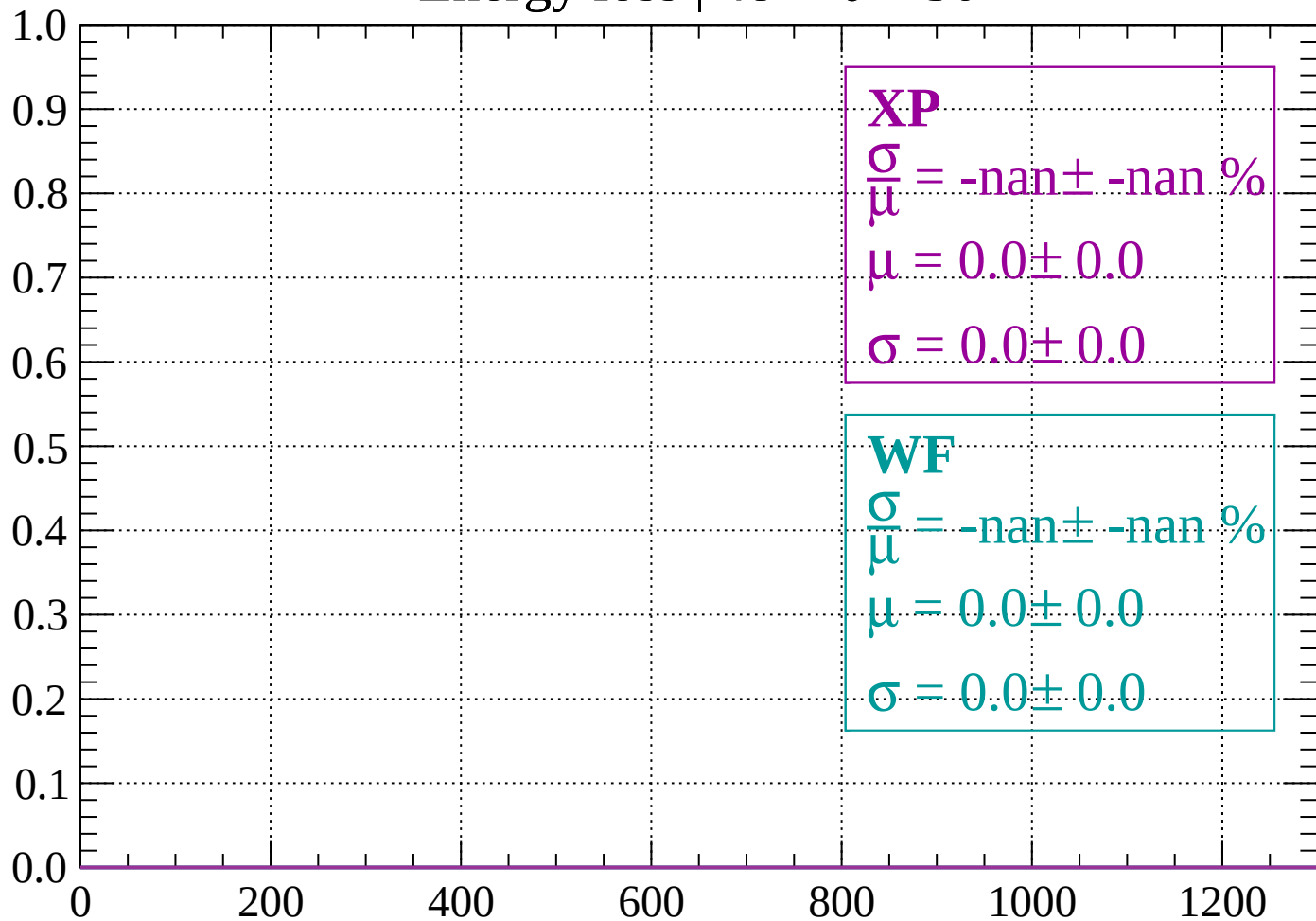
Count



dE/dx (ADC counts/cm)

Energy loss | $48 < \theta < 50$

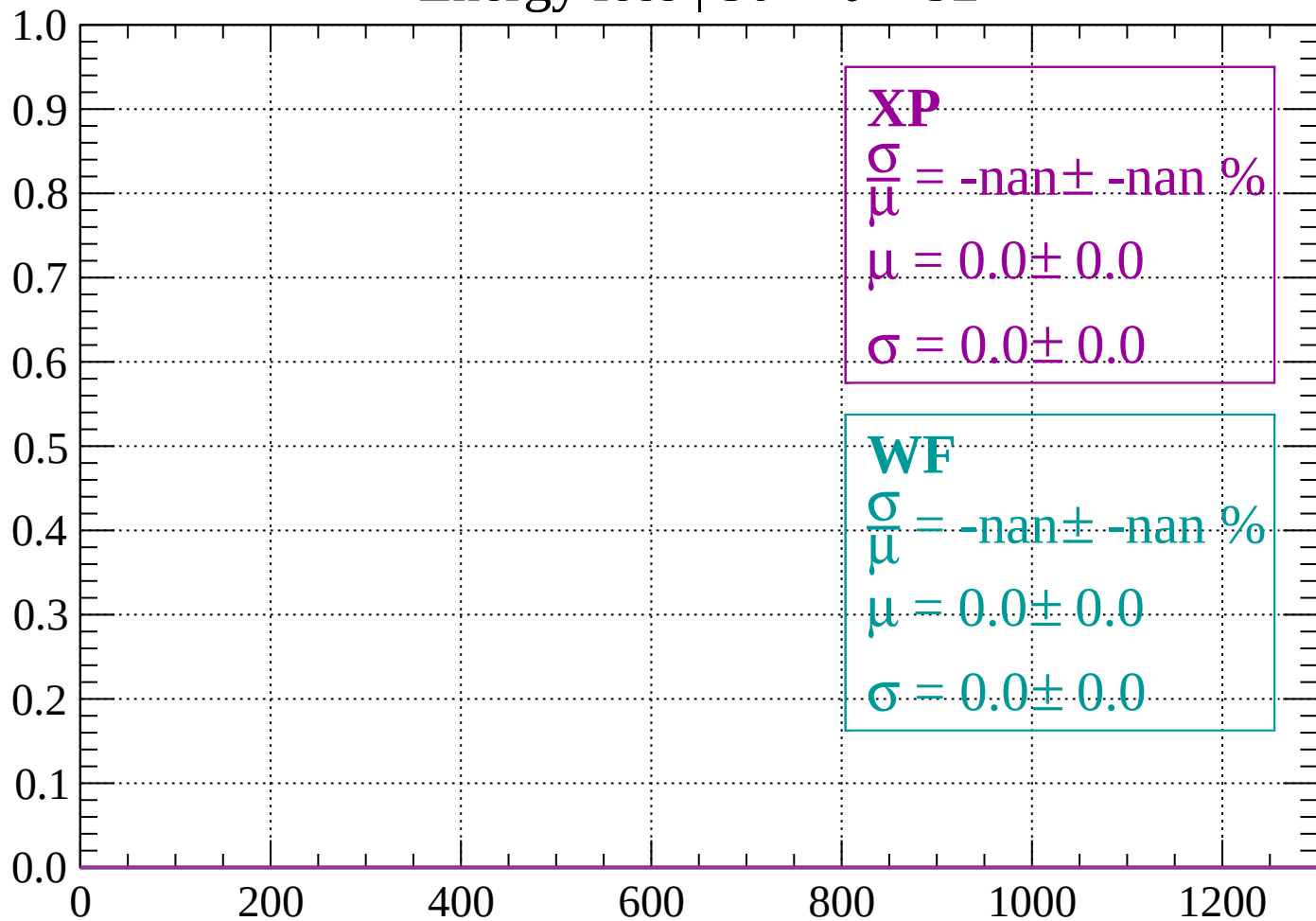
Count



dE/dx (ADC counts/cm)

Energy loss | $50 < \theta < 52$

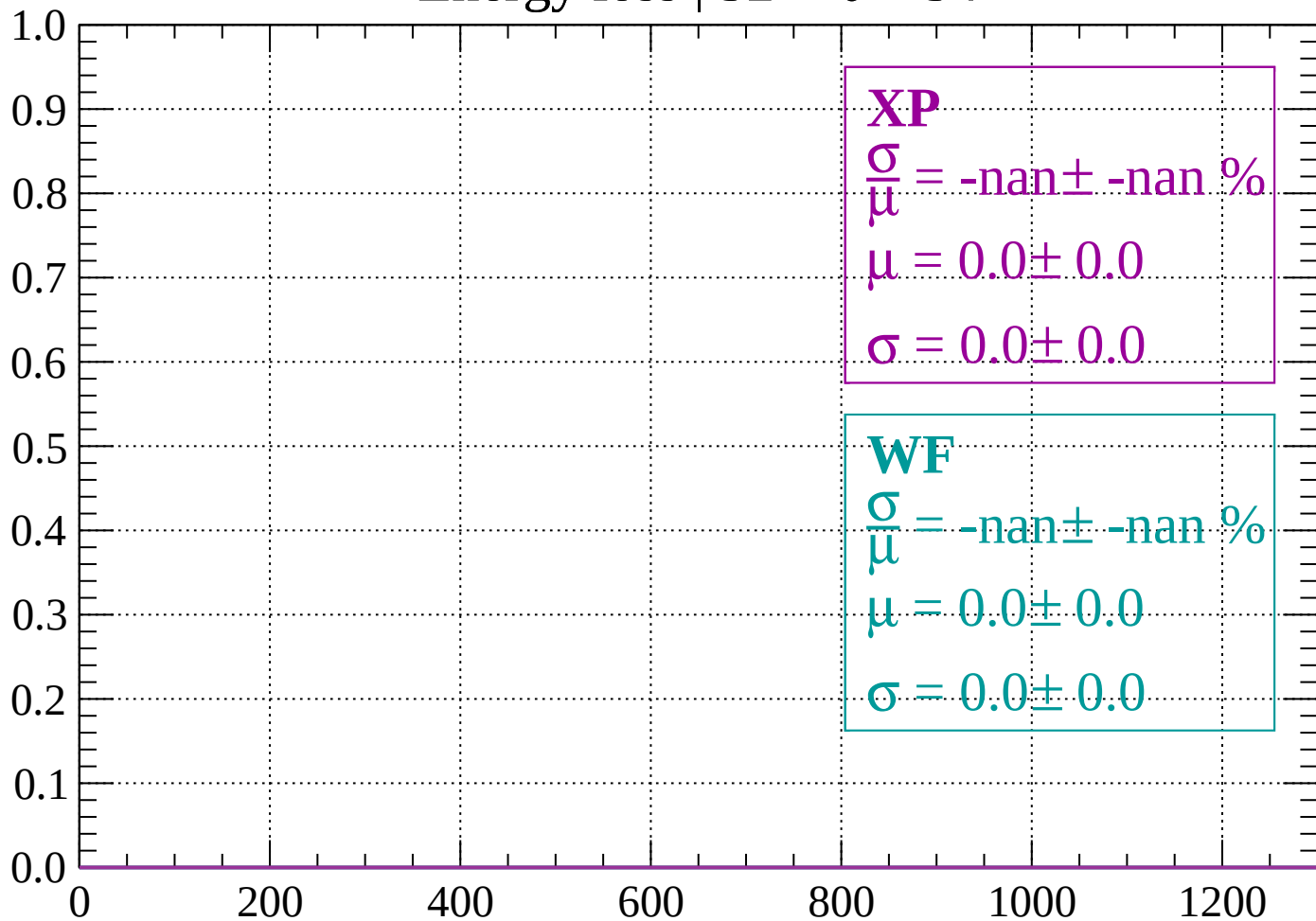
Count



dE/dx (ADC counts/cm)

Energy loss | $52 < \theta < 54$

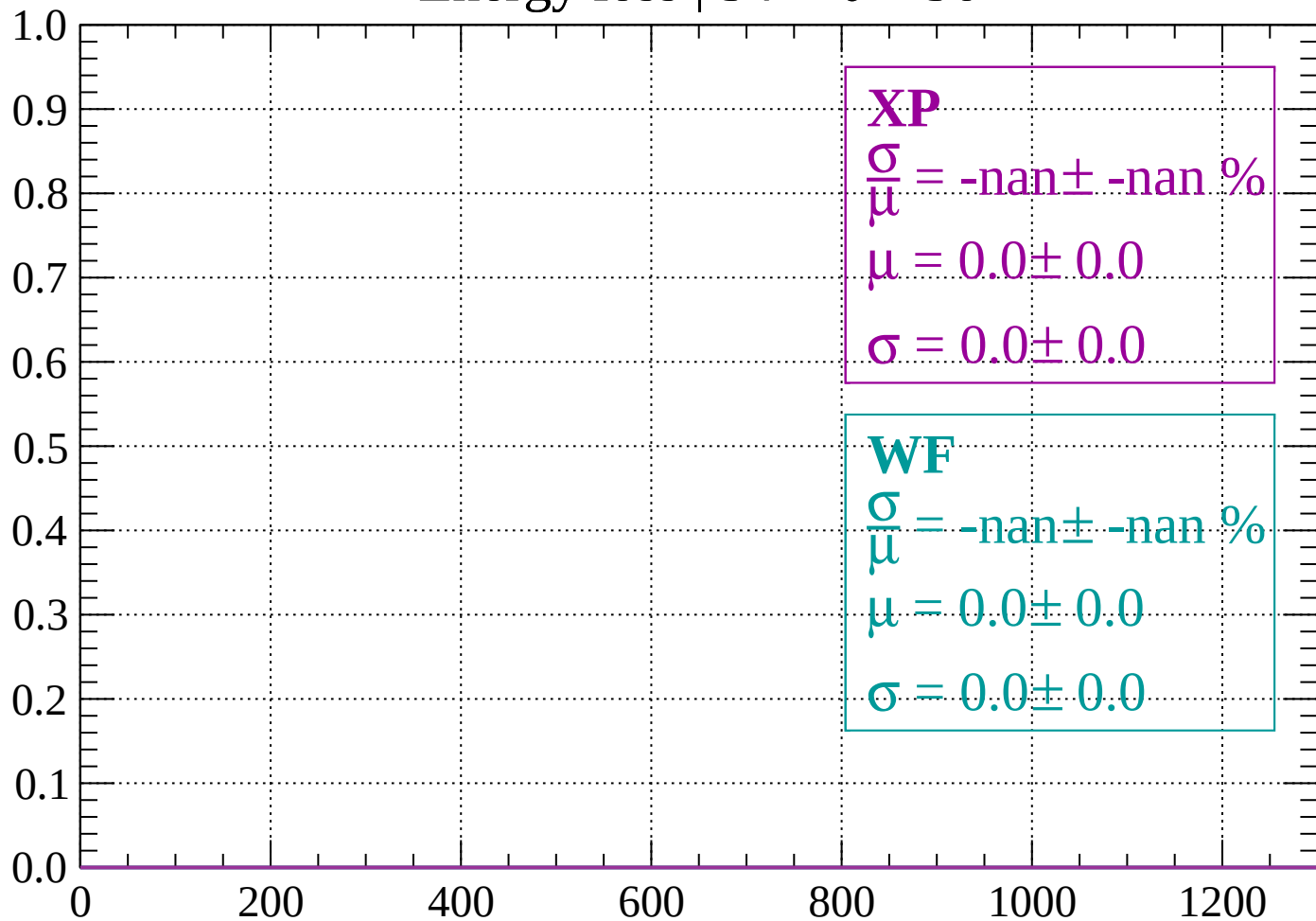
Count



dE/dx (ADC counts/cm)

Energy loss | $54 < \theta < 56$

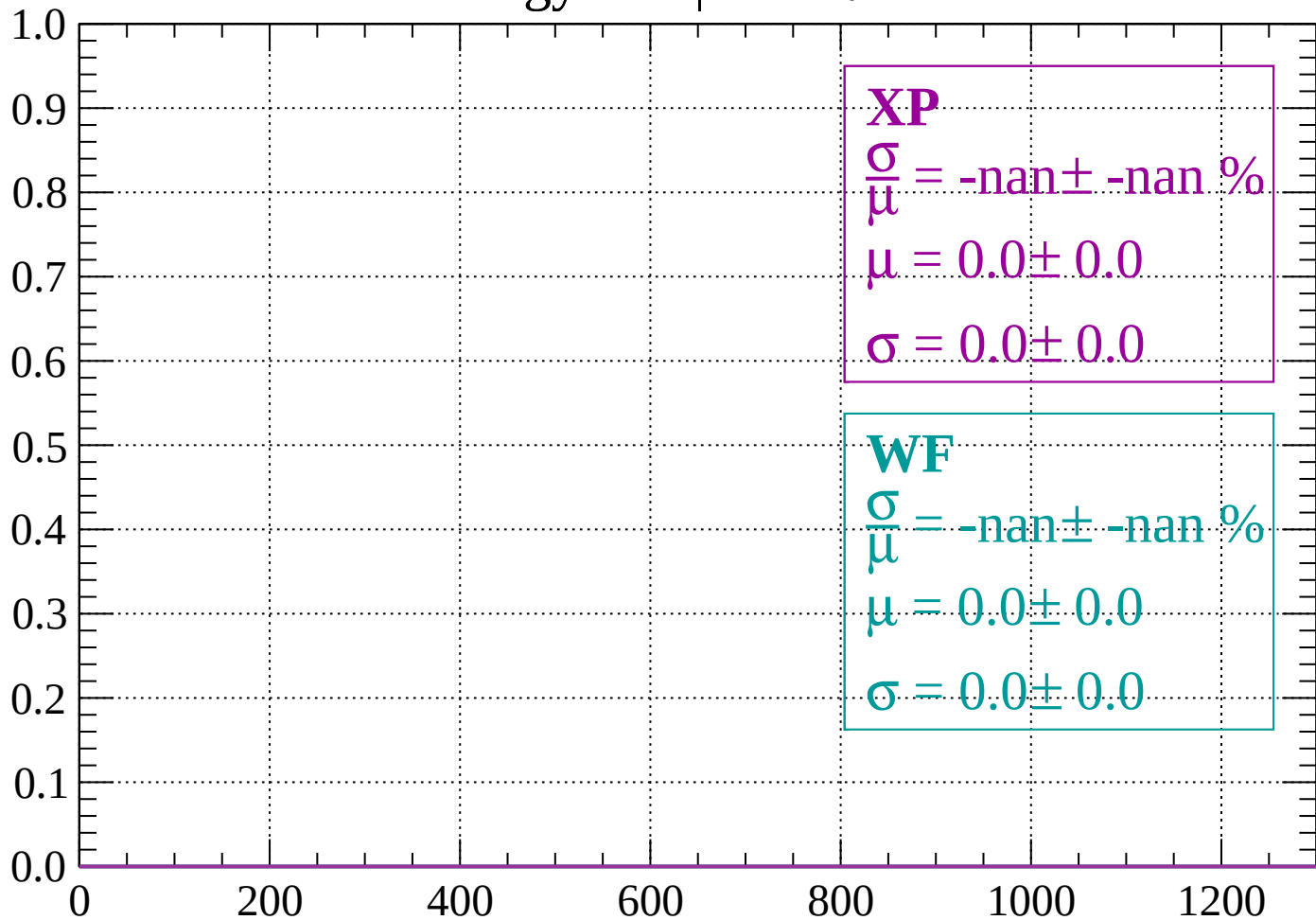
Count



dE/dx (ADC counts/cm)

Energy loss | $56 < \theta < 58$

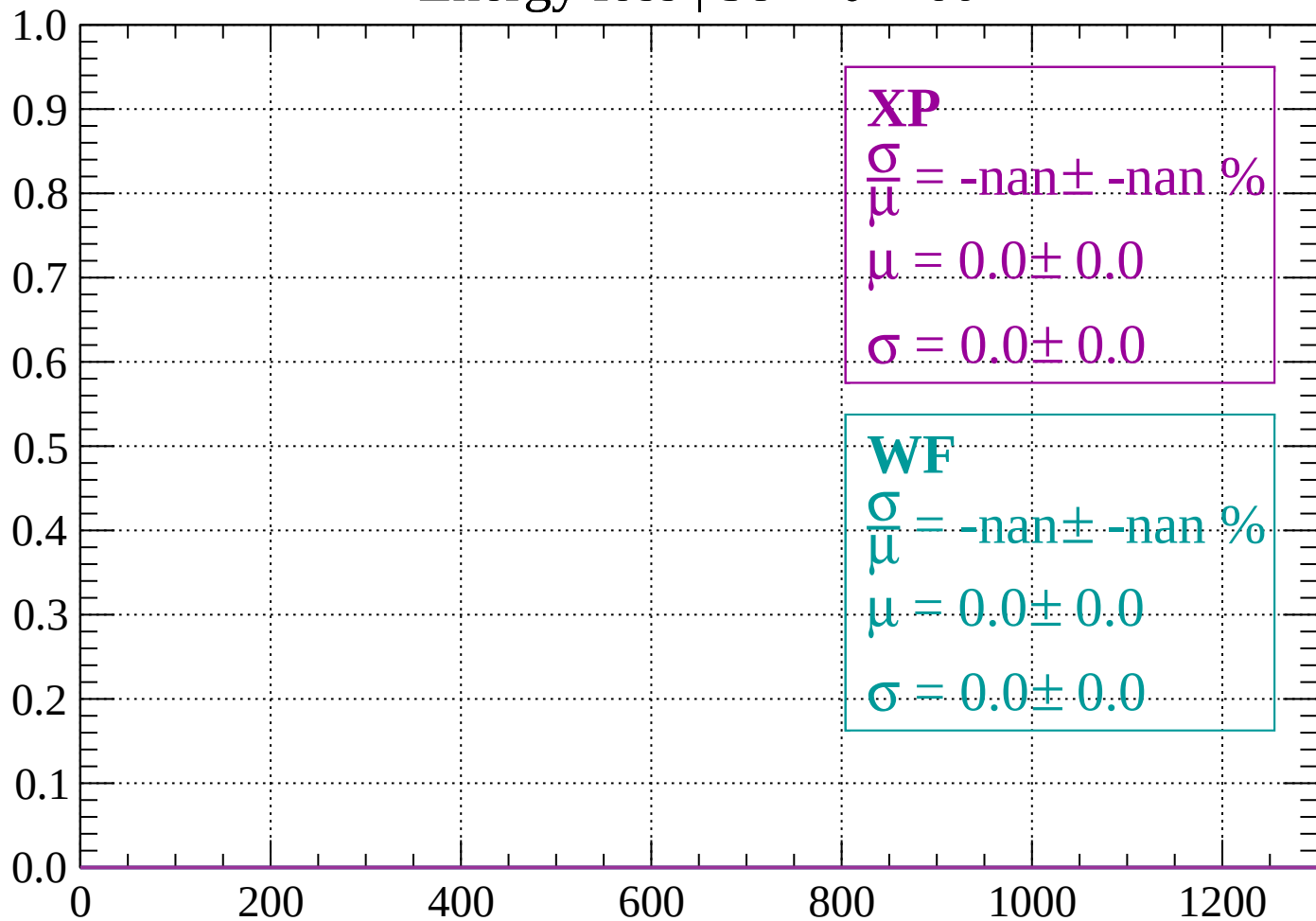
Count



dE/dx (ADC counts/cm)

Energy loss | $58 < \theta < 60$

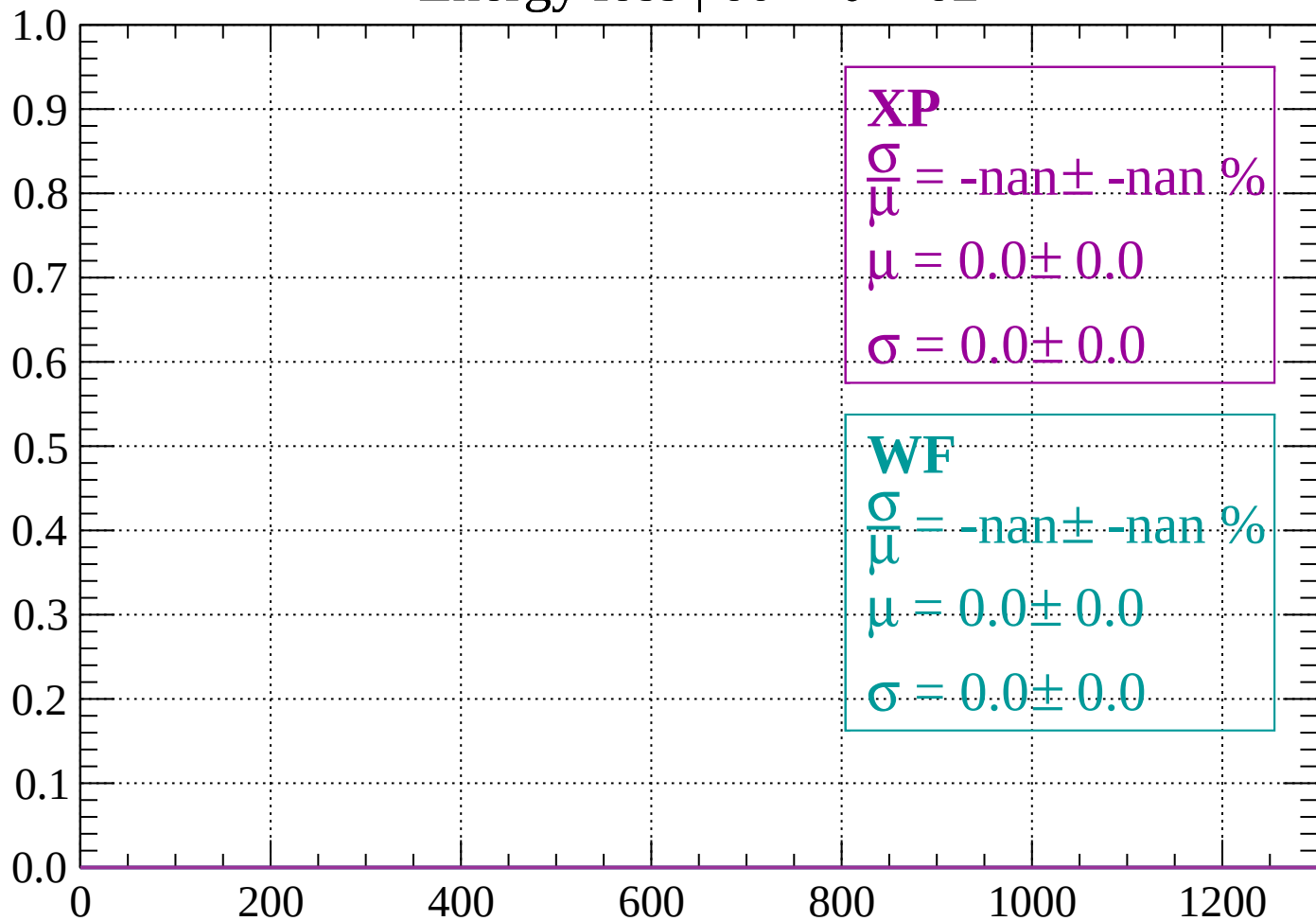
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 62$

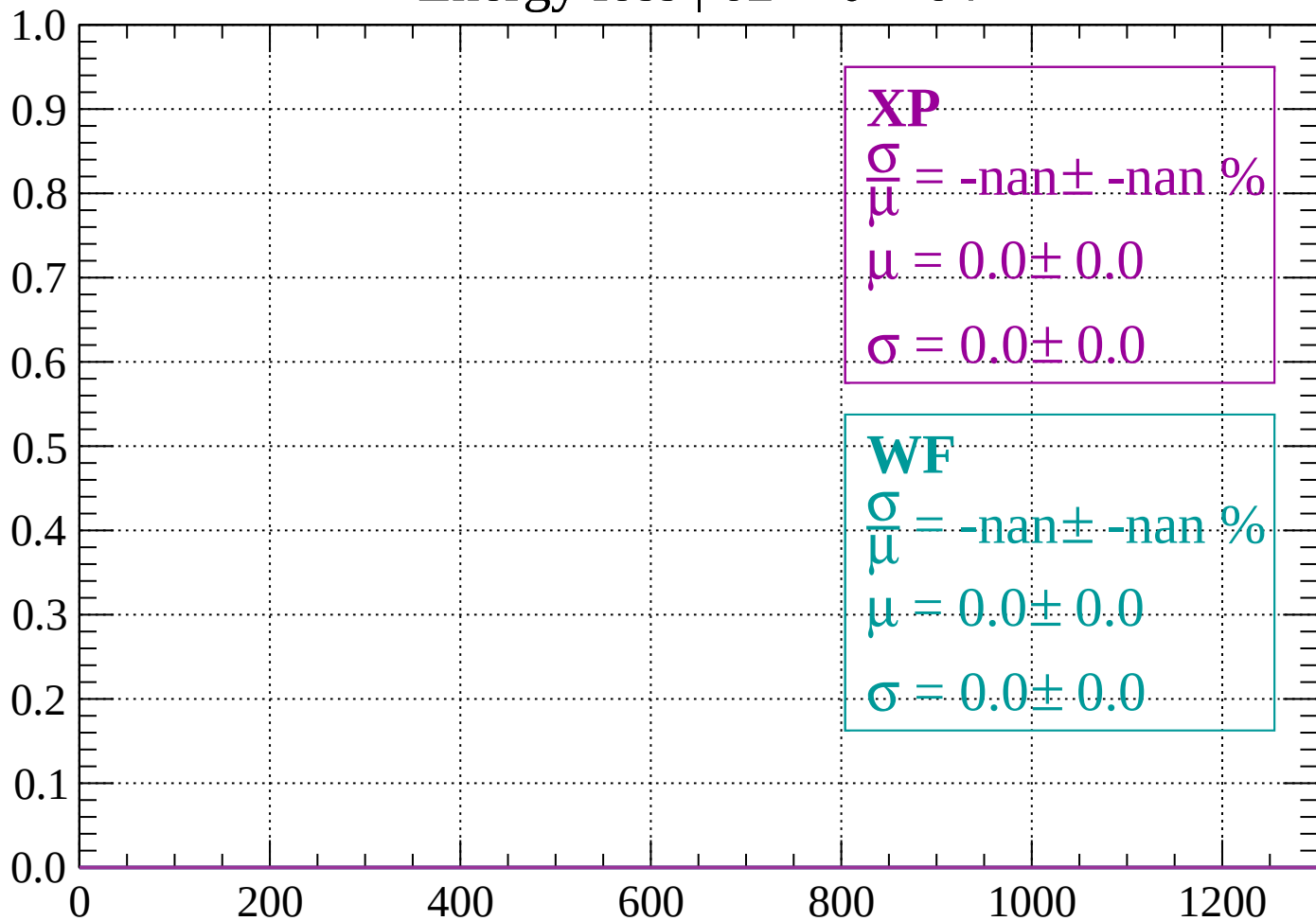
Count



dE/dx (ADC counts/cm)

Energy loss | $62 < \theta < 64$

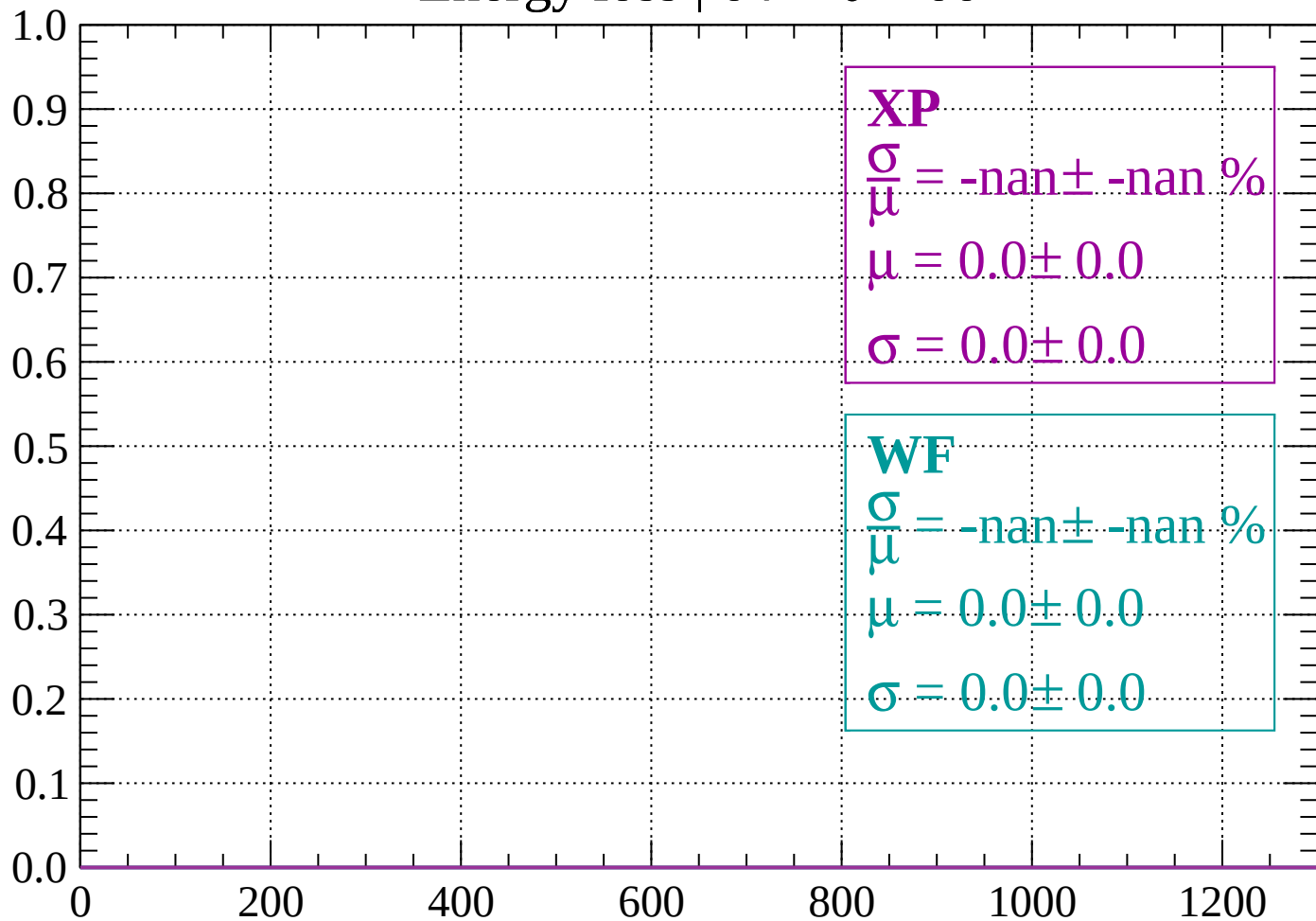
Count



dE/dx (ADC counts/cm)

Energy loss | $64 < \theta < 66$

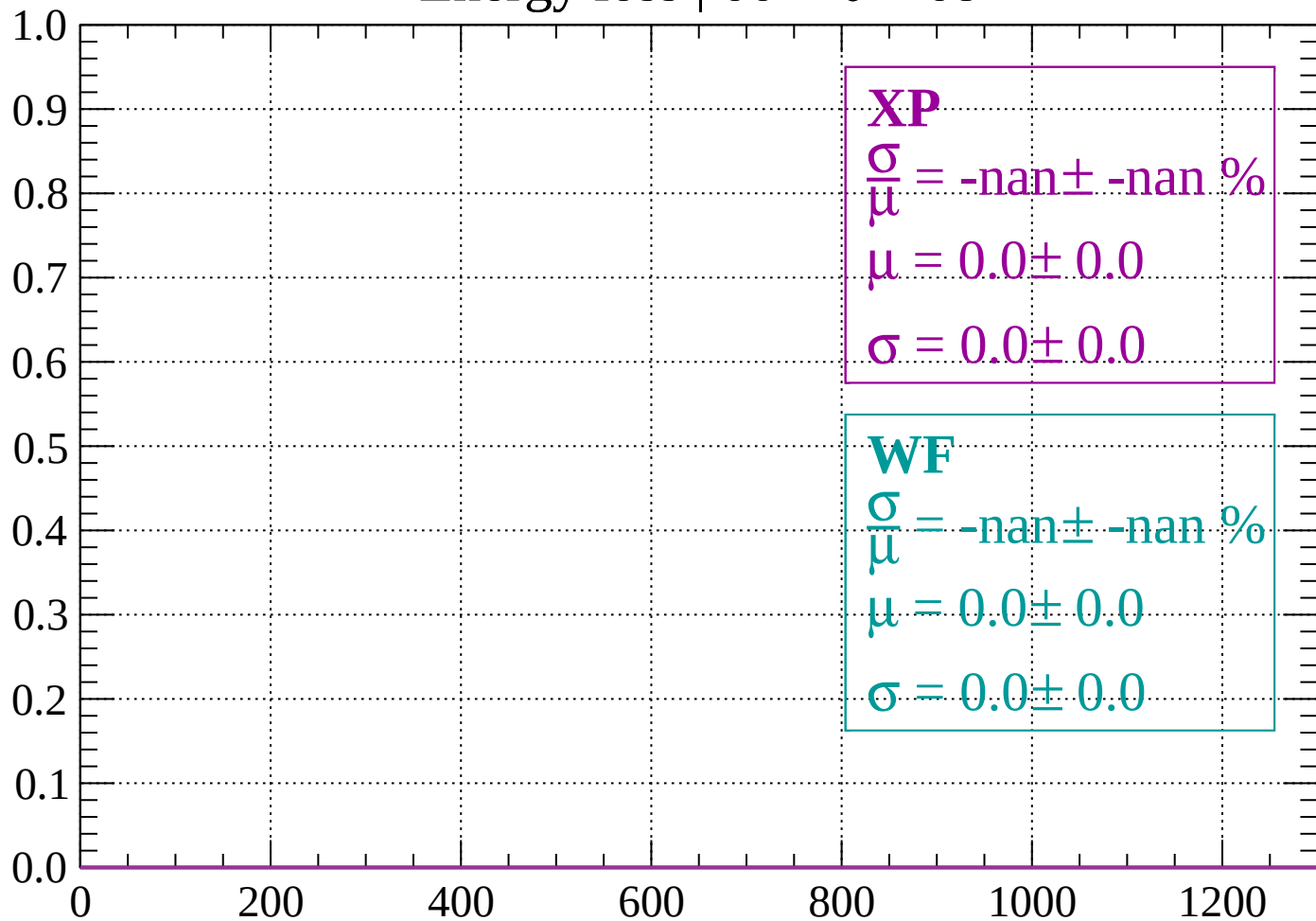
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 68$

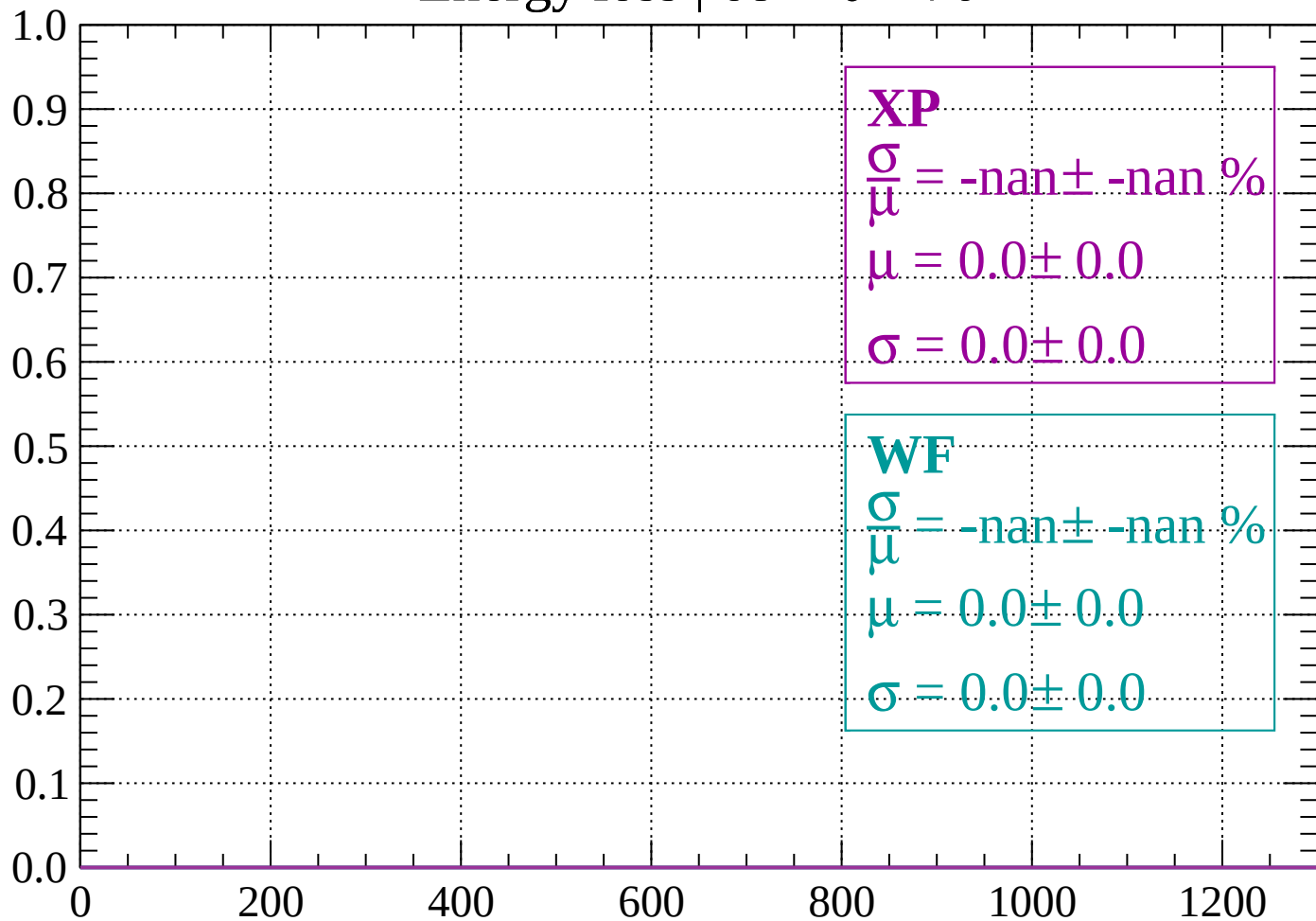
Count



dE/dx (ADC counts/cm)

Energy loss | $68 < \theta < 70$

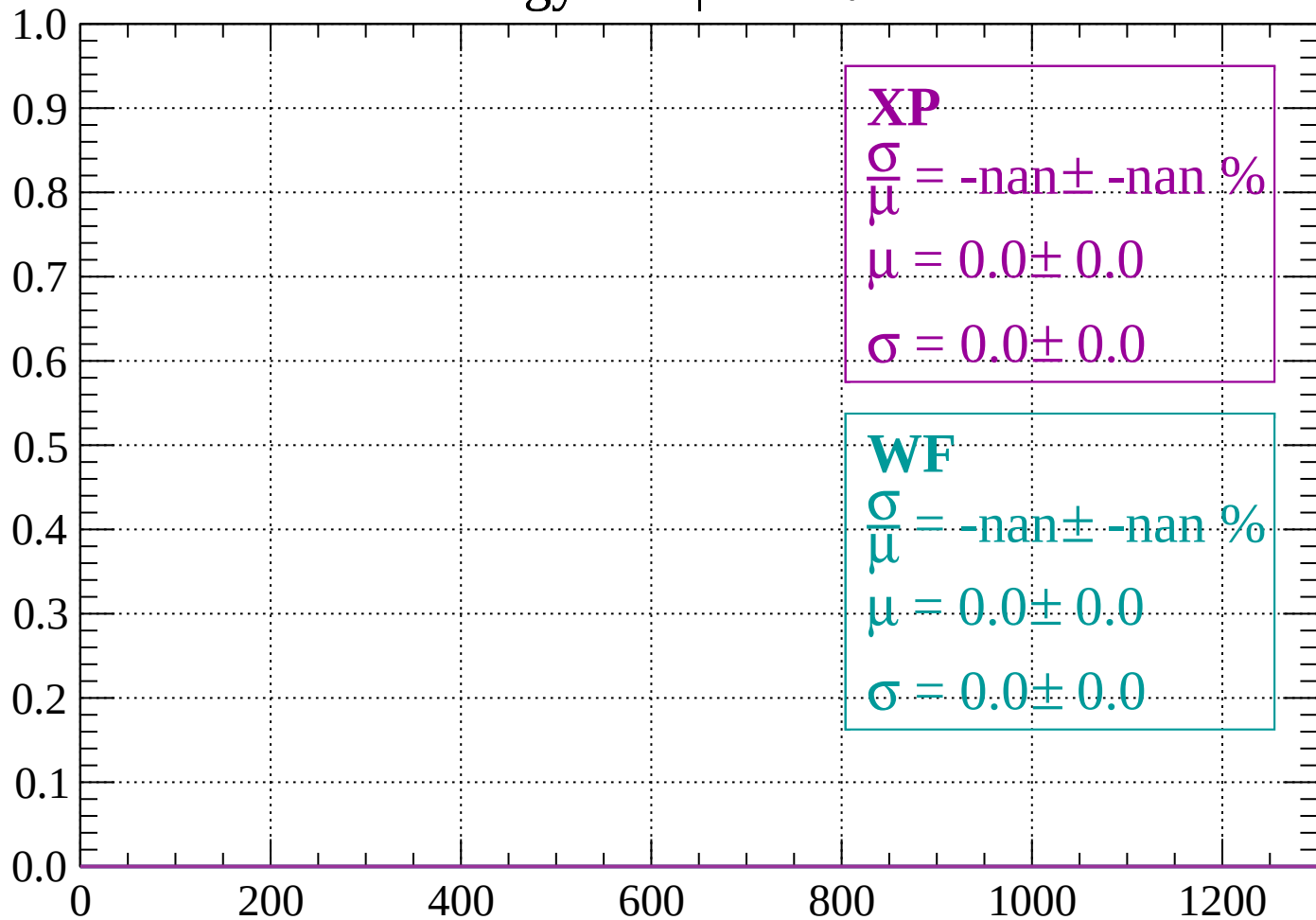
Count



dE/dx (ADC counts/cm)

Energy loss | $70 < \theta < 72$

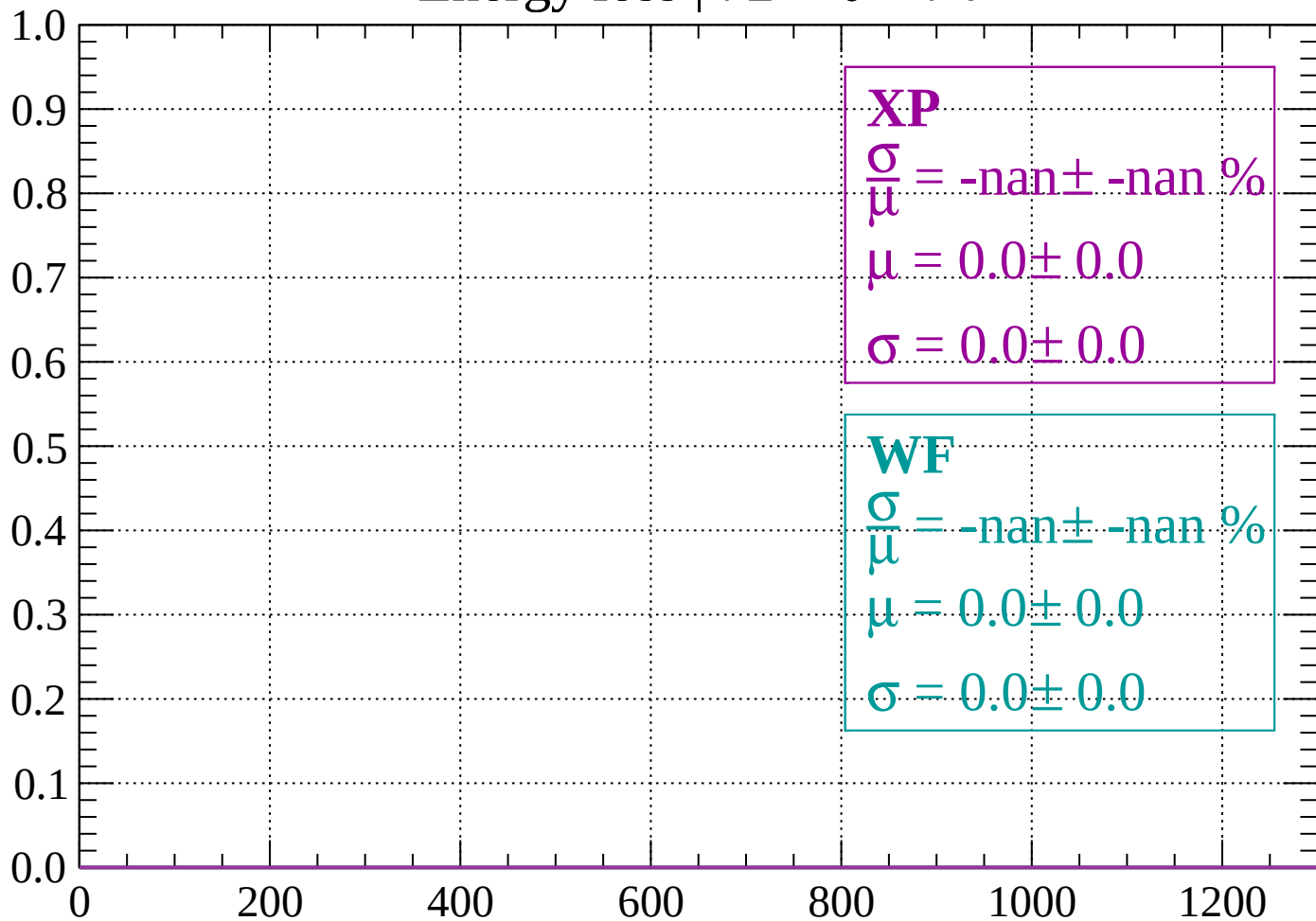
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 74$

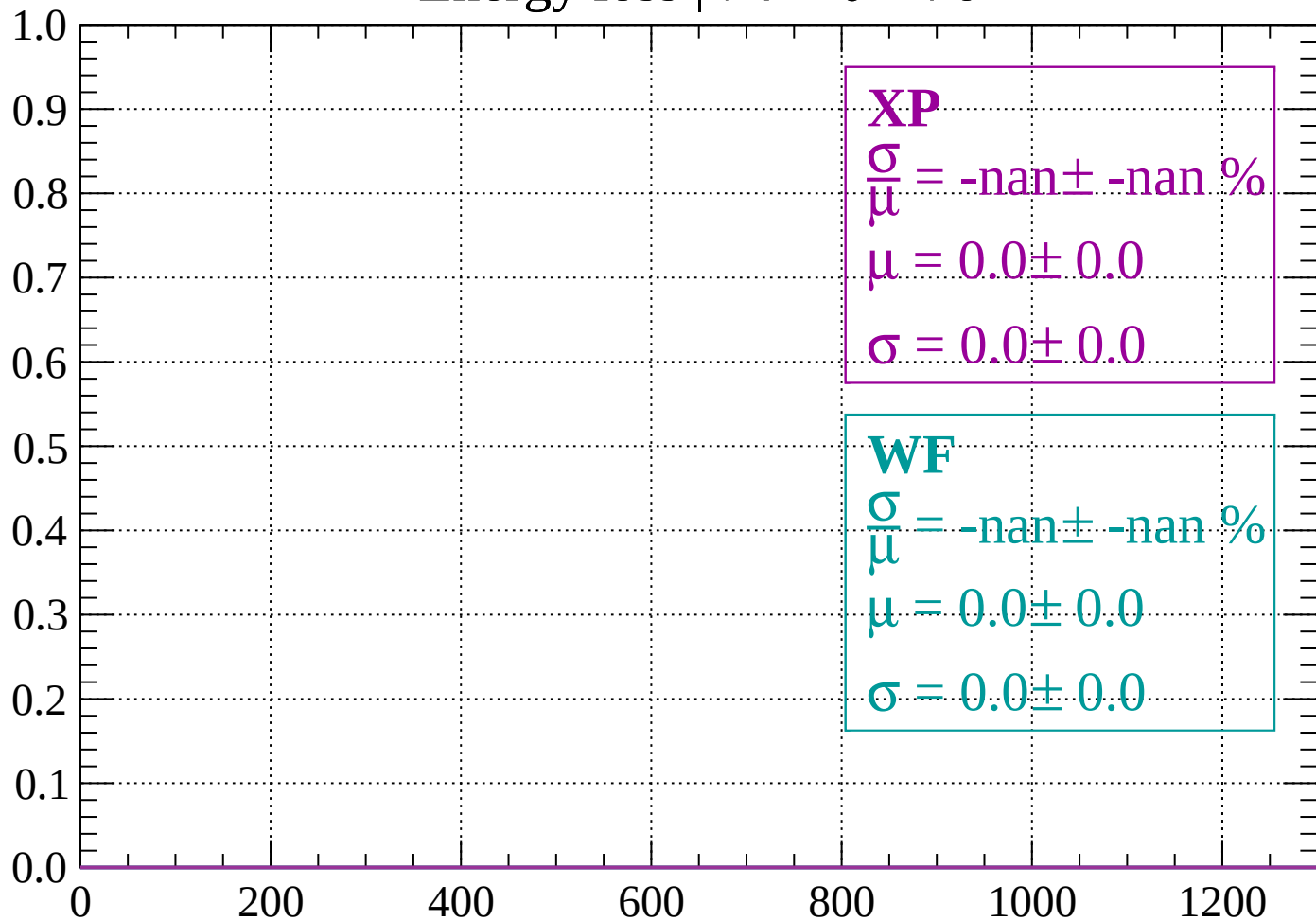
Count



dE/dx (ADC counts/cm)

Energy loss | $74 < \theta < 76$

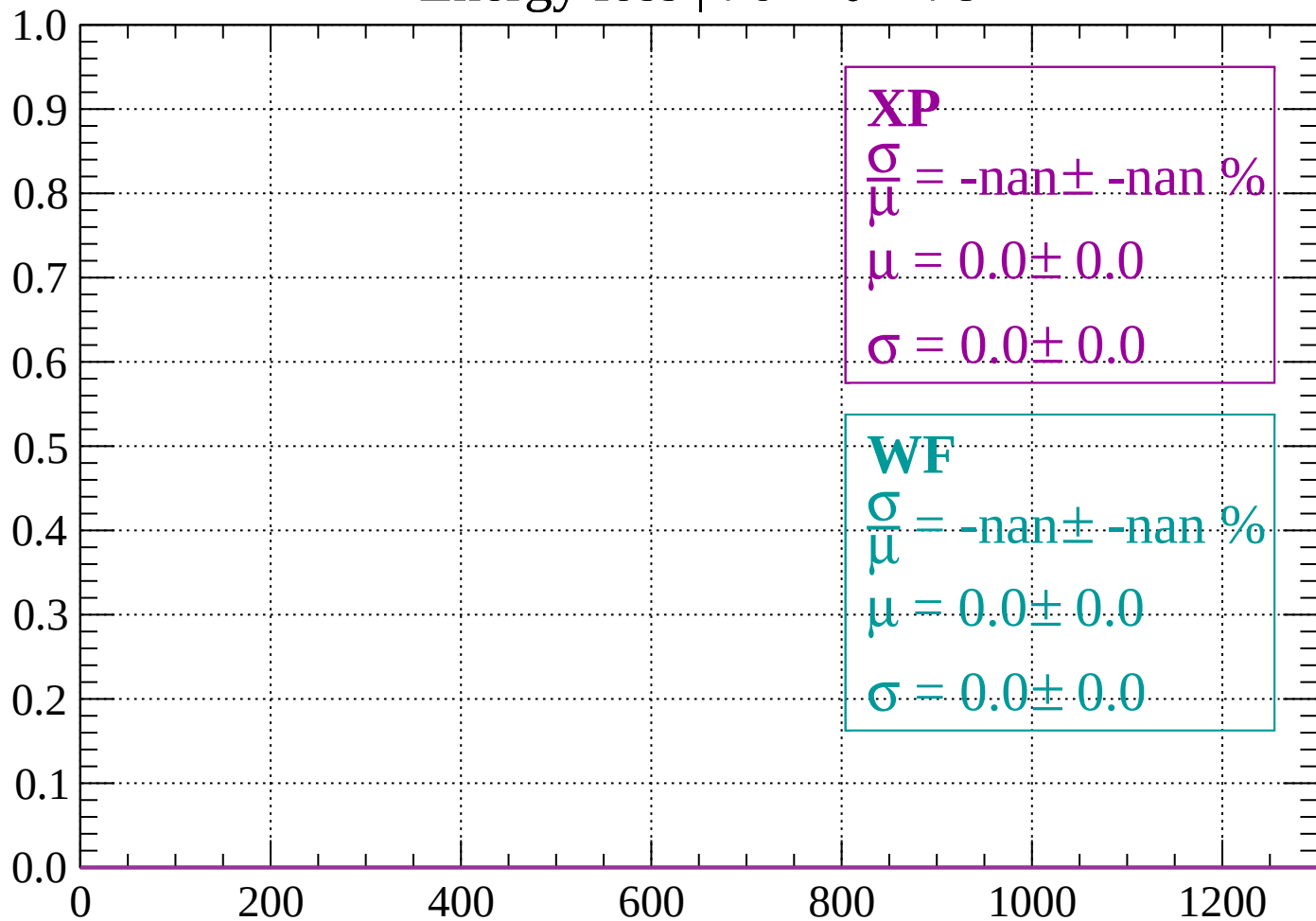
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

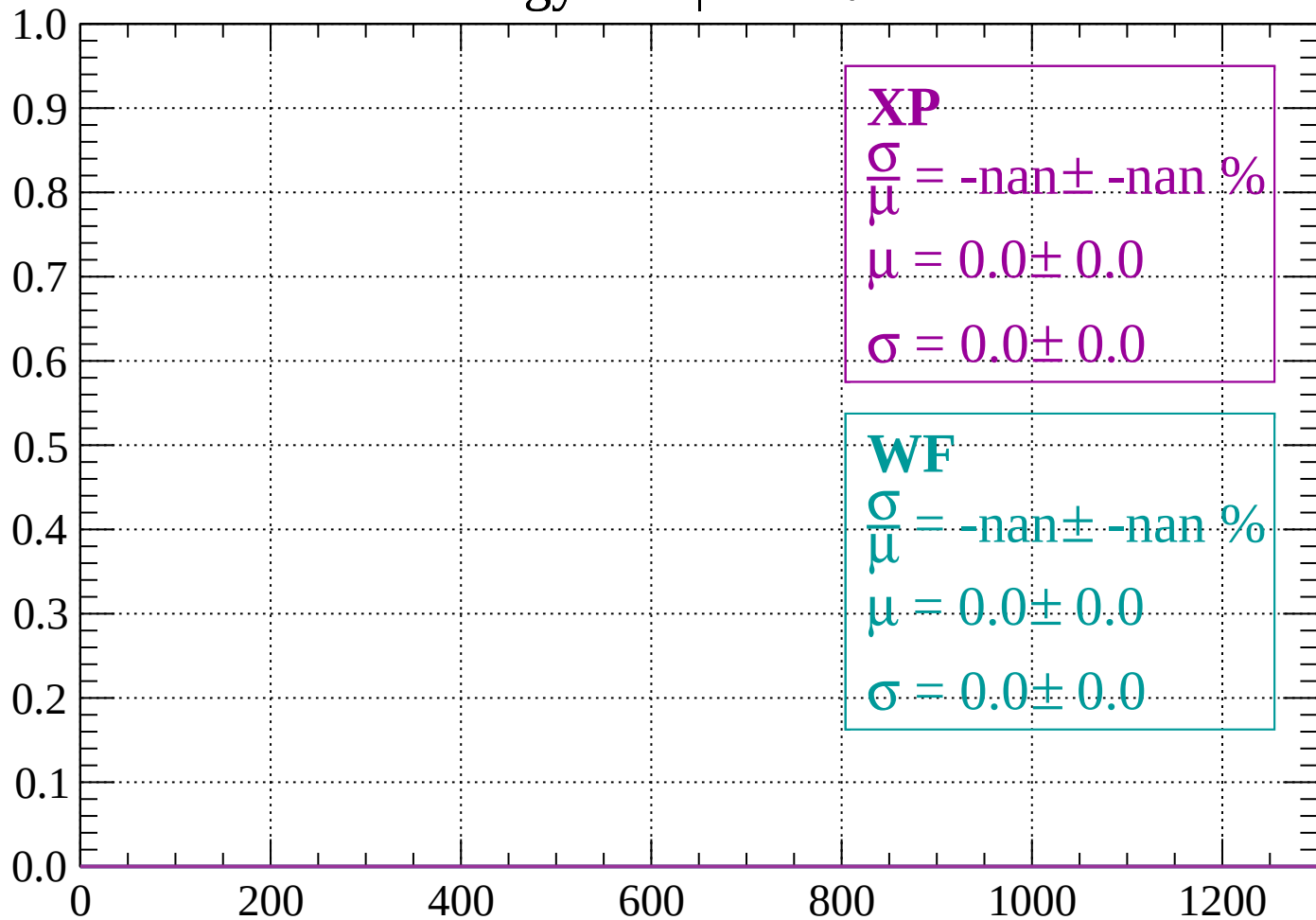
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

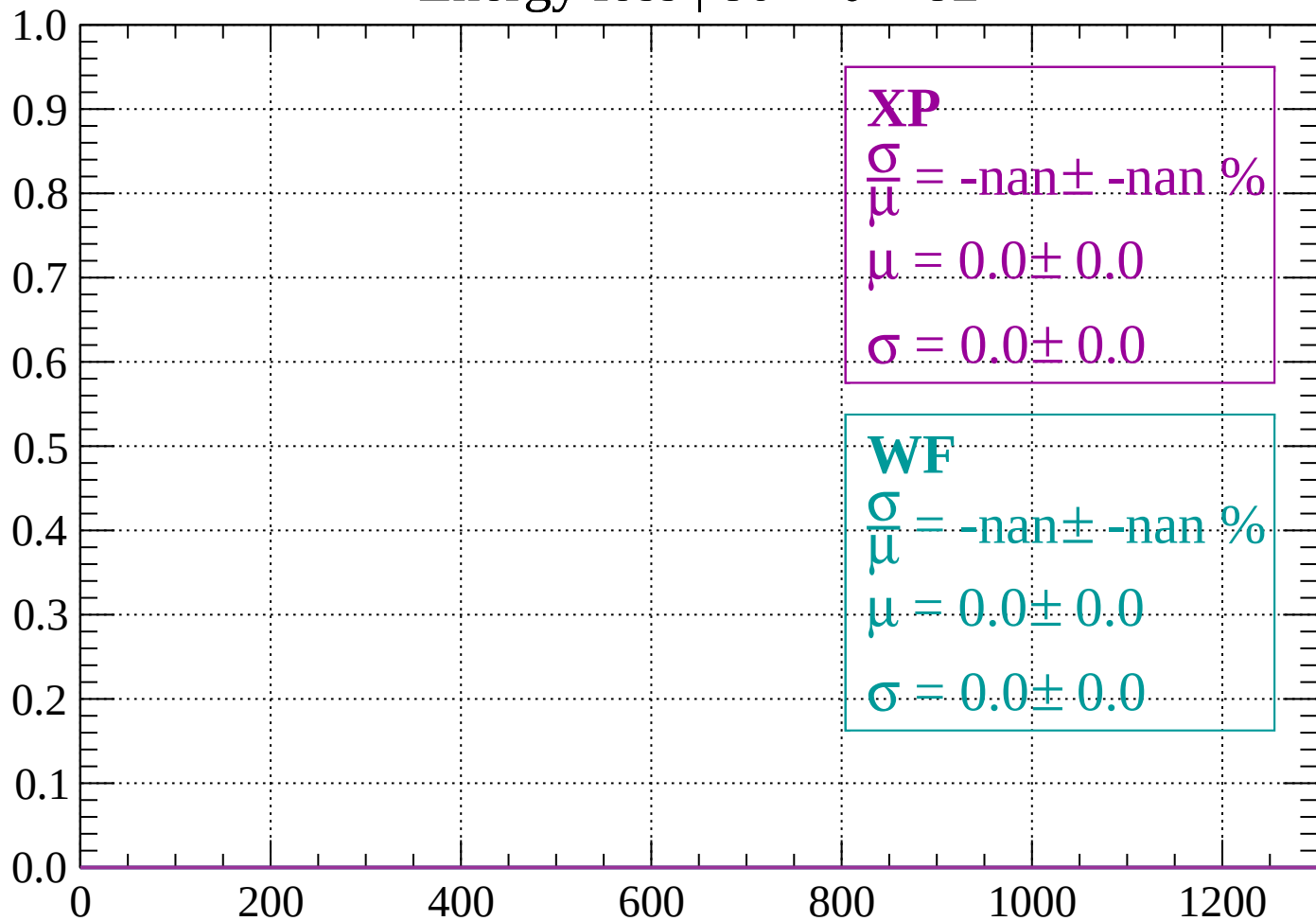
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

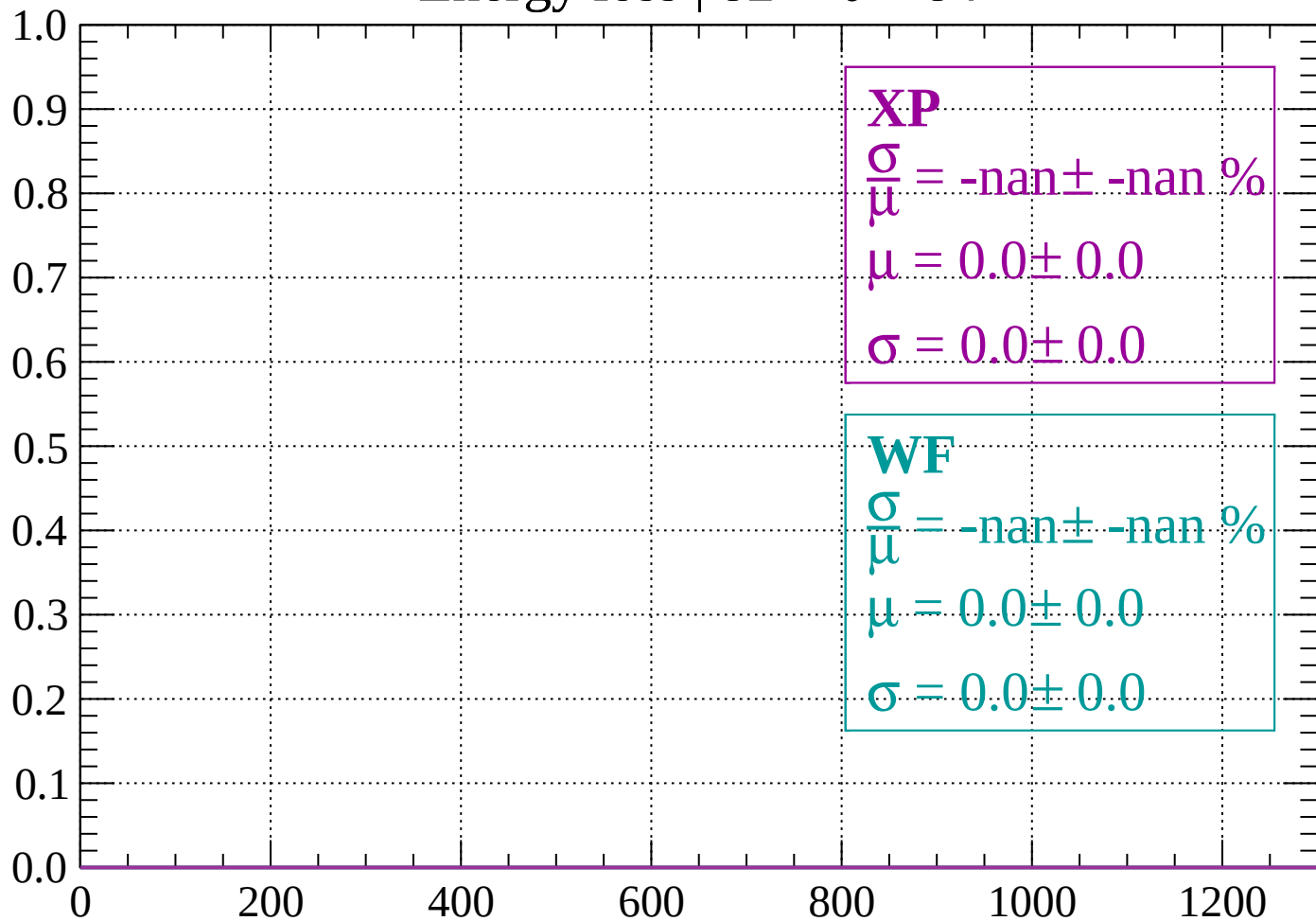
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

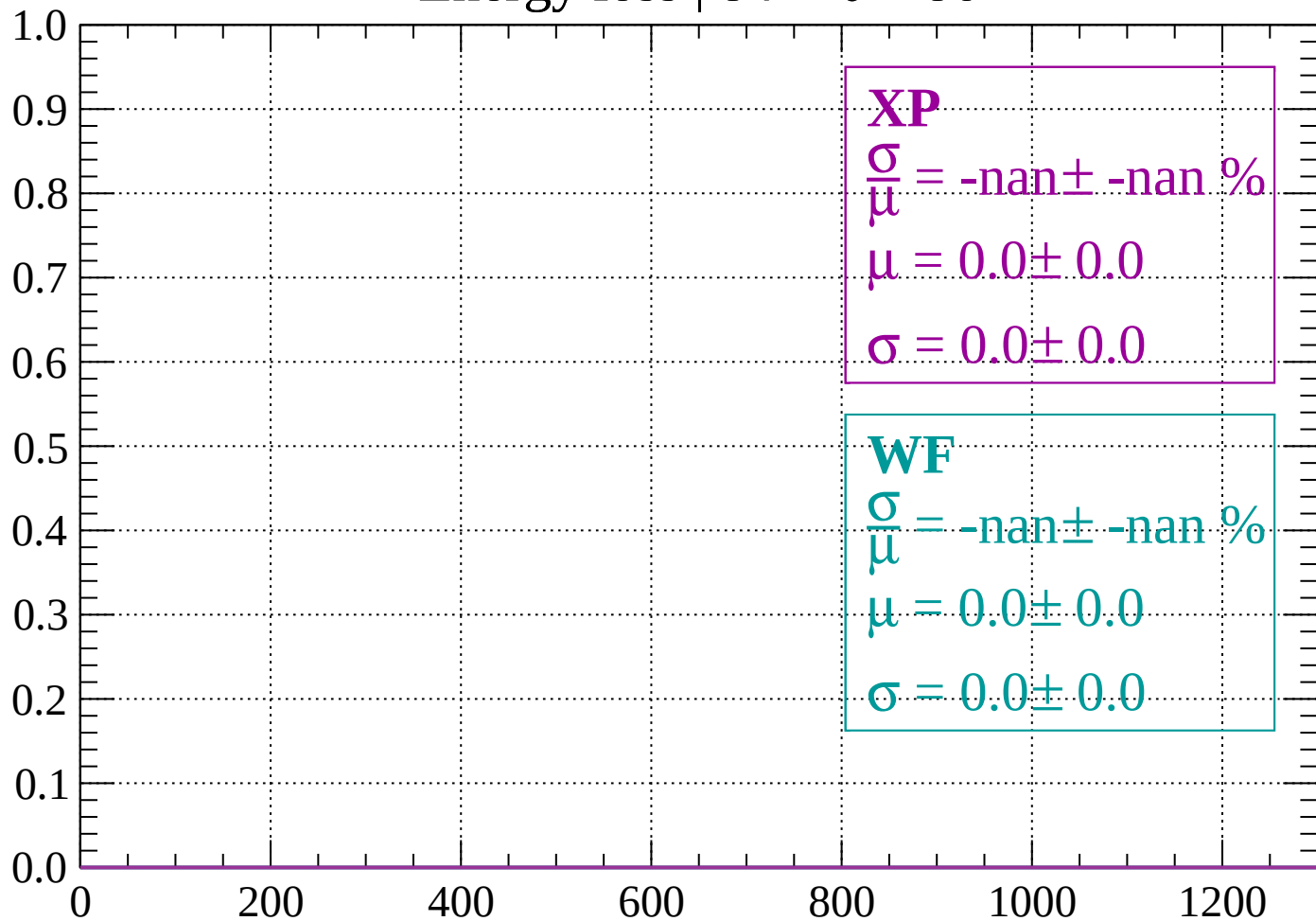
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

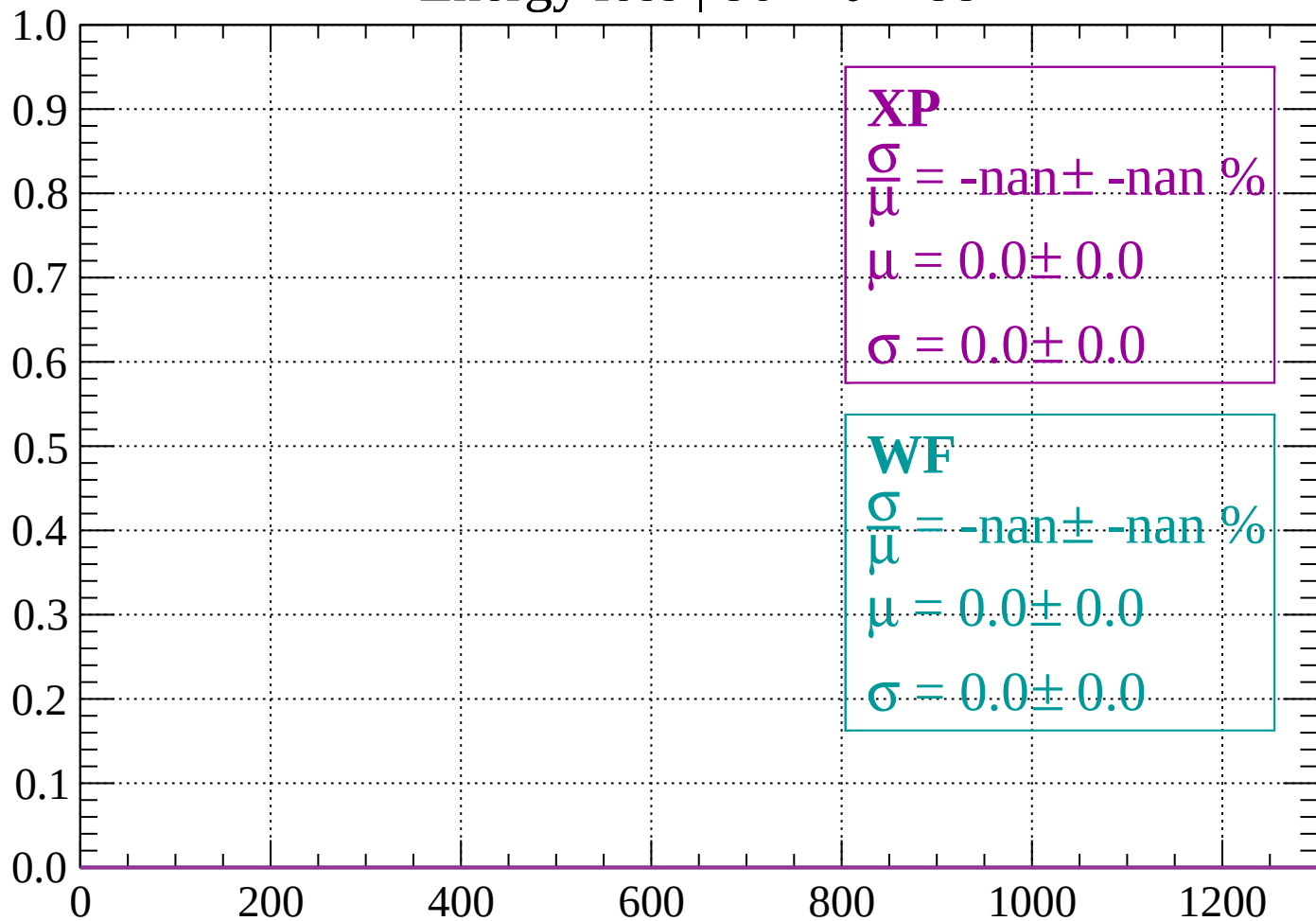
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

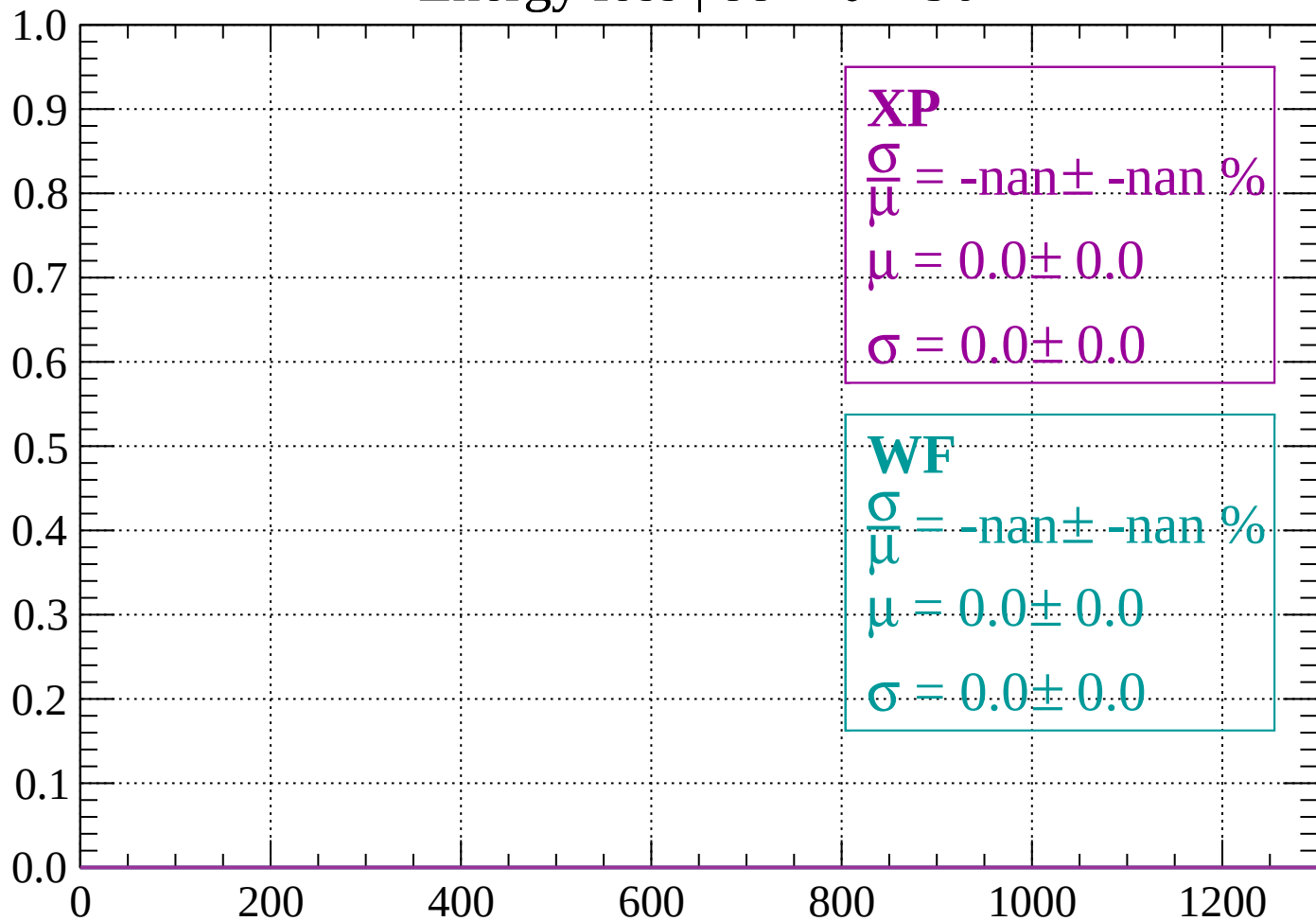
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

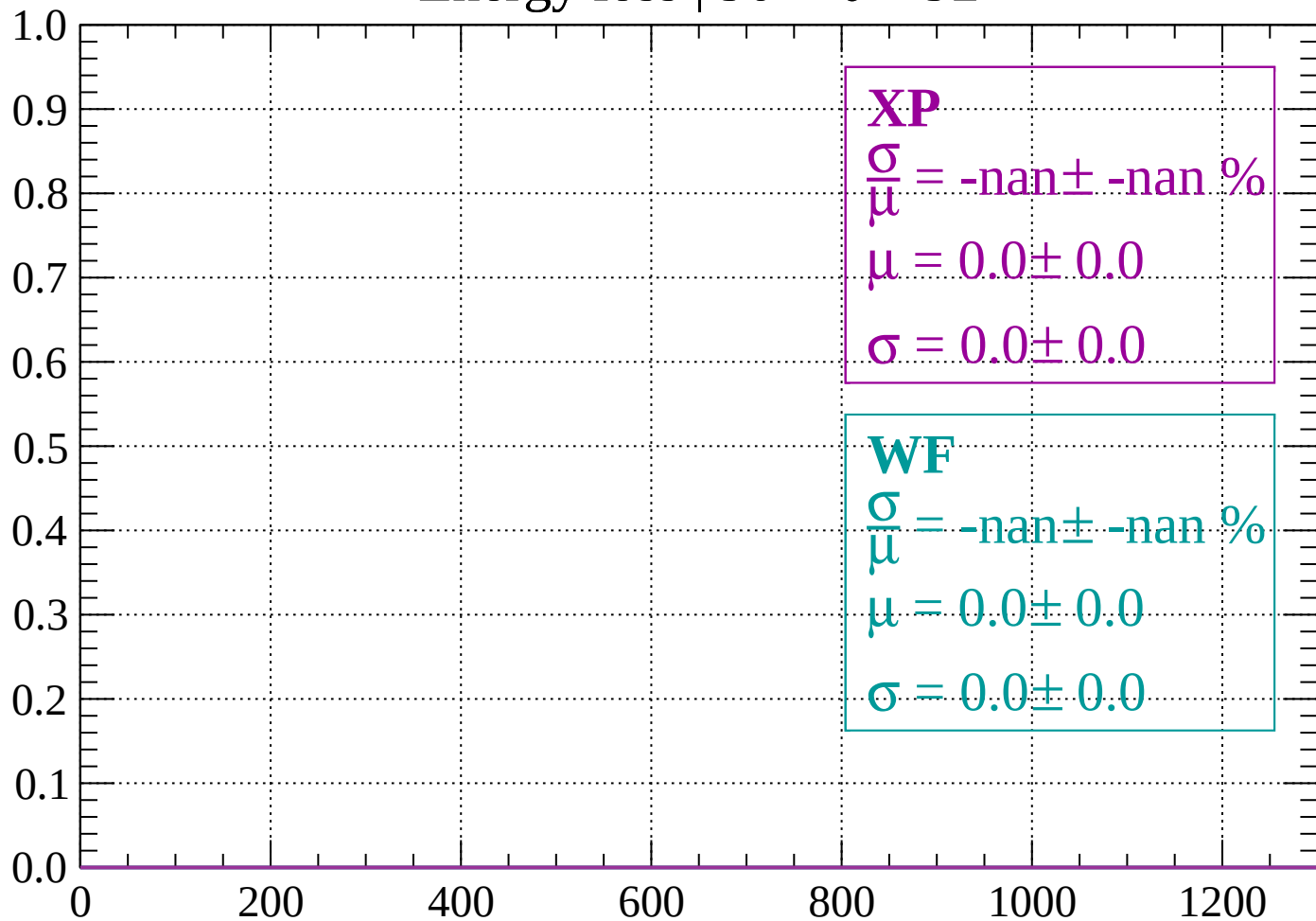
Count



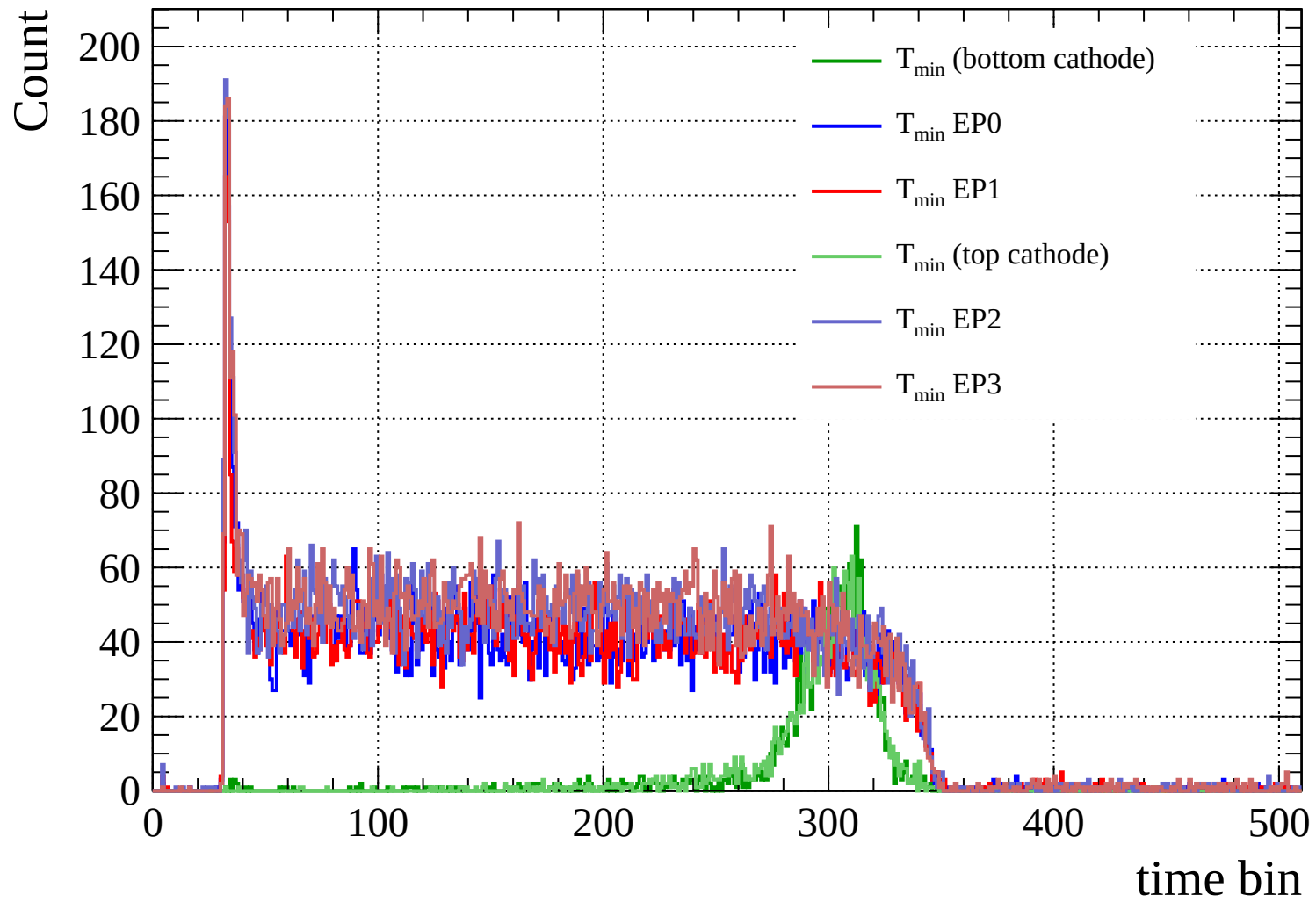
dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)



Count

